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BASIC ELECTRICAL ENGINEERING

BACHELOR OF ENGINEERING

POKHARA & PURBANCHAL UNIVERSITY

Civil, Computer, Electrical and Electronics
Electronics and Communication, I.T., Biomedical

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CHAPTER 1

INTRODUCTION

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1.1 ROLE OF ELECTRICITY IN MODERN SOCIETY 1
1.2 ENERGY SOURCES AND PRODUCTION 2
1.3 GENERATION, TRANSMISSION AND DISTRIBUTION OF ELECTRICAL ENERGY 3

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1.1 ROLE OF ELECTRICITY IN MODERN SOCIETY

Electricity is one of the most important blessing that science has given to mankind. It has also become a part of modern life and one cannot think of a world without it. Electricity has many uses in our daily life. It is used for lighting rooms, working fans and domestic appliances like using mobile charger, A/C, electric heater, television etc. All these provide comfort to people. In factories, large machine are worked with the help of electricity. Essential items like food, cloth, paper and many other things are the product of electricity.

Modern means of transportation and communication have been revolutionized by it. Electric trains and battery cars are quick means of travel. Electricity also provides means of amusement, radio, television and cinema, which are the most popular forms of entertainment are the result of electricity. Modern equipment like computers and robots have also been developed because of electricity. The modern age is the age of machinery. The true object of substituting human labour by mechanical

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labour is to find greater leisure for man. Machines must be driven by natural power and the most pervasive of all sources of natural power is electricity, mechanical, hydro, wind followed by electronic now- a- days. Electricity is something that people cannot live without in modern days. Without it, life will be so much difficult and slow. People need to learn how to value electricity and learn how to produce it from renewable sources. Electrical power is the backbone of modern industrial society.

Electricity has changed a way of farming. In fact, it has mechanized farming. It has enabled as to distribute the water of the river to canals and irrigate dry and barren lands and also used in cutting of crops in modern way. It has also allowed to become healthier without electricity, hospital would have significantly less medical equipment available to help people with medical problems. Hence, it saves lives and allows people to live longer. Electricity has an enormous impact on most lives. Its absence threatens the stability of even the most advanced civilizations.

Hence, the greatest scientific achievement of the 19th century is the discovery of electricity. The 20th century is making use of electricity and is so extensive that it has almost changed the face of earth in this century.

1.2 ENERGY SOURCES AND PRODUCTION

Energy is the capacity of a physical system to perform work. Energy exist in several forms such as heat, kinetic, mechanical energy, potential energy, electrical energy or other forms. Energy sources is classified as Renewable and non-renewable sources.

Renewable energy is derived from natural processes that are replenished constantly such as solar, wind, ocean, hydropower, geothermal resources etc.

A non-renewable resource is a resource that does not renew itself at a sufficient rate for sustainable economic extraction in meaningful human time-frames.

Difference between renewable and non-renewable energy sources:

SN	Renewable energy sources	Non-renewable energy sources
i)	These resources are renewed or replenished by nature in a short span of time.	These resources may not be replenished by nature or take very long time to be formed again.
ii)	Renewable resources can be divided into continuous or flow.	Non-renewable resources can be divided into recyclable or non-recyclable.
iii)	*Example*: Solar energy, wind energy, biomass, hydropower energy etc.	*Example*: Fossils fuels like coal and petroleum.
iv)	It has low carbon emission and hence environment friendly.	It has high carbon emission and hence not environment friendly.
v)	Low cost	High cost
vi)	These are the free gifts of nature.	These are not the free gifts of nature.
vii)	Renewable resources are inexhaustible.	Non-renewable resources are exhaustible.

Electricity production by energy sources – 2015

Oil – 0.3%
Coal – 8.3%
Natural gas – 7.6%
Nuclear power – 33.7%
Waste fuel – 1.2%
Bio-mass – 16.2%
Hydro power – 25.1%
Wind power – 3.5%
Peat – 4.1%

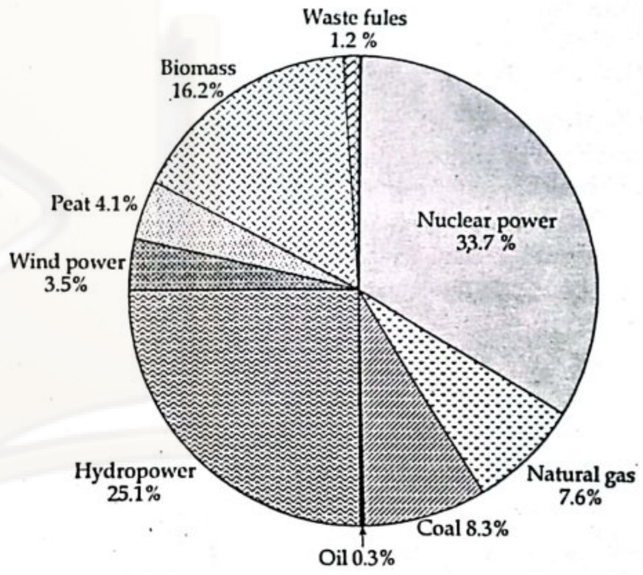

Fig: Pie-chart of electricity production in 2015

1.3 GENERATION, TRANSMISSION AND DISTRIBUTION OF ELECTRICAL ENERGY

The electrical energy is produced at generating stations and through the transmission network, it is transmitted to the consumers. Between the generating stations and the distribution stations, three different levels of

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voltage *i.e.*, transmission, sub-transmission and distribution level of voltage are used. The high voltage is required for long distance transmission and low voltage is required for utility purpose. The voltage level goes on decreasing from the transmission system to the distribution system.

a) Generation

The place where electric power produced by the parallel connected three phase alternators is called generating station (*i.e.*, power plant). The normal generating voltage is 11 KV / 11.5 KV / 12 KV or 13 KV. But, economically it is good to step up the produced voltage from (11 KV / 11.5 KV / 12 KV / 13 KV) to (66 KV / 132 KV / 220 KV) by step up transformer. The generated power is to be transmitted to the locations where it is demanded like industries, cities which are known as load.

b) Transmission

Electric power transmission is the bulk movement of electrical energy from a generating site such as a power plant to an electrical substation. The interconnected lines which facilitate this movement are known as a transmission network. Electric power can be transmitted by underground power cables or overhead power cables.

c) Distribution

Electric power distribution is the final stage in the delivery of electric power; it carries electricity from the transmission system to the individual consumers. Circuit breakers and switchgear enables the substation to be disconnected from the transmission grid or for the distribution lines to be connected. Distribution line generally consists of:

- Feeders
- Distributors
- Service mains

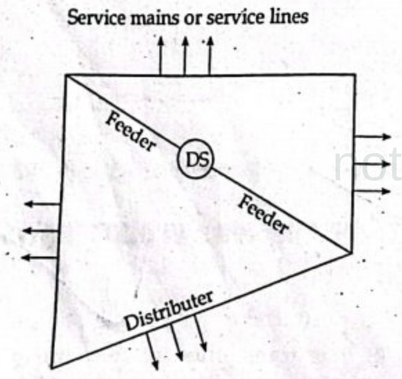

Fig: Elements of a distribution system

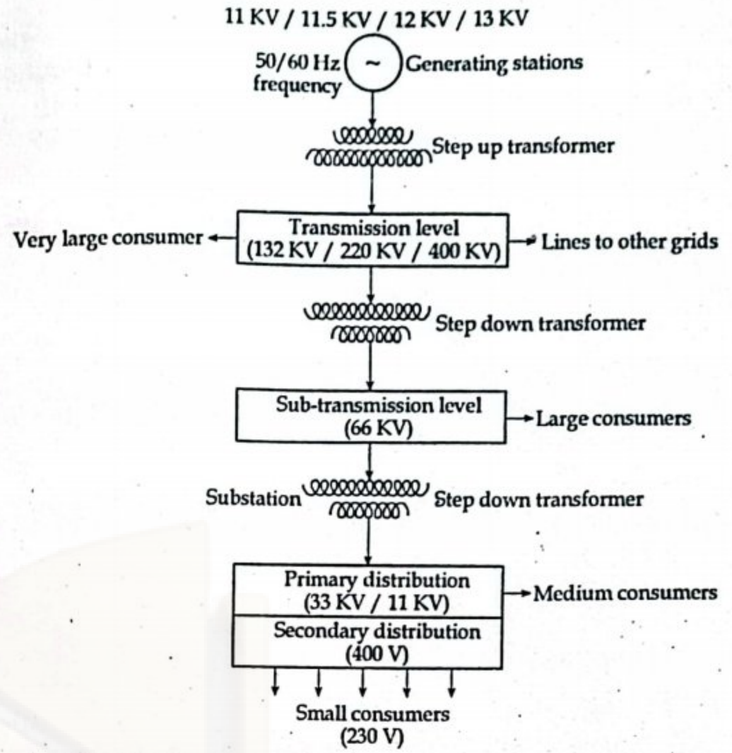

Fig: Typical AC power supply system scheme in single line diagram

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BOARD EXAM SOLVED PROBLEMS

1. Explain in brief about generation, transmission and distribution of electrical power with the help of single line diagram. [12/F, 13/F, 13/S, 15/F, 17/S, 18/S, 19/S]

Solution: See the definition part 1.3.

2. Explain present energy scenario and role of electricity in context of Nepal. [18/F]

Solution: Write yourself regarding "present energy scenario of Nepal" and role of electricity in Nepal.

3. Throw light on role of electricity in modern society. [14/S, 15/S, 16/S, 17/F]

Solution: See the definition part 1.1.

4. Explain why transformer is needed to transfer power over long distance? [13/S]

Solution:

We know;

Power loss = Current × Voltage

$P = I \times V = I \times I \times R$

$\therefore P = I^2R$

Transformer is needed to transfer power over long distance because the transformer boost the low voltage to high voltage and thus less current flows which ultimately reduce the power loss in transmission system. Hence, to reduce I^2R losses in long transmission line it is needed to transform low voltage to high voltage which is done by transformer.

5. What are the sources of electricity? [15/S]

Solution:

Sources of electricity are:

i) Fossil fuels [coal, oil, natural gas]
ii) Renewable energy [Hydroelectric, wind, solar, geothermal]
iii) Nuclear power
iv) Other sources [Bio fuels and Biomass]

6. Write short notes on generation, transmission and distribution of electricity. [19/F]

***Solution:** See the definition part 1.3.*

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CHAPTER 2

DC CIRCUIT ANALYSIS

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2.1 CIRCUITS CONCEPT (LUMPED AND DISTRIBUTED PARAMETERS)

If the component used in system has identical values of physical parameters (current, voltage etc) throughout its area and can be considered as a single point (node) in the system, then it is called as lumped parameter system.

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For example: Normal components like resistor, capacitor in low frequency applications etc.

If the component used in system has different values of physical parameters (current, voltages etc) throughout its area and cannot be considered as a single point (node) in the system, then it is called as distributed parameter system.

For example: Transmission lines, microwave tubes which normally used in high frequency system.

Lumped circuit analysis is substantially simpler than distributed circuit analysis.

2.2 LINEAR AND NON-LINEAR PARAMETER

Linear parameter-Resistor, Inductor and capacitor

Non-linear parameter-diode, transistor, inductor (when the core is saturated)

Linear elements are the elements in which the constituent relation, the relation between voltage and current, is a linear function. They obey the superposition principle.

Non-linear elements are the elements in which the relation between voltage and current is a non-linear function. They does not obey the superposition principle.

2.3 ACTIVE AND PASSIVE CIRCUITS

Active circuits are circuits with energy supplying components in them whereas passive circuit can only consume energy. Active devices are incapable of providing power gain. Active devices can control the energy flow within the circuit whereas passive device cannot control it.

Example of active components: diode, transistors.

Example of passive components: resistors, capacitors, inductors and transformers.

2.4 CIRCUIT ELEMENTS (RESISTANCE, INDUCTANCE AND CAPACITANCE)

a) Resistance is the property of a material by virtue of which it opposes the flow of electrons or current through it. The unit of resistance is ohm (Ω). The resistance of a conducting wire is found to:

i) Be directly proportional to its length (L),
ii) Inversely proportional to its cross section area (a)
iii) Depends upon temperature
iv) Depends upon the nature of the conducting material.

i.e., $R \propto \frac{L}{a}$ $\qquad R = \rho \cdot \frac{L}{a}$

where, ρ (rho) is a constant of material called specific resistance or resistivity. The unit of resistivity is ohm-meter.

$$\rho = \frac{R \cdot a}{L} = \text{Ohm} \times \frac{\text{Meter}^2}{\text{Meter}} = \text{Ohm - Meter}$$

Reciprocal of resistance is called conductance (G). Thus,

$$G = \frac{1}{R} = \frac{\rho \cdot a}{L}$$

and, Its unit is ohm^{-1} or mho.

b) Capacitance is the ratio of the change in an electric charge in a system to the corresponding change in its electric potential. The SI unit of capacitance is Farad (F).

$$C = \frac{dQ}{dV}$$

where, dQ = Change in an electric charge, C
dV = Change in it's electric potential, V

Also, Q = CV

and, $V = \frac{Q}{C}$

c) Inductance is the property of an electrical conductor by which a change in electric current through it induces an electromotive force (emf). The SI unit of inductance is Henry.

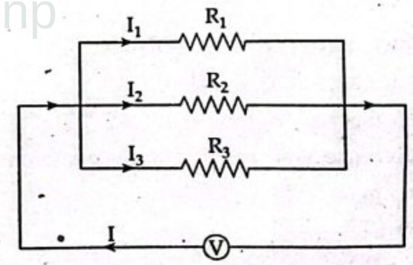
$$V = L \cdot \frac{di}{dt}$$

where, V = Instantaneous voltage across the inductor
L = Inductance
$\frac{di}{dt}$ = Instantaneous rate of current change.

2.5 SERIES AND PARALLEL COMBINATION OF RESISTANCE

a) Combination of resistor in parallel

Consider three resistors R_1, R_2, R_3 which are connected in parallel. The charge splits into three branch and flows through R_1, R_2 and R_3.



Current, $I = I_1 + I_2 + I_3$

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Potential difference applied to $R_1 = V = I_1R_1$

Potential difference applied to $R_2 = V = I_2R_2$

Potential difference applied to $R_3 = V = I_3R_3$.

Thus,

$$I = I_1 + I_2 + I_3$$
$$= \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$
$$= V\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)$$

If this parallel combination is replaced by an equivalent resistance, R_{eq}, then,

$$I = \frac{V}{R_{eq}}$$

or, $$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

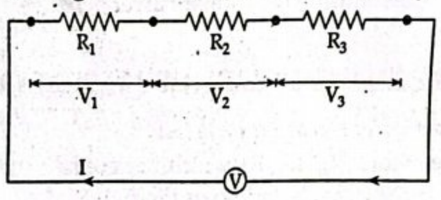
Thus, for n-number of resistors in parallel,

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots\ldots\ldots + \frac{1}{R_n}$$

So, in a parallel connection, the total resistance of a circuit is determined by adding the reciprocal of the resistance. In parallel combination, the total resistance or the equivalent resistance is always less than the smallest resistor present in the circuit.

b) Combination of resistors in series

Consider three resistor R_1, R_2, R_3 which are connected in series. Here the charge first flows through R_1 and enters to R_2 and finally reaches R_3.



By ohm's law,

Potential difference across $R_1 = V_1 = IR_1$

Potential difference across $R_2 = V_2 = IR_2$

Potential difference across $R_3 = V_3 = IR_3$

Thus, the potential difference V across this series connection of resistors is,

$$V = V_1 + V_2 + V_3$$
$$= IR_1 + IR_2 + IR_3$$
$$= I(R_1 + R_2 + R_3)$$

Hence, in case of series connection, the equivalent resistance, $R_{eq} = \frac{V}{I}$

$\therefore \quad R_{eq} = R_1 + R_2 + R_3$

For n number of resistors, connected in series, the equivalent resistance $R_{eq} = R_1 + R_2 + R_3 + \ldots\ldots\ldots + R_n$. Thus, in a series connection, the total resistance of a circuit is determined by adding the resistance of each individual resistors.

2.6 STAR-DELTA TRANSFORMATION

It is a method of obtaining equivalent delta of a star connection.

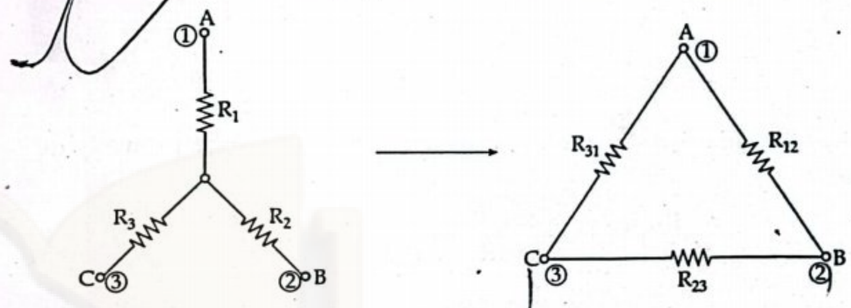


Fig: Star-delta transformation.

Consider a star circuit ABC made up of three resistors R_1, R_2 and R_3 as shown in figure. Suppose its equivalent delta circuit be represented by R_{12}, R_{13} and R_{31}. Total resistance between B and C must be same for both star and delta connection so,

$$R_2 + R_3 = R_{23} \parallel (R_{31} + R_{12}) = \frac{R_{23} \times R_{31} + R_{23} \times R_{12}}{R_{23} + R_{31} + R_{12}} \quad (1)$$

Similarly, we get;

$$R_3 + R_1 = R_{31} \parallel (R_{12} + R_{23}) = \frac{R_{31} \times R_{12} + R_{31} \times R_{23}}{R_{31} + R_{12} + R_{23}} \quad (2)$$

and, $$R_1 + R_2 = R_{12} \parallel (R_{31} + R_{23}) = \frac{R_{12} \times R_{31} + R_{12} \times R_{23}}{R_{12} + R_{31} + R_{23}} \quad (3)$$

Adding (1), (2) and (3); we get,

$$2(R_1 + R_2 + R_3) = \frac{2(R_{23} \times R_{31} + R_{31} \times R_{12} + R_{12} \times R_{23})}{R_{12} + R_{23} + R_{31}}$$

or, $$R_1 + R_2 + R_3 = \frac{R_{23} \times R_{31} + R_{31} \times R_{12} + R_{12} \times R_{23}}{R_{12} + R_{23} + R_{31}}$$

Subtracting (1), (2) and (3) from (4) turn by turn,

$$\therefore \quad R_1 = \frac{R_{31} \times R_{12}}{R_{12} + R_{23} + R_{31}} \quad (5)$$

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$$\therefore \quad R_2 = \frac{R_{12} \times R_{23}}{R_{12} + R_{23} + R_{31}} \tag{6}$$

$$\therefore \quad R_3 = \frac{R_{23} \times R_{31}}{R_{12} + R_{23} + R_{31}} \tag{7}$$

Dividing (5) by (6);

$$\frac{R_1}{R_2} = \frac{R_{31}}{R_{23}}$$

$$\text{or,} \quad R_{31} = R_1 \times \frac{R_{23}}{R_2} \tag{8}$$

Similarly, we get

$$\frac{R_3}{R_1} = \frac{R_{23}}{R_{12}}$$

$$\text{or,} \quad R_{12} = R_1 \times \frac{R_{23}}{R_3} \tag{9}$$

Putting the values of R_{31}, and R_{12} from equation (8) and (9) respectively in equation (1);

$$\therefore \quad R_2 + R_3 = \frac{R_{23}(R_1R_{23}/R_2) + R_{23}(R_1R_{23}/R_3)}{R_{23} + (R_1R_{23}/R_2) + (R_1R_{23}/R_3)}$$

$$\text{or,} \quad R_2 + R_3 = \frac{(R_1R_{23}/R_2) + (R_1R_{23}/R_3)}{1 + (R_1/R_2) + (R_1/R_3)}$$

$$\text{or,} \quad R_2 + R_3 = \frac{R_1R_3R_{23} + R_1R_2R_{23}}{R_2R_3 + R_1R_3 + R_1R_2}$$

$$\text{or,} \quad R_2 + R_3 = \frac{R_1R_{23}(R_2 + R_3)}{R_2R_3 + R_1R_3 + R_1R_2}$$

$$\therefore \quad R_1R_{23} = R_2R_3 + R_1R_3 + R_1R_2$$

$$\text{or,} \quad R_{23} = \frac{R_2R_3 + R_1R_3 + R_1R_2}{R_1}$$

$$\therefore \quad R_{23} = R_2 + R_3 + \frac{R_2R_3}{R_1}$$

Similarly,

$$\therefore \quad R_{31} = R_3 + R_1 + \frac{R_3R_1}{R_2}$$

$$\therefore \quad R_{12} = R_1 + R_2 + \frac{R_1R_2}{R_3}$$

Thus, in star-delta transformation, the equivalent delta resistance between any two terminals is the sum of star resistances between the involved terminals plus the ratio of the product of these two star resistance and the remaining star resistance.

2.7 DELTA-STAR TRANSFORMATION

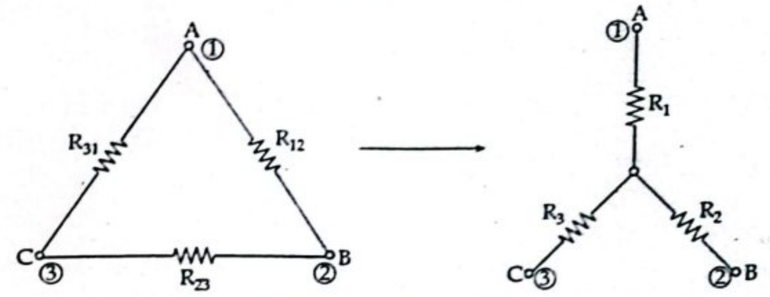

Fig: Delta-star connection

Consider a delta circuit ABC made up of three resistors R_{12}, R_{23} and R_{31} as shown in figure. Its equivalent star circuit be represented by R_1, R_2 and R_3. The resistance between terminals A and B in delta connection is,

$$= \frac{R_{12} \times (R_{23} + R_{31})}{R_{12} + (R_{23} + R_{31})} \tag{1}$$

and, The resistance between the same terminal A and B in the star connection,

$$= R_1 + R_2 \tag{2}$$

Since the terminals are same, so

$$R_1 + R_2 = \frac{R_{12} \times (R_{23} + R_{31})}{R_{12} + (R_{23} + R_{31})} \tag{3}$$

Similarly by solving for terminals B and C and C and A, we get,

$$R_2 + R_3 = \frac{R_{23} \times (R_{31} + R_{12})}{R_{23} + (R_{31} + R_{31})} \tag{4}$$

$$\text{and,} \quad R_3 + R_1 = \frac{R_{31} \times (R_{12} + R_{23})}{R_{31} + (R_{12} + R_{23})} \tag{5}$$

On solving (3), (4) and (5), we get the delta star transformation as;

$$\therefore \quad R_1 = \frac{R_{12} \times R_{31}}{R_{12} + R_{23} + R_{31}}$$

$$\therefore \quad R_2 = \frac{R_{23} \times R_{32}}{R_{12} + R_{23} + R_{31}}$$

$$\therefore \quad R_3 = \frac{R_{31} \times R_{23}}{R_{12} + R_{23} + R_{31}}$$

Thus, in delta- star transformation, the resistance of any arm of the star is equal to the product of the resistances of the two delta sides meeting it divided by the sum of the three delta resistances.

2.7 CONCEPT OF POWER, ENERGY AND ITS CALCULATIONS

Electric power is the rate per unit time at which electrical energy is transferred by an electric circuit. The SI unit of power is the watt, one joule per second.

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$$P = \text{Work done per unit time} = \frac{VQ}{t} = V \cdot I$$

where, Q = Electric charge in coulombs
t = Time in seconds
I = Electric current in amperes
V = Electric potential or voltage in volts

Also,

$$P = V \cdot I = IR \cdot I$$

$$\therefore \quad P = I^2R$$

and, $P = V \cdot I = V \cdot \frac{V}{R} = \frac{V^2}{R}$

Electrical energy is a form of energy resulting from the flow of electric charge. Energy is the ability to do work or apply force to move an object. The unit of energy is Joule.

∴ Electrical energy = Power × Time.

2.8 SHORT AND OPEN CIRCUIT

Any circuit which is not complete is considered as on open circuit. Electricity does not flow from an active energy sources to the desired endpoint of the circuit since the path is not closed one or since the path is open.

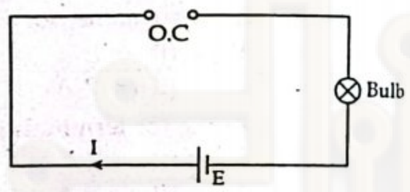


Fig: Open circuit

A short circuit is an electrical circuit that allows a current to travel along an unintended path with no or a very low electrical impedance. This results in an excessive amount of current flowing into the circuit. The electrical opposite of a short circuit is an "open circuit" which is an infinite resistance between two nodes.

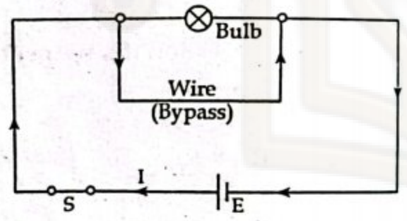
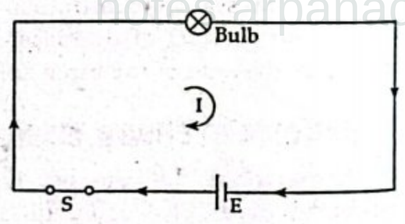


Fig: Short circuit

Closed circuit is an electric circuit providing an uninterrupted endless path for the flow of the current.



A complete circuit which is not performing any actual work can still be considered as a closed circuit. *For example;* a circuit connected to a dead battery may not perform any work, but it is still a closed circuit.

2.9 IDEAL AND NON-IDEAL SOURCES

Ideal sources provides the desired current and voltage regardless of the load connected to it. An ideal voltage source has zero series impedance and no internal volt drop. So, it can deliver constant voltage and infinite current. An ideal current source has no parallel impedance so all current appears at the terminals. So, it can deliver constant current and infinite voltage.

Non-ideal sources has some internal resistance and cannot provide desired current and voltage regardless of the load connected to it.

a) Ideal voltage source have zero internal resistance. Practically an ideal voltage source cannot be obtained.

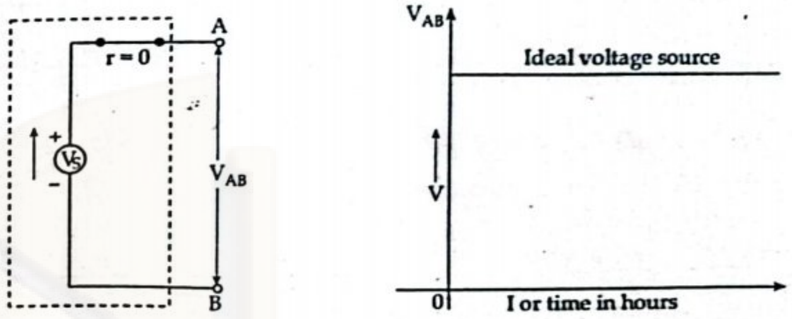


Fig: Ideal voltage source and it's graph.

Sources having some amount of internal resistance are known as practical voltage source. Due to this internal resistance, voltage drop takes place and it causes the terminal voltage to reduce.

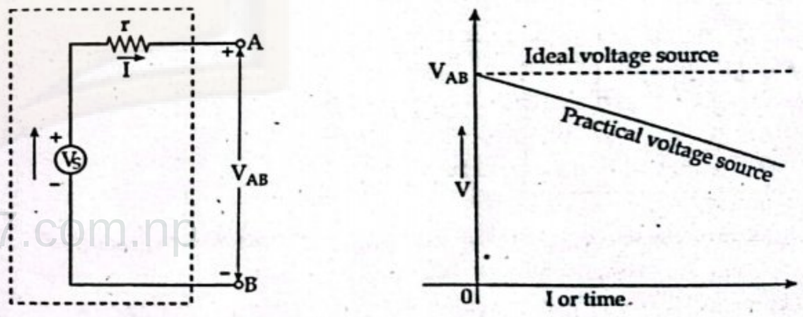


Fig: Practical voltage source and it's graph.

b) The ideal current source have infinite internal resistance. An ideal current source is a two terminal circuit element which supplies the same current to any load resistance connected across it's terminals.

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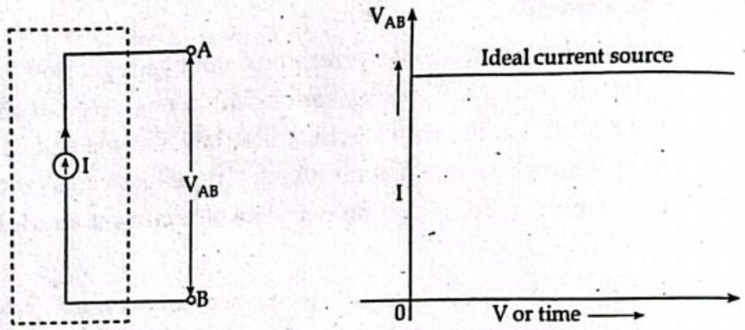
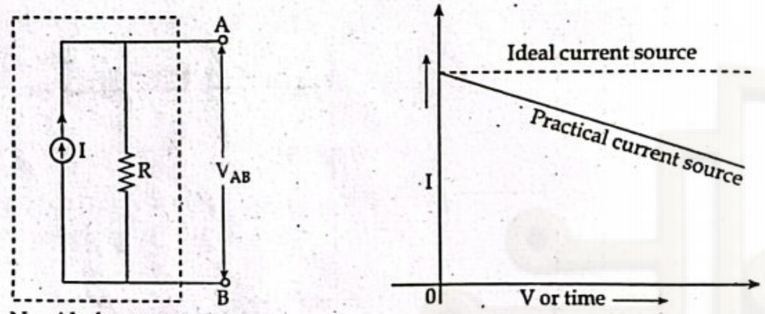


Fig: Ideal current source and it's graph.

A practical current source is represented as an ideal current source connected with the resistance in parallel.

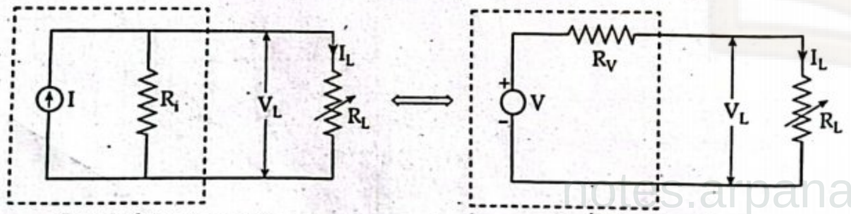


Non-ideal current sources

Fig: Practical current source and it's graph.

2.10 SOURCE CONVERSION

Practical voltage source and practical current sources are identical when load current and terminal voltage are identical.



Practical current source Practical voltage source

Load current $(I_L) = \frac{I \cdot R_i}{R_L + R_i}$ (1)

Terminal voltage $(V_L) = IR_i$ (3)

$I_L = \frac{V}{R_V + R_L}$ (2)

$V_L = V$ (4)

We have, both sources must be identical,

$$\frac{V}{R_V + R_L} = I \cdot \frac{R_i}{R_i + R_L}$$

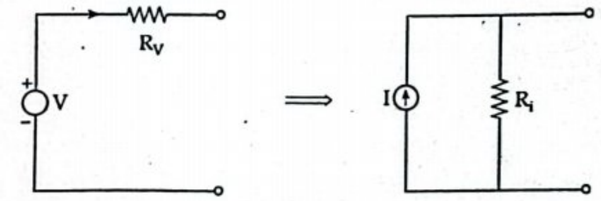
From (3) and (4);

or $\frac{V}{R_V + R_L} = \frac{V}{R_i + R_L}$

or, $R_V + R_L = R_i + R_L$

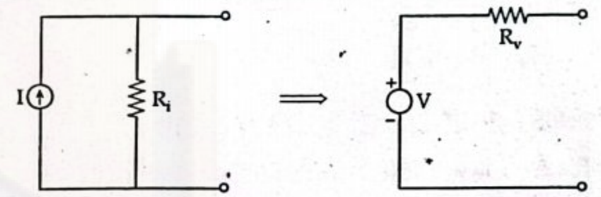
$\therefore R_V = R_L$

If practical voltage source is transferred into the practical current source,



$I = \frac{V}{R_V}$ and $R_i = R_V$

If practical current source is transferred into the practical voltage source



$V = IR_i$ $R_i = R_v$

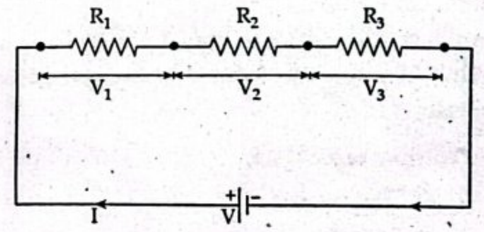
2.11 VOLTAGE DIVIDER AND CURRENT DIVIDER FORMULA

Here;

$V = V_1 + V_2 + V_3 = I(R_1 + R_2 + R_3)$

$V = I \times R_{eq}$

or, $I = \frac{V}{R_{eq}}$



By Ohm's Law,

$V_1 = IR_1 = V\frac{R_1}{R_{eq}}$

$V_2 = IR_2 = V\frac{R_2}{R_{eq}}$

$V_3 = IR_3 = V\frac{R_3}{R_{eq}}$

$\therefore V_n = IR_n = V\frac{R_n}{R_{eq}}$

This is known as voltage divider formula.

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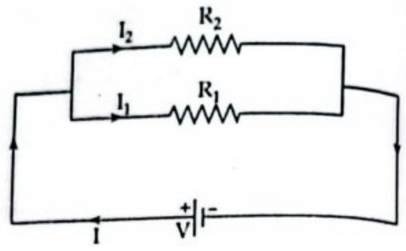
Here;



$$I = I_1 + I_2 = \frac{V}{R_1} + \frac{V}{R_2}$$

or, $$I = V\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

or, $$I = V\left(\frac{R_2 + R_1}{R_1 \cdot R_2}\right)$$

$$\therefore \quad V = I \times \left(\frac{R_1 \cdot R_2}{R_1 + R_2}\right)$$

By Ohm's Law,

$$I_1 = \frac{V}{R_1} = I \times \frac{(R_1 \times R_2)}{R_1 (R_1 + R_2)} = I \cdot \frac{R_2}{R_1 + R_2}$$

$$I_2 = \frac{V}{R_2} = I \cdot \frac{R_2}{R_1 + R_2}$$

,

$$\therefore \quad I_b = \frac{\text{Total current} \times \text{Opposite resistance}}{\text{Sum of total divided resistance}}$$

This is known as current divider rule.

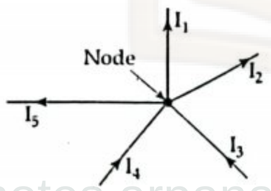
2.12 KIRCHOFF'S CURRENT AND VOLTAGE LAWS

a) Kirchoff's first law or Kirchoff's current law (KCL)

The law is based on the conservation of charge where the charge (measured in coulombs) is the product of the current (in amperes) and the time (in seconds). According to it, "Whenever two or more conductors meet at a point, then sum of the current flowing towards the junction point is equal to sum of the currents flowing away from that point. *i.e.*, $\Sigma I = 0$ at a junction point".

Sign convection:

All incoming current may be taken as positive while all outgoing currents taken as negative.



Example:

According to this law,

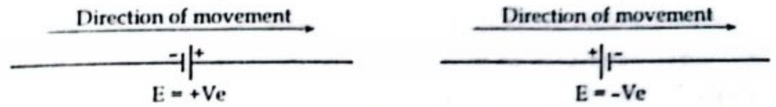
$$I_3 + I_4 - I_5 - I_1 - I_2 = 0$$

b) Kirchoff's second law or Kirchoff's mesh or voltage law (KVL)

The law is based on the conservation of energy. According to it, "the algebraic sum of the products of current and resistance of each conductor in any closed path of mesh in a network plus the algebraic sum of the voltages in that path is zero" *i.e.*, $\Sigma IR + \Sigma V = 0$.

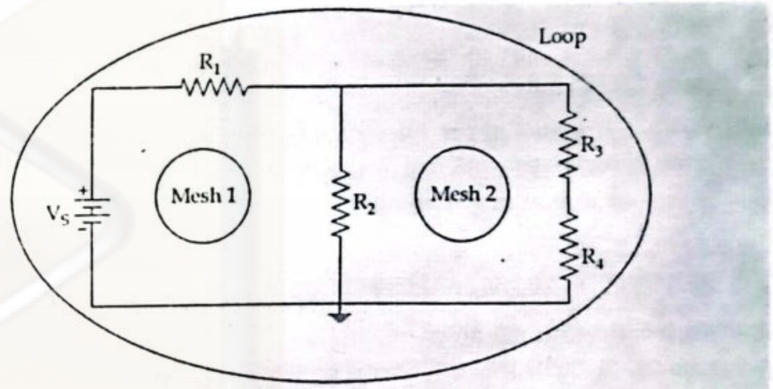
Sign convention:

1. Give positive sign to all rise in voltage and negative sign to voltage drops. Thus, if we move from negative (-ve) terminal of a battery to positive (+ve) terminal, a positive sign should be given, since there is a rise in voltage on the other hand, if we go from positive (+ve) terminal to negative terminal, a negative sign should be given, since there is a drop in voltage.



2. For a conductor of resistance R, if the direction of current is same/ opposite as the direction of movement, then voltage should be taken negative/positive, since current flows from higher voltage to lower one and hence there is a voltage drop/ voltage rise while crossing the conducting element.

NOTE:



i) A mesh is a loop in an electric circuit which does not contain any other loops within it.

ii) A loop is an electric circuit in a closed path.

2.13 NODAL METHOD AND MESH METHOD OF NETWORK ANALYSIS (WITHOUT DEPENDENT SOURCE)

a) Mesh analysis or loop analysis

The mesh or loop analysis is based on KVL. Here the different meshes are assigned continuous paths so that they do not split at a junction into branch currents. In this method, loop voltage equations are written by KVL in terms of unknown loop currents.

Procedure:

i) Convert all current sources to voltage sources.

ii) Draw all loop currents in a clockwise direction and identify them

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iii) Identify all resistor voltage drops as + to - in the direction of the loop current and assume these drops to be positive.
iv) Identify all voltage sources according to their correct polarity.
v) Write the equations for the voltage drop around each loop in turn, by equating the sum of the voltage drops to zero.
vi) Solve the equations to find the unknown currents and/or voltage drops.

b) Nodal analysis

Nodal analysis is based on KCL. This method has the advantage that a minimum number of equations are needed to determine the unknown quantities. For the application of this method, one of the nodes in the network is regarded as the reference or datum node or zero potential node. The number of simultaneous equations to be solved becomes (n - 1), where n is the number of independent nodes.

Procedure:

i) Convert all voltage sources to current sources and redraw the circuit diagram.
ii) Identify all nodes and choose a reference node (usually, the common node is the reference node).
iii) Write the equation for the currents flowing into and out of each node, with the exception of the reference node.
iv) Solve the equation to determine the node voltage and the required branch currents.

2.14 NETWORK THEOREM (*i.e.* SUPERPOSITION, THEVENIN'S, NORTON'S)

A. Superposition theorem

According to it, "In a linear bilateral network containing several sources, the current through or voltage across any branch in the network equals the algebraic sum of the currents or voltages of each individual sources considered separately with all other sources made inoperative *i.e.*, replaced by resistances equal to their internal resistances.

It may be noted here that while removing the voltage source it should be replaced by its internal resistance (if any) or by a short circuit and while removing the current source it should be replaced by an open circuit. Superposition theorem is applicable only to linear networks (both AC and DC) where current is linearly related to voltage as per ohm's law.

Procedure:

i) Select one source and replace all other sources by their internal impedances.
ii) Determine the level and direction of the current that flows through the desired branch as a result of the single source acting alone.

iii) Repeat steps (i) and (ii) using each source in turn until the branch current components have been calculated for all sources.
iv) Algebraically sum the component currents to obtain the actual branch current (s).

B. Thevenin's Theorem

Statement: The current flowing through a resistance R_L connected across any two terminals (say A and B) of a network containing one or more source of current/voltage (*i.e.*, active network) is given by,

$$I = \frac{V_{th}}{R_L + R_{th}}$$

where,

V_{th} = Open circuit voltage between A and B *i.e.*, with R_L disconnected
R_{th} = Equivalent resistance of the network with R_L removed and the sources of current/voltage, if any, replaced by their respective internal resistances.

Limitations:

i) This theorem cannot be applied to a network which contains non-linear impedance.
ii) This theorem cannot calculate the power consumed internally in the circuit or the efficiency of the circuit.

Procedure:

i) Calculate the open circuit voltage (V_{th}) across the network terminals.
ii) Redraw the network with each independent source replaced by its internal resistance. This is called "deactivation of the sources".
iii) Calculate the resistance (R_{th}) of the redrawn network as seen from the output terminals.

C. Norton's theorem

According to this theorem, any two terminal of active network containing voltage sources and resistances when viewed from its output terminals is equivalent to a constant current source and an internal (parallel) resistance. The constant current source (known as Norton's equivalent current source) is of the magnitude of the short circuit current at the terminals. The internal resistance is the equivalent resistance of the network looking back into the terminals with all the sources replaced by their internal resistance.

Procedure:

i) Calculate the short-circuit current I_N at the network terminals.
ii) Redraw the network with each source replaced by its internal resistance.

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iii) Calculate the resistance R_N of the redrawn network as seen from the output terminals.

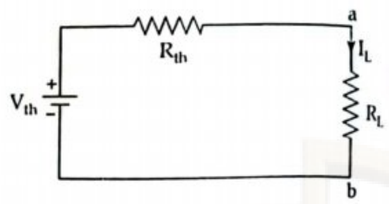
iv) Draw Norton's equivalent circuit.

2.15 MAXIMUM POWER TRANSFER THEOREM (MPTT)

This theorem states that, "maximum power output from a network is obtained when the load resistance (R_L) is equal to Thevenin's equivalent resistance of the network as seen from the load terminals." This theorem is used to find R_L such that the power dissipated in R_L is maximum.

Proof:

Figure shows a simple resistive network in which a load resistance R_L is connected across terminal 'a' and 'b' of the network.



Here;

$$\text{Load current, } I_L = \frac{V_{th}}{R_{th} + R_L}$$

Therefore, Power delivered to load,

$$P_L = I_L^2 \cdot R_L$$

$$= \left(\frac{V_{th}}{R_{th} + R_L}\right)^2 \cdot R_L$$

$$= \frac{V_{th}^2 \cdot R_L}{R_{th}^2 + 2R_{th} \cdot R_L + R_L^2}$$

$$= \frac{V_{th}^2}{(R_{th}^2 / R_L) + 2R_{th} + R_L}$$

If $R_L = 0$ (*i.e.*, short circuit), then $P_L = 0$ and if R_L = infinite (*i.e.* open circuit), $I_L = 0$ and hence $P_L = 0$.

Between these two extreme conditions, P_L always possesses finite and positive values. Hence, for a particular value of R_L, the power dissipated (P_L) must be maximum. This occurs when RHS of above equation is maximum or when denominator of RHS of above equation is minimum, since V_{th} is constant.

Let, $\frac{R_{th}^2}{R_L} + 2R_{th} + R_L = r$

For minimum value of r,

$\frac{dr}{dR_L} = 0$ and $\frac{d^2r}{dR_L^2}$ is positive. *i.e.*,

or, $\frac{d}{dR_L}\left(\frac{R_{th}^2}{R_L} + 2R_{th} + R_L\right) = -\frac{R_{th}^2}{R_L^2} + 0 + 1 = 0$

$\therefore \quad \frac{R_{th}^2}{R_L^2} = 1$

or, $R_L = R_{th}$

Also,

$$\frac{d^2r}{dR_L^2} = \frac{d}{dt}\left(-\frac{R_{th}^2}{R_L^2} + 1\right) = 2\frac{R_{th}^2}{R_L^3} = 2\frac{R_{th}^2}{R_{th}^3} \qquad [\because R_L = R_{th}]$$

$$= \frac{2}{R_{th}}$$

Therefore, $\frac{d^2r}{dR_L^2}$ is positive, thereby showing that $R_L = R_{th}$ is a condition for minimum value of r or maximum value of R_L.

Now,

$$P_{Lmax} = \frac{V_{th}^2}{\left(\frac{R_{th}^2}{R_L}\right) + 2R_{th} + R_L} = \frac{V_{th}^2 \cdot R_L}{R_{th}^2 + 2R_{th} \cdot R_L + R_L^2}$$

$$= \frac{V_{th}^2 R_{th}}{(R_{th} + R_{th})^2} \quad (\because R_L = R_{th})$$

$$= \frac{V_{th}^2 R_{th}}{4R_{th}^2}$$

$\therefore \quad P_{Lmax} = \frac{V_{th}^2}{4R_{th}} = P_{th}$

and, $P_{total} = P_{th} + P_{Lmax} = \frac{V_{th}^2}{4R_{th}} + \frac{V_{th}^2}{4R_{th}} = \frac{2V_{th}^2}{4R_{th}} = \frac{V_{th}^2}{2R_{th}}$

and, Efficiency under maximum power transfer condition is

$\therefore \quad \eta = \frac{P_{th}}{P_{total}}$

or, $\eta = \frac{\left(\frac{V_{th}^2}{4R_{th}}\right)}{\left(\frac{V_{th}^2}{2R_{th}}\right)} = \frac{1}{2} = 0.50$

$\therefore \quad \eta_{max} = 50\%.$

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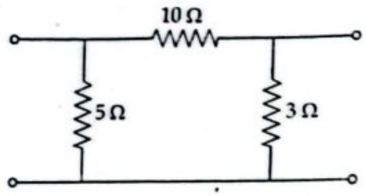
SOME SOLVED EXAMPLES

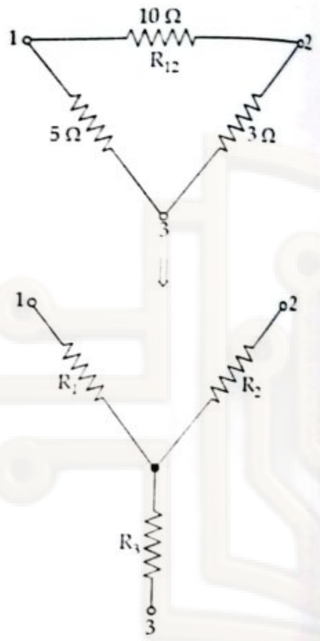
1. Calculate the equivalent resistance of following network:

A.



Solution:

Redrawing given figure. Here, R_1, R_2 and R_3 in star combination represent the equivalent of the given delta network.



$$R_1 = \frac{R_{12} \times R_{31}}{R_{12} + R_{23} + R_{31}}$$

$$= \frac{10 \times 5}{10 + 5 + 3}$$

$$= 2.78$$

$$R_2 = \frac{R_{23} \times R_{12}}{R_{12} + R_{23} + R_{31}}$$

$$= \frac{3 \times 10}{10 + 3 + 5}$$

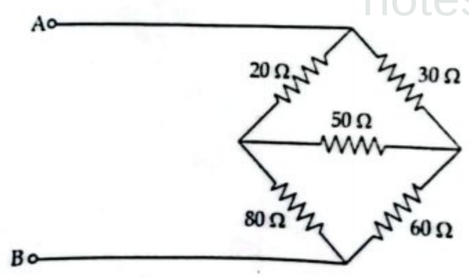
$$= 1.67\ \Omega$$

$$R_3 = \frac{R_{23} \times R_{31}}{R_{12} + R_{23} + R_{31}}$$

$$= \frac{3 \times 5}{10 + 3 + 5}$$

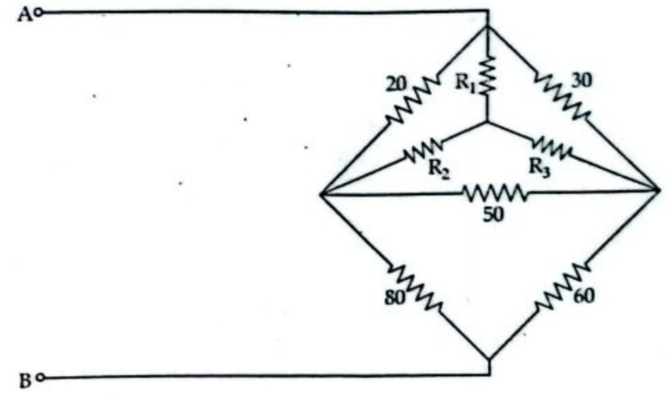
$$= 0.83\ \Omega.$$

B.



Solution:

Converting the upper delta network into a star network as,

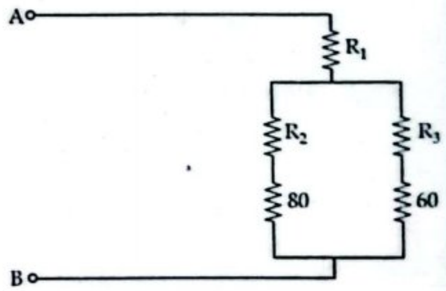


so, $R_1 = \frac{20 \times 30}{20 + 30 + 50} = 6\ \Omega$

$R_3 = \frac{20 \times 50}{20 + 30 + 50} = 10\ \Omega$

$R_3 = \frac{30 \times 50}{20 + 30 + 50} = 15\ \Omega$

Next, reorient the network as,



Here,

$$R_{eq} = R_1 + \frac{(R_2 + 80) \times (R_3 + 60)}{(R_2 + 80) + (R_3 + 60)}$$

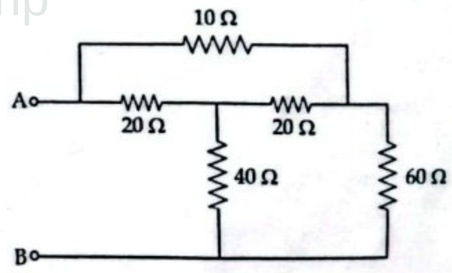
$$= 6 + \frac{(10 + 80) \times (15 + 60)}{(10 + 80) + (15 + 60)}$$

$$= 6 + \frac{(90 \times 75)}{(90 + 75)}$$

$$= 46.9\ \Omega$$

Hence, $R_{AB} = 46.9\ \Omega$.

C.



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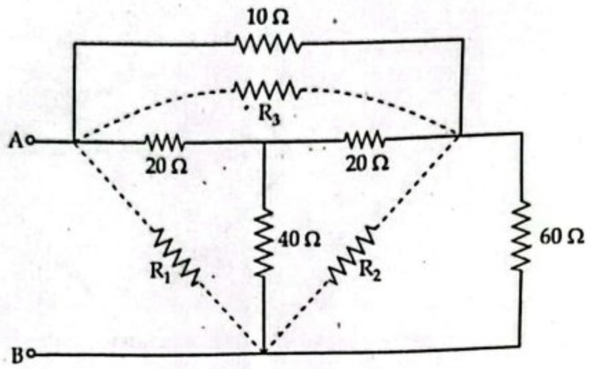
Solution:

First convert the star connected network using 20 Ω, 20 Ω and 40 Ω resistors to an equivalent delta network.

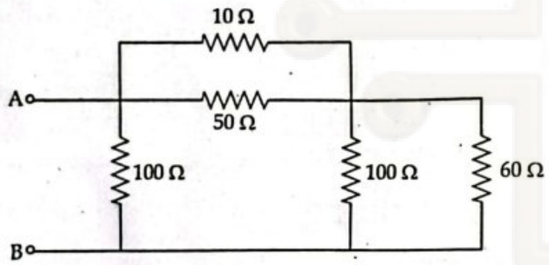


Here; $R_1 = \frac{20 \times 40 + 20 \times 40 + 20 \times 20}{20} = 100\ \Omega$

$R_2 = \frac{20 \times 40 + 20 \times 40 + 20 \times 20}{20} = 100\ \Omega$

$R_3 = \frac{20 \times 40 + 20 \times 40 + 20 \times 20}{40} = 50\ \Omega$

Then, final form of network,

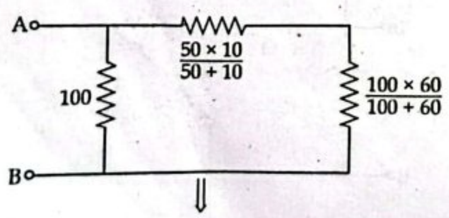


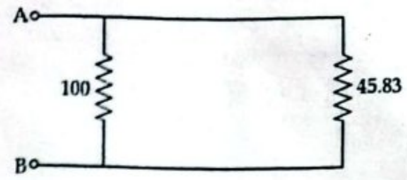
Thus, the equivalent resistance between terminals A and B is,

$$R_{eq} = 10 \parallel \left[\frac{50 \times 10}{50 + 10} + \frac{1000 \times 60}{100 + 60}\right] = 100 \parallel \left[\frac{50}{60} + \frac{600}{16}\right]$$

$$= 100 \parallel 45.83 = \frac{100 \times 45.83}{100 + 45.83}$$

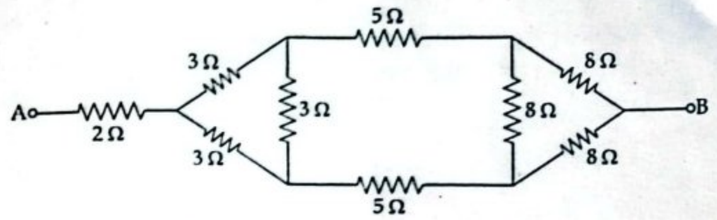
$$= 31.43\ \Omega$$





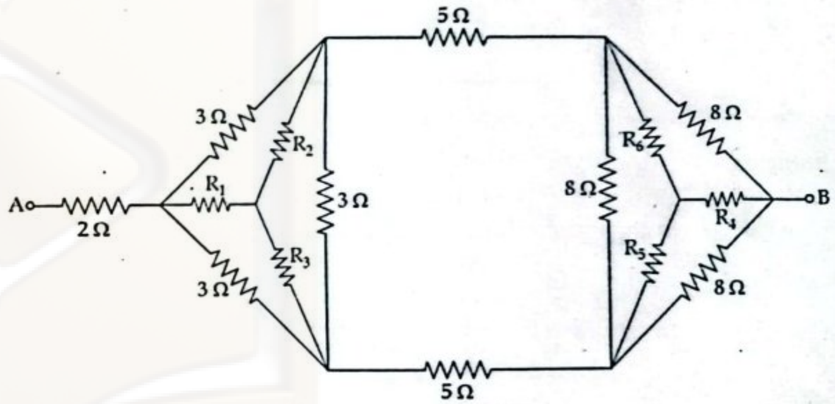
Hence, $R_{AB} = 31.43\ \Omega$.

D.



Solution:

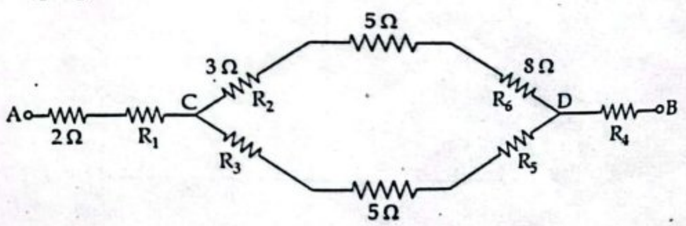
Convert the two delta networks formed in the given circuit to equivalent star networks.



Here; $R_1 = R_2 = R_3 = \frac{3 \times 3}{3 + 3 + 3} = 1\ \Omega$

$R_4 = R_5 = R_6 = \frac{8 \times 8}{8 + 8 + 8} = 2.67\ \Omega$

Redrawing figure,



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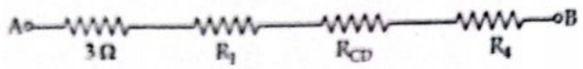
Now,

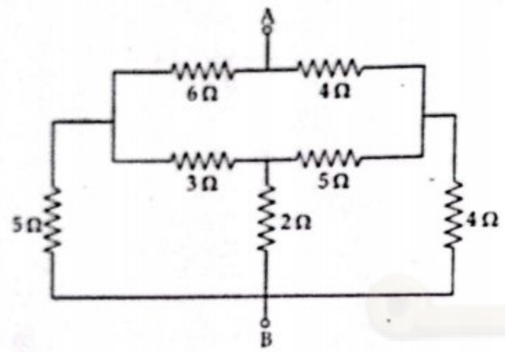
$$R_{CD} = (R_2 + 5 + R_4) \parallel (R_3 + R_5 + 5)$$
$$= (1 + 5 + 2.67) \parallel (1 + 2.67 + 5)$$
$$= 8.67 \parallel 8.67$$
$$= 4.335\ \Omega$$

Hence,



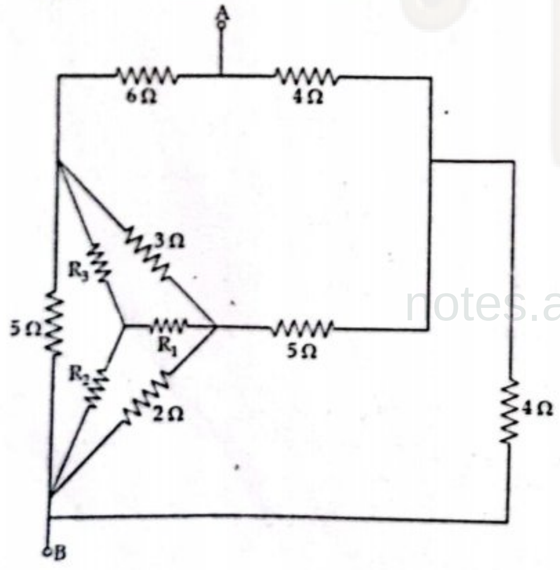
$$\therefore \quad R_{eq} = 2 + R_1 + R_{CD} + R_4$$
$$= 2 + 1 + 4.335 + 2.67 = 10\ \Omega$$

E.



Solution:

Redrawing figure to represent a star equivalent to the part of the given network containing resistances 3 Ω, 2 Ω and 5 Ω.



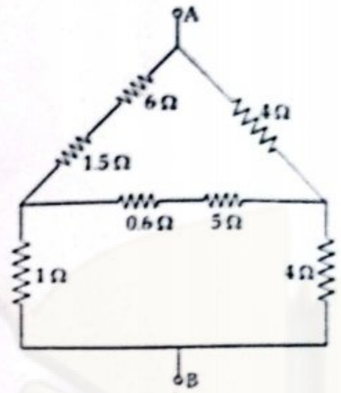
From figure,

$$R_1 = \frac{3 \times 2}{3 + 2 + 5} = 0.6\ \Omega$$

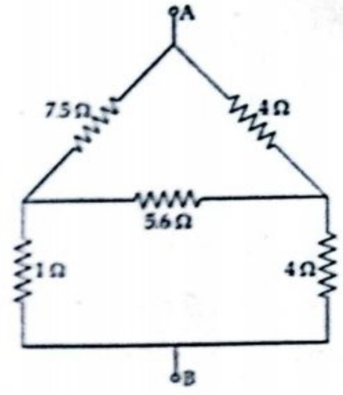
$$R_2 = \frac{2 \times 5}{2 + 5 + 3} = 1\ \Omega$$

$$R_3 = \frac{3 \times 5}{3 + 5 + 2} = 1.5\ \Omega$$

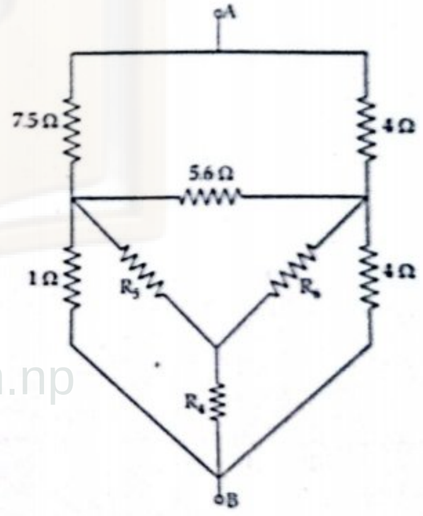
Now, the circuit becomes,



Simplifying,



Converting delta connected resistances 5.6 Ω, 4 Ω and 1 Ω into star equivalent.



Here;

$$R_4 = \frac{1 \times 4}{1 + 4 + 5.6} = 0.377\ \Omega$$

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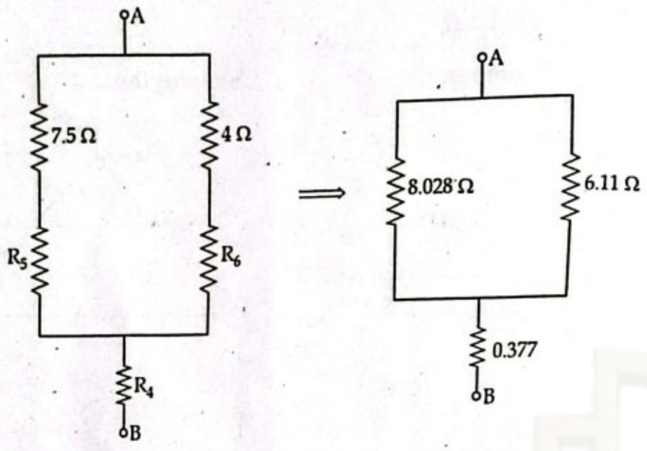
$$R_5 = \frac{1 \times 5.6}{1 + 4 + 5.6} = 0.528\ \Omega$$

$$R_6 = \frac{4 \times 5.6}{1 + 4 + 5.6} = 2.11\ \Omega$$

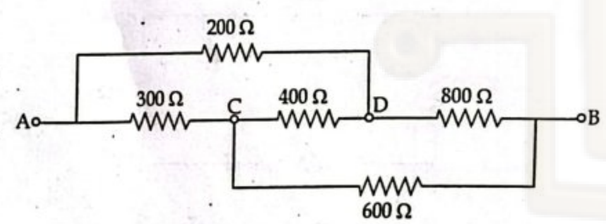
Now, the network becomes,



Hence,

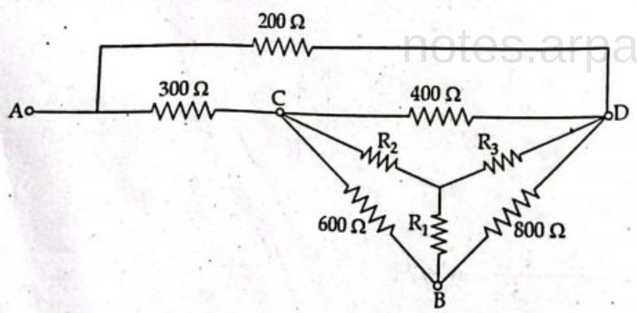
$$R_{AB} = (6.11 \parallel 8.028) + 0.377$$
$$= 3.469 + 0.377$$
$$= 3.847\ \Omega.$$

F.



Solution:

Converting delta BCD into star,



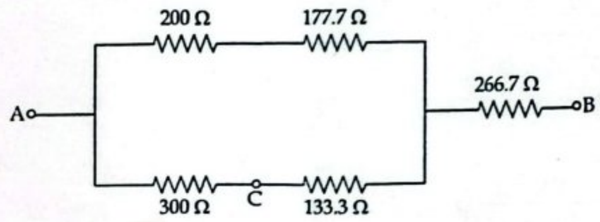
Here;

$$R_1 = \frac{600 \times 800}{600 + 800 + 400} = 266.7\ \Omega$$

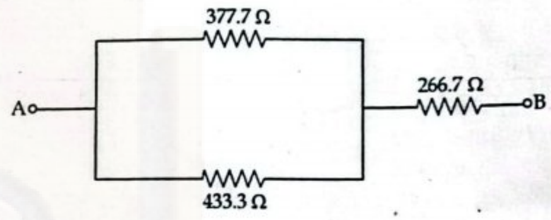
$$R_2 = \frac{600 \times 400}{600 + 800 + 400} = 133.3\ \Omega$$

$$R_3 = \frac{400 \times 800}{600 + 800 + 400} = 177.7\ \Omega$$

Then the circuit becomes,



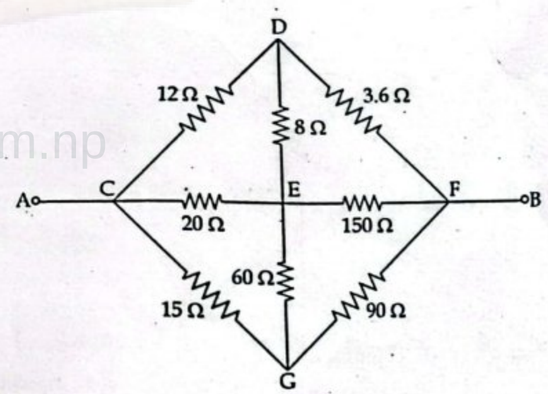
Adding series resistance,



Hence,

$$R_{AB} = (377.7 \parallel 433.3) + 266.7$$
$$= \frac{377.7 + 433.3}{377.7 + 433.3} + 266.7$$
$$= 201.8 + 266.7$$
$$= 468.50\ \Omega$$

G.



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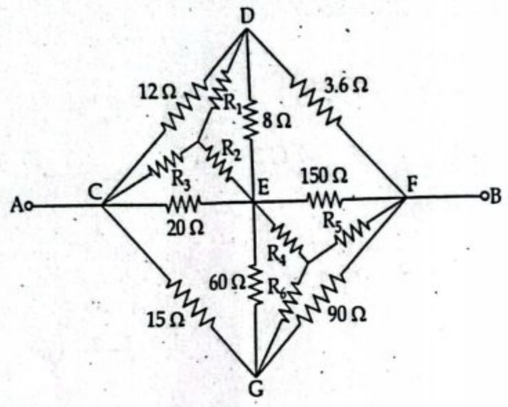
Solution:

Converting delta CDE and EFG into equivalent stars, we get;

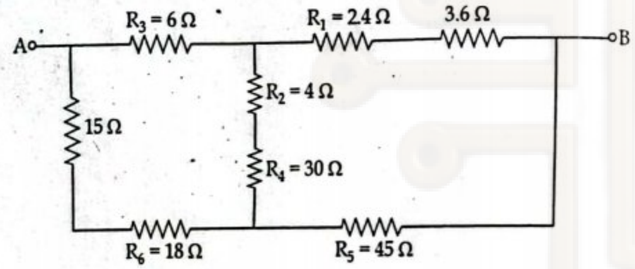


$$R_1 = \frac{12 \times 8}{12 + 8 + 40} = 2.4\ \Omega \qquad R_2 = \frac{20 \times 8}{12 + 8 + 40} = 4\ \Omega$$

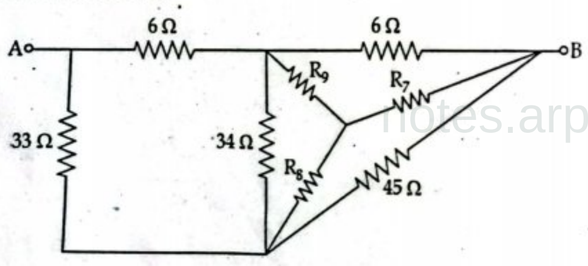
$$R_3 = \frac{12 \times 20}{12 + 8 + 40} = 6\ \Omega \qquad R_4 = \frac{150 \times 60}{150 + 90 + 60} = 300\ \Omega$$

$$R_5 = \frac{150 \times 90}{150 + 90 + 60} = 45\ \Omega \qquad R_6 = \frac{60 \times 90}{150 + 90 + 60} = 18\ \Omega$$

Then, circuit reduces to



On adding series resistance 15 Ω and 18 Ω, 2.4 Ω and 3.6 Ω and 4 Ω and 30 Ω, the circuit reduces to,



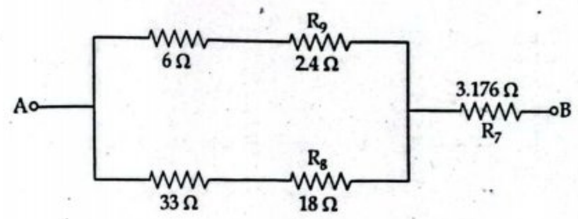
Here;

$$R_7 = \frac{6 \times 45}{6 + 45 + 34} = 3.1765\ \Omega$$

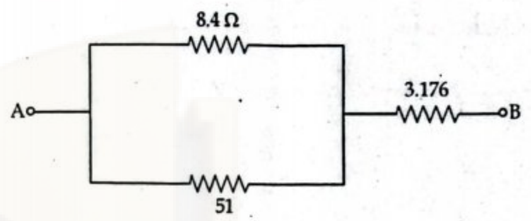
$$R_8 = \frac{45 \times 34}{6 + 45 + 34} = 18\ \Omega$$

$$R_9 = \frac{34 \times 6}{6 + 45 + 85} = 2.5\ \Omega$$

Then,



On adding series resistance,



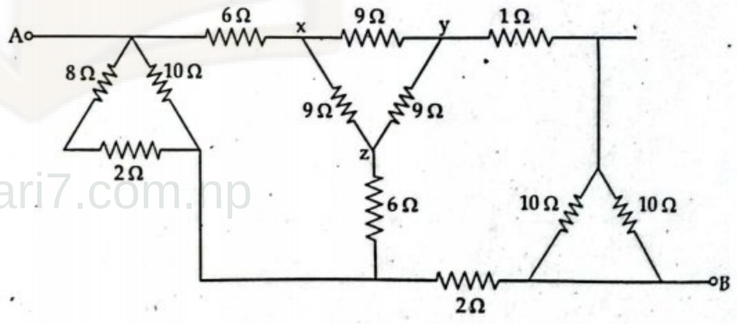
Hence,

$$\therefore \quad R_{AB} = (8.4 \| 51) + 3.176$$

$$= \frac{8.4 \times 51}{8.4 + 51} + 3.176$$

$$= 10.3886\ \Omega.$$

H.



Solution:

Adding parallel resistance (2 + 8) Ω ‖ 10 Ω and 10 Ω ‖ 10 Ω on sides and converting 9 Ω each delta into star,

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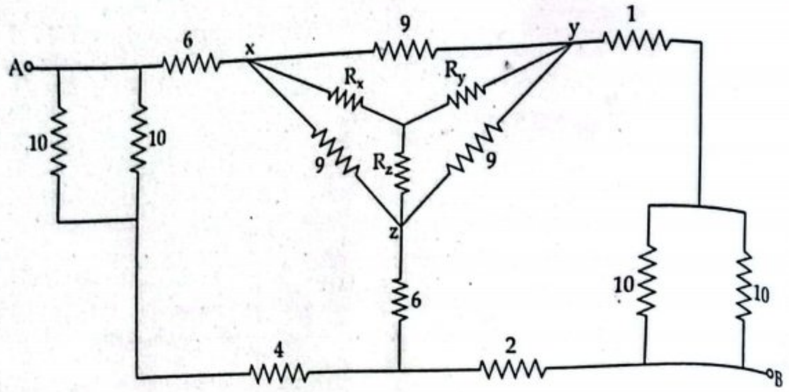
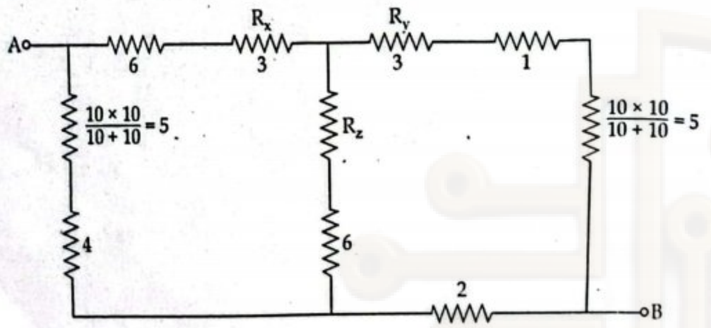


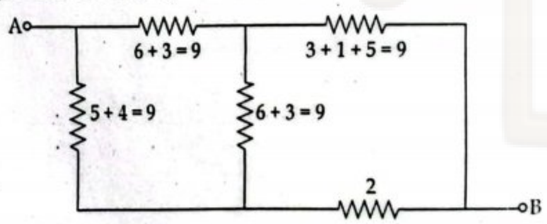
Here;

$$R_x = R_y = R_z = \frac{9 \times 9}{9+9+9} = 3\,\Omega$$

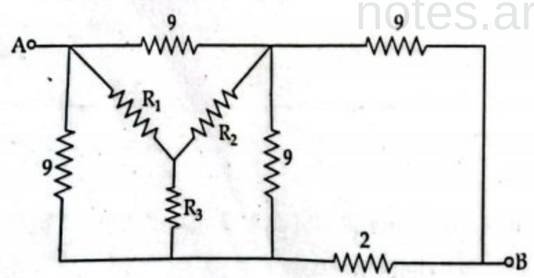
Then, circuit reduces to



On adding series resistance,

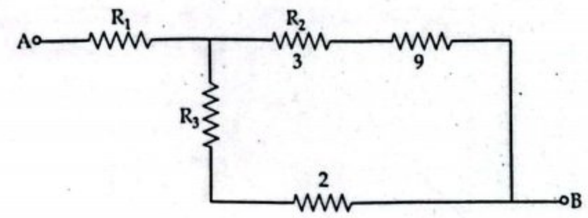


Hence, converting 9 Ω each delta in star,

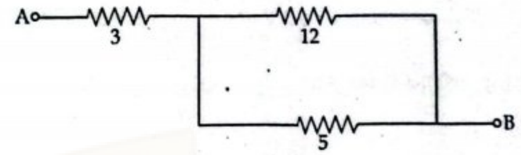


$$R_1 = R_2 = R_3 = \frac{9 \times 9}{9+9+9} = 3\,\Omega$$

so,



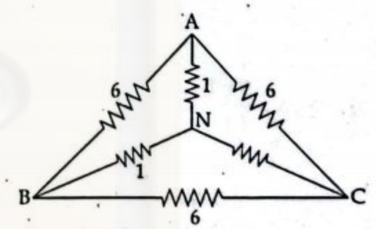
Adding series 3 and 9 Ω, and 3 and 2Ω,



Hence,

$$R_{AB} = 3 + (12 \parallel 5) = 3 + \frac{12 \times 5}{12+5} = 6.33\,\Omega.$$

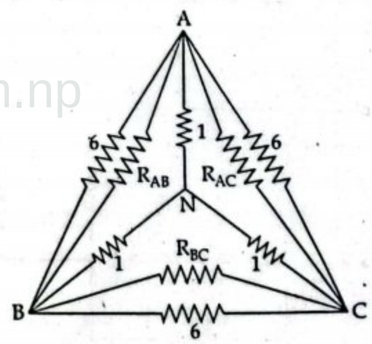
I.



Solution:

Converting the inner star into delta of resistance in each branch of

$$\frac{(1 \times 1 + 1 \times 1 + 1 \times 1)}{1 \times 1} = 3\,\Omega$$



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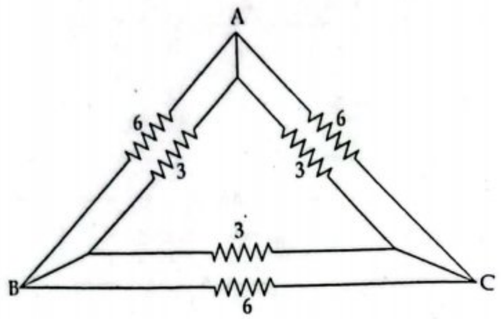
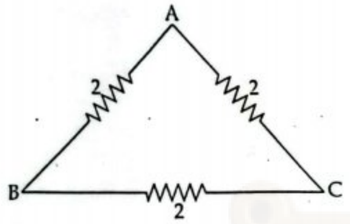
Here;

$R_{AC} = R_{BC} = R_{AB} = 3\ \Omega$

Hence, the circuit reduces to,



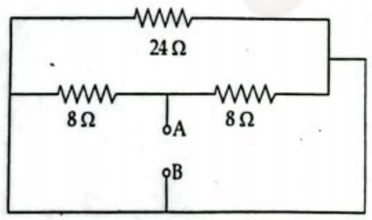
Then combining $6\ \Omega \parallel 3\ \Omega$ to get,



Hence,

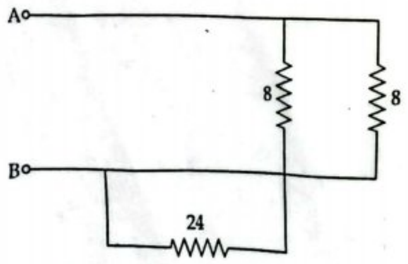
$\therefore \quad R_{BC} = 2 \parallel (2+2) = 2 \parallel 4 = \dfrac{2 \times 4}{2+4} = 1.33\ \Omega.$

J.



Solution:

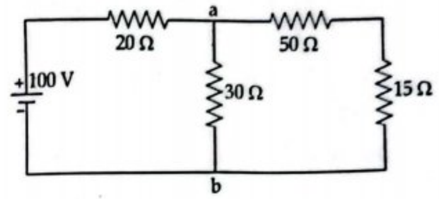
It can be redraw as,



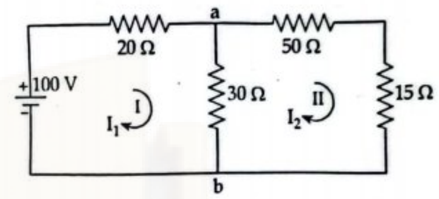
Hence, $R_{AB} = (8 \parallel 8)\ \Omega = \dfrac{8 \times 8}{8+8} = 4\ \Omega.$

[∵ Both terminals of 24 Ω are short circuited to terminal B]

2. Using KVL, determine total current drawn from the source and also current in 15 Ω resistance.



Solution:



Applying KVL to meshes I and II,

or, $20\ I_1 + 30\ (I_1 - I_2) - 100 = 0$

or, $50\ I_1 - 30\ I_2 = 100$

or, $5\ I_1 - 3\ I_2 = 10$ **(1)**

and, $(50 + 15)\ I_2 + 30\ (I_2 - I_1) = 0$

or, $-30\ I_1 + 95\ I_2 = 0$

or, $I_2 = 30\dfrac{I_1}{95} = 6\dfrac{I_1}{19}$ **(2)**

Solving **(1)** and **(2)**, we get;

$5\ I_1 - 3 \times 6\dfrac{I_1}{19} = 10$

or, $95\ I_1 - 18\ I_1 = 190$

or, $I_1 = \dfrac{190}{77}$

$\therefore \quad I_1 = 2.4675\ A$ **(3)**

From **(2)** and **(3)**;

$I_2 = \left(\dfrac{6}{19}\right) I_1 = \dfrac{6}{19} \times 2.4675$

$\therefore \quad I_2 = 0.7792\ A$

$\therefore \quad I_1 = 2.4675\ A.$

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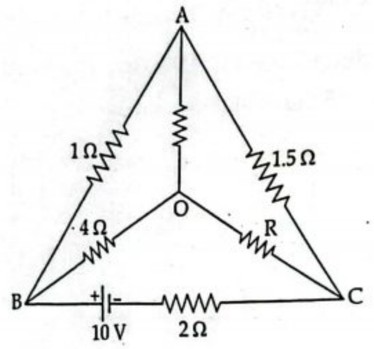
3. Find the value of R and the current flowing through it in following network, when the current in branch OA is zero.

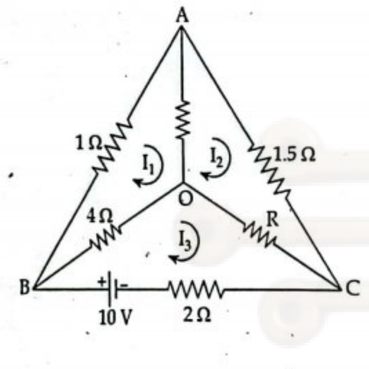


Solution:

Applying KVL to meshes I, II and III, we get,

or, $1 \times I_1 + 0 + 4(I_1 - I_3) = 0$

or, $5 I_1 = 4 I_3$



or, $I_3 = 1.25 I_1$ (1)

Current in OA branch, $(I_1 - I_2) = 0$

$\therefore \quad I_1 = I_2$ (2)

and, $1.5 I_2 + R(I_2 - I_3) + 0 = 0$

or, $I_2 (1.5 + R) = R I_3$

or, $(1.5 + R) I_1 = R (1.2 I_1)$

or $R = \dfrac{1.5}{0.25}$

$\therefore \quad R = 6\ \Omega$ (4)

Also,

$4(I_3 - I_1) + R(I_3 - I_2) + 2I_3 - 10 = 0$

or, $-4 I_1 - R I_2 + (6 + R) I_3 = 10$

or, $-4 I_1 - 6 I_1 + (6 + 6) \times 1.25 I_1 = 10$

or, $-4 I_1 - 6 I_1 + 15 I_1 = 10$

or, $5 I_1 = 10$

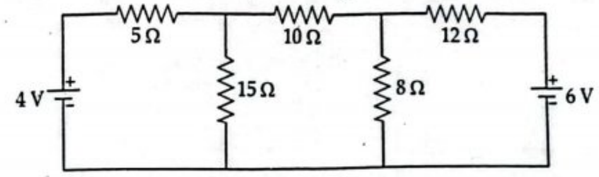
or, $I_1 = I_2 = 2$ A

and, $I_3 = 2 \times 1.25$ A $= 2.5$ A

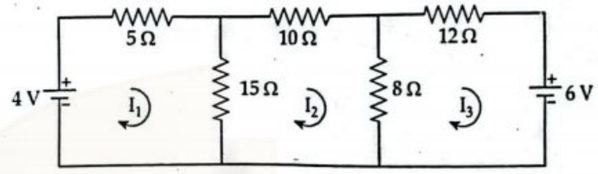
Hence, current flowing through R is,

$= (I_3 - I_2) = 2.5 - 2 = 0.5$ A.

4. A network is arranged as shown in figure. Determine the value of the current in the 8 Ω resistor using mesh equations.



Solution:



Using mesh equations,

Loop 1:

$4 - 5I_1 - 15(I_1 - I_2) = 0$

or, $20 I_1 - 15 I_2 = 4$ (1)

Loop 2:

$-10 I_2 - 8(I_2 - I_3) - 15(I_2 - I_1) = 0$

or, $15 I_1 - 33 I_2 + 8 I_3 = 0$ (2)

Loop 3:

$-12 I_3 - 6 - 8(I_3 - I_2) = 0$

or, $8 I_2 - 20 I_3 = 6$

Using Cramer's rule for evaluating the network,

NOTE:

Cramer's rule:

If there are a set of number of simultaneous equations of the form:

$a_{11} x_1 + a_{12} x_2 + \ldots\ldots + a_{1n} x_n = k_1$

$a_{12} x_1 + a_{22} x_2 + \ldots\ldots + a_{2n} x_n = k_2$

$\ldots\ldots + \ldots\ldots + \ldots\ldots + \ldots\ldots = \ldots\ldots$

$\ldots\ldots + \ldots\ldots + \ldots\ldots + \ldots\ldots = \ldots\ldots$

$a_{n1} x_1 + a_{n2} x_2 + \ldots\ldots + a_{nx} x_n = k_n$

Then by Cramer's rule, the solution of these simultaneous equations is given by,

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$$x_1 = \frac{D_1}{\Delta}$$

$$x_2 = \frac{D_2}{\Delta}$$

.....

.....

$$x_n = \frac{D_n}{\Delta}$$

where, Δ = Determinant = $\begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix}$

The value of D_j is obtained by replacing the j^{th} column of the system of determined by the column $k_1, k_2, \dots\dots, k_n$. Thus,

$$D_1 = \begin{vmatrix} k_1 & a_{12} & \dots & a_{1n} \\ k_2 & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \\ k_n & a_{n2} & \dots & a_{nn} \end{vmatrix}$$

$$D_2 = \begin{vmatrix} a_{11} & k_1 & \dots & a_{1n} \\ a_{21} & k_2 & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \\ a_{n1} & k_n & \dots & a_{nn} \end{vmatrix}, \text{ etc.}$$

Cramer's rule can be used for solving simultaneous equations if the number of unknown are more than two. This gives quick results.

so, $\Delta = \begin{vmatrix} 20 & -15 & 0 \\ 15 & -33 & 8 \\ 0 & 8 & -20 \end{vmatrix} = 7{,}420$

$\Delta_1 = \begin{vmatrix} 4 & -15 & 0 \\ 0 & -33 & 8 \\ 6 & 8 & -20 \end{vmatrix} = 1{,}664$

$\Delta_2 = \begin{vmatrix} 20 & 4 & 0 \\ 15 & 0 & 8 \\ 0 & 6 & -20 \end{vmatrix}$

$\Delta_3 = \begin{vmatrix} 20 & -15 & 4 \\ 15 & -33 & 0 \\ 0 & 8 & 6 \end{vmatrix} = -2{,}130$

$$\left[\because \begin{bmatrix} 20 & -15 & 0 \\ 15 & -33 & 8 \\ 0 & 8 & -20 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 6 \end{bmatrix} \right]$$

Hence,

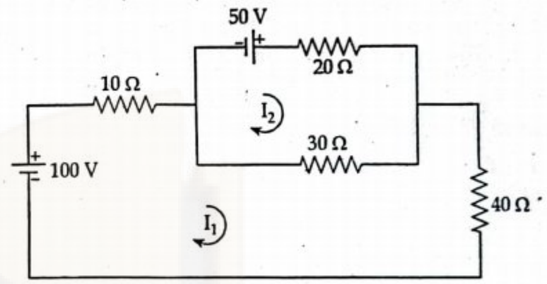
$$I_1 = \frac{\Delta_1}{\Delta} = \frac{1{,}664}{7{,}420} = 0.224 \text{ A}$$

$$I_2 = \frac{\Delta_2}{\Delta} = \frac{240}{7{,}420} = 0.032 \text{ A}$$

$$I_3 = \frac{\Delta_3}{\Delta} = -\frac{2{,}130}{7{,}420} = -0.2870 \text{ A}$$

$\therefore$ Current in the 8 Ω resistor $= I_2 - I_3$
$= 0.032 + 0.287$
$= 0.319$ A

5. Find the current through 30 Ω branch by mesh analysis.



Solution:

Applying KVL to mesh I and II,

or, $100 - 10 I_1 - 30 (I_1 - I_2) - 40 I_2 = 0$

or, $-80 I_1 + 30 I_2 = -100$

or, $+8 I_1 - 3 I_2 = 10$ (1)

and, $50 - 20 I_2 - 30 (I_2 - I_1) = 0$

or, $30 I_1 - 50 I_2 = -50$

or, $3 I_1 - 5 I_2 = -5$ (2)

Solving equation (1) and (2);

$40 I_1 - 15 I_2 = 50$

$-9 I_1 + 15 I_2 = 15$

$31 I_1 = 65$

$\therefore I_1 = 2.0967$ A

and, $6.29 - 5 I_2 = -5$

or, $I_2 = \frac{11.29}{5} = 2.258$ A

Hence, current in 30 Ω branch $= I_2 - I_1 = 2.2587 - 2.0967$ A
$= 0.162$ A

6. Find the current supplied by each battery and the power in the 10 ohms for the circuit below:

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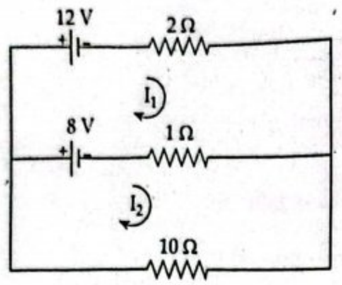


Solution:

Applying KVL to two loops, we get,

or, $-12 - 2I_1 - 1 \times (I_1 - I_2) + 8 = 0$

or, $-3I_1 + I_2 = 4$ (1)

and, $-8 - 1 \times (I_2 - I_1) - 10 I_2 = 0$

or, $I_1 - 11 I_2 = 8$ (2)

Solving equation (1) and (2);

$$-3 I_1 + I_2 = 4$$
$$\underline{3 I_1 - 33 I_2 = 24}$$
$$-32 I_2 = 28$$

or, $I_2 = \frac{28}{32}$

$\therefore \quad I_2 = -0.875$ A

Hence,

$I_1 = 8 + 11 I_2 = 8 + 11 \times (-0.875)$

$\therefore \quad I_1 = -1.625$ A

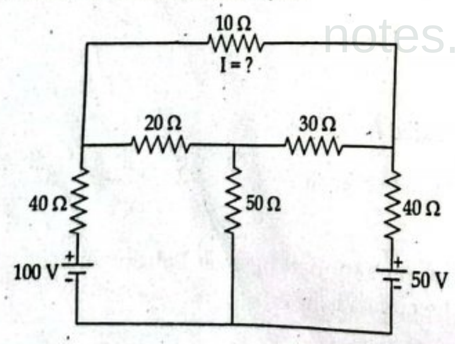
Hence, current supplied by 12 V battery $= -I_1 = 1.625$ A

Current supplied by 8 V battery $= -I_2 = 0.875$ A

Therefore, Power in 10 Ω resistor,

$P = I_2^2 \times 10 = (0.875)^2 \times 10 = 7.656$ W.

7. Determine the current I by applying Maxwell's loop current method in the network shown below.



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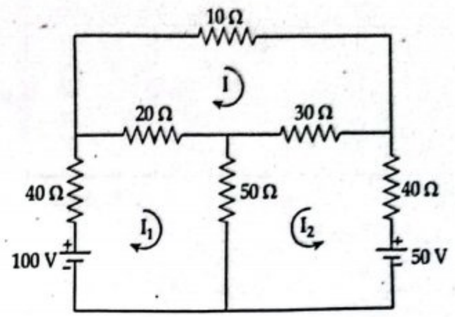
Solution:

Let, the direction of current flowing be as shown in figure



Applying KVL,

$40 I_1 + 20 (I_1 - I) + 50 (I_1 + I_2) = 100$

or, $4 I_1 + 2 I_1 - 2I + 5I_1 + 5 I_2 = 10$

or, $11 I_1 + 5 I_2 - 2 I = 10$ (1)

and, $40 I_2 + 30 (I_2 + I) + 50 (I_1 + I_2) = 50$

or, $5 I_1 + 12 I_2 + 3 I = 5$ (2)

and, $10 I + 30 (I + I_2) + 20 (I - I_1) = 0$

or, $-2 I_1 + 3 I_2 + 6 I = 0$ (3)

Solving equation (1), (2) and (3), using Cramer's rule,

$$\begin{bmatrix} 11 & 5 & -2 \\ 5 & 12 & 3 \\ -2 & 3 & 6 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \\ 0 \end{bmatrix}$$

$$\Delta = \begin{bmatrix} 11 & 5 & -2 \\ 5 & 12 & 3 \\ -2 & 3 & 6 \end{bmatrix} = 435$$

$$\Delta_1 = \begin{vmatrix} 10 & 5 & -2 \\ 5 & 12 & 3 \\ 0 & 3 & 6 \end{vmatrix} = 450$$

$$\Delta_2 = \begin{vmatrix} 11 & 10 & -2 \\ 5 & 5 & 3 \\ -2 & 0 & 6 \end{vmatrix} = -50$$

$$\Delta_3 = \begin{vmatrix} 11 & 5 & 10 \\ 5 & 12 & 5 \\ -2 & 3 & 0 \end{vmatrix} = 175$$

Then,

$\therefore \quad I = \frac{\Delta_3}{\Delta} = \frac{175}{435} = 0.4023$ A

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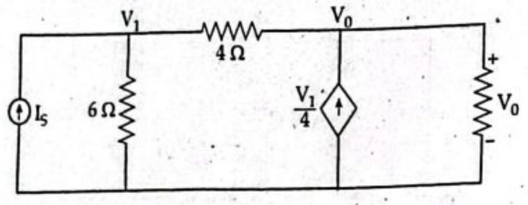
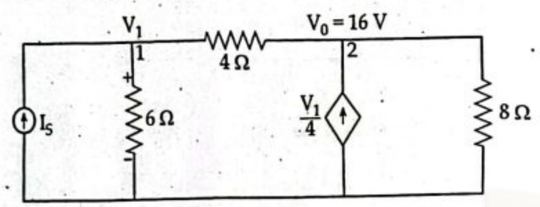
8. Apply KCL to determine current I_s in the electric circuit shown below. Take $V_0 = 16$ V.



Solution:

Applying KCL at node 1 and 2,



Here,

$$I_s = \frac{V_1}{6} + \frac{V_1 - V_0}{4} = V_1\left[\frac{1}{6} + \frac{16}{4}\right] - V_0 = \frac{5V_1}{12} - \frac{16}{4} = 5\frac{V_1}{12} - 4 \quad (1)$$

and, $\frac{1}{4}V_1 = \frac{V_0 - V_1}{4} + \frac{V_0}{8}$

or, $V_1 = V_0 - V_1 + \frac{1}{2}V_0 = \frac{3}{2}V_0 - V_1$

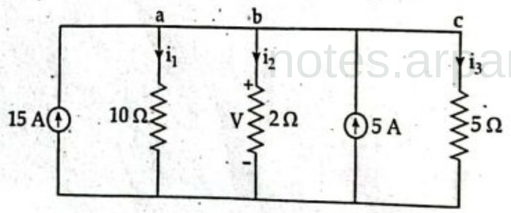
or, $2V_1 = \frac{3}{2}V_0 \Rightarrow \frac{3}{2} \times 16 = 24$

$\therefore \quad V_1 = 12 \quad (2)$

From (1) and (2);

$$\therefore \quad I_s\left(\frac{5 \times 12}{12} - 4\right) = 5 - 4 = 1 \text{ A}$$

9. Find V and magnitude of the unknown currents through 10 Ω, 2 Ω and 5 Ω resistance.



Solution:

Applying KCL at node 'a',

$$-15 + i_1 + i_2 - 5 + i_3 = 0$$

or, $i_1 + i_2 + i_3 = 20 \quad (1)$

From ohm's law,

$$i_2 = \frac{V}{2}$$

$$i_1 = \frac{V}{10}$$

$$i_3 = \frac{V}{5}$$

Hence, from equation (1);

or, $\frac{V}{10} + \frac{V}{2} + \frac{V}{5} = 20$

or, $V + 5V + 2V = 200$

$\therefore \quad V = 25$ V

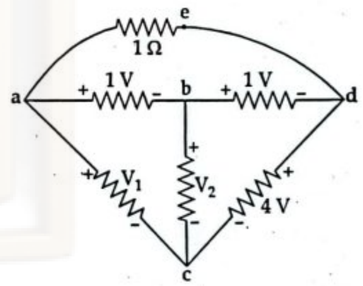
Hence,

$\therefore \quad i_1 = \frac{V}{10} = \frac{25}{10} = 2.5$ A

$\therefore \quad i_2 = \frac{V}{2} = \frac{25}{2} = 12.5$ A

$\therefore \quad i_3 = \frac{V}{5} = \frac{25}{5} = 5$ A

10. Find V_1 and V_2 using KVL.



Solution:

In loop abca, from KVL,

$$1 + V_2 - V_1 = 0$$

or, $V_2 - V_1 = -1$

or, $V_1 - V_2 = 1 \quad (1)$

In loop 'acdb', using KVL,

or, $-V_2 + 1 + 4 = 0$

$\therefore \quad V_2 = 5$

Hence,

$V_1 = 6$ V.

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Using nodal analysis at node 2,

or, $\frac{V_2 - V_1}{10} + \frac{V_2}{90} + \frac{V_2 - V_1}{15} = 0$

or, $\frac{-V_1}{15} + V_2\left(\frac{1}{10} + \frac{1}{15} + \frac{1}{90}\right) V_2 = 20$

or, $\frac{-1}{15} V_1 + \frac{16}{90} V_2 = 20$ (2)

On solving (1) and (2);

$\therefore \quad V_1 = 264.88\ V$

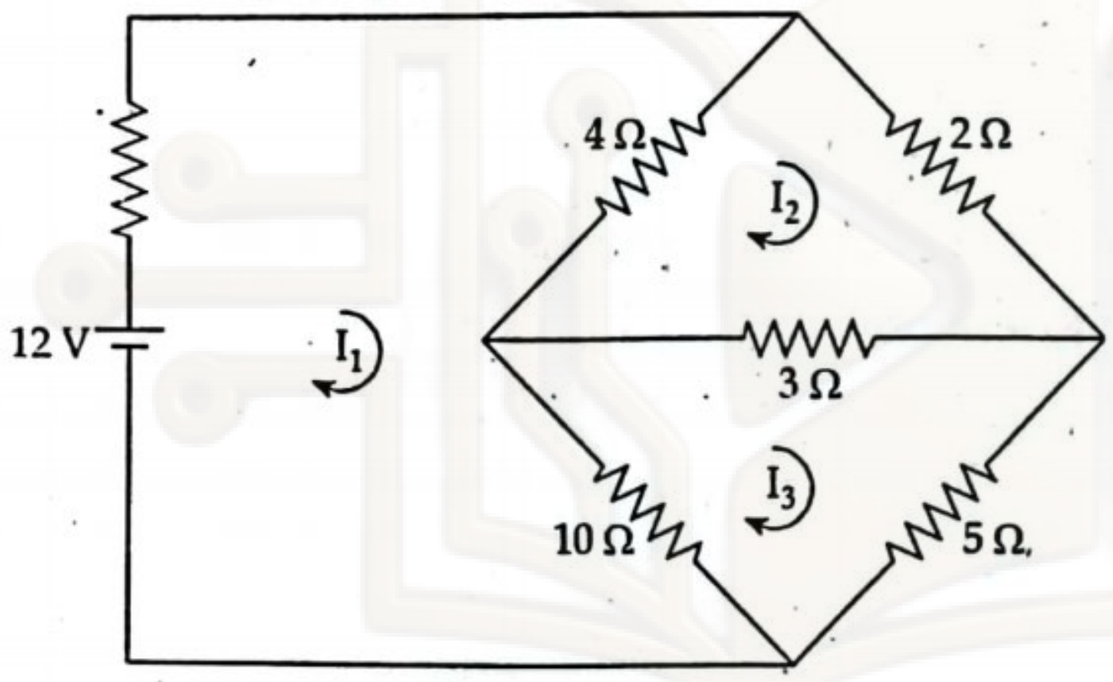
$\therefore \quad V_2 = 211.33\ V$

Now, current in 15 Ω resistor,

$$I_{15} = \left(\frac{V_1 - V_2}{15}\right) = \left(\frac{264.88 - 211.33}{15}\right) = 3.57\ A.$$

The direction of current is from node 1 to node 2.

13. Find the currents in 2 Ω, 3 Ω, 4 Ω, 5 Ω and 10 Ω resistances in the below circuit using loop method.



Solution:

Loop 1:

$$-12 + 6\,I_1 + (I_1 - I_2)\,4 + (I_1 - I_2)\,10 = 0$$

or, $10\,I_1 - 2\,I_2 - 5\,I_3 = 6$ (1)

Loop 2:

$$2\,I_2 + (I_2 - I_3)\,3 + (I_2 - I_1)\,4 = 0$$

or, $4\,I_1 - 9\,I_2 + 3\,I_3 = 0$ (2)

Loop 3:

$$5\,I_3 + (I_3 - I_1)\,10 + (I_3 - I_2)\,3 = 0$$

or, $10\,I_1 + 3\,I_2 - 18\,I_3 = 0$ (3)

Using Cramer's rule,

$$\begin{bmatrix} 10 & -2 & -5 \\ 4 & -9 & 3 \\ 10 & 3 & -18 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$$

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$$\Delta = \begin{vmatrix} 10 & -2 & -5 \\ 4 & -9 & 3 \\ 10 & 3 & -18 \end{vmatrix} = 816$$

$$\Delta_1 = \begin{vmatrix} 6 & -2 & -5 \\ 0 & -9 & 3 \\ 0 & 3 & -18 \end{vmatrix} = 918$$

$$\Delta_2 = \begin{vmatrix} 10 & 6 & -5 \\ 4 & 0 & 3 \\ 10 & 0 & -18 \end{vmatrix} = 612$$

$$\Delta_3 = \begin{vmatrix} 10 & -2 & 6 \\ 4 & -9 & 0 \\ 10 & 3 & 0 \end{vmatrix} = 612$$

Hence,

$$\therefore \quad I_1 = \frac{\Delta_1}{\Delta} = \frac{918}{816} = 1.125 \text{ A}$$

$$\therefore \quad I_2 = \frac{\Delta_2}{\Delta} = \frac{612}{816} = 0.75 \text{ A}$$

$$\therefore \quad I_3 = \frac{\Delta_3}{\Delta} = \frac{612}{816} = 0.75 \text{ A}$$

Thus,

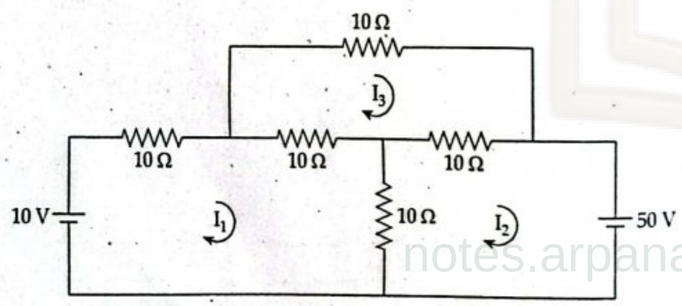
Current in 2 Ω and 5 Ω resistors is 0.75 A each.

Current in 4 Ω resistor = $I_1 - I_2 = 0.375$ A

Current 10 Ω resistor = $I_1 - I_3 = 0.375$ A

Current in 3 Ω resistor = $I_2 - I_3 = 0$ A.

14. Find the mesh currents using mesh current method.



Solution:

Applying KVL,

Loop 1:

$$-100 + 10(I_1 - I_2) + 10(I_1 - I_3) + 10 I_1 = 0$$

or, $I_1(10 + 10 + 10) - 10 I_2 - 10 I_3 = 100$

or, $30 I_1 - 10 I_2 - 10 I_3 = 100$ (1)

Loop 2:

$$50 + 10(I_1 - I_3) + 10(I_2 - I_1) = 0$$

or, $-10 I_1 + 20 I_2 - 10 I_3 = -50$ (2)

Loop 3:

$$10 I_3 + 10(I_3 - I_1) + 10(I_3 - I_2) = 0$$

or, $-10 I_1 - 10 I_2 + 30 I_3 = 0$ (3)

Writing above equation matrix form,

$$\begin{bmatrix} 30 & -10 & -10 \\ -10 & 20 & -10 \\ -10 & -10 & 30 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 100 \\ -50 \\ 0 \end{bmatrix}$$

Hence,

$$\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 30 & -10 & -10 \\ -10 & 20 & -10 \\ -10 & -10 & 30 \end{bmatrix}^{-1} \begin{bmatrix} 100 \\ -50 \\ 0 \end{bmatrix}$$

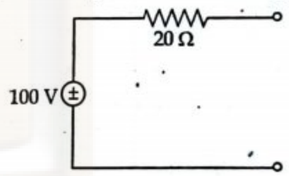
or,
$$\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \frac{1}{8{,}000} \begin{bmatrix} 500 & 400 & 300 \\ 400 & 800 & 400 \\ 300 & 400 & 500 \end{bmatrix} \begin{bmatrix} 100 \\ -50 \\ 0 \end{bmatrix}$$

$$\therefore \quad I_1 = \frac{500 \times 100 - 400 \times 50}{8{,}000} = 3.75 \text{ A}$$

$$\therefore \quad I_2 = \frac{400 \times 100 - 50 \times 800}{8{,}000} = 0 \text{ A}$$

$$\therefore \quad I_3 = \frac{300 \times 100 - 400 \times 50}{8{,}000} = 1.25 \text{ A}.$$

15. Convert the constant voltage source into equivalent current source.



Solution:

The current supplies by the 100 V source when a short circuit is placed across the output terminals is,

$$I = \frac{100}{20} = 5 \text{ A}.$$

so, The value of equivalent constant current source is 5 A.

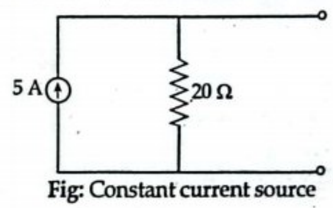


Fig: Constant current source

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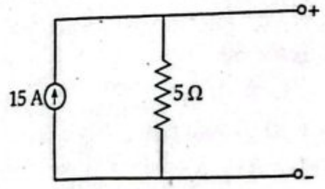
16. Convert the constant current source into equivalent voltage source.

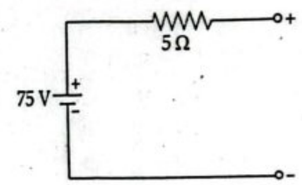


Solution:

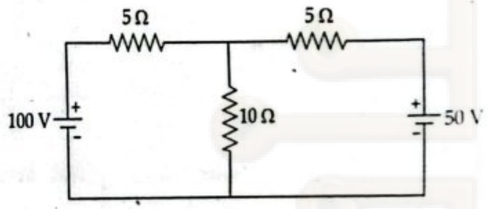
The value of the equivalent constant voltage source is given as

$$V = IR = 15 \times 5 = 75 \text{ V}.$$

Hence, the equivalent network with the voltage source is shown below:

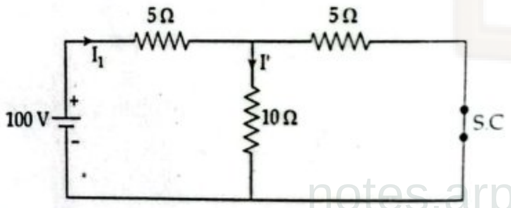


17. Compute the current in the 10 Ω resistor using superposition theorem.



Solution:

Considering the 100 V source acting alone, the direction of currents supplied by the source is shown in figure below.



Here,

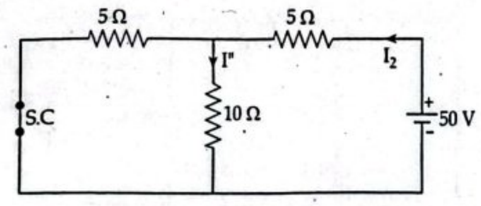
$$\therefore I_1 = \frac{100}{5 + (10 \parallel 5)} = \frac{100}{5 + 3.334}$$

$$\therefore I_1 = 12 \text{ A}.$$

Hence, current through 10 Ω resistor,

$$I' = I_1 \times \frac{5}{5 + 10} = \frac{12 \times 5}{5 + 10} = 4 \text{ A}$$

Now, considering a 50 V source acting alone, the direction of currents supplied by the source is shown below.



Here;

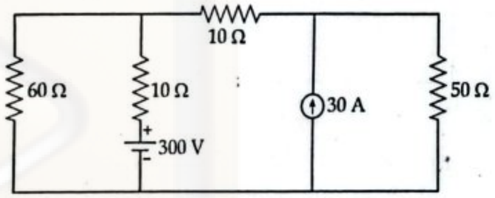
$$I_2 = \frac{50}{5 + (10 \parallel 5)} = 6 \text{ A}$$

Current through 10 Ω resistor is,

$$I'' = I_2 \times \frac{5}{5 + 10} = \frac{6 \times 5}{15} = 2 \text{ A}$$

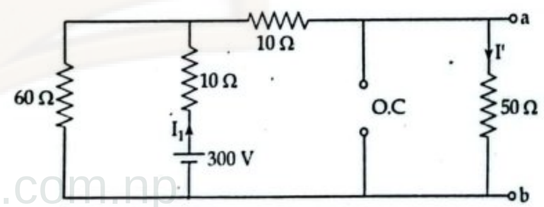
When both the sources are acting simultaneously, the current through 10 Ω resistor is given by (I' + I") *i.e.*, (4 + 2) A = 6 A.

18. Find the current through 50 Ω resistor using superposition theorem.



Solution:

Considering the voltage source acting alone and removing the current source,



Here, the total current supplied by the voltage source is,

$$I_1 = \frac{300}{10 + 60 \parallel (10 + 50)} = \frac{300}{10 + 60 \parallel 60} = \frac{300}{10 + 30} = 7.5 \text{ A}$$

Hence, current through 5 Ω resistor due to voltage source acting alone is,

$$I' = 7.5 \times \frac{60}{60 + 10 + 50} = 3.75 \text{ A (from a to b)}$$

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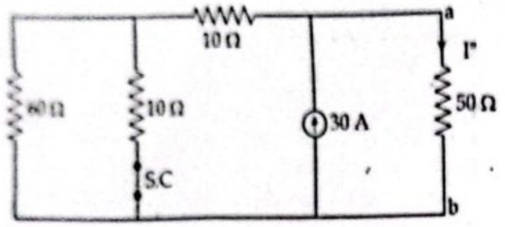
Next, removing the voltage source and considering the current source acting alone,



Current through 50 Ω resistor is,

$$I'' = 30 \times \frac{[10 + (60 \parallel 10)]}{50 + 10 + (60 \parallel 10)}$$

$$= 30 \times \frac{(10 + 8.571)}{(60 + 8.571)}$$

$$= 8.125 \text{ A}$$

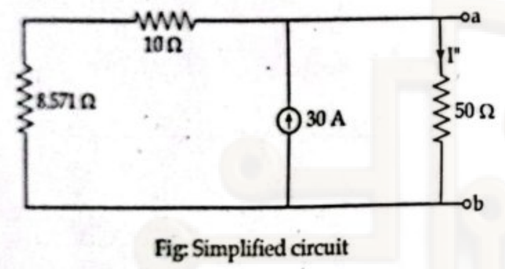


Fig: Simplified circuit

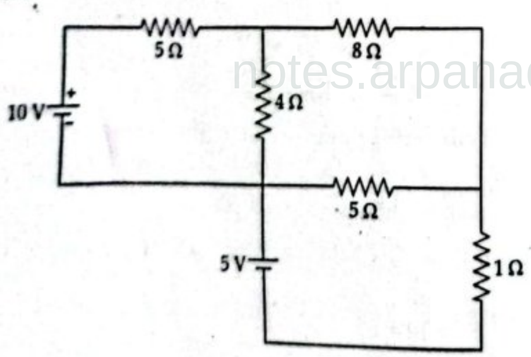
According to superposition theorem, when both the sources are acting simultaneously, the current through the 50 Ω resistor is,

$$= I' + I''$$

$$= 3.75 + 8.125$$

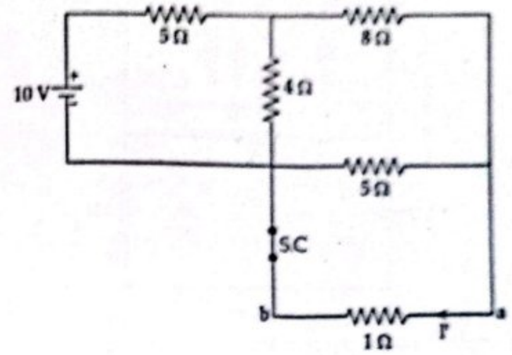
$$= 11.875 \text{ (from a to b)}$$

19. Find the current through 1 Ω resistor applying the superposition theorem.

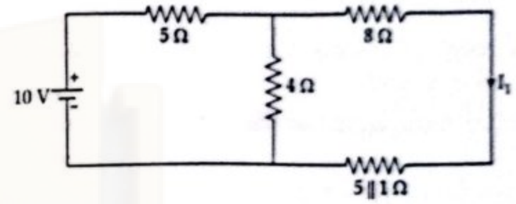


Solution:

Consider 10 V source acting alone,



It can be simplified as,



Hence,

$$I_1 = \left(\frac{10}{5 + \frac{4 \times (8 \parallel 5/6)}{4 + 8 + (5/6)}} \times \frac{4}{4 + 8 + (5/6)} \right)$$

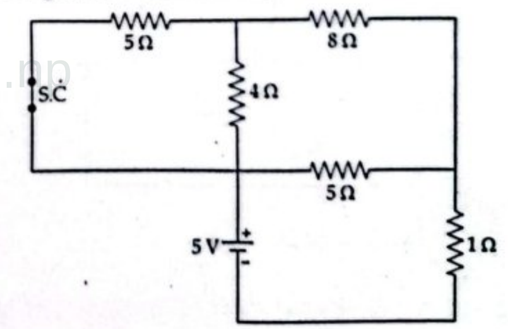
$$= 0.40 \text{ A}$$

Hence, current through 1 Ω resistor is,

$$I' = 0.4 \times \frac{5}{5 + 1}$$

$$= 0.33 \text{ A (from a to b)}$$

Next considering the 5 V source alone,



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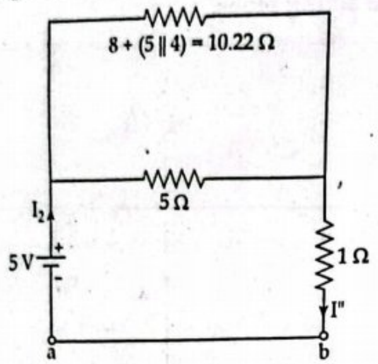
Simplifying above figure,



The current supplied by 5 V source is,

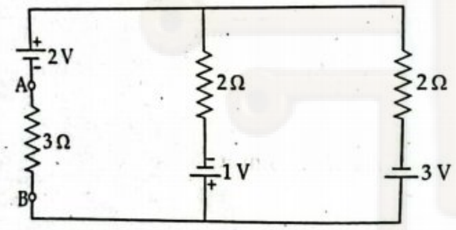
$$I_2 = \frac{5}{5 + 10.22} = 1.147 \text{ A}$$

The current through 1 Ω resistor due to the 5 V source acting along is then $I'' = I_2 = 1.147$ (from a to b).

Hence, according to superposition theorem, current through 1 Ω resistor is,

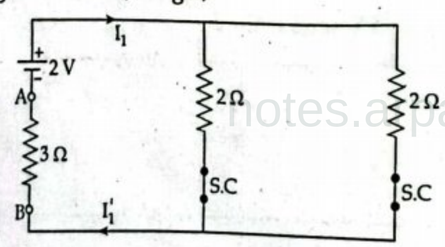
$= I' + I'' = 0.333 + 1.147 = 1.48$ A.

20. Find the current through 3 Ω resistor of the circuit shown below.



Solution:

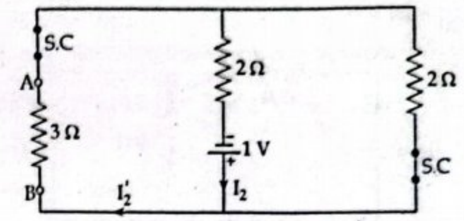
Short circuiting 1 V and 3 V, we get,



$$\therefore \quad R_1 = 3 + \frac{2 \times 2}{2 + 2} = 4\ \Omega$$

$$I_1 = I_1' = \frac{2}{4} = 0.5 \text{ A (from B to A)}$$

Now, short circuiting 2 V and 3 V sources,



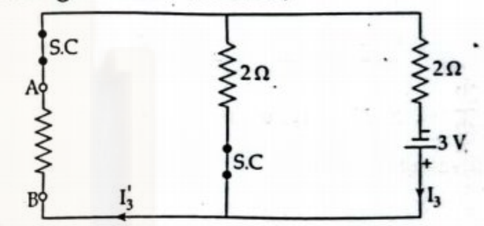
Here;

$$R_2 = 2 + \frac{2 \times 3}{2 + 3} = 3.2\ \Omega$$

$$I_2 = \frac{1}{3.2} = 0.3125 \text{ A}$$

$$I_2' = 0.3125 \times \frac{2}{2 + 3} = 0.125 \text{ A (B to A)}$$

and, Short circuiting 2 V and 1 V sources,



$$R_3 = 2 + \frac{2 \times 3}{2 + 3} = 3.2\ \Omega$$

$$I_3 = \frac{3}{3.2} = 0.9375 \text{ A}$$

$$\therefore \quad I_3' = 0.9375 \times \frac{2}{(2 + 3)} = 0.375 \text{ A (from B to A)}$$

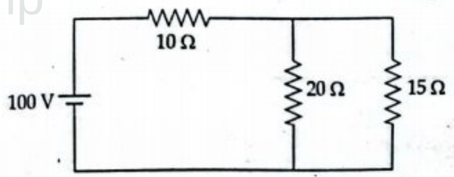
Then, Current through 3 Ω resistor,

$I = I_1 + I_2 + I_3$

$= 0.5 + 0.125 + 0.375$

$= 1$ A (from B to A).

21. Using Thevenin's theorem, find the current through 15 Ω resistor.



Solution:

Removing the 15 Ω resistor. The open circuit voltage across a and b is

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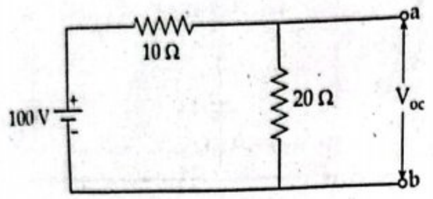
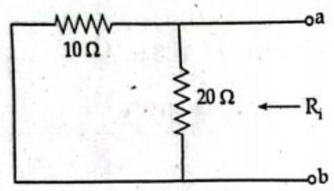
$$V_{oc} = \frac{100}{10+20} \times 20\ V = \frac{200}{3}\ V$$



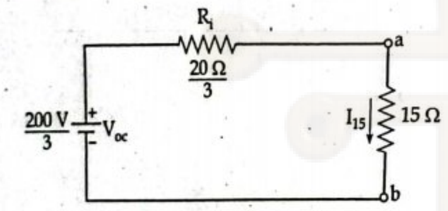
Thevenin's equation voltage, $V_{th} = V_{oc} = \frac{200}{3}\ V$

Next, removing the source, the internal resistance of the network as viewed from the open circuited terminals.



$R_i = \frac{10 \times 20}{10+20}$ *i.e.*, thevenin's equivalent resistance is $R_{th} = R_i = \frac{20}{3}\ \Omega$.

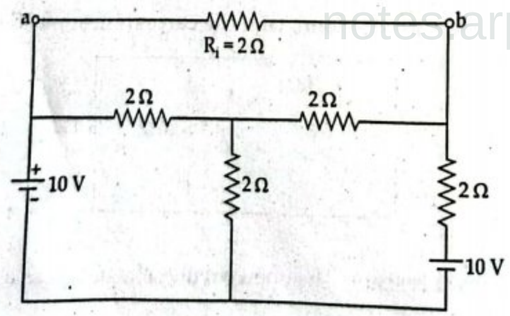
Thevenin's equivalent circuit is,



Now, the current through 15 Ω resistor is,

$$\therefore \quad I_{15} = \frac{200/3}{(20/3)+15} = 3.077\ A$$

22. Find the current in R_L of the circuit shown below using thevenin's theorem.



Solution:

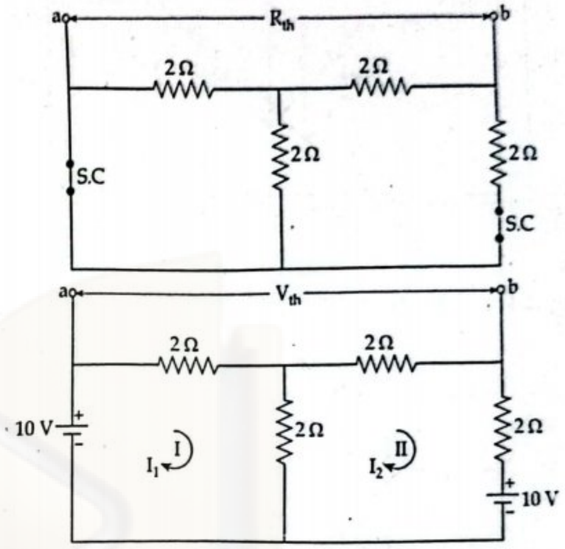
Calculation of R_{th}:

By removing R_L and short circuiting voltage sources. We get;

$$\therefore \quad R_{th} = [(2 \parallel 2) + 2] \parallel \Omega = \left[\frac{3 \times 2}{2+2} + 2\right] \parallel 2 = 3 \parallel 2 = \frac{3 \times 2}{3+2} = 1.2\ \Omega$$

Calculation of V_{th}:

By removing R_L, we get;



Applying KVL to loops I and II, we get,

$$-2\ I_1 - 2(I_1 - I_2) + 10 = 0$$

or, $4\ I_1 - 2\ I_2 = -10$

or, $2\ I_1 - I_2 = 5$ (1)

and, $-(2+2)\ I_2 - 10 - 2\ (I_1 - I_2) = 0$

or, $-I_1 + 6\ I_2 = -10$ (2)

On solving,

$$2\ I_1 - I_2 = 5$$

$$-2\ I_1 + 6\ I_2 = -10$$

$$5I_2 = -5$$

$\therefore \quad I_2 = -1\ A$

and, $2I_1 = 5 - 1$

$\therefore \quad I_1 = 2\ A$

$\therefore \quad V_{th} = 2I_1 + 2I_2 = 2\ (2 - 1) = 2V$

Hence, Current through R_L,

$$I = \frac{V_{th}}{R_{th} + R_L} = \frac{2}{1.2+2}$$

$\therefore \quad I = 0.625\ A$

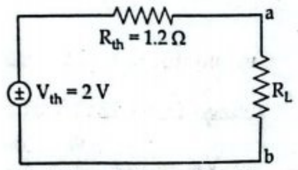


Fig: Thevenin's equation ckt.

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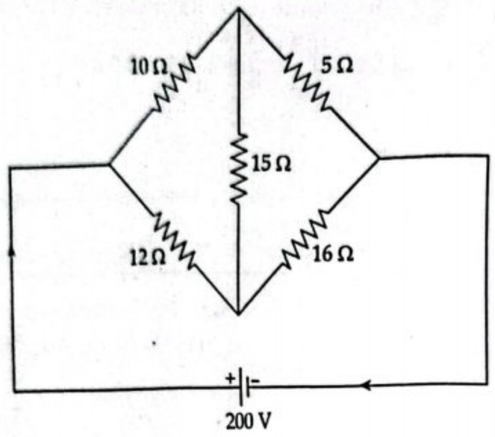
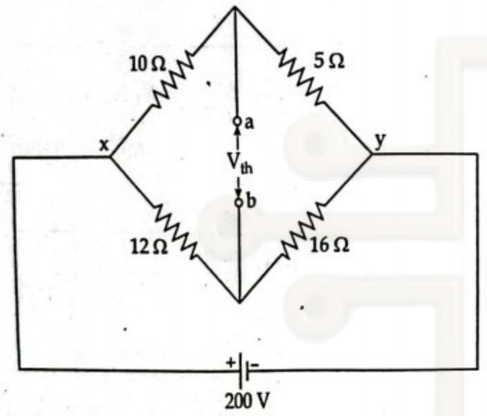
23. Find the current through 15 Ω resistor for the network shown using thevenin's theorem.



Solution:

Removing 15 Ω resistor, the open circuit voltage across its terminals is,



The current through 10 Ω resistor is, $\frac{200}{10+5} = \frac{200}{15}$ A

Voltage across 10 Ω resistor,

$$V_{xa} = 10 \times \frac{200}{15} = \frac{2,000}{15}$$

Current through 12 Ω resistor, is $\frac{200}{12+16} = \frac{200}{28}$ A

Voltage across 12 Ω resistor is,

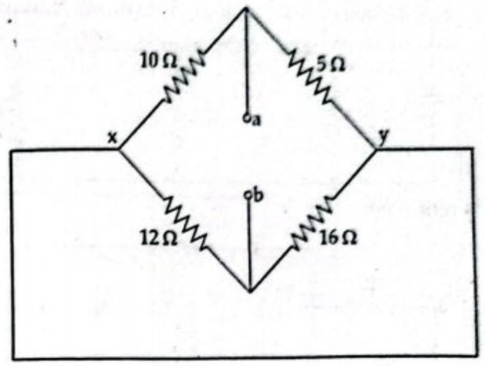
$$V_{xb} = 12 \times \frac{200}{28} = \frac{2,400}{28} \text{ V}$$

$$V_{ab} = V_{xb} - V_{xa} = \frac{2,400}{28} - \frac{2,000}{15} = -47.62 \text{ V}$$

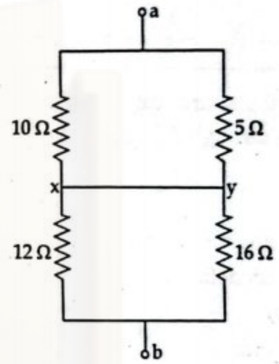
Hence, b is at higher potential with respect to a.

Thus, $V_{th} = V_{ba} = 47.62$ V

Deactivating the voltage source, Thevenin's equivalent resistance can be obtained as,



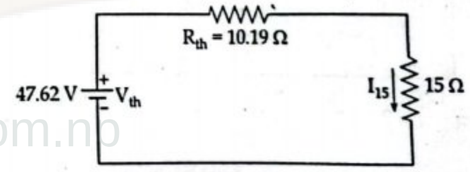
Simplifying,



The resistance between a and b is,

$$R_{th} = (10 \parallel 5) + (12 \parallel 16) = \frac{10 \times 5}{10+5} + \frac{12 \times 16}{12+16}$$

$$\therefore \quad R_i = R_{th} = 10.19 \ \Omega$$



Therefore, current through 15 Ω resistor is,

$$I_{15} = \frac{V_{th}}{R_{th} + R_L}$$

$$= \frac{47.62}{10.19 + 15}$$

= 1.89 A. (from terminal b to a)

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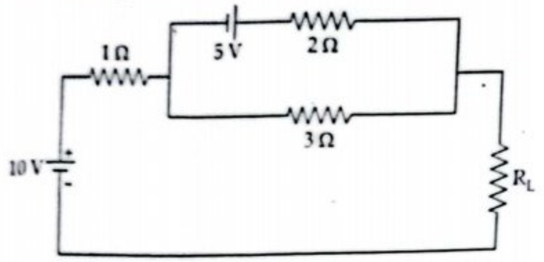
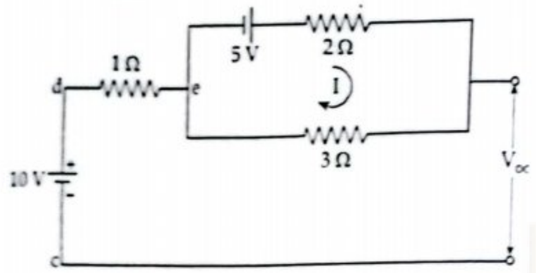
24. Find thevenin's equivalent circuit of the network across R_L.



Solution: R_L is removed:



Under this condition, 10 V source cannot deliver any current. Current due to 5 V source circulates through 2 Ω and 3 Ω resistor.

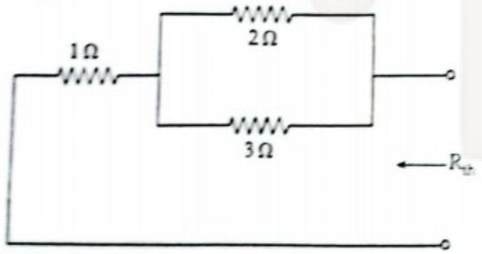
$$\therefore \quad I = \frac{5}{2+3} = 1 \text{ A}$$

Voltage across 3 Ω resistor = 3 × 1 = 3 V

Applying KVL in loop abcdefa,

$$V_{th} = 10 + 3 = 13 \text{ V}$$

Now, Deactivating the sources,



$$\therefore \quad R_{th} = 1 + \frac{3 \times 2}{3+2} = 2.2\ \Omega$$

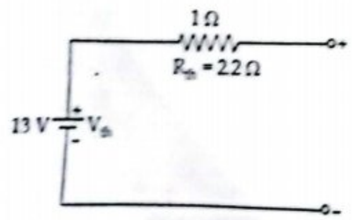
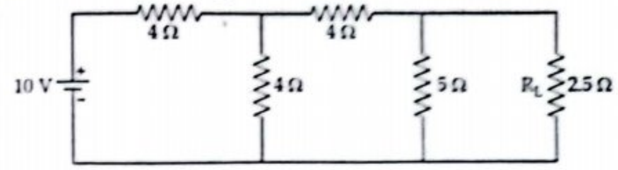


Fig: Thevenin's equivalent circuit.

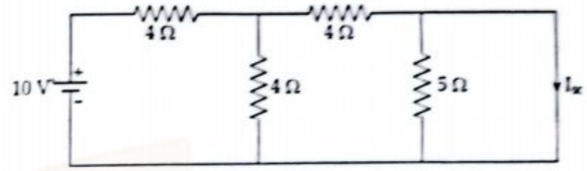
$$\therefore \quad I_R = \frac{V_{th}}{R_{th} + R_L} = \frac{13}{2.2 + R_L}$$

25. Find the current through R_L using Norton's theorem.



Solution:

R_L is removed and terminals are short circuited.



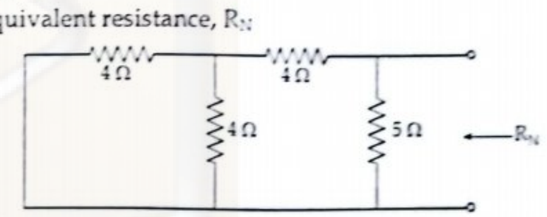
Short circuit current is,

$$I_{sc} = \left(\frac{10}{4 + \frac{4 \times 4}{4+4}} \right) = 1.67 \text{ A}$$

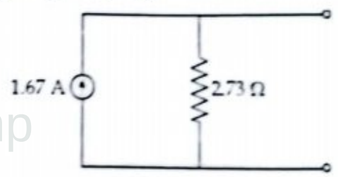
Therefore, Norton's equivalent current,

$I_N = 1.67$ A

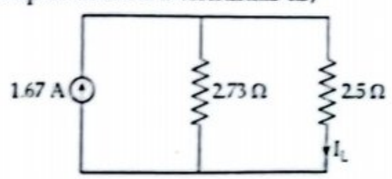
Norton's equivalent resistance, R_N



$$R_N = \left[\left(\frac{4 \times 4}{4+4} + 4 \right) \| 5 \right] = \left(\frac{6 \times 5}{6+5} \right) = 2.73\ \Omega$$



Replacing R_L across open circuited terminals as,



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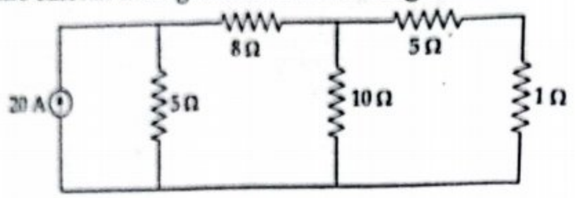
Now,

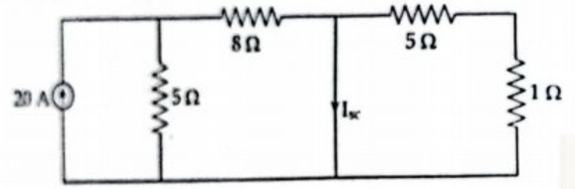
$$\therefore \quad I_L = 1.67 \times \left(\frac{2.73}{2.73 + 2.5}\right) = 0.87 \text{ A}$$

26. Find the current through 10 Ω resistor using Norton's theorem.



Solution:

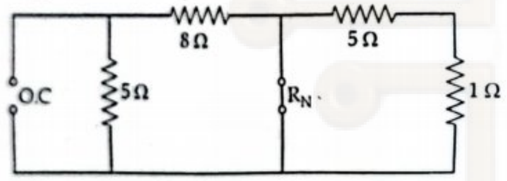
Remove 10 Ω resistor and short circuit terminals.



Current through the short circuited path is,

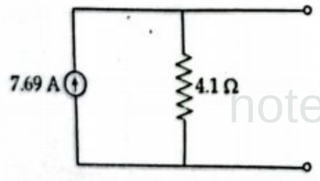
$$I_{sc} = 20 \times \frac{5}{5 + 8} = 7.69 \text{ A}$$

Hence, Norton's equivalent current, $I_N = 7.69$ A. Norton's equivalent resistance as seen from the open circuited terminals of the network is obtained as,

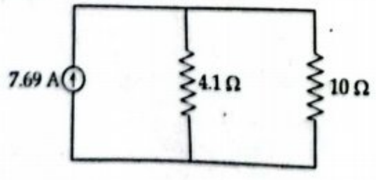


$$R_N = (8 + 5) \parallel (5 + 1) = \frac{13 \times 6}{13 + 6} = 4.1\ \Omega$$

Norton's equivalent circuit is,



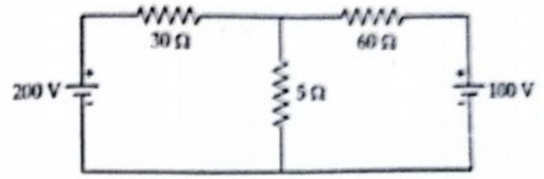
Hence, current through the 10 Ω resistor is,



Now,

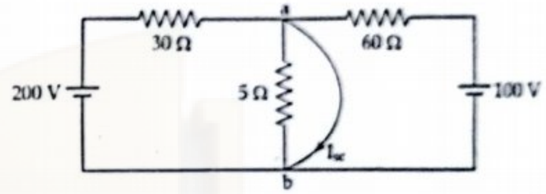
$$\therefore \quad I_L = 7.69 \times \left(\frac{4.1}{4.1 + 10}\right) = 2.236 \text{ A}$$

27. Find current through 5 Ω resistor in the circuit below using Norton's theorem.



Solution:

5 Ω resistor is short circuited.

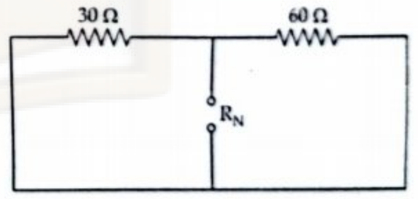


$I_{sc} = \frac{200}{30} + \frac{100}{60} = 8.33$ A from a to b.

$\therefore \quad I_{sc} = I_N = 8.33$ A

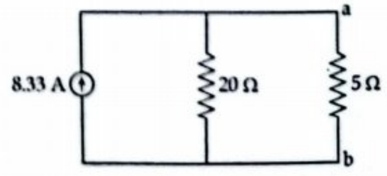
Now,

The Norton equivalent resistance is obtained by removing all sources and looking from open circuited terminals a and b.



$$R_N = \frac{30 \times 60}{30 + 60} = 20\ \Omega$$

Norton's equivalent circuit is,



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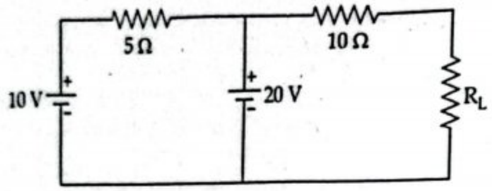
Hence, current through 5 Ω resistor is,

$$\therefore \quad I = 8.33 \times \frac{20}{20+5} = 6.664 \text{ A (from a to b).}$$

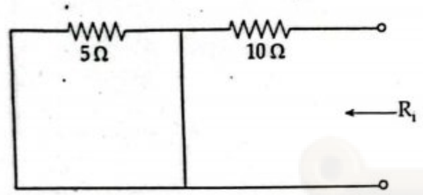
28. Calculate the value of R_L which will abstract the maximum power from the circuit shown below. Also find maximum power.



Solution:

Removing all the sources and open circuiting the terminals of R_L the internal resistance R_i of the network is found as 10 Ω.

i.e., $R_i = 10\ \Omega$



Therefore, for maximum power transfer,

$R_i = R_L = 10\ \Omega$

Again, for $R_L = 10\ \Omega$, the total current through R_L due to both sources is given by,

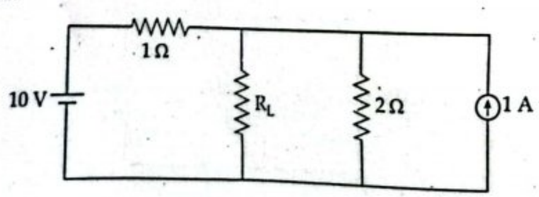
$$I = \frac{20}{\frac{5 \times (10+10)}{5+(10+10)}} \times \frac{5}{5+(10+10)} = 1 \text{ A}$$

Therefore, the current due to 10 V source circulates through 5 Ω resistor and 20 V source only.

and, Maximum power across load

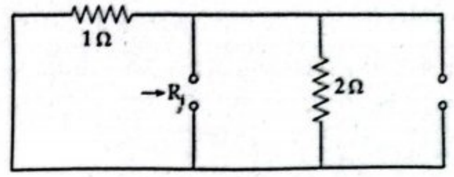
$$= I^2 \cdot R_L = I^2 \times 10 = 10 \text{ W}$$

29. Calculate the value of R_L which will absorb maximum power from the circuit shown below. Also calculate the value of this maximum power.



Solution:

Let, R removed and internal resistance of the network is calculated looking from the open circuited terminals after removing all the sources as shown in figure.



Here;

$$R_i = \frac{1 \times 2}{1+2} = \frac{2}{3}\ \Omega$$

i.e., $R = R_i = \frac{2}{3} = 0.667\ \Omega$ for maximum power transfer.

The current through R due to both sources acting simultaneously is,

$$I = \left[\frac{10}{1 + \frac{0.667 \times 2}{0.667+2}} \times \left(\frac{2}{2+0.667} \right) + 1 \times \frac{\left(\frac{2 \times 1}{2+1} \right)}{0.667 + \left(\frac{2 \times 1}{2+1} \right)} \right]$$

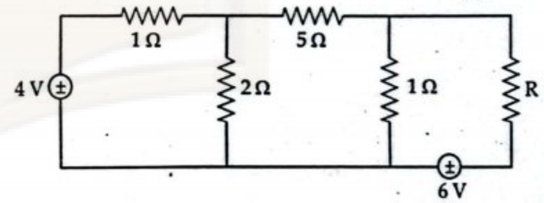
$$= 4.998 + 0.5$$

$$= 5.5 \text{ A}$$

Therefore, Maximum power,

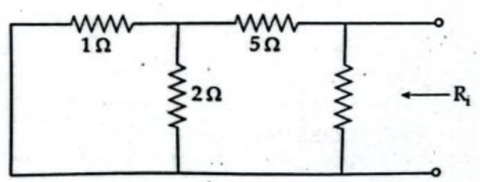
$$P_{max} = I^2 R_L = 5.5^2 \times 0.667 = 20 \text{ W.}$$

30. Find the value of R in the circuit such that the maximum power transfer takes place. What is the amount of this power?



Solution:

Deactivating all the sources, internal resistance R_i of the network is found as,



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Now,

$$R_t = [(2 \parallel 1) + 5] \parallel 1 = \frac{5.67 \times 1}{5.67 + 1} = 0.85\ \Omega$$

According to MPTT, the maximum power takes place across R when

$$R = R_t = 0.85\ \Omega$$

The current through R due to 4V source acting alone is,

$$I_1 = \left[\frac{4}{\left[\left(\frac{0.85 \times 1}{0.85 + 1} \parallel 2\right)\right] + 1} \times \frac{2}{2 + 5 + \frac{1 \times 0.85}{1 + 0.85}} \times \frac{1}{1 + 0.85}\right]$$

$$= \frac{4}{\left(\frac{5.46 \times 2}{5.46 + 2} + 1\right)} \times \frac{2}{13.8} = 0.235\ \text{A}$$

The current through R due to 6V source acting alone is

$$I_2 = \frac{6}{\left[\left(\frac{1 \times 2}{1 + 2} + 5\right) \parallel 1\right] + 0.85} = \frac{6}{\frac{5.67 \times 1}{5.67 + 1} + 0.85} = 3.53\ \text{A}.$$

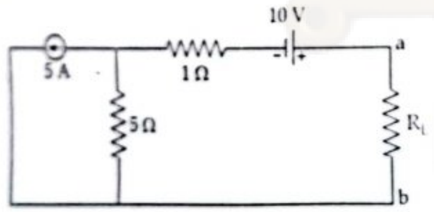
According to superposition, the current through R when both the sources are acting simultaneously is

$$I = I_1 + I_2 = 0.235 + 2.53 = 3.765\ \text{A}$$

Thus, the maximum power is

$$I^2R = (3.765)^2 \times 0.85 = 12\ \text{W}.$$

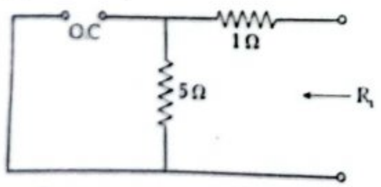
31. Find the value of R_L for which the power transfer across R_L is maximum.



Solution:

Deactivating the sources, the internal resistance of the network is found out looking back from the open circuited terminals of R_L.

$\therefore$ $R_t = 1 + 5 = 6\ \Omega$



Power transfer across R_L is maximum when,

$$R_t = R_L = 6\ \Omega$$

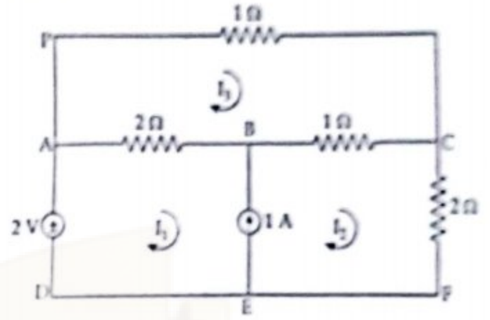
Current through (R_L) is,

$$I = \left[\frac{10}{6 + 5 + 1} - 5 \times \frac{5}{6 + 5 + 1}\right] = -1.25\ \text{(from a to b)}$$

Therefore, Value of maximum power,

$$P_{max} = I^2 \cdot R_L = 1.25^2 \times 6 = 9.375\ \text{W}$$

32. From the circuit shown, use loop analysis to determine the loop current I_1, I_2, I_3.



Solution:

From figure, the current source of 1A is equivalent to $(I_2 - I_1)$.

i.e. $I_2 - I_1 = 1\ \text{A}$ **(1)**

From loop ABCFA,

$$2(I_3 - I_1) + 1(I_3 - I_2) + 1 \times I_3 = 0$$

or, $-2I_1 - I_2 + 4I_3 = 0$ **(2)**

From loop ABCFEDA,

or $2(I_1 - I_3) + 1(I_2 - I_3) + 2I_2 - 2 = 0$

or, $2I_1 + 3I_2 - 3I_3 = 2$ **(3)**

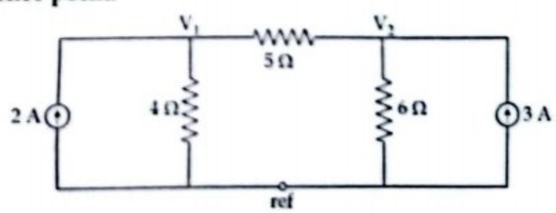
On solving,

$\therefore$ $I_1 = \frac{-1}{11}\ \text{A}$

$\therefore$ $I_2 = \frac{10}{11}\ \text{A}$

$\therefore$ $I_3 = \frac{2}{11}\ \text{A}$

33. Determine the voltage at nodes 1 and 2 with respect to the reference point.



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Solution:

Applying nodal analysis, at node (1);

$$\frac{V_1}{4}+\frac{V_1-V_2}{5}-2=0$$

or, $9V_1-4V_2=40$ (1)

Applying nodal analysis at node (2);

$$\frac{V_2-V_1}{5}+\frac{V_2}{6}-3=0$$

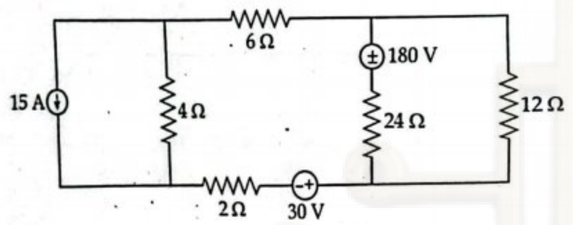
or, $-6V_1+11V_2=90$ (2)

On solving,

$\therefore \quad V_1=10.66\ V$

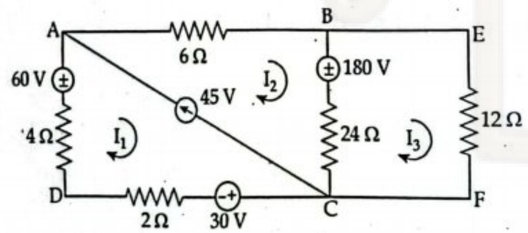
$\therefore \quad V_2=14\ V$

34. Using loop equations, obtain the current in the 12 Ω resistor of the network shown below.



Solution:

First replace 15 A current source by an equivalent voltage source; the corresponding figure,



Applying KVL in loop ABCDA,

$$6I_2+180+24(I_2-I_3)+30+6I_1+60=0$$

or, $I_1+5I_2-4I_3=-45$ (1)

and, $I_2-I_1=45$ (2)

Applying KVL in loop BEFCB,

or, $12I_3+24(I_3-I_2)-180=0$

or, $-2I_2+3I_3=15$ (3)

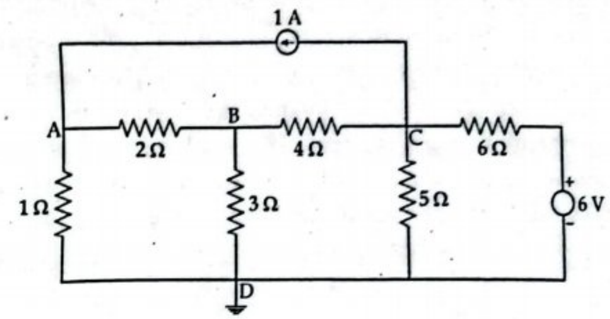
Solving equation (1), (2) and (3)

$\therefore \quad I_2=6A$

$\therefore \quad I_3=9A$

$\therefore \quad I_1=-39\ A$

35. Use nodal analysis to determine the power in the 2 Ω and 4 Ω resistor.



Solution:

At node A,

$$\frac{V_A-V_B}{2}+V_A-1=0$$

or, $3V_A-V_B=2$ (1)

At node B,

$$-\frac{V_A-V_B}{2}+\frac{V_B-V_C}{4}+\frac{V_B}{3}=0$$

or, $6V_B-6V_A+3V_B-3V_C+4V_B=0$

or, $-6V_A+13V_B-3V_C=0$ (2)

At node C,

$$\frac{V_C-V_B}{4}+\frac{V_C}{4}+\frac{V_C-6}{6}+1=0$$

or, $15V_C-15V_B+12V_C+10V_C-60+60=0$

or, $-15V_B-37V_C=0$

Solving equation (1), (2) and (3), we get;

$\therefore \quad V_A=0.8031\ V$

$\therefore \quad V_B=0.4088\ V$

$\therefore \quad V_C=0.1658\ V$

Now,

$$\text{Power in 2 Ω resistor}=\frac{(V_A-V_B)^2}{2}=\frac{(0.8031-0.4088)^2}{2}=0.078\ W$$

$$\text{and, Power in 4 Ω resistor}=\frac{(V_B-V_C)^2}{4}=\frac{(0.4088-0.1658)^2}{4}=0.01476\ W.$$

$\left(\because P=\frac{V^2}{R}\right)$.

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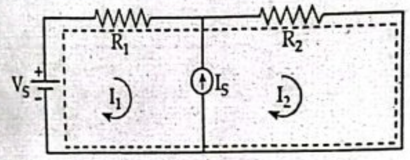
BOARD EXAM SOLVED PROBLEMS

NOTE:

Supermesh and Supernode concept

A supermesh occurs when a current source is contained between two essential meshes. The circuit is first treated as if the current sources is not there. This leads to one equation that incorporates two mesh currents. Once this equation is formed, an equation is needed that relates the two mesh current with the current source. This will be an equation where the current source is equal to one of the mesh currents minus the other. The following is a simple example of dealing with a supermesh.

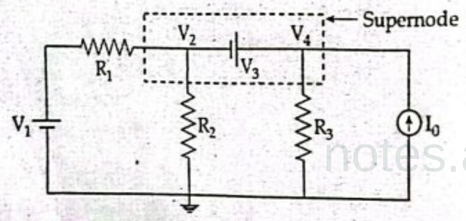


Loop 1 and loop 2 forms supermesh, applying KVL on supermesh,

$$+V_s - I_1R_1 - I_2R_2 = 0 \quad (1)$$

and, Current, $I_s = I_2 - I_1$ (2)

A supernode is the combination of two extraordinary nodes (excluding the reference node) between which a voltage source exists. The voltage source may be of the independent or dependent type and the voltage source may include elements in parallel with it but not in series with it. If one of the two nodes of a supernode is a reference node, it is called a quasi supernode. If a supernode contains a resistor in parallel with the voltage source, the resistor will exercise no influence on the currents and voltages in the other parts of the circuit, and therefore, it may be ignored altogether.



Here,

$$\frac{V_2 - V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_4}{R_3} = 0 \quad (1)$$

and, $V_4 - V_2 = V_3$ (2)

On solving equation (1) and (2); we can get required voltages. V_2 and V_4 and other parameters.

1. Define active and passive elements. **[13/F, 16/F]**

Solution:

Active elements are those elements that requires external source for their operation. In other words, those elements which produce energy in the form of voltage or current is called as active elements. Example: Diode, transistor etc.

Passive elements are those elements that does not requires external source for their operation. In other words, those elements which store or maintain energy in the form of voltage or current is called as passive elements.

Example: Capacitor, inductor, resistor.

2. State Thevenin's theorem. **[13/F, 16/F, 17/S]**

Solution: See the definition part 2.14 'B'.

3. State maximum power transfer theorem. Prove that the maximum power transferred to the load is $\frac{V_{th}^2}{4R_{th}}$. **[13/F, 14/F, 15/ F, 16/S, 17/F, 18/F, 19/S]**

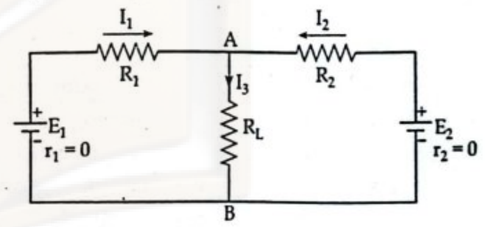
Solution: See the definition part 2.15.

4. State and explain Norton's theorem. **[13/S, 18/S, 19/S]**

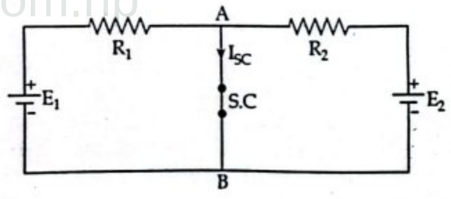
Solution: See the definition part 2.14 'C' and.

Illustration:

Let, we want to find voltage and current, across the load resistance R_L.



When a short circuit is placed between A and B, then currents between A and B, due to emf sources E_1 and E_2 are $\frac{E_1}{R_1}$ and $\frac{E_2}{R_2}$ respectively.



$$\therefore \quad I_{SC} = \frac{E_1}{R_1} + \frac{E_2}{R_2}$$

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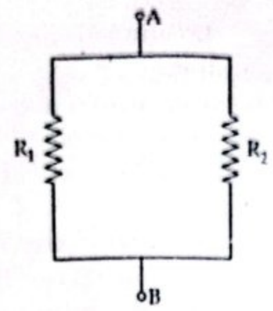
Therefore, Internal resistance of the network as viewed from A and B,

Thus,

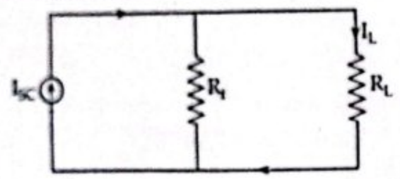
$$R_t = R_1 \parallel R_2 = \left(\frac{R_1 \times R_2}{R_1 + R_2}\right)$$

$$\therefore \quad I_L = I_{sc} \times \left(\frac{R_t}{R_t + R_L}\right)$$



Hence,

Norton's equivalent circuit is,



5. Write short notes on ideal and practical sources. [13/S, 14/F, 16/F, 16/S, 17/F]

Solution: See the definition part 2.9.

6. Write short notes on star/ delta transformation. [15/F, 17/S, 18/F]

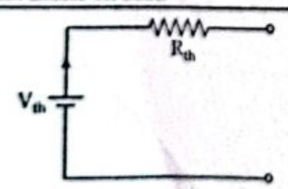
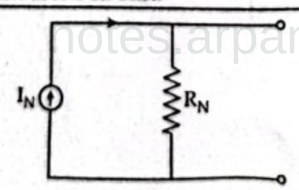
Solution: See the definition part 2.6.

7. Illustrate about the voltage divider and current divider circuit. [16/S]

Solution: See the definition part 2.11.

8. Differentiate between Thevenin's theorem and Norton's theorem. [17/F, 18/F]

Solution:

SN	Thevenin's theorem	Norton theorem
a)	It represents the circuits in terms of voltage source in series with its internal resistance with respect to load.	It represents the circuit in terms of current source in parallel with its internal resistance with respect to load.
b)	It is a voltage form of an equivalent circuit.	It is a current form of an equivalent circuit.
c)	 Thevenin's equivalent.	 Norton's equivalent circuit.

9. Explain KCL and KVL. [18/F]

Solution: See the definition part 2.12.

10. State superposition theorem. [15/S, 18/S]

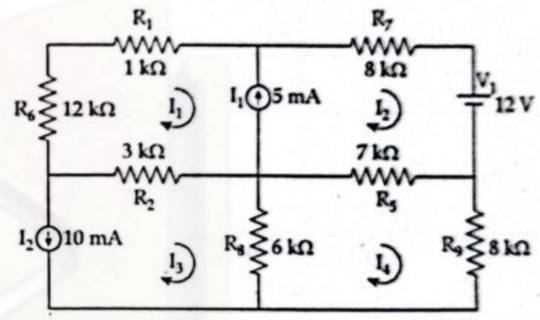
Solution: See the definition part 2.14 'A'.

11. Compare mesh analysis and nodal analysis. [17/F]

Solution:

SN	Mesh analysis	Nodal analysis
a)	Mesh analysis is the application of kirchoff's voltage law (KCL) to solve for mesh currents.	Nodal analysis is the application of kirchoff's current law (KVL) to solve for voltages at each node.
b)	Mesh analysis is best suited when energy sources are voltage sources rather than current sources.	It is best suited when there are current sources.

12. Determine the voltage across 3 kΩ resistor using mesh analysis for the circuit shown below. [14/F]



Solution:

Applying KVL in mesh-4

or, $-7(I_4 - I_2) - 8I_4 - 6(I_4 - I_3) = 0$

or, $-7I_4 + 7I_2 - 8I_4 - 6I_4 + 6I_3 = 0$

or, $7I_2 + 6I_3 - 21I_4 = 0$ (1)

From mesh 1 and 2,

$I_2 - I_2 = 5$

$\therefore \quad -I_1 + I_2 = 5$ (2)

Applying KVL in supermesh (mesh 1 and 2)

or, $-I_1 - 8I = 2 - 12 - 7(I_2 - I_4) - 3(I_1 - I_3) - I_2 I_1 = 0$

or, $-16I_1 - 15I_2 + 3I_3 + 7I_4 = 12$ (3)

From mesh – 3,

$I_3 = -10$ (4)

Hence, putting value of I_3 in equation (3);

or, $-16I_1 - 15I_2 + 3(-10) + 7I_4 = 12$

$\therefore \quad -16I_1 - 15I_2 + 7I_4 = 42$ (5)

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Also, from equation (1);

or, $7I_2 + 6(-10) - 21I_4 = 0$

$\therefore \quad 7I_2 - 21I_4 = 60$ (6)

Solving equation (2), (5) and (6),

$\therefore \quad I_1 = -3.046$ mA

$\therefore \quad I_2 = 1.953$ mA

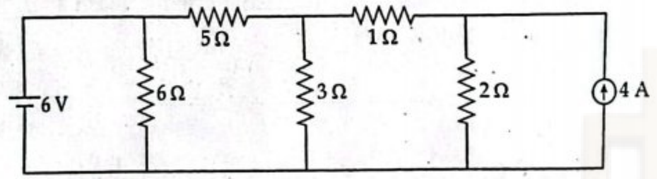
$\therefore \quad I_4 = -2.205$ mA

Hence, voltage across 3 kΩ $= (I_3 - I_1) \times 3$

$= [-10 - (-3.046)] \times 3$

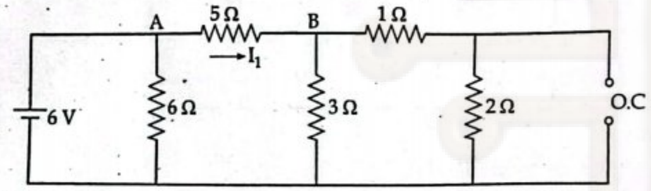
$= -20.86$ V.

13. Applying superposition theorem, calculate the current in 5 Ω resistor of circuit given below. [14/S]

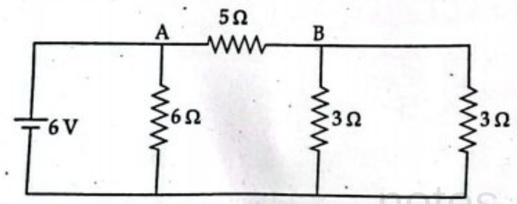


Solution:

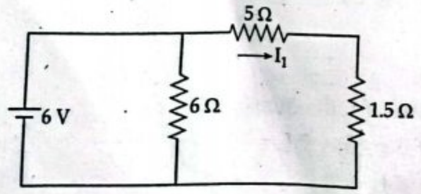
Considering 6 V source only *i.e.* open circuit 4 A source then,



Simplifying, 1 Ω and 2 Ω are in series and resultant is parallel to 3 Ω so,



$$R_{eq} = [(3 \parallel 3) + 5] \parallel 6 = \left[\frac{3 \times 3}{3+3} + 5\right] \parallel 6 = 6.5 \parallel 6 = \frac{6.5 \times 6}{6.5 + 6} = 3.12\ \Omega$$

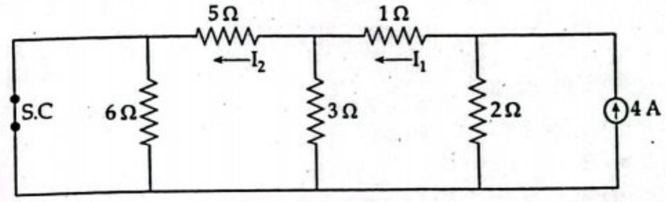


Now,

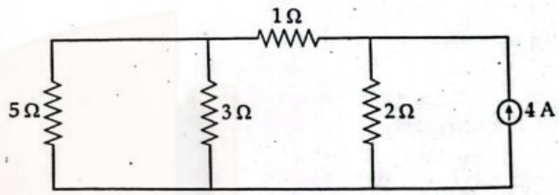
$$I = \frac{V_1}{R_{eq}} = \frac{6}{3.12} = 1.92\ \Omega$$

$$\therefore \quad I_1 = I \times \frac{6}{6+6.5} = \frac{1.92 \times 6}{6+6.5} = 0.92 \text{ (from A to B).}$$

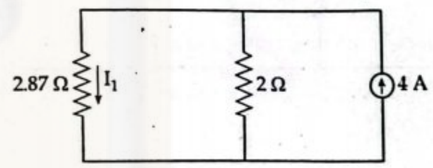
Considering 4 A source only, (short circuiting voltage source).



6 Ω resistor is short circuited. So, 5 Ω and 0 Ω are in series and the resultant is parallel with 3 Ω.



$$\therefore \quad R_{eq} = [(3 \parallel 5) + 1] \parallel 2 = \left[\frac{3 \times 5}{3+5} + 1\right] \parallel 2 = 2.87 \parallel 2 = \frac{2.87 \times 2}{2.87 + 2} = 1.178\ \Omega$$

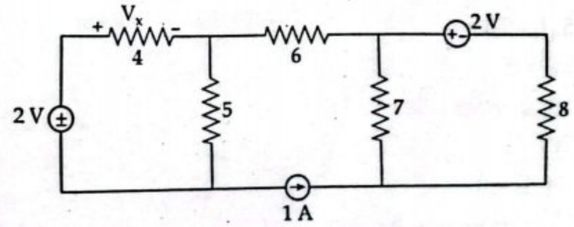


$$I_1 = 4 \times \frac{2}{2 + 2.87} = 1.64 \text{ A}$$

and, $I_2 = \frac{I' \times 3}{5+3} = \frac{1.64 \times 3}{5+3} = 0.61$ A (from B to A) $= -0.61$ A (from A to B)

$\therefore$ Net current through 5 Ω $= I_1 + I_2 = 0.92 + (-0.61)$ A $= 0.31$ A

14. Use superposition theorem to calculate V_x for the circuit shown below (All resistors are in ohm). [16/F]



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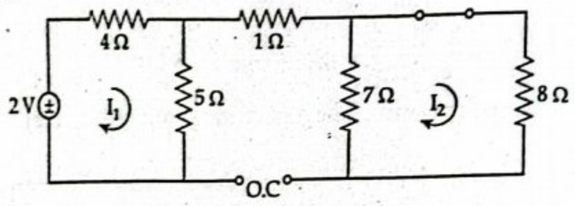
Solution:

Considering 2 V source only,



Applying KVL in mesh-1,

or, $-4I_1 - 5I_1 + 2 = 0$

or, $-9I_1 = -2$

$\therefore \quad I_1 = 0.22$ A

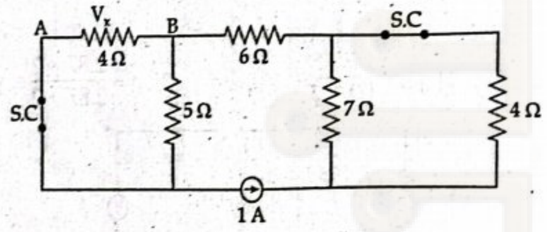
Applying KVL in mesh - 2,

or, $-8I_2 - 7I_2 = 0$

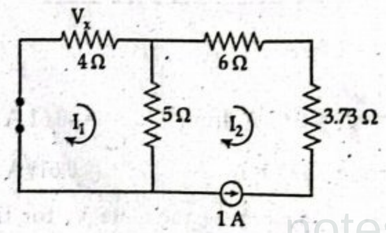
or, $-15I_2 = 0$

$\therefore \quad I_2 = 0$

Considering 1 A source only,



On simplifying,



Applying KVL in mesh - 1,

or, $-4I_1 - 5(I_1 - I_2) = 0$

$\therefore \quad -9I_1 + 5I_2 = 0$ (1)

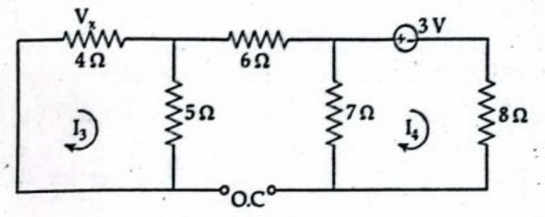
Applying KVL in mesh - 2;

$I_2 = -1$ A

so, $-9I_1 + 5(-1) = 0$

$\therefore \quad I_1 = -0.55$ A (from B to A)

Considering 3 V source only,



Applying KVL,

or, $-4I_3 - 5I_3 = 0$

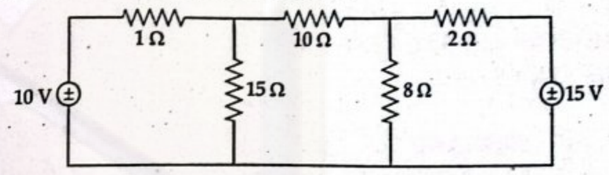
$\therefore \quad I_3 = 0$ [from A to B]

$\therefore$ Net current through 4 Ω $= I_1 + I_2 + I_3 = 0.22 - 0.55 + 0$

$\therefore \quad I = -0.33$ A [from A to B]

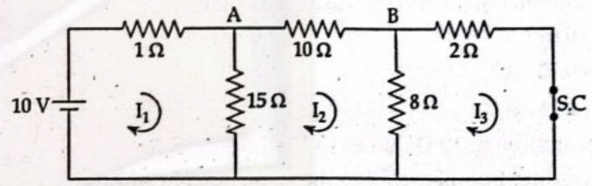
$\therefore \quad V_x = I \times R_x = -0.33 \times 4 = -1.32$ V.

15. Use superposition theorem to calculate the current through 10 Ω resistor of the circuit shown below. [13/F]



Solution:

Considering 10 V source only and short circuiting 15 V source,



Applying KVL,

Loop 1:

or, $-I_1 - 15(I_1 - I_2) + 10 = 0$

or, $-16I_1 + 15I_2 = -10$ (1)

Loop 2:

or, $-10I_2 - 8(I_2 - I_3) - 15(I_2 - I_1) = 0$

or, $15I_1 - 33I_2 + 8I_3 = 0$ (2)

Loop 3:

or, $-2I_3 - 8(I_3 - I_2) = 0$

or, $8I_2 - 10I_3 = 0$ (3)

Solving equation (1), (2), and (3)

$\therefore \quad I_1 = 1.326$ A

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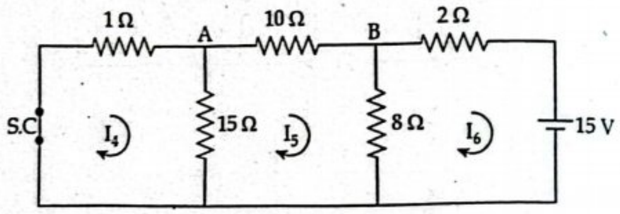
$\therefore$ $I_2 = 0.748$ A

$\therefore$ $I_3 = 0.598$ A

Hence,

Current through 10 Ω due to 10 V source = $I_2 = 0.748$ A

Considering 15 V source only, *i.e.*, short circuiting 10 V source,



Applying KVL:

Loop 1:

or, $-I_4 - 15(I_4 - I_5) = 0$

or, $-16 I_4 + 15 I_5 = 0$

Loop 2:

or, $-10 I_5 - 8(I_5 - I_6) - 15(I_5 - I_4) = 0$

or, $-10 I_5 - 8 I_5 + 8 I_6 - 15 I_5 + 15 I_4 = 0$

or, $15 I_4 - 33 I_5 + 8 I_6 = 0$

Loop 3:

or, $-2 I_6 - 15 - 8(I_6 - I_5) = 0$

or, $-2 I_6 - 15 - 8 I_6 + 8 I_5 = 0$

or, $8 I_5 - 10 I_6 = 15$

Solving equation (1), (2) and (3), (using calculator)

$\therefore$ $I_4 = -0.897$ A

$\therefore$ $I_5 = -0.957$ A

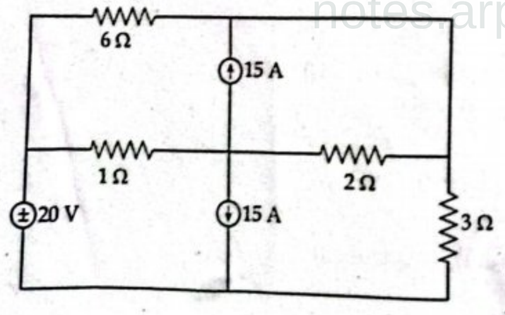
$\therefore$ $I_6 = -2.265$ A

$\therefore$ Current through 10 Ω due to 15 V = $I_5 = -0.95$ A

Hence, according to superposition theorem, current through

10 Ω = $I_2 + I_5 = 0.74 - 0.95 = -0.21$ A.

16. Calculate current through 3 Ω of the circuit shown below using superposition theorem. [13]



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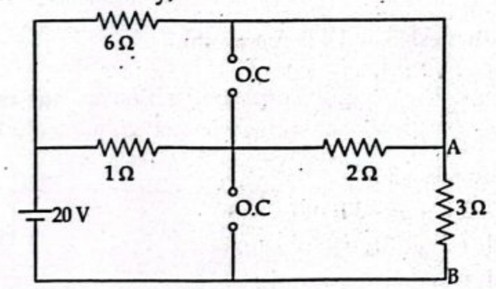
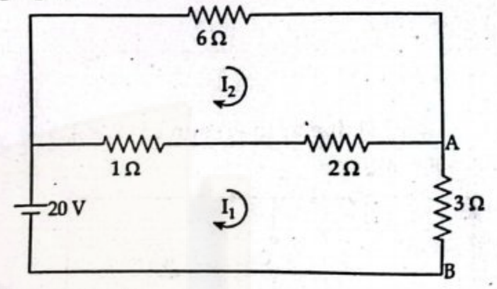
Solution:

Considering 20 V source only,



On simplifying figure,



Applying KVL, on mesh-1:

or, $-(I_1 - I_2) - 2(I_1 - I_2) - 3I_1 + 20 = 0$

or, $-I_1 + I_2 - 2I_1 + 2I_2 - 3I_1 = -20$

or, $-6 I_1 + 3 I_2 = -20$ (1)

On mesh-2:

or, $-6 I_2 - 2(I_2 - I_1) - 1(I_2 - I_1) = 0$

or, $-6 I_2 - 2 I_2 + 2 I_1 - I_2 + I_1 = 0$

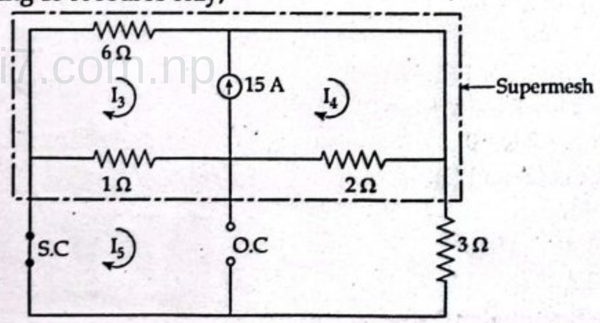
or, $3 I_1 - 9 I_2 = 0$ (2)

On solving,

$\therefore$ $I_1 = 4$ A $\quad\quad$ $\therefore$ $I_2 = 1.33$ A

Hence, current through 3 Ω resistor due to 20 V source = $I_1 = 4$ A

Considering 15 A source only,



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Here;

$I_4 - I_3 = 15$ A

Applying KVL in mesh 3 and 4 (supermesh),

or, $-6 I_3 - 2 (I_4 - I_5) - 1 (I_3 - I_5) = 0$

or, $-6 I_3 - 2 I_4 + 2I_5 - I_3 + I_5 = 0$

or, $-7 I_3 - 2 I_4 + I_5 = 0$

Applying KVL in mesh 3,

or, $-1 (I_5 - I_3) - 2(I_5 - I_4) - 3 I_5 = 0$

or, $-I_5 + I_3 - 2I_5 + 2 I_4 - 3I_5 = 0$

or, $I_3 + 2I_4 - 6I_5 = 0$

On solving equation (1), (2) and (3),

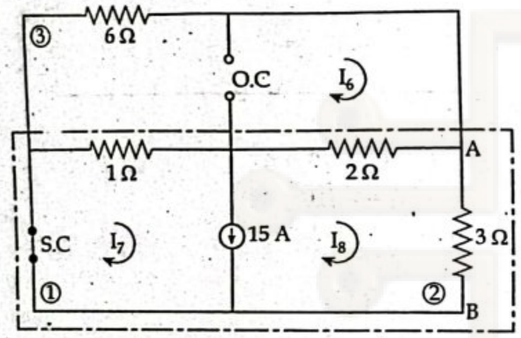
$\therefore I_3 = -2$ A

$\therefore I_4 = 13$ A

$\therefore I = 4$ A

Hence, current through 3 Ω due to 15 A source is $I_5 = 4$ A

Considering 15 A source only,



Here;

$I_7 - I_8 = 15$ A

Applying KVL is supermesh,

or, $-1(I_7 - I_6) - 2(I_8 - I_6) - 3I_8 = 0$

or, $-I_7 + I_6 - 2 I_8 + 2 I_6 - 3I_8 = 0$

or, $3 I_6 - I_7 - 5 I_8 = 0$

On loop 3:

or, $-6 I_6 - 2(I_6 - I_8) - 1 (I_6 - I_7) = 0$

or, $-6 I_6 - 2 I_6 + 2 I_8 - I_6 + I_7 = 0$

or, $-9 I_6 + I_7 + 2 I_8 = 0$

On solving (1), (2) and (3);

$\therefore I_6 = 1$ A

$\therefore I_7 = 13$ A

$\therefore I_8 = -2$ A

$\therefore$ Current through 3 Ω = $I_8 = -2$ A

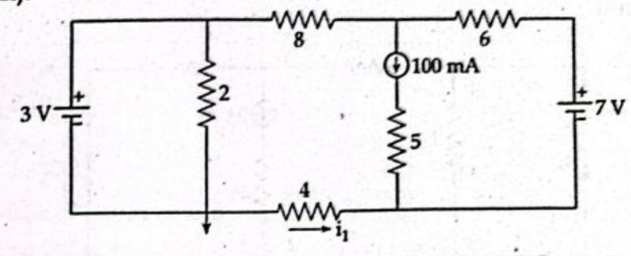
Now,

$\therefore$ Current through 3 Ω $= I_1 + I_5 + I_8$

$= 4 + 4 - 2$

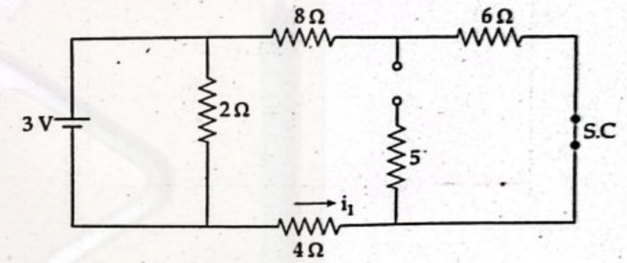
$= 6$ A (from A to B)

17. Solve for the current i_1 (through the 4 Ω resistor) in the circuit shown below using superposition theorem. (All resistors are in ohm). [17/S]

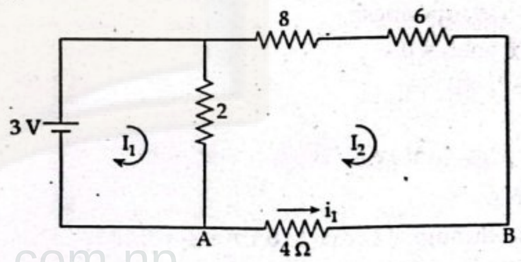


Solution:

Considering 3 V source only, *i.e.*, short circuiting 7 V source and open circuiting 100 mA source,



On simplifying,



Applying KVL,

Loop 1:

or, $3 - 2 (I_1 - I_2) = 0$

or, $-2I_1 + 2I_2 = -3$ (1)

Loop 2:

or, $-14 I_2 - 4 I_2 - 2(I_2 - I_1) = 0$

or, $-18 I_2 - 2 I_2 + 2 I_1 = 0$

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or, $2I_1 - 20I_2 = 0$

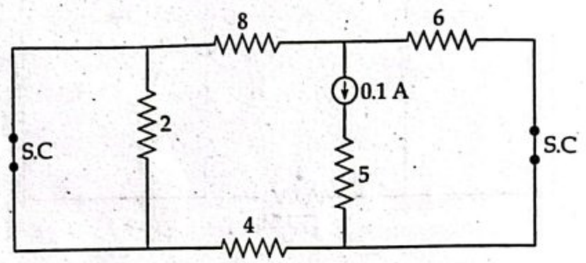
On solving,

$\therefore \quad I_1 = 1.66$ A

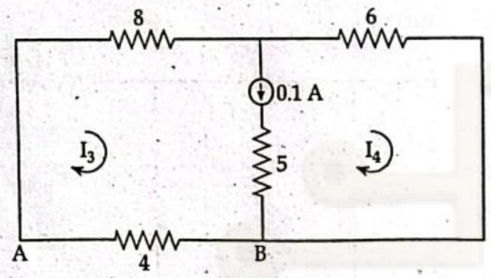
$\therefore \quad I_2 = 0.16$ A (from B to A)

Hence, current through 4 Ω due to 3 V only = $I_2 = 0.16$ A.

Considering current source of 100 mA only, replacing voltage source by short circuit,



On simplifying,



Here;

$I_3 - I_4 = 0.1$ A

Applying KVL is supermesh,

or, $-8I_3 - 6I_4 - 4I_3 = 0$

or, $-12I_3 - 6I_4 = 0$

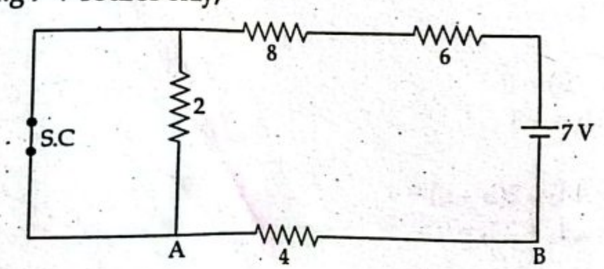
On solving,

$\therefore \quad I_3 = 0.03$ A (from B to A)

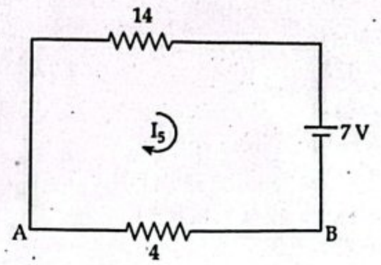
$\therefore \quad I_4 = -0.06$ A

Hence, current through 4 Ω due to 0.1 A = $I_3 = 0.03$ A

Considering 7 V source only,



On simplifying,



Applying KVL,

or, $-14I_5 - 7 - 4I_5 = 0$

or, $-18I_5 = 7$

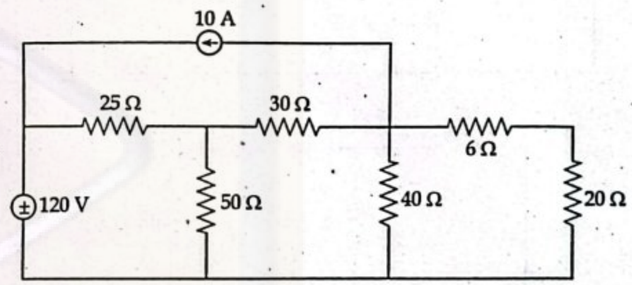
$\therefore \quad I_5 = -0.38$ A (from A to B)

$\therefore \quad$ Current through 4 Ω due to 7 V only = $I_5 = -0.38$ A

Hence,

Net current through 4 Ω = $I_2 + I_3 + I_5 = 0.16 + 0.03 + 0.38 = 0.57$ A.

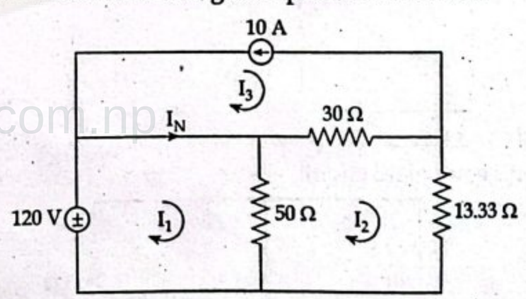
18. Find the current across 25 Ω resistance by using Norton's theorem. [18/F]



Solution:

$$40\ \Omega \parallel 20 = \frac{40 \times 20}{40 + 20} = 13.33\ \Omega$$

Removing R_L and short circuiting the open load terminal



Here;

$I_3 = -10$ A

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Applying KVL in mesh-1:

$120 - 50(I_1 - I_2) = 0$

or, $120 - 50 I_1 + 50 I_2 = 0$

or, $-50 I_1 + 50 I_2 = -120$

Applying KVL in mesh-2:

or, $-30(I_2 - I_3) - 13.33 I_2 - 50(I_2 - I_1) = 0$

or, $-30 I_2 + 30 I_3 - 13.33 I_2 - 50 I_2 + 50 I_1 = 0$

or, $50 I_1 - 30 I_2 - 13.33 I_2 - 50 I_2 + 30 I_3 = 0$

or, $50 I_1 - 93.33 I_2 + 30 I_3 = 0$

or, $50 I_1 - 93.33 I_2 + 30 \times (-10) = 0$

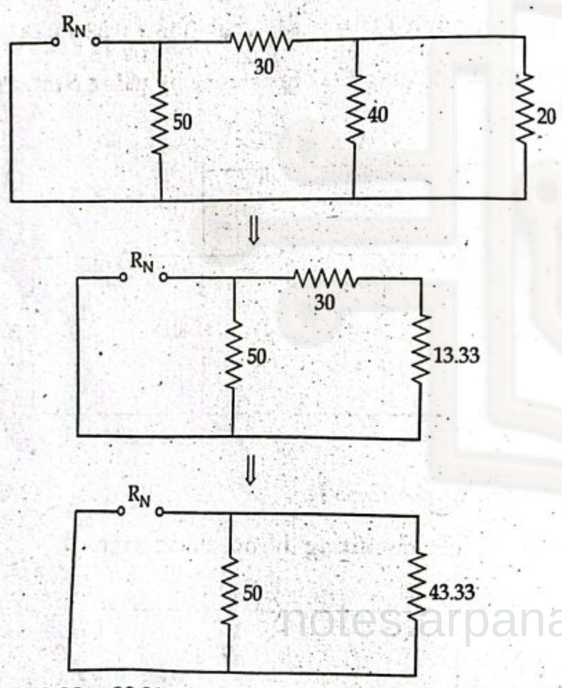
or, $50 I_1 - 93.33 I_2 = 300$

On solving,

$\therefore \quad I_1 = -1.75$ A $\qquad \therefore \quad I_2 = -4.15$ A

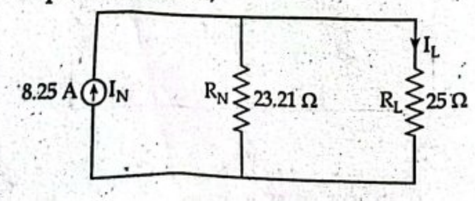
Hence, Norton current, $I_N = I_1 - I_3 = -1.75 - (-10) = 8.25$ A

Replacing R_L and removing all sources by their internal resistance,



$R_N = 50 \parallel 43.33 = 23.21\ \Omega$

Hence, Norton's equivalent circuit,

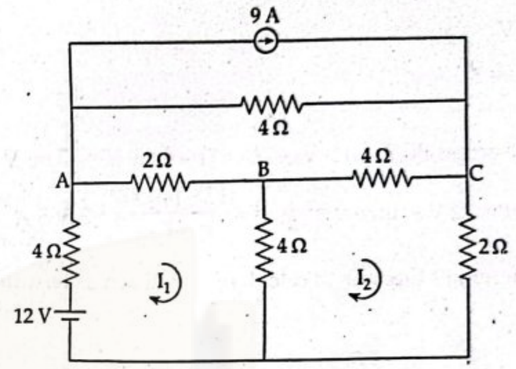


$\therefore \quad I_L = I_N \times \dfrac{R_N}{R_N + R_L}$

$= 8.25 \times \dfrac{23.21}{23.21 + 25}$

$\therefore \quad I_L = 3.971$ A.

19. Use nodal analysis to determine the voltage across BC and the current in the 12 V as shown in figure below. [13/S, 15/S]



Solution:

At node A,

$$\left(\frac{V_A - 12}{4}\right) + \left(\frac{V_A - V_B}{2}\right) + \left(\frac{V_A - V_C}{4}\right) + 8 = 0$$

or, $\dfrac{V_A}{4} - 3 + \dfrac{V_A}{2} - \dfrac{V_B}{2} + \dfrac{V_A}{4} - \dfrac{V_C}{4} + 8 = 0$

or, $V_A\left(\dfrac{1}{4} + \dfrac{1}{4} + \dfrac{1}{2}\right) - \dfrac{V_B}{2} - \dfrac{V_C}{4} = -5$

or, $V_A - \dfrac{V_B}{2} - \dfrac{V_C}{4} = -5$

or, $4V_A - 2V_B - V_C = -10$ (1)

At node B,

$$\left(\frac{V_B - V_A}{2}\right) + \left(\frac{V_B - V_C}{4}\right) + \left(\frac{V_B}{4}\right) = 0$$

or, $V_B\left(\dfrac{1}{2} + \dfrac{1}{4} + \dfrac{1}{4}\right) - \dfrac{V_A}{2} - \dfrac{V_C}{4} = 0$

or, $V_B \times 1 - \dfrac{V_A}{2} - \dfrac{V_C}{4} = 0$

or, $-0.5 V_A + V_B - 0.25 V_C = 0$ (2)

At node C:

$$-9 + \left(\frac{V_C - V_A}{4}\right) + \left(\frac{V_C - V_B}{4}\right) + \frac{V_C}{2} = 0$$

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or, $-9 + V_C\left(\frac{1}{4}+\frac{1}{4}+\frac{1}{2}\right) - \frac{V_A}{4} - \frac{V_B}{4} = 0$

or, $-9 + V_C - \frac{V_A}{4} - \frac{V_B}{4} = 0$

or, $-0.25\,V_A - 0.25\,V_B + V_C = 9$

On solving equation (1), (2) and (3);

$\therefore \quad V_A = -2.34$ V

$\therefore \quad V_B = 1$ V

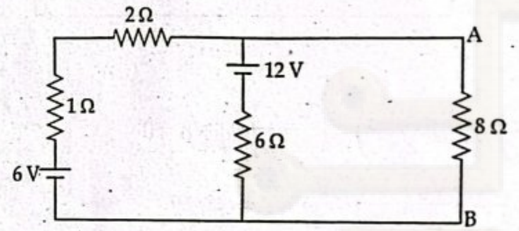
$\therefore \quad V_C = 8.66$ V

Then,

Voltage across BC (4 Ω) = $V_B - V_C = (1 - 8.66)$ V = –7.66 V

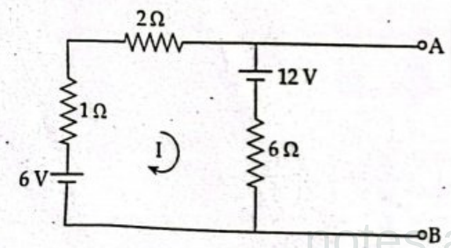
and, Current in 12 V source = $\frac{12 - V_A}{4} = \frac{12 - (-2.34)}{4} = 3.585$ A.

20. Use Therenin's theorem to calculate the pd across terminals A and B [15/S]



Solution:

Removing the 8 Ω resistance of AB



Applying KVL,

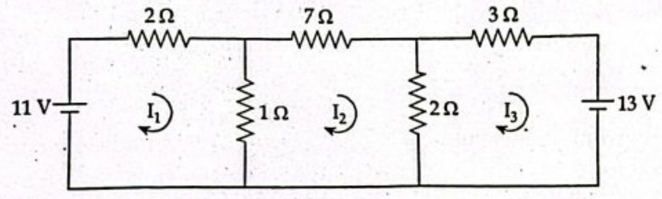
or, $6 - I - 2I - 12 - 6I = 0$

or, $I = \frac{-2}{3}$ A

and, $V_{AB} = (V_A - V_B) = -12 - 6 \times -\frac{2}{3} = -12 - 2 \times 2 = -8$ V

Hence, pd across AB is –8 V.

21. Calculate the mesh currents of the circuit shown below. [15/S]



Solution:

In loop 1:

$11 - 2I_1 - 1(I_1 - I_2) = 0$

or, $11 - 2I_1 - I_1 + I_2 = 0$

or, $3I_1 - I_2 = 11$ (1)

In loop-2:

$-7I_2 - 2(I_2 - I_3) - (I_2 - I_1) = 0$

or, $-7I_2 - 2I_2 + 2I_3 - I_2 + I_1 = 0$

or, $-10I_2 + I_1 + 2I_3 = 0$

or, $-I_1 + 10I_2 - 2I_3 = 0$ (2)

In loop-3:

$-13 - 2(I_3 - I_2) - 3I_3 = 0$

or, $-13 - 2I_3 + 2I_2 - 3I_3 = 0$

or, $-2I_2 + 5I_3 = -13$ (3)

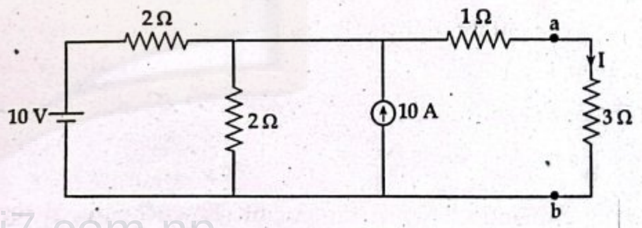
On solving equation (1), (2) and (3);

$\therefore \quad I_1 = 3.61$ A

$\therefore \quad I_2 = -0.17$ A

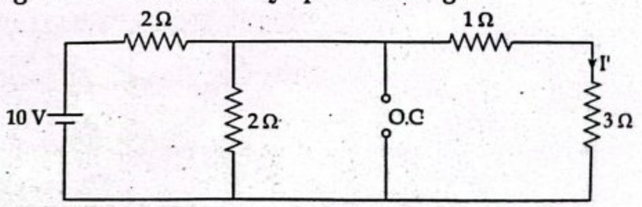
$\therefore \quad I_3 = -2.67$ A

22. Use superposition theorem to calculate power consumed by 3 Ω resistor for the circuit shown below. [15/S]



Solution:

Removing current source 10 A by open circuiting it



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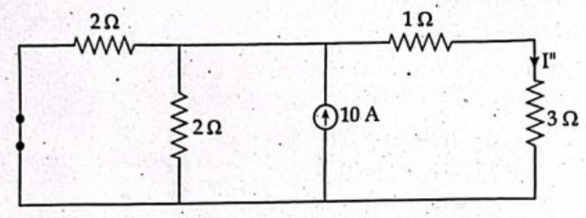
$$R_{eq} = 2 + [2 \parallel (1+3)] = 2 + \frac{2 \times 4}{2+4} = 3.34\ \Omega.$$

Total current, $I_T = \frac{V}{Req} = \frac{10}{3.34} = 3$

Using current divider rule,

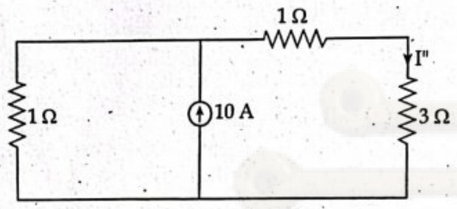
$$I' = \frac{2}{3+2+1} \times I_T = 1\ A$$

and, Removing voltage source by short circuiting it,



$R_{eq} = (2 \parallel 2) = 1$

New circuit becomes,



Now, using current divider rule,

$$I'' = \frac{1}{1+1+3} \times 10 = 2\ A$$

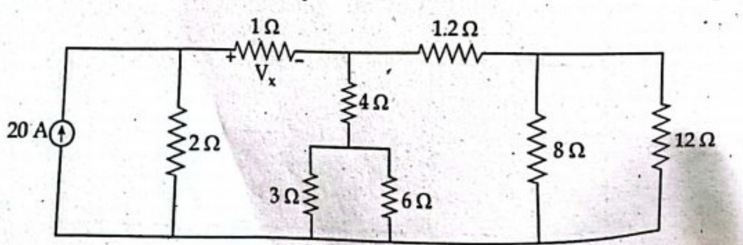
Hence, total current through 3 Ω resistor,

$$I_{3\Omega} = I' + I'' = 1 + 2 = 3\ A$$

so, Total power consumed,

$$P_3 = I_3^2 \times 3 = 3^2 \times 3 = 27\ \Omega.$$

23. Using Kirchoff's law, in the circuit shown below, determine V_x and the power absorbed by 12 Ω resistor. [18/S]



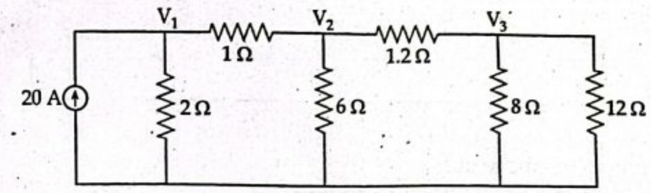
Solution:

Here;

$(3 \parallel 6)\ \Omega = 2\ \Omega$

and, $(4+2)\ \Omega = 6\ \Omega$

Then, circuit becomes,



At V_1, using KCL,

$$\text{or, } -20 + \frac{V_1}{2} + \frac{V_1 - V_2}{1} = 0$$

$$\text{or, } 1.5\ V_1 - V_2 = 20 \quad (1)$$

At V_2,

$$\frac{V_2 - V_1}{1} + \frac{V_2}{6} + \frac{V_2 - V_3}{1.2} = 0$$

$$\text{or, } V_2 - V_1 + \frac{V_2}{6} + \frac{V_2}{1.2} - \frac{V_3}{1.2} = 0$$

$$\text{or, } V_2 - V_1 + 0.1667\ V_2 + 0.834\ V_2 - 0.834\ V_3 = 0$$

$$\text{or, } 2\ V_2 - V_1 - 0.834\ V_3 = 0 \quad (1)$$

At V_3,

$$\frac{V_3 - V_2}{1.2} + \frac{V_3}{8} + \frac{V_3}{12} = 0$$

$$\text{or, } 0.834\ V_3 - 0.834\ V_2 + 0.125\ V_3 + 0.0834\ V_3 = 0$$

$$\text{or, } 1.042\ V_3 - 0.834\ V_2 = 0 \quad (3)$$

On solving equation (1), (2) and (3),

$$\therefore\ V_1 = 26.68\ V$$

$$\therefore\ V_2 = 20.02\ V$$

$$\therefore\ V_3 = 16.00\ V$$

so, $V_x = V_1 - V_2 = 26.68 - 20.02 = 6.66\ V$

Thus, Power absorbed by 12 Ω resistor,

$$P = \frac{V_3^2}{12} = \frac{16^2}{12} = 21.33\ \text{watts.}$$

24. Find the equivalent resistance of the given network seen from the source. [13/F]

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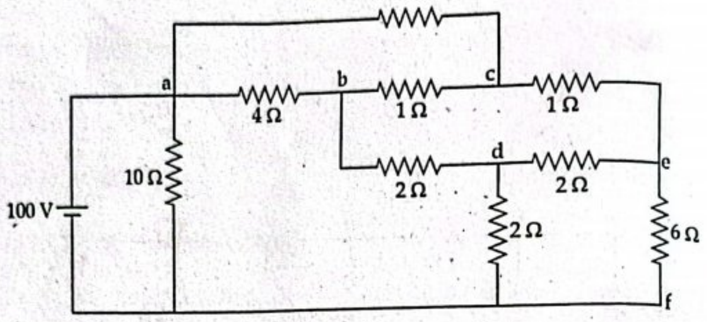


Solution:

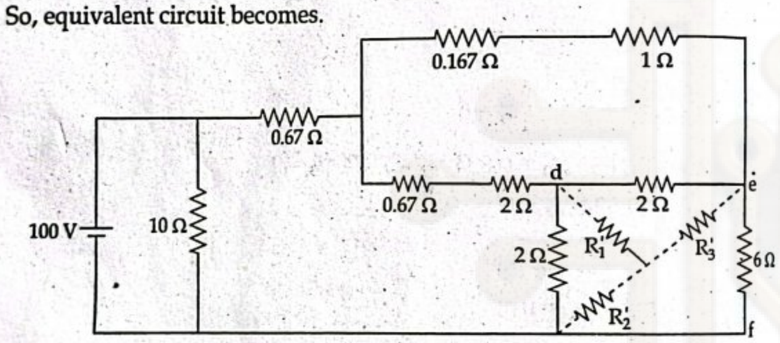
Converting delta abc to star

$$R_1 = \frac{1 \times 4}{1+4+1} = 0.67\ \Omega$$

$$R_2 = \frac{1 \times 4}{1+4+1} = 0.67\ \Omega$$

$$R_3 = \frac{1 \times 1}{1+4+1} = 0.167\ \Omega$$

So, equivalent circuit becomes.



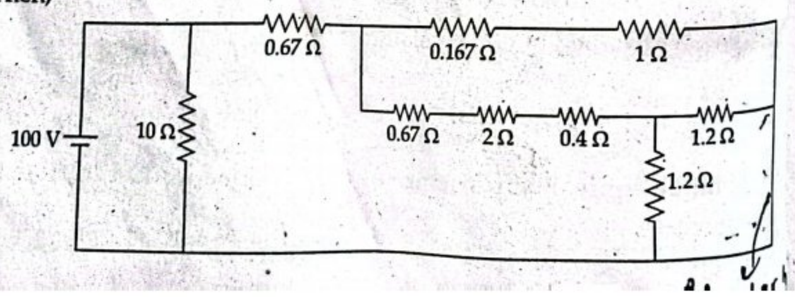
Converting delta def to star

$$R_1' = \frac{2 \times 2}{6+2+2} = 0.4\ \Omega$$

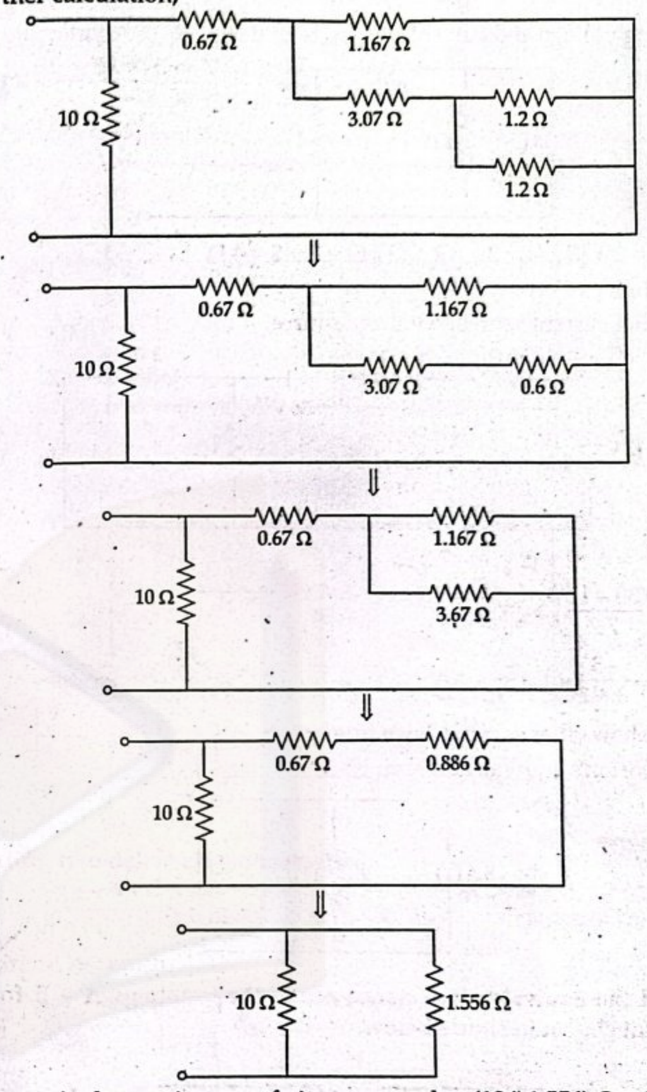
$$R_2' = \frac{2 \times 6}{6+2+2} = 1.2\ \Omega$$

$$R_3' = \frac{6 \times 2.}{6+2+2} = 1.2\ \Omega$$

Then,

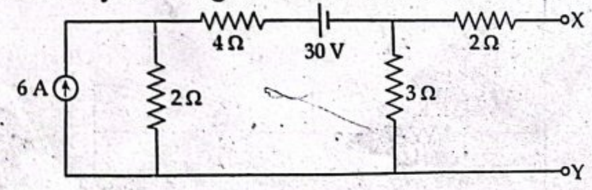


On further calculation,



Hence, equivalent resistance of given network = (10 || 1.556) Ω = 1.346 Ω.

25. Find the Norton's equivalent for the network to the left of terminals x – y in the figure shown below. [13/S]



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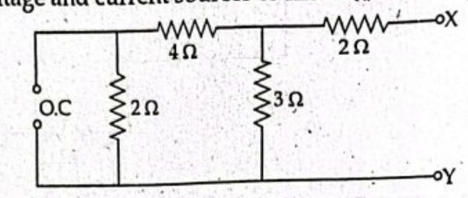
Solution:

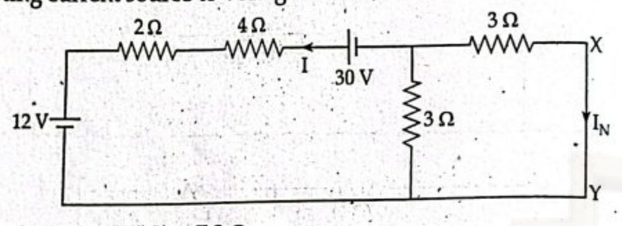
Removing voltage and current sources to find R_N,



$$R_N = 2 + [3 \parallel (4+2)] = 2 + (3 \parallel 6) = 2 + 2 = 4\ \Omega$$

To find I_N:

Converting current source to voltage source,



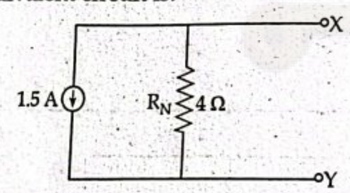
$$R_{eq} = 4 + 2 + (2 \parallel 3) = 7.2\ \Omega$$

so, $I = \left(\frac{30-12}{7.2}\right) = 2.5\ A$

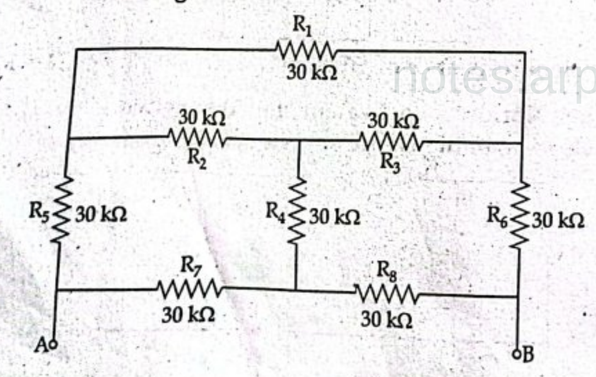
Now, $I_N = \frac{-3}{3+2} \times I = \frac{-3}{5} \times 2.5$

–ve sign shows that current flows from y to x.

Hence, Norton's equivalent circuit is:



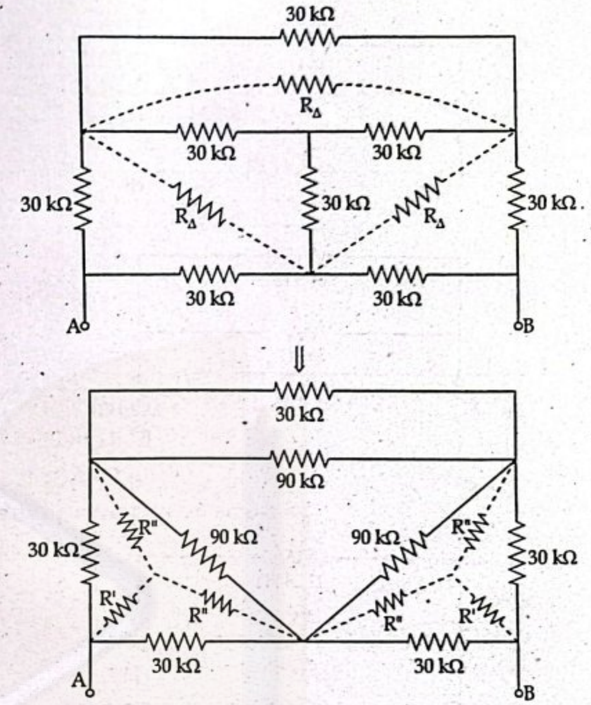
26. Find the equivalent resistance across the terminals A – B for the circuit shown in figure below. [14/F]



Solution:

Converting star of R_2, R_3 and R_4 to delta,

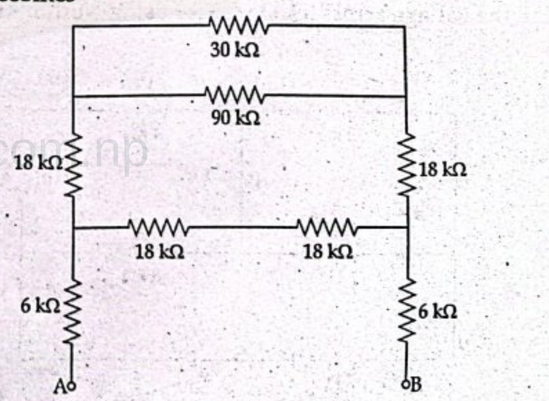
$$R_\Delta = \frac{30 \times 30 + 30 \times 30 + 30 \times 30}{30 + 30 + 30} = 90\ k\Omega$$



Converting two deltas to star equivalent,

$$R' = \frac{30 \times 30}{30 + 30 + 90} = 6\ k\Omega \qquad \text{and,}\quad R'' = \frac{30 \times 90}{30 + 30 + 90} = 18\ k\Omega$$

Thus, circuit becomes



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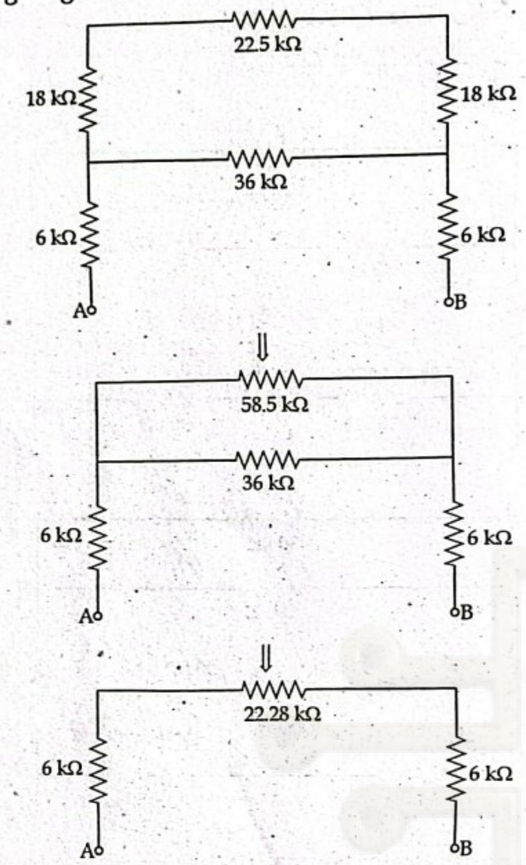
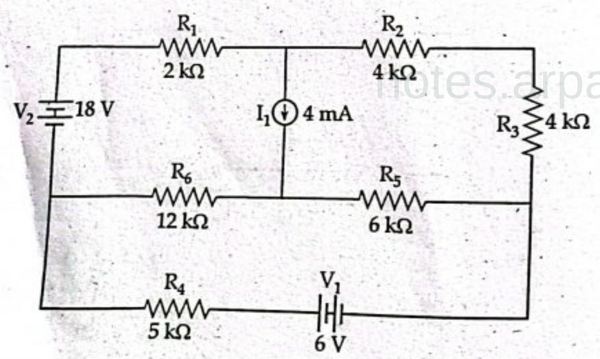
On simplifying diagram,



Hence, equivalent resistance,

$R_{AB} = (6 + 22.28 + 6)\ k\Omega = 34.28\ k\Omega.$

27. Find the voltage across 6 k Ω resistor using Norton's theorem. [14/F



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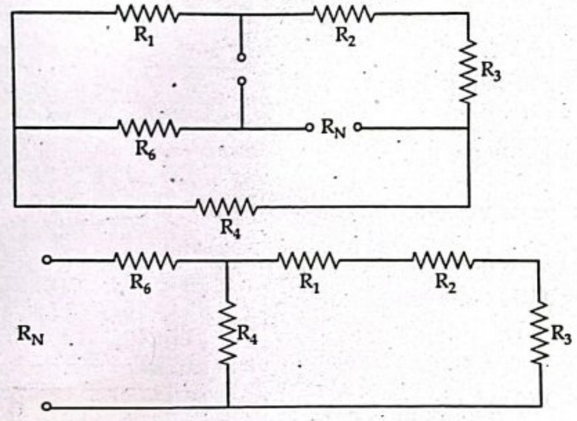
Solution:

Removing R_S resistor as it is a load and also removing all current and voltage source.

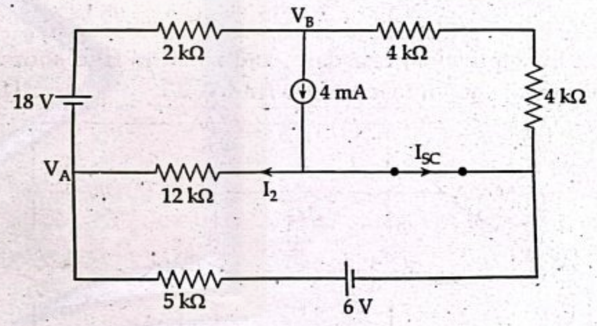


$R_N = R_6 + [R_4 \parallel (R_1 + R_2 + R_3)] = 12 + [5 \parallel (2 + 4 + 4)]$

$= 12 + 5 \parallel 10 = 12 + 3.33$

$\therefore \quad R_N = 15.33\ \Omega$

To find short circuit current across R_S,



At node-A:

$$\frac{V_A - 6}{5} + \frac{V_A - V_B - (-18)}{2} + \frac{V_A - 0}{12} = 0$$

or, $$V_A\left(\frac{1}{5} + \frac{1}{2} + \frac{1}{12}\right) - \frac{V_B}{2} = \frac{6}{5} - 9$$

or, $$V_A \times \frac{47}{60} - \frac{V_B}{2} = \frac{-39}{5}$$

or, $47\ V_A - 30\ V_B = -468$ (1)

At node B:

$$4 + \frac{V_B - V_A - 18}{2} + \frac{V_B}{4 + 4} = 0$$

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or, $4 - 9 + V_B\left(\frac{1}{2}+\frac{1}{8}\right) - \frac{V_A}{2} = 0$

or, $V_B \times \frac{5}{8} - \frac{V_A}{2} = 5$

or, $-4\,V_A + 5V_B = 40$

On solving equation (1) and (2),

$\therefore \quad V_B = 0.07\ V$

$\therefore \quad V_A = -9.91\ V$

Now, Current through R_6 is $\frac{0-(-9.91)}{12} = 0.83\ mA$

$\therefore \quad I_2 = 0.83\ mA$

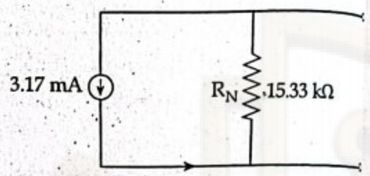
Applying KCL, we have,

$I_1 = I_2 + I_{SC}$

or, $I_{SC} = I_1 - I_2 = 4 - 0.83$

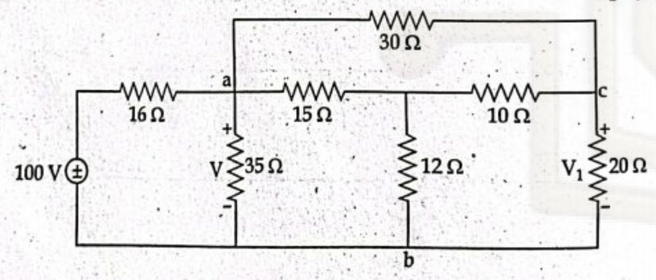
$\therefore \quad I_{SC} = 3.17\ mA$

Hence, Norton's equivalent circuit is,

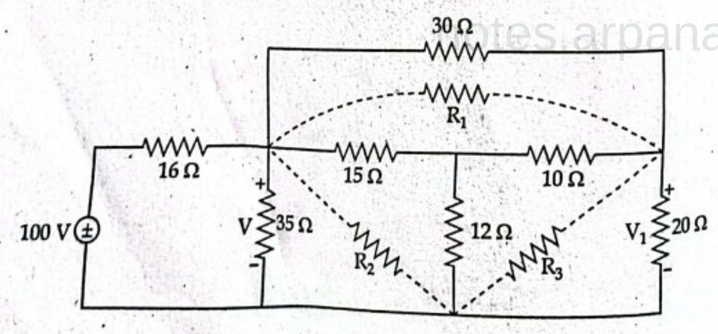


Thus, voltage across 6 k Ω = 3.17 mA × 15.33 k Ω = 48.65 V.

28. Obtain the equivalent resistance and use it to find source curren for the circuit shown below. Also find V. [14/S, 16/F



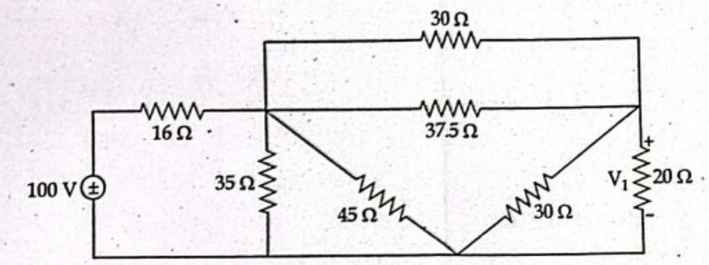
Converting star abc to delta equivalent,



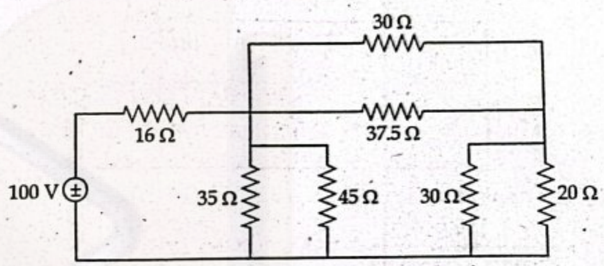
$R_1 = \frac{15\times10 + 10\times12 + 15\times12}{12} = 37.5\ \Omega$

$R_2 = \frac{15\times10 + 10\times12 + 15\times12}{10} = 45\ \Omega$

$R_3 = \frac{15\times10 + 10\times12 + 15\times12}{15} = 30\ \Omega$



On simplifying above diagram,



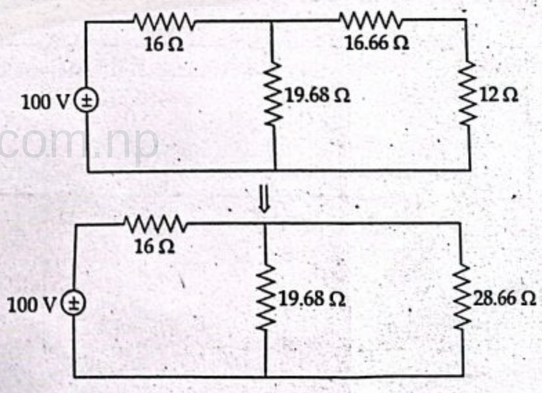
Here;

$(35 \parallel 45)\ \Omega = 19.68\ \Omega$

$(30 \parallel 20)\ \Omega = 12\ \Omega$

$(30 \parallel 37.5)\ \Omega = 16.66\ \Omega$

Thus,



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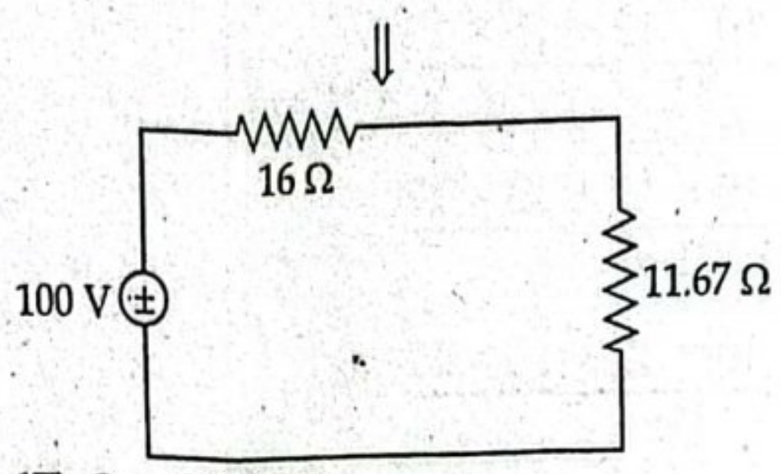


so, $R_{eq} = (16 + 11.67)\ \Omega = 27.67\ \Omega$

so, Source current,

$$I = \frac{V}{R_{eq}} = \frac{100}{27.67} = 3.614\ A$$

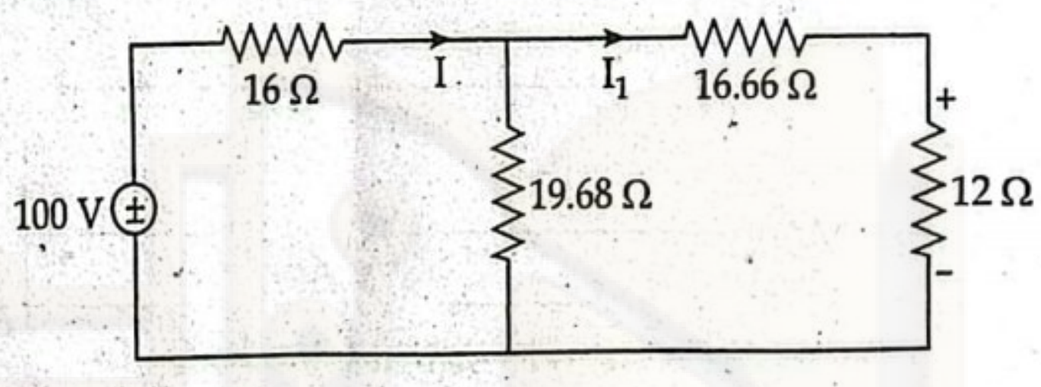
Now,

$$100 - V = I \times 16$$

or, $V = 100 - 3.61 \times 16$

$\therefore \quad V = 42.18\ V$

From the circuit,



Using current divider rule,

$$I_1 = I \times \frac{19.68}{19.68 + 12 + 16.67} = 3.61 \times 0.41 = 1.47\ A$$

Thus, voltage across 20 Ω is,

$$V - V_1 = I_1 \times 16.67$$

or, $V_1 = 42.28 - 1.47 \times 16.67$

$\therefore \quad V_1 = 17.68\ V$

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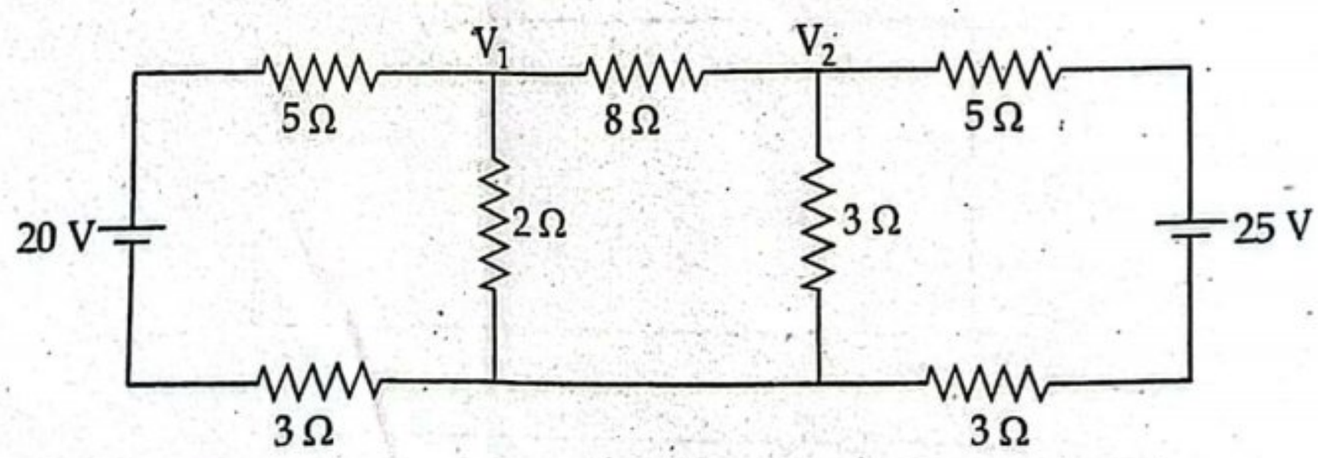
29. Find the current across 8 Ω resistor using nodal analysis. [1



Solution:

Using KCL, at node-V_1:

$$\frac{V_1 - 20}{5 + 3} + \frac{V_1}{2} + \frac{V_1 - V_2}{8} = 0$$

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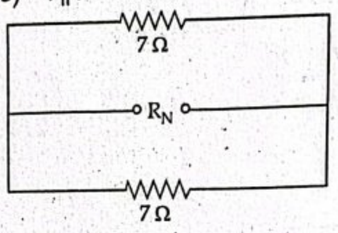
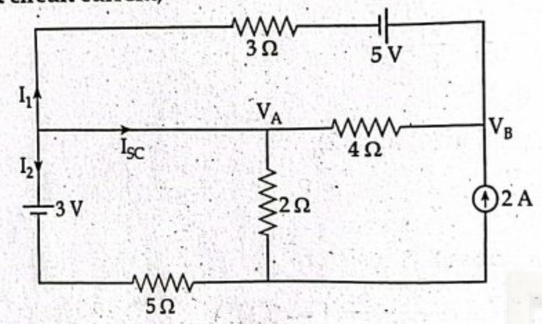
$R_N = (3+4) \| (2+5) = 7 \| 7 = 3.50\ \Omega$



To find short circuit current,



At node-V_A:

$$\frac{V_A-3}{5}+\frac{V_A-V_B+5}{3}+\frac{V_A}{2}+\frac{V_A-V_B}{4}=0$$

or, $V_A\left(\frac{1}{5}+\frac{1}{3}+\frac{1}{2}+\frac{1}{4}\right)-V_B\left(\frac{1}{3}+\frac{1}{4}\right)=\frac{3}{5}-\frac{5}{3}$

or, $\frac{77}{60}V_A-7\frac{V_B}{12}=\frac{-16}{15}$

or, $77\,V_A-35\,V_B=-64$

At node-V_B:

or, $\frac{V_B-V_A}{4}+\frac{V_B-V_A-5}{3}-2=0$

or $V_B\left(\frac{1}{4}+\frac{1}{3}\right)-V_A\left(\frac{1}{4}+\frac{1}{3}\right)=2+\frac{5}{3}$

or, $\frac{7}{12}V_B-\frac{7}{12}V_A=\frac{11}{3}$

or, $-7\,V_A+7\,V_B=44$

Solving **(1)** and **(2)**, we get;

$\therefore\ V_A = 3.71\ V$

$\therefore\ V_B = 10\ V$

Now,

$$I_2=\frac{3.71}{5}=0.742\ A$$

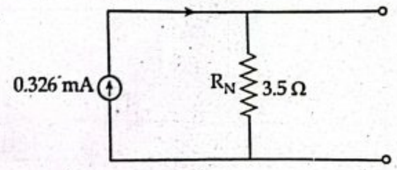
Thus, $I_1=\frac{3.71+5-10}{3}=-0.417$

so, $I_{SC}+I_1+I_2=0$

or, $I_{SC}=-0.742-(-0.417)$

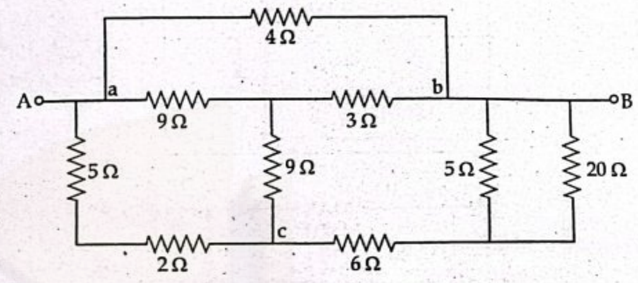
$\therefore\ I_{SC}=-0.326\ A$

Thus, Norton's equivalent circuit is,



Hence, voltage across 1 Ω, $V_{1\Omega} = 0.326 \times 3.5 = 1.1410\ V$.

31. Find the equivalent resistance R_{AB} for the network given below. [16/S

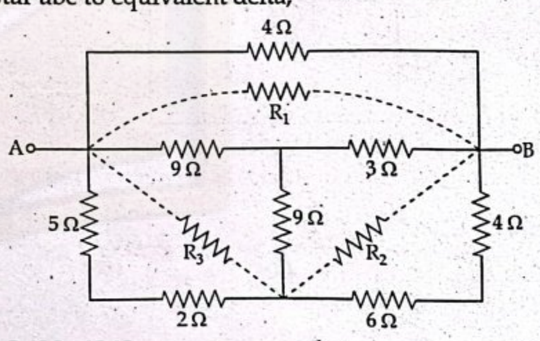


Solution:

Here;

$5\ \Omega \| 20\ \Omega = 4\ \Omega$

Converting star abc to equivalent delta,



Here;

$$R_1=\frac{9\times3+3\times9+9\times9}{9}=15\ \Omega$$

$$R_2=\frac{9\times3+9\times9+3\times9}{9}=15\ \Omega$$

$$R_3=\frac{9\times3+9\times3+9\times9}{3}=45\ \Omega$$

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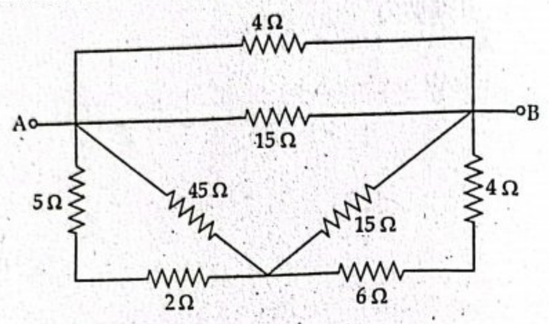
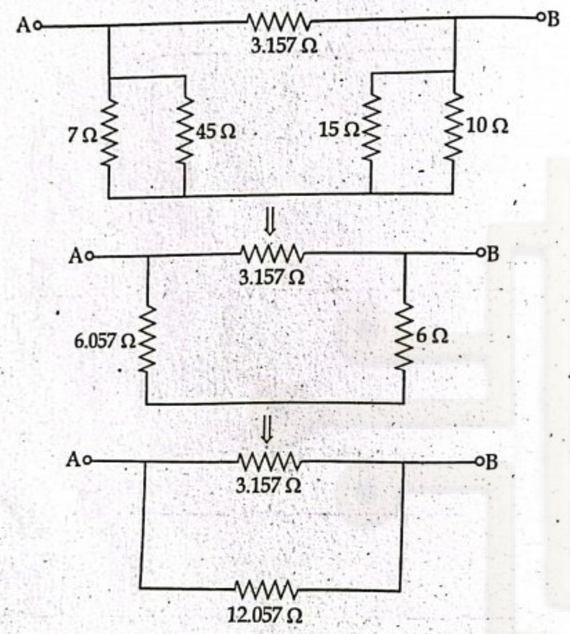
Thus, equivalent circuit becomes,



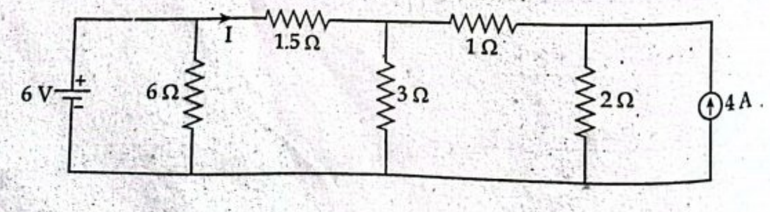
Adding series resistance and simplifying,



Hence,

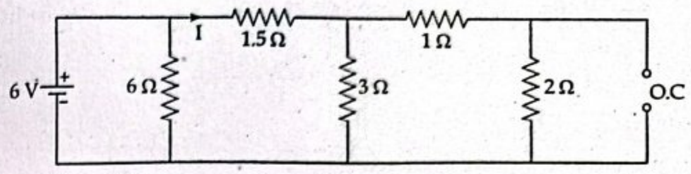
$$R_{AB} = 3.157 \parallel 12.057$$
$$= \frac{3.157 \times 12.057}{3.157 + 12.057}$$
$$= 2.50\ \Omega.$$

32. Find the current I using superposition theorem. [16/5]

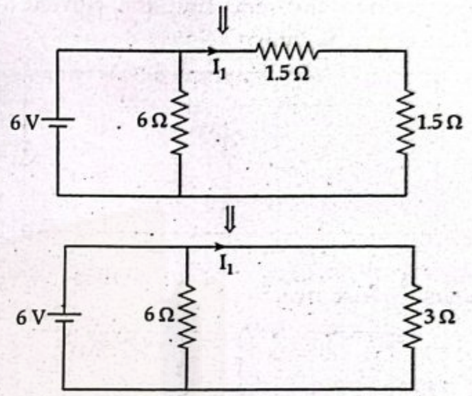


Solution:

First removing current source,

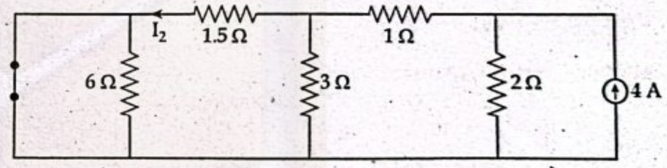


On simplifying figure,

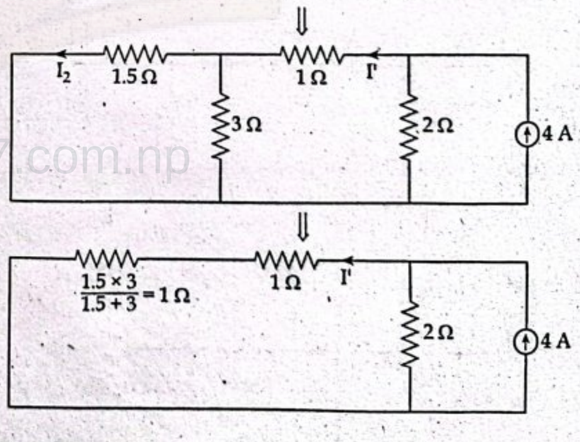


so, $I_1 = \frac{6}{3} = 2$ A

Now removing voltage source,



The 6 Ω resistor can be removed as it is short circuited.



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Using current divider rule,

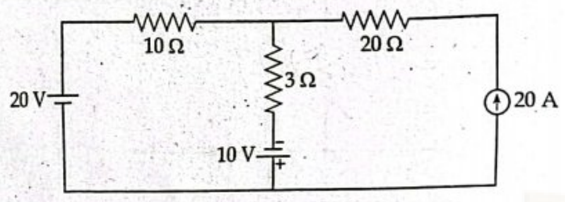
$$I' = \frac{2}{2+1+1} \times 4 = 2 \text{ A}$$

Now,

$$I_2 = \frac{3}{3+1.5} \times I' = \frac{3}{4.5} \times 2 = 1.3 \text{ A}$$

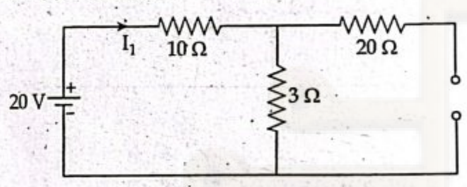
Hence, $I = I_1 - I_2 = 2 - 1.33 \text{ A} = 0.66 \text{ A}$.

33. Using superposition theorem, find the current through 10Ω resistor of the circuit as shown below. [1]



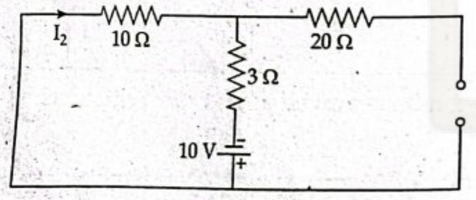
Solution:

Removing 10 V and 20 A source.



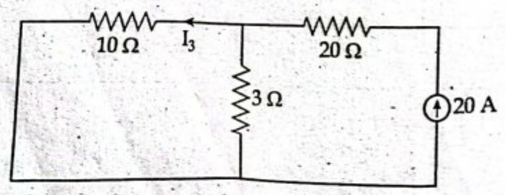
$$I_1 = \frac{20}{10+3} = 1.54 \text{ A}$$

Removing 20 V and 20 A source,



$$I_2 = \frac{10}{10+3} = 0.77 \text{ A}$$

and, Removing 20 V and 10 V source,

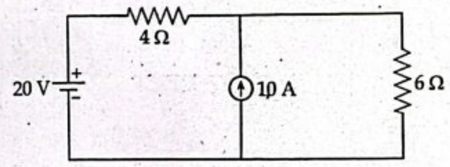


$$I_3 = 20 \times \frac{3}{13} = 4.62 \text{ A}$$

According to superposition theorem,

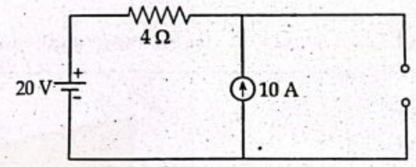
$I = I_3 - I_2 - I_1 = 4.62 - 0.77 - 1.54 = 2.31$ A towards 20 V source.

34. Find the voltage across 6 Ω resistance by using Norton's theorem. **[17/F]**

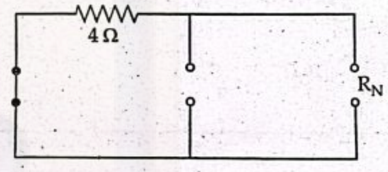


Solution:

Removing 6 Ω resistor, to find equivalent circuit

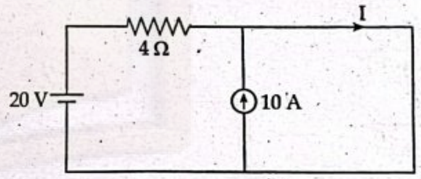


Removing current source to find equivalent resistance, and short circuiting voltage source,



$\therefore \quad R_N = 4 \Omega$

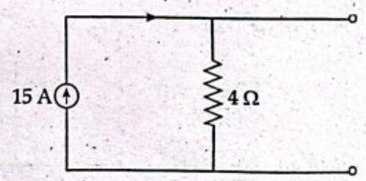
To find short circuit current,



Using KCL,

$$I_N = \frac{20-0}{4} + 10 = 15 \text{ A}$$

Thus, Norton's equivalent circuit is,



and, voltage across 6 Ω resistance = 15 × 4 = 60 V.

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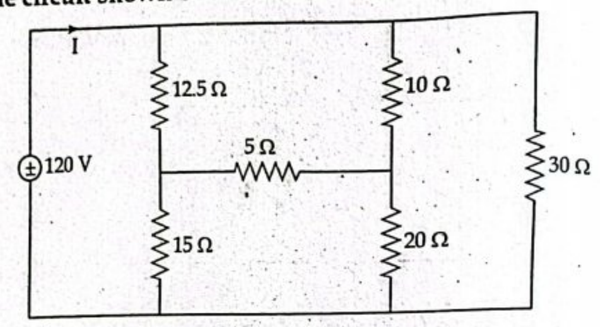
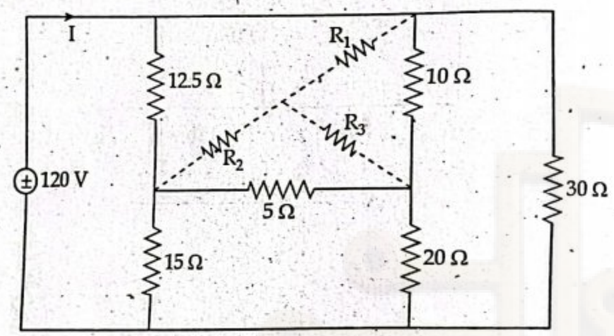
35. Obtain the equivalent resistance and use it to find source current for the circuit shown below.



Solution:

Converting delta of 12.5 Ω, 5 Ω and 10 Ω to equivalent star,



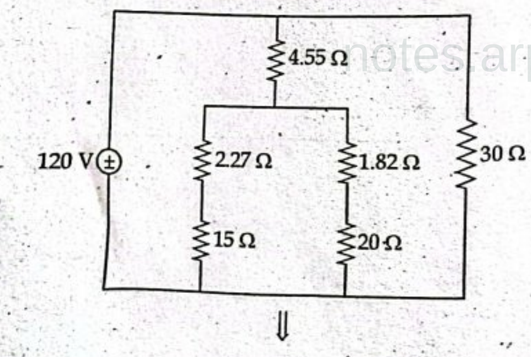
Here;

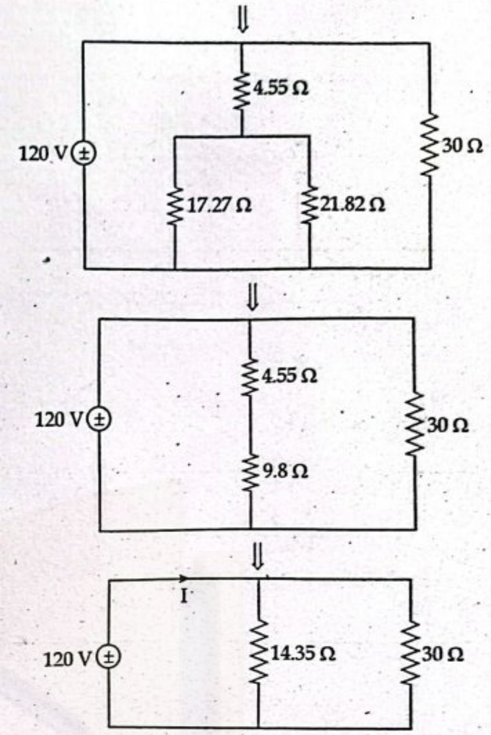
$$R_1 = \frac{10 \times 12.5}{12.5 + 5 + 10} = 4.55\ \Omega$$

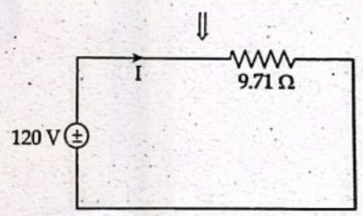
$$R_2 = \frac{12.5 \times 5}{12.5 + 5 + 10} = 2.27\ \Omega$$

$$R_3 = \frac{10 \times 5}{12.5 + 5 + 10} = 1.82\ \Omega$$

Then circuit becomes,





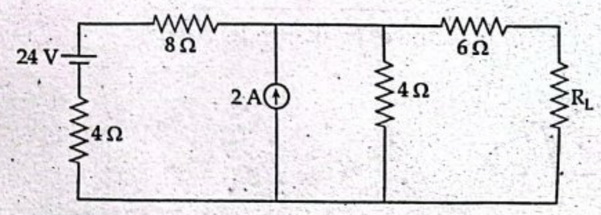


Hence,

$R_{eq} = 9.71\ \Omega$

so, Source current, $I = \frac{120}{9.71} = 12.4$ A.

36. Calculate the value of R_L and power dissipated such that maximum power delivered across load resistor. [18/F]



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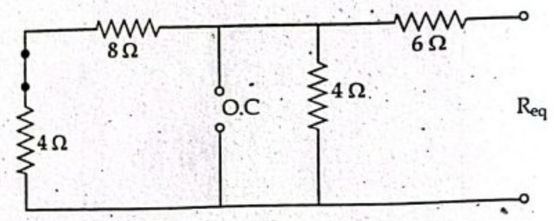
Solution:

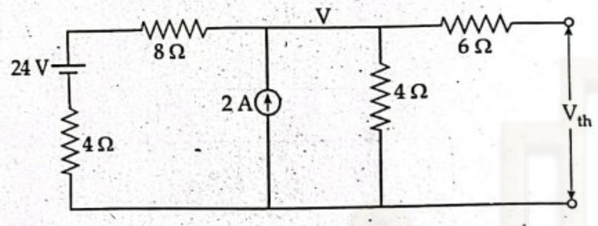
To find R_{eq}:



$$R_{eq} = [6 + 4 \parallel (8 + 4)]$$
$$= [6 + 4 \parallel 12]$$
$$= 6 + 3$$

$\therefore \quad R_{th} = 9\ \Omega = R_{eq}$

To find thevenin's voltage across R_L,



At V, using KCL,

$$\text{or,} \quad \frac{V - 24}{8+4} + \frac{V}{4} + (-2) = 0$$

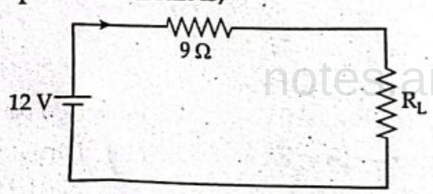
$$\text{or,} \quad \frac{V}{12} - 2 - 2 + \frac{V}{4} = 0$$

$$\text{or,} \quad \frac{V}{12}(1 + 3) = 4$$

$$\text{or,} \quad V = \frac{4}{4} \times 12$$

$\therefore \quad V_{th} = 12\ V$

Hence, thevenin's equivalent circuit is,



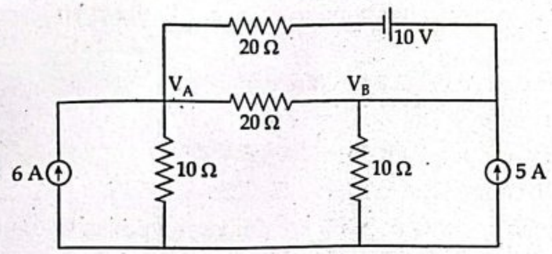
We know, for maximum power across R_L, $R_L = R_{eq}$

Thus, $R_L = 9\ \Omega$

and, Power dissipated,

$$P = \frac{V_{th}^2}{4\,Req} = \frac{12^2}{4 \times 9} = 4\ W.$$

37. Using node voltage method, find the current through each 10 Ω resistors of the circuit as shown below. [17/S, 19/S]



Solution:

At node A,

$$\text{or,} \quad -6 + \frac{V_A}{10} + \frac{V_A - V_B}{20} + \frac{V_A - V_B - (-10)}{20} = 0$$

$$\text{or,} \quad -6 + V_A\left(\frac{1}{10} + \frac{1}{20} + \frac{1}{20}\right) - V_B\left(\frac{1}{20} + \frac{1}{2}\right) + \frac{1}{2} = 0$$

$$\text{or,} \quad 4\frac{V_A}{20} - 2\frac{V_B}{20} = \frac{11}{2}$$

$$\text{or,} \quad 4V_A - 2V_B = 110 \qquad \textbf{(1)}$$

and, At node B,

$$\frac{V_B - V_A}{20} + \frac{V_B - V_A - 10}{20} + \frac{V_B}{10} + (-5) = 0$$

$$\text{or,} \quad V_B\left(\frac{1}{20} + \frac{1}{20} + \frac{1}{10}\right) - V_A\left(\frac{1}{20} + \frac{1}{20}\right) = 5 + \frac{1}{2}$$

$$\text{or,} \quad 4\frac{V_B}{20} - 2\frac{V_A}{20} = \frac{11}{2}$$

$$\text{or,} \quad -2V_A + 4V_B = 110 \qquad \textbf{(2)}$$

Solving **(1)** and **(2)**,

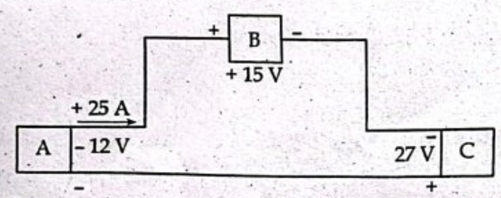
$\therefore \quad V_A = 55\ V$

$\therefore \quad V_B = 55\ V$

Thus, current through 10 Ω resistor is,

$$\therefore \quad I = \frac{V_A}{10} = \frac{55}{10} = 5.50\ A$$

38. For the circuit shown below. [17/S]



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i) Determine which components are absorbing power and which are delivering power.

ii) Is conservation of power satisfied? Explain your answer.

Solution:

Since the components A, B and C are connected in series,

Power in component A, $P_A = 12 \times 25 = 300$ W

Power in component B, $P_B = 15 \times 25 = 375$ W

Power in component C, $P_C = -27 \times 25 = -675$ W

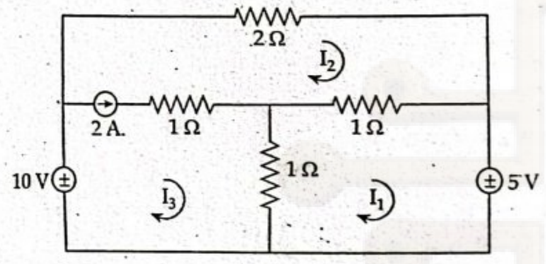
i) Hence, components A and B are delivering power and component C is absorbing power.

ii) Yes, conservation of power is satisfied since $P_A + P_B + P_C = 0$

i.e., $300 + 375 - 675 = 0$

i.e., For a closed loop, the algebraic sum of all the power around any closed loop in a circuit is equal to zero.

39. By using supermesh concept, calculate voltage across 2 Ω resistor in the circuit shown below. [13/F]



Solution:

Applying KVL, on supermesh 2 and 3,

or, $-2I_2 - 1(I_2 - I_1) - 1(I_3 - I_1) + 10 = 0$

or, $-2I_2 - I_2 + I_1 - I_3 + I_1 + 10 = 0$

or, $-3I_2 + 2I_1 - I_3 = -10$ (1)

From supermesh,

$I_3 - I_2 = 2A$ (2)

Applying KVL on loop-2,

or, $-5 - 1(I_1 - I_3) - 1(I_1 - I_2) = 0$

or, $-5 - 2I_1 + I_2 + I_3 = 0$

or, $-2I_1 + I_2 + I_3 = 5$ (3)

On solving equation (1), (2) and (3); we get,

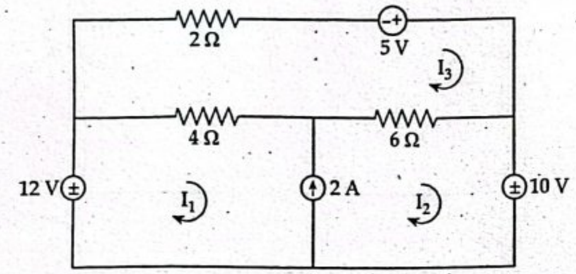
$\therefore I_1 = 1$ A

$\therefore I_2 = 2.5$ A

$\therefore I_3 = 4.5$ A

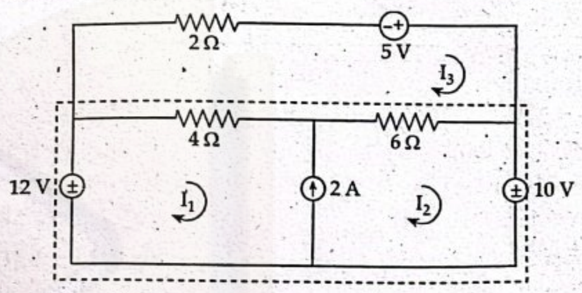
Thus, voltage across 2 Ω resistor $= I_2 \times 2 = 2.5 \times 2 = 5$ V.

40. Solve the circuit in figure. Use mesh analysis to determine the mesh current I_1, I_2 and I_3 and evaluate power developed in 10 V source. [14/S]



Supermesh is formed between loop 1 and loop 2 as 2 A current is common between loop 1 and loop 2.

so, Applying KVL on supermesh,



$12 - 4(I_1 - I_3) - 6(I_2 - I_3) - 10 = 0$

or, $12 - 4I_1 + 4I_3 - 6I_2 + 6I_3 - 10 = 0$

or, $-4I_1 + 10I_3 - 6I_2 = -2$ (2)

Applying KVL on loop-3;

or, $-2I_3 + 5 - 6(I_3 - I_2) - 4(I_3 - I_1) = 0$

or, $-2I_3 + 5 - 6I_3 + 6I_2 - 4I_3 + 4I_1 = 0$

or, $-12I_3 + 6I_2 + 4I_1 = -5$ (3)

On solving equation (1), (2) and (3);

$\therefore I_1 = 2.5$ A

$\therefore I_2 = 4.5$ A

$\therefore I_3 = 3.5$ A

Thus, power developed in 10 V source,

$P_{10V} = I_2 \times V_{10} = 4.5 \times 10 = 45$ W.

41. Use supermesh concept to calculate the mesh currents of the circuit shown below. [15/F]

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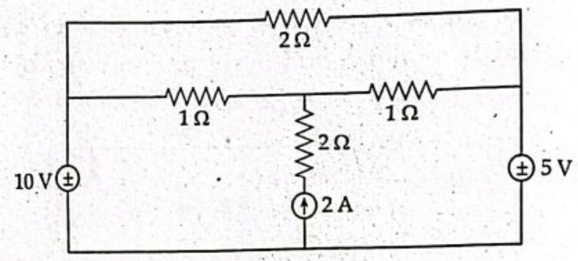
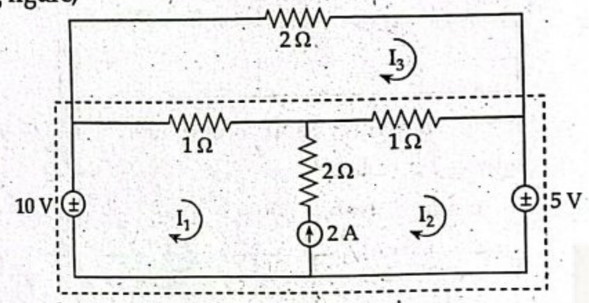


Solution:

Redrawing figure,



From supermesh,

$I_2 - I_1 = 2$ A

Applying KVL to loop-3,

$-2I_3 - 1(I_3 - I_2) - 1(I_3 - I_1) = 0$

or, $-2I_3 - I_3 + I_2 - I_3 + I_1 = 0$

or $-5I_3 + I_2 + I_1 = 0$

or, $I_1 + I_2 - 5I_3 = 0$

Applying KVL to loop loop-2 and loop-1,

or, $10 - 1(I_1 - I_3) - 1(I_2 - I_3) + 5 = 0$

or, $10 - I_1 + I_3 - I_2 + I_3 + 5 = 0$

or, $-I_1 - I_2 + 2I_3 = -5$

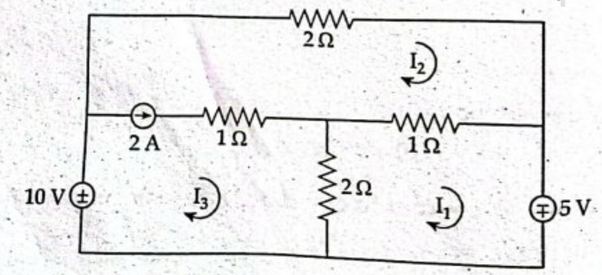
On solving equation (1), (2) and (3); we get,

$\therefore$ $I_1 = 3.166$ A

$\therefore$ $I_2 = 5.166$ A

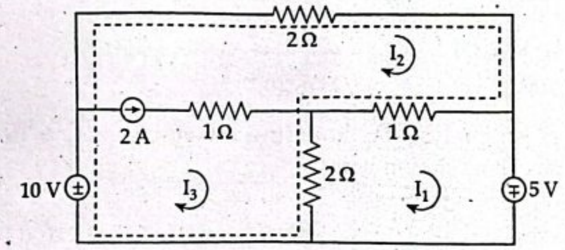
$\therefore$ $I_3 = 1.666$ A.

42. Find the mesh current I_1, I_2 and I_3 in the circuit given below usi the concept of mesh analysis. [16



Solution:

Supermesh is formed between loop-2 and loop 3 because 2 A current is common between loop-2 and loop-3.



so, $I_3 - I_2 = 2$ A (1)

Applying KVL to loop – 1,

$5 - 2(I_1 - I_3) - 1(I_1 - I_2) = 0$

or, $-2I_1 + 2I_3 - I_1 + I_2 = -5$

or, $-3I_1 + I_2 + 2I_3 = -5$ (2)

Applying KVL to supermesh 3 and 2,

or, $-2I_2 - 1(I_2 - I_1) - 2(I_3 - I_1) + 10 = 0$

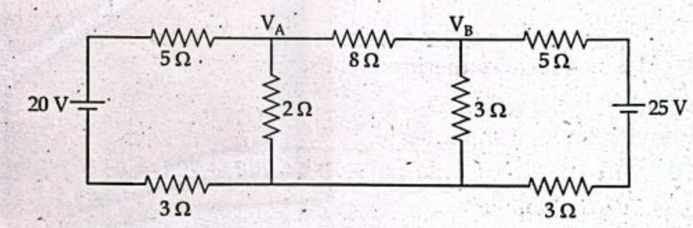
or, $-2I_2 - I_2 + I_1 - 2I_3 + 2I_1 + 10 = 0$

or, $3I_1 - 3I_2 - 2I_3 = -10$ (3)

On solving equation (1), (2) and (3);

$\therefore$ $I_1 = 10.5$ A, $\therefore$ $I_2 = 7.5$ A, $\therefore$ $I_3 = 9.5$ A

43. Find the node voltage at each nodes using nodal analysis for the given circuit. [18/F]



Solution:

Applying KCL at V_A

$$\frac{V_A - 20}{5+3} + \frac{V_A - V_B}{8} + \frac{V_A}{2} = 0$$

or, $$\frac{V_A - 20}{8} + \frac{V_A - V_B}{8} + \frac{V_A}{2} = 0$$

or, $V_A - 20 + V_A - V_B + 4V_A = 0$

or, $5V_A - V_B = 20$ (1)

and, Applying KCL at V_B,

$$\frac{V_B - V_A}{8} + \frac{V_B}{3} + \frac{V_B - 25}{5+3} = 0$$

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or, $\frac{V_B - V_A}{8} + \frac{V_B}{3} + \frac{V_B - 25}{8} = 0$

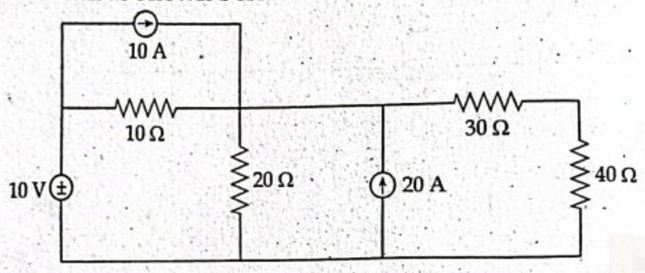
or, $3V_B - 3V_A + 8V_B + 3V_B - 75 = 0$

or, $14V_B - 3V_A = 75$

On solving (1) and (2);

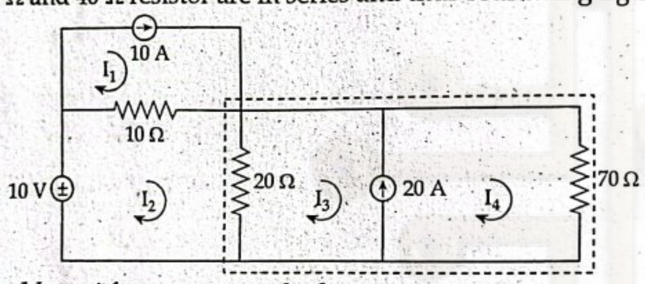
$\therefore \quad V_A = 5.29\ V \qquad \therefore \quad V_B = 6.49\ V.$

44. Using mesh analysis method, find the voltage across 10 Ω resistor of the circuit as shown below. [18/S]



Solution:

Here, 30 Ω and 40 Ω resistor are in series and thus redrawing figure,



Loop 3 and loop 4 forms supermesh, thus,

$I_4 - I_3 = 10\ A$ (1)

and, Applying KVL on supermesh,

$-70I_4 - (I_3 - I_2)20 = 0$ or, $-70I_4 - 20I_3 + 20I_2 = 0$ (2)

Applying KVL on loop-2;

$10 - 10(I_2 - I_1) - 20(I_2 - I_3) = 0$

or, $10 - 10I_2 + 10I_1 - 20I_2 + 20I_3 = 0$

or, $10 - 30I_2 + 10I_1 + 20I_3 = 0$

or, $I_1 - 3I_2 + 2I_3 = -1$ (3)

Applying KVL on loop-1,

$I_1 = 10\ A$ (4)

so, Equation (3) becomes

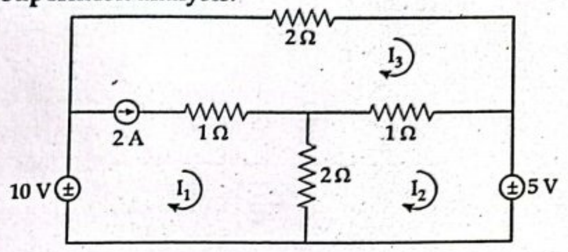
or, $-3I_2 + 2I_3 = -11$ (5)

$\therefore \quad I_2 = -7.86\ A, \qquad \therefore \quad I_3 = -17.30\ A, \qquad \therefore \quad I_4 = 2.695\ A$

Thus, voltage across 10 Ω resistor

$= 10 \times (I_2 - I_1) = 10 \times (-7.86 - 10) = -178.60\ V.$

45. Calculate the power absorbed or delivered by 2 A current source using supermesh analysis. [16/F]



Solution:

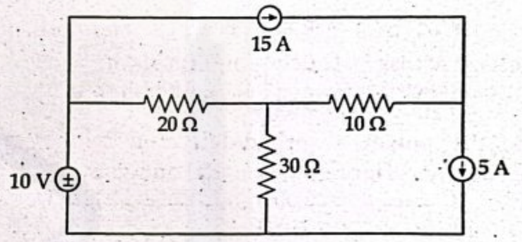
See question number (42). Thus,

$\therefore \quad I_1 = 9.5\ A, \qquad \therefore \quad I_2 = 10.5\ A, \qquad \therefore \quad I_3 = 7.5\ A$

Hence power delivered by 2 A current source

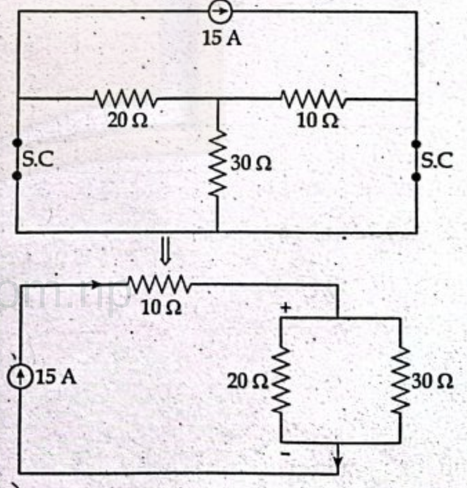
$= (I_1^2 - I_3)^2 \times R = (9.5 - 7.5)^2 \times 1 = 2^2 \times 1 = 4$ watt.

46. Find voltage drop across 20 Ω resistor using superposition theorem. [18/S]



Solution:

Taking 15 A current source only,



$I_{20} = \left(\frac{30}{20 + 30}\right) \times 15 = 9\ A$

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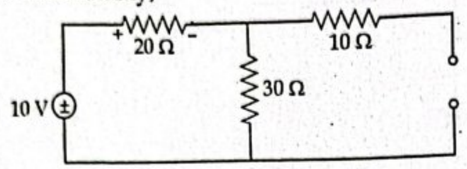
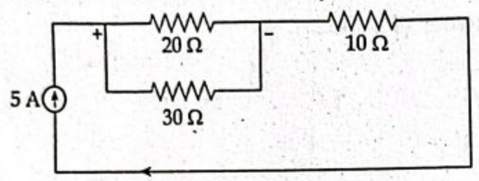
∴ Voltage across it, $V_1 = 9 \times 20 = 180$ V

Now, Taking 10 V source only,



Voltage across 20 Ω, $V_2 = \left(\frac{20}{20+30}\right) \times 10 = 4$ V

Again, taking 5 A current source,

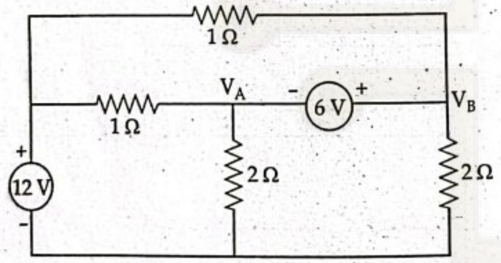


Current across 20 Ω, $I = \left(\frac{30}{30+2}\right) \times 5 = 3$ A

Voltage across 20 Ω, $V_3 = 20 \times 3 = 60$ V

Now, Total voltage across 20 Ω using superposition Theorem, $V_{20} = V_1 - V_2 - V_3 = (180 - 4 - 60)$ V $= 116$ V.

47. Calculate the power absorbed/delivered by 6V source for t network shown in figure using nodal analysis. [14



Solution:

Using supernode at node A and B,

$$\frac{V_A - 12}{1} + \frac{V_A}{2} + \frac{V_B - 12}{1} + \frac{V_B}{2} = 0$$

or, $2V_A - 24 + V_A + 2V_B - 24 + V_B = 0$

or, $2V_A + 3V_A = 48$

or, $V_A + V_B = 16$

Also from supernode,

$V_B - V_A = 6$ V

On solving equation (1) and (2), we get;

∴ $V_A = 5$ V

∴ $V_B = 11$ V

Now, at B using KCL,

$$I + \left(\frac{11 - 12}{1}\right) + \frac{11}{2} = 0$$

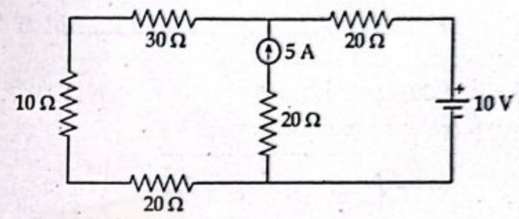
or, $I = -5.5 + 1$

∴ $I = -4.5$ A

This shows that current is flowing from node B to node A.

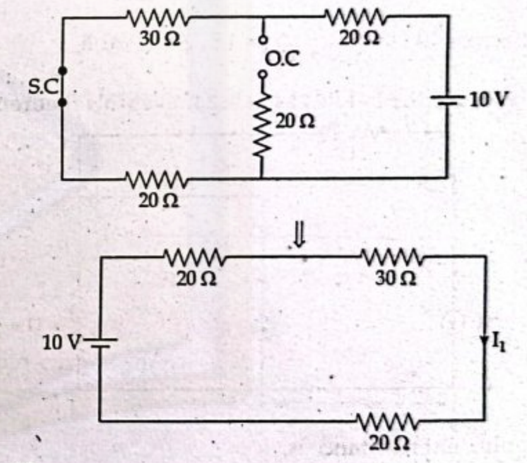
so, Power absorbed by 6 V $= 6 \times 4.5 = 27$ W.

48. Find the current across 10 Ω resistance by using Norton's theorem. [17/S]



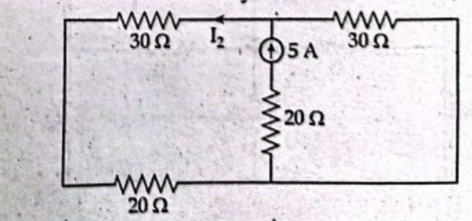
Solution:

Using superposition theorem, taking 10 V source only,



$$I_1 = \frac{10}{20 + 30 + 20} = 0.143 \text{ A}$$

Now, taking 5 A current source only,



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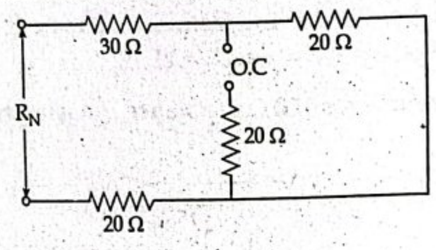
$$I_2 = \frac{20}{20+50} \times 5 = 1.429 \text{ A}$$

so, Total short current,

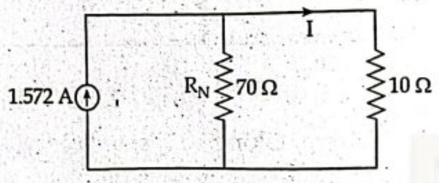
$$I = I_1 + I_2 = 0.143 + 1.429 = 1.572 \text{ A}$$

Then, Total resistance seen across 10 Ω is,



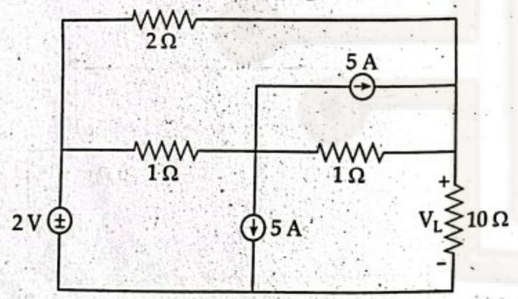
$$R_N = (30 + 20 + 20)\ \Omega = 70\ \Omega$$

so, Norton's equivalent circuit becomes,



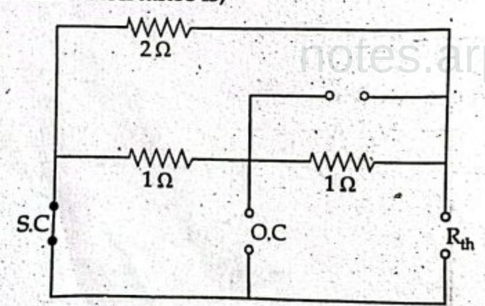
Now, current across $10\ \Omega = \left(\frac{10}{70+10}\right) \times 1.572 = 1.376$ A.

49. Calculate V_L for the below ckt using thevenin's theorem. [16]



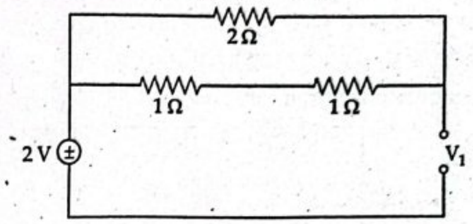
Solution:

Thevenin's equivalent resistance is,



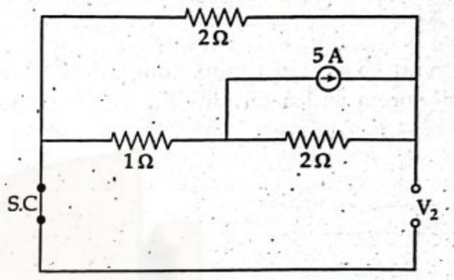
$$R_{th} = 2 \parallel (1 + 1) = 1\ \Omega$$

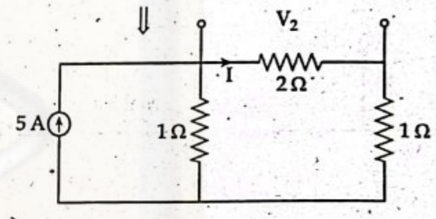
Using superposition theorem, taking 2 V only,



$\therefore$ $V_1 = 2V$

and, Taking 5 A current source only,

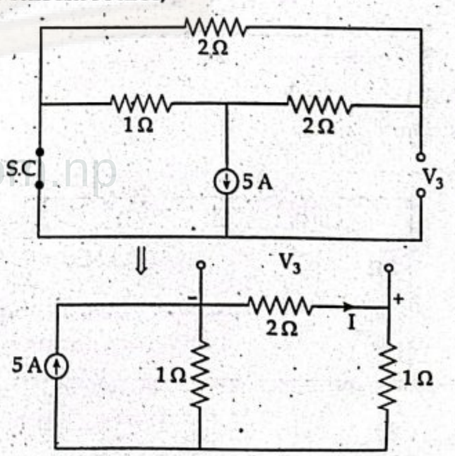




$$I = \left(\frac{1}{1+2+1}\right) \times 5 = 1.25 \text{ A}$$

so, $V_2 = 1.25 \times 2 = 2.5$ V

Now, taking 5 A current source,



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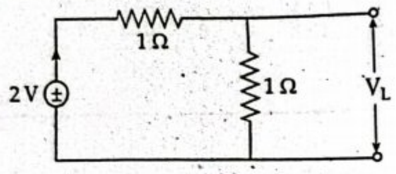
so, $I = \left(\frac{1}{1+2+1}\right) \times 5 = 1.25\ A$

Thus, $V_3 = -I \times 2 = -1.25 \times 2 = -2.5\ V$

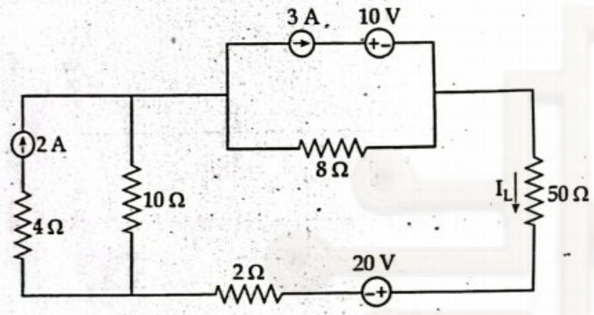
Thus, $V_{th} = V_1 + V_2 + V_3 = 2 + 2.5 - 2.5 = 2\ V$

so, Thevenin's equivalent circuit is,



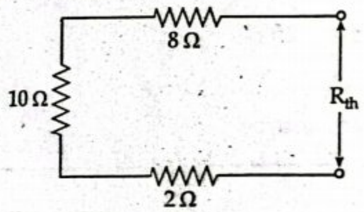
Thus, $V_L = \left(\frac{1}{1+1} \times 2\right) = 1\ V.$

50. For the circuit shown in figure, compute the load current using thevenin theorem and determine the value of the load for which power transfer is maximum. [19



Solution:

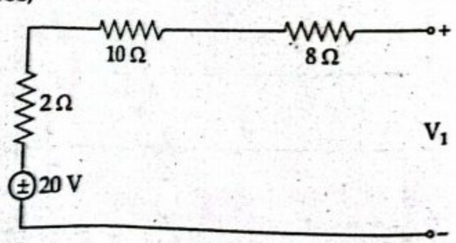
Thevenin's equivalent resistance is,



$\therefore\ R_{th} = 8\ \Omega + 10\ \Omega + 2\ \Omega = 20\ \Omega$

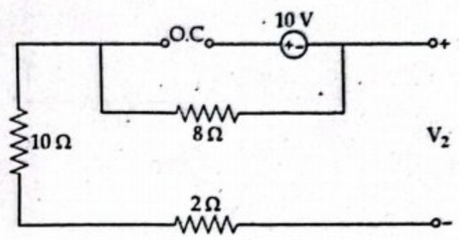
Using superposition theorem,

Taking 20 V source,



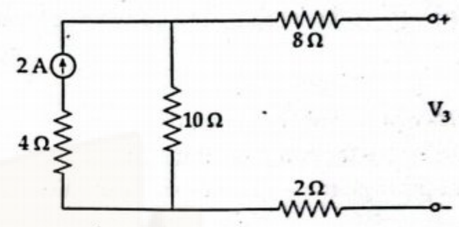
$V_1 = -20\ V$

Now, taking 10 V source only,



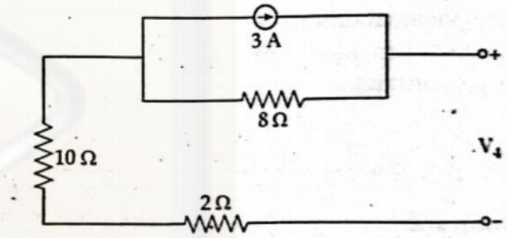
$\therefore\ V_2 = 0\ V$

and, Taking 2 A source only,



$\therefore\ V_3 = 2 \times 10\ V = 20\ V$

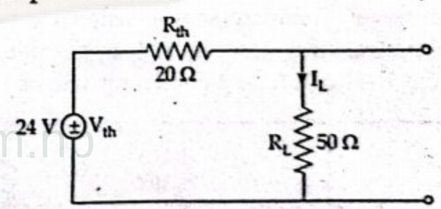
Taking 3 A source only,



$\therefore\ V_4 = 3 \times 8 = 24\ V$

so, $V_{th} = V_1 + V_2 + V_3 + V_4 = -20 + 0 + 20 + 24 = 24\ V$

Hence, thevenin's equation ckt is,



so, $I_L = \left(\frac{24}{20+50}\right) = 0.343\ A$

For maximum power transfer, value of load should be 20 Ω according to maximum power transfer theorem.

51. State Ohm's law and write its limitations. Obtain the equivalent resistance of the given circuit. [19/F]

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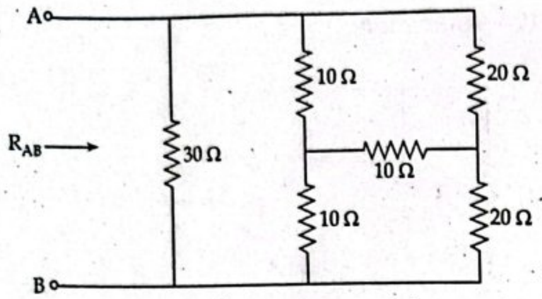


Solution:

Ohm's law states that the potential difference across an ideal conductor is proportional to the current through it. The constant of proportionality is called resistance.

i.e., $V \propto I \quad \therefore \quad V = IR$

where, R = Resistance of component in ohms

V = Voltage across the component measured in volts

I = Current through the component measured in ampere

OR,

Ohm's law state that the current flowing through a conductor is directly proportional to the potential difference between its ends provided its temperature remains constant.

i.e., $I \propto V$

or, $I = \frac{1}{R} V$ or, $R = \frac{V}{I}$

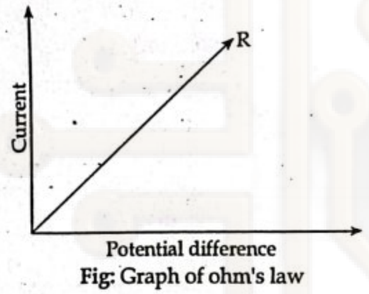


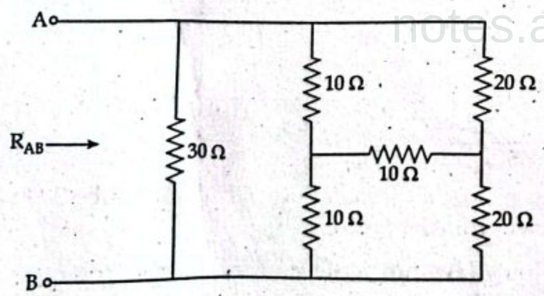
Fig: Graph of ohm's law

Limitations of ohm's are;

i) It doesn't apply to all non-metallic conductors.

ii) It also doesn't apply to non-linear devices such as zener diode vacuum tubes, discharge tubes, electrolytes etc.

iii) It is true for metal conductors at constant temperature. If the temperature change, this law is not applicable. For changing temperature, the V – I graph for a conductor will be non-linear.



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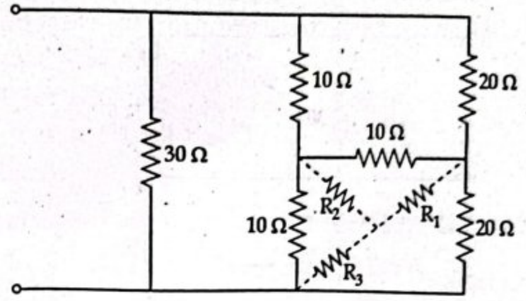
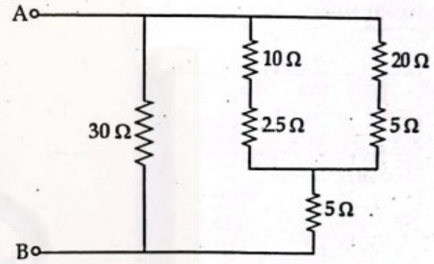
Transforming delta to star connection



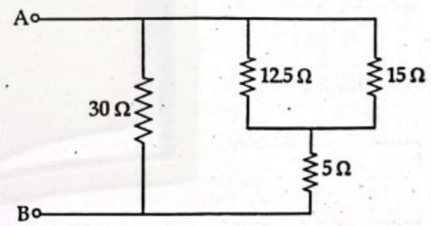
$$R_1 = \frac{10 \times 20}{10 + 20 + 10} = 5\ \Omega \qquad R_2 = \frac{10 \times 10}{10 + 20 + 10} = 2.5\ \Omega$$

$$R_3 = \frac{10 \times 20}{10 + 20 + 10} = 5\ \Omega$$

Redrawing the figure with star connection,



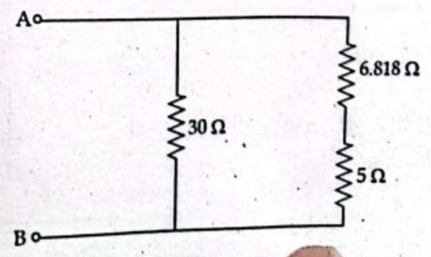
Adding series resistance,



Here, 12.5 Ω and 15 Ω are in parallel,

$$\frac{12.5 \times 15}{12.5 + 15} = 6.818\ \Omega$$

Thus,



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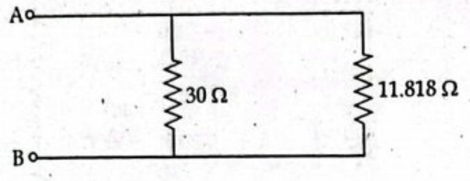
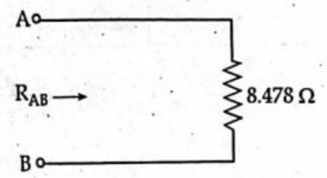
6.818 Ω and 5 Ω are in series = 6.818 Ω + 5 Ω = 11.818 Ω

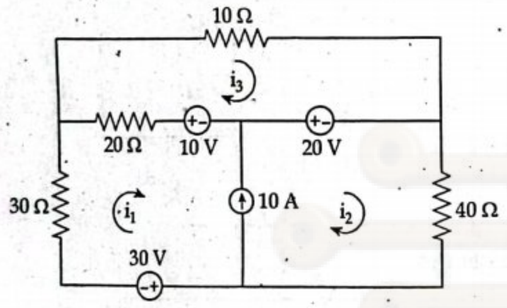


Hence, 30 Ω || 11.818 Ω = 8.478 Ω



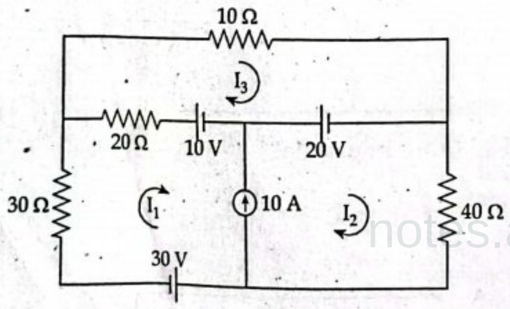
Thus equivalence resistance is 8.478 Ω.

52. Find i_1, i_2, i_3 current using mesh analysis method. [19/F]



Solution:

Here the loop I and II forms supermesh.



Applying KVL to supermesh,

or, $-30 I_1 - 20 (I_1 - I_3) - 10 - 20 - 40 I_2 - 30 = 0$

or, $-30 I_1 - 20 I_1 + 20 I_3 - 10 - 20 - 40 I_2 - 30 = 0$

or, $-50 I_1 - 40 I_2 + 20 I_3 = 60$ (1)

and, $I_2 - I_1 = 10$

or, $-I_1 + I_2 = 10$ (2)

Applying KVL to mesh III,

or, $-10 I_3 + 20 + 10 - 20 (I_3 - I_1) = 0$

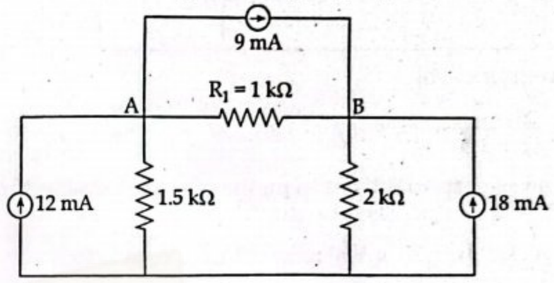
or, $-50 I_3 + 20 I_1 + 30 = 0$

or, $20 I_1 - 50 I_3 = -30$ (3)

On solving equation (1), (2) and (3);

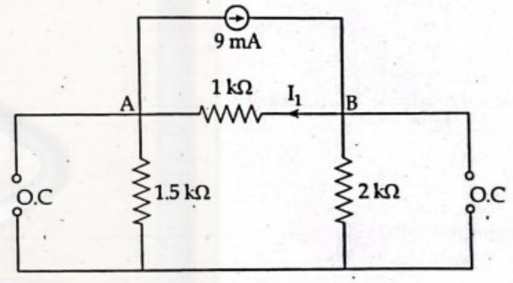
$\therefore$ $I_1 = -7.212$ A, $\therefore$ $I_2 = 2.787$ A, $\therefore$ $I_3 = -9.454$ A.

53. Find the current through R_1 using superposition theorem. [19/F]



Solution:

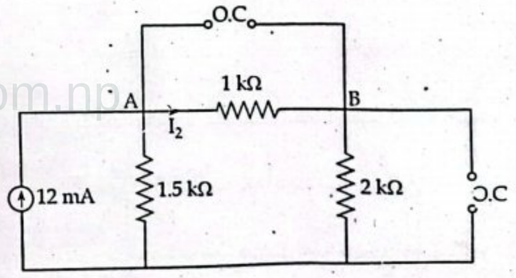
Taking 9 mA current source only and open circuiting other current sources,



Using current divider rule,

$$I_1 = 9 \times \frac{(2 + 1.5)}{(2 + 1.5) + 1} = 7 \text{ mA}$$

Now, taking 12 mA current source only and open circuiting other current sources,



Using current divider rule,

$$I_2 = 12 \times \frac{1.5}{(1.5 + 1 + 2)} = 4 \text{ mA}$$

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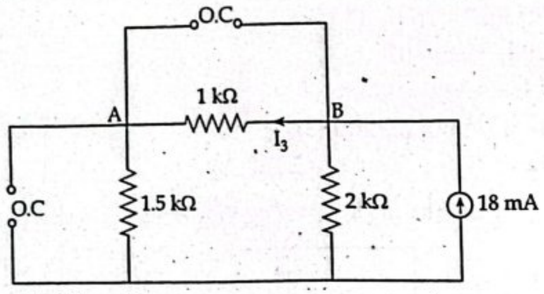
and taking 18 mA current source only



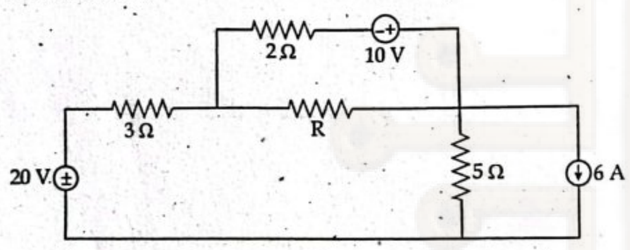
Using current divider rule

$$I_3 = 18 \times \frac{2}{(2+1.5+1)} = 8 \text{ mA}$$

Thus, applying superposition theorem, the current flowing through 1 kΩ resistor is

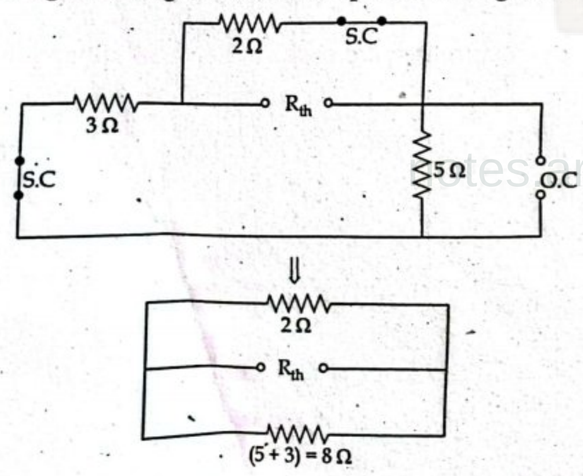
$$I_{R_1} = I_1 - I_2 + I_3 = (7 - 4 + 8) \text{ mA} = 11 \text{ mA from B to A}$$

54. Find the maximum power that can be delivered to the resistor R in the below circuit. [19/F]



Solution:

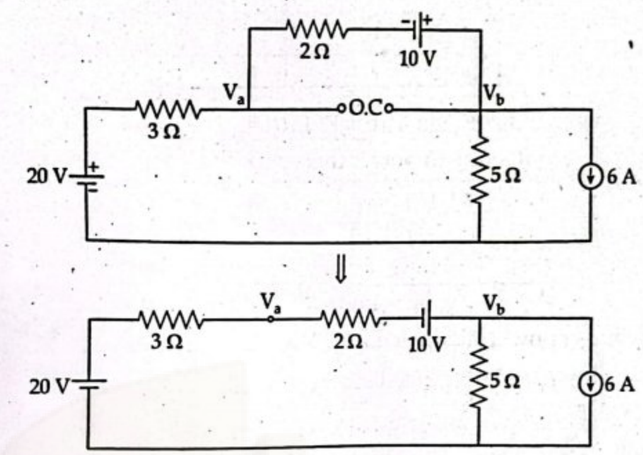
Removing the resistor R and removing all other sources to find the R^{th} *i.e.*, short circuiting the voltage sources, and open circuiting current sources.



Thus,

$$R_{th} = 2\,\Omega \parallel (3+5)\,\Omega = 2 \parallel 8 = \frac{2 \times 8}{2+8} = 1.60\,\Omega$$

To find voltage across R,



Using KCL at V_A,

$$\frac{V_b - V_a - 10}{2} + \frac{V_b}{5} + 6 = 0$$

or, $$\frac{V_b - V_a}{2} - 5 + \frac{V_b}{5} + 6 = 0$$

or, $$\frac{5V_b - 5V_a + 2V_b}{10} = -1$$

or, $-5V_a + 7V_b = -10$ (1)

Using KCL at V_b

$$\frac{V_a - V_b + 10}{2} + \frac{V_a - 20}{3} = 0$$

or, $$\frac{V_a - V_b}{2} + 5 + \frac{V_a}{3} - \frac{20}{3} = 0$$

or, $$\frac{3V_a - 3V_b + 2V_a}{6} = \frac{20}{3} - 5$$

or, $5V_a - 3V_b = (6.66 - 5) \times 6$

or, $5V_a - 3V_b = 10$ (2)

On solving equation (1) and (2);

$\therefore \quad V_a = 2\text{ V}$

$\therefore \quad V_b = 0\text{ V}$

Thus, voltage across R,

$$V_{th} = V_a - V_b = 2 - 0 = 2\text{V}$$

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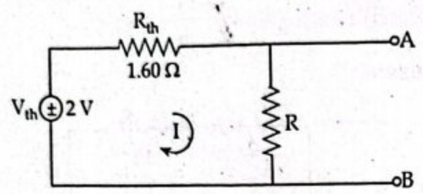
Thevenin's equivalent circuit is,



Maximum power is dissipated across R if $R = R_{th}$
(According to maximum power theorem)

Hence,

$R = 1.60\ \Omega$

so, Current, $I = \frac{V_{th}}{R + R_{th}} = \frac{2}{1.60 + 1.60} = 0.625$ A

Thus, maximum power dissipated across R,

$P_{max} = I^2R = (0.625)^2 \times 1.60 = 0.625$ watt

$\therefore\ P_{max} = 625$ mW.

55. Write short notes on color coding of resistance. [19/F]

Solution:

There are many different types of resistor available which can be used in both electrical and electronic circuits to control the flow of current or to produce a voltage drop in many different ways. But in order to do this the actual resistor needs to have some form of resistive or resistance value. Resistors are available in a range of different resistance value from fractions of an ohm to millions of ohms.

Obviously it would be impractical to have available resistors of every, possible value. Resistors are manufactured in what are are called preferred values with their resistance value printed onto their body in colored ink. The resistance value, tolerance and wattage rating are generally printed into the body of the resistor as numbers or letters when the resistors body is big enough to read the print, such as large power resistor. But when the resistor is small, these specifications must be shown in some other manner as the print would be too small to read. So to overcome this, small resistors use colored painted bands to indicate both their resistive value and their tolerance with the physical size of the resistor indicating its wattage rating. These colored painted bands produce a system of identification generally known as resistors color code.

The resistor color code markings are always read one band at a time starting from the left to the right with the larger width tolerance band oriented to the right side indicating its tolerance. By matching the color of the first band with its associated number in the digit column of the colour chart below, the first digit is indentified and this represent the first digit of the resistance value.

The resistor color code table

Colour	Digit	Multiplier	Tolerance
Black	0	1	± 1%
Brown	1	10	± 2%
Red	2	100	
Orange	3	1,000	
Yellow	4	10,000	
Green	5	1,00,000	± 0.5%
Blue	6	10,00,000	± 0.25%
Violet	7	1,00,00,000	± 0.1 %
Grey	8		± 0.05%
White	9		
Gold			± 5%
Silver			± 10%
None			± 20%

The 4 band color code is the most common and reliable color coding method to find the resistance value of a resistor. The A, B, C and tolerance are the 4 color bands which represent the actual value of a resistor.

Resistance = $AB \times 10^C \pm$ Tolerance % (Ω)

Example,

If color band A is green, B is violet, C is blue and tolerance is silver.

Then,

Band A = Green = 5

Band B = Violet = 7

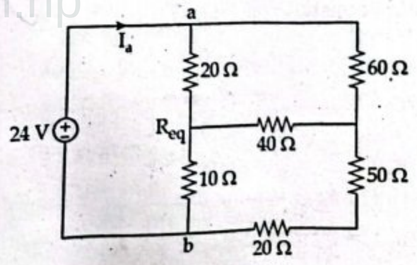
Band C = Blue = 6

Tolerance = Silver = ± 10%

so, Resistance = $AB \times 10^C \pm$ Tolerance % = $57 \times 10^6 \pm 10\%$

= 57 Megaohm ± 10%

56. Obtain the equivalent resistance and use it to find the source current for the circuit shown below. [19/S]



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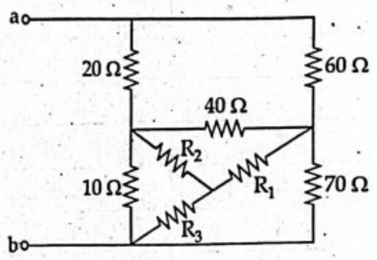
Solution:

Adding 20 Ω and 50 Ω = 20 Ω + 50 Ω = 70 Ω

Then circuit becomes,



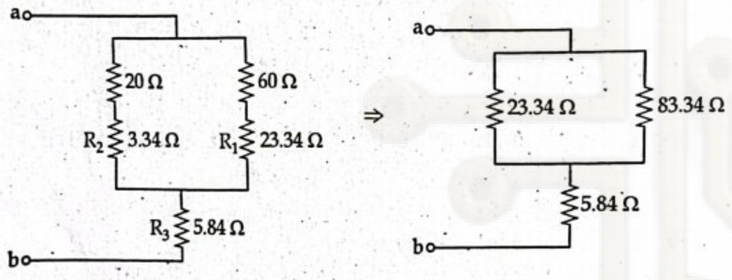
Transforming delta to star connection,

Here, $R_1 = \frac{40 \times 70}{40 + 70 + 10} = 23.34\ \Omega$

$R_2 = \frac{40 \times 10}{40 + 10 + 70} = 3.34\ \Omega$

$R_3 = \frac{10 \times 70}{10 + 70 + 40} = 5.84\ \Omega$

Redrawing the figure with star connection,



23.34 Ω and 83.34 Ω are in parallel.

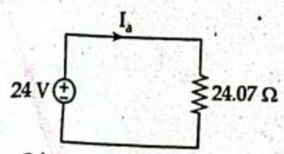
$= \frac{23.34 \times 83.34}{23.34 + 83.34} = 18.23\ \Omega$

Then,

Hence, $R_{eq} = 18.23 + 5.84\ \Omega = 24.07\ \Omega$

Therefore, equivalent resistance is 24.07 Ω

Now,



∴ Source current, $I_a = \frac{24}{24.07} = 0.997\ A \simeq 1\ A.$

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PURBANCHAL UNIVERSITY EXAMINATION QUESTION SOLUTIONS

1. State and verify reciprocity theorem with numerical example.

Solution:

The reciprocity theorem states that "The current I in any branch of a network due to a single voltage source E anywhere else in the network will equal the current through the branch in which current I was originally measured." The reciprocity theorem is thus applicable only to single source networks. In other words, the location of the voltage source and the resulting current may be interchanged without a change in current. The precaution is to be taken such that the polarity of the voltage source has the same correspondence with the direction of the branch current in that position.

Numerical proof:

In the network of figure 'a', the current I due to voltage source E was determined.

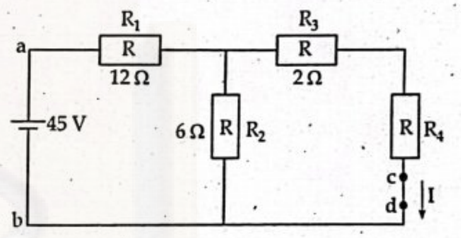

Fig: 'a'

If the position of each is interchanged as shown in figure 'b', the current I will be of the same value as indicated.

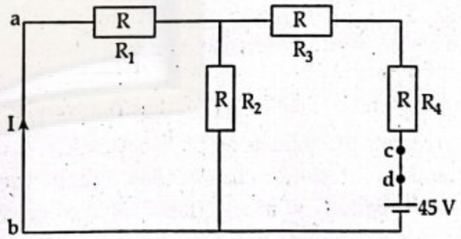

Fig: 'b'

To demonstrate the validity of this statement and the theorem, consider the network of fig 'a', with the values of the elements assigned as shown there,

Total resistance,

$R_T = R_1 + R_2 \parallel (R_3 + R_4) = 12 + 6 \parallel (2 + 4) = 12 + 6 \parallel 6 = 12 + 3$

∴ $R_T = 15\ \Omega$

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and, $I_T = \frac{E}{R_T} = \frac{45}{15} = 3\ A$

and, $I = 3 \times \frac{6}{6+6} = 1.5\ A$

For the network of figure 'b' which corresponds to that of figure 'a',

$R_T' = R_4 + R_3 + R_1 \| R_2 = 4 + 2 + 12 \| 6 = 6 + 4$

$R_T' = 10\Omega$

Thus, $I_T' = \frac{E}{R_T'} = \frac{45}{10} = 4.5\ A$

$I' = 4.5 \times \frac{6}{12+6} = 1.5\ A$

Thus, I' = I and I' equal which verifies reciprocity theorem.

2. Explain about different types of independent source and describe with neat sketch.

Those voltage or current sources which do not depend on any other quantity in the circuit are called independent sources.

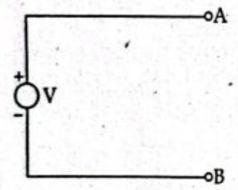


Fig: An ideal independent dc voltage source

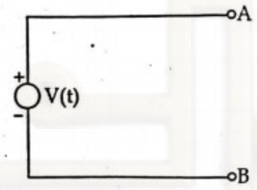


Fig: Time varying voltage source

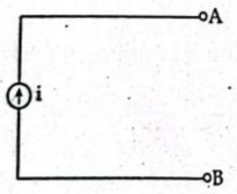


Fig: Ideal constant current source

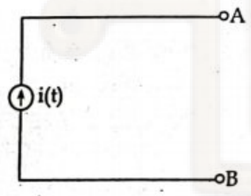


Fig: Time varying current source

Independent sources actually exists as physical entities such as a battery, a DC generator and an alternator etc. An ideal independent source is an active element that provides a specifies voltage or current that is completely independent of other circuit elements. In other words, an ideal independent voltage source delivers to the circuit whatever current is necessary to maintain its terminal voltage. Similarly, as ideal independent current source is an active element that provides a specified current completely independent of the voltage across the source. That is the current source delivers to the circuit whatever voltage is necessary to maintain.

3. State and prove maximum power transfer theorem.

Solution: See the definition part 2.15.

4. State and explain Kirchoff's voltage law.

Solution: See the definition part 2.12 and example number 2.

5. Write short notes on dependent and independent voltages sources.

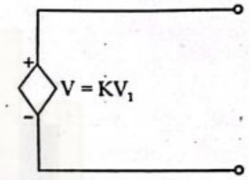
Solution:

Dependent source is a voltage source or a current source whose value depends on a voltage or current elsewhere in the network. An ideal dependent or controlled source is an active element in which the source is an active element in which the source quantity is controlled by another voltage or current. Dependent sources are usually designated by diamond-shaped symbols.

There are four possible types of dependent source,

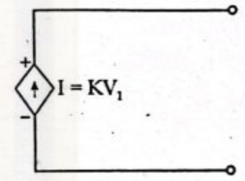
i) A voltage controlled voltage sources

It produces a voltage as a function of voltages elsewhere in the given circuit.



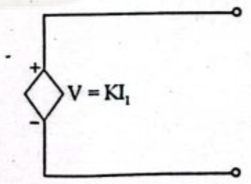
ii) Current controlled current source

It produces a current as a function of currents elsewhere in the given circuit.



iii) Current dependent voltage sources

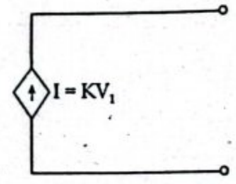
It produces a voltage as a function of current elsewhere in the given circuit.



iv) Voltage dependent current source

It produces a current as a function of voltage elsewhere in the given circuit.

Here, K is constant and V_1 and I_1 are the voltage and current respectively. Dependent sources are useful in modeling elements such as transistors, operational amplifiers and integrated circuit.



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6. Write short notes on temperature coefficient of resistance.

Solution:

Temperature coefficient of resistance is the measure of change in electrical resistance of any substance per degree of temperature change.

If suffix 1 indicates the initial condition and suffix 2 the final condition, we can write, from the definition of the temperature coefficient of resistance,

$$\alpha_1 = \frac{\text{Change in resistance/Change in temperature}}{\text{Resistance at the reference temperature}}$$

$$= \frac{(R_2 - R_1)/(t_1 - t_1)}{R_1} \quad (1)$$

where, α_1 is the temperature coefficient of the material at t_1°C

Similarly, at temperature t_2 °C, the temperature coefficient of resistance being expressed as α_2,

$$\alpha_2 = \frac{(R_2 - R_1)/(t_2 - t_1)}{R_2} \quad (2)$$

It is usual to specify the reference temperature as 0°C while the temperature coefficient of resistance is α_0. If t °C is the rise in temperature and the resistance changes from R_0 (at 0°C) to R_t and t °C, we can write,

$$\alpha_0 = \frac{(R_t - R_0)/(t - 0)}{R_0}$$

$$\text{or,} \quad \alpha_0 = \frac{R_t - R_0}{R_0 t}$$

$$\text{or,} \quad (R_t - R_0) = R_0 \alpha_0 t$$

$$\text{or,} \quad R_t = R_0 + R_0 \alpha_0 t$$

$$\therefore \quad R_t = R_0 (1 + \alpha_0 t)$$

Again from equation (1) and (2);

$$\frac{\alpha_2}{\alpha_1} = \frac{R_1}{R_2} = \frac{R_1}{R_1 [1 + \alpha_1 (t_2 - t_1)]}$$

$$\text{or,} \quad \frac{\alpha_2}{\alpha_1} = \frac{1}{1 + \alpha_1 (t_2 - t_1)}$$

$$\text{or,} \quad \alpha_2 = \frac{\alpha_1}{1 + \alpha_1 (t_2 - t_1)}$$

$$\therefore \quad \alpha_2 = \frac{1}{\frac{1}{\alpha_1} + (t_2 - t_1)}$$

In general, if the reference temperature is 0°C for t °C rise in temperature,

$$\alpha_t = \frac{1}{\frac{1}{\alpha_0} + (t - 0)}$$

$$\therefore \quad \alpha_t = \frac{1}{\frac{1}{\alpha_0} + t}$$

The unit of α is Ω/Ω °C *i.e.*, per °C

Hence, $\alpha_0 = \frac{R_t - R_0}{R_0 \Delta t}$

If $R_0 = 1\Omega$ and $\Delta t = 1$°C; then

$\alpha = R_t - R_0$

If the resistance of a conductor is R_2 at t_2 °C and R_1 at t_1 °C such that $t_2 > t_1$, Then,

$R_2 = R_1 [1 + \alpha_1 (t_2 - t_1)]$.

7. State and explain Norton's theorem with an appropriate example.

Solution: See the definition part 2.14. C and example number 25.

8. A coil has a resistance of 3.146 Ω at 40°C and 3.767Ω at 100°C. Find the resistance at 0°C and temperature coefficient of resistance at 40°C.

Solution:

$R_{40} = 3.146\ \Omega$ at $t = 40$°C

$R_{100} = 3.767\ \Omega$ at $t = 100$°C

We know,

$R_t = R_0 (1 + \alpha t)$

$$\text{or,} \quad \frac{R_{100}}{R_{40}} = \frac{R_0 (1 + \alpha_0 t_{100})}{R_0 (1 + \alpha_0 t_{40})}$$

$$\text{or,} \quad \frac{3.767}{3.146} = \frac{1 + 100\,\alpha_0}{1 + 40\,\alpha_0}$$

$$\text{or,} \quad 1.1974 = \frac{1 + 100\,\alpha_0}{1 + 40\,\alpha_0}$$

$$\text{or,} \quad 1.1974 + 47.896\,\alpha_0 = 1 + 100\,\alpha_0$$

$$\text{or,} \quad 0.1974 = 52.104\,\alpha_0$$

$$\text{or,} \quad \alpha_0 = \frac{0.1974}{52.104}$$

$$\therefore \quad \alpha_0 = 0.003788 \text{ per °C}$$

Now,

At 100°C,

$$3.767 = R_0 (1 + 0.003786 \times 100)$$

$$\text{or,} \quad \frac{3.767}{1.3788} = R_0$$

$$\therefore \quad R_0 = 2.732\ \Omega$$

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and, Temperature coefficient of resistance at 40°C is,

$$\alpha_{40} = \frac{\alpha_0}{1 + 40 \times \alpha_0}$$

or, $$\alpha_{40} = \frac{0.003786}{1 + 40 \times 0.003786}$$

$\therefore$ $\alpha_{40} = 0.003288$ per °C.

9. Define specific resistance. A coil consists of 200 turns of copper wire having a cross sectional area of 0.8 mm². The mean length per turn is 80 cm and resistivity of copper is 0.02 μΩ – m. Find the resistance of the coil and power absorbed by coil when connected across 110 V dc supply.

Solution:

Specific resistance of a conductor is the resistance per unit length and unit area of cross section. The specific electrical resistance of a conductor depends on its material and the temperature.

$$\rho = R \cdot \frac{A}{L}$$

where, ρ = Specific resistance

R = Resistance, Ω

L = Length of conductor, m

A = Cross section area, m²

Number of turns of copper coil, N = 200

Area of copper wire, $A = 0.8\ mm^2 = 8 \times 10^{-7}\ m^2$

Resistivity of copper, ρ = 0.02 μΩ – m

Mean length per turn, L/N = 80 cm = 0.8 m

Resistance of coil, R =?

We know,

$$R = \rho \frac{L}{A}$$

or, $$R = \rho \times 0.8 \times \frac{N}{A} \qquad \left[\because \frac{L}{N} = 0.8\ m\right]$$

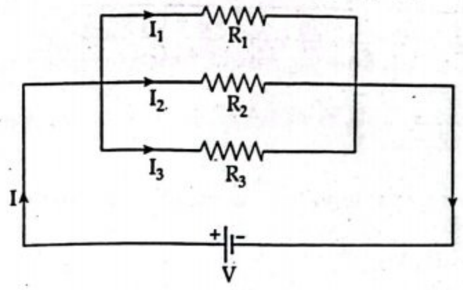
or, $$R = 0.02 \times 0.8 \times \frac{200}{8 \times 10^{-7}}$$

$\therefore$ $R = 4 \times 10^6\ \Omega = 4$ Mega ohm

Now, Power absorbed by copper coil,

$$P = VI = \frac{V^2}{R} = \frac{110^2}{4 \times 10^6} = 3.025 \times 10^{-3}\ \text{watt} = 3.025\ \text{mW}.$$

10. Derive a formula to express branch currents in terms of total current and conductance present in the circuit. Assume that there are three branches having resistance R_1, R_2, R_3.



Solution:

From above parallel circuit,

$$I = I_1 + I_2 + I_3$$

We know, from ohm's law,

$$V = IR$$

so, $$V_1 = I_1 R_1$$

$$V_2 = I_2 R_2$$

$$V_3 = I_3 R_3$$

Since in parallel circuit voltage is same and current in each branch is different. *i.e.*, $V = V_1 = V_2 = V_3$.

We get,

$$\frac{V}{R_{eq}} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

or, $$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

or, $$G_{eq} = G_1 + G_2 + G_3$$

Here, G_1, G_2 and G_3 are conductance of the resistors R_1, R_2 and R_3 respectively.

Thus, $$R_{eq} = \frac{R_1 R_2 R_3}{R_1R_2 + R_2R_3 + R_1R_3}$$

Applying current divider rule,

$$I_1 = \frac{V}{R_1} = \frac{I\ Req}{R_1} = \frac{I}{R_1} \times \frac{R_1 R_2 R_3}{R_1R_2 + R_2R_3 + R_1R_3} = I \times \frac{R_2 R_3}{R_1R_2 + R_2R_3 + R_1R_3}$$

Similarly,

$$I_2 = I \times \frac{R_1 R_3}{R_1R_2 + R_2R_3 + R_3R_1}$$

$$I_3 = I \times \frac{R_1 R_2}{R_1R_2 + R_2R_3 + R_1 R_3}$$

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In terms of conductance,

$$I_1 = I \times \frac{\frac{1}{G_2} \times \frac{1}{G_3}}{\frac{1}{G_1}\cdot\frac{1}{G_2} + \frac{1}{G_2}\cdot\frac{1}{G_3} + \frac{1}{G_1}\cdot\frac{1}{G_3}} = \frac{I \times \frac{1}{G_2 G_3}}{\frac{G_3 + G_1 + G_2}{G_1 G_2 G_3}}$$

$$\therefore \quad I_1 = I \times \frac{G_1}{G_1 + G_2 + G_3}$$

Similarly,

$$\therefore \quad I_2 = I \times \frac{G_2}{G_1 + G_2 + G_3}$$

$$\therefore \quad I_3 = I \times \frac{G_3}{G_1 + G_2 + G_3}.$$

11. A current of 78 A is divided into three branches. The lengths of the wires in three branches are in ratio 8, 12, 16. The wires are of same materials and cross sectional area. Calculate current in each branch.

Solution:

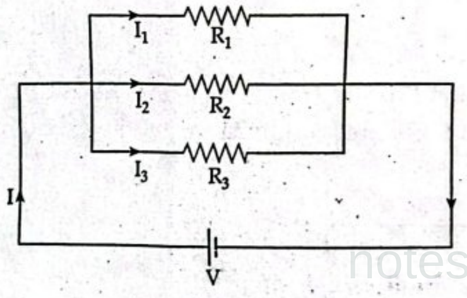
Ratio of lengths of wire in three branch = 8: 12: 16

We know,

Resistance, $R = \rho \cdot \frac{L}{A}$

Since, wires are of same material and have same cross sectional area.

$R \propto L$



i.e., $R_1 : R_2 : R_3 = L_1 : L_2 : L_3 = 8 : 12 : 16$

Thus,

$$I_1 = I \times \frac{R_2 R_3}{R_1R_2 + R_2R_3 + R_1R_3} = 78 \times \frac{12 \times 16}{8 \times 12 + 12 \times 16 + 8 \times 16}$$

$$\therefore \quad I_1 = 78 \times \frac{192}{416}$$

$$\therefore \quad I_1 = 36 \text{ A}$$

Now,

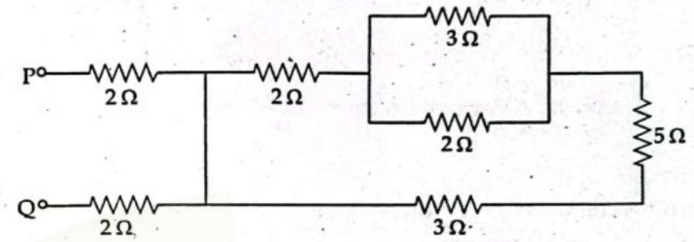
$$I_2 = I \times \frac{R_1 R_3}{R_1R_2 + R_2R_3 + R_1R_3} = 78 \times \frac{8 \times 16}{416}$$

$$\therefore \quad I_2 = 24 \text{ A}$$

and, $I_3 = I \times \frac{R_2R_1}{R_1R_2 + R_2R_3 + R_1R_3} = 78 \times \frac{8 \times 12}{416} = 18 \text{ A}.$

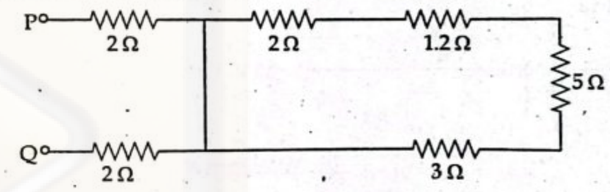
Hence current in each branch are 36 A : 24 A : 18 A having length ratio 8 : 12 : 16 of wires.

12. Find the equivalent resistance of the circuit across P-Q shown below.



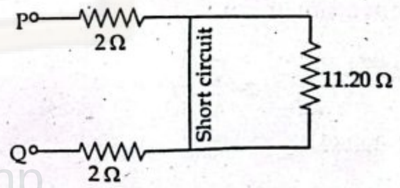
Solution:

Here, 2 Ω and 3 Ω are in parallel,

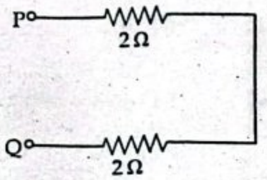


$$3\ \Omega \parallel 2\ \Omega = \frac{3 \times 2}{3 + 2} = 1.20\ \Omega$$

Also, 2 Ω, 1.2 Ω, 5 Ω and 3 Ω are in series connection, so adding those resistance,



Due to short circuit present in parallel with 11.20 Ω, there will be zero resistance, thus,



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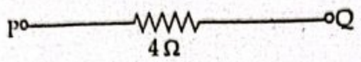
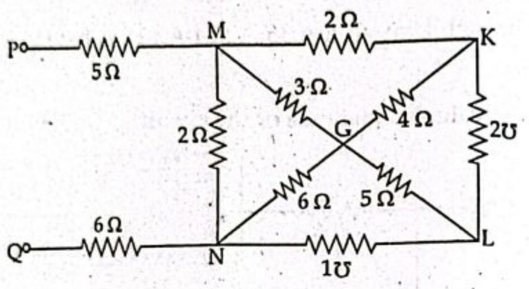
Thus, equivalent resistance between P and Q is $2\,\Omega + 2\,\Omega = 4\,\Omega$.



13. Use star delta conversion to find resistance between terminal P of the circuit shown below.

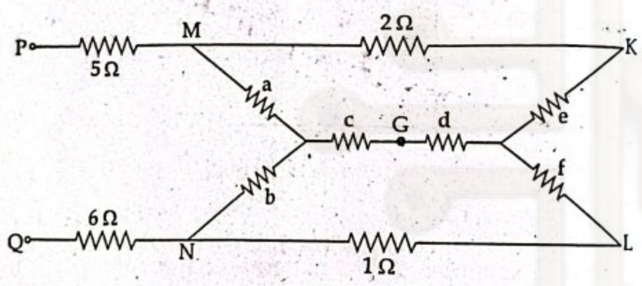


Solution:

Transforming delta circuit MGN and KGL into star circuit. By converting 1 ℧ and 2 ℧ to resistance,

$2\,℧ = 0.5\,\Omega$

$1\,℧ = 1\,\Omega$



Using delta to star conversion formula,

$$a = \frac{2 \times 3}{2+3+6} = 0.55\,\Omega$$

$$b = \frac{2 \times 6}{2+3+6} = 1.090\,\Omega$$

$$c = \frac{3 \times 6}{2+3+6} = 1.64\,\Omega$$

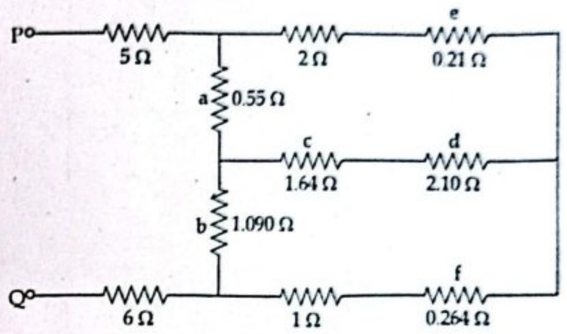
$$d = \frac{4 \times 5}{4+5+0.5} = 2.10\,\Omega$$

$$e = \frac{4 \times 0.5}{4+5+0.5} = 0.21\,\Omega$$

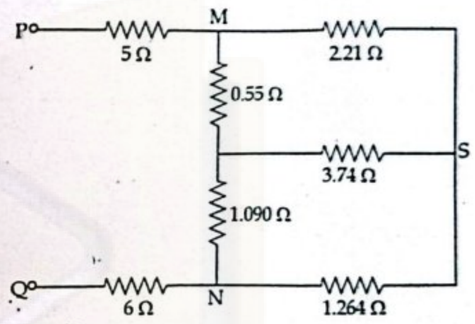
$$f = \frac{5 \times 0.5}{4+5+0.5} = 0.264\,\Omega$$

In above circuit, c and d are in series also, 2 Ω and e and 1 Ω and f are in series.

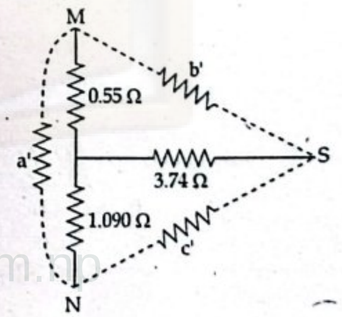
so,



On adding series resistance



Transforming MNS star into equivalent delta



$$a' = 0.55 + 1.090 + \frac{0.55 \times 1.090}{3.74} = 1.80\,\Omega$$

$$b' = 0.55 + 3.74 + \frac{0.55 \times 3.74}{1.090} = 6.18\,\Omega$$

$$c' = 1.090 + 3.74 + \frac{1.090 \times 3.74}{0.55} = 12.24\,\Omega$$

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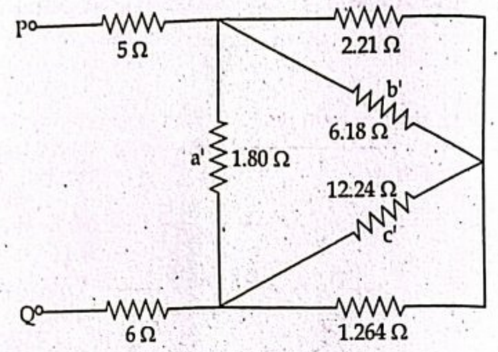
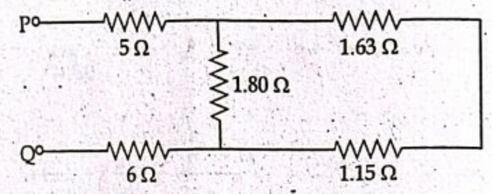
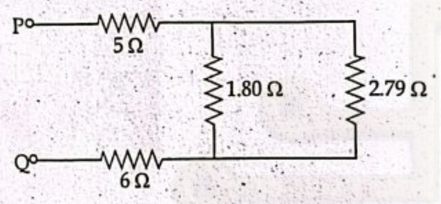
Now,



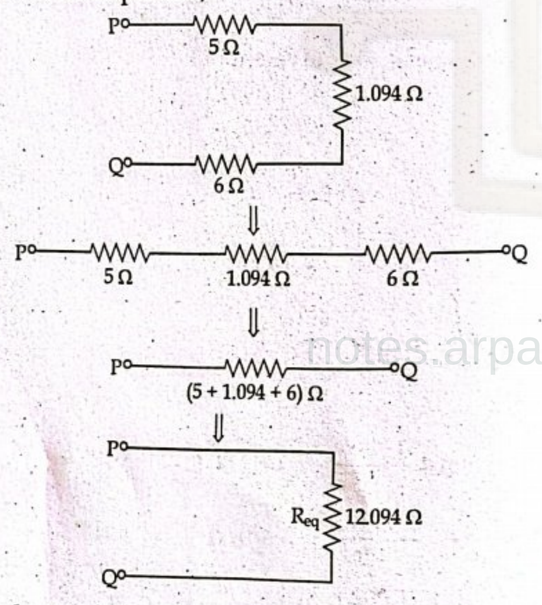
Here, 2.21 || 6.18 and 12.24 || 1.264 Ω



1.63 Ω and 1.15 Ω and in series, so

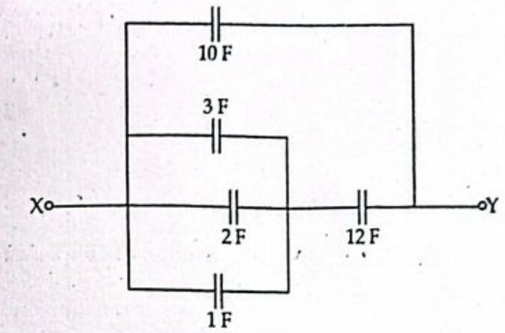


1.80 Ω and 2.79 Ω are in parallel,



Hence, equivalent resistance between P and Q is 12.094 Ω.

14. Find the equivalent capacitance of the circuit across X-Y shown below.



Solution

For n number of capacitance to series,

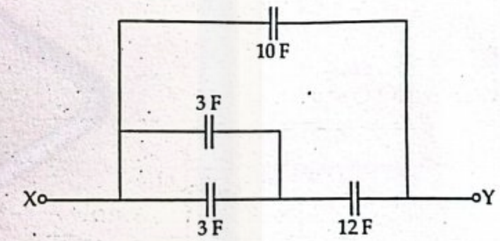
$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots\ldots + \frac{1}{C_n}$$

For n number of capacitance in parallel

$$C = C_1 + C_2 + \ldots\ldots + C_n$$

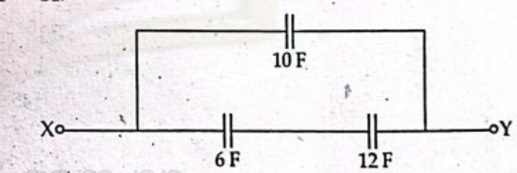
Here, 1F and 2F are in parallel.

so, 1F + 2F = 3F



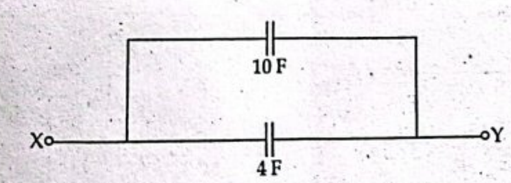
Also, 3F and 3F are in parallel.

so, 3F + 3F = 6F



Here, 6F and 12F are in series.

so, $\frac{6 \times 12}{6 + 12}$ F = 4F



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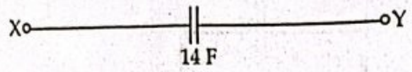
Now,

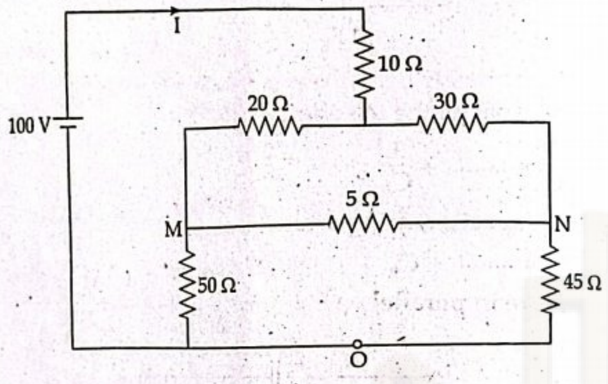
4F and 10F are in parallel, so adding 4F and 10F.

i.e., 4F + 10F = 14F



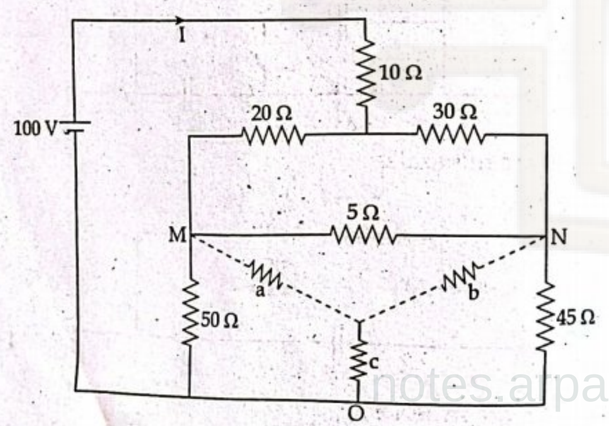
Thus, equivalent capacitance is 14F.

15. Find current drawn from the source voltage of 100 V in the figure



Solution:

Transforming delta MNO to star,

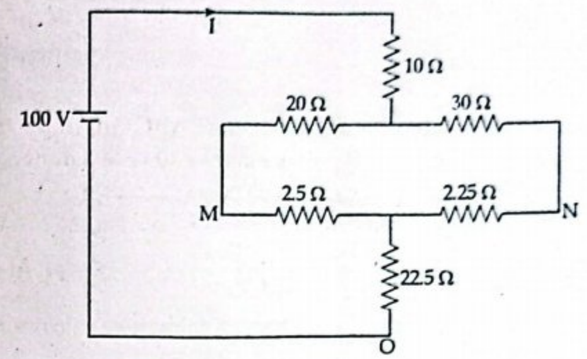


$$a = \frac{50 \times 5}{50 + 5 + 45} = 2.5\ \Omega$$

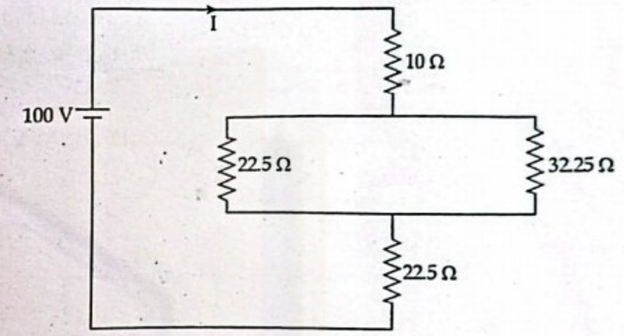
$$b = \frac{5 \times 45}{50 + 5 + 45} = 2.25\ \Omega$$

$$c = \frac{50 \times 45}{50 + 5 + 45} = 22.5\ \Omega$$

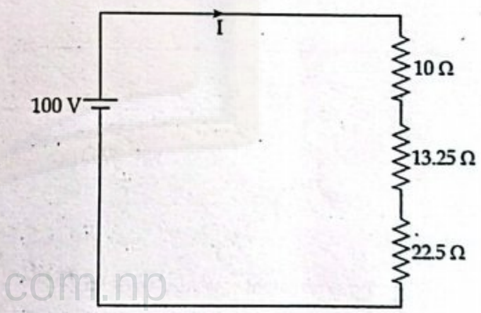
Thus,



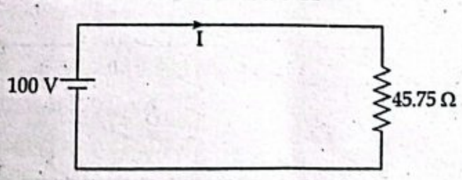
Adding series resistance,



Also, $22.5\ \Omega \parallel 32.25\ \Omega = \frac{22.5 \times 32.25}{22.5 + 32.25} = 13.25\ \Omega$



Thus, $R_{eq} = 10\ \Omega + 13.25\ \Omega + 22.5\ \Omega = 45.75\ \Omega$



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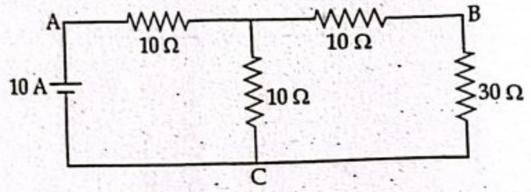
and, Current drawn from the source is,

$$I = \frac{V}{R} = \frac{100}{45.75}$$

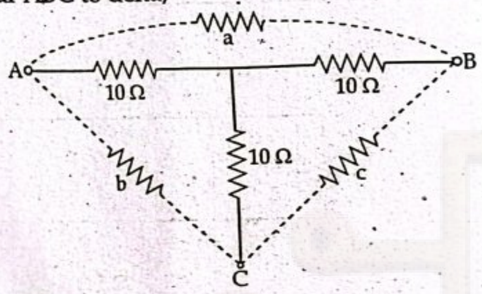
$\therefore$ $I = 2.19$ A.

16. In the circuit below, transform star ABC to delta and apply thevenins theorem to find voltage across 30 Ω resistance.



Solution:

Transforming star ABC to delta,



$$a = 10 + 10 + \frac{10 \times 10}{10} = 30\ \Omega$$

$a = b = c = 30\ \Omega$

Thus,

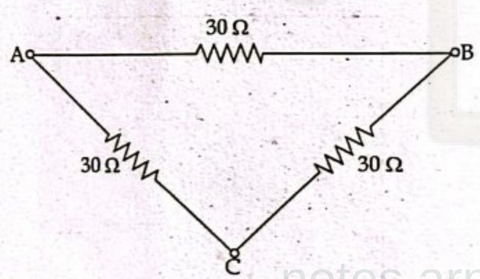
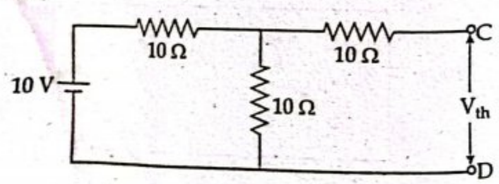


Fig: Equivalent star ABC in delta

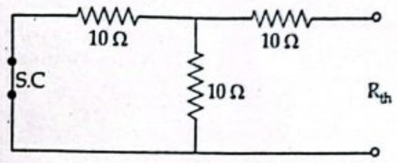
Thevenin's theorem:

Open 30 Ω resistor,



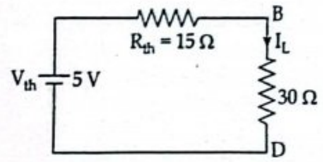
$$V_{th} = V \times \frac{10}{10 + 10} = \frac{10 \times 10}{10 + 10} = 5\ V$$

Short circuit voltage sources to get R_{th},



$$R_{th} = (10 \parallel 10)\ \Omega + 10\ \Omega = \frac{10 \times 10}{10 + 10} + 10 = 5 + 10 = 15\ \Omega$$

Hence, thevenin's equivalent circuit is,



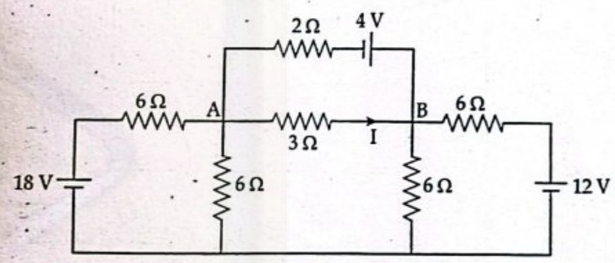
$$I_L = \frac{V_{th}}{R_{th} + R_L} = \frac{5}{15 + 30}$$

$\therefore$ $I_L = 0.1111$ A

and, Voltage across 30 Ω = $I_L \times 30$

$= 0.1111 \times 30 = 3.333$ V.

17. Find I in the circuit using nodal analysis.



Solution:

At node A,

$$\frac{V_A}{6} + \frac{V_A - 18}{6} + \frac{V_A - V_B}{3} + \frac{V_A + 4 - V_B}{2} = 0$$

or, $\frac{V_A}{6} + \frac{V_A}{6} - 3 + \frac{V_A}{3} + \frac{V_A}{2} + 2 - \frac{V_B}{3} - \frac{V_B}{2} = 0$

or, $1.1667\ V_A - 0.1667\ V_B - 1 = 0$

or, $1.1667\ V_A - 0.1667\ V_B = 1$ (1)

At node B,

$$\frac{V_B}{6} + \frac{V_B - 12}{6} + \frac{V_B - 4 - V_A}{2} + \frac{V_B - V_A}{3} = 0$$

or, $1.1667\ V_B - 4 - 0.5\ V_A - 0.333\ V_A = 0$

or, $-0.833\ V_A + 1.1667\ V_B = 4$ (2)

On solving equation (1) and (2);

$\therefore$ $V_A = 1.50$ V

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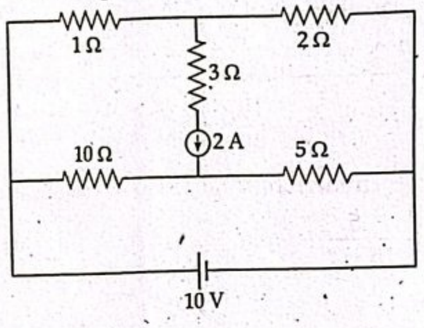
$\therefore \quad V_B = 4.50\ V$

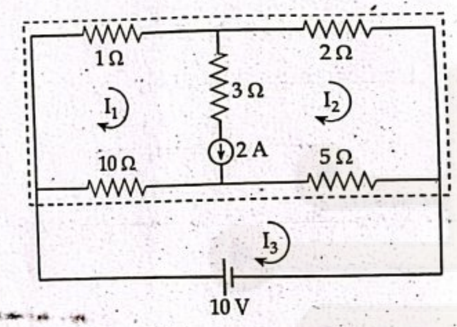
Hence,

$$I = \frac{V_A - V_B}{3} = \frac{1.50 - 4.50}{3} = -1\ A$$

18. Find the current through 5 Ω resistor using mesh analysis.



Solution:



Here loop 1 and loop 2 forms supermesh because 2 A current is comm between loop.1 and loop 2.

$$I_1 - I_2 = 2\ A$$

Applying KVL to supermesh,

$$-I_1 - 2I_2 - 5(I_2 - I_3) - 10(I_1 - I_3) = 0$$

or, $-I_1 - 2I_2 - 5I_2 + 5I_3 - 10I_1 + 10I_3 = 0$

or, $-11I_1 - 7I_2 + 15I_3 = 0$

Applying KVL to loop III,

$$10 - 10(I_3 - I_1) - 5(I_3 - I_2) = 0$$

or, $10 - 10I_3 + 10I_1 - 5I_3 + 5I_2 = 0$

or, $10 - 15I_3 + 10I_1 + 5I_2 = 0$

or, $10I_1 + 5I_2 - 15I_3 = -10$

On solving equation (1), (2) and (3);

$\therefore \quad I_1 = 4.67\ A$

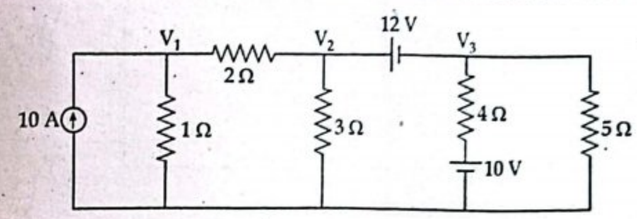
$\therefore \quad I_2 = 2.67\ A$

$\therefore \quad I_3 = 4.67\ A$

Hence current through 5 Ω resistor,

$$I_{5\Omega} = I_3 - I_2 = (4.67 - 2.67)\ A = 2\ A.$$

19. Calculate the current in 3 Ω resistor for the circuit shown below.



Solution:

V_2 and V_3 forms supernode.

so, At node V_1:

$$-10 + \frac{V_1}{1} + \frac{V_1 - V_2}{2} = 0$$

or, $V_1 + \frac{V_1}{2} - \frac{V_2}{2} = 0$

or, $1.5\ V_1 - 0.5\ V_2 = 10$ (1)

From supernode,

$$V_2 - V_3 = 12$$ (2)

Applying KCL at supernode,

$$\frac{V_2 - V_1}{2} + \frac{V_2}{3} + \frac{V_3}{5} + \frac{V_3 - 10}{4} = 0$$

or, $\frac{V_2}{2} - \frac{V_1}{2} + \frac{V_2}{3} + \frac{V_3}{5} + \frac{V_3}{4} - \frac{10}{4} = 0$

or, $0.834\ V_2 - 0.5\ V_1 + 0.45\ V_3 = 2.50$ (3)

On solving equation (1), (2) and (3);

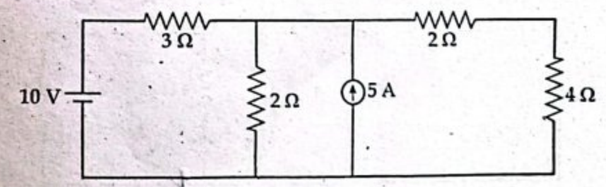
$\therefore \quad V_1 = 10.0179\ V$

$\therefore \quad V_2 = 10.05\ V$

$\therefore \quad V_3 = -1.95\ V$

and, Current in 3 Ω resistor $= \frac{V_2}{3} = \frac{10.05}{3} = 3.35\ A.$

20. Using Thevenin's theorem, calculate the current flowing through 4 Ω resistor.



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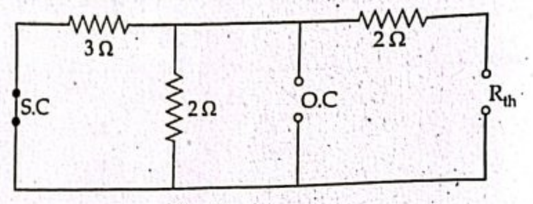
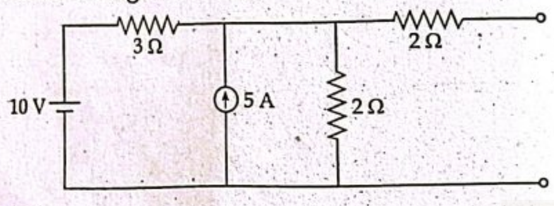
Solution:

To find equivalent resistance; open circuit current source, short circuit voltage source and remove 4 Ω.



$$R_{th} = [2 + (2 \parallel 3)]\ \Omega = 2 + \frac{2 \times 3}{2+3} = 3.20\ \Omega$$

To find Thevenin's voltage across R_L,



At V, using KCL,

$$\frac{V-10}{3} + \frac{V}{2} - 2 = 0$$

or, $\frac{V}{3} - \frac{10}{3} + \frac{V}{2} = 2$

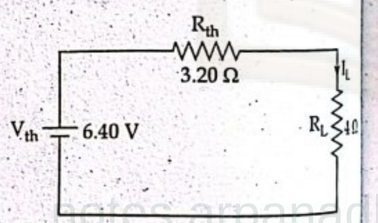
or, $0.333\ V + 0.5\ V = 2 + 2.333$

or, $0.833\ V = 5.333$

or, $V = \frac{5.333}{0.833}$

$\therefore\ V_{th} = 6.40\ V$

Then, Thevenin's equivalent circuit is,

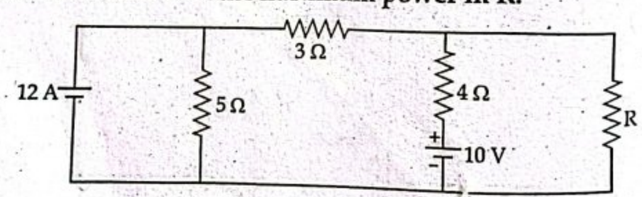


$$I_L = \frac{V_{th}}{R_{th} + R_L} = \frac{6.40}{3.20 + 4}$$

$\therefore\ I_L = 0.888\ A$

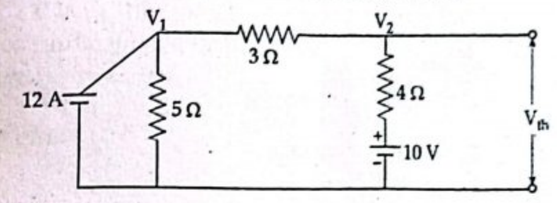
Hence, current through 4 Ω resistor is 0.888 A.

21. Find the value of 'R' in the network necessary to obtain maximum power in it. Also find the maximum power in R.



Solution:

Using Thevenin theorem to solve the network, for V_{th},



Applying KCL,

Using the concept of supernode; if the voltage source is connected between the reference and a non-reference node, we simply set the voltage at the non-reference node equal to the voltage of the voltage source.

so, $V_1 = 12\ V$

At node V_2,

$$\frac{V_2 - 10}{4} + \frac{V_2 - V_1}{3} = 0$$

or, $\frac{V_2 - 10}{4} + \frac{V_2 - 12}{3} = 0$

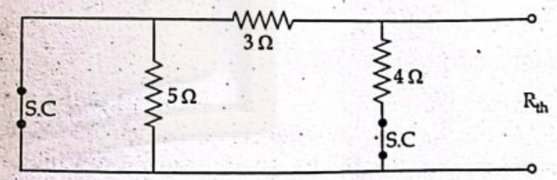
or, $\frac{V_2}{4} - \frac{10}{4} + \frac{V_2}{3} - \frac{12}{3} = 0$

or, $0.25\ V_2 + 0.333\ V_2 = 4 + 2.50$

or, $V_2 = \frac{6.50}{0.583}$

$\therefore\ V_2 = V_{th} = 11.15\ V$

To get R_{th},



$$R_{th} = 3\ \Omega \parallel 4\ \Omega = \frac{3 \times 4}{3+4} = 1.714\ \Omega$$

For maximum power,

$R_{th} = R_L$

so, $R_L = 1.714\ \Omega$

Thus, value of R is 1.714 Ω.

Now,

$$\text{Maximum power, } P_{max} = \frac{V_{th}^2}{4\,R_{th}} = \frac{(11.15)^2}{4 \times 1.714} = 18.14 \text{ watt.}$$

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ADDITIONAL QUESTION SOLUTIONS

1. A particular metal filament has a resistance of 10 Ω at 0°C. At 20°C the resistance becomes 12 Ω. Calculate the temperature coefficient of resistance of that filament at 20°C. What is the temperature coefficient at 0°C?

Solution:

$R_0 = 10\ \Omega$

$R_t = 12\ \Omega$

We know,

$$\alpha_t = \frac{R_t - R_0}{R_t \cdot t}$$

or, $R_0 = R_t (1 - \alpha_t t)$

Here, $t = 20°C\ [= (t_2 - t_1) = (20 - 0)°C]$

Hence, $10 = 12 (1 - \alpha_t \cdot 20)$

or, $10 = 12 - 12 \times 20 \times \alpha_t$

or, $\alpha_t = \frac{12 - 10}{12 \times 10}$

$\therefore \quad \alpha_t = 0.00833/°C$

i.e., $\alpha_{20} = 0.00833/°C$

Also,

$R_t = R_0 (1 + \alpha_0 t)$

or, $12 = 10 (1 + \alpha_0 \times 20)$

or, $\alpha_0 = \left(\frac{12}{10} - 1\right) \times \frac{1}{20}$

$\therefore \quad \alpha_0 = 0.001/°C$

i.e., temperature coefficient at 0°C is 0.01/°C.

2. A given conductor has a resistor of 5 W. What is the resistance of another conductor of same material, which has one-half the diameter and five times the length of the given conductor?

Solution:

We know,

$$R_1 = \rho \frac{L_1}{a_1} \text{ and}$$

$$R_2 = \rho \frac{L_2}{a_2}$$

$$\therefore \quad \frac{R_2}{R_1} = \frac{L_2}{L_1} \times \frac{a_1}{a_2}$$

Here, $a_1 = \pi \frac{d_1^2}{4}$; $a_2 = \pi \frac{d_2^2}{4}$

Also, $d_2 = \frac{d_1}{2}$ (given)

$$\therefore \quad \frac{a_1}{a_2} = \frac{d_1^2}{d_2^2} = \left(\frac{d_1}{d_2}\right)^2 = \left(\frac{d_1}{d_1/2}\right)^2 = 4$$

Also, $L_2 = 5L_1$ or $\frac{L_2}{L_1} = 5$

Hence, $\frac{R_2}{R_1} = \frac{L_2}{L_1} \times \frac{a_1}{a_2} = 5 \times 4 = 20$

or, $R_2 = R_1 \times 20$

$\therefore \quad R_2 = 5 \times 20 = 100\ \Omega.$

3. Obtain the ratio of powers lost in two resistor when

i) Each being connected across the same voltage.

ii) Each resistor is to carry the same current

Assume same length and material for each resistor but the first one having a diameter twice that of other. The cross section of each of the resistor wire is circular.

Solution:

i) Voltage applied remains same for each resistor.

$$P_1 = \frac{V^2}{R_1} = \frac{V^2}{\frac{\rho L}{a_1}} = \frac{\pi}{4} \frac{V^2 d_1^2}{\rho L}$$

Similarly,

$$P_2 = \frac{V^2}{R_2} = \frac{\pi}{4} \frac{V^2 d_2^2}{\rho L}$$

Since, $d_1 = 2d_2$

We have,

$$\therefore \quad \frac{P_1}{P_2} = \frac{d_1^2}{d_2^2} = \frac{4 d_2^2}{d_2^2} = 4$$

ii) Each resistor carries same current

$$P_1 = I^2 R_1 = I^2 \times \rho \times \frac{L}{a_1} = \frac{4}{\pi} L^2 \cdot \frac{\rho L}{d_1^2}$$

Similarly, $P_2 = I^2 R_2 = \frac{4}{\pi} L^2 \frac{\rho L}{d_2^2}$

$\because \quad d_1 = 2\, d_2$

Now,

$$\therefore \quad \frac{P_1}{P_2} = \frac{d_2^2}{d_1^2} = \frac{d_2^2}{4 d_2^2} = \frac{1}{4}$$

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4. A coil has a resistance of 18 ohm at 20°C and 22 ohm at 50°C. Find the rise in the temperature when resistance becomes 24 ohm. The room temperature is 18°C.

Solution:

$R_1 = 18\ \Omega$

$t_1 = 20°C$

$R_2 = 22\ \Omega$

$t_2 = 50°C$

Now,

$R_2 = R_1 [1 + \alpha_1 (t_2 - t_1)]$

or, $22 = 18 [1 + \alpha_1 (50 - 20)]$

or, $\frac{22}{18} = 1 + 30\,\alpha_1$

or, $1.222 - 1 = 30\,\alpha_1$

or, $\alpha_1 = \frac{0.222}{30}$

$\therefore \quad \alpha_1 = 0.00740/°C$

and, $R_3 = 24\ \Omega$

so, $R_3 = R_1 [1 + 0.00740 (t_3 - 20)]$

or, $24 = 18 [1 + 0.00740 (t_3 - 20)]$

or, $\frac{24}{18} - 1 = 0.00740 (t_3 - 20)$

or, $t_3 - 20 = \frac{0.3333}{0.00740}$

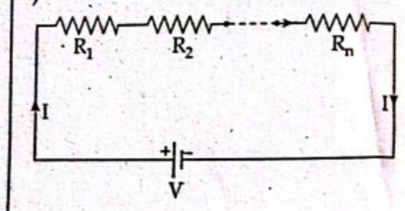
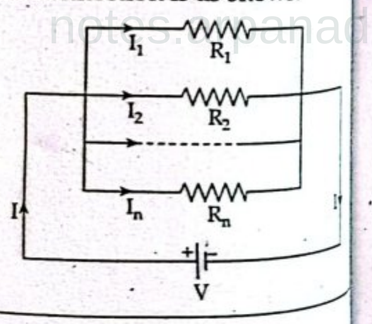
or, $t_3 = 20 + 45$

$\therefore \quad t_3 = 65°C$

so, room temperature is 18°C and temperature rise is $(65 - 18)° C = 47°C$

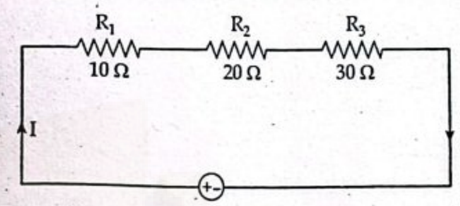
5. Comparison of series and parallel circuit.

Solution:

Series circuit	Parallel circuit
a) The connection is as shown 	The connection is as shown 

b) The same current flows through each resistance.	The same voltage exists across all the resistance in parallel.
c) The voltage across each resistance is different.	The current through each resistance is different.
d) Equivalent resistance is $R_{eq} = R_1 + R_2 + \ldots\ldots + R_n$	Equivalent resistance is $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots\ldots + \frac{1}{R_n}$
e) The equivalent resistance is the largest than each of the resistance in series.	The equivalent resistance is the smaller than the smallest of all the resistance in parallel.

6. Find the voltage across the three resistance.



Solution:

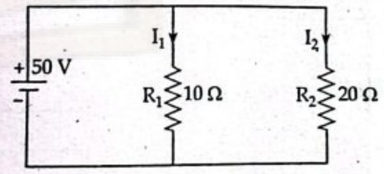
$$I = \frac{V}{R_1 + R_2 + R_3} = \frac{60}{10 + 20 + 30} = 1\ A$$

so, Voltage across $10\ \Omega = I R_1 = \frac{V}{R_1 + R_2 + R_3} \times R_1 = 1 \times 10 = 10\ V$

Voltage across $20\ \Omega = I R_2 = \frac{V}{R_1 + R_2 + R_3} \times R_2 = 1 \times 20 = 20\ V$

Voltage across $30\ \Omega = \frac{V}{R_1 + R_2 + R_3} \times R_3 = 1 \times 30 = 30\ V.$

7. Find the magnitudes of total current, current through R_1 and R_2.



Solution:

Equivalent resistance, $R_{eq} = \frac{10 \times 20}{10 + 20} = 6.67\ \Omega$

so, Total current, $I_T = \frac{V}{R_{eq}} = \frac{50}{6.67} = 7.5\ A$

As per the current divider rule,

$$I_1 = I_T \times \left(\frac{R_2}{R_1 + R_2}\right) = 7.5 \times \left(\frac{20}{10 + 20}\right) = 5\ A$$

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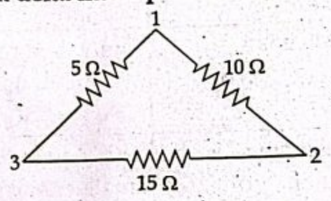
and, $I_2 = I_T \times \left(\frac{R_1}{R_1 + R_2}\right) = 7.5 \times \left(\frac{10}{10+20}\right) = 2.5$ A.

8. Convert the given delta into equivalent star



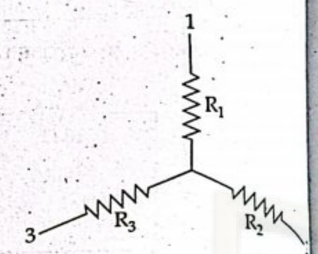
Solution:

Its equivalent star is shown below:

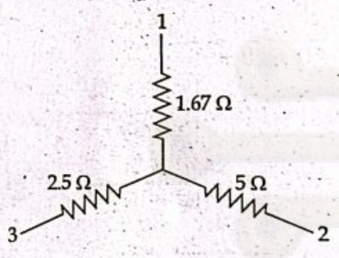
$$R_1 = \frac{5 \times 10}{5+10+15} = 1.67\ \Omega$$

$$R_2 = \frac{15 \times 10}{5+10+15} = 5\ \Omega$$

$$R_3 = \frac{5 \times 15}{5+10+15} = 2.5\ \Omega$$



9. Convert the given star into equivalent delta.



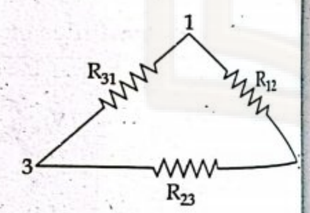
Solution:

Its equivalent delta is shown below:

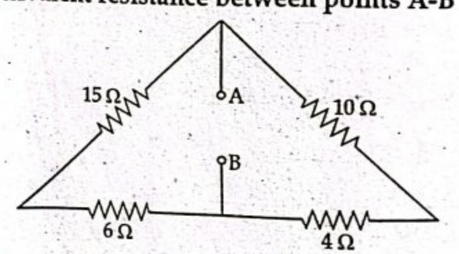
$$R_{12} = 1.67 + 5 + \frac{1.67 \times 5}{2.5} = 10\ \Omega$$

$$R_{23} = 5 + 2.5 + \frac{5 \times 2.5}{1.67} = 15\ \Omega$$

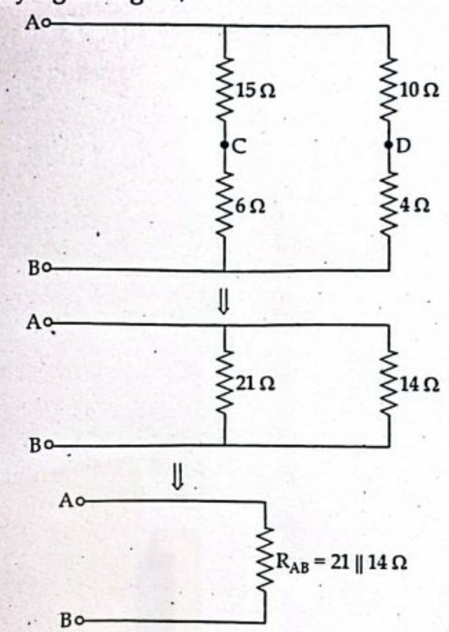
$$R_{31} = 2.5 + 1.67 + \frac{2.5 \times 1.67}{5} = 5\ \Omega$$



10. Find the equivalent resistance between points A-B

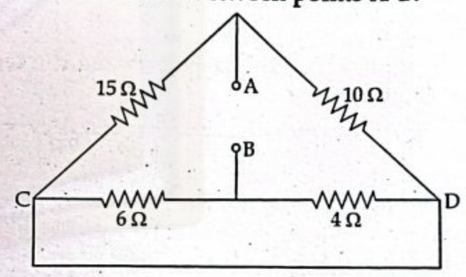


Solution: Simplifying the figure,



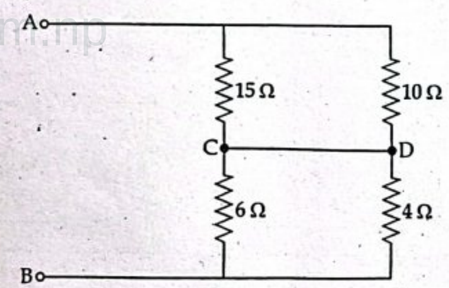
$$\therefore \quad R_{AB} = \frac{21 \times 14}{21+14} = 8.40\ \Omega.$$

11. Find equivalent resistance between points A-B.



Solution:

Redraw the circuit,



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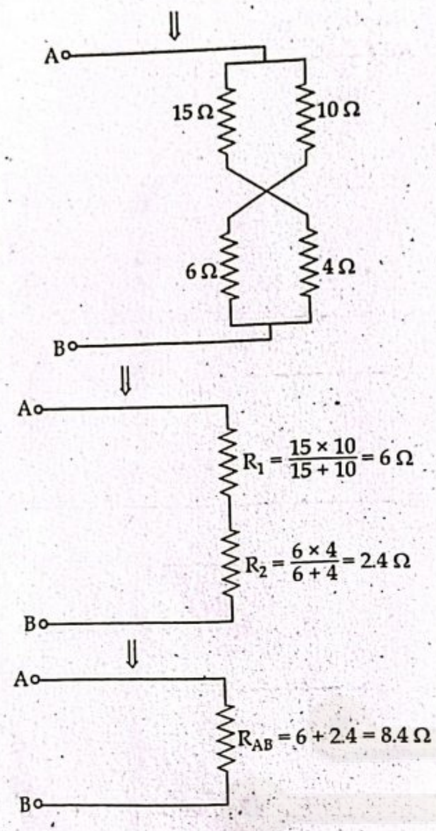
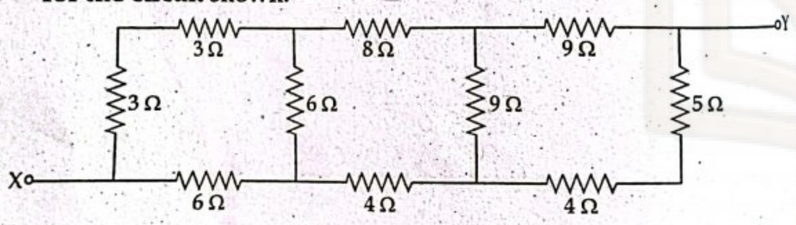


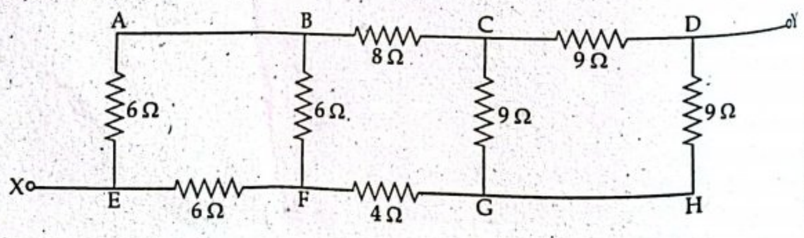
Thus, $R_{AB} = 8.4\ \Omega$.

12. Calculate the equivalent resistance between the terminals X and Y for the circuit shown.



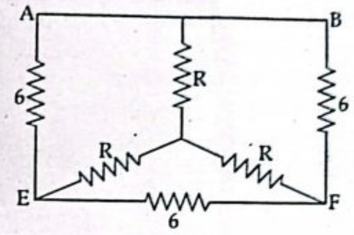
Solution:

On the left side 3 Ω and 3 Ω are in series while on the right side 5 Ω and 4 Ω are in series.

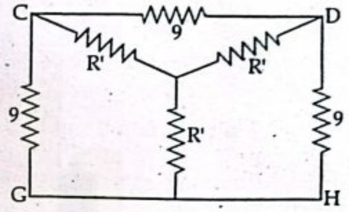


Convert ΔABEF to equivalent star and ΔCDHG to equivalent star. As all the resistance of ΔABEF are equal so,

$$R = \frac{6 \times 6}{6 + 6 + 6} = 2\ \Omega$$

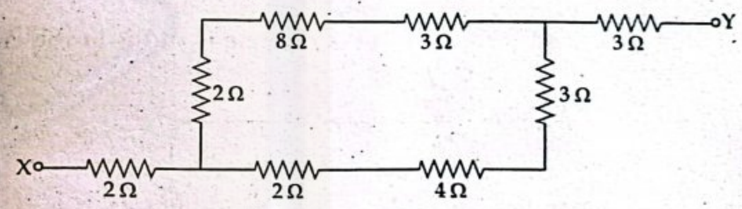


Similarly in equivalent star of CDGH, each resistance is equal to R'

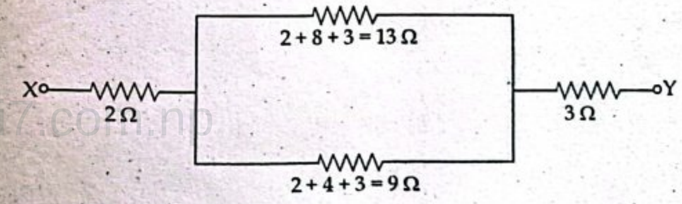


$$R' = \frac{9 \times 9}{9 + 9 + 9} = 3\ \Omega$$

Thus circuit reduces to,

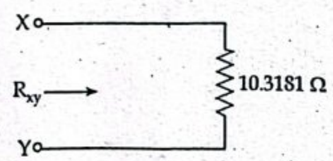


The resistances 2 Ω, 8 Ω and 3 Ω are in series. Similarly, the resistance 2 Ω, 4 Ω and 3 Ω are in series,



Thus,

$$R_{xy} = 2\ \Omega + 9 \parallel 13\ \Omega + 3\ \Omega$$
$$= 2 + \frac{9 \times 13}{9 + 13} + 3$$
$$= 10.3181\ \Omega$$



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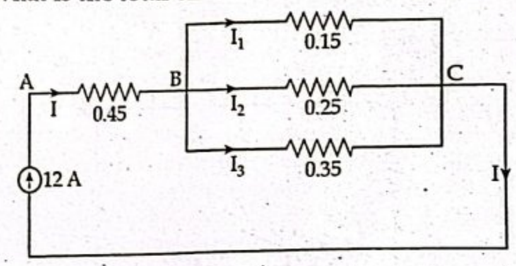
13. Calculate the values of different currents for the circuit shown below. What is the total circuit conductance and resistance.



Solution:

We know,

$$I_1 = I \times \frac{G_1}{G_1 + G_2 + G_3} = 12 \times \frac{0.1}{0.1 + 0.2 + 0.3} = 2 \text{ A}$$

$$I_2 = I \times \frac{G_2}{G_1 + G_2 + G_3} = 12 \times \frac{0.2}{0.6} = 4 \text{ A}$$

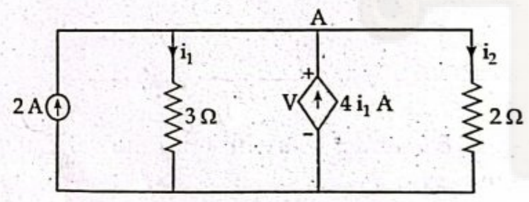
$$I_3 = I \times \frac{G_3}{G_1 + G_2 + G_3} = 12 \times \frac{0.3}{0.6} = 6 \text{ A}$$

Thus, $G_{BC} = 0.1 + 0.2 + 0.3 = 0.6 \text{ S}$

and, $\frac{1}{G_{AC}} = \frac{1}{G_{AB}} + \frac{1}{G_{BC}} = \frac{1}{0.4} + \frac{1}{0.6} = \frac{25}{6} \text{S}^{-1}$

Hence, $R_{AC} = \frac{1}{G_{AC}} = \frac{25}{6} \Omega = 4.167 \Omega$.

14. Using KCL, find the values of the currents i_1 and i_2 in the circuit below.



Solution:

Applying KCL at node A,

$$2 - i_1 + 4 i_1 - i_2 = 0$$

or, $-3 i_1 + i_2 = 2$

By ohms law,

$$i_1 = \frac{V}{3} \text{ and } i_2 = \frac{V}{2}$$

Thus,

$$-3\frac{V}{3} + \frac{V}{2} = 2$$

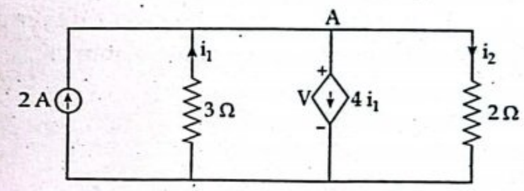
or, $V = -4 \text{ V}$

Hence, $i_1 = -\frac{4}{3} \text{A}$

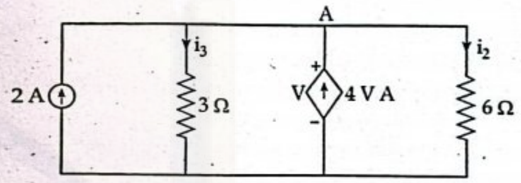
and, $i_2 = -\frac{4}{2} = -2 \text{ A}$.

The value of dependent current sources $= 4 i_1 = 4 \times \left(\frac{-4}{3}\right) = \frac{-16}{3} \text{ A}$

Since i_1 and i_2 come out to be negative, it means that they flow upwards and not downwards as preassumed. Similarly, the current of the dependent source flow downwards as shown in figure below.



15. By applying KCL, obtain the values of V, i_1 and i_2 in the circuit which contains a voltage dependent current source.



Solution:

Applying KCL at node A,

$$2 - i_1 + 4 \text{ V} - i_2 = 0$$

or, $i_1 + i_2 - 4 \text{ V} = 2$

Now,

$$i_1 = \frac{V}{3} \text{ and } i_2 = \frac{V}{6}$$

Now,

$$\frac{V}{3} + \frac{V}{6} - 4 \text{ V} = 2$$

or, $V = \frac{-4}{7} \text{ V}$

$\therefore \quad i_1 = \frac{4}{21} \text{ A}$

and, $i_2 = \frac{2}{21} \text{ A}$

and, $4 \text{ V} = 4 \times \frac{-4}{7} = \frac{-16}{7} \text{ V}$

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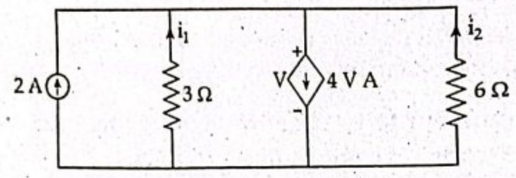
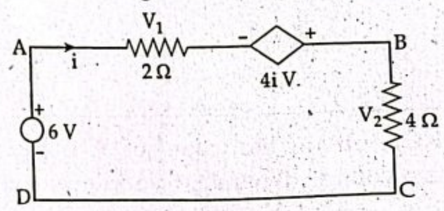
Since i_1 and i_2 come out to be negative and value of current source is also negative, their directions of flow are opposite to those preassumed. Actual current directions are shown in figure below.



16. Apply KVL, to find the values of current i and the voltage drops V_1 and V_2 in the circuit which contains a current dependent voltage source. What is the voltage of the dependent source?



Solution:

Applying KVL,

$$-V_1 + 4_i - V_2 + 6 = 0$$

or $V_1 - 4_i + V_2 = 0$

Now,

$V_1 = 2_i$

and, $V_2 = 4_i$

so, $2_i - 4_i + 4_i = 6$

or, $2_i = 6$

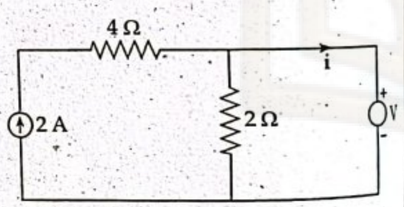
$\therefore \quad i = 3$ A

Hence, $V_1 = 2 \times 3 = 6$ V

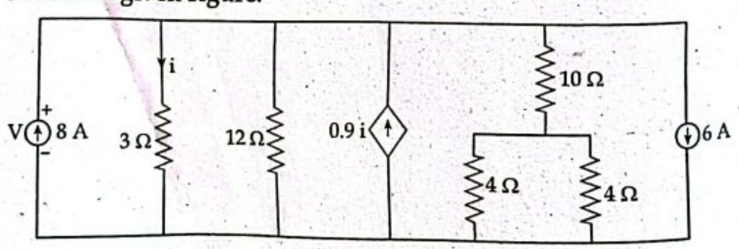
and, $V_2 = 4 \times 3 = 12$ V

Voltage of the dependent source

$= 4_i = 4 \times 3 = 12$ V.

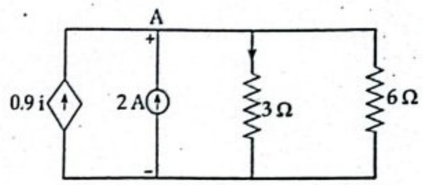


17. Find the current i and also the power and voltage of the dependent source in given figure.



Solution:

The two current sources can be combined into a single source of 8 – 6 = 2 A. The two parallel resistance when combined have a value of 2 Ω which being in series with the 10 Ω resistance, gives the branch resistance of 10 + 2 = 12 Ω. This 12 Ω resistance when combined with the other 12 Ω resistance gives a combination resistance of 6 Ω. The simplified circuit is shown in figure.



Applying KCL at node A,

$$0.9\,i + 2 - i - \frac{V}{6} = 0$$

or, $0.6\,i = 12 - V$

also, $V = 3\,i$

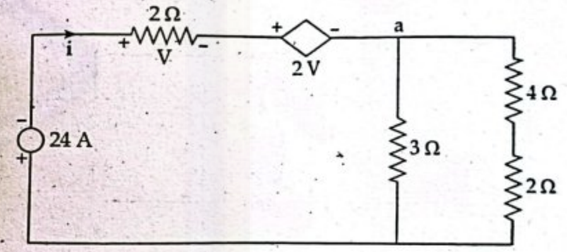
or, $i = \frac{10}{3}$ A

Hence, $V = \frac{10}{3} \times 3 = 10$ V

The power by the current source

$$= V \times 0.9\,i = 10 \times 0.9 \times \frac{10}{3} = 30 \text{ watt}$$

18. Find the current i in the circuit.



Solution:

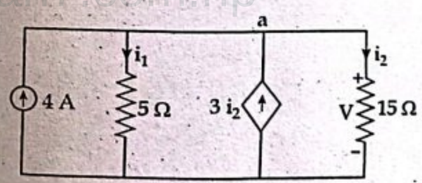
The equivalent resistance of the two parallel paths across point a is $3 \parallel (4 + 2) = 2\ \Omega$. Now applying KVL to the closed loop, we get

or, $24 - V - 2\,V - 2_i = 0$. Since $V = 2_i$, we get

or, $24 - 2_i - 2 \times 2_i - 2_i = 0$

$\therefore \quad i = 3$ A

19.



Determine the value of current i_2 and voltage drop across 15 Ω resistor.

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Solution:

Applying KCL to node a,

$$4 - i + 3i_2 - i_2 = 0$$

or, $4 - i_1 + 2i_2 = 0$

Applying ohms law, we get

$$i_1 = \frac{V}{5}$$

and $i_2 = \frac{V}{15}$

Then, $4 - \frac{V}{5} + 2 \times \frac{V}{15} = 0$

or, $4 - 0.20\,V + 0.1334\,V = 0$

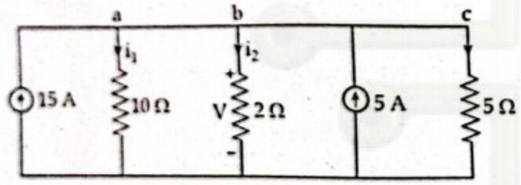
or, $4 - 0.0667\,V = 0$

or, $V = \frac{4}{0.0667}$

$\therefore$ $V = 60\,V$

and, $i_2 = \frac{V}{15} = \frac{60}{15} = 4\,A$

20. Find the magnitude and directions of the unknown currents through 10 Ω, 2 Ω and 5 Ω resistor. Also find V.



Solution:

Applying KCL at node 'a',

$$-15 + i_1 + i_2 - 5 + i_3 = 0$$

or, $i_1 + i_2 + i_3 = 20$

From ohm's law,

$$i_2 = \frac{V}{2}$$

$$i_1 = \frac{V}{10}$$

$$i_3 = \frac{V}{5}$$

From equation (1), we have;

$$\frac{V}{10} + \frac{V}{2} + \frac{V}{5} = 20$$

or, $V + 5V + 2V = 200$

$\therefore$ $V = 25\,V$

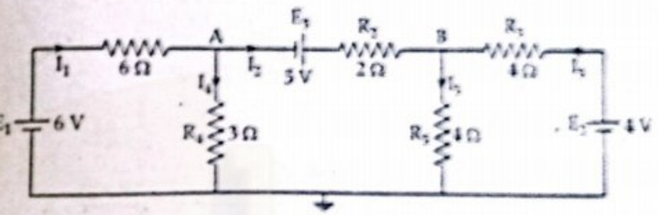
Hence,

$\therefore$ $i_1 = \frac{V}{10} = \frac{5}{10} = 2.5\,A$

$\therefore$ $i_2 = \frac{25}{2} = 12.5\,A$

$\therefore$ $i_3 = \frac{25}{5} = 5\,A$

21. Find the branch currents in the circuit by using nodal analysis and loop analysis.



Solution:

Using nodal analysis,

At node A,

$$V_A\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_4}\right) - \frac{E_1}{R_1} - \frac{V_B}{R_2} + \frac{E_3}{R_2} = 0$$

or, $V_A\left(\frac{1}{6} + \frac{1}{2} + \frac{1}{3}\right) - \frac{6}{6} - \frac{V_B}{2} + \frac{5}{2} = 0$

or, $2V_A - V_B = -3$ (1)

At node B,

$$V_B\left(\frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_5}\right) - \frac{E_2}{R_3} - \frac{V_A}{R_2} - \frac{E_3}{R_2} = 0$$

or, $V_B\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{4}\right) - \frac{V_A}{2} - \frac{5}{2} = 0$

$\therefore$ $V_B - \frac{V_A}{2} = 5$ (2)

Solving equation (1) and (2), we get;

$\therefore$ $V_A = \frac{4}{3}V$

$\therefore$ $V_B = \frac{17}{3}V$

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Hence,

$$I_1 = \frac{E_1 - V_A}{R_1} = \frac{6 - \frac{4}{3}}{6} = \frac{7}{9}\text{ A}$$

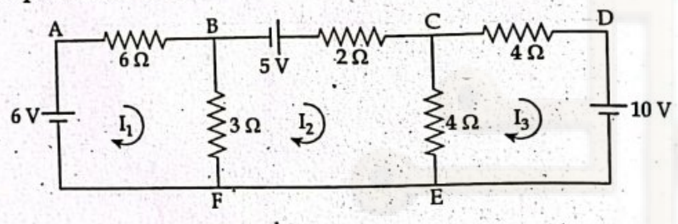
$$I_2 = \frac{V_A - E_3 - V_B}{R_2} = \frac{\left(\frac{4}{3}\right) - 5 - \left(\frac{17}{3}\right)}{2} = \frac{1}{3}\text{ A}$$

$$I_3 = \frac{E_2 - V_B}{R_3} = \frac{10 - \frac{17}{3}}{4} = \frac{13}{12}\text{ A}$$

$$I_4 = \frac{V_A}{R_4} = \frac{\frac{4}{3}}{3} = \frac{4}{9}\text{ A}$$

$$I_5 = \frac{V_B}{R_5} = \frac{\frac{17}{3}}{4} = \frac{17}{12}\text{ A}$$

Using loop current method



Loop ABFA,

$$-6I_1 - 3(I_1 + I_2) + 6 = 0$$

or, $3I_1 - I_2 = 2$ (1)

Loop BCEFB

$$5 - 2I_2 - 4(I_2 - I_3) - 3(I_2 - I_1) = 0$$

or, $3I_1 - 9I_2 + 4I_3 = -5$ (2)

Loop CDEC

$$-4I_3 - 10 - 4(I_3 - I_2) = 0$$

or, $2I_2 - 4I_3 = 5$ (3)

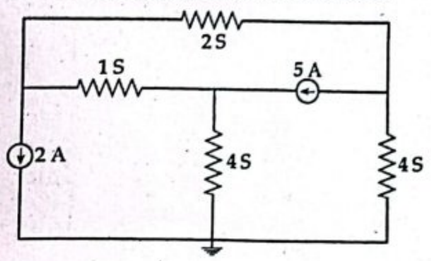
On solving equation (1), (2) and (3);

$$\therefore \quad I_1 = \frac{7}{9}\text{ A}$$

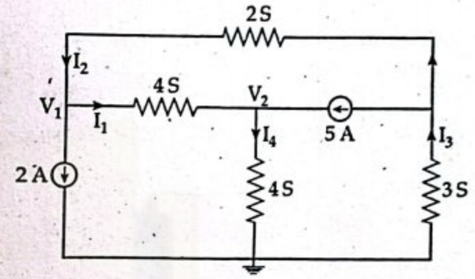
$$\therefore \quad I_2 = \frac{1}{3}\text{ A}$$

$$\therefore \quad I_3 = \frac{-13}{12}\text{ A.}$$

22. Using nodal analysis, find the different branch currents in the circuit. All branch conductance are in siemens.



Let the various branch currents be as shown in figure below.



1st node:

$$V_1(1 + 2) - V_2 \times 1 - V_3 \times 2 = -2$$

or, $3V_1 - V_2 - 2V_3 = -2$ (1)

2nd node:

$$V_2(1 + 4) - V_1 \times 1 = 5$$

or, $V_1 - 5V_2 = -5$ (2)

3rd node:

$$V_3(2 + 3) - V_1 \times 2 = -5$$

or, $2V_1 - 5V_3 = 5$ (3)

Solving for the different voltage,

$$\therefore \quad V_1 = -\frac{3}{2}\text{ V}$$

$$\therefore \quad V_2 = \frac{7}{10}\text{ V}$$

$$\therefore \quad V_3 = -\frac{8}{5}\text{ V}$$

and, $I_1 = (V_1 - V_2) \times 1 = (-1.5 - 0.7) \times 1 = -2.2\text{ A}$

$I_2 = (V_3 - V_1) \times 2 = [-1.6 - (-1.5)] \times 2 = -0.2\text{ A}$

$I_4 = V_2 \times 4 = 4 \times \frac{7}{10} = 2.8\text{ A}$

$I_3 = 2 + 2.8 = 4.8\text{ A.}$

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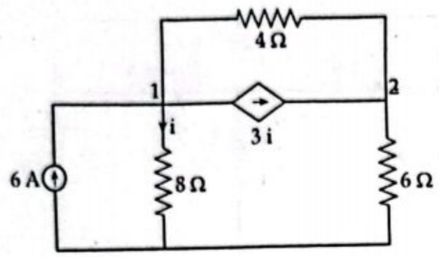
23. Use nodal analysis to determine the value of current i in the following figure.



Solution:

Applying KCL at node 1,

$$6 = \frac{V_1 - V_2}{4} + \frac{V_1}{8} + 3_i$$

Since, $i = \frac{V_1}{8}$,

$$6 = \frac{V_1 - V_2}{4} + \frac{V_1}{8} + 3 \times \frac{V_1}{8}$$

or, $3V_1 - V_2 = 24$ (1)

and, Applying KCL at node 2, we get

$$\frac{V_1 - V_2}{4} + 3_i = \frac{V_2}{6}$$

or, $\frac{V_1 - V_2}{4} + 3 \times \frac{V_1}{8} = \frac{V_2}{6}$

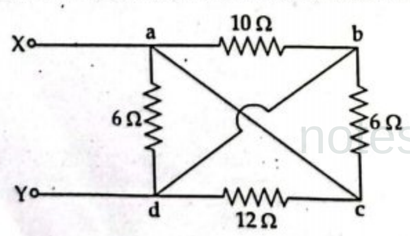
or, $3V_1 = 2V_2$ (2)

From equation (1) and (2);

$\therefore \quad V_1 = 16\ V$

$\therefore \quad i = \frac{16}{8} A = 2\ A.$

24. Find the equivalent resistance across terminals Z and Y.



Solution:

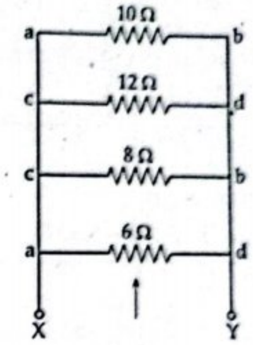
Points 'X' and 'a' are at same potential.

Points 'Y' and 'd' are at same potential.

Points 'a' and 'c' are at same potential

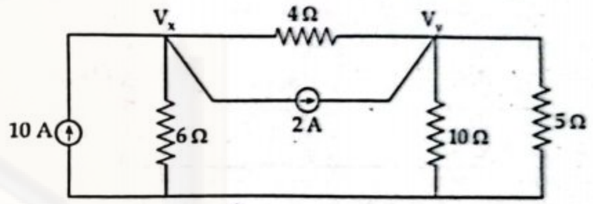
Points 'd' and 'b' are at same potential

All resistances are in parallel.



$$\therefore \quad R_{xy} = \frac{1}{\frac{1}{6} + \frac{1}{8} + \frac{1}{12} + \frac{1}{10}} = 2.1053\ \Omega$$

25. Find the node voltage, V_x and V_y using nodal analysis.



Solution:

At node x,

$$-10 + \frac{V_x}{6} + 2 + \frac{V_x - V_y}{4} = 0$$

or, $V_x\left(\frac{1}{4} + \frac{1}{6}\right) - \frac{V_y}{4} = 8$

or, $5V_x - 3V_y = 96$ (1)

Applying nodal analysis at 'y',

or, $-2 + \frac{V_y - V_x}{4} + \frac{V_y}{10} + \frac{V_y}{5} = 0$

or, $-\frac{V_x}{4} + V_y\left(\frac{1}{4} + \frac{1}{5} + \frac{1}{10}\right) = 2$

or, $5V_x - 11V_y = -40$ (2)

On solving equation (1) and (2); we get,

$\therefore \quad V_x = 29.4\ V$

$\therefore \quad V_y = 17\ V$

26. Verify the application of reciprocity theorem in the network shown in figure below.

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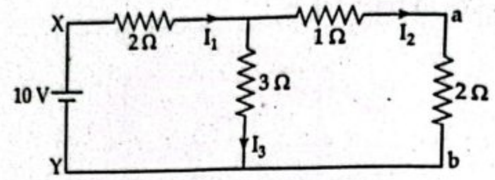


Solution:

Equivalent resistance across X–Y is,

$R_{x-y} = [(2+1) \parallel 3] + 2 = 3 \parallel 3 + 2 = 3.50\ \Omega$

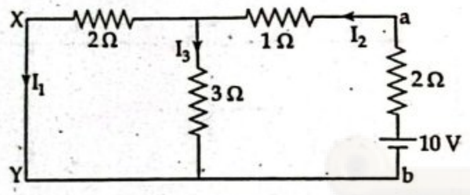
Using current divider rule,

$$I_1 = \frac{10}{3.5} = 2.86\ A$$

$$I_2 = 2.86 \times \frac{3}{3+3} = 1.43\ A$$

$I_3 = 2.86 - 1.43 = 1.43\ A$

Now replacing the 10 V source in branch ab



Equivalent resistance across ab, $R_{ab} = (2 \parallel 3) + 1 + 1 = 4.2\ \Omega$

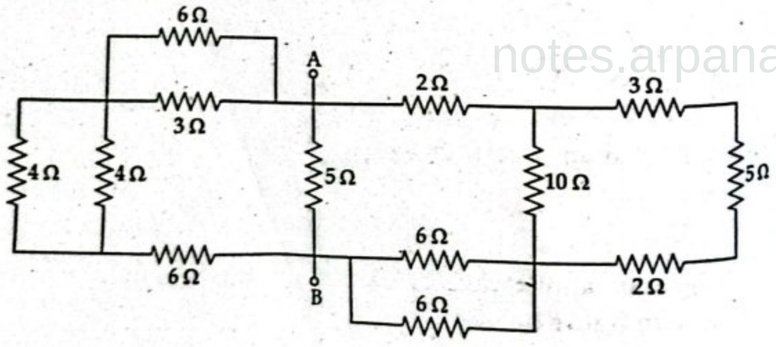
Now,

$$I_2 = \frac{10}{4.2} = 2.381\ A$$

$$I_2 = I_2 \times \frac{3}{2+3} = 1.43\ A$$

Hence, when the source is in ab branch, the x–y branch current becomes 1.43 A and when the source is in x–y branch the current in branch ab is 1.43 A.

27. Find the equivalent resistance across the terminals A and B.



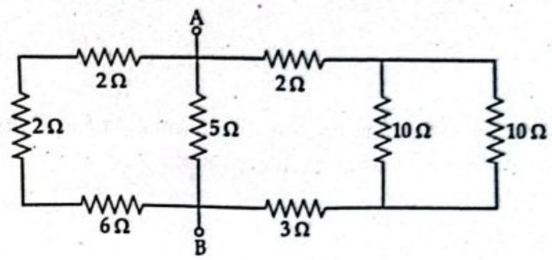
Solution:

Here;

$4\ \Omega \parallel 4\ \Omega = 2\ \Omega, 3\ \Omega, 2\ \Omega, 5\ \Omega$ are in series

i.e., $(2+3+5)\ \Omega = 10\ \Omega, 5\ \Omega \parallel 6\ \Omega = \frac{6 \times 6}{6+6} = 3\ \Omega$

and, $3\ \Omega \parallel 6\ \Omega = 2\ \Omega$, so



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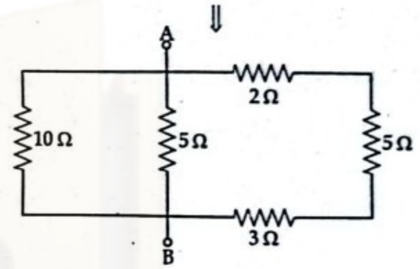

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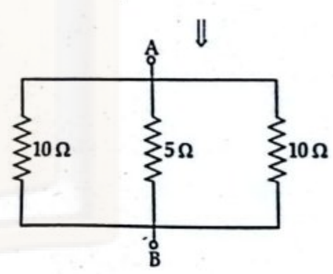
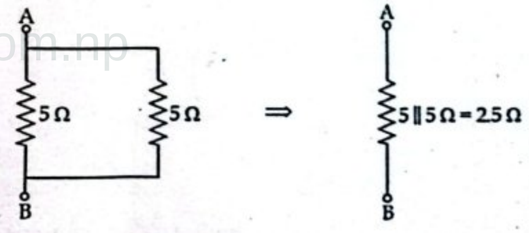


Here, $10\ \Omega \parallel 10\ \Omega = 5\ \Omega$



Thus, $R_{AB} = 2.5\ \Omega$.

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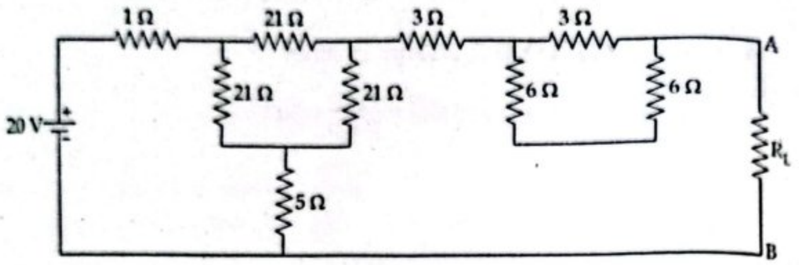
28. Determine R_L for maximum power transfer to the load.



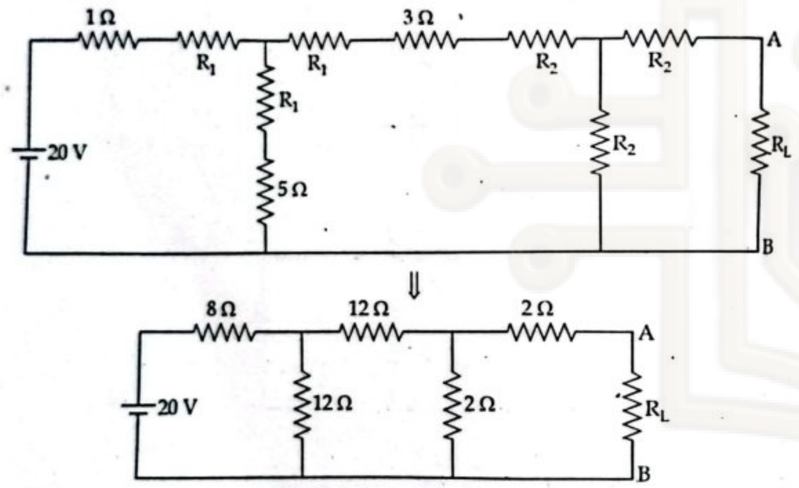
Solution:

The two delta networks, one formed by 3 numbers of 6 Ω, 3 Ω and 21 Ω are first converted into equivalent star network.

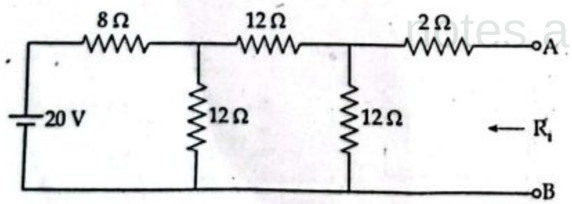
$$R_1 = \frac{21 \times 21}{21 + 21 + 21} = 7\ \Omega$$

$$R_2 = \frac{6 \times 6}{6 + 6 + 6} = 2\ \Omega$$

The corresponding network is shown below,



To find R_L

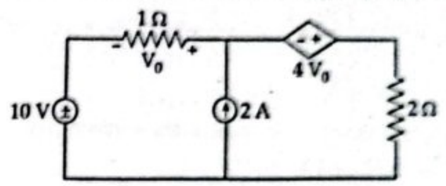


For maximum power transfer to the load. R_L, the value of R_L should be equal to R_i which is equal to the internal resistance of the network.. R_i can be by removing the source and open circuiting terminals AB.

$$R_L = R_i = \left[\left(\frac{8 \times 12}{8 + 12} + 12\right) \| 12\right] + 2 = \frac{16.8 \times 12}{16.8 + 12} + 2$$

$\therefore$ $R_L = 9\ \Omega$

29. Find power loss in the 2 Ω resistor using superposition theorem.



Solution:

Considering the 10 V source acting alone,

or, $10 + V_0 + 4\ V_0 - 2_i = 0$

or, $5\ V_0 - 2_i + 10 = 0$

Now,

$1 \times i = -V_0$

Hence,

$5\ (-i) - 2_i + 10 = 0$

or $-7_i + 10 = 0$

or $i = \frac{10}{7}$

$\therefore$ $i = 1.43$ A.

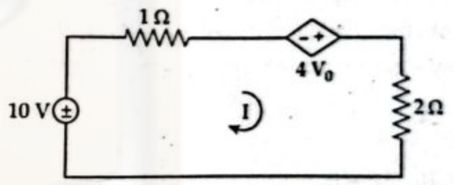
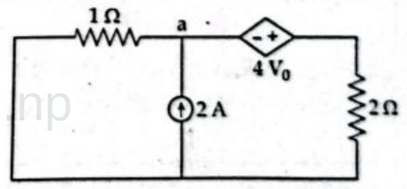


Fig: 10 V source acting alone

Considering 2 A source acting alone and applying KCL at node a,



or, $2 - \frac{V_a}{1} - \frac{V_a + 4\ V_0}{2} = 0$

or, $4 - 2\ V_a - V_a - 4\ V_0 = 0$

or, $4 - 3\ V_a - 4\ V_0 = 0$

From given figure, $V_a = V_0$

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Hence, $4 - 7V_0 = 0$

$\therefore \quad V_0 = \frac{4}{7} = 0.57\ V$

Current through 2 Ω resistor

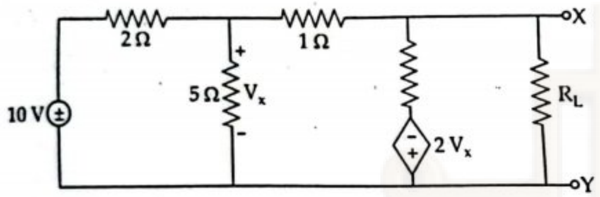
$$= \frac{V_a + 4V_0}{2} = \frac{V_0 + 4V_0}{2} = \frac{5}{2}V_0 = \frac{5}{2} \times 0.57$$

$\therefore$ i' = 1.425 A

Now, applying superposition theorem, the current through 2 Ω resistor is thus (1.43 + 1.425) A *i.e.*, 2.855 A.

Thus, power loss in 2 Ω resistor = $2.855^2 \times 2 = 16.31$ W.

30. Using maximum power transfer theorem, find the value of the load resistance R_L so that the maximum power is transferred across R_L in the circuit shown.



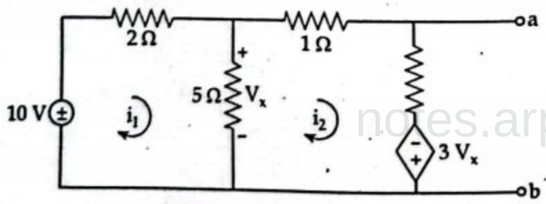
Solution:

Let us remove R_L and open circuit terminals xy to find out the internal resistance R_i of the circuit. According to maximum power transfer theorem, the maximum power will be transformed through R_L When $R_L = R_i$

Appling KVL,

$$10 - 2i_1 - 5(i_1 - i_2) = 0$$

or, $7i_1 - 5i_2 = 10$ (1)



and, $3V_x - 5(i_2 - i_1) - i_2 \times 1 - 2i_2 = 0$

or, $3 \times 5(i_1 - i_2) - 5(i_2 - i_1) - 3i_2 = 0$

or, $20i_1 - 23i_2 = 0$

or, $i_1 = \frac{23}{20}i_2$ (2)

Thus, using equation (2) in (1);

or, $7 \times \frac{23}{20}i_2 - 5i_2 = 10$

or, $i_2 = 3.28$ A

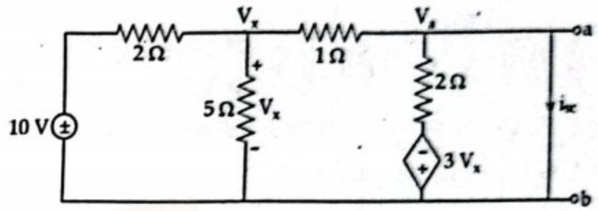
and, $i_1 = 3.772$ A

Thus, $V_x = 5(3.772 - 3.28) = 2.46$ V

Now,

$$V_{ab} = -3V_x + 2i_2 = -3 \times 2.46 + 2 \times 3.28 = -0.82\ V$$

Let us now short circuit the xy terminals.



At node 'a',

$$\frac{V_a - V_x}{1} + \frac{V_a + 3V_x}{2} + i_{sc} = 0$$

As a – b are shorted, $V_a = 0$

Hence, $-V_x + 1.5V_x + i_{sc} = 0$

i.e., $i_{sc} = -0.5V_x$

Now,

V_x = Voltage across 5 Ω resistor.

Current through 5 Ω resistor is I_5 Ω

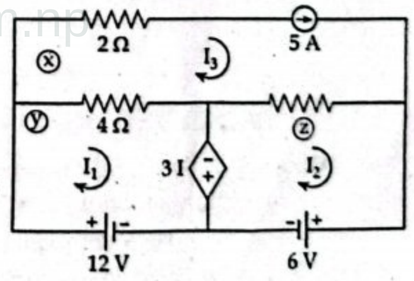
$$= \left(\frac{10}{2 + \frac{5 \times 1}{5 + 1}}\right) \times \left(\frac{1}{5 + 1}\right) = \frac{60}{12 + 5} \times \frac{1}{6}$$

Hence, $V_x = 5 \times 0.588 = 2.94$ V

and, $i_{sc} = -0.5 \times 2.94 = -1.47$ A

Thus, $R_L = R_i = \frac{-0.82}{-1.47} = 0.558\ \Omega$.

31. Obtain the values of I_1, I_2 and I_3 in the circuit shown below.



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Solution:

In loop (x), circulating current will be driven by 5 A constant current source and hence $I_3 = 5$ A

In loop (y),

$$(I_1 - I_3)\,4 + 3\,I - 12 = 0 \quad (1)$$

But $I = I_2 - I_3$ in loop (z)

so, $(I_1 - I_3)4 + 3\,(I_2 - I_3) - 12 = 0$

or, $4\,I_1 + 3\,I_2 - 7\,I_3 = 12$

Since, $I_3 = 5$ A,

$$4\,I_1 + 3\,I_2 - 7 \times 5 = 12$$

or, $4\,I_1 + 3\,I_2 = 47$ (2)

For loop (z),

$$1 \times I + 6 - 3\,I = 0$$

Using $I = 3$ A, $I = I_2 - I_1$, we get,

$$I_2 - I_3 = 3$$

or, $I_2 = I_3 + 3 = 8$ A

Thus, from (2),

$$4I_1 = 47 - 3\,I_2 = 47 - 3 \times 8 = 23 \text{ A}$$

or, $I_1 = \frac{23}{4} = 5.75$ A

Thus finally,

$\therefore \quad I_1 = 5.75$ A

$\therefore \quad I_2 = 8$ A

$\therefore \quad I_3 = 5$ A.

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CHAPTER **3**

SINGLE PHASE AC CIRCUIT ANALYSIS

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3.1 GENERATION OF EMF BY ELECTROMAGNETIC INDUCTION

Consider a rectangular coil having N number of turns and area A m^2, which is rotating in a uniform magnetic field with an angular velocity ω radian/s. If in t-seconds, the coil rotates through an angle $\theta = \omega t$ from the x-axis, the component of the flux perpendicular to the plane of the coil

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is $\phi = \phi_m \cdot \cos \omega t$ (where ϕ_m = maximum flux density perpendicular to the axis of rotation; when the plane of the coil coincides with the x-axis).

Now, according to faraday's law of electromagnetic induction, the induced emf in the coil is equal to the rate of change of flux linkages of the coil. Consequently, the instantaneous induced emf (e) at time, when $\theta = \omega t$, is given by,

$$e = -\frac{d}{dt}(-N\phi) \text{ volts.}$$

Again, lenz's law states that, "when a circuit and a magnetic field move relatively to each other, the electric current induced in the circuit will have magnetic field opposing the motion".

so, $$e = -\frac{d}{dt}(N\phi_m \cos \omega t)$$

$$= -\omega N \phi_m \frac{d}{dt}(\cos \omega t)$$

$$= \omega N \phi_m \sin \omega t \text{ volts}$$

When $\theta = 90°$, $e = \omega N \phi_m = E_m$

where, E_m = maximum value of the instantaneous induced emf.

Now, if f be the frequency of rotation of coil in hertz and B_m the maximum flux density in wb/m².

$$e = E_m \cdot \sin \theta = (2\pi f N B_m A) \sin \theta \qquad [\because B_m \cdot A = \phi_m]$$

Let, i be the instantaneous value of the current in the coil.

$\therefore \quad i = I_m \cdot \sin \omega t$

where, I_m = Maximum value of current

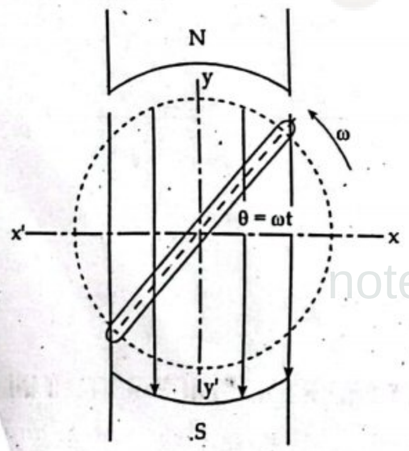

Fig: Generation of alternating emf

As both the induced emf and induced current varies sinusoidally, hence the emf or current can be plotted against time.

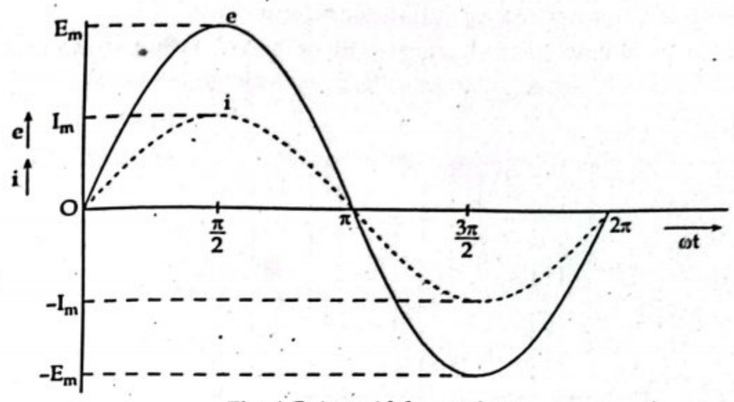

Fig: AC sinusoidal wave form

3.2 GENERATION OF ALTERNATING VOLTAGE

Alternating voltage can be generated by rotating coil in a magnetic field or by rotating a magnetic field within a stationary coil. The value of the voltage generated depends, in each case, upon the number of turns in the coil, strength of the field and the speed at which the coil or magnetic field rotates.

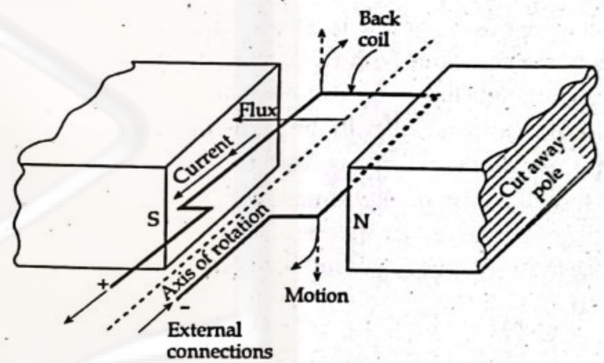

Fig: Rotating a coil in a stationary magnetic field.

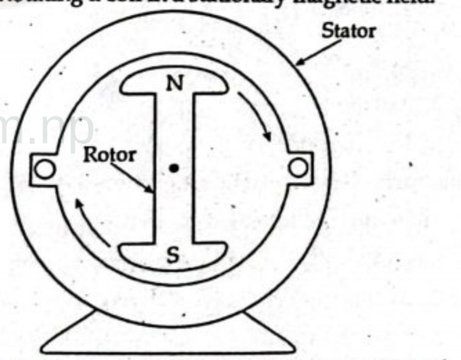

Fig: Rotating a magnetic field within a stationary coil.

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Equations of the alternating voltages and currents

Consider a rectangular coil, having N turns and rotating in a uniform magnetic field, with an angular velocity of ω radian/second as shown in figure below.

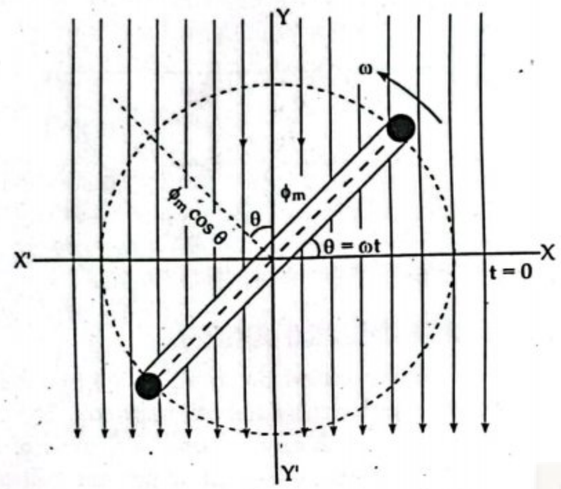


Fig: A coil rotating in a magnetic field

Let time be measured from the x-axis. Maximum flux ϕ_m is linked with the coil, when its plane coincides with the x-axis. In time t seconds, this coil rotates through an angle $\theta = \omega t$. In this deflected position, the component of the flux which is perpendicular to the plane of the coil, is $\phi = \phi_m \cos \omega t$. Hence, flux linkages of the coil at any time are $N\phi = N\phi_m . \cos \omega t$.

According to faraday's law of electromagnetic induction, the emf induced in the coil is given by the rate of change of flux linkages of the coil. Hence, the value of the induced emf at this induced emf is,

$$e = -\frac{d}{dt}(N\phi)$$

$$= -N\frac{d}{dt}(\phi_m \cdot \cos \omega t)$$

$$= -N\phi_m(-\sin \omega t) \cdot \omega$$

$$= \omega N \phi_m \sin \omega t$$

$$\therefore \quad e = \omega N \phi_m \cdot \sin \theta \text{ volts}$$

when the coil has turned through 90° *i.e.*, when $\theta = 90°$, then $\sin \theta = 1$, hence e has the maximum value, E_m

$$\therefore \quad E_m = \omega N \phi_m = \omega N B_m \cdot A = 2\pi f N B_m A \text{ volts}$$

where, B_m = Maximum flux density
A = Area of the coil
f = Frequency

Hence,

$$e = E_m \sin \theta = E_m \sin \omega t$$

Similarly, the equation of induced alternating current is,

$$I = I_m \sin \omega t$$

Provided that the coil circuit has been closed through a resistive load.

Since, $\omega = 2\pi f$,

$$e = E_m \sin 2\pi f t$$

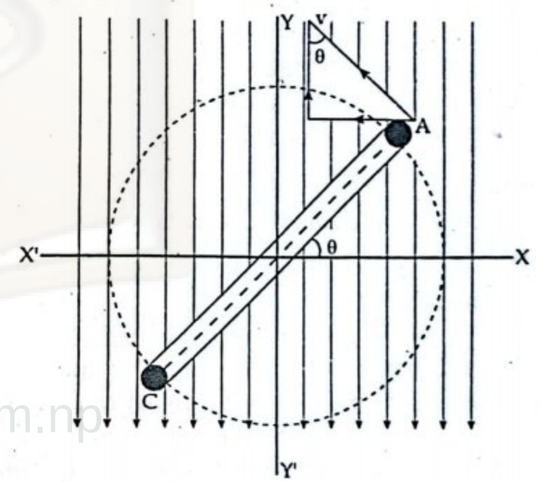
$$\therefore \quad e = E_m \sin\left(\frac{2\pi}{T}\right) t$$

and, $i = I_m \sin\left(\frac{2\pi}{T}\right) t$

where, f = Frequency of rotation of the coil
t = Time period of the alternating voltage or current.

Alternative method for the equations of alternating voltages and currents

Figure below shows a rectangular coil AC having N turns and rotating in a magnetic field of flux density B wb/m². Let the length of each of its sides A and C be L meter and their peripheral velocity v m/s. Let, angle be measured from the horizontal position, the two sides A and C moves parallel to the lines of the magnetic flux. Hence, no flux is cut and so no emf is generated in the coil.



When the coil has turned through angle θ, its velocity can be resolved into two mutually perpendicular components (i) v cos θ component parallel to the direction of the magnetic flux and (ii) v sinθ component perpendicular to the direction of the magnetic flux. The emf is generated due entirely to the perpendicular component *i.e.*, v sin θ.

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Hence, the emf generated in one side of the coil which contains N conductors is given by,

$$e = N B L v \sin\theta$$

Total emf generated in both sides of the coil is,

$$e = 2 BNLv \sin\theta \text{ volts}$$

Now, e has maximum value of E_m when $\theta = 90°$

Hence,

$$E_m = 2BNLv$$

and, $e = E_m \sin\theta$ volts

If b = Width of the coil in meters,

f = Frequency of rotation of coil in Hz

Then, $v = \pi bf$

Thus,

$$E_m = 2BNL \times \pi bf$$

$$\therefore \quad E_m = 2\pi fNBA \text{ volts.} \qquad [\because A = L \times b]$$

3.3 SINUSOIDAL FUNCTIONS-TERMINOLOGY

An alternating quantity (voltage or current) is one which changes continuously in magnitude and alternates in direction at regular intervals of time. An alternating quantity that varies sinusoid ally (*i.e.*, according to the sine of angle θ) is called the sinusoidal quantity. Sinusoidal quantities are expressed as,

$$e = E_{max} \sin\theta = E_{max} \sin\omega t = E_{max} \sin 2\pi ft$$

and, $i = I_{max} \sin\theta = I_{max} \sin\omega t = I_{max} \sin 2\pi ft$

The important AC terms are defined below:

a) Wave form:

The shape of the curve of the voltage or current when plotted against time as abscissa (base) is called the wave form.

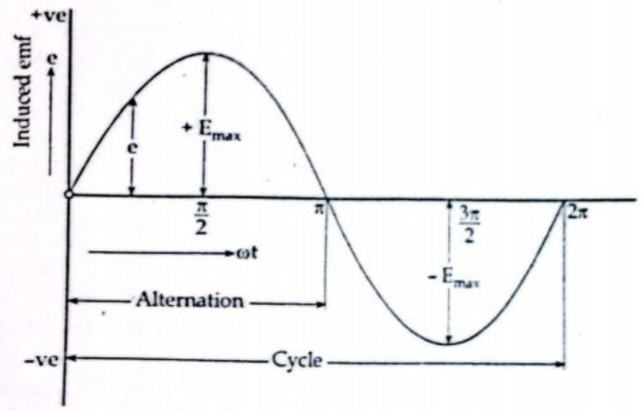
b) Instantaneous value

The value of alternating quantity (emf, voltage or current) at any particular instant is called the instantaneous value and is designated by a small italic letter (e for emf, v for voltage and i for current). The instantaneous values of an alternating quantity can be determined either from the curve or from an equation of the alternating quantity.

c) Alternation and cycle

When a periodic wave, such as sinusoidal wave goes through one complete set of positive and negative values, it is said to have completed one cycle. One cycle corresponds to 360° or 2π radians.

Alternation is one half of cycle and corresponds to 180° or π radians.



d) Time period or periodic time (T)

The time taken in seconds by an alternating quantity to complete one cycle is known as time period.

e) Frequency (f)

Frequency is the number of cycles that occurs in one second. Thus, frequency of an alternating quantity is reciprocal of the time period. The unit of frequency is hertz (Hz) or cycles/second.

$$f = \frac{1}{T}$$

or, $T = \frac{1}{f}$

In Nepal, the standard frequency for power supply is 50 Hz.

f) Amplitude

The maximum value, positive or negative which is an alternating quantity attains during one cycle is called the amplitude of the alternating quantity.

g) Phase

Phase of an alternating quantity is the fraction of the time period or cycle that has elapsed since it last passed from the chosen zero position on origin. The phase at time t from chosen origin is given by t/T, where T is the time period of alternating quantity.

h) Phase angle (ϕ)

Phase angle is equivalent of phase in radians or degree. Thus, phase angle, ϕ is $\frac{2\pi t}{T}$.

Maximum value of an alternating quantity is at a phase angle of $\frac{\pi}{2}$ radian or 90°.

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i) Phase difference

Phase difference between two alternating quantities is the fractional part of a period through which time of one alternating quantity has advanced over another alternating quantity. Alternatively, it is "the angular displacement between two alternating quantities.

i) Two alternating quantities are in phase when each pass through their zero value at the same instant, and also attain their maximum values at the same instant in a given cycle.

ii) Two alternating quantities are out of phase, if they reach their maximum or minimum values at different times, but always have an equal phase angle between them. In the curve shown below, curve A is leading curve B by a phase angle ϕ. Thus, if curve B is advanced by ϕ radians, then the two curve would coincide. A plus sign (+) is used, when the phase difference denotes 'lead'; while 0 minus sign (–) denotes 'lag'

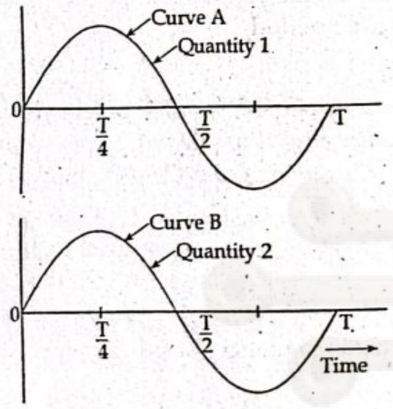

Fig: Two AC quantities in phase

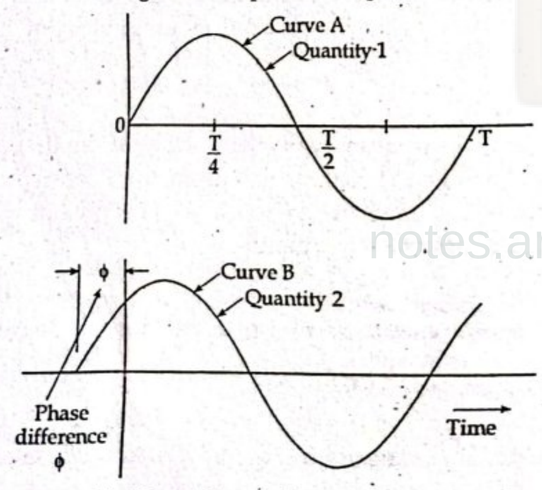

Fig: Two AC quantities out of phase

j) Average or mean value

Average value is arithmetic sum of all the values divided by the total numbers of value used to obtain the sum. The average value of any cycle of wave form is area under the wave form divided by time period (length of cycle). The average of a function f(t) over a specific interval of time t_1 and t_2 is given by,

$$F_{av} = \frac{1}{t_2 - t_1}\int_{t_1}^{t_2} f(t)\,dt$$

Any function whose cycle is repeated continuously, irrespective of its wave shape is called periodic function and its average value is given by,

$$F_{av} = \frac{1}{T}\int_0^t f(t)\,dt$$

Since the area under a sine wave over one complete cycle is zero, because positive and negative areas are equal and opposite, so in such a case, average value of half sinusoidal is calculated as,

$$F_{av} = \frac{1}{\pi}\int_0^{\pi} F_{max}\cdot \sin\theta\, d\theta$$

The average value of an alternating current can also be defined as the equivalent direct current which transforms the same charge as transferred by that alternating current during the same time.

$$I_{av} = \frac{1}{T}\int_0^T i(t)\,dt$$

The average value of current (I_{av}) of an AC sinusoidal wave can be computed by using integration method. The standard equation for alternating current is;

$$\therefore\quad I_{av} = \int_0^{\pi}\frac{i\,d\theta}{\pi - 0}$$

$$= \frac{I_m}{\pi}\int_0^{\pi}\sin\theta\,d\theta$$

$$= \frac{I_m}{\pi}[-\cos\theta]_0^{\pi}$$

$$= \frac{I_m}{\pi}[1 - (-1)]$$

$$= 2\frac{I_m}{\pi}$$

$$= 0.637\, I_m$$

Hence, average value of AC sinusoidal current

$(I_{av}) = 0.63 \times$ Maximum value of current (I_{max})

Also,

$$I_{av} = \frac{i_1 + i_2 + i_3 + \ldots\ldots + i_n}{n} = \frac{\text{Area of one alternation}}{\text{Length of base over one alternation}}$$

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k) Effective or Rms value of alternating quantity

It is the square root of average values of square of alternating quantity over a time period.

i.e., $F_{rms} = \frac{1}{T}\int_0^T f^2(t)\,dt$

If alternating quantity is current, then

$$I_{rms} = \frac{1}{T}\int_0^T i^2(t)\,dt$$

or, I_{eff} or $I_{rms} = \sqrt{\frac{i_1^2 + i_2^2 + i_3^2 + \ldots\ldots + i_n^2}{n}}$

Thus, RMS or effective value of AC current is equivalent steady current or direct current which when allowed to flow through a given circuit for a given time produces the same amount of heat as produced by the alternating current, when allowed to flow through the same circuit for the same time.

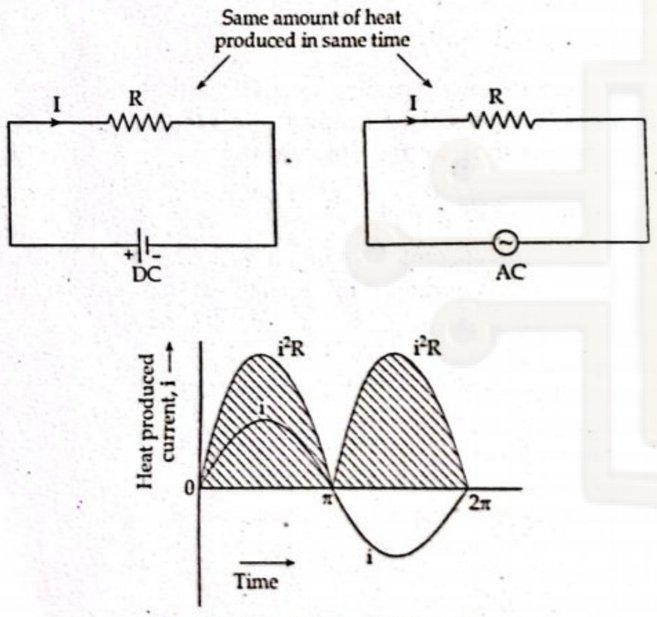

Fig: Heat produced by DC and AC

The heat produced by a direct current (I) or its equivalent RMS values of the alternating current (i) is proportional to I^2 or i^2. So, the area under the curve i^2 versus 2π represents the total heat produced by an alternating current.

∴ RMS value of current,

$$I_{rms} = \sqrt{\int_0^{2\pi} \frac{i^2 \cdot d\theta}{(2\pi - 0)}}$$

$$= \sqrt{\int_0^{2\pi} \frac{I_m^2 \sin^2\theta\,d\theta}{2\pi}} \qquad [\because i = I_m \sin\theta]$$

$$= \sqrt{\frac{I_m^2}{2\pi}\int_0^{\pi} \sin^2\theta\,d\theta}$$

$$= \sqrt{\frac{I_m^2}{4\pi}\int_0^{\pi} (1 - \cos 2\theta)\,d\theta}$$

$$= \sqrt{\frac{I_m^2}{4\pi}\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{2\pi}} \quad \left[\because \cos 2\theta = 1 - \sin^2\theta \text{ and } \sin^2\theta = \frac{1 - \cos 2\theta}{2}\right]$$

$$= \sqrt{\frac{I_m^2}{4\pi} \times 2\pi}$$

$$= \frac{I_m}{\sqrt{2}}$$

∴ $I_{rms} = 0.707\, I_m$

Hence, RMS value of AC sinusoidal current,

I_{rms} = 0.707 × Maximum value of current (I_m).

l) Form factor (K_f)

It is the ratio of rms value to the average value of an alternating quantity.

$$K_f = \frac{\text{rms value}}{\text{average value}}$$

For sinusoidal wave,

Form factor, $K_f = \frac{0.707\, I_m}{0.637\, I_m} = 1.11.$

m) Crest or peak or amplitude factor (Ka)

It is the ratio of the peak or maximum value to the rms value of an alternating quantity.

Thus, for a sine wave,

$$K_a = \frac{I_m}{I_{rms}} = \frac{I_m}{0.707\, I_m} = 1.414.$$

Example 1

An alternating emf of frequency 50 Hz, has an amplitude of 100 V. Write down the equation for the instantaneous value. Also, find the instantaneous value of the emf after 1/600 second.

Solution:

The instantaneous equation for the emf is,

$$e = 100 \sin(2\pi ft) = 100 \sin(2\pi \times 50\, t) = 100 \sin(100\,\pi t)$$

At $t = \frac{1}{600}$ sec

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$$e = 100 \sin\left(100\pi \times \frac{1}{600}\right) = 100 \sin\left(100 \times \frac{180°}{600}\right) = 100 \sin(30°)$$

$\therefore \quad e = 50$ V.

Example 2

An alternating voltage is given by the equation

$$V = 282.84 \sin\left(377t + \frac{\pi}{6}\right).$$

Find the (a) rms value (b) frequency (c) time period.

Solution:

$V_m = 282.84$ V

a) $V_{rms} = \frac{282.84}{\sqrt{2}} = 200$ V

b) $\omega = 377$ rad/s

$$f = \frac{377}{2\pi} = \frac{377}{2 \times 3.14} = 60 \text{ Hz}$$

c) $T = \frac{1}{f} = \frac{1}{60}$

$\therefore T = 0.167$ sec.

Example 3

If the form factor of a current wave form is 2 and the amplitude factor is 2.5, find the average of the current if the maximum value of the current is 500 A.

Solution:

$K_f = 2$.

$K_a = 2.5$

$I_m = 500$ A

Hence,

$$K_f = 2 = \frac{I_{rms}}{I_{av}}$$

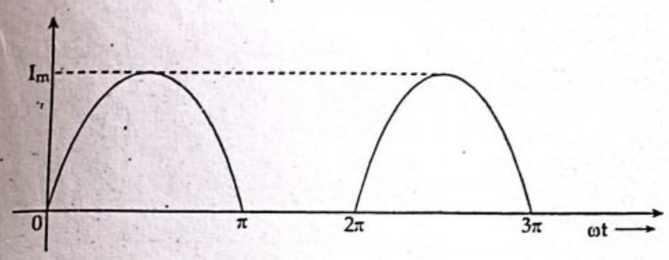
and, $K_a = 2.5 = \frac{I_m}{I_{rms}} = \frac{500}{I_{rms}}$

so, $I_{rms} = \frac{500}{2.5} = 200$ A

$\therefore \quad I_{av} = \frac{I_{rms}}{2} = \frac{200}{2} = 100$ A

Example 4

Find the average and rms value of the wave form shown below *i.e.*, half wave rectifier.



Solution:

$$I_{av} = \frac{1}{2\pi}\int_0^{\pi} I_m \cdot \sin \omega t \, d(\omega t)$$

$$= \frac{I_m}{2\pi}[-\cos \omega t]_0^{\pi}$$

$$= \frac{I_m}{\pi}$$

$\therefore \quad I_{av} = 0.318 \, I_m$

$$\text{and, } I_{rms} = \sqrt{\frac{1}{2\pi}\int_0^{\pi} I_m^2 \sin^2 \omega t \, d(\omega t)}$$

$$= \sqrt{\frac{I_m^2}{4\pi}\int_0^{\pi} (1 - \cos 2\omega t) \, d(\omega t)}$$

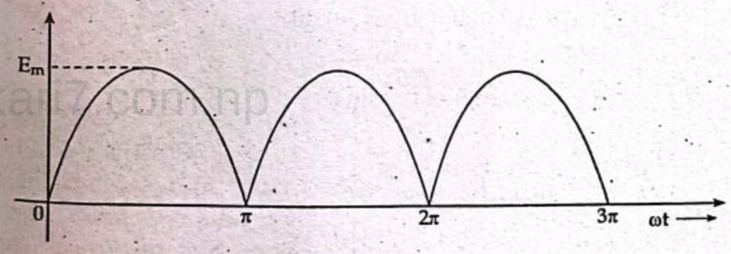
$$= \frac{I_m}{2}\sqrt{\frac{1}{\pi}\left[\omega t - \frac{\sin^2 \omega t}{2}\right]_0^{\pi}}$$

$$= \frac{I_m}{2}\sqrt{\frac{1}{\pi} \cdot \pi}$$

$$= 0.5 \, I_m.$$

Example 5

Find the rms and average value of the wave form shown below; *i.e.*, full wave rectifier.



Solution:

$$E_{rms} = \sqrt{\frac{1}{\pi}\int_0^{\pi} E_m^2 \cdot \sin^2 \omega t \, d(\omega t)}$$

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$$= \sqrt{\frac{E_m^2}{2\pi}\int_0^{\pi}(1-\cos 2\omega t)\,d(\omega t)}$$

$$= \frac{E_m}{\sqrt{2\pi}}\sqrt{\left[\omega t - \frac{\sin 2\omega t}{2}\right]_0^{\pi}}$$

$$= \frac{E_m}{2\pi}\times\sqrt{\pi}$$

$$= \frac{E_m}{\sqrt{2}}$$

$\therefore \quad E_{rms} = 0.707\ E_m$

and, $E_{av} = \frac{1}{\pi}\int_0^{\pi} E_m \sin\omega t\, d(\omega t)$

$$= \frac{E_m}{\pi}[-\cos\omega t]_0^{\pi}$$

$$= \frac{E_m}{\pi}\times 2$$

$$= 0.637\ E_m$$

Example 6

Find the average and rms value of the current,

$i(t) = 10 + 10 \sin 314t$ A

Solution:

$i = 10\,(1 + \sin 314t) = 10\,(1 + \sin\theta)$ [$\because \theta = 314\,t$]

$\therefore \quad i^2 = 100 + 200\sin\theta + 100\sin^2\theta$

$= 100 + 200\sin\theta + 50\,(1 - \cos 2\theta)$

$= 100 + 200\sin\theta - 50\cos 2\theta$

$\therefore \quad I_{rms}^2 = \frac{1}{2\pi}\int_0^{2\pi} i^2\, d\theta$

$= \frac{1}{2\pi}\int_0^{2\pi}(150 + 200\sin\theta - 50\cos 2\theta)\, d\theta$

$= \frac{1}{2\pi}\left[150\,\theta - 200\cos\theta - \left(\frac{50}{2}\right)\sin 2\theta\right]_0^{2\pi}$

$= \frac{1}{2\pi}[150\times 2\pi - 0 - 0]$

or, $I_{rms}^2 = 150$

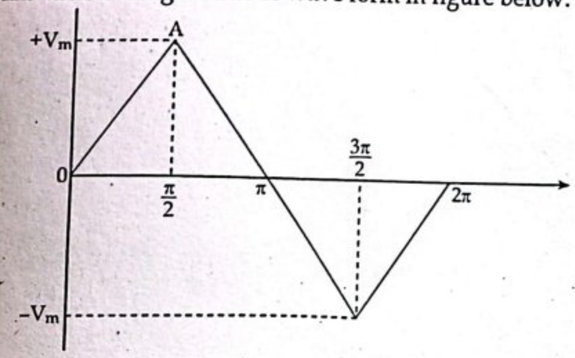
or, $I_{rms} = \sqrt{150}$

$\therefore \quad I_{rms} = 12.247$ A

Hence, $I_{av} = 10$ A.

Example 7

Find the rms and average value of wave form in figure below.



Solution:

$V_{av} = \frac{\text{Area under the wave for the first quarter-sycle}}{\text{Base}}$

$= \left(\frac{1}{2}\right)\frac{V_m\times(\pi/2)}{(\pi/2)} = 0.5\ V_m.$

Considering again first quarter cycle, represented by line OA, then equation for instantaneous voltage is,

$V = m.\theta$

where, m = slope line OA = $(V_m/\pi/2)$

$\therefore \quad V = 2\frac{V_m}{\pi}\theta$

$\therefore$ Mean square value $= \frac{\text{Area under squared wave}}{\text{Base}}$

$= \frac{2}{\pi}\int_0^{2\pi} V^2\, d\theta = \frac{2}{\pi}\int_0^{2\pi} 4\frac{V_m^2\cdot\theta^2}{\pi^2}\cdot D\theta$

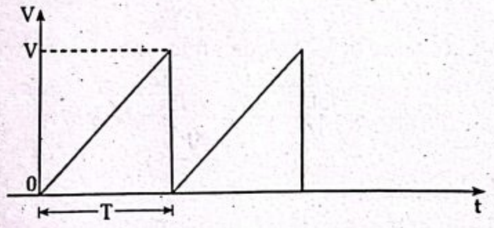
or, $V_{rms}^2 = \frac{V_m^2}{3}$

or, $V_{rms} = \sqrt{\frac{V_m^2}{3}}$

$\therefore \quad V_{rms} = 0.577\ V_m.$

Example 8

Calculate the RMS value of a triangular wave in which voltage rises uniformly from zero to V and completes the cycle by falling instantly back to zero. Find also the form factor and peak factor.



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Solution:

$V_m = V$

$V_{av} = \frac{0 + V}{2} = 0.5\ V$

so, $V = \frac{V}{T} \times t$

i) $V_{rms}^2 = \frac{1}{T}\int_0^T \left(\frac{V}{T}\right)^2 t^2\, dt = \frac{1}{T} \cdot \frac{V^2}{T^2} \cdot \frac{T^3}{3}$

or, $V_{rms}^2 = \frac{V^2}{3}$

$\therefore \quad V_{rms} = 0.5774\ V$

ii) $K_f = \frac{V_{rms}}{V_{av}} = \frac{0.5774\ V}{0.5\ V} = 1.1557$

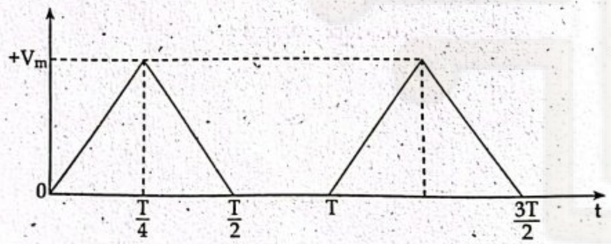
iii) $K_a = \frac{V_m}{V_{rms}} = \frac{V}{0.5774\ V} = 1.732$

NOTE:

Area under squared triangular wave $= \frac{1}{3} \times (\text{Altitude})^2 \times \text{Base}$.

Example 9

Find the average and rms values of the wave shape given below over one cycle.



Solution:

Average value,

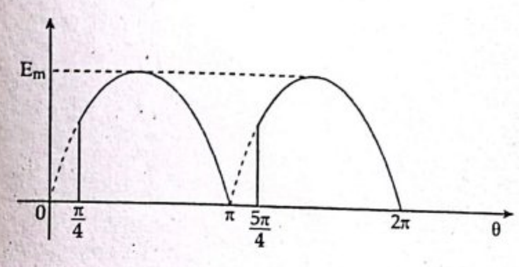
$V_{av} = \frac{\text{Area under the wave from o to T}}{T} = \frac{1/2 \times 10\ V \times T/2}{T} = 2.5\ V$

Mean value of V^2 = Area under the squared wave $= \frac{1}{3} \times 10^2 \times \frac{T}{2} = \frac{50\ T}{3}$

$\therefore \quad V_{rms} = \sqrt{\frac{50\ T}{3} \times \frac{1}{T}} = \sqrt{\frac{50}{3}} = 4.08\ V.$

Example 10

Calculate the average and rms value of a full rectified sine wave of waveform.



Solution:

$V_{av} = \frac{\text{Area under half cycle}}{\text{Half period}}$

$= \int_{\pi/4}^{\pi} E_m \frac{\sin\theta}{\pi}\, d\theta$

$= E_m \frac{(\cos \pi/4 - \cos \pi)}{\pi}$

$= E_m \times \frac{1.767}{\pi}$

$\therefore \quad V_{av} = 0.5434\ E_m$

and, $E_{rms} = \left[\frac{1}{\pi}\int_{\pi/4}^{\pi} E_m^2 \sin^2\theta\, d\theta\right]^{\frac{1}{2}}$

$= \left[\frac{E_m^2}{2\pi}\int_{\pi/4}^{\pi} (1 - \cos 2\theta)\, d\theta\right]^{\frac{1}{2}}$

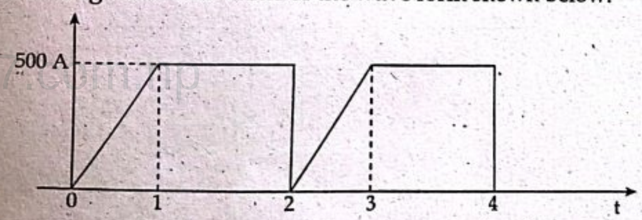
$= \left[\frac{E_m^2}{2\pi}\left(\theta - \frac{\sin 2\theta}{2}\right)_{\pi/4}^{\pi}\right]^{\frac{1}{2}}$

$= \left[\frac{E_m^2}{2\pi}\left(\frac{3\pi}{4} + \frac{1}{2}\right)^{\frac{1}{2}}\right]$

$= 0.674\ E_m$

Example 11

Find the average and rms value of the wave form shown below.



Solution:

$I_{av} = \frac{1}{2}\int_0^2 i\, dt = \frac{1}{2}\left[\int_0^1 i_1\, dt + \int_0^2 i_2\, dt\right]$

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where, $i_1 = \frac{500t}{1} + 0 = 500t$, $\quad i_2 = 500$

Hence,

$$I_{av} = \frac{1}{2}\left[\int_0^1 500t\,dt + \int_1^2 500\,dt\right]$$

$$= \frac{1}{2}\left[500\left[\frac{t^2}{2}\right]_0^1 + 500\,[t]_1^2\right]$$

$$= \frac{1}{2}\left[500 \times \frac{1}{2} + 500\right]$$

$$= \frac{1}{2} \times 750$$

$$= 375\ A$$

and, $I_{rms} = \sqrt{\frac{1}{2}\left[\int_0^1 i_1^2\,dt + \int_0^2 i_2^2\,dt\right]}$

$$= \sqrt{\frac{1}{2}\left[\int_0^1 (500\,t)^2\,dt + \int_0^1 (500)^2\,dt\right]}$$

$$= 500\sqrt{\left[\frac{t^3}{3}\right]_0^1 + [t]_1^2}$$

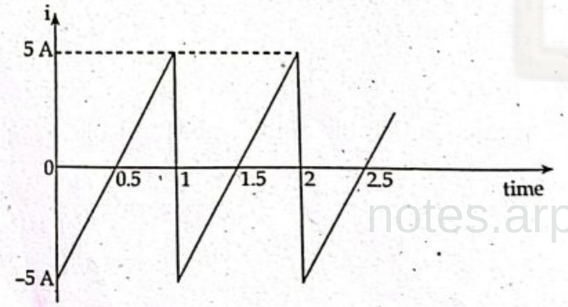
$$= 500\sqrt{\frac{1}{3} + 1}$$

$$= 500 \times \frac{2}{\sqrt{3}}$$

$\therefore \quad I_{rms} = 577.35$ A.

Example 12

Find the average and rms value of the wave form shown below.



Solution:

The average value of the wave form over a full cycle (t = 0 to t = 1) is 0.

Considering half cycle (t = 0.5 to t = 1).

$$I_{av} = \frac{1}{0.5}\int_{0.5}^{2} i\,dt$$

Now,

$$i = \frac{5}{0.5}t - 0.5 = 10\,t - 5$$

Hence,

$$I_{av} = \frac{1}{0.5}\int_{0.5}^{1} (10\,t - 5)\,dt$$

$$= 2\left[\frac{10\,t}{2} - 5t\right]_{0.5}^{1}$$

$$= 2\,[5 \times 1^2 - 5 \times 1 - 5 \times (0.5)^2 + 5 \times 0.5]$$

$$= 2\,[2.5 - 1.25]$$

$$= 2.5\ A$$

and,

$$I_{rms} = \sqrt{\frac{1}{0.5}\int_{0.5}^{1} (10\,t - 5)^2\,dt}$$

$$= \sqrt{\frac{1}{0.5}\int_{0.5}^{1} (100\,t^2 + 25 - 100\,t)\,dt}$$

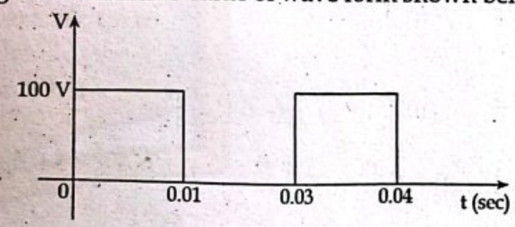
$$= \sqrt{2\left[100\frac{t^3}{3} - 50\frac{t^2}{2} + 25\,t\right]_{0.5}^{1}}$$

$$= \sqrt{2\left[\frac{100}{3} \times 1^3 - 50 \times \frac{1^2}{2} + 25 \times 1 - \frac{100}{3}(0.5)^3 + 50 \times 0.5^2 \times \frac{1}{2} - 25 \times 0.5\right]}$$

$= 7.90$ A.

Example 13

Obtain average and effective value of wave form shown below.



Solution:

$$V_{av} = \frac{1}{T}\int_0^{T/3} V\,dt$$

$$= \frac{1}{0.03}\int_0^{0.01} 100\,dt$$

$$= \frac{1}{0.03} \times [t]_0^{0.01}$$

$$= 33.33\ V$$

and, $V_{rms} = \sqrt{\frac{1}{T}\int_0^{T/3} V^2\,dt} = \sqrt{\frac{1}{0.03} \times 100^2 \times [t]_0^{0.01}} = 57.74$ V.

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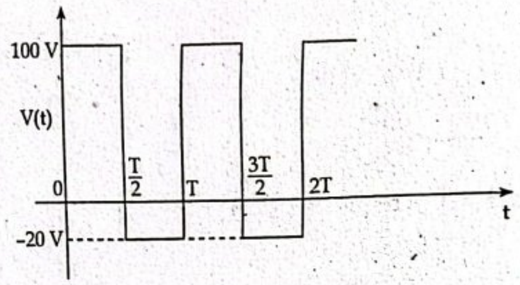
Example 14

Determine the average and rms value of the wave form shown in figure below.



Solution:

Here;

$V(t) = 100$ for $0 \le t \le T/2 = -20$ for $T/2 \le t \le T$

Now,

$$V_{av} = \frac{1}{T}\int_0^T V(t)\,dt$$

$$= \frac{1}{T}\left[\int_0^{T/2} 100\,dt + \int_{T/2}^{T} -20\,dt\right]$$

$$= \frac{100}{T}[t]_0^{T/2} + \frac{(-20)}{T}[T]_{T/2}^{T}$$

$$= 50 - 10$$

$$= 40 \text{ V}$$

and, $V_{rms} = \left[\frac{1}{T}\int_0^T V^2(t)\,dt\right]^{\frac{1}{2}}$

$$= \left[\frac{1}{T}\int_0^{T/2} (100)^2\,dt + \int_{T/2}^{T} (-20)^2\,dt\right]^{\frac{1}{2}}$$

$$= \left[\frac{10^4}{T}[t]_0^{T/2} + \frac{400}{T}[t]_{T/2}^{T}\right]$$

$$= [5 \times 10^3 + 200]^{1/2}$$

$$= (5{,}200)^{1/2}$$

$\therefore$ $V_{rms} = 72.12$ V.

3.4 PHASE ALGEBRA

Alternating quantities have varying magnitude and direction. So, they are represented by a rotating vector. A phasor is a vector rotating at a constant angular velocity. AC circuits cannot be solved by use of simple algebra since geometrical relations are to be taken into considerations. Though phasor diagram method is quite satisfactory for the fairly simple

circuits but become complicated when a circuit or network is made up of several branches. AC circuits can be conveniently solved by use of complex algebra. The results obtained by this method are of the same order of accuracy as those obtained by trigonometrical methods although calculations are usually much simple.

There are four forms or methods of representing the vector or phasor quantities Viz;

i) Symbolic form

A phasor is resolved into two components at right angles to each other and the component lying y-axis is designated by + j or – j according to it's direction. The horizontal component is called the in-phase or active component while the vertical component is called the quardrature or reactive component.

Note:

j is an operator which turns phasor through an angle of 90° in counter clockwise direction, each time that it is applied and has a numerical value of $\sqrt{-1}$, so that

$$j^2 = \left(\sqrt{-1}\right)^2 = -1$$

$$j^3 = \left(\sqrt{-1}\right)^3 = -\sqrt{-1}$$

$$j^4 = \left(\sqrt{-1}\right)^4 = (-1)^2 = 1$$

Representation = $a \pm jb$

$$\theta = \tan^{-1}\left(\frac{\pm b}{a}\right)$$

ii) Trigonometric form

x-component of $v = V\cos\theta$

y-component of $v = V\sin\theta$

Hence,

$$v = V\cos\theta + jV\sin\theta = V(\cos\theta + j\sin\theta)$$

In general,

$$v = V(\cos\theta \pm j\sin\theta)$$

iii) Polar form

In general,

$$v = V\angle \pm \theta$$

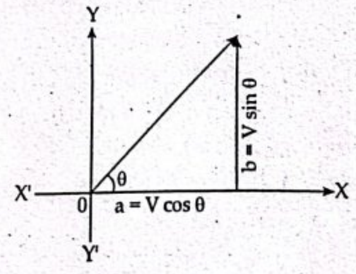
iv) Exponential form

By Euler's equation.

$$E^{j\theta} = \cos\theta + j\sin\theta$$

Hence, In general,

$$V = V \cdot e^{\pm j\theta}$$



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NOTE:

i) Addition of phasor

If $V_1 = a_1 + jb_1$

$V_2 = a_2 + jb_2$

The resultant voltage is,

$$V = V_1 + V_2$$
$$= a_1 + jb_1 + a_2 - jb_2$$
$$= (a_1 + a_2) + j(b_1 - b_2)$$

Magnitude of resultant voltage, $V = \sqrt{(a_1 + a_2)^2 + (b_1 - b_2)^2}$

Argument or phase angle of resultant voltage, θ

$$= \tan^{-1}\left(\frac{b_1 - b_2}{a_1 + a_2}\right)$$

ii) Subtraction of phasors

$$\text{Magnitude} = \sqrt{(a_1 - a_2)^2 + (b_1 + b_2)^2}$$

$$\text{Argument, } \theta = \tan^{-1}\left(\frac{b_1 + b_2}{a_1 - a_2}\right)$$

iii) Multiplication of phasors

Consider two phasors,

$$V_1 = a_1 + jb_1 = V_1 \angle\theta_1 = V_1 e^{j\theta_1}$$
$$V_2 = (a_2 - jb_2) = V_2 \angle -\theta_2 = V_2 e^{-j\theta_2}$$

Now, multiplication of phasors:

$$V_1 \times V_2 = (a_1 + jb_1) \times (a_2 - jb_2)$$
$$= a_1 a_2 - ja_1 b_2 + ja_2 b_1 - j^2 b_1 b_2$$
$$= (a_1 a_2 + b_1 b_2) + j(a_2 b_1 - a_1 b_2)$$

$$\text{Magnitude} = \sqrt{(a_1 a_2 + b_1 b_2)^2 + (a_2 b_1 - a_1 b_2)^2}$$

$$\text{Argument, } \theta = \tan^{-1}\left(\frac{a_2 b_1 - a_1 b_2}{a_1 a_2 + b_1 b_2}\right)$$

Hence,

$$V_1 = V_1 \angle\theta_1 = V_1 e^{j\theta_1} = V_1(\cos\theta_1 + j\sin\theta_1)$$
$$V_2 = V_2 \angle -\theta_2 = V_2 e^{-j\theta_2} = V_2(\cos\theta_2 - j\sin\theta = 2)$$
$$\therefore\ V_1 V_2 = [V_1(\cos\theta_1 + j\sin\theta_1)][V_2(\cos\theta_2 - j\sin\theta_2)]$$
$$= V_1V_2[\cos\theta_1\cos\theta_2 - j\cos\theta_1\sin\theta_2 + j\sin\theta_1\cos\theta_2 - j^2\sin\theta_1\sin\theta_2]$$
$$= V_1V_2[\cos\theta_1\cos\theta_2 + \sin\theta\sin\theta_2 + j(\sin\theta_1\cos\theta_2 - \sin\theta_2\cos\theta_1]$$
$$= V_1V_2[\cos(\theta_1 - \theta_2) + j\sin(\theta_1 - \theta_2)]$$

Thus, in polar form,

$$V_1 V_2 = V_1 V_2 \angle\theta_1 - \theta_2$$

In exponential form,

$$V_1 V_2 = V_1 V_2 \cdot e^{j(\theta_1 - \theta_2)}$$

iv) Division of phasors

$$\frac{V_1}{V_2} = \frac{a_1 + jb_1}{a_2 - jb_2}$$

Multiplying both numerator and denominator by conjugate of $(a_2 + jb_2)$, we have,

$$\therefore\ \frac{V_1}{V_2} = \frac{a_1 + jb_1}{a_2 - jb_2} \times \frac{a_2 + jb_2}{a_2 + jb_2}$$
$$= \frac{a_1 a_2 - b_1 b_2}{a_2^2 + b_2^2} + j\frac{a_1 b_2 + a_2 b_1}{a_2^2 + b_2^2}$$

In polar and exponential form:

$$\frac{V_1}{V_2} = \frac{V_1(\cos\theta_1 + j\sin\theta_1)}{V_2(\cos\theta_2 + j\sin\theta_2)} = \frac{V_1 \angle\theta_1}{V_2 \angle -\theta_2} = \frac{V_1 e^{j\theta_1}}{V_2 e^{-j\theta_2}}$$

Now,

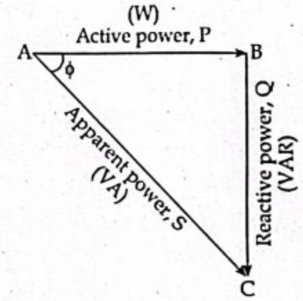
$$\frac{V_1}{V_2} = [\cos(\theta_1 + \theta_2) + j\sin(\theta_1 + \theta_2)]$$

$$\therefore\ \frac{V_1}{V_2} = \frac{V_1}{V_2}\angle\theta_1 - (\theta_2) = \frac{V_1}{V_2}\angle\theta_1 + \theta_2$$

and, $\frac{V_1}{V_2} = \frac{V_1}{V_2} e^{j(\theta_1 + \theta_2)}$

3.5 POWER TRIANGLE

Power triangle is a right angled triangle whose sides represent the active, reactive and apparent power. Base, perpendicular and hypotenous of this right angled triangle denotes the active, reactive and apparent power respectively.



Apparent power (S) is the product of rms values of the applied voltage and circuit current. It is also known as wattles of idle component of power. Thus,

$$S = VI = (IZ) \cdot I = I^2 \cdot Z \text{ volt} \cdot \text{Amp (VA)}.$$

Active power (P) is the power which is actually dissipated in the circuit resistance. It is also known as wattful component of power or true power.

$$P = I^2 R = VI\cos\phi \text{ watts.}$$

Reactive power (Q) is the power developed in the inductive reactance of the circuit, thus,

$$Q = I^2 \cdot X_L$$
$$= I^2 \cdot Z\sin\phi$$
$$= I(IZ)\sin\phi$$
$$= VI\sin\phi \text{ volt-amp reactive (VAR)}$$

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These powers are related as,

$$S^2 = P^2 + Q^2$$

or, $S = \sqrt{P^2 + Q^2}$

Power factor ($\cos\phi$) is the cosine of the angle of lead or lag. It is also known as ratio of active power (P) to the apparent power (S).

i.e., $\cos\phi = \frac{P}{S}$

$\therefore$ $p.f. = \frac{P}{VI}$

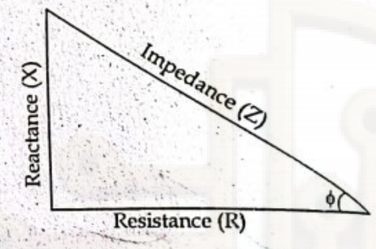
or, Active power (P) = Apparent power (S) × p.f.

Also,

$\therefore$ $(VA)^2 = (W)^2 + (VAR)^2$.

3.6 IMPEDANCE TRIANGLE AND ADMITTANCE TRIANGLE

A triangle formed by the three sides namely the resistance, the reactance and the impedance is called the impedance triangle.



Impedance triangle is used calculate the impedance when resistance (R), Inductance (L) and capacitance (C) all are present in the circuit and the total reactance (X) is the difference between the inductive reactance (X_L) and capacitive reactance (X_C).

The effective resistance of an AC circuit is called impedance (Z) and is formed through the combination of resistance and reactance. It's unit is ohm.

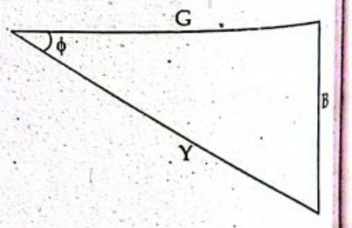
Admittance (y) is the reciprocal of impedance (Z) of an AC circuit. Thus, admittance,

$$Y = \frac{1}{Z} = \frac{I}{E} = \frac{\text{rms value of current}}{\text{rms value of voltage}}$$

Admittance also has two components, namely (i) x-component, conductance (G) and (ii) the y-component, susceptance (B).

Conductance (G) = $Y\cos\phi$

$$= \frac{1}{Z}\cdot\frac{R}{Z} = \frac{R}{Z^2}$$



and, Susceptance (B) = $Y\sin\phi = \frac{1}{Z}\cdot\frac{X}{Z} = \frac{X}{Z^2}$

$\therefore$ Admittance (Y) = $\sqrt{G^2 + B^2}$

The unit of G, B and Y are mho or ohm^{-1} or simons.

3.7 POWER FACTOR AND SIGNIFICANCE OF POWER FACTOR

Power factor is the cosine of the angle of lead or lag. It is also the ratio of active power (P) to the apparent power (S). Thus,

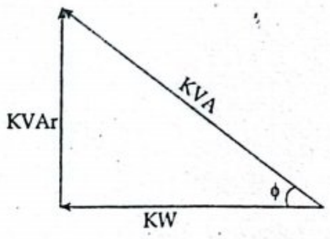
$$p.f. = \cos\phi = \frac{P}{VI}$$

or, Active power (P) = apparent power (S) × p.f.

Power factor is also equal to the ratio of resistance (R) to the impedance (Z) of the circuit.

i.e., $p.f. = \frac{\text{Resistance (R) in ohms}}{\text{Impedance (Z) in ohms}} = 0 \text{ to } 1$

Also, power factor is the relation between the KW and the KVA drawn by an electrical load where the KW is the actual load power and KVA is the apparent load power.



$$p.f. = \frac{KW}{KVA} = \cos\phi$$

Importance of power factor (p.f.)

The p.f. of an AC system should be as high as possible so that current and applied voltage are brought as near in phase as possible. A power factor of less than 1 indicates the voltage and current are not in phase, reducing the instantaneous product of the two.

Low power factor means poor electrical efficiency. The lower the power factor, the higher the apparent power drawn from the distribution network. Low power factor possesses the following drawbacks.

i) It reduces KWH, since the true power is less.
ii) It reduces the torque of the consumer motor.
iii) It causes fall in terminal voltage.
iv) It causes more power loss in the line, thereby reducing the line efficiency.
v) It causes poor voltage regulation.
vi) It limits the output of generators and transformers, since higher currents drawn out from these results in the increase in temperature.
vii) The current needed to obtain a given power is very high. This in-turn results in the increased resistive losses, there by resulting in decreased efficiency.
viii) Over loading of cable, switchagear, busbar etc.

3.8 STEADY STATE RESPONSE OF CIRCUITS

a) Series R–L circuit

Consider a resistor of R ohms and an inductiye coil of inductance L henries are connected across AC circuit. Let, E = rms value of applied

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voltage, I = rms value of resultant current, $V_R = (iR)$ = voltage drop across R in phase with I and $V_L = IX_L$ = voltage drop across L at right angle to I

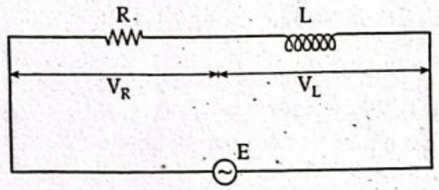

Fig: R-L series AC circuit

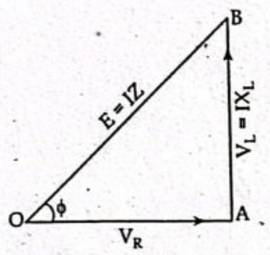

Fig: Voltage triangle

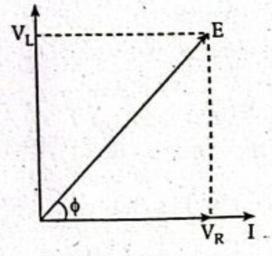

Fig: Phasor diagram

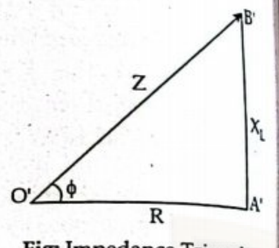

Fig: Impedance Triangle

Here, $OA = V_R$

$OB = V_L$

The applied voltage (E) is the vector sum of these two,

$$E = \sqrt{V_R^2 + V_L^2} = \sqrt{(IR)^2 + (IX_L)^2} = I\sqrt{(R^2 + X_L)^2}$$

or, $$I = \frac{E}{\sqrt{R^2 + X_L^2}}$$

$$\therefore \quad I = \frac{E}{Z} \qquad (1)$$

The quantity $\sqrt{R^2 + X_L^2}$ is called impedance (Z) of the circuit.

$$\therefore \quad Z^2 = R^2 + Z_L^2$$

or, $(\text{Impedance})^2 = (\text{Resistance})^2 + (\text{Reactance})^2$

From (1) and (2);

$E = IZ$

$V_R = IR$

$V_L = IX_L = I\omega L$

Here, E leads the vector I by an angle ϕ or current (I) lags the voltage (E) by an angle ϕ, such that,

$$\tan\phi = \frac{X_L}{R} = \frac{\omega L}{R} = \frac{\text{Reactance}}{\text{Resistance}}$$

or, $$\phi = \tan^{-1}\left(\frac{X_L}{R}\right)$$

and, Power $= E \times I\cos\phi = E_{rms} \times I_{rms} \times \cos\phi$

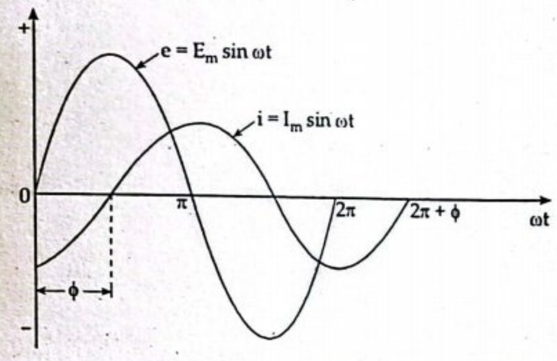

Fig: Voltage and current waves in R-L series AC circuit

b) Series R–C AC circuit

Consider an AC circuit in which a resistor of R ohms and a capacitor of capacitance C Farad are connected in series.

Let, E = rms value of applied voltage

I = rms value of resultant current

$V_R = IR$ = Voltage drop across R in phase with I

$V_C = IX_C$ = Voltage drop across C lagging I by $\frac{\pi}{2}$ radians.

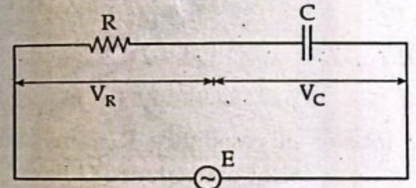

Fig: Series R-C circuit

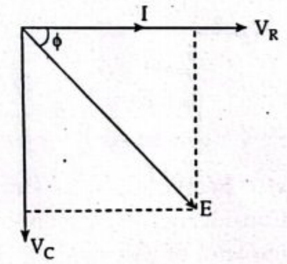

Fig: Phasor diagram

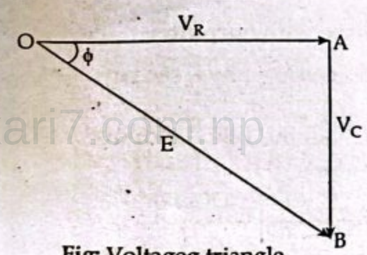

Fig: Voltageg triangle

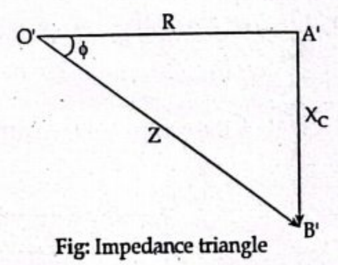

Fig: Impedance triangle

Since the capacitive reactance (X_C) is regarded as negative. So, V_C is shown along the negative direction of y-axis (*i.e.*, down wards) in the voltage triangle OAB.

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$$E^2 = V_R^2 + V_C^2 = (IR)^2 + (-IX_C)^2 = I^2(R^2 + X_C^2)$$

or, $$I = \frac{E}{\sqrt{R^2 + X_C^2}}$$

$$\therefore \quad I = \frac{E}{Z}$$

where Z is called impedance of the circuit. In this circuit, current (I) leads the voltage (E) by an angle ϕ.

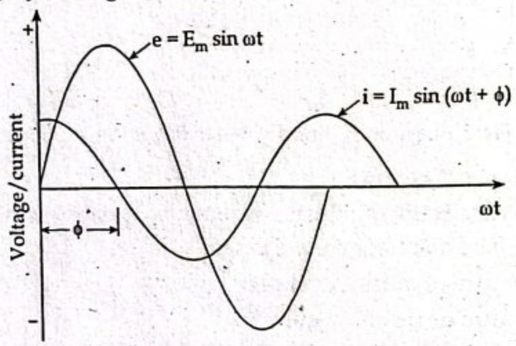


Fig: Voltage and current waves in series R-C AC circuit

$$\tan\phi = -\frac{X_C}{R}$$

Therefore, corresponding instantaneous values of voltage,

$e = E_m \sin\omega t$

and, $i = I_m \sin(\omega t + \phi)$

Power consumed, $P = e \cdot i = E_{rms} \cdot I_{rms} \cos\phi$.

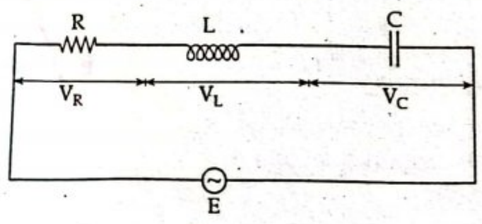
c) Series R–L–C AC circuit

Consider an AC circuit containing a resistor of resistance R ohms, an inductor of inductance L Henries and a capacitor of capacitance C Farad, all connected in series across AC supply of rms voltage E volts and rms current I amperes.

Let, $V_R = IR$ = Voltage drop across R in phase with current, I

$V_L = IX_L = I\omega_L$ = Voltage drop across L leading I by $\frac{\pi}{2}$ radians.

$V_C = IX_C = \frac{I}{\omega_C}$ = Voltage drop across C lagging I by $\frac{\pi}{2}$ radians.



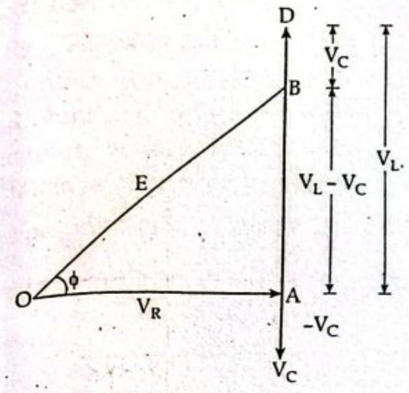


Fig: Voltage triangle.

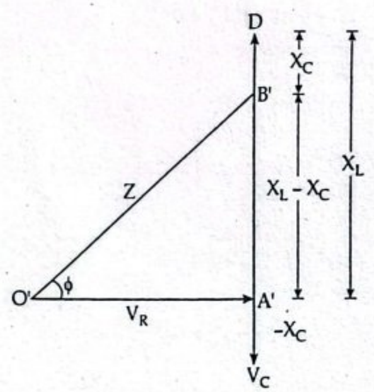


Fig: Impedance triangle

Here;

$OA = V_R$

$OB = E$

$AB = V_L - V_C$

V_L and V_C are out of phase by π radians with each other *i.e.*, they are directly in opposition to each other. Here, V_L has been assumed to be greater in magnitude then V_C, so the net reactive drop = AB,

$V_L - V_C = I(X_L - X_C)$

But the applied voltage (E), represented by OB, is vector sum of OA and OB,

i.e., $$E = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$= \sqrt{(IR)^2 + (IX_L - IX_C)^2}$$

$$= I\sqrt{R^2 + (X_L - X_C)^2}$$

or, $$I = \frac{E}{\sqrt{R^2 + (X_L - X_C)^2}}$$

or, $$I = \frac{E}{\sqrt{R^2 + X^2}}$$

$$\therefore \quad I = \frac{E}{Z}$$

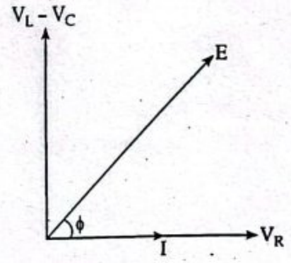


Fig: Phasor diagram

where,

Z = Impedance of the circuit

X = Net reactance = $X_L - X_C$

$\therefore$ Phase angle, $\tan\phi = \frac{X_L - X_C}{R} = \frac{X}{R} = \frac{\text{Net reactance}}{\text{Resistance}}$

Therefore, Voltage and the resulting current are given by,

$e = E_m \sin\omega t$

$i = I_m \sin(\omega t \pm \phi)$

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When current leads (*i.e.*, $X_C > X_L$), then positive (+) sign is to be used and When current lags (*i.e.*, $X_L > X_C$), then negative (–) sign is to be used.

$$\therefore \quad \text{Power factor, } \cos\phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + (X_L \pm X_C)^2}}$$

$$\therefore \quad \text{Power consumed, } P = EI\cos\phi$$

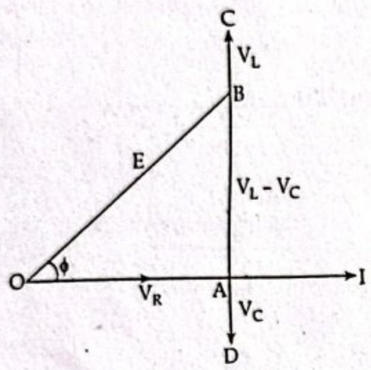

Fig: Voltage triangle for lagging p.f.

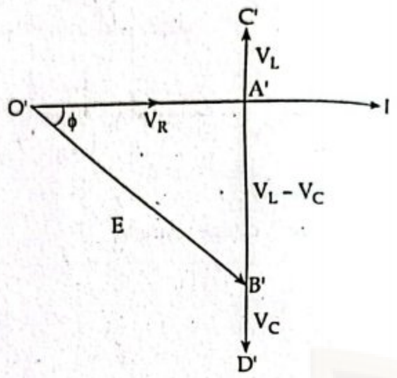

Fig: Voltage triangle for leading p.f.

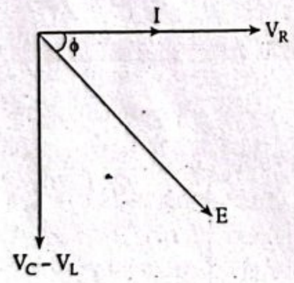

Fig: Phasor diagram

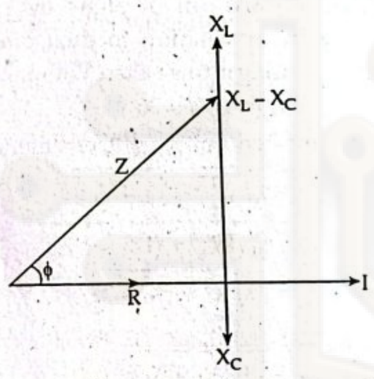

Fig: Impedance triangle for lagging p.f.

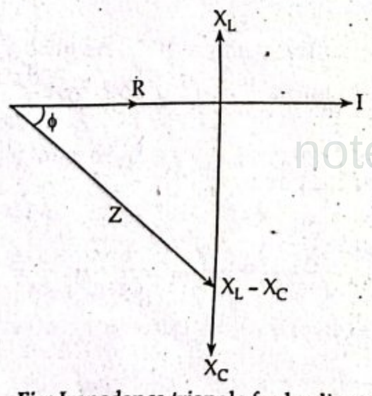

Fig: Impedance triangle for leading p.f.

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d) Parallel R–L circuit

Consider a parallel circuit consisting resistor of R ohms and inductor of L Henries. For parallel circuit, voltage remains same. So, voltage is taken as reference phasor.

For inductance, V_L leads I_L by $\frac{\pi}{2}$ radians.

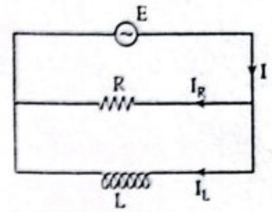

Fig: Parallel R–L circuit

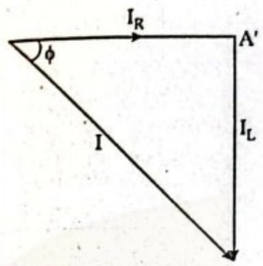

Fig: Current triangle

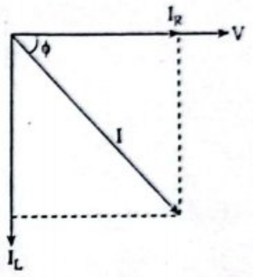

Fig: Phasor diagram

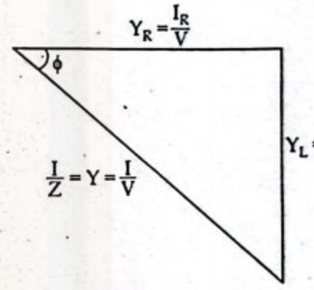

Fig: Admittance triangle

Hence,

$$I = I_R - jI_L$$

or, $$I = \sqrt{(I_r)^2 + (I_L)^2}$$

or, $$\frac{I}{V} = \frac{I_R}{V} - j\frac{I_L}{V}$$

or, $$Y = Y_R - j\,Y_L$$

where, $\frac{I}{V}$ = admittance of parallel circuit

From admittance triangle,

$$Y = \sqrt{Y_R^2 + (Y_L)^2}$$

and, $$\tan = -\frac{Y_L}{Y_R}$$

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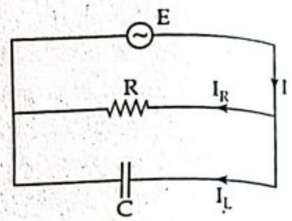
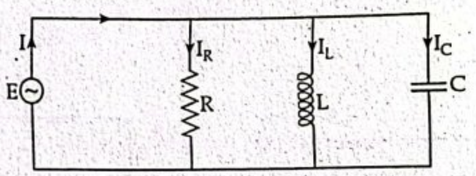
e) Parallel R-C circuit

Consider a parallel circuit consisting resistor of R ohms and capacitor of F Farads, with a voltage source of E volts.

Similarly as preceding derivation, solve this.



f) R-L-C parallel circuit



In parallel RLC circuit, supply voltage E is common to all three component. But the current flowing through each branch is different. The total current drawn from the supply will not be mathematical sum of their individual branch currents but their vector sum.

Here;

$$I_C^2 = I_R^2 + (I_L - I_C)^2$$

or, $$I = \sqrt{I_R^2 + (I_L - I_C)^2}$$

or, $$I = \sqrt{\left(\frac{V}{R}\right)^2 + \left(\frac{V}{X_L} - \frac{V}{X_C}\right)^2}$$

$$\therefore \quad I = \frac{V}{Z}$$

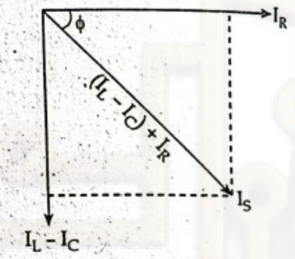


Fig: Current triangle

where, $$I_R = \frac{V}{R} \Rightarrow R = \frac{V}{I_R}$$

$$I_L = \frac{V}{X_L} \Rightarrow X_L = \frac{V}{I_L}$$

$$I_C = \frac{V}{X_C} \Rightarrow X_C = \frac{V}{I_C}$$

Now,

$$Z = \frac{1}{\sqrt{\left(\frac{1}{R}\right)^2 + \left(\frac{1}{X_L} - \frac{1}{X_C}\right)^2}}$$

Also,

$$Y = \sqrt{G^2 + (B_L - B_C)^2}$$

where, $$Y = \frac{1}{Z},\ G = \frac{1}{R},\ B_2 = \frac{1}{\omega L},\ B_C = \omega C$$

Admittance: $$Y = \sqrt{\left(\frac{1}{R}\right)^2 + \left(\frac{1}{\omega L} - \omega C\right)^2}$$

Impedance: $$Z = \frac{1}{Y} = \frac{1}{\sqrt{\left(\frac{1}{R}\right)^2 + \left(\frac{1}{\omega L} - \omega C\right)^2}}$$

and, $$\tan\phi = \frac{B}{G}$$

$$\cos\phi = \frac{G}{Y}.$$

3.8 INSTANTANEOUS POWER AND AVERAGE POWER

Let us consider a single phase power circuit in which current lags behind the voltage by an angle ϕ.

Let, the instantaneous electrical potential difference,

$$V = V_m \sin\omega t$$

The instantaneous current, $i = I_m \sin(\omega t - \phi)$

Then, the instantaneous power of the circuit is,

$$\begin{aligned} P &= i \cdot V \\ &= V_m \sin\omega t \cdot I_m \sin(\omega t - \phi) \\ &= \frac{1}{2} V_m I_m \cdot 2 \cdot \sin\omega t \sin(\omega t - \phi) \\ &= \frac{1}{2} V_m I_m [\cos\omega t \cdot \cos(\omega t - \phi) + \sin\omega t \cdot \sin(\omega t - \phi) - \cos\omega t \\ &\qquad - \cos(\omega t - \phi) - \sin\omega t \cdot \sin(\omega t - \phi)] \\ &= \frac{1}{2} V_m I_m [\cos\{\omega t - (\omega t - \phi)\} - \cos\{\omega t + (\omega t - \phi)\}] \end{aligned}$$

$$\therefore \quad P = \frac{1}{2} V_m I_m [\cos\phi - \cos(2\omega t - \phi)]$$

Instantaneous power is the power at any instant of time.

It varies with time like voltage and current, power is also continuously changing with time, so the average power is given by,

$$P = \frac{1}{T}\int_0^T p \cdot dt$$

By convention, P always means average power,

Also,

$$P = \frac{1}{2\pi}\int_0^{2\pi} p \cdot d\theta = \frac{1}{2\pi}\int_0^{2\pi} Vi\, d\theta$$

3.9 RESONANCE IN SERIES AND PARALLEL RLC CIRCUIT

An AC circuit is said to be in resonance when the circuit current is in phase with the applied voltage. So the power factor of the circuit becomes

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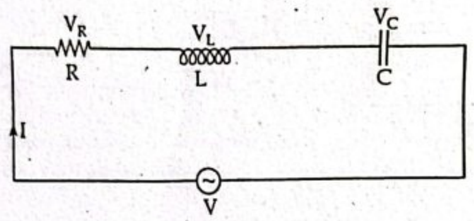
unity at resonance and the impedance of the circuit consists of only resistance.



In the RLC series circuit, the impedance is given by,

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

The current in the circuit is,

$$I = \frac{V}{Z} = \frac{V}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

Power factor, $\cos\phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$

At resonance, $Z = R$

or, $\omega L - \frac{1}{\omega C} = 0$

or, $\omega = \frac{1}{\sqrt{LC}}$

If f_0 be the resonant frequency, then,

or, $2\pi f_0 = \frac{1}{\sqrt{LC}}$

$\therefore \quad 2\pi f_0 = \frac{1}{2\pi\sqrt{LC}}$

The p.f. at resonance, $\cos\phi = \frac{R}{Z} = \frac{R}{R} = 1$

and, The current, $I = \frac{V}{Z} = \frac{V}{R}$

Properties of series resonant circuits

i) The circuit impedance Z is minimum and equal to the circuit resistance R.

ii) The power factor is unity.

iii) The resonant frequency is $f_0 = \frac{1}{2\pi\sqrt{LC}}$

iv) The power dissipated is maximum, *i.e.*, $P_0 = \frac{V^2}{R}$

v) The voltage across inductor is equal and opposite to the voltage across capacitor.

vi) The circuit impedance, Z is minimum and equal to the circuit resistance, R.

Since the circuit current is maximum at resonance, it produces large voltage drop across L and C. But as these voltages are equal and opposite to each other so the net voltage across L and C is zero however large the current is flowing. If R is not present, then the circuit would act like a short circuit at resonant frequency. Hence, a series circuit is sometimes called an acceptor circuit and the series resonance is often referred to as the voltage resonance.

Resonance in parallel circuit

Let us consider a circuit where a capacitance C is connected in parallel with an inductive coil of resistance R and inductive reactance X_L as shown in figure.

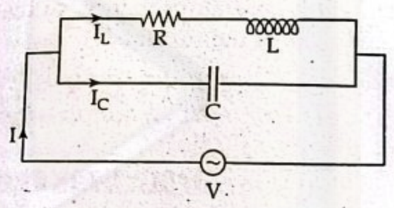


Fig: AC parallel circuit.

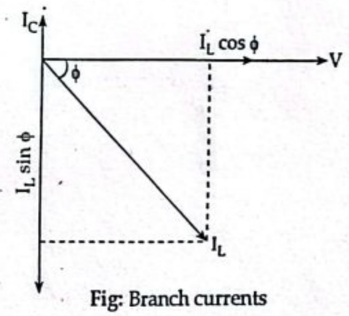


Fig: Branch currents

I_L = current through the coil

I_C = current through the capacitor

I = total current

Under resonance, as the p.f. is unity, the reactive component of the total current is zero, the reactive component of the current ($I_C - I_L \sin\phi$) = 0, where ϕ is the power factor angle of the coil. Hence,

$$I_C = I_L \sin\phi$$

or, $\frac{V}{X_C} = \frac{V}{Z_L} \times \frac{X_L}{Z_L}$

where, Z_L = Impedance of the coil = $\sqrt{R^2 + X_L^2}$

or, $X_C X_L = Z_L^2$

or, $\frac{\omega L}{\omega C} = Z_L^2 = R^2 + (\omega L)^2$

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or, $\omega^2 L^2 = \frac{L}{C} - R^2$

at resonance, $\omega_0 = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$

and, Resonant frequency, $f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$

If resistance is neglected, then

$$\omega_0 = \frac{1}{\frac{1}{\sqrt{LC}}}$$

and, $f_0 = \frac{1}{2\pi\sqrt{LC}}$

Thus, if resistance is neglected the resonant frequency of the parallel circuit is equal to that of series circuit. Also, at resonance the net susceptance is zero. Net susceptance $= \left(\omega C - \frac{1}{\omega L}\right) = 0$

$\therefore \quad \omega_0 = \frac{1}{\sqrt{LC}}$

As the net reactive component of the current is zero at resonance, so the supply current I is equal to the active component of the current.

so, $I = I_L \cos\phi = \frac{V}{Z_L} \cdot \frac{R}{Z_L} = \frac{VR}{Z_L^2} = \frac{VR}{L/C} \qquad \left(\because Z_L^2 = X_C \cdot X_L = \frac{L}{C}\right)$

or, $I = \frac{V}{(L/RC)}$

Thus, at resonance, the net impedance is given by L/RC and is known as the dynamic impedance of the parallel circuit at resonance. This impedance is resistive only. The current is minimum at resonance as its reactive part is zero and thus, (L/RC) represents the maximum impedance of the circuit. It is called a rejector circuit.

Properties of parallel resonant circuit

i) Resonant frequency, $f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$

ii) The power factor is unity at resonance

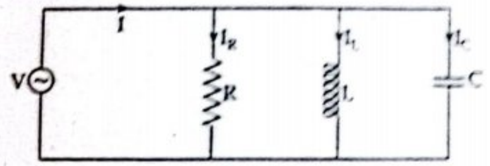
iii) Net susceptance is zero at resonance *i.e.* $\left(\omega C - \frac{1}{\omega L}\right) = 0$

iv) Line current is minimum at resonance or $I = \frac{V}{(L/RC)}$

v) At resonance, the net reactive component of the line current is zero and the circuit current is equal to the active component of the total current. *i.e.*, $I = I_L \cos\phi$

This resonance is often referred as current resonance because the current circulating between two branches is many times greater than the line current taken from the supply.

Parallel resonance in RLC circuit



In the figure, the resonance occurs when the net susceptance is zero.

Admittance, $Y = G + jB$

$$= \frac{1}{R} + j\omega C - j\frac{1}{\omega L}$$

At resonance,

Net susceptance $\left(\omega C - \frac{1}{\omega L}\right) = 0$

or, $\omega_0 = \frac{1}{\sqrt{LC}}$

$\therefore \quad f_0 = \frac{1}{2\pi\sqrt{LC}}$

At resonant frequency (f_0), the admittance is minimum so the impedance is maximum and the current is minimum.

3.10 BANDWIDTH

The bandwidth of a given circuit is given by the band of frequencies which lies between two points on either side of resonant frequency where the current is $\frac{1}{\sqrt{2}}$ times of the current at resonance and hence the power is half of the power at resonance.

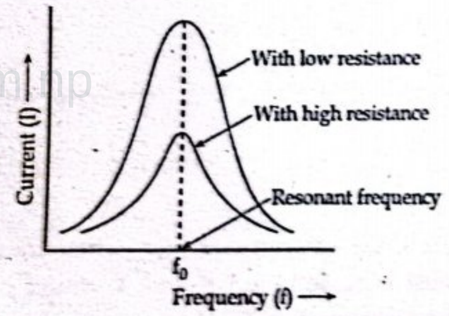


Fig: Variation of resonance curve with resistance (R) of the circuit

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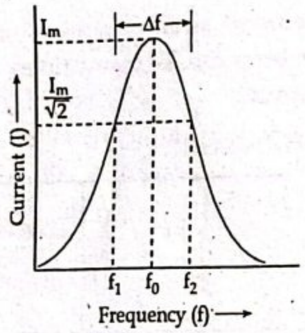

Fig: Bandwidth of a series resonant circuit

The magnitude of band-width = $\Delta f = f_2 - f_1$

Now, power dissipated at resonant frequency (f_0),

$$P_0 = I_m^2 \cdot R$$

$$\therefore \quad P_1 = P_2 = \left(\frac{I_m}{\sqrt{2}}\right)^2 . R = \frac{I_m^2 R}{2} = \frac{\pi}{2} \qquad (1)$$

Thus, f_1 and f_2 are half power points. Hence, bandwidth is the range of frequency within which the power dissipated in the resistance is greater than or equal to half the power dissipated at resonance. Now, for R-L- C series circuit,

$$I = \frac{E}{[R^2 + (X_L - X_C)^2]^{1/2}}$$

$$= \frac{E/R}{\left[1 + \left(\frac{X_L - X_C}{R}\right)^2\right]^{\frac{1}{2}}}$$

$$= \frac{I_m}{\left[1 + \left(\frac{X_L - X_C}{R}\right)^2\right]^{\frac{1}{2}}} \qquad (2)$$

and, Current passing at half power points, I_1 and I_2 are given by

$$I_1 = I_2 = \frac{I_m}{\sqrt{2}} \qquad (3)$$

Comparing equation (2) and (3), we get at half power points,

or, $\frac{(X_L - X_C)}{R} = 1$

or, $X_L - X_C = R$

But at resonance, $(X_L - X_C) = 0$, consequently, when frequency increase from f_0 to f_2, X_L must increase by 0.5 R and X_C must correspondingly decrease by 0.5 R to satisfy equation (2), Thus,

$$2\pi f_2 L - 2\pi f_0 L = 0.5\ R$$

or, $f_2 - f_1 = \frac{R}{4\pi L}$ (4)

Similarly, when frequency decreases from f_0 to f_1, X_C must increase by 0.5 R, and X_C must correspondingly decrease by 0.5 R, thus,

$$2\pi f_0 L - 2\pi f_1 L = 0.5\ R$$

or, $f_0 - f_1 = \frac{R}{4\pi L}$ (5)

On adding equation (4) and (5); we get,

$$\Delta f = f_2 - f_1 = \frac{R}{4\pi L} + \frac{R}{4\pi L}$$

$$\therefore \quad \Delta f = \frac{R}{2\pi L} = (f_2 - f_1)\ \text{Hz}$$

The ratio of the bandwidth to the resonant frequency is defined as the selectivity of the circuit. Thus, selectivity = $\frac{(f_2 - f_1)}{f_0}$.

Thus, narrower the bandwidth, higher is the selectivity property of the circuit.

3.11 QUALITY FACTOR OR Q-FACTOR

Q-factor is the voltage magnification factor at resonance. It is also defined as the ratio of resonance frequency to bandwidth. It is a measure of selectivity or sharpness of tuning of series RLC circuit.

Now,

$$E = I_m \cdot R$$

and, $I_m X_L = I_m X_c$

$$\therefore \quad \text{Q-factor} = \frac{I_m X_L}{I_m \cdot R} = \frac{X_L}{R} = \frac{2\pi f_0 L}{R} = \frac{2\pi L f_0}{R}$$

$$= \frac{2\pi L}{R} \times \frac{1}{2\pi\sqrt{LC}}$$

$$\therefore \quad \text{Q-factor} = \frac{1}{R}\sqrt{\frac{L}{C}} \qquad \left[\because f_0 = \frac{1}{2\pi\sqrt{LC}}\right]$$

Thus, Q-factor can be increased by decreasing R or by increasing L/C ratio.

Also, Q-factor is current magnification factor at resonance of RLC parallel circuit.

i.e., $\text{Q-factor} = \frac{\text{Current in the capacitor or the coil, } I_C \text{ at resonance}}{\text{Total current } (I) \text{ in the circuit at resonance}}$

$$= \frac{I_C}{I} = \frac{I_L \sin\phi_L}{I}$$

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$$= \frac{I_L \sin\phi_L}{I_L \cos\phi_L} \qquad [\because I = I_L \cos\phi_L]$$

$$= \tan\phi_L$$

But, $\tan\phi_L = \frac{2\pi fL}{R} = 2\pi L \cdot \frac{f}{R}$

and, $f_0 = \frac{1}{2\pi\sqrt{LC}}$ (∴ assuming R to be very small)

Hence, Q-factor $= \frac{2\pi L}{R} \times \frac{1}{2\pi\sqrt{LC}} = \frac{1}{R}\sqrt{\frac{L}{C}}$

The Q-factor of a resonant circuit is a measure of the goodness or quality of a resonant circuit. A higher value for this figure of merit corresponds to a more narrow bandwidth which is desire able in many applications.

Example 15

A capacitor and a 50 Ω resistor are connected in series to an alternating current supply. The voltage across the capacitor is 200 V rms and across the resistor is 150 V rms. Determine rms value of supply voltage, peak value of the voltage across the capacitor assuming sinusoidal wave form, power used in the resistor.

Solution:

$R = 50\ \Omega$

$V_R = 150$ V

$V_C = 200$ V

∴ Current, $I = \frac{V_R}{R} = \frac{150}{50} = 3$A

i) Supply voltage $= \sqrt{V_R^2 + V_C^2} = \sqrt{(150)^2 + (200)^2} = 250$ V

ii) Peak value of the voltage across capacitor

$= \sqrt{2} \cdot V_{crms} = \sqrt{2} \times 200 = 282.8$ V

iii) Power used in the resistor

$= I^2R = (3)^2 \times 50 = 450$ W

Example 16

A resistance of 10 Ω is connected in series with an inductance of 0.05 H and a capacitance of 300 μF to a 100 V AC supply. Calculate the value and phase angle of the current when the frequency is 25 Hz.

Solution:

$f = 25$ Hz, $R = 10\ \Omega$, $L = 0.05$ H, $C = 300 \times 10^6$ F, $V = 100$ V

Hence, $X_L = 2\pi fL = 2\pi \times 25 \times 0.05 = 7.85\ \Omega$

and, $X_C = \frac{1}{2\pi fC} = \frac{10^6}{2 \times 3.14 \times 25 \times 300} = 21.23\ \Omega$

∴ Impedance, $Z = \sqrt{R^2 + (X_C - X_L)^2} = \sqrt{10^2 + (21.23 - 7.85)^2} = 16.7\ \Omega$

and, Net reactance, $X = X_C - X_L = 13.38\ \Omega$.

As $X_C > X_L$, so the current is leading.

If θ be the angle of lead, then,

$$\theta = \tan^{-1}\left(\frac{X}{R}\right) = \tan^{-1}\left(\frac{13.38}{10}\right) = 53.23°$$

Current, $I = \frac{V}{Z} = \frac{100}{16.67} = 5.988$ A.

Example 17

A 4 KW load takes a current of 20 A from a 240 V AC supply. Calculate the KVA and KVAr of the load.

Solution:

$V = 240$ V.

$I = 20$ A

$P = 4$ KW

If cos θ be the power factor, then

$VI \cos\theta = P$

or, $240 \times 20 \times \cos\theta = 4{,}000$

or, $\cos\theta = \frac{4{,}000}{240 \times 20}$

∴ $\cos\theta = 0.833$

and, $\sin\theta = \sqrt{1 - \cos^2\theta} = \sqrt{1 - (0.833)^2} = 0.553$

∴ $KVA = VI = \frac{240 \times 20}{10^3} = 4.8$ KVA

∴ $KVAr = VI \sin\theta = \frac{240 \times 20 \times 0.553}{10^3} = 2.65$ KVAr.

Example 18

A series connected RLC circuit has R = 15 Ω, L = 40 mA and C = 40 μF. Find the resonant frequency and under resonant condition, calculate the current, power, voltage drop across various elements if the applied voltage is 75 V.

Solution:

Resonant frequency,

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2 \times 3.14\sqrt{0.04 \times 0.40 \times 10^{-6}}} = 125.88\ \text{Hz}$$

Current at resonance condition,

$$I_0 = \frac{V}{R} = \frac{75}{15} = 5\ \text{A}$$

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Power = $VI_0 = 75 \times 5 = 375$ W

Voltage drop across R = $I_0R = 5 \times 15 = 75$ V

Voltage drop across L = $I_0 X_L = 5 \times 2\pi \times 125.88 \times 0.04 = 158.1$ V

$\therefore$ Voltage drop across C = $I_0 X_C = I_0 \frac{1}{2\pi fC} = \frac{5 \times 10^6}{2\pi \times 125.88 \times 40} = 158.1$ V

Example 19

Convert following time domain voltages into phasor domain

a) $100 \sin\left(\omega t + \frac{\pi}{3}\right)$

Solution:

$\frac{100}{\sqrt{2}} \angle 60° = 70.7 \angle 60°$

b) $100 \sin \omega t$

Solution:

$\frac{100}{\sqrt{2}} \angle 0° = 70.7 \angle 0°$

c) $170 \sin\left(\omega t - \frac{2\pi}{9}\right)$

Solution:

$\frac{170}{\sqrt{2}} \angle -40° = 120 \angle -40°$

Example 20

A two element series circuit is connected across an AC source V = 300 cos (314 t + 20°) volts. The current is drawn 15 cos (314 t – 10°) Amp. Determine circuit impedance magnitude and phase angle. What is the average power drawn?

Solution:

$V = 300 \cos (314 t + 20°)$ [$\because \cos\theta = \sin(\theta + 90°)$]

$\therefore$ $V = 300 \sin (314t + 110°)$

In polar form,

$V = \frac{300}{\sqrt{2}} \angle 110°$

$i = 15 \cos (314 t - 10°)$

$\therefore$ $i = 15 \sin (314 t + 80°)$

$I = \frac{15}{\sqrt{2}} \angle 80°$

Now,

$$Z = \left(\frac{\frac{300}{\sqrt{2}} \angle 110°}{\frac{15}{\sqrt{2}} \angle 80°}\right) = 20 \angle 30°$$

Hence the angle between voltage and current is 30° and current lags the voltage by 30°.

Now,

$\therefore$ $P_{av} = \frac{V_m I_m}{2} \cos\phi = \frac{1}{2} \times 15 \times 300 \times \cos 30° = 1{,}949.85$ watt.

Example 21

The voltage across a circuit is given by (300 + j 60) V and the current through it is (10 – j 5)A. determine the active power, reactive power and apparent power.

Solution:

$V = 300 + j\,60 = \sqrt{(300)^2 + (60)^2} \angle \tan^{-1}\left(\frac{60}{300}\right) = 305.94 \angle 11.31°$ V

$I = 10 - j\,5 = \sqrt{10^2 + 5^2} \angle -\tan^{-1}\left(\frac{5}{10}\right) = 11.18 \angle -26.56°$A

Angle between voltage and current, $\theta = 11.31° - (-26.56°) = 37.87°$

and current is lagging with respect to the voltage

Thus,

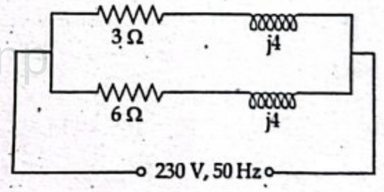
i) Active power = $VI \cos\theta = 305.94 \times 11.18 \cos(37.87°) = 2700$ W

ii) Reactive power = $VI \sin\theta = 305.94 \times 11.18 \times \sin(37.87)$
= 2,099.68 VAR

iii) Apparent power = $VI = 305.94 \times 11.18 = 3{,}420.40$ VA

Example 22

For the circuit shown below, find the current and power drawn from the source.



Solution:

$Z_1 = 3 + 4j = 5 \angle 53.13° \Omega$

$Z_2 = 6 + 8j = 10 \angle 53.13° \Omega$

$\therefore$ $Z_1 + Z_2 = 9 + 12j = 15 \angle 53.13° \Omega$

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Since, $Z_1 \parallel Z_2$,

$$Z = \frac{Z_1 \times Z_2}{Z_1 + Z_2} = \frac{5 \times 10 \angle 106.26°}{15 \angle 53.13°}$$

$\therefore \quad Z = 3.33 \angle 53.13°$

Now,

$\therefore$ Current drawn, $I = \frac{V}{Z} = \frac{230 \angle 0°}{3.33 \angle 53.13°} = 69 \angle -53.13°$ A

and, Power drawn $= VI \cos \phi = 230 \times 69 \times \cos (53.13) = 9.522$ KW

Example 23

A series R-L-C circuit has R = 10 Ω, L = 0.1 H, C = 8 μF. Determine,

i) Resonant frequency

ii) Q-factor at resonance

iii) Half power frequencies.

Solution:

Given that,

$R = 10\ \Omega$

$L = 0.1$ H

$C = 8\ \mu F = 8 \times 10^{-6}$ F

i) For a series RLC circuit, resonant frequency, f_r is given by,

$$f_r = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.1 \times 8 \times 10^{-6}}} = 177.94 \text{ Hz}$$

ii) $\text{Q-factor} = \frac{\omega L}{R} = \frac{2\pi f_r L}{R} = \frac{2 \times 3.14 \times 177.94 \times 0.1}{10} = 11.17$

iii) Bandwidth, $BW = f_2 - f_1$

so, $f_1 = f_r - \frac{BW}{2}$

$$= f_r - \frac{BW}{4\pi L}$$

$$= 177.94 - \frac{10}{4 \times 3.14 \times 0.1}$$

$$= 170 \text{ Hz}$$

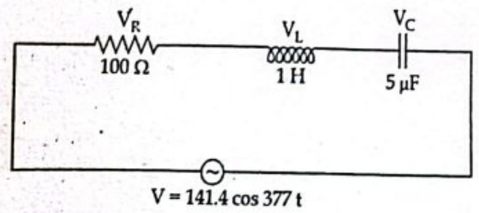
and, $f_2 = f_r + \frac{BW}{2} = 177.94 + 7.95 = 185.89$ Hz

Example 24

A series RLC circuit of a 100 Ω resistance, 1 H inductance and 5 μF capacitance. A voltage V = 141.4 cos 377 t volts is applied on the circuit. Determine the current and voltages V_R, V_L and V_C.

Solution:

$V_{max} = 141.4$ V



$$V_{rms} = \frac{V_{max}}{\sqrt{2}} = \frac{141.4}{\sqrt{2}} = 100 \text{ V}$$

$R = 100\ \Omega$

$C = 5\ \mu F$

$L = 1$ H

$\therefore \quad X_L = 2\pi fL = 2\pi\ 50 \times 1 = 377\ \Omega$

$\therefore \quad X_C = \frac{1}{2\pi fC} = \frac{1}{377 \times 5 \times 10^{-6}}$

Since $X_C > X_L$, so the reactance X is capacitive

$\therefore \quad X = X_C - X_L = 530.5 - 377 = 153.5\ \Omega$

and, $Z = \sqrt{R^2 + X^2} = \sqrt{100^2 + 153.5^2} = 183.2\ \Omega$

$\therefore$ Current, $I = \frac{V}{Z} = \frac{100}{183.2} = 0.5458$ V

i) $V_R = IR = 0.5458 \times 100 = 54.58$ V

ii) $V_R = IX_L = 0.5458 \times 377 = 205.76$ V

iii) $V = C = IX_C = 0.5458 \times 530.5 = 289.55$ V

Example 25

A circuit is composed of a resistance of 8 Ω, and a series capacitive reactance of 6 Ω. A voltage, V(t) = 141 sin 314t is applied to the circuit.

i) Find complex impedance and draw impedance triangle.

ii) Calculate rms and instantaneous value of current.

iii) Find the equation for the voltage appearing across the capacitor.

iv) Calculate power delivered to the circuit.

v) Find the value of the capacitor in Farads.

Solution:

$R = 8\ \Omega$

$X_C = 6\ \Omega$

$V_m = 141$ V

$$V = \frac{141}{\sqrt{2}} = 100 \text{ V}$$

$\omega = 314 = 2\pi f$

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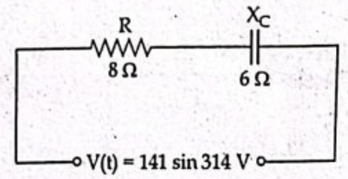
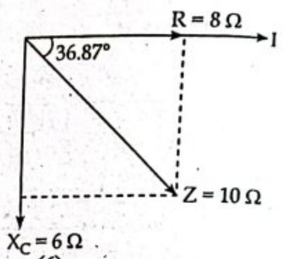




i) Impedance $(Z) = 8 - j6\ \Omega = \sqrt{8^2 + 6^2}\ \angle \tan^{-1}\left(\frac{6}{8}\right) = 10\ \angle -36.87°\ \Omega$

ii) Rms current, $I = \frac{V}{Z} = \frac{100}{10} = 10$ A

$I_m = 10\sqrt{2} = 14.1$ A

∴ Instantaneous equation of current is,

$i(t) = 14.1 \sin (314\ t + 36.87°)$

iii) Power, $P = VI \cos \phi = 100 \times 10 \times \cos 36.87° = 800$ W

iv) Voltage across capacitor,

$V_C = IX_C = 100 \times 6 = 60$ V

∴ $V_{C(max)} = 60\sqrt{2} = 84.98$ V

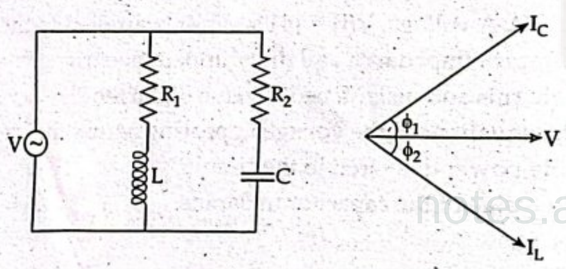
Therefore, equation for the voltage appearing across the capacitor is:

$V_C = 84.85 \sin (314\ t - 90°)$

v) Capacitance, $C = \frac{1}{\omega X_C} = \frac{1}{314 \times 6} = 530.8\ \mu F.$

Example 26

Find condition that the current in the two parallel branches of the circuit shown below shall remain in quadrature, when both R_1 and R_2 are varied simultaneously.



Solution:

Let, ω, I_L and I_C be respectively the angular frequency, the current through R_1 – L branch and the current through R_2 – C branch. Let, I_2 lags V by ϕ_1 and I_C leads V by ϕ_2, then,

$$\tan \phi_1 = \frac{\omega L}{R_1}$$

$$\tan \phi_2 = \frac{1}{\omega C R_2}$$

Since, I_C and I_L are in quadrature, so $\phi_1 + \phi_2 = 90°$,

Now,

$$\tan (\phi_1 + \phi_2) = \frac{\tan \phi_1 + \tan \phi_2}{1 - \tan \phi_1 \cdot \tan \phi_2} = \tan 90° = \infty$$

or, $1 - \tan \phi_1 \tan \phi_2 = 0$

or, $\tan \phi_1 \cdot \tan \phi_2 = 1$

or, $\frac{\omega L}{R_1} \times \frac{1}{\omega C R_2} = 1$

or, $R_1 R_2 = \frac{1}{C}$

Which is the required condition for the two branch current I_L and I_2 to remain in quadratore, when both R_1 and R_2 are varies simultaneously.

Example 27

The voltage and current of R–L–C series circuit are V = 141.4 sin (314t + 45°) V and i = 28.28 sin (314 t – 15°) A, find rms value of voltage and current, power factor, power consumption of the circuit, time periods of AC supply and resistance of the circuit.

Solution:

$$V = \frac{V_m}{\sqrt{2}} = \frac{141.4}{\sqrt{2}} = 100\ V$$

i) $I = \frac{I_m}{\sqrt{2}} = \frac{28.28}{\sqrt{2}} = 20$ A

ii) $V = 141.4 \sin (314\ t + 45°)$

$= 141.4\ [\cos 45° + j\ 45°]$

$= 100 + j\ 100$

$= 141.4\ \angle 45°$

and, $i = 28.28$ in $(314\ t - 15°)$

$= 28.28\ (\cos 15° - \sin 15°)$

$= 27.31 - j\ 7.31$

$= 28.28\ \angle -15°$

iii) Power factor, $\cos \phi = \cos (15 + 45°)$

$= \cos 60°$

$= 0.50$ (lag)

and, Power $= VI \cos \phi = 100 \times 20 \times 0.5 = 1{,}000$ W

iv) $2\pi f = 314$

or, $f = \frac{314}{2\pi} = 50$ Hz

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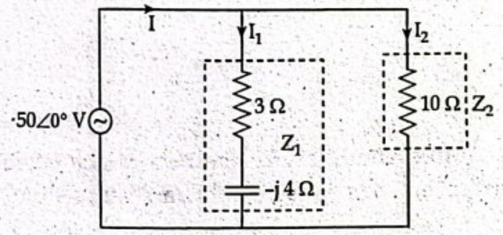
$\therefore$ Time period, $T = \frac{1}{f} = \frac{1}{500} = 0.025$

v) Impedance, $Z = \frac{V}{I} = \frac{141.4\angle 45°}{28.28\angle 15°} = 5\angle 60°$

and, resistance, $R = Z\cos\phi = 5\cos 60° = 5 \times 0.5 = 2.5\ \Omega$.

Example 28

For the parallel circuit shown below, find the branch currents and total current.



Solution:

$$Z_{eq} = Z_1 \parallel Z_2 = \frac{Z_1 \times Z_2}{Z_1 + Z_2}$$

$$= \frac{(3 - j4) \times 10}{(3 - j4) + 10}$$

$$= \frac{30 - j40}{13 - j4}$$

$$= \frac{(10 - j40)(13 + j4)}{13^2 + 42}$$

$$= \frac{550 - j400}{185}$$

$$= 2.973 - j2.162\ \Omega$$

or, $Z_{eq} = \sqrt{(2.973)^2 + (2.162)^2}\angle \tan^{-1}\left(\frac{-2.162}{2.973}\right)$

$\therefore$ $Z_{eq} = 3.676\angle -36.03°\ \Omega$

Now,

i) Current, $I = \frac{50\angle 0°}{3.676\angle -36.03°} = 13.60\angle 36.03°$ A

ii) Current, $I_1 = \frac{Z_2}{Z_2 + Z_1} \times I$

$$= 13.6\angle 36.03° \times \frac{10\angle 0°}{13 - j4}$$

$$= \frac{136\angle 36.03°}{13.6\angle -17.10°}$$

$$= 10\angle -53.13°\ A$$

iii) $I_2 = \frac{Z_1 \times I}{Z_1 + Z_2} = \frac{(3 - j4) \times 13.6\angle 36.03°}{13.6\angle -13.10°} = \frac{5\angle -53.13° \times 13.6\angle 36.83°}{13.6\angle -17.10°}$

$= 5\angle 0°$ A

Example 29

A circuit consists of the following in parallel, a resistor of 500 Ω, an inductance of 2 H and a capacitor of 10 μF. In this circuit, a 200 V, 50 Hz source is applied. Calculate the total current drawn from the supply, complex power, active power, reactive power and p.f. of the circuit.

Solution:

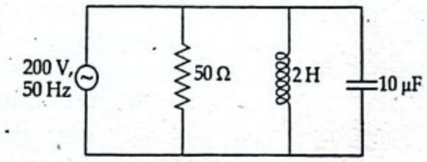
$R = 50\ \Omega$

$X_L = 2\pi \times 50 \times 2$

$= 628.3\ \Omega$

$X_C = \frac{1}{2\pi \times 50 \times 10^{-6}}$

$= 318.3\ \Omega$



$\therefore$ $X_L \parallel X_C = j628.3 \parallel -j318.3 = j\frac{628.3 \times (-j318.3)}{j628.3 - j318.3} = -645.122j$

$\therefore$ $Z_{total} = 50 \parallel (-645.122j) = \frac{50 \times -645.122j}{50 - 645.122j} = 49.70 - 3.85j$

$= 49.85\angle -4.31°$

i) $I_{total} = \frac{V}{Z_{total}} = \frac{200\angle 0°}{49.85\angle -4.31°} = 4.01\angle 4.31°$

ii) Complex power $= VI_{total}$

$= 200 \times 4.01\ (\cos 4.31° + j \sin 4.31°)$

$= (799.73 + j60.27)$ watt

iii) Active power = 799.73 W

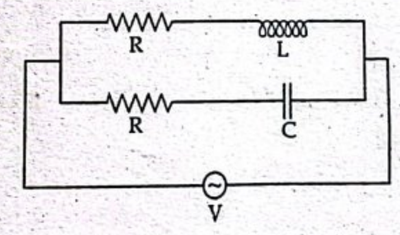
iv) Reactive power = 61.95 W

v) p.f. of circuit = $\cos\angle -4.31° = 0.997$.

Example 30

Prove that the impedance of a parallel AC circuit containing R and L in one branch and R and C in the other branch is equal to R when $R^2 = \frac{L}{C}$.

Solution:



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Now,

$$\text{Impedance, } Z = \frac{(R + j\omega L)(R - j/\omega C)}{2R + j\left(\omega L - \frac{1}{\omega C}\right)} = \frac{R^2 + L/C + j(R\omega L - R/\omega C)}{2R + j\left(\omega L - \frac{1}{\omega C}\right)}$$

Since, $\frac{L}{C} = R^2$

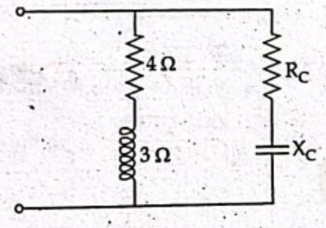
We have,

$$Z = \frac{R^2 + R^2 + jR(\omega L - 1/\omega C)}{2R + j\left(\omega L - \frac{1}{\omega C}\right)} = R \times \left[\frac{2R + j(\omega L - 1/\omega C)}{2R + j\left(\omega L - \frac{1}{\omega C}\right)}\right]$$

$\therefore \quad Z = R$

Example 31

For the network shown below, below, find the maximum value of R_C beyond which the circuit will not resonant.



Solution:

Net admittance of the circuit,

$$y = \frac{1}{4 + j3} + \frac{1}{R_C - jX_C}$$

Under resonance condition,

$$\text{or} \quad \frac{-3}{16 + 9} + \frac{X_C}{R_C^2 - X_C^2} = 0$$

$$\text{or,} \quad 3X_C^2 + 3R_C^2 = 25\,X_C$$

$$\text{or,} \quad 3X_C^2 - 25\,X_C + 3R_C^2 = 0$$

$$\text{or,} \quad X_C = \frac{25 \pm \sqrt{25^2 - 4 \times 3 \times 3\,R_C^2}}{3 \times 2}$$

If X_C is real, then $(4 \times 3 \times 3R_C^2)$ should be less than or equal to 25^2

$$\text{so,} \quad R_C = \sqrt{\frac{25^2}{4 \times 3 \times 3}}$$

$\therefore \quad R_C = 4.167\ \Omega.$

BOARD EXAM SOLVED PROBLEMS

1. **Write short notes on power factor and its significance.** **[13/F, 13/S, 14/F, 15/S, 16/F, 16/S, 17/F, 18/S, 19/S]**

Solution: See the definition part 3.7.

2. **Write short notes on quality factor of RLC series circuit.** **[14/F]**

Solution: See the definition part 3.11.

3. **Write short notes on quality factor and bandwidth.** **[14/S, 16/F]**

Solution: See the definition part 3.10 and 3.11.

4. **Show that "the form factor of a sinusoidal quantity is 1.11", with necessary derivation.** **[15/S]**

Solution: See the definition part 3.3. J, 3.3. k and 3.3. L.

5. **Write the difference between series resonance and parallel resonance. Show that the bandwidth for the series resonant circuit is the ratio of resonant frequency and quality factor.** **[17/F, 19/S]**

Solution:

SN	Series Resonance	Parallel resonance
i)	Impedance of the circuit is minimum.	Impedance of the circuit is maximum.
ii)	Current drawn is maximum.	Current drawn is minimum.
iii)	It is known as acceptor circuit.	It is known as rejecter circuit.
iv)	It magnifies voltage	It magnifies current.
v)	Effective impedance is R.	Effective impedance is L/RC
vi)	Admittance is maximum.	Admittance is minimum.
vii)	Also known as voltage resonance.	Also known as current resonance.

See the definition part 3.10

and, $\Delta f = \frac{R}{2\pi L}$

We have,

$$\text{Q-factor} = \frac{2\pi L\, f_0}{R}$$

so, $\Delta f = \frac{f_0}{Q}$

Hence, Bandwidth for the series resonant circuit is the ratio of resonant frequency and quality factor.

6. **What are the advantages of AC over DC?** **[17/F]**

Solution:

The advantages of AC over DC are;

i) The generation of AC is cheaper than DC.

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ii) AC can be easily be converted into DC with the help of rectifiers.
iii) AC machines are simple, robust and do not require much attention for their repairs and maintenance during their use.
iv) AC transmission and distribution is more economical as line material can be saved by transmitting power at higher voltages.
v) The AC generators have higher efficiency than DC.
vi) Wide range of voltages are obtained by the use of transformer.
vii) The loss of energy during transmission is negligible for AC.

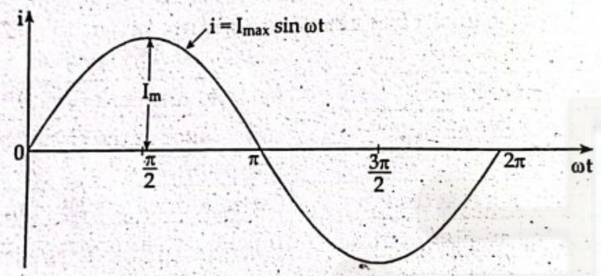
7. Explain resonance in parallel RLC circuit. **[16/S, 18/F]**

Solution: See the definition part 3.9.

8. Show that, "The form factor of a sinusoidal quantity is 1.11", with necessary derivation. **[15/S]**

Solution:

Alternative method:



The half average values is intended, since average value of sine wave over a complete cycle is zero.

Instantaneous value of sinusoidal current is,

$$I = I_{max} \sin \omega t \quad (1)$$

Consider first half cycle, when ωt varies from 0 to π,

$$I_{avg} = \frac{\text{Area of first half cycle}}{\pi} = \frac{1}{\pi}\int_0^{\pi} i\, d(\omega t)$$

$$= \frac{1}{\pi}\int_0^{\pi} I_{max} \cdot \sin \omega t\, d(\omega t) = \frac{I_{max}}{\pi}[-\cos \omega t]_0^{\pi}$$

or, $$I_{avg} = \frac{2}{\pi} I_{max}$$

$\therefore$ $$I_{avg} = 0.637\, I_{max}$$

and, $$I_{max}^2 = \frac{\text{Area of first half cycle of } i^2}{\pi}$$

$$= \frac{1}{\pi}\int_0^{\pi} i^2\, d(\omega t)$$

$$= \frac{1}{\pi}\int_0^{\pi} I_{max}^2 \cdot \sin^2 \omega t\, d(\omega t)$$

$$= \frac{I_{max}^2}{2\pi}\int_0^{\pi} (1 - \cos 2\omega t)\, d(\omega t)$$

$$= \frac{I_{max}^2}{2\pi}\left[\omega t - \frac{1}{2}\sin 2\omega t\right]_0^{\pi}$$

or, $$I_{rms} = \frac{I_{max}^2}{2\pi} \times \pi$$

or, $$I_{rms} = \sqrt{\frac{I_{max}^2}{2}}$$

$\therefore$ $$I_{rms} = \frac{I_{max}}{\sqrt{2}} = 0.707\, I_{max}$$

Thus, form factor $= \frac{I_{rms}}{I_{avg}} = \frac{V_{rms}}{V_{avg}} = \frac{I_{max} \times 0.707}{I_{max} \times 0.637}$

$\therefore$ Form factor = 1.11

9. Calculate the average values, rms (effective) valve and form factor of the output of half wave rectifier when the input to the rectifier is a purely sinusoidal alternating current. **[17/S, 19/S]**

Solution: See the solution of example No. 4.

10. What is a phasor? Determine the current- voltage relationship for passive elements in phasor domain as well as time domain. **[18/F]**

Solution:

A phasor is a complex number representing a sinusoidal function whose amplitude, angular frequency and initial phase are time invariant. Also, phasor is a complex number that represents magnitude and angle for a sine wave.

Current-voltage relationship for:

i) **Resistor**

If the voltage and current are represented in phasor form as,

$$v(t) = V e^{j\omega t}$$

$$i(t) = I e^{j\omega t}$$

Hence, $V e^{j\omega t} = RI e^{j\omega t}$

$\therefore$ $V = IR$

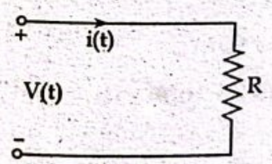


Fig: Time domain

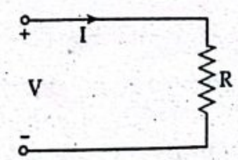


Fig: Phasor domain

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ii) Inductor:

The voltage current relationship for inductor is,

$$V(t) = L\frac{di(t)}{dt}$$

or, $V e^{j\omega t} = L \cdot \frac{d}{dt}[I e^{j\omega t}]$

$\therefore \quad V e^{j\omega t} = L \cdot (j\omega) I e^{j\omega t}$

Dividing by $e^{j\omega t}$ in both sides and rearranging,

or, $V = j\omega L \cdot I$

In the phasor or frequency domain, the inductor's phasor voltage is proportional to its phasor current.

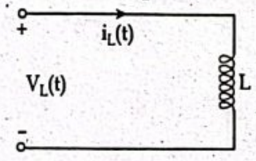


Fig: Time domain

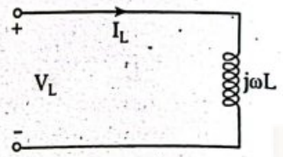


Fig: Frequency domain

iii) Capacitor:

The voltage current relationship for capacitor is, $i(t) = C \cdot \frac{dv(t)}{dt}$

or, $I e^{j\omega t} = C\frac{d[Ve^{j\omega t}]}{dt} = C(j\omega) Ve^{j\omega t}$ representation,

Dividing by $e^{j\omega t}$

or, $I = j\omega C \cdot V$

$\therefore \quad V = \frac{1}{j\omega C} \cdot I$

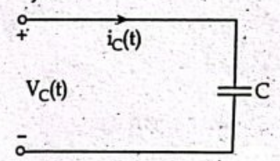


Fig: Time domain

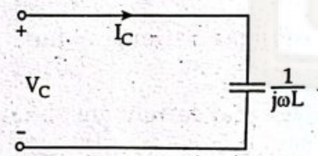


Fig: Frequency domain

The factor of $\frac{1}{j}$ in the voltage-current relationship introduces a 90° phase shift between inductor voltage and current. Since, $\frac{1}{j} = e^{-j90°} = 1\angle -90°$, the voltage across a capacitor lags the current by 90°.

11. Define resonance in electric circuit. Explain the variation of inductive reactance, capacitive reactance, impedance and current with frequency in RLC series circuit. Also, find the resonant frequency in RLC series circuit. [13/F]

Solution: See the definition part 3.9.

Figure below represents variation of resistance (R), inductive reactance (X_L), capacitive reactance (X_C), impedance (Z) and current (I).

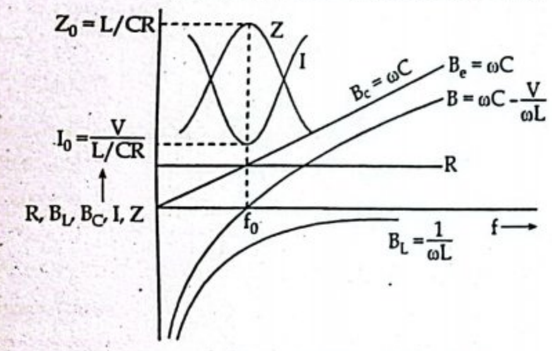
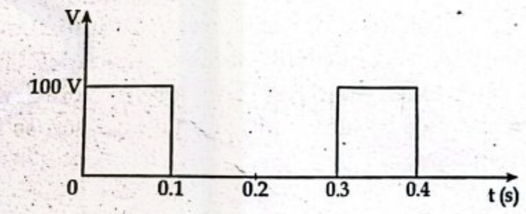


Fig: Variation of R, X_L, X_C, Z, I with frequency.

As R is independent of frequency so it is a straight line. Inductive reactance, X_L is directly proportional to the frequency so it is a straight line passing through origin. Capacitive reactance, X_C is inversely proportional to frequency and its characteristic is a rectangular hyperbola. At resonant frequency, Z is minimum and I is maximum.

12. Compute the average and effective values of the square voltage wave shown in figure below. [13/F]



Solution:

For $0 < t < 1$, *i.e.*, time interval 0 to 0.1 sec,

$V = 20$ V

For $0.1 < t < 0.3$

$V = 0$ V

Now,

$$V_{avg} = \frac{1}{T}\int_0^T V\,dt = \frac{1}{0.3}\int_0^{0.1} 20\,dt = \frac{1}{0.3} \times (20 \times 0.1)$$

$\therefore \quad V_{avg} = 6.67$ V

and, $V_{rms}^2 = \frac{1}{T}\int_0^T V^2\,dt = \frac{1}{0.3}\int_0^{0.1} 20^2\,dt = \frac{1}{0.3}(400 \times 0.1)$

or, $V_{rms}^2 = 133.3$ V

$\therefore \quad V_{rms} = 11.55$ V

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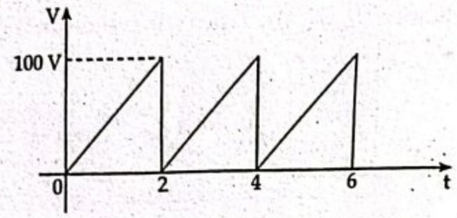
13. Calculate the mean value and effective value of the wave shown. [17/F]



Solution:

$$V_{avg} = \frac{1}{2}\int_0^2 \left(\frac{100}{2}\right) t\, dt = 25\left[\frac{t^2}{2}\right]_0^2 = 25\left[\frac{2^2}{2} - 0\right] = 50 \text{ V}$$

Alternatively,

$$V_{avg} = \left(\frac{0+V}{2}\right) = \left(\frac{0+100}{2}\right) = 50 \text{ V}$$

and, $$V_{rms}^2 = \frac{1}{2}\int_0^2 \left(\frac{100}{2}\right)^2 t^2\, dt = \frac{1}{2} \times \frac{100^2}{2^2}\left[\frac{t^3}{3}\right]_0^2 = \frac{1250}{3} \times [2^3 - 0^3]$$

or, $V_{rms}^2 = 3{,}333.33$

or, $V_{rms} = \sqrt{3{,}333.33}$

$\therefore$ $V_{rms} = 57.73$ V

14. A series R-L-C circuit has 10 Ω resistance, 0.01μ inductance and 100 μF capacitor. A 100 V supply with variable frequency is applied to this circuit and tunes until resonance occurs. Find the quality factor, current drawn by the circuit, and voltage across inductor at resonance condition. [13/S]

Solution:

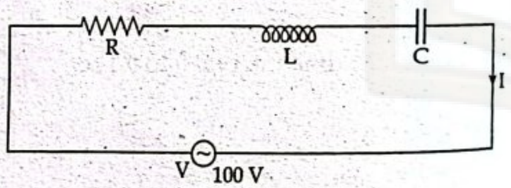
Given that;

$V = 100$ V

$R = 10\ \Omega$

$L = 0.01$ H

$C = 100\ \mu F = 100 \times 10^{-6}$ F



i) Resonant frequency, $$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.01 \times 100 \times 10^{-6}}} = 159.159 \text{ Hz}$$

ii) Current at resonant frequency, $$I_0 = \frac{V}{R} = \frac{100}{10} = 10 \text{ A}$$

iii) Q-factor, $$Q = \frac{2\pi f_0 L}{R} = \frac{2\pi \times 159.154 \times 0.01}{10} = 0.9999$$

iv) Bandwidth, $$(B.W) = \frac{f_0}{Q} = \frac{159.154}{0.9999} = 159.154 \text{ Hz}$$

v) Voltage across resistor, $V_R = I_0 \cdot R = 10 \times 10 = 100$ V

Voltage across inductor,

$$V_L = I_0 X_L = 10 \times 2\pi f_0 L = 10 \times 2\pi \times 159.154 \times 0.01 = 100 \text{ V}.$$

15. A RL series circuit has a resistor of 25 Ω and inductance of 40 mH is connected to 230 V, 50 Hz supply, calculate. [14/S, 15/S]

i) Current
ii) Power factor
iii) Power consumed
iv) Impedance
v) Input current
vi) Voltage across resistor and inductor

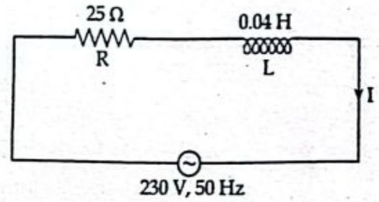
Solution:

$R = 25\ \Omega$

$L = 40 \text{ mH} = 0.04$ H

$V = 230$ V

$F = 50$ Hz



i) Reactance of coil,

$$X_L = 2\pi fL = 2\pi \times 50 \times 0.04 = 12.57\ \Omega$$

ii) Impedance of circuit, $$Z = \sqrt{R^2 + X_L^2} = \sqrt{25^2 + 12.57^2} = 27.98\ \Omega$$

iii) Current in the circuit, $$I = \frac{V}{Z} = \frac{230}{27.98} = 8.22 \text{ A}$$

iv) Voltage across resistance,

$$V_R = IR = 8.22 \times 25 = 205.50 \text{ V}$$

v) Power factor, $$\cos\phi = \frac{R}{Z} = \frac{25}{27.98} = 0.893$$

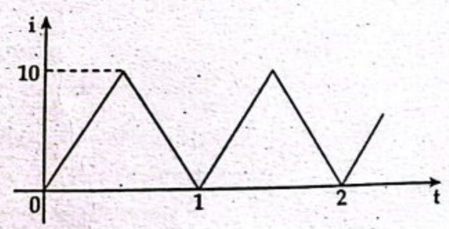
vi) Voltage across inductor,

$$V_L = IX_L = 8.22 \times 12.57 = 103.32 \text{ V}.$$

vii) Power consumed,

$$P = VI \cos\phi = 230 \times 8.22 \times 0.893 = 1{,}688.30 \text{ W}.$$

16. Find the average, rms value and form factor of the given triangular wave form. [14/F]



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Solution:

$$I_{avg} = \left[\frac{\text{Area under the wave from 0 to T}}{T}\right]$$

$$= \left(\frac{1}{2}\right) \times \frac{\text{Base} \times \text{Height}}{T}$$

$$= \frac{1}{2} \times 1 \times \frac{10}{1}$$

$\therefore \quad I_{avg} = 5 \text{ A}$

and, $I_{rms}^2 = \frac{1}{3} \times (\text{Altitude})^2 \times \text{Base} = \frac{1}{3} \times 10^2 \times 1$

or, $I_{rms}^2 = 33.33$

or, $I_{rms} = \sqrt{33.33}$

$\therefore \quad I_{rms} = 5.773 \text{ A}$

and, Form factor $= \frac{I_{rms}}{I_{avg}} = \frac{5.773}{5} = 1.154.$

17. A coil of resistance 40 Ω and inductive reactance of 25 Ω is connected in series to 220 V, 50 Hz supply. Calculate. [13/F]

i) Active and reactive components

ii) Total power of the circuit.

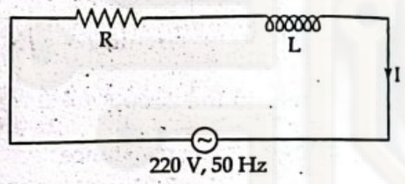
Solution:

$V = 220 \text{ V}$

$F = 50 \text{ Hz}$

$R = 40 \Omega$

$X_L = 25 \Omega$



i) Impedance, $Z = \sqrt{R^2 + X_L^2} = \sqrt{40^2 + 25^2} = 47.1699 \ \Omega$

ii) Current, $I = \frac{V}{Z} = \frac{220}{47.169} = 4.663 \text{ A}$

iii) Total power of circuit (P) $= VI \cos\phi = 220 \times 4.663 \times \left(\frac{40}{47.1699}\right)$

$\left(\because \cos\phi = \frac{R}{Z} = \frac{40}{47.1699} = 0.8479\right)$

$\sin\phi = \sqrt{1 - \cos^2\phi} = \sqrt{1 - (0.8479)^2} = 0.53$

iv) $Z = \sqrt{R^2 + X_L^2} \angle \tan^{-1}\left(\frac{X_L}{R}\right)$

$= \sqrt{40^2 + 25^2} \angle \tan^{-1}\left(\frac{25}{40}\right)$

$= 47.1699 \angle +32°$

Thus,

$$I = \frac{V}{Z} = \frac{220 \angle 0°}{47.1699 \angle 32°} = 4.663 \angle 32°$$

Hence, active component of current,

$I_a = I \cos\phi = 4.663 \angle -32° \times 0.8479 = 3.953 \angle -32° \text{ A}$

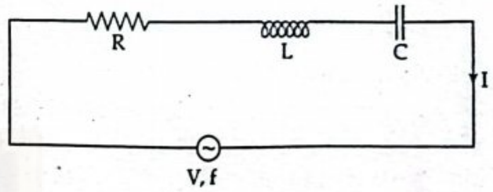
Reactive component of current,

$Ir = I \sin\phi = 4.663 \angle -32° \times 0.53 = 2.471 \angle -32° \text{ A}$

$\therefore$ Total current, $I = \sqrt{I_a^2 + I_r^2} = \sqrt{3.953^2 + 2.471^2} = 4.663 \text{ A}.$

18. In an R-L-C series circuit, current supplied by the single phase AC source is 15 ∠–38° A. Determine the value of all three kinds of power if R = 100 ohm, X_L = 35 ohm and X_C = 25 ohm. If inductance is 8H, find the resonant frequency. [14/F]

Solution:



$R = 100 \ \Omega$

$X_L = 35 \ \Omega$

$X_C = 25 \ \Omega$

$I = 15 \angle -38° \text{ A}$

We have,

$Z = \sqrt{R^2 + (X_L - X_C)^2} \angle \tan^{-1}((X_L - X_C)/R)$

$= \sqrt{100^2 + (35 - 25)^2} \angle \tan^{-1}(10/100)$

$\therefore \quad Z = 100.4988 \angle 5.710 \ \Omega$

Now,

$$I = \frac{V}{Z}$$

or, $V = Z.I = 100.4988 \angle 5.710° \times 15 \angle -38°$

$\therefore \quad V = 1{,}507.48 \angle -32.29° \text{ V}$

and, Power factor, $\cos\phi = \cos(-38°) = 0.788$

so, $\sin\phi = \sqrt{1 - 0.788^2} = 0.616$

Now,

i) Active power

$P = IV \cos\phi = 15 \times 1{,}507.48 \times 0.788 = 17{,}818.41 \text{ W} = 17.818 \text{ KW}$

ii) Apparent power

$S = IV = 15 \times 1507.48 = 22.612 \text{ KW}$

iii) Reactive power

$Q = IV \sin\phi = 15 \times 1{,}507.48 \times 0.616 = 13.929 \text{ KW}.$

19. Two impedances (60 + j15)Ω and (30 – j10)Ω are connected in series across a 220 V, 50 Hz supply. Find current, active power, reactive power, apparent power and power factor of the circuit. [15/F, 16/S]

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Solution:

$Z_1 = 60 + j15$

$Z_2 = 30 - j10$

Now,

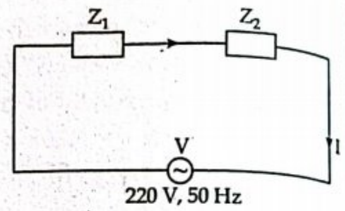
$Z_1 + Z_2 = 60 + j\,15 + 30 - j\,10$

$= 90 - 5j$

$= 90.138 \angle -3.179°$

$V = 220$ V

$F = 50$ Hz



i) Hence, current, $I = \frac{V}{Z} = \frac{220 \angle 0°}{90.138 \angle -3.179° \text{ (capacitive)}}$

$= 2.45 \angle 3.179°$ A

Power factor, $\cos\phi = \cos(3.179°) = 2.45 \angle 3.179°$ A

and, $\sin\phi = \sqrt{1 - 0.9984^2} = 0.0554$

ii) Active power, $P = VI\cos\phi$

$= 220 \times 2.45 \times 0.9984$

$= 538.137$ W

iii) Apparent power, $S = VI$

$= 220 \times 2.45$

$= 539$ VA

iv) Reactive power, $Q = VI\sin\phi$

$= 220 \times 2.45 \times 0.0554$

$= 29.86$ VAr.

20. A series circuit consists of a non-inductive resistance of 5 Ω and inductive reactance of 10 Ω. When connected to a single phase AC supply, it draws a current $i(t) = 27.89 \sin(628.3t - 45°)$ A. Find;

i) The voltage applied to the series circuit in the form $V(t) = Vm \sin(\omega t + 0°)$

ii) The inductance

iii) Power drawn by circuit [16/F]

Solution:

$R = 5\ \Omega$

$X_L = 10\ \Omega$

Given that;

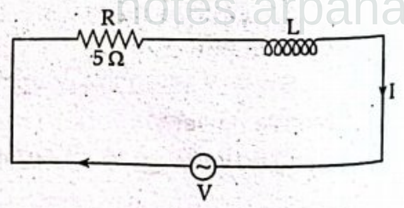
$i(t) = 27.89 \sin(628.3\,t - 45°)$

Comparing with $i(t) = I_m \sin(\omega t - \phi)$

Hence,

$I_m = 27.89 \angle 45°$ A

or, $\omega = 628.3$



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or, $2\pi f = 628.3$

$\therefore f = 100$ Hz

Also,

$X_L = 2\pi fL$

or, $10 = 2\pi \times 100 \times L$

$\therefore L = 0.01592$ H

Now,

Impedance, $Z = \sqrt{R^2 + X_L^2}$

$= \sqrt{5^2 + 10^2} \angle \tan^{-1}(10/5)$

$= 11.180 \angle 63.434°\ \Omega$

Current, $i = \frac{V}{Z}$

or, $V = iZ = 27.89 \angle -45° \times 11.180 \angle 63.434°$

$\therefore V = 311.810 \angle 18.434°$

Thus,

$V = 311.810 \sin(\omega t + 18.434°)$ V

i) $V = 311.810 \sin(628.3\,t + 18.434°)$ V

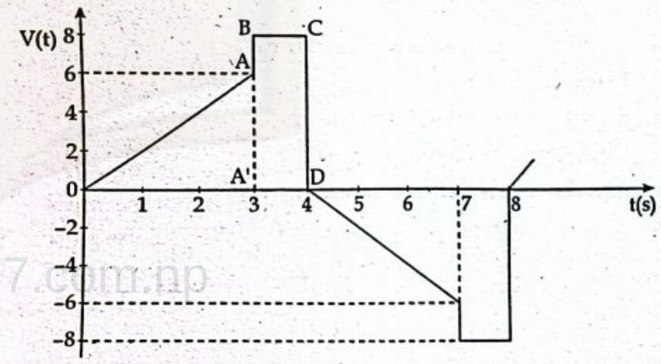
ii) Power drawn (P) $= VI\cos\phi$

$= 311.810 \times 27.89 \times \cos\left(\frac{5}{11.180}\right)$

$= 8{,}696.115$ W

iii) Inductance, $L = 0.01592$ henry.

21. A voltage wave has the variations as shown below,



Find the average and effective values of the voltage. If voltage is applied to a 50 Ω resistors, calculate power dissipated in watts. [16/F]

Solution:

Period = 8 sec, since the wave is symmetrical, so the average value can be calculated over one half cycle.

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Thus,

i) $V_{avg} = \frac{\text{Area under one - Half cycle}}{\text{Half-period}}$

$= \frac{\text{Area of (Triangle OAA' + Rectangle A'BCD)}}{\text{Half-period}}$

$= \frac{(1/2) \times 6 \times 3 + 8 \times (4-3)}{4}$

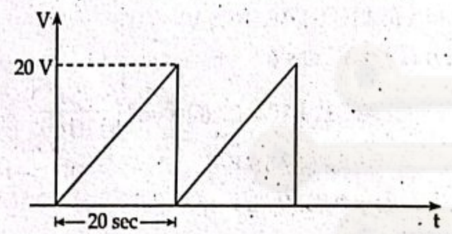
and, $V_{rms} = \sqrt{\frac{(1/3) \times 6^2 \times 3 + 8^2 \times 1}{4}}$

$= \sqrt{\frac{100}{4}} = 5 \text{ V}$

ii) Power dissipated,

$= I^2R = \frac{V_{rms}^2}{R} = \frac{5^2}{5} = 5$ watt.

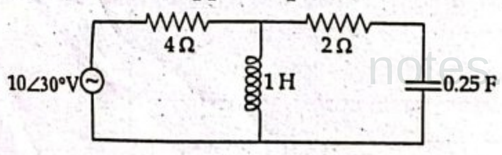
22. Find the average value, rms value and form factor for a wave given below. [16/S]



Solution: See the solution of example No. 8.

23. For the circuit given below, determine. [17/F]

i) Circuit impedance
ii) Supply current and its phase angle
iii) Circuit power factor
iv) Active, reactive and apparent power

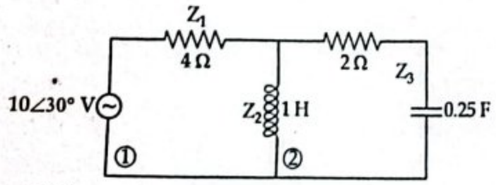


Solution:

$Z_1 = 4\ \Omega$

$Z_2 = 2\pi \times 50 \times 1 = 100\pi\ \Omega$

$Z_3 = \sqrt{2^2 + \left(\frac{1}{2\pi \times 50 \times 0.25}\right)^2} = 2\ \Omega$



Current in branch-1:

$I_1 = \frac{V}{Z_2} = \frac{10\ \angle 30°}{100\ \angle 90°} = 0.1\ \angle -60°$

Current in branch-2,

$I_2 = \frac{V}{Z_3} = \frac{10\ \angle 30°}{2\ \angle -0.3647} = 5\ \angle 30.3647°$

i) Total impedance, $Z = Z_2 \parallel Z_3 + Z_1 = Z_1 + \frac{Z_2 \times Z_3}{Z_2 + Z_3}$

$= 4 + \frac{100\pi \times 2}{2 + 100\pi}$

$= 5.9873\ \angle 0.00002°\ \Omega$

ii) Supply current, $I = \frac{V}{Z} = \frac{10\ \angle 30°}{5.9873\ \angle 0.00002°} = 1.6702\ \angle 30°$ A

Phase angle = 30°

iii) Circuit power factor, p.f. = $\cos\phi = \frac{Z_1}{Z} = \frac{4}{5.9873} = 0.6680$

iv) Active power = $VI \cos\phi = 10 \times 1.6702 \times 0.6680 = 11.158$ W

∴ Apparent power = $VI = 10 \times 1.6702 = 16.702$ VA

and, $\sin\phi = \sqrt{1 - 0.6680^2} = 0.7441$

∴ Reactive power = $VI \sin\phi = 10 \times 1.6702 \times 0.7441 = 12.427$ VAr.

24. A series RLC circuit consists of a 100 Ω resistor, an inductor of 0.318 H and a capacitor of unknown value. When this is energized by $V = 230\sqrt{2} \sin \omega t$ volts supply, the current was found to be $i = 2.3\sqrt{2} \sin \omega t$ amperes. Find, [17/S]

i) Value of capacitance
ii) Voltage across inductor
iii) Total power consumed

(Assume, ω = 314.15 rad/sec).

Solution:

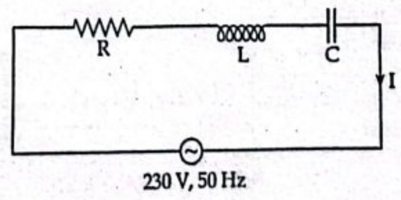
$R = 100\ \Omega$

$L = 0.318$ H

$C = ?$

$V = 230\sqrt{2} \sin \omega t$ V

$I = 2.3\sqrt{2} \sin \omega t$ A



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$$\therefore \quad V_{rms} = \frac{230\sqrt{2}}{\sqrt{2}} = 230 \text{ V}$$

$$\therefore \quad I_{rms} = \frac{2.3\sqrt{2}}{\sqrt{2}} = 2.3 \text{ A}$$

We know,

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Here;

$X_C = ?$

$X_L = 2\pi fL = \omega L = 314.15 \times 0.318 = 99.899\ \Omega \approx 100\ \Omega$

so, $Z = \sqrt{100^2 + (100 - X_C)^2}$

or, $\frac{V}{I} = \sqrt{100^2 + (100 - X_C)^2}$

or, $\frac{230}{2.3} = \sqrt{100^2 + (100 - X_C)^2}$

or, $100 = 100^2 + (100 - X_C)^2$

or, $0 = (100 - X_C)^2$

$\therefore \quad X_C = 100\ \Omega$

Hence,

$$X_C = \frac{1}{2\pi fC} = \frac{1}{\omega C}$$

or, $100 = \frac{1}{314.15\, X_C}$

i) $C = 314.15 \times 10^{-5}$ F

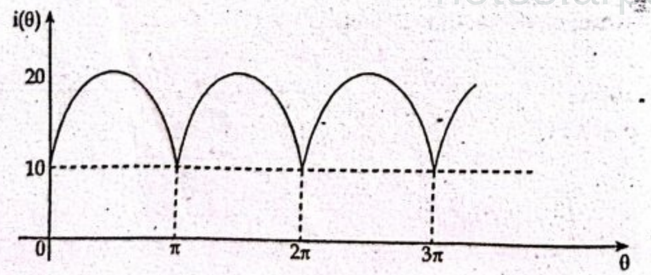
Hence, value of capacitance is 3.183×10^{-5} F

ii) Voltage across inductor, $V_L = IX_L = 2.3 \times 100 = 230$ V

iii) Total power consumed, $P = VI \cos\phi = 230 \times 2.3 \times \left(\frac{100}{100}\right) = 529$ watt.

25. Find the average and rms value of the given wave form. Also, find the form factor. [17/F]

Solution:



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Time period, $T = \pi$

For one complete cycle, $i(\theta) = 10 + 10\sin\theta$

so, $I_{avg} = \frac{1}{T}\int_0^T i(\theta)\, d\theta$

$= \frac{1}{\pi}\int_0^{\pi} (10 + 10\sin\theta)\, d\theta$

$= \frac{1}{\pi}[10\theta + 10(-\cos\theta)]_0^{\pi}$

$= \frac{10}{\pi}[\pi - (\cos\pi - \cos 0)]$ [∵ Calculate in radian value]

$= \frac{10}{\pi}[\pi - (-1 - 1)]$

$= \frac{10}{\pi}[\pi + 2]$

$\therefore \quad I_{avg} = 16.366$ A

and, $I_{rms}^2 = \frac{1}{T}\int_0^T i^2(\theta)\, d\theta$

$= \frac{1}{\pi}\int_0^{\pi} (10 + 10\sin\theta)^2\, d\theta$

$= \frac{10^2}{\pi}\left[\int_0^{\pi} (1 + \sin\theta)^2\, d\theta\right]$

$= \frac{100}{\pi}\left[\int_0^{\pi} (1 \cdot d\theta + 2\sin\theta\, d\theta + \sin^2\theta\, d\theta)\right]$

$= \frac{100}{\pi}\left[[\theta]_0^{\pi} + [2(-\cos\theta)]_0^{\pi} + \int_0^{\pi} \frac{(1 - \cos 2\theta)}{2} d\theta\right]$

$= \frac{100}{\pi}\left[(\pi - 0) - 2(\cos\pi - \cos 0) + \frac{1}{2}\int_0^{\pi} d\theta - \frac{1}{2}\int_0^{\pi} \cos 2\theta\, d\theta\right]$

$= \frac{100}{\pi}\left[\pi - 2 \times (-2) + \frac{1}{2} \times (\pi - 0) - \frac{1}{2}\left[\frac{\sin 2\theta}{2}\right]_0^{\pi}\right]$

$= \frac{100}{\pi}\left[\pi + 4 + \frac{1}{2}\pi - \frac{1}{4}(\sin 2\pi - \sin 0)\right]$

$= \frac{100}{\pi}\left[\pi + 4 + \frac{1}{2}\pi - \frac{1}{4} \times 0\right]$

or, $I_{rms}^2 = \frac{100}{\pi} \times 8.71238 = 277.323$

$\therefore \quad I_{rms} = 16.653$ A

Hence,

$\therefore$ Form factor $= \frac{I_{rms}}{I_{avg}} = \frac{16.653}{16.366} = 1.0775.$

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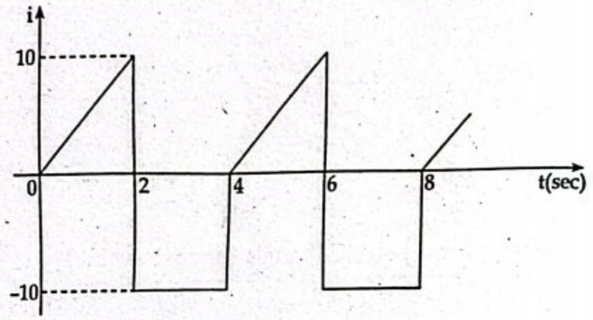
26. Determine the average value, rms value and form factor of the given triangular waveform. [18/S]



Solution:

Time period = 4 sec

Here'

$$i(t) = \begin{cases} 5t \text{ for } 0 \le t \le 2 \\ -10 \text{ for } 2 \le t \le 4 \end{cases}$$

Now,

$$I_{avg} = \frac{1}{T}\int_0^T i\,dt$$

$$= \frac{1}{4}\left[\int_0^2 (5t)\,dt + \int_2^4 (-10)\,dt\right]$$

$$= \frac{1}{4}\left[\left[5\frac{t^2}{2}\right]_0^2 - 10\,[t]_2^4\right]$$

$$= \frac{1}{4}[2.5\,(2^2 - 0) - 10\,(4 - 2)]$$

$$= \frac{1}{4}[10 - 10 \times 2]$$

$$= -2.5 \text{ A}$$

and, $I_{rms}^2 = \frac{1}{T}\int_0^T i^2\,dt$

$$= \frac{1}{4}\left[\int_0^2 (5t)^2\,dt + \int_2^4 (-10)^2\,dt\right]$$

$$= \frac{1}{4}\left[25 \times \left[\frac{t^3}{3}\right]_0^2 + 100 \times [t]_2^4\right]$$

$$= \frac{1}{4}\left[\frac{25}{3} \times 2^3 + 100 \times (4 - 2)\right]$$

or, $I_{rms}^2 = 66.667$

$\therefore$ $I_{rms} = 8.165$ A

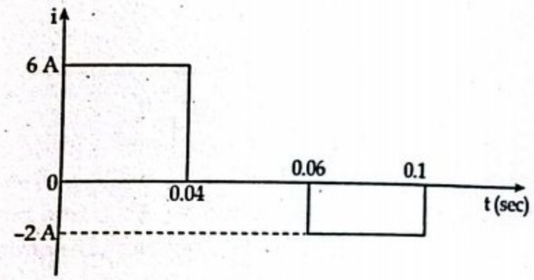
and, Form factor $= \frac{I_{rms}}{I_{avg}} = \frac{8.165}{-2.5} = -3.266.$

27. A varying current with a periodic wave form as shown in figure below flows through an 8 Ω resistor. Determine. [13/S]

i) average value

ii) rms value

iii) heat dissipated in 5 minute



Solution:

Time period (T) = 0.1 sec

Here;

$$i(t) = \begin{cases} 6 \text{ A for } 0 \le t \le 0.04 \\ 0 \text{ A for } 0.04 \le t \le 0.06 \\ -2 \text{ A for } 0.06 \le t \le 0.1 \end{cases}$$

Now,

$$I_{avg} = \frac{1}{T}\int_0^t i(t)\,dt$$

$$= \frac{1}{0.1}\left[\int_0^{0.04} 6\,dt + \int_{0.04}^{0.06} 0\,dt - 2\int_{0.04}^{0.1} dt\right]$$

$$= 10\,[6\,(0.04 - 0) + 0 - 2\,(0.1 - 0.06)]$$

$$= 10 \times 0.16$$

i) $I_{avg} = 1.60$ A

and, $I_{rms}^2 = \frac{1}{T}\int_0^T i^2\,dt$

$$= \frac{1}{0.1}\left[\int_0^{0.04} (6)^2\,dt + \int_{0.04}^{0.06} 0^2\,dt + \int_{0.06}^{0.1} (-2)^2\,dt\right]$$

$$= 10\left[(6^2\,[t]_0^{0.04} + 0 + 4\,[t]_{0.06}^{0.1}\right]$$

$$= 10\,[36 \times 0.04 + 0 + 4 \times (0.1 - 0.06)]$$

$$= 10 \times 1.60$$

$$= 1.60$$

or, $I_{rms} = \sqrt{1.60}$

ii) $I_{rms} = 1.264$ A

Then, form factor $= K_f = \frac{I_{rms}}{I_{avg}} = \frac{1.264}{1.60} = 0.79$

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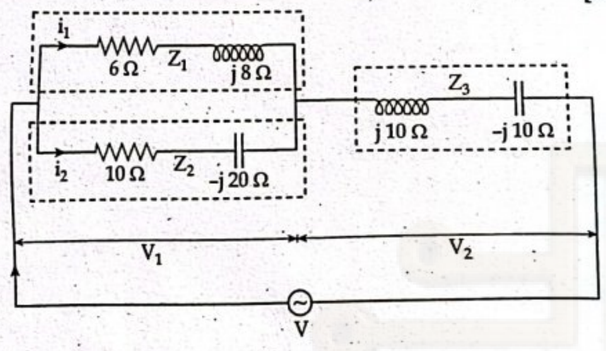
iii) Heat dissipated in 5 minute,

$$H = I^2 Rt = (1.264)^2 \times 8 \times (5 \times 60) = 3{,}834.47 \text{ Joule.}$$

NOTE:

Heat produced/dissipated due to flow of current = I^2RT
where,
I = rms value of current in Ampere
t = Time in seconds
R = Resistance in ohms.

28. In the circuit given below, total current, i = (20 + j0) A. calculate branch current and i_1 and i_2 and V_1 and V_2. Also, find power factor of entire circuit and active and reactive power in the entire circuit. [14/F, 18/S]



Solution:

$i = (20 + j0)A = 20 \angle 0°A$

$Z_1 = R + jX_L = (6 + j8)\Omega$

$Z_2 = R - jX_C = (10 - j10)\Omega$

$Z_3 = jX_L - jX_C = (j10 - j10Q) = 0\Omega$

i) $i_1 = i \times \dfrac{Z_2}{Z_1 + Z_2}$

$= (20 + j\,0) \times \dfrac{(10 - j\,20)}{(6 + j8) + (10 - j20)}$

$= \dfrac{200 - 400j}{16 - 12j}$

$= (20 - j10)\ A$

$= 22.36 \angle -26.56°A$

$i_2 = i \times \dfrac{Z_1}{Z_1 + Z_2}$

$= (20 + j0) \times \dfrac{(6 + j\,8)}{(6 + j8) + (10 - j20)}$

$= \dfrac{120 + 160j}{16 - 12j}$

$= j10\ A$

$= 10 \angle 90°\ A$

ii) $Z_{eq} = Z_1 \parallel Z_2$

$= \dfrac{Z_1 \times Z_2}{Z_1 + Z_2}$

$= \dfrac{(6 + j8) \times (10 - j20)}{(6 + j\,8) + (10 - j20)}$

$= 10 + j15$

$= 11.18 \angle 26.56°$

and, $Z = Zeq + Z_3$

$= (10 + j15) + (j10 - j10)$

$= 10 + j15\Omega$

Hence,

$V = i \cdot Z$

$= 20 \angle 0° \times 11.18 \angle 26.56°$

$= 223.6 \angle 26.56°$

and, $V_1 = i_1 Z_1 = i_2 Z_2$ [$\because Z_1$ and Z_2 are in parallel]

$= (20 - j10) \times (6 + j8)$

$= 200 + j100$

$= 223.6 \angle 26.56°$

and, $V_2 = iZ_3$

$= (20 + j\,0) \times 0$

$= 0\ V$

iii) Power factor, p.f. $= \dfrac{R}{Z} = \dfrac{10}{11.10} = 0.89$

iv) Active power $= VI \cos\phi = 223.6 \times 20 \times 0.89 = 3{,}980\ W$

$\therefore$ Reactive power $= VI \sin\phi$

$= 223.6 \times 20 \times \sin(26.56)$

$= 2{,}000\ VAr$

$\therefore$ Apparent power $= VI = 223.6 \times 20 = 4{,}472\ VA.$

29. Show that the bandwidth of RLC resonance AC series circuit is $\omega = R/L$ rad/sec, where symbols have their usual meanings. [18/S]

Solution: See the definition part 3.10 and derive upto

$$\Delta f = \frac{R}{2\pi L}$$

But, $\omega = 2\pi\,\Delta f$

so, $\omega = \dfrac{R}{L}$ rad/sec.

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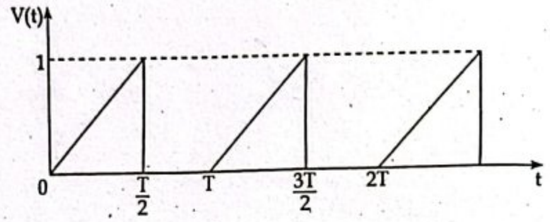
30. Find the average and rms voltage of the wave form shown in figure. [14/S]



Solution:

Here;

$$V(t) = \begin{cases} 0 \text{ for } T/2 \text{ to } T \\ \frac{1}{2} \text{ for } 0 \text{ to } T/2 \end{cases}$$

Therefore, Average voltage,

$V_{avg} = 0.5\ V$

and, $V_{rms}^2 = \dfrac{\frac{1}{2} \times 1 \times \frac{T}{2} + 0}{T}$

$V_{rms} = \sqrt{\frac{1}{4}}$

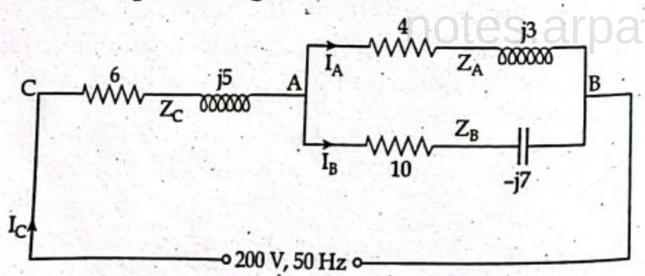
$\therefore \quad V_{rms} = 0.5\ V$

and, $V_{avg} = \dfrac{\frac{1}{2} \times 1 \times \frac{T}{2} + 0}{T} = \dfrac{\frac{T}{4}}{T}$

$\therefore \quad V_{avg} = \frac{1}{4} V = 0.25\ V$

31. In a series-parallel circuit, two impedance are connected with series impedance as shown in figure below. If the voltage applied to the circuit is 200 V, 50 Hz. Calculate [15/S, 17/S]

i) Current I_A, I_B and I_C
ii) Voltage across AB
iii) Active and reactive power of the entire circuit.
iv) Draw the phasor diagram.



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Solution:

$V = 200 \angle 0°\ V$

$f = 50\ Hz$

$Z_C = R + jX_L = 6 + j5\ \Omega$

$Z_A = 4 + j3\Omega$

$Z_B = 10 - j7\ \Omega$

so, $Z_A \parallel Z_B = \dfrac{Z_A \times Z_B}{Z_A + Z_B} = \dfrac{(4 + j3) \times (10 - j7)}{4 + j3 + 10 - j7} = \dfrac{25 + j30}{14 - j4} = 1.0849 + 2.452j$

$\therefore \quad Z_{AB} = 2.682 \angle 66.13°$

Now,

Total impedance of circuit, $Z = Z_C + Z_A \parallel Z_B$

$= 6 + j5 + 1.0849 + 2.452j$

$= 7.0849 + 7.452j\ \Omega$

$\therefore \quad Z = 10.282 \angle 46.44\ \Omega$

i) $I_C = \dfrac{V}{Z} = \dfrac{200 \angle 0°}{10.282 \angle 46.44°} = 19.451 \angle -46.44°\ A$

and, Voltage across parallel branch AB,

$V_{AB} = IZ_{AB}$

$= 19.451 \angle -46.44° \times 2.682 \angle 66.13°$

$= 49.117 + 17,576j$

$\therefore \quad V_{AB} = 52.167 \angle 19.69°\ V$

Current flowing through Z_A is I_A,

$I_A = \dfrac{V_{AB}}{Z_A} = \dfrac{52.167 \angle 19.69°}{4 + j3}$ or $49.117 + 17.576j$

$= 9.967 - 3.081j$

$= 10.434 \angle -17.18°\ A$

and, $I_B = \dfrac{V_{AB}}{Z_B} = \dfrac{49.117 + 17.576\,j}{10 - j7} = 2.470 + 3.487j = 4.273 \angle 54.68°$

and, $I_C = 19.451 \angle -46.44°\ A$

ii) Voltage across AB, $V_{AB} = 52.167 \angle 19.69°V$

iii) Active power, $P = I_c^2 \times R = (19.451)^2 \times 7.0849$

$\therefore \quad P = 2,680.52$ watt.

Reactive power, $Q = I_c^2 \times X = (19.451)^2 \times 7.452 = 2,819.40$ VAR

Apparent power, $S = \sqrt{P^2 + Q^2}$

$= \sqrt{(2,680.52)^2 + (2,819.40)^2}$

$= 3,890.27$ VA

and, Power factor, p.f. $= \dfrac{P}{S} = \dfrac{2,680.52}{3,890.27} = 0.6890$

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Alternatively,

Power factor = cos (– 46.44) = 0.6890

iv)

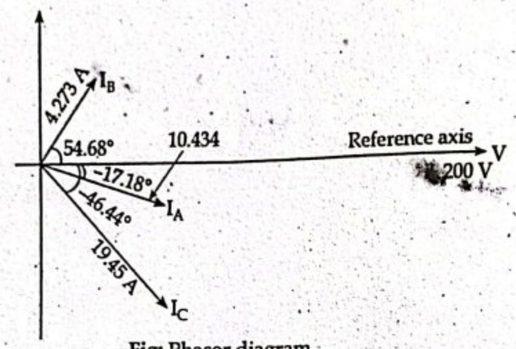


Fig: Phasor diagram

32. In a series parallel circuit, two impedances are in parallel with series impedance as shown in figure below. The voltage across impedance Z_C is (200 + j0) V. Calculate

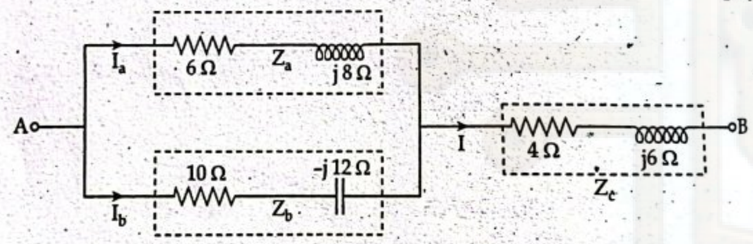
i) Each branch currents, both magnitude and phase

ii) Draw phasor diagram

iii) Active and reactive power of entire circuit

iv) Supply voltage across AB and power factor of whole circuit.

[15/F]



Solution:

$Z_a = (6 + j8)\ \Omega$

$Z_b = (10 - j12)\ \Omega$

$Z_c = (4 + j6)\ \Omega$

$V_c = (200 + j0)$ V

Now,

i) $I = \frac{V_c}{Z_c} = \frac{(200 + j0)}{4 + j6} = 15.38 - j23.07 = 27.73 \angle -56.30°$ A

so, $I_a = \frac{I \times Z_b}{Z_a + Z_b}$

$= \frac{(27.73 \angle 56.30°)(10 - j12)}{(6 + j8) + (10 - j12)}$

$= -1.13 - j\,26.244$

$= 26.26 \angle 92.45°$ A

and, $I_b = \frac{I \times Z_a}{Z_a + Z_b}$

$= \frac{(15.38 - j23.07) \times (6 + j8)}{(6 + j8 + 10 - j12)}$

$= 16.51 + j3.166$

$= 16.81 \angle 10.86°$ A

Thus,

ii) $V_{AB} = I_b \times Z_b + V_c$

$= 16.81 \angle 10.86° \times (10 - j\,12) + (200 + j\,0)$

$= 203.095 - j166.44 + 200$

$= 403 - j166.44$

$= 436.104 \angle -22.43°$ V

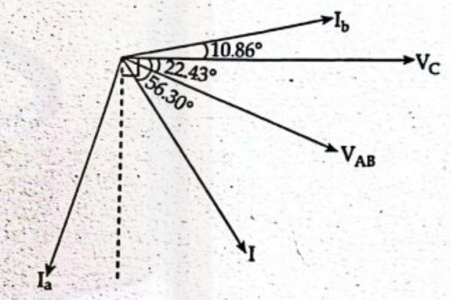
and, Power factor, p.f. = cos (56.30° – 22.430°) = 0.830

iii) Active power, P = VI cos ϕ = 436.104 × 27.73 × 0.830 = 10,037.28 W

Reactive power, Q = VI sin ϕ = 436.104 × 27.73 × 0.55 = 6,651.17 VAr

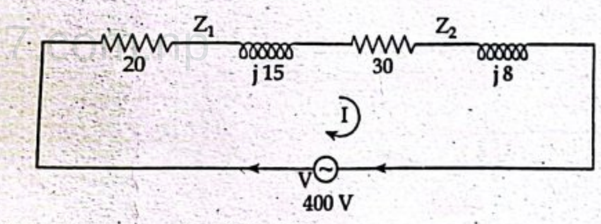
Apparent power, S = VI = 436.104 × 27.73 = 12,091 VA.

iv) Phasor diagram



33. Two impedances (20 + j5) Ω and (30 + j8) are connected in series across a 400 V, 60 Hz supply. Find current, active power, reactive power, apparent power and power factor of the whole circuit. [18/F]

Solution:



$Z_1 = (20 + j15)\ \Omega = 25 \angle 36.87°\ \Omega$

$Z_2 = (30 + j8)\ \Omega = 31 \angle 14.93°\ \Omega$

$V = 400$ V

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f = 60 Hz

$Z_t = Z_1 + Z_2 = 25 \angle 36.87° + 31 \angle 14.93° = 50 + 23j = 55 \angle 24.702° \Omega$

Now,

i) Current, $I = \frac{V}{Z} = \frac{400 \angle 0°}{55 \angle 24.702°} = 7.272 \angle -24.70° \text{ A}$

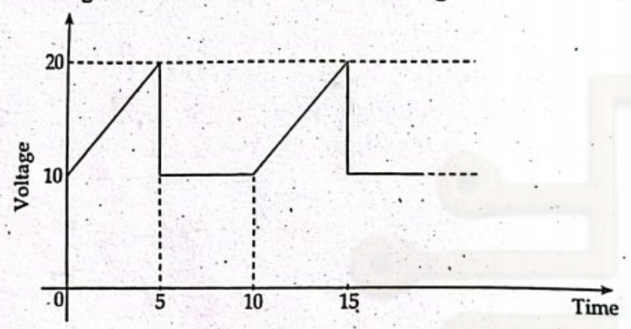
ii) Active power, $P = VI \cos\phi$
$= 400 \times 7.272 \times \cos(24.70°)$
$= 2{,}642.66 \text{ W}$

iii) Reactive power, $Q = VI \sin\phi$
$= 400 \times 7.272 \times \sin(24.70°)$
$= 1{,}215.50 \text{ VAr}$

iv) Power factor of circuit, p.f. = $\cos\phi = \cos(24.70°) = 0.9085$

v) Apparent power, $S = VI = 400 \times 7.272 = 2{,}908.80$ VA.

34. Find average, rms value and form factor of given wave form. [15/F]



Solution:

Time period, T = 10 sec

$$V_{avg} = \frac{\text{Area of rectangle} + \text{Area of triangle}}{T}$$

$$= \frac{10 \times 10 + 1/2 \times 5 \times (20 - 10)}{10}$$

$$= \frac{100 + 25}{10}$$

$$= 12.5 \text{ V}$$

and, $V_{rms} = \sqrt{\frac{10 \times 10^2 + 1/2 \times 5 \times (20^2 - 10^2)}{10}}$

$$= \sqrt{\frac{1{,}000 + 1/2 \times 5 \times 300}{10}}$$

$$= \sqrt{\frac{1{,}750}{10}}$$

$$= 13.228 \text{ V}$$

Thus, Form factor, $K_f = \frac{V_{rms}}{V_{av}} = \frac{13.228}{12.5} = 1.06.$

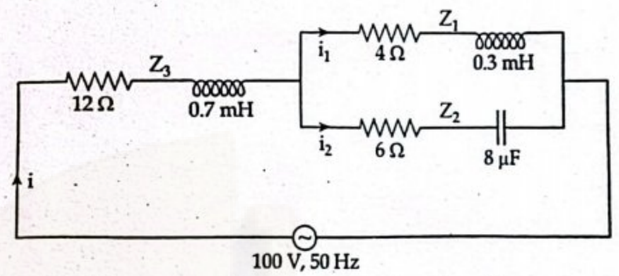
35. Show the peak factor of sine wave is equal to 1.41. [19/F]

Solution: See the solution of Q. No. 7.

We know,

Peak of crest factor, $K_a = \frac{I_m}{I_{rms}} = \frac{Im}{0.707\, I_m} = 1.41$

36. Determine the total impedance and current in each branch of circuit below. [19/F]



Solution:

Supply voltage,

$V = 100 \angle 0° \text{ V}$

Supply frequency,

f = 50 Hz

$Z_1 = R_1 + j\omega L_1$

$= 4 + j2\pi \times 50 \times 0.3 \times 10^{-3}$

$= (4 + j\,0.09424)\ \Omega$

$= 4 \angle 1.349° \ \Omega$

$Z_2 = R_2 - \frac{j}{\omega C_2}$

$= \frac{j}{2\pi \times 50 \times 8 \times 10^{-6}}$

$= (6 - j\,397.88)\ \Omega$

$= 397.93 \angle -89.13° \ \Omega$

$Z_3 = R_3 + j\omega L_3$

$= 12 + j2\pi \times 50 \times 0.7 \times 10^{-3}$

$= (12 + j\,0.2199)\ \Omega$

$= 12 \angle 1.0498° \ \Omega$

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Hence, total impedance,

$$Z = (Z_1 \parallel Z_2) + Z_3$$

$$= (4\angle 1.349°) \parallel (397.93\angle -89.13°) + 12\angle 1.0498°$$

$$= \frac{4\angle 1.349° \times 397.93\angle -89.13°}{4\angle 1.349° + 397.93\angle -89.13°} + 12\angle 1.0498°$$

$$= \frac{1{,}591.72\angle -87.78°}{397.916\angle -88.55°} + 12\angle 1.0498°$$

$$= 4\angle 0.774° + 12\angle 1.0498°$$

$$\therefore \quad Z = 16\angle 0.98°\ \Omega$$

Now, Current in circuit,

$$i = \frac{V}{Z} = \frac{100\angle 0°}{16\angle 0.98°}$$

$$i = i_3 = 6.25\angle -0.980° A$$

Current in branch-1,

$$i_1 = \frac{V_{1\text{-}2}}{Z_1} = i \times \frac{Z_{1\text{-}2}}{Z_1} = \frac{6.25\angle -0.98° \times 4\angle 0.774°}{4\angle 1.349°}$$

$$\therefore \quad i_1 = 6.250\angle -1.55°\ A$$

and current in branch-2

$$i_2 = \frac{V_{1\text{-}2}}{Z_2} = \frac{6.25\angle -0.980° \times 4\angle 0.774°}{397.93\angle -89.13°}$$

$$\therefore \quad i_2 = 0.0628\angle 88.924°\ A$$

and current in branch-3,

$$i_3 = i = 6.25\angle -0.98°\ A.$$

ADDITIONAL QUESTION SOLUTIONS

1. A 50 Hz alternating voltage of 150 V rms is applied independently to

i) Resistance of 10 Ω

ii) Inductance of 0.2 H

iii) Capacitance of 50 F

Find the expression for the instantaneous current in each case. Draw the phasor diagram in each case.

Solution:

i) Resistance of 10 Ω

$$V = V_m \sin \omega t$$

$$V_m = \sqrt{2}\, V_{rms} = \sqrt{2} \times 150 = 212.13\ V$$

$$I_m = \frac{V_m}{R} = \frac{212.13}{10} = 21.213\ A$$

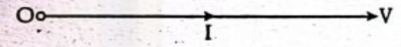
In pure resistive circuit, current is in phase with voltage.

i.e., $\phi = 0°$

$$i = I_m \sin \omega t = I_m \sin (2\pi ft)$$

$$\therefore \quad i = 21.213 \sin (100\ \pi t)\ A$$

The phasor diagram is shown below,



ii) L = 0.2 H

Inductive reactance,

$$X_L = \omega L = 2\pi fL = 2\pi \times 50 \times 0.2 = 62.83\ \Omega$$

$$\therefore \quad I_m = \frac{V_m}{X_L} = \frac{212.13}{62.83} = 3.37\ A$$

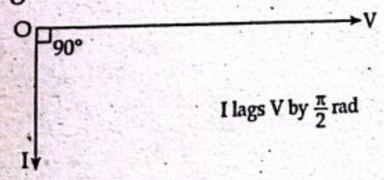
In pure inductive circuit, current lags voltage by 90°,

$$f = -90° = \frac{\pi}{2}\ \text{radian}$$

or, $i = I_m \sin (\omega t - \phi)$

$$\therefore \quad i = 3.37 \sin \left(100\ \pi t - \frac{\pi}{2}\right) A$$

The phasor diagram is shown below,



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iii) $C = 50\ \mu F$

Capacitive reactance, $X_C = \frac{1}{\omega C} = \frac{1}{2\pi fC}$

$$= \frac{1}{2\pi \times 50 \times 50 \times 10^{-6}}$$

$$= 63.66\ \Omega$$

$$I_m = \frac{V_m}{X_c} = \frac{212.13}{63.66} = 3.33\ A$$

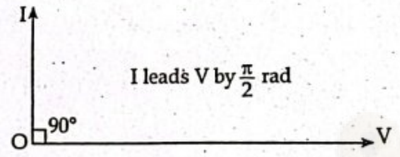
In pure capacitive circuit, current leads voltage by 90°,

$$\phi = 90° = \frac{\pi}{2}\ rad$$

or, $i = I_m \sin(\omega t + \phi)$

$$\therefore \quad i = 3.33 \sin\left(100\pi t + \frac{\pi}{2}\right) A.$$

The phasor diagram is shown below.

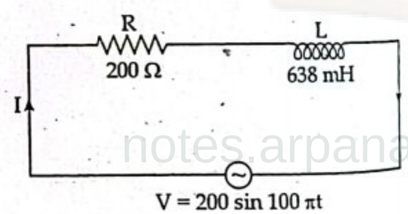


2. A voltage e = 200 sin 100 πt is applied to a load having R = 200 Ω in series with L = 638 mH.
Estimate
i) Expression for current in $i = I_m \sin(\omega t \pm \phi)$ form
ii) Power consumed by the load
iii) Reactive power of the load
iv) Voltage across R and L.

Solution:

$e_m = 200\ V$

$$V = \frac{200}{\sqrt{2}} = 141.421\ V\ rms$$



$\omega = 100\pi$

or, $2\pi f = 100\pi$

$\therefore$ Frequency, $f = 50\ Hz$

$X_L = \omega L = 100\pi \times 638 \times 10^{-3} = 200.433\ \Omega$

$Z = R + jX_L = 200 + j\,200.433\ \Omega = 283.149 \angle 45.06°\ \Omega$

$$\therefore \quad I = \frac{V}{2} = \frac{141.421 \angle 0°}{283.149 \angle 45.06°}$$

$\therefore \quad I = 0.5 \angle -45.06°\ A$

and, $I_m = \sqrt{2} \times 0.5 = 0.7071\ A$

$\phi = -45.06°$

i) $i = I_m \sin(\omega t - \phi) = 0.7071 \sin(100\pi t - 45.06°)$

ii) $P = IV \cos\phi = 141.421 \times 0.5 \times \cos(45.06°) = 49.9474$ watt

iii) $Q = VI \sin\phi = 141.421 \times 0.5 \times \sin(45.06°) = 50$ VAR

iv) $V_R = IR = 0.5 \times 200 = 100\ V$

v) $V_L = IXL = 0.5 \times 200.433 = 100.21\ V$

3. A series RLC circuit is connected to 230 V ac supply. The current drawn from the circuit at the resonance is 25 A. The voltage drop across the capacitor is 4,000 V at the series resonance. Calculate the resistance, inductance if capacitance is 5 μF, also calculate the resonant frequency.

Solution:

At resonance,

$$R = \frac{V}{I} = \frac{230}{25} = 9.2\ \Omega$$

Voltage drop across capacitor, $V_c = I X_c$

or, $4{,}000 = 25\ X_c$

$\therefore \quad X_c = 160\ \Omega$

At resonant frequency, f = fr

and, $X_c = \frac{1}{2\pi frC}$

or, $160 = \frac{1}{2\pi fr \times 10^{-6}}$

$\therefore \quad fr = 198.943\ Hz$

At resonance, $2\pi fr\ L = 160$

or, $L = \frac{160}{2\pi \times 198.943}$

$\therefore \quad L = 0.128\ H.$

4. Two impedances Z_1 and Z_2 are connected in parallel across applied voltage of (100 + j 200) volts. The total power supplied to the circuit is 5 KW. The first branch takes a leading current of 16 A and has a resistance of 5 ohm while the second branch takes a lagging current of 0.8 power factor calculate,
i) Current in second branch
ii) Total current
iii) Circuit constants

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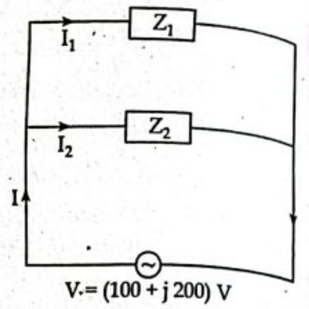
Solution:



$V = (100 + j\,200)$ V

$V = 100 + j\,200 = 223.60 \angle 63.43°$ volts

$I_1 = 16$ A

$R = 5\ \Omega$

$\cos \phi_2 = 0.8$

Now,

$$|Z_1| = \frac{V}{I_1} = \frac{223.60}{16} = 13.975\ \Omega$$

$R_1 = Z_1 \cos \phi_1$

or, $\cos \phi_1 = \frac{R_1}{Z_1} = \frac{5}{13.975} = 0.3577$

$\therefore \quad \phi_1 = -69.03°$

Hence, $\sin \phi_1 = 0.9338$

$X_1 = Z_1 \sin \phi_1 = 13.975 \times 0.9338 = 13.04\ \Omega$

Power consumed in Z_1, $P_1 = I_1^2 R_1 = 16^2 \times 5 = 1{,}280$ W

Power consumed in Z_2, $P_2 = I_2^2 R_2$ = Total power supplied $- I_1^2 R_1$

$\therefore \quad P_2 = 5{,}000 - 1{,}280 = 3{,}720$ W

Power consumed in Z_2, $P_2 = V I_2 \cos \phi_2$

$$\therefore \quad I_2 = \frac{P_2}{V \cos \phi_2} = \frac{3{,}720}{223.60 \times 0.8} = 20.79 \text{ A}$$

Now,

$$P_2 = I_2^2 R_2$$

or, $$R_2 = \frac{P_2}{I_2^2} = \frac{3720}{(20.79)^2} = 8.60\ \Omega$$

Thus, $\cos \phi_2 = 0.8$

$\sin \phi_2 = 0.6$

$$\therefore \quad X_2 = |Z_2| \sin \phi_2 = \frac{V}{I_2} \text{Sin}\, \phi_2 = \frac{223.60}{20.79} \times 0.6 = 6.4531\ \Omega$$

$Z_2 = R_2 + jX_2 = 8.60 + j\,6.4531 = 10.75 \angle 36.88°\ \Omega$

$$I_2 = \frac{V}{Z_2} = \frac{223.60 \angle 63.43°}{10.75 \angle 36.88°} = 20.79 \angle 26.55° \text{ A}$$

Similarly,

$Z_1 = R_1 - j\,X_1 = 5 - j\,13.04 = 13.965 \angle -69.02°\ \Omega$

$$I_1 = \frac{V}{Z_1} = \frac{223.60 \angle 63.43°}{13.965 \angle -69.02°} = 16 \angle 132.45° \text{ A}$$

Hence,

Total current,

$I = I_1 + I_2 = 16 \angle 132.45° + 20.79 \angle 26.55° = 7.8 + j\,21.09°$

i) $I = 22.48 \angle 69.70°$ A

ii) Current in branch -2, $I_2 = 20.79$ A

iii) Circuit constants are;

$R_1 = 5\ \Omega$; $X_1 = 13.04\ \Omega$

$R_2 = 8.60\ \Omega$; $X_2 = 6.453\ \Omega$.

5. A room heater of 2 KW, 125 V rating is to be operated on 230 V, 50 Hz ac supply. Calculate the value of inductance that must be connected in series with the heater so that heater will not get damaged due to over voltage.

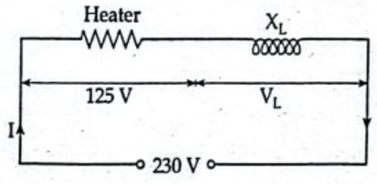
Solution:

Heater is to be assumed as purely resistive load.

$P = 2$ KW $= 2{,}000$ W

$V_R = 125$ V

$$I_R = \frac{P}{V_R} = \frac{2{,}000}{125} = 16 \text{ A} = I$$



$V = \sqrt{V_R^2 + V_L^2}$

or, $230 = \sqrt{(125)^2 + (V_L)^2}$

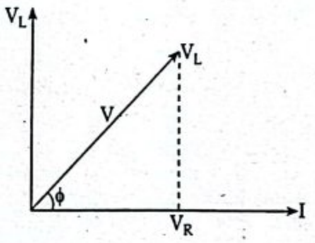
or, $52{,}900 = (125)^2 + V_L^2$

or, $V_L^2 = 37{,}275$

$\therefore \quad V_L = 193.067$ V

Now, $V_L = IX_L$

$$\text{or,} \quad X_L = \frac{V_L}{I} = \frac{193.067}{16} = 12.066\ \Omega$$



and, $X_L = 2\pi fL$

$$\text{or,} \quad \frac{12.066}{2\pi \times 50}$$

$\therefore \quad L = 0.0384$ H

Hence an inductor of 0.0384 H much be connected in series with the heater so that it will be get damaged.

6. A series R-C circuit connected across 200 V, 50 Hz ac supply draws a current of 20 A. When the frequency of the supply is increased to 100 Hz, the current increases to 23.4082 A. Calculate the value of resistance and capacitance of the circuit.

Solution:

When frequency is 50 Hz, *i.e.*, $f_1 = 50$ Hz

$$Z_1 = \frac{V}{I_1} = \frac{200}{20} = 10\ \Omega$$

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When frequency is 100 Hz, *i.e.* $f_2 = 100$ Hz

$$Z_2 = \frac{V}{I_2} = \frac{200}{23.4082} = 8.544\ \Omega$$

We know,

$$Z_1 = \sqrt{R^2 + X_{c1}^2} \quad (1)$$

and, $Z_2 = \sqrt{R^2 + (X_{c2})^2}$ (2)

so, $10^2 = R^2 + (X_{c1})^2$ (3)

$(8.544)^2 = R^2 + (X_{c2})^2$ (4)

Subtracting (3) from (4);

$\therefore \quad (X_{c1})^2 - (X_{c2})^2 = 27$

or, $\left(\frac{1}{2\pi f_1 C}\right)^2 - \left(\frac{1}{2\pi f_2 C}\right)^2 = 27$

or, $\frac{1}{4\pi^2 \times 50^2 \times C^2} - \frac{1}{4\pi^2 \times 100^2 \times C^2} = 27$

or, $\frac{1 \times 7.6 \times 10^{-6}}{C^2} = 27$

or, $C^2 = 2.814 \times 10^{-7}$

$\therefore \quad C = 5.305 \times 10^{-4}$ F

Hence,

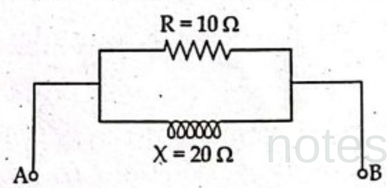
$$R^2 = 10^2 - (X_{c1})^2 = 10^2 - \frac{1}{4\pi^2 \times 50^2 \times (5.305 \times 10^{-4})^2}$$

or, $R^2 = 100 - 36$

or, $R^2 = 64$

$\therefore \quad R = 8\ \Omega.$

7. A circuit is shown below. Draw its equivalent admittance circuit. Also calculate admittance, conductance and suspectance.



Solution:

The impedance of branch 1,

$$Z_1 = R + j\,0 = 10 + j\,0$$

$\therefore \quad Z_1 = 10\ \angle 0°\ \Omega$

The impedance of branch 2,

$$Z_2 = 0 + j\,X = 0 + j\,20 = 20\ \angle 90°\ \Omega$$

Admittance, $Y_1 = \frac{1}{Z_1} = \frac{1}{10\ \angle 0°} = 0.1\ \angle 0°$ Siemens

Admittance, $Y_2 = \frac{1}{Z_2} = \frac{1}{20\ \angle 90°} = 0.05\ \angle -90°$ Siemens

$Y_1 = 0.1\ \angle 0° = 0.1 + j\,0$ Siemens

$Y_2 = 0.05\ \angle -90° = 0 - j\,0.05$ Siemens

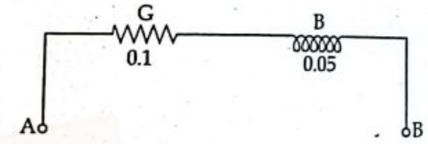
$Y = Y_1 + Y_2 = 0.1 + j\,0 + 0 - j\,0.05 = 0.1118\ \angle -26.56°$

$\therefore$ Conductance, $G = 0.1$

$\therefore$ Suspectance, $B = 0.05$

$\therefore$ Admittance, $Y = 0.1118$ Siemens

Equivalent admittance circuit is,



8. Two circuits having same numerical value of ohmic impedance are connected in parallel. The power factor of one circuit having impedance Z_1 is 0.8 and other impedance Z_2 is 0.6. What is the power factor of te combination, when,

i) Both impedance are inductive?

ii) Z_1 is inductive and Z_2 is capacitive?

Solution:

$|Z_1| = |Z_2| = Z\ \Omega$

$\cos \phi_1 = 0.8$ and $\sin \phi_1 = 0.6$

$\cos \phi_2 = 0.6$ and $\sin \phi_2 = 0.8$

i) When both impedance are inductive,

$$Z_1 = |Z_1|\ [\cos \phi_1 + j \sin \phi_1] = Z\ (0.8 + j\,0.6)$$

$$Z_2 = |Z_2|\ [\cos \phi_2 + j \sin \phi_2] = Z\ (0.6 + j\,0.8)$$

Since, $Z_1 \parallel Z_2$

$$Z_T = \frac{Z_1 \times Z_2}{Z_1 + Z_2}$$

$$= \frac{Z^2\ (0.8 + j\,0.6) \times (0.6 + j\,0.8)}{Z[0.8 + 0.6 + j\,0.6 + j\,0.8]}$$

$$= Z\left[\frac{1\ \angle 36.869° \times 1\ \angle 53.13°}{1.4 + j\,1.4}\right]$$

$$= Z\left[\frac{1\ \angle 90°}{1.9798\ \angle 45°}\right]$$

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$\therefore \quad Z_T = \frac{Z}{1.9748} \angle 45°$

$\therefore$ Power factor of combination = cos 45° = 0.7071 lagging.

ii) Z_1 is inductive and Z_2 is capacitive

$Z_1 = Z(0.8 + j\,0.6) = Z[1 \angle 36.869°]$

$Z_2 = Z(0.6 - j\,0.8) = Z[1 \angle -53.13°]$

While j term is negative due to capacitive

Thus, $Z_T = \frac{Z_1 Z_2}{Z_1 + Z_2}$

$= Z^2 \frac{[1 \angle 36.869° \times 1 \angle -53.13°]}{Z[0.8 + j0.6 + 0.6 - j\,0.8]}$

$= Z\left[\frac{1 \angle -16.261°}{1.4 - j\,0.2}\right]$

$= Z\left[\frac{1 \angle -16.261°}{1.4142 \angle -8.13°}\right]$

$\therefore \quad Z_T = \frac{Z}{1.4142} \angle -8.1308°$

$\therefore$ Pf combination = cos (−8.1308°) = 0.9899 leading.

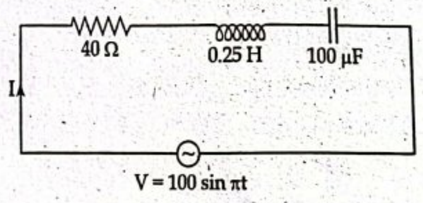
9. An emf given of V = 100 sin πt is impressed across a circuit consists of resistance of 40 Ω in series with 100 μF capacitor and 0.25 H inductor. Find,

i) RMS value of current

ii) Power consumed

iii) Power factor

Solution:



Comparing voltage with $V = V_m \sin \omega t$

$\therefore \quad V_m = 100$ V

$\omega = 100\,\pi$ red/sec

$\therefore \quad V = \frac{V_m}{\sqrt{2}} = \frac{100}{\sqrt{2}} = 70.7106$ V

Now,

$X_L = 2\pi fL = \omega L = 100\,\pi \times 0.25 = 78.5398\ \Omega$

$X_C = \frac{1}{2\pi fC} = \frac{1}{\omega C} = \frac{1}{100\pi \times 100 \times 10^{-6}} = 31.8309\ \Omega$

Thus,

$Z_T = R + j\,X_L - j\,X_C$

$= 40 + j\,78.5398 - j\,31.8309$

$= 40 + j\,46.7089$

$= 61.4957 \angle 49.4243°\ \Omega$

In polar form,

$V = 70.7106 \angle 0°$ V

$\therefore \quad I = \frac{V}{Z_T} = \frac{70.7106 \angle 0°}{61.4957 \angle 49.4243°} = 1.1498 \angle -49.4243°$ A

i) RMS value of current = 1.1498 A

ii) Power consumed $= IV \cos\phi = I^2R$

$= 1.1498 \times 70.7106 \times \cos(-49.4243°)$

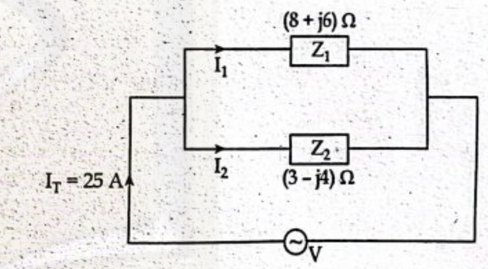
$= 52.8837$ W

iii) Power factor $= \cos(-49.4243°)$

$= 0.6504$ lagging since $X_L > X_C$

10. Two impedances (8 + j 6) Ω and (3 − 4 j) Ω are connected in parallel. If the total current drawn by the combination is 25 A, find the current and power drawn by each impedance.

Solution:



$Z_1 = 8 + j6\ \Omega = 10 \angle 36.869°\ \Omega$

$Z_2 = 3 - j4\ \Omega = 5 \angle -53.1301°\ \Omega$

i) Branch currents

Using current divider rule,

$I_1 = I_T \times \frac{Z_2}{Z_1 + Z_2}$

$I_2 = I_T \times \frac{Z_1}{Z_1 + Z_2}$

$Z_1 + Z_2 = 8 + j6 + 3 - j4 = 11 + j2 = 11.1803 \angle 10.3048°\ \Omega$

so, $I_1 = 25 \angle 0° \times \frac{5 \angle -53.1301°}{11.1803 \angle 10.3048°} = 11.1803 \angle -63.4349°$ A

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and, $I_2 = 25\angle 0° \times \frac{10\angle 36.869°}{11.1803\angle 10.3048°} = 22.3607\angle 26.5642°\ A$

ii) Power consumed

$P_1 = I_1^2 R_1 = (11.1803)^2 \times 8 = 1{,}000\ W$

$P_2 = I_2^2 R_2 = (22.3607)^2 \times 3 = 1{,}500\ W$

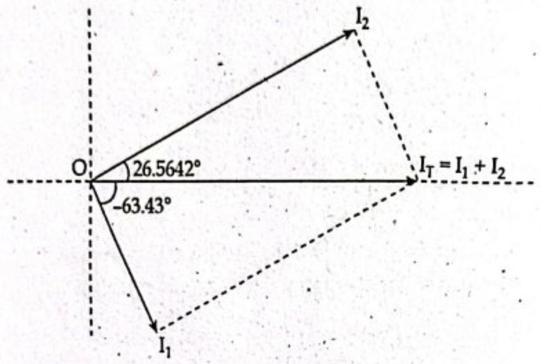


Fig: Phasor diagram

11. An alternating voltage, $V = 141.4 \sin\left(157.08 + \frac{\pi}{12}\right)$ volts is applied to a circuit and an ac ammeter, wattmeter and power factor meter are connected to measure the respective quantities. Reading of ammeter is 5 Amp and that of power factor meter is 0.5 lagging. Find,

i) Expression for the instantaneous value of current

ii) The wattmeter reading

iii) Impedance of the circuit in rectangular form

Solution:

Comparing given voltage with $V = V_m \sin(\omega t + \theta)$

$V_m = 141.4\ V$

$\omega = 157.08\ rad/sec$

$\phi = \frac{\pi}{12} = 15°$

$\therefore\ V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{141.4}{\sqrt{2}} = 100\ V$

$f = \frac{\omega}{2\pi} = \frac{157.08}{2\pi} = 25\ Hz$

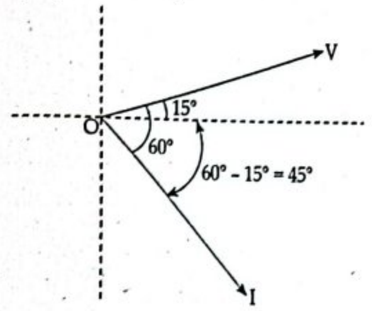
Hence,

$V_{rms} = 100\angle 15°\ V$

$I_{rms} = 5\ A$

$\cos\phi = 0.5$ lagging

I lags V by $\phi = \cos^{-1}(0.5) = 60°$ and phase of V is 15°



Thus, phase of current = –45° *i.e.*, $\frac{\pi}{4}$ rad and

$I_m = \sqrt{2}\, I = 5\sqrt{2}\ A$

i) The expression for instantaneous value of current

$$i = I_m \sin(\omega t + \theta) = 5\sqrt{2} \sin\left(157.08\, t - \frac{\pi}{4}\right) A$$

ii) $V = 100\angle 15°\ V$

$I = 5\angle -45°\ A$

$\therefore\ Z = \frac{V}{I} = \frac{100\angle 15°}{5\angle -45°} = 20\angle 60°\ \Omega$

$\therefore\ Z = (10 + j\,17.3205)\ \Omega$

12. When an inductive coil is connected to a dc supply at 240 V, the current in it is 16 A. When the same coil is connected to an ac supply at 240 V, 50 Hz, the current is 12.27 A. Calculate

i) Resistance

ii) Impedance

iii) Reactance iv inductance of the coil.

Solution:

For a dc supply, $f = 0$, hence, $X_L = 2\pi fL = 0\ \Omega$

Hence only resistance of coil is effective

$\therefore\ I_{dc} = \frac{V}{R}$

or, $16 = \frac{240}{R}$

i) Resistance, $R = 15\ \Omega$

ii) When connected to ac supply,

$Z = R + j\,2\pi fL$

$I_{ac} = 12.27\ A$

$|Z| = \frac{|V|}{|I_{ac}|} = \frac{240}{12.27} = 19.56\ \Omega$

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$I_{ac} = 12.27$ A

$$|Z| = \frac{|V|}{|I_{ac}|} = \frac{240}{12.27} = 19.56\ \Omega$$

Hence, impedance, $Z = 19.56\ \Omega$

iii) $|Z| = \sqrt{R^2 + X_L^2}$

or, $19.56 = \sqrt{(15)^2 + X_L^2}$

or, $X_L^2 = 19.56^2 - 15^2$

or, $X_L = \sqrt{157.5936}$

$\therefore X_L = 12.5536\ \Omega$

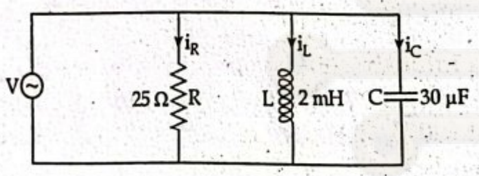
Hence, reactance, $X_L = 12.5536\ \Omega$

iv) $X_L = 2\pi fL$

$$L = \frac{12.5536}{2\pi \times 50} = 0.04\ \text{H}$$

Thus inductance of the coil is 0.04 H.

13. The voltage applied across 3-branched circuit shown below is given by $V = 100 \sin\left(5{,}000\,t + \frac{\pi}{4}\right)$. Calculate the branch current and total current.



Solution:

Total instantaneous current is the vector sum of the three branch currents,

$$i_r = i_R + i_L + i_C$$

where,

$$i_R = \frac{V}{R} = \frac{100}{25}\sin\left(5{,}000\,t + \frac{\pi}{4}\right)$$

$$i_L = \frac{1}{L}\int v\,dt = \frac{1{,}000}{2}\int 100 \sin\left(1{,}000\,t + \frac{\pi}{4}\right) dt$$

$$= \frac{10^3}{2} \times 100\left[-\frac{\cos\left(5{,}000\,t + \frac{\pi}{4}\right)}{5{,}000}\right]$$

$$= -10\cos\left(5{,}000\,t + \frac{\pi}{4}\right)$$

$$i_C = C\frac{dV}{dt}$$

$$= C\frac{d}{dt}\left[100 \sin\left(5{,}000\,t + \frac{\pi}{4}\right)\right]$$

$$= 30 \times 10^{-6} \times 100 \times 5{,}000 \times \cos\left(5{,}000\,t + \frac{\pi}{4}\right)$$

$$= 15\cos\left(5{,}000\,t + \frac{\pi}{4}\right)$$

Thus,

$$i_r = 4\sin\left(5{,}000\,t + \frac{\pi}{4}\right) + 5\cos\left(5{,}000\,t + \frac{\pi}{4}\right).$$

14. A single phase 7.46 KW motor is supplied from a 400 V, 50 Hz ac mains. If its efficiency is 85% and power factor 0.8 lagging. Calculate,

i) KVA input

ii) Reactive components of input current

iii) KVAr

Solution:

$$\text{Efficiency} = \frac{\text{Output in watts}}{\text{Input in watts}}$$

or, $0.85 = \frac{7.46 \times 1{,}000}{VI\cos(0.8)}$

or, $VI = \frac{7{,}460}{0.85 \times 0.8}$

$\therefore VI = 10{,}970$ VA

i) $\text{Input} = \frac{1{,}0970}{1{,}000} = 10.97$ KVA

ii) Input current,

$$I = \frac{\text{Volt ampere}}{\text{Volt}} = \frac{10{,}970}{400} = 27.43\ \text{A}$$

Active component of current,

$= I\cos\phi = 27.43 \times 0.8 = 21.94$ A

Reactive component of current

$= I\sin\phi = 27.43 \times 0.6 = 16.46$ A

ii) KVAr = KVA × $\sin\phi$ = 10.97 × 0.6 = 6.58 KVAr.

15. Draw the phasor diagram for each of the following combinations.

i) R and L in series and combination in parallel with C.

ii) R, L and C in series when ac voltage source is connected to it; ($X_C > X_L$)

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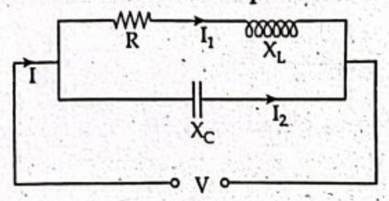
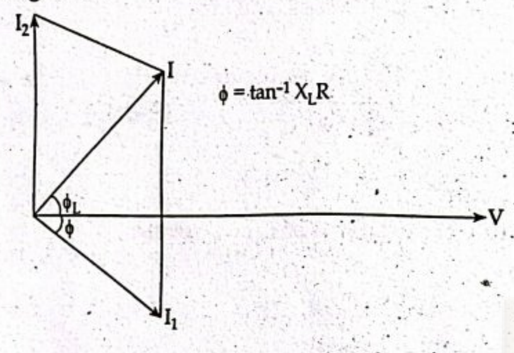
Solution:

i) R and L in series and combination in parallel with C.



Phasor diagram is,



ii)

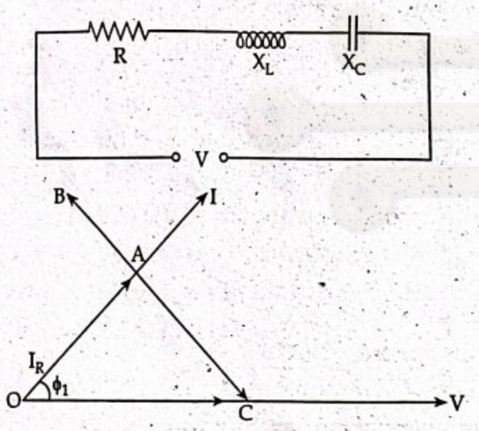


Fig: Phasor diagram

$OA = IR$; $AB = IX_L$; $AC = I(X_C - X_L)$

$BC = I X_C$

$$f = \tan^{-1}\frac{(X_C - X_L)}{R}$$

16. In a circuit, the applied voltage is found to lag the current by 30°.

i) Is the power factor lagging or leading?

ii) What is the value of power factor?

iii) Is the circuit inductive or capacitive?

Solution:

i) Power factor is leading because current leads the voltage.

ii) Power factor = cos 30° = 0.86 leading.

iii) The circuit is capacitive.

17. An R – L – C series circuit consists of a resistance of 1000 Ω, an inductance of 100 mH and a capacitance of 10 p.f. If a voltage of 100 V is applied across the terminal, find (1) resonance frequency (2) Q- factor (3) half power points.

Solution:

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi \times \sqrt{10^{-1} \times 10^{-11}}} = \frac{10^6}{2\pi} = 159 \text{ KHz}$$

$\therefore$ Resonant frequency, $f_0 = 159$ KHz

ii) Q-factor, $Q = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{1}{1{,}000}\sqrt{\frac{10^{-1}}{10^{-11}}} = 100$

iii) Half power points,

$$f_1 = f_0 - \frac{R}{4\pi L} = 159 \times 10^3 - \frac{1{,}000}{4\pi \times 10^{-1}} = 158.3 \text{ KHz}$$

$$f_2 = f_0 + \frac{R}{4\pi L} = 159 \times 10^3 + \frac{1{,}000}{4\pi \times 10^{-1}} = 159.8 \text{ KHz}.$$

18. A series R-L-C circuit consists of R = 1,000 Ω, L = 100 mH and C = 10 p.f. The applied voltage across the circuit is 100 V.

i) Find the resonant frequency of the circuit.

ii) Find the quality factor at the resonant frequency.

iii) Calculate the bandwidth of the circuit.

iv) At what angular frequencies do the half power points occur?

Solution:

i) $$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{100 \times 10^{-3} \times 10 \times 10^{-12}}} = 159.15 \text{ KHz}$$

ii) Q-factor, $Q_0 = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{1}{1{,}000}\sqrt{\frac{100 \times 10^{-3}}{10 \times 10^{-12}}} = 100$

iii) $$B_{HP} = \frac{R}{2\pi L} = \frac{1{,}000}{2\pi \times 100 \times 10^{-3}} = 1{,}591.5 \text{ Hz}$$

Also, $B_{HP} = \frac{f_0}{Q_0} = \frac{159.15}{100} = 1{,}591.5$ KHz

iv) $\omega_1 = \omega_0\left(1 - \frac{1}{2Q_0}\right)$

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$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\omega_2 = \omega_0\left(1 + \frac{1}{2Q_0}\right)$$

$$\text{so, } \omega_0 = \frac{1}{\sqrt{100 \times 10^{-3} \times 10 \times 10^{-12}}} = 1 \times 10^6$$

Hence,

$$\omega_1 = 1 \times 10^6 \times \left(1 - \frac{1}{2 \times 100}\right) = 9{,}95{,}000 \text{ red/sec}$$

$$\omega_2 = 1 \times 10^6\left(1 + \frac{1}{2 \times 100}\right) = 1{,}00{,}5000 \text{ rad/sec}$$

Thus,

Bandwidth, B.W = $\omega_2 - \omega_1$ = 1,005,000 – 9,95,000 = 10,000 rad/sec = 10 KHz.

19. A coil of inductance 9 H and resistance 50 Ω in series with a capacitor is supplied at constant voltage from a variable frequency source. If the maximum current of 1 A occurs at 75 Hz. Find the frequency when the current is 0.5 A.

Solution:

Here;

$$N = \frac{I_0}{I} = \frac{1}{0.5} = 2$$

$$Q = \frac{\omega_0 L}{R} = 2\pi \times 75 \times \frac{9}{50} = 84.8$$

Let f_1 and f_2 be the frequencies at which current falls to half its maximum value at resonance frequency. Then,

$$\frac{f_0}{f_1} - \frac{f_1}{f_0} = \frac{\sqrt{3}}{Q}$$

or $$\frac{75}{f_1} - \frac{f_1}{75} = \frac{\sqrt{3}}{84.4}$$

or $$\frac{(75^2 - f_1^2)}{75 f_1} = 0.02$$

or $$f_1^2 + 1.5 f_1 - 5{,}625 = 0$$

$\therefore$ $f_1 = 74.25$ Hz

Also, $$\frac{f_2}{75} - \frac{75}{f_2} = \frac{\sqrt{3}}{84.4}$$

or $$f_2^2 - 1.5 f_2 - 5{,}625 = 0$$

Thus, $f_1 = 74.25$ Hz and $f_2 = 75.75$ Hz.

20. Two circuits, the impedance of which are given by $Z_1 = 10 + j\,15$ Ω and $Z_2 = 6 - j\,8$ ohm are connected in parallel. If the total current supplied is 15 A. What is the power taken by each branch? Find also the p.f. of individual circuit and of combination. Draw vector diagram.

Solution:

Let, $I = 15 \angle 0°$

$Z_1 = 10 + j\,15 = 18 \angle 57°$

$Z_2 = 6 - j\,8 = 10 \angle -53.1°$

Total impedance,

$$Z' = \frac{Z_1 Z_2}{Z_1 + Z_2} = \frac{(10 + j\,15)(6 - j\,8)}{16 + j\,7}$$

$$= 10.3 \angle -20.4°$$

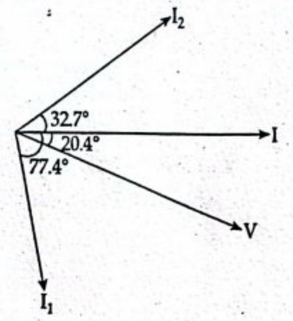
Applied voltage,

$$V = IZ = 15 \angle 0° \times 10.3 \angle -20.4°$$

$$= 154.5 \angle -20.4°$$

$$I_1 = \frac{V}{Z_1} = \frac{154.5 \angle -20.4°}{18 \angle 57°} = 8.58 \angle -77.4°$$

$$I_2 = \frac{V}{Z_2} = \frac{154.5 \angle -20.4°}{10 \angle -53.1°} = 15.45 \angle 32.7°$$



From phasor diagram, I_1 lags V by (77.4° – 20.4°)

= 57° and I_2 leads it by (32.7° + 20.4°) = 53.1°

Now,

$$P_1 = I_1^2 R_1 = 8.58^2 \times 10 = 736 \text{ W}$$

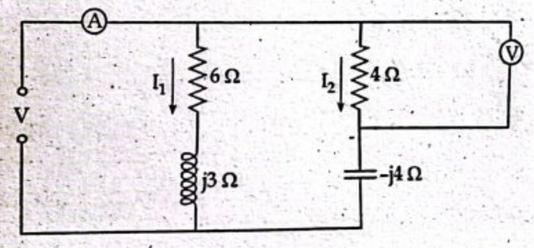
Power factor = cos (57°) = 0.544 (lag)

$$P_2 = I_2^2 R_2 = 15.45^2 \times 6 = 1{,}432 \text{ W}$$

Power factor = cos (53.1°) = 0.6

Combined power factor = cos (20.4°) = 0.937 (lead).

21. If the voltmeter in figure below below reads 60 V, find the reading of the ammeter.



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Solution:

$$I_2 = \frac{60}{4} = 15 \text{ A}$$

Taking it as reference quality,

$$I_2 = 15 \angle 0°$$

Applied voltage is,

$$V = 15 \angle 0° \times (4 - j4) = 84.8 \angle -45°$$

$$I_1 = \frac{84.8 \angle -45°}{(6 + j3)}$$

$$= 84.8 \angle -45° \div 6.7 \angle 26.6$$

$$= 12.6 \angle -71.6°$$

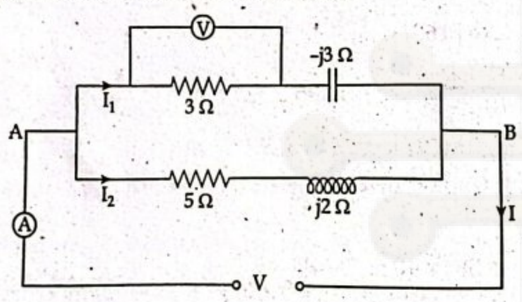
$$= 4 - j\,12$$

$$I = I_1 + I_2 = (15 + j\,0) + (4 - j\,12) = 19 - j\,12$$

$$\therefore \quad I = 22.47 \angle -32.3°$$

Hence ammeter reads 22.47 A.

22. Find the reading of the ammeter when the voltmeter across 3 ohm resistor in the circuit reads 45 V.



Solution:

$$I_1 = \frac{45}{3} = 15 \text{ A}$$

Taking it as reference, $I_1 = 3 \angle 0°$

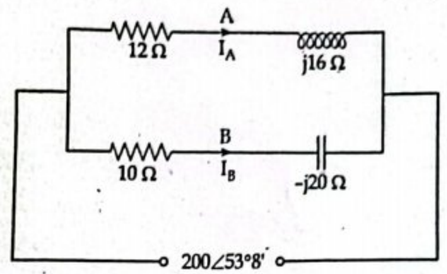
Now, $Z_1 = 3 - j\,3 = 4.24 \angle -45°$

Hence, $V = I_1 Z_1 = 15 \angle 0° \times 4.24 \angle -45° = 63.6 \angle -45°$

$$I_2 = \frac{V}{Z_2} = \frac{63.6 \angle -45°}{5 + j2} = \frac{63.6 \angle -45°}{5.4 \angle 21.8°} = 4.64 - j\,10.8 = 11.77 \angle -66.8°$$

so, $I = I_1 + I_2 = 19.64 - j10.8 = 22.4 \angle 28.8°$ A.

23. A voltage of 200 ∠53° 8' is applied across two impedance in parallel. The values of impedance are 12 + j 16 and 10 – j 20 ohm. Determine the KVA, KVAr and KW in each branch and power factor of the whole circuit.



Solution:

$$Y_A = \frac{1}{12 + j\,16} = 0.03 - j\,0.04 \text{ mho}$$

$$Y_B = \frac{1}{10 - j\,20} = 0.02 + j\,0.04 \text{ mho}$$

Now,

$$V = 200 \angle 5°8' = 200 (\cos 53°8' + j \sin 53.8')$$

$$= 200 (0.6 + j\,0.8)$$

$$= 120 + j\,160 \text{ volt}$$

$$I_A = VY_A$$

$$= (120 + j\,160)(0.03 - j\,0.04)$$

$$= (10 + j\,0) \text{ A along reference axis}$$

$$I_B = VY_B$$

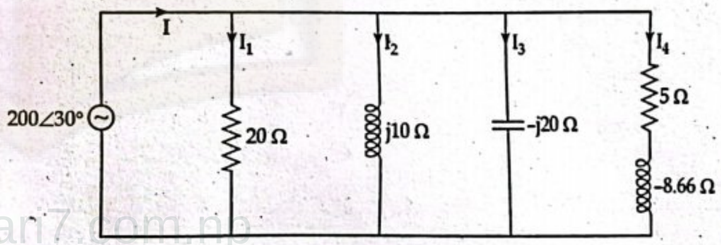
$$= (120 + j\,160)(0.02 + j\,0.04)$$

$$= -4 + j\,8 \text{ A (leading)}$$

24. Calculate,

i) Total current

ii) Equivalent impedance for the four branched circuit



Solution:

$$Y_1 = \frac{1}{20} = 0.05 \text{ S}$$

$$Y_2 = \frac{1}{j\,10} = -j\,0.1 \text{ S}$$

$$Y_3 = \frac{1}{-j\,20} = j\,0.05 \text{ S}$$

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$$Y_4 = \frac{1}{5 - j\,8.66} = \frac{1}{10\angle 60°} = (0.05 - j\,0.0866)\ S$$

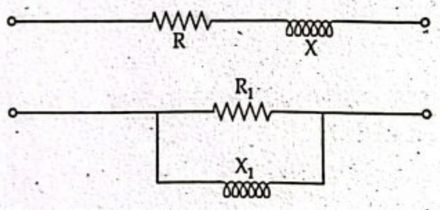
Now,

$$Y = Y_1 + Y_2 + Y_3 + Y_4 = (0.1 - j\,0.1360)\ S = 0.169\angle -53.8°\ S$$

i) Total current, $I = YV = 200\angle 30° \times 0.169\angle -53.8° = 33.8\angle -23.8°$ A

ii) Impedance, $Z = \frac{1}{Y} = \frac{1}{0.169\angle -53.8°} = 5.9\angle 53.8°\ \Omega$.

25. The series and parallel circuit shown below have the same impedance and the same power factor. If R = 3 Ω and X = 4 Ω, find the value of R_1 and X_1. Also find the impedance and power factor.



Solution:

Series circuit

$$Y_S = \frac{1}{R + jX} = \frac{R - jX}{R^2 + jX^2} = \frac{R}{R^2 + X^2} - \frac{jX}{R^2 + X^2}$$

Parallel circuit

$$Y_P = \frac{1}{R + j0} = \frac{1}{0 + jX_1} = \frac{1}{R_1} + \frac{1}{jX_1} = \frac{1}{R_1} - \frac{j}{X_1}$$

$$\therefore \quad \frac{R}{R^2 + X^2} = -j\frac{X}{R^2 + X^2} = \frac{1}{R_1} - \frac{j}{X_1}$$

or, $R_1 = R + \frac{X^2}{R}$

and, $X_1 = X + \frac{R^2}{X}$

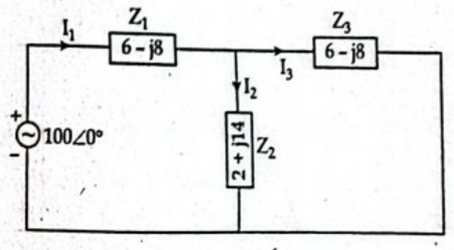
$$\therefore \quad R_1 = 3 + \left(\frac{16}{3}\right) = 8.33\ \Omega$$

$$X_1 = 4 + \left(\frac{9}{4}\right) = 6.25\ \Omega$$

Impedance, $Z = 3 + j\,4\ \Omega = 5\angle 53.1°\ \Omega$

Power factor = $\cos 53.1° = 0.6$ (lag).

26. Determine the average power delivered to each of the three boxed networks in the circuit shown below.



Solution:

$Z_1 = 6 - j\,8 = 10\angle -53.13°\ \Omega$

$Z_2 = 2 + j\,14 = 14.14\angle 81.87°\ \Omega$

$Z_3 = 6 - j\,8 = 10\angle -53.13°\ \Omega$

$$Z_{23} = \frac{Z_2 Z_3}{Z_2 + Z_3} = 14.14\angle -8.13° = 14 - j\,2$$

Drop across two parallel impedance,

$$V_{23} = 100 \times \frac{14 - j\,2}{(6 - j\,8) + (14 - j\,2)} = 63.2\angle 18.43° = (60 + j\,20)\ V$$

$$V_1 = 100 \times \frac{10}{6 - j\,8 + (14 - j\,2)} = 47.7\angle -26.57° = (40 - j\,20)\ V$$

Now,

$$I_1 = \frac{44.7\angle -26.57°}{10\angle -53.13°} = 4.47\angle 26.56°\ A$$

$$I_2 = \frac{63.2\angle 18.43°}{14.14\angle 81.87°} = 4.47\angle -63.44°\ A$$

$$I_3 = \frac{63.2\angle 18.43°}{10\angle -53.13°} = 6.32\angle 71.56°\ A$$

Also,

$P_1 = V_1 I_1 \cos\phi_1 = 44.7 \times 4.47 \times \cos(53.13°) = 120\ W$

$P_2 = V_2 I_2 \cos\phi_2 = 63.2 \times 4.47 \times \cos(81.87°) = 40\ W$

$P_3 = V_3 I_3 \cos\phi_3 = 63.2 \times 6.32 \times \cos(53.13°) = 240\ W$

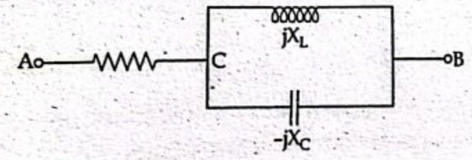
Total average power,

$P = P_1 + P_2 + P_3 = 120 + 40 + 240 = 400$ W.

27. Draw admittance triangle between the terminals AB. Labelling its side with appropriate values and units in case of:

i) $X_L = 4\ \Omega$ and $X_C = 8\ \Omega$

ii) $X_L = 10\ \Omega$ and $X_C = 5\ \Omega$



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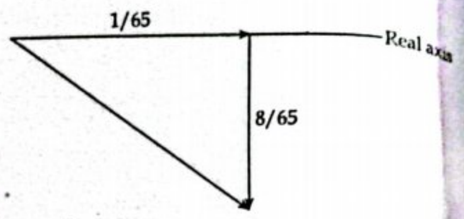
Solution:

i) $X_L = 4\ \Omega, X_C = 8\ \Omega$

$$Z_{CB} = \frac{j X_L (-j X_C)}{j (X_L - X_C)} = j8$$

$Z_{AB} = 1 + j8$ ohms.

$$y_{AB} = \frac{1}{Z_{AB}} = \frac{1}{65} - j\frac{8}{65} \text{ mho}$$

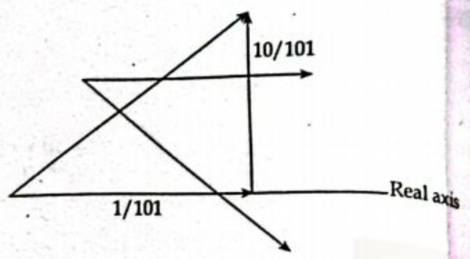


ii) $X_L = 10\ \Omega, X_C = 5\ \Omega$

$$Z_{CB} = j\frac{10 \times (-j5)}{j5} = -j10$$

$Z_{AB} = 1 - j10$ ohms

$$y_{AB} = \frac{1}{101} + j\frac{10}{101} \text{ mho}$$



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CHAPTER 4

POLY-PHASE AC CIRCUIT ANALYSIS

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4.1 CONCEPT OF A BALANCED THREE PHASE SUPPLY

A three phase electric system may be considered as three single phase system displaced from each other by 120°.

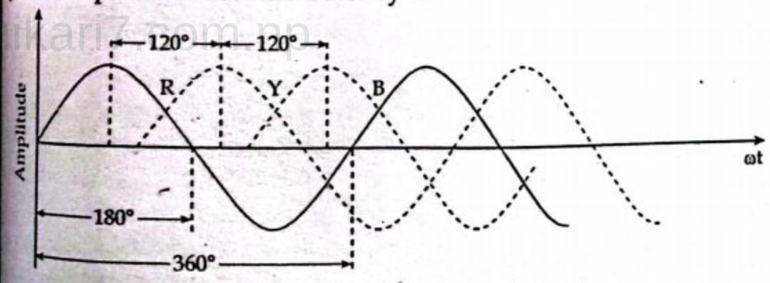

Fig: Graphical representation of 3-phase system

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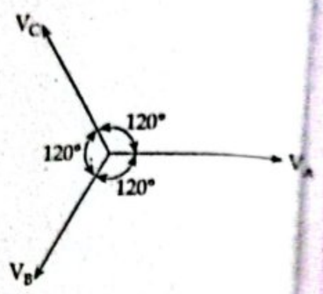
If $|V_A| = |V_B| = |V_C| = |V_{ph}|$ and if the phase difference between V_A and V_B, V_B and V_C, V_C and V_A is equal to 120° as shown in figure, then the supply is said to be balanced or symmetrical.



4.2 GENERATION OF THREE-PHASE EMF

When three identical coils are placed with their axes at 120° displaced from each other and rotated in a uniform magnetic field, a sinusoidal voltage is generated across the coil. Figure below shows three set of coils RR', YY' and BB' displaced from each other by 120° and rotating in an anticlockwise direction with angular velocity ω in a uniform magnetic field. Since the three coils are identical, the generated voltages have the same magnitude. The generated voltages in the coils are given by,

$V_R = V_m \sin \omega t$

$V_Y = V_m \sin (\omega t - 120°)$

$V_B = V_m \sin (\omega t - 240°)$

$= V_m \sin (\omega t + 120°)$

[Here, voltage generated in coil R is taken as reference. So, V_Y lags V_R lags V_R by 120° and V_B lags V_R by 240°]

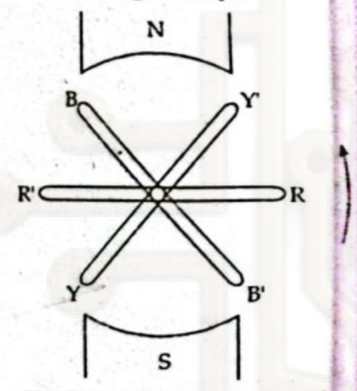


Fig: Three-phase emf generation

In polar form,

$V_R = |V| \angle 0°$

$V_y = |V| \angle -120°$

$V_B = |V| \angle -240°$

$= |V| \angle 120°$

The three phase may be numbered a, b, c or 1, 2, 3 or R, Y, B as customary. The phase sequence is usually RYB. Vector ratio is usually anticlockwise

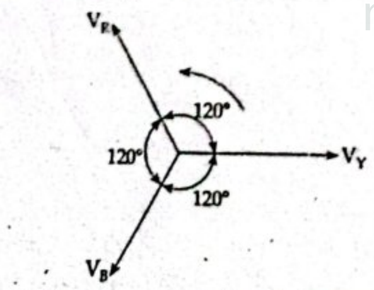


Fig: Vector representation of phase voltages.

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4.3 DIFFERENCE BETWEEN SINGLE PHASE AND THREE PHASE SYSTEM

SN	Single phase system	Three phase system
i)	It requires more copper for transmitting the same power.	It requires less copper for transmission of same power.
ii)	Single phase generators cannot be operated in parallel.	Poly-phase generators can be worked in parallel.
iii)	Single phase motors are less costly.	Poly phase motors are more costly.
iv)	For a given size frame, the output of a single phase machine is low.	For a given size of frame, the output of a poly-phase machine is high.
v)	Efficiency of a single phase generator is low.	Efficiency of a poly-phase generator is high.
vi)	Transmission and distribution of single phase power is costly and less efficient.	Transmission and distribution of poly phase power is cheaper and more efficient.
vii)	Single phase motors require more space for conveying the same power.	Poly phase motors are more compact.
viii)	Single phase motors have fluctuating torque, which is a series disadvantage in case of large machines.	Poly phase motors have uniform torque.
ix)	Power in it may fall to zero	Power in it never falls to zero.
x)	Single phase motors possess no starting torque, so they must be provided with additional starting devices.	Poly phase motors can be made self-starting.
xi)	Single phase generator, though smaller in size, is more costly in use.	Poly phase generator though larger in size, is more economical in operation.

4.4 STAR AND DELTA CONNECTED SUPPLY AND LOAD CIRCUITS

a) Star connection (3-phase and 4 wire system or Y connection)

In this, one end of each winding (W_1, W_2 and W_3) is brought to the common point, 0, called star- point or neutral wire point; while the other ends are joined to the live wires. When the load across the system is balanced (*i.e.*, when the emf, currents and phase angles are equal for each phase), the neutral wire carries the three currents exactly equal in magnitude but at 120° phase difference with each other in such a way that

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the vector sum of $I_R + I_Y + I_B = 0$. In other words, in a balanced star connected system, the neutral wire carries no current. The voltage and current induced in each winding are called phase voltage (V_{ph}) and phase current (I_{ph}) respectively; while the available voltage and current between any pair terminals are called line voltage (V_L) and line current (I_L) respectively.

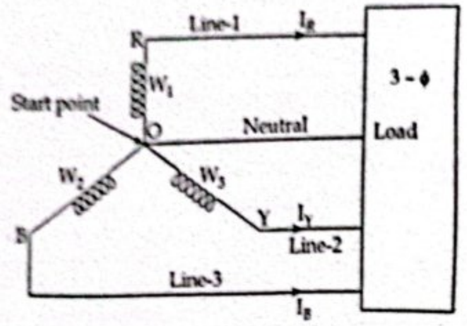

Fig: 3-phase load using a star connection.

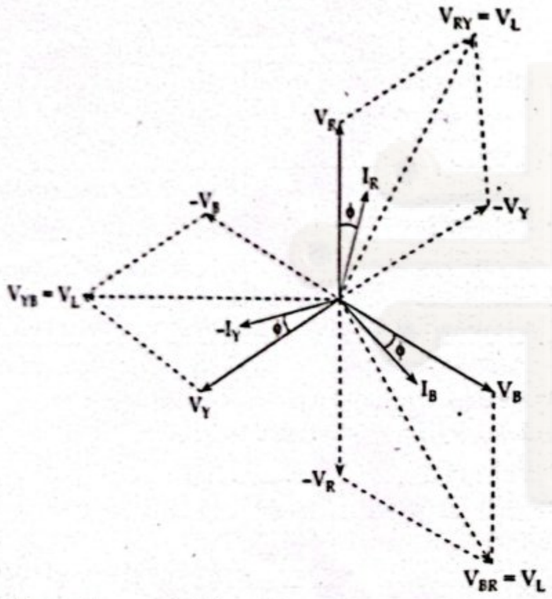

Fig: Phasor diagram of a balanced three phase star- connected load

Let, V_R, V_y and V_B be the three phase voltages and I_R, I_y and I_B be the phase currents lagging behind their respective voltage by constant angle in a balanced load. Then, current between lines = line current.

Now,

V_L = Phasor difference between two phase voltages.

or, $V_{RY} = V_R - V_y = (V_R + V_y)\cos 30° = 2\,V_{ph}\cos 30° = 2\,V_{ph} \times \frac{\sqrt{3}}{2}$

$\therefore\ V_{RY} = \sqrt{3}\,V_{ph}$

Similarly,

$V_{YB} = V_{BR} = \sqrt{3}\,V_{ph}$

i.e., Line voltages (V_L) each are equal to $\sqrt{3}$ times the phase voltage.

or, $V_L = \sqrt{3}\,V_{ph}$.

But, line current (I_L) = Phase current (I_{ph})

$\therefore$ Total power (P) = 3 × Phase power

$$= 3 \times V_{ph} \times I_{ph} \times \cos\phi$$

$$= 3 \times \frac{V_L}{\sqrt{3}} \times I_L \times \cos\phi$$

$\therefore\ P = \sqrt{3}\,V_L\,I_L\cos\phi$

NOTE:

i) Line voltages are 120° apart

ii) Line voltages are 30° ahead of their respective phase voltages

iii) The angle between the line currents and the corresponding line voltages is $(30° + \phi)$ assuming current lagging by an angle $\phi°$ (for lagging loads).

Similarly,

Total reactive power, $Q = \sqrt{3}\,V_L\,I_L\sin\phi$

so, Total apparent power or complex power of three phase is,

$S = \sqrt{P^2 + Q^2} = \sqrt{3}\,V_L\,I_L$

Also, $S = P + jQ$

b) Delta (Δ) or mesh connection

In this system, the 'starting' end of one phase is joined to the 'finishing' end of other phase and so on *i.e.* the dissimilar ends of the three phases are joined together in series to form a closed mesh. Let, the system is balanced, then

Voltage between lines (V_L) = phase voltage (V_{ph})

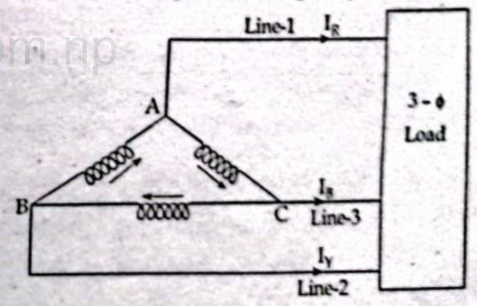

Fig: 3-phase delta connected load.

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and, Line current, I_L = Phasor difference of two phase currents

$$= I_R - I_Y$$
$$= 2I_{ph} \cos 30°$$
$$= 2I_{ph} \times \frac{\sqrt{3}}{2}$$
$$= \sqrt{3}\, I_{ph}$$

i.e., The line currents (I_L) are each equal to $\sqrt{3}$ times the phase current.

$|I_R| = |I_Y| = |I_B| = \sqrt{3} I_{ph}$

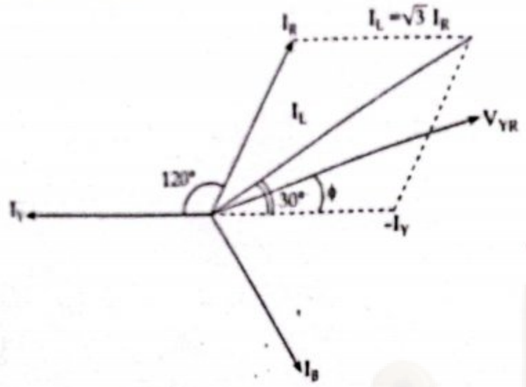


Fig: Phasor diagram for line current (I_L)

∴ Total power, P = 3 × Phase power

$$= 3\, V_{ph}\, I_{ph} \cos \phi$$
$$= 3\, V_{ph} \left(\frac{I_L}{\sqrt{3}}\right) \cos \phi$$
$$= \sqrt{3}\, V_L\, I_L \cos \phi$$

Which is the same as in case of star connected load.

and, Total reactive power, $Q = \sqrt{3}\, V_L\, I_L \sin \phi$

Total apparent power, $S = P + jQ$

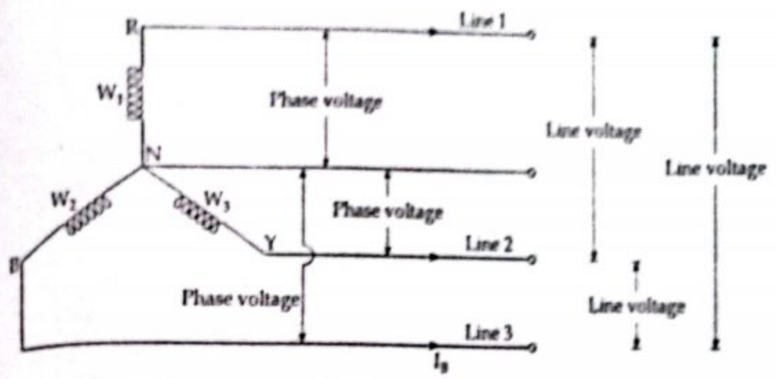
Also,

$$S = \sqrt{P^2 + Q^2} = \sqrt{3}\, V_L\, I_L$$

so, Power factor, $\cos \phi = \frac{P}{\sqrt{3} \cdot V_L \cdot I_L}$

NOTE:

The current flowing through each phase is called phase current (I_{ph}) and the current flowing through each line conductor is called line (I_L). The voltage across each phase is called phase voltage (E_{ph}) and the voltage across two line conductors is called line voltage (E_L).



4.4.1 DIFFERENCE BETWEEN STAR CONNECTION AND DELTA CONNECTION

SN	Star connection (y)	Delta connection (Δ)
a)	$V_L = \sqrt{3}\, V_{ph}$ and $I_L = I_{ph}$	$V_L = V_{ph}$ and $I_L = \sqrt{3}\, I_{ph}$
b)	Line voltages are 120° apart	Same
c)	Line voltages are 30° ahead of the respective phase voltages.	Line currents are 30° behind the respective phase currents.
d)	Used mostly for high voltage 3-phase motors.	Used mostly for small and low voltage 3-phase motors.
e)	This type of connections are not suitable for rotary converters.	It is the only connection suitable for rotary converters.
f)	Star connected transformers work less satisfactorily.	Delta connected transformers work more satisfactorily.
g)	In a balanced system, the resultant potential at the neutral point is zero.	In a balanced system, the resultant current in the closed circuit is zero.
h)	System can be arranged to suit both lighting and power circuits simultaneously.	System can be arranged only to suit either lighting or power.
i)	There is a neutral or star point.	No neutral point.
j)	Low insulation required as phase voltage is low.	Heavy insulation is required as phase voltage is equal to line voltage.
k)	3-phase, 4-wire system	3-phase, 3-wire system
l)	In power transmission, star connected system is general and typical to be used.	In power distribution and industries, delta connection is general and typical to be used.

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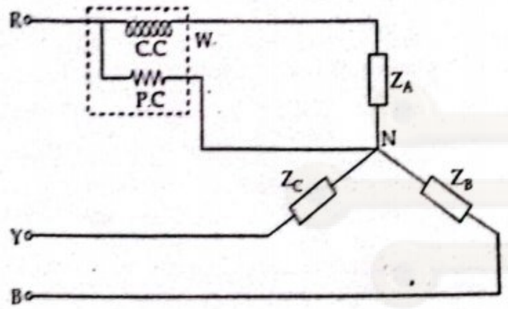
4.5 THREE PHASE POWER AND ITS MEASUREMENT BY SINGLE AND TWO WATTMETER METHOD

For measuring power in a 3-phase circuit, one wattmeter method, two wattmeter method and three wattmeter method are available. Wattmeter is an instrument to measure electric power in watts. A wattmeter provides accurate measurement of true power, reactive power, power factor and many others over a broad range of wave shapes, frequencies and power factor.

a) One or single wattmeter method

A single phase wattmeter consists of two coils namely the current coil and the pressure coil. The current coil is connected in series with the line and thus carries the line current. The pressure coil is connected in parallel with the line. The wattmeter reading gives the power per phase. In this method, a single wattmeter is used to obtain the two readings which are obtained by two wattmeter by the two wattmeter method.



The objection to this method is that without additional measurements, one never can be sure that the three voltages are equal. It should be kept in mind that this method is not of as much universal application as the two wattmeter method because it is restricted to fairly balanced loads only. However, it may be conveniently applied, for instance, when it is desired to find the power input to a factory motor in order to check the load upon the motor.

b) Two wattmeter method

In this method, the current coils of the two wattmeters are inserted in any two lines and the potential coil of each joined to the third line. It can be proved that the sum of the instantaneous powers indicated by W_1 and W_2 gives the instantaneous power absorbed by a three loads, L_1, L_2 and L_3.

Let us consider the connection diagram for star connected load.

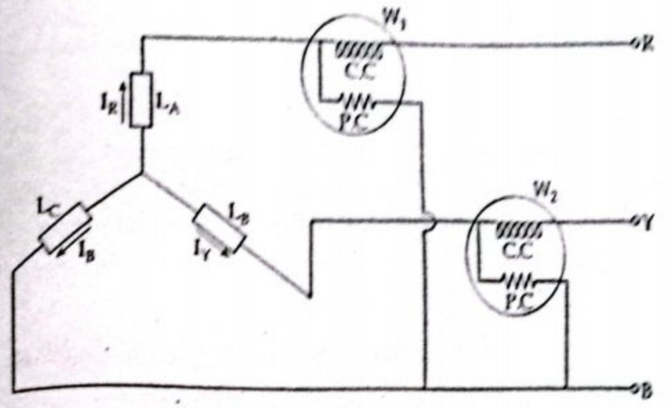


Fig: Two wattmeter method for measuring power in star connected circuit

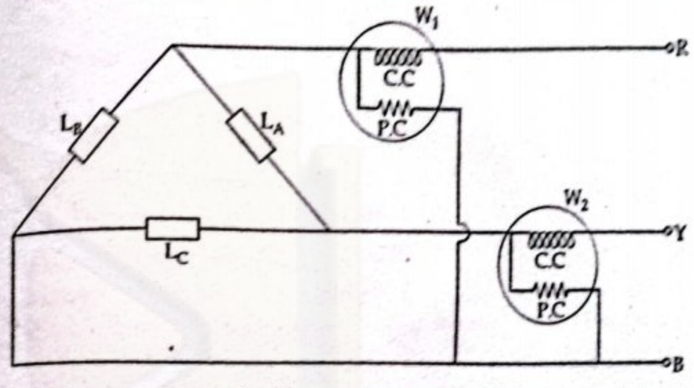


Fig: Two wattmeter method for measuring power in delta connected circuit.

For wattmeter W_1;

The instantaneous current through its current coil is i_R and instantaneous voltage across. Its pressure coil is $e_{RB} = e_R - e_B$

$\therefore$ Instantaneous power read by it = $i_R (e_R - e_B)$ (1)

For wattmeter W_2;

The instantaneous current through its current coil is i_Y and the voltage across its pressure coil is $e_{YB} = e_Y - e_B$

$\therefore$ Instantaneous power read by it = $i_Y (e_Y - e_B)$ (2)

Now, total instantaneous power read by two watlmeters, W_1 and W_2.

i.e., W_2, *i.e.*, $W_1 + W_2 = i_R (e_R - e_B) + i_Y (e_Y - e_B)$

$= i_R e_R + i_Y e_Y - e_B (i_R + i_Y)$ (3)

But, $i_R + i_Y + i_B = 0$ (From Kirchoff's point law)

or, $i_R + i_Y = - i_B$ (4)

From equation (3) and (4),

$W_1 + W_2 = i_R e_R + i_Y e_Y + i_B e_B$

$= P_R + P_Y + P_C$

= Total instantaneous power

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$\therefore \quad W_1 + W_2 = \sqrt{3}\, EL \cos\phi$

where, P_R, P_Y and P_B are the respective power absorbed by load currents i_R, i_Y and i_B.

The proof is true whether the load is balanced or unbalanced. If the load is Y-connected, it should have no neutral connection (*i.e.*, 3 – ϕ, 3 – wire connected) and if it as a neutral connection (*i.e.*, 3 – ϕ, 4 –wire connected) then it should be exactly balanced so that in each case there is no neutral current, i_N otherwise kirchoff's current law will give $i_N + i_R + i_Y + i_B = 0$.

Power factor angle

The phasor diagram for the balanced star-connected 3- phase load is shown in figure below.

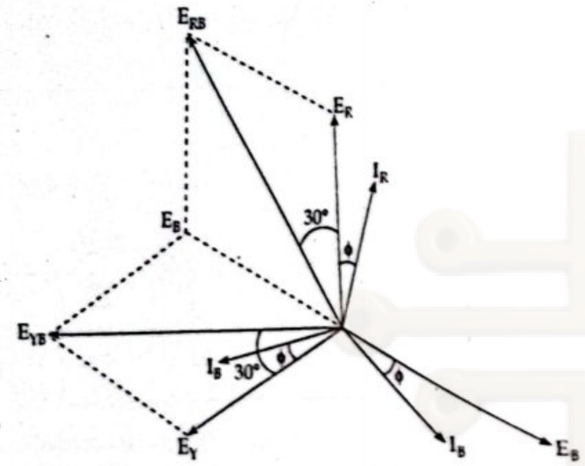

Fig: Phasor diagram for two wattmeter method of measuring power in a 3-phase circuit.

For wattmeter W_1, the rms current through its current coil is I_R and the rms voltage across its pressure coil is E_{RB}.

For wattmeter W_2, the rms current through its current coil is I_Y and the rms voltage across its pressure coil is E_{YB}

However, the magnitude of each voltage is $\sqrt{3}$ E, where E is the line to neutral voltage.

From the phasor diagram,

i) The phase angle between E_{RB} and I_R is $(30° + \phi)$

ii) The phase angle between E_{YB} and I_Y is $(30° - \phi)$

where, ϕ is the power factor angle of load.

$\therefore \quad W_1 + W_2 = \sqrt{3}\, E\, I_R \cos(30° + \phi) + \sqrt{3}\, EI_Y \cos(30° - \phi)$

But, $I_R = I_Y = I_B = I$

so, $W_1 + W_2 = \sqrt{3}\, EI\, [\cos(30° + \phi) + \cos(30° - \phi)]$

$$= \sqrt{3}\, EI \times 2 \times \frac{\sqrt{3}}{2} \times \cos\phi$$

$$= 3\, EI \cos\phi \qquad (5)$$

Now,

$$W_1 - W_2 = \sqrt{3}\, EI\, [\cos(30° + \phi) - \cos(30° - \phi)]$$

$$= \sqrt{3}\, EI\, [-2 \sin 30° \cdot \sin\phi]$$

$$= \sqrt{3}\, EI\, (-2)\left(\frac{1}{2}\right) \sin\phi$$

$$= -\sqrt{3}\, EI \sin\phi \qquad (6)$$

From equations (5) and (6),

$$\frac{W_1 - W_2}{W_1 + W_2} = -\frac{\sqrt{3}\, EI \sin\phi}{3\, EI \cos\phi} = -\frac{\tan\phi}{\sqrt{3}}$$

or, $\tan\phi = \sqrt{3}\, \frac{(W_2 - W_1)}{W_1 + W_2}$

i) If $\phi = 0$, $W_1 = W_2 = V_L I_L \cos 30°$

ii) If $\phi \angle 60°$, then both W_1 and W_2 are positive and total power equals $(W_1 + W_2)$

iii) If $\phi = 60°$, then $W_2 = 0$ and W_1 = +ve

iv) If $\phi > 60°$, *i.e.*, p.f. of load is less than 0.5, then W_1 is negative. In such a case, the connections of either the current coil or the pressure coil are made reversed and then readings are taken. Under the condition, W_1 reading is taken as negative while calculating the power or power factor.

ϕ	0°	60°	90°
cos ϕ	1	0.5	0
W_1	+Ve	+Ve	+Ve
W_2	+ Ve	0	-Ve
	$W_1 = W_2$		$W_1 = W_2$

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SOME SOLVED EXAMPLES

NOTE:

The quantities given in the question is always a line quantities, unless, it is stated other (phase).

1. Three similar coils each of resistance 20 Ω and inductance 0.5H are connected in (i) star (ii) delta to a 3-phase, 50 Hz, 400 V supply, calculate the line current and the total power absorbed.

Solution:

$R_{ph} = 20\ \Omega$

$X_L = 2\pi fL = 2\pi \times 50 \times 0.5 = 157\ \Omega$

$Z_{ph} = \sqrt{20^2 + 157^2} = 158.3\ \Omega$

$\cos\phi = \frac{R}{Z} = \frac{20}{158.3} = 0.1264$ (lag)

a) In star connection

$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 230.9\ V$

$I_L = \frac{V_{ph}}{Z_{ph}} = I_{ph} = \frac{230.9}{158.3} = 1.457\ A$

$P = \sqrt{3}\, V_L I_L \cos\phi = \sqrt{3} \times 400 \times 1.459 \times 0.1264 = 127.8\ W.$

b) Delta connection,

$V_{ph} = V_L = 400\ V$

$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{400}{158.3} = 2.528\ A$

$I_L = \sqrt{3}\, I_{ph} = \sqrt{3} \times 2.528\ A = 4.38\ A$

and, $P = \sqrt{3}\, V_L I_L \cos\phi = \sqrt{3} \times 400 \times 4.38 \times 0.1264 = 383.6\ W$

2. A balanced star-connected load of (8 + j6) Ω per phase is connected to a three phase 230 V supply. Find the line current, p.f., active reactive and total volt amps.

Solution:

$Z_{ph} = (8 + j6)\ \Omega = 10\ \angle 36.87°$

$V_L = 230\ V$

For Y connection,

$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{230}{\sqrt{3}} = 132.79\ V$

i) $I_L = I_{ph} = \frac{V_{ph}}{R_{ph}} = \frac{132.79}{10\ \angle 36.87°} = 13.28\ \angle 36.87°\ A$

ii) Power factor = cos (+ 36.87°) = 0.80 (lag)

iii) Active power $= \sqrt{3}\, V_L I_L \cos\phi$
$= \sqrt{3} \times 230 \times 13.28 \times 0.8$
$= 4{,}232\ W$

iv) Reactive power $= \sqrt{3}\, V_L I_L \sin\phi$
$= \sqrt{3} \times 230 \times 13.28 \times 0.6$
$= 3{,}174\ VAr$

v) Total volt-amp $= \sqrt{3}\, V_L I_L$
$= \sqrt{3} \times 230 \times 13.28$
$= 5{,}290\ VA$

3. Three equal star-connected inductors take 8 KW at a power factor 0.8 when connected across a 460 V, 3-phase, 3-wire supply. Find the circuit constants of the load per phase.

Solution:

$P = \sqrt{3}\, V_L I_L \cos\phi$

or, $8{,}000 = \sqrt{3} \times 460 \times I_L \times 0.8$

$\therefore\ I_L = 12.55\ A = I_{ph}$

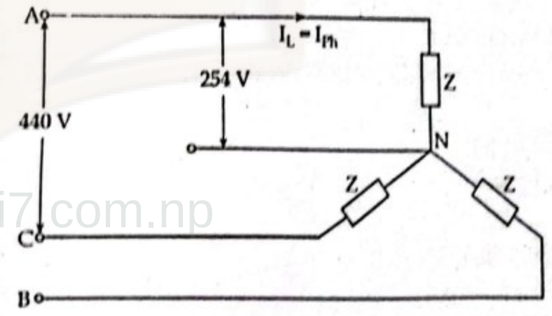
and, $V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{460}{\sqrt{3}} = 265\ V$

$I_{ph} = \frac{V_{ph}}{Z_{ph}}$

or, $Z_{ph} = \frac{V_{ph}}{I_{ph}} = \frac{265}{12.55} = 21.1\ \Omega$

$\therefore\ R_{ph} = Z_{ph} \cos\phi = 21.1 \times 0.8 = 16.9\ \Omega$

$\therefore\ X_{ph} = Z_{ph} \sin\phi = 21.1 \times 0.6 = 12.66$



4. Each phase of a star connected load consists of a nonreactive resistance of 100 Ω in parallel with a capacitance of 31.8 μF. Calculate the line current, power absorbed, the total KVA and the power factor when connected to a 416 V, 3-phase, 50 Hz supply.

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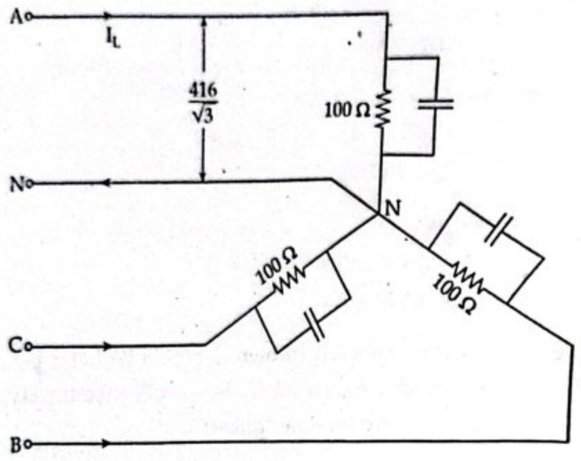
Solution:



$$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{416}{\sqrt{3}} = 240\angle 0° = (240 + j0)\ V$$

Admittance of each phase,

$$Y_{ph} = \frac{1}{R} + jX_C$$

$$= \frac{1}{100} + j\,2\pi \times 50 \times 31.8 \times 10^{-6}$$

$$= 0.01 + j\,0.01$$

$$\therefore \quad I_{ph} = V_{ph} \cdot Y_{ph}$$

$$= 240\,(0.01 + j0.01)$$

$$= (2.4 + j2.4)\ A$$

$$= 3.39\angle 45°\ A$$

Since, $I_{ph} = I_L$

$$\therefore \quad I_L = 3.39\angle 45°\ A$$

Power factor, $= \cos\phi = \cos 45° = 0.707$ (lead)

Now,

$$V_{ph} = (240 + j0)$$

$$I_{ph} = 2.4 + j2.4$$

so, $P = V_{ph} \cdot I_{ph}$

$$= (240 + j0)\,(2.4 + j2.4)$$

$$= 240 \times 2.4 - j2.4 \times 240$$

$$= 576 - j576$$

$$= 814.4\angle -45°\ \text{per phase}$$

Hence,

Total power = 3 × 576 = 1.728 KW

Total volt amper = 814.4 × 3 = 2.433 KVA.

5 **The load connected to a three phase supply contains three similar impedances connected in star. The line currents are 50 A and the KVA and KW inputs are 50 and 27 respectively. Find the line and phase voltages, KVAr input and the resistance and reactance of each coil.**

Solution:

$I_L = 50A$, KW = 27

KVA = 50

$$KVAr = \sqrt{kVA^2 - kW^2} = \sqrt{50^2 - 27^2} = 42$$

As the load is star connected,

Phase current, $I_{ph} = I_L = 50$ A.

Now, $\sqrt{3}\,V_L\,I_L \cos\theta = 27 \times 10^3$ [∵ Active power formula]

or, $$\cos\theta = \frac{27 \times 10^3}{\sqrt{3} \times V_L \cdot I_L} \quad (1)$$

and, $\sqrt{3}\,V_L\,I_L \sin\theta = 42 \times 10^3$ [∵ reactive power formula]

or, $$\sin\theta = \frac{42 \times 10^3}{\sqrt{3} \times V_L\,I_L} \quad (2)$$

Dividing equation (2) by (1);

or, $$\tan\theta = \left(\frac{42}{27}\right) = \frac{X}{R}$$

or, $$\frac{X}{R} = 1.56$$

$$\therefore \quad X = 1.56R$$

Again,

$$\sqrt{3}\,V_L\,I_L = 50 \times 10^3$$

or, $$V_L = \frac{50 \times 10^3}{\sqrt{3} \times 50}$$

$$\therefore \quad V_L = 577.35\ V$$

$$\therefore \text{ Phase voltage, } V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{577.37}{\sqrt{3}} = 333.33\ V$$

If Z be the impedance per phase,

$$Z = \frac{V_{ph}}{I_{ph}}$$

or, $R^2 + X^2 = (6.67)^2$

or, $R^2 + (1.56\,R)^2 = 44.49$

or, $3.4336\,R^2 = 44.49$

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or, $R^2 = 12.9572$

$\therefore$ $R = 3.6\ \Omega$

and, $X = 1.56 \times R = 1.56 \times 3.6 = 5.62\ \Omega$.

6. Two wattmeters connected to measure the input to a balanced three phase circuit indicate 2,500 W and 500 W respectively. Find the power factor of the circuit (i) when both reading are positive and (ii) when the latter reading is obtained after reversing the connections to the current coil.

Solution:

$W_1 = 2{,}500$ W

$W_2 = 500$ W

i) When both readings are positive,

$$P = W_1 + W_2 = 2{,}500 + 500 = 3{,}000 \text{ W}$$

$$\therefore \quad \tan\phi = \sqrt{3}\left(\frac{W_1 - W_2}{W_1 + W_2}\right) = \sqrt{3}\left(\frac{2{,}000}{3{,}000}\right) = 1.1547$$

so, $\phi = \tan^{-1}(1.1547) = 49.11°$

and, Power factor, $\cos\phi = \cos(49.11°) = 0.6547$

ii) When W_2 is negative,

$$P = W_1 + W_2 = 2500 + (-500) = 2{,}000 \text{ W}$$

$$\therefore \quad \tan\phi = \sqrt{3}\left(\frac{W_1 - W_2}{W_1 + W_2}\right) = \sqrt{3}\left(\frac{3{,}000}{2{,}000}\right) = 2.598$$

or, $\phi' = \tan^{-1}(2.598) = 68.95°$

and, Power factor, $\cos\phi' = \cos(68.95°) = 0.3592$.

7. The power in a 3-phase circuit is measured by two watt meters. If the input power is 100 KW, and power factor is 0.66 (lagging), what will be the reading of each watt meter? And for what power factor will one of the watt meters read zero?

Solution:

$\cos\phi = 0.66$

or, $\phi = \cos^{-1}(0.66)$

$\therefore$ $\phi = 48.7°$

and, $\tan\phi = 1.1383$

$$\therefore \quad \tan\phi = \sqrt{3}\frac{(W_1 - W_2)}{(W_1 + W_2)} = \sqrt{3}\frac{(W_1 - W_2)}{100 \text{ kW}}$$

or, $W_1 - W_2 = 65.7$ KW (1)

and, $W_1 + W_2 = 100$ KW (2)

On solving (1) and (2),

or, $W_1 = 82.85$ KW

and, $W_2 = 17.15$ KW

If $W_2 = 0$, then $\tan\phi = \sqrt{3}\left(\frac{W_1}{W_1}\right) = \sqrt{3} = \tan 60°$

so, Power factor, $\cos\phi' \cos(60°) = 0.50$.

8. Three equal impedances, each (16 + j12) ohms are connected in star, and a 400 V, 3- phase AC supply is applied to load. Find (i) phase and line currents (ii) power factor of the load (iii) the readings of the wattmeters, when two wattmeter method is used to measure the power.

Solution:

$V_L = 100$ V

$Z_{ph} = (16 + j12)\ \Omega = 20\ \angle 36.87°$ A

$$V_{ph} = \frac{400}{\sqrt{3}} = 230.9 \text{ V}$$

$$\therefore \quad I_L = I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{230.9}{20}\angle 36.87° = 11.53\ \angle -36.87° \text{ A}$$

i) Line current, $I_L = I_{ph} = 11.53$ A

ii) Power factor, $\cos\phi = \cos(36.87°) = 0.8$ (lag)

iii) Total power, $P = \sqrt{3}\, V_L\, I_L \cos\phi$

$= \sqrt{3} \times 400 \times 11.53 \times 0.8$

$= 4800$ W

Now,

$\tan\phi = \tan 36.87° = 0.75$

or, $0.75 = \sqrt{3}\left(\frac{W_1 - W_2}{W_1 + W_2}\right)$

or, $0.75 = \sqrt{3}\left(\frac{W_1 - W_2}{4{,}800}\right)$

or, $W_1 - W_2 = 4{,}800 \times \frac{0.75}{\sqrt{3}}$

$\therefore$ $W_1 - W_2 = 2{,}078$ W (1)

and, $W_1 + W_2 = 4{,}800$ W (2)

From (1) and (2);

$\therefore$ $W_1 = 3{,}439$ W

$\therefore$ $W_2 = 1{,}361$ W

9. A balanced, star connected load is supplied from a symmetrical 3-phase, 400 V system. The current in each phase is 30A, and lags by 30° behind the voltage. Find (i) impedance in each phase (ii) the total power drawn. Draw the phasor diagram also.

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Solution:

For star- connected load,

$V_L = 400$ V

$$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 230.94 \text{ V}$$

$I_{ph} = 30$ A (lagging V_{ph} by 30°)

i) $$Z_{ph} = \frac{V_{ph}}{I_{ph}} = \frac{230.94}{30} = 7{,}698\ \Omega$$

ii) Total power,

$$P = 3V_{ph} I_{ph} \cos\phi$$
$$= 3 \times 230.94 \times 30 \times \cos 30°$$
$$= 3 \times 230.94 \times 30 \times 0.866$$
$$= 18{,}000 \text{ W}$$

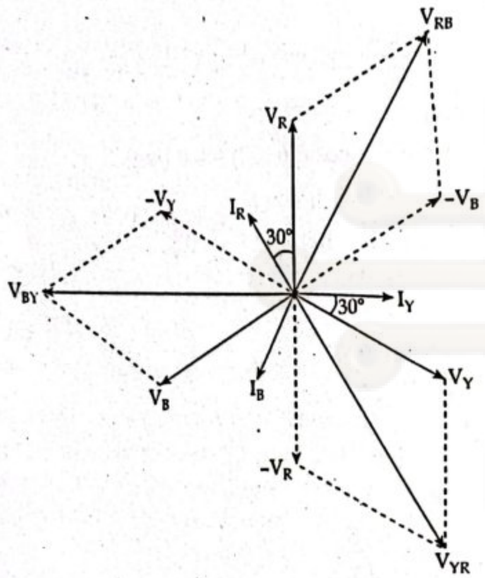

Fig: Phasor diagram

BOARD EXAM SOLVED PROBLEMS

1. Write short notes on advantages of 3 – ϕ system. **[13/F, 15/S, 18/F, 18/S]**

Solution:

i) For a given size of frame, a poly-phase generator or motor has greater output than that of a single-phase.

ii) Efficiency of a poly-phase generator is higher than that of a single phase generator.

iii) Poly phase motors are more compact whereas single phase motors require more space for conveying the same power.

iv) Power in a poly-phase system never falls to zero whereas it may fall to zero in a single phase system.

v) Poly-phase motors possess uniform torque where as single phase motors (except commutator motors) possess pulsating torque.

vii) A poly-phase transmission line requires much less conductor than a single phase line.

viii) A poly-phase generators works in parallel without any difficulty.

2. Explain with a neat diagram the two wattmeter methods for the measurement of three phase power and determine watt meter reading when it is connected to resistive load.

OR

Write short notes on two wattmeter method of power measurement. **[13/S, 14/F, 15/F, 16/F, 16/S, 17/F, 19/S]**

Solution: See the definition part 3.5. b.

3. Explain how the three phase voltage is generated. **[16/S, 18/F]**

Solution: See the definition part 4.2.

4. Difference between single phase and three phase system. **[16/S, 17/S]**

Solution: See the definition part 3.3.

5. List out the advantages and disadvantages of star connected supply. **[17/S]**

Solution:

Advantages:

i) Phase to earth voltage is $\left(\frac{1}{\sqrt{3}}\right)$ times the phase to phase voltage. The size of insulation required is reduced. Hence the cost is also reduced.

ii) Protection relays can easily detect a ground fault in the grounded star connected side. Ground fault detection is easy.

iii) By grounding star connected winding, arcing ground problem is eliminated. *i.e.*, shifting of neutral can be reduced.

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iv) Both phase and line voltages are available for four wire supply. This is useful for low voltage distribution where both three phase and single phase loads have to be supplied.

v) Large unbalanced loads can be balanced.

Disadvantages

i) The speed of star connected motors are slow as they receive $\frac{1}{\sqrt{3}}$ voltage.

ii) Construction cost is more expensive (4-conductors).

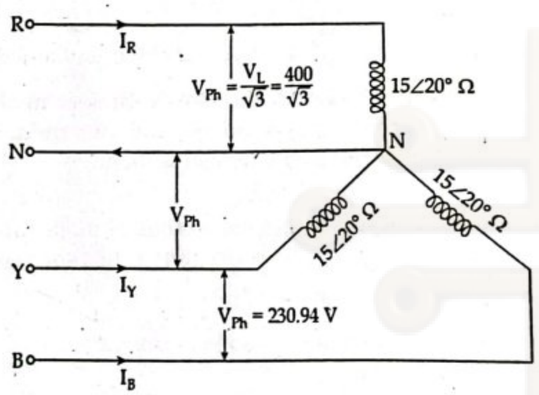
6. Three similar coils having impedance of 15 ∠20° Ω are connected in star to 400 V, 3 – ϕ, 50 Hz supply. Calculate [17/F]

i) Line and phase current

ii) Power factor

iii) Total 3 – ϕ power

Solution:



Impedance, $Z_{ph} = 15 \angle 20° \Omega$

Frequency, f = 50 Hz

Phase voltage, $V_{ph} = 400$ V

In star connection,

$$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 230.94 \text{ V}$$

i) Phase current,

$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{230 \angle 0°}{15 \angle 20°} = 15.34 \angle -20° \text{ A}$$

Line current, $I_L = I_{ph} = 15.34 \angle -20°$ A

ii) Total 3-phase power,

$$P = \sqrt{3}\, V_L I_L \cos\phi = \sqrt{3} \times 400 \times 15.34 \times 0.9396$$
$$= 9985.94 \text{ W} = 9.9859 \text{ KW}.$$

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7. A three phase balanced star connected load with 6 + j8 ohm per phase is supplied by 400 V, 3-phase source. Find the line current and phase currents and the total power dissipated in the load. [14/F, 16/S]

Solution:

Impedance, $Z_{ph} = 6 + j8$ ohm $= 10 \angle 53.13° \Omega$

$V_L = 400$ V

$$V_{ph} = \frac{400}{\sqrt{3}} = 230.94 \text{ V}$$

i) Line current (I_L) = Phase current (I_{ph})

$$= \frac{V_{ph}}{Z_{ph}} = \frac{230.94}{10 \angle 53.13°} = 23.094 \angle -53.13° \text{ A}$$

ii) Total power dissipated, P =?

We have,

Power factor, $\cos\phi = \cos(-53.13°) = 0.60$

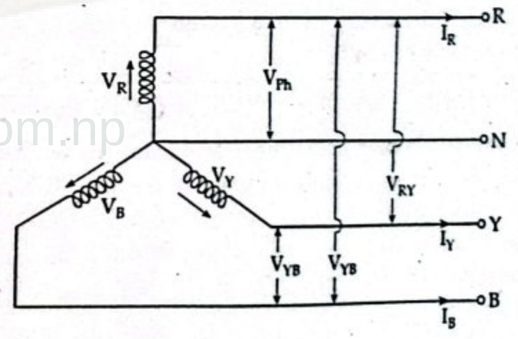
so, $P = \sqrt{3}\, V_L I_L \cos\phi = \sqrt{3} \times 400 \times 23.094 \times 0.60 = 95648.235$ W

$\therefore$ P = 95.6482 KW.

8. Derive the relationship between line and phase voltage in three phase balanced star connection. [14/F]

Solution:

Suppose due to load impedance, the current lags the applied voltage in each phase of the system by an angle ϕ. Considering the system is balanced, the magnitude of current and voltage of each phase is same. In the balanced star system, magnitude of phase voltage in each phase is V_{ph}. Therefore, $V_R = V_y = V_B = V_{ph}$ and in star connection, line current is same as phase current *i.e.*, $I_R = I_y = I_B = I_L$. Let us say the voltage across R and y terminal of the star connected circuit is V_{RY}, the voltage across B and R terminal of star connected circuit is V_{BR} and voltage across y and B terminal be V_{yB}.



From figure,

$$V_{RY} = V_R + (-V_Y)$$

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$V_{YB} = V_Y + (-V_B)$

$V_{BR} = V_B + (-V_R)$

Now, As angle between V_R and V_Y is 120° (electrical), the angle between V_R and $-V_Y$ is 180° – 120° = 60° (electrical).

so, $V_L = |V_{RY}| = \sqrt{V_R^2 + V_Y^2 + 2V_R V_Y \cos 60°} = \sqrt{V_{ph}^2 + V_{ph}^2 + 2V_{ph} \cdot \frac{1}{2}}$

$\therefore \quad V_L = \sqrt{3}\, V_{ph}$

Thus, for star connected system, line voltage = $\sqrt{3}$ × Phase voltage.

9. 3 similar coils each having a resistance of 20 Ω and an inductance of 0.05 H are connected in star to a 3-phase, 50 Hz supply with 400V between lines, calculate the total power absorbed and the line current in each case. Find the magnitude of current flowing in the neutral wire. [12/F, 16/ F]

Solution:

Resistance (R) = 20 Ω

Inductance (L) = 0.05 H

Inductive reactance (X_L) = $2\pi fL = 2\pi \times 50 \times 0.05 = 15.707$

Impedance (Z) = $\sqrt{R^2 + X_L^2} = \sqrt{20^2 + (15.707)^2} \angle \tan^{-1}(15.707/20)$

$= 25.44 \angle 38.144°\ \Omega$

Line voltage, (V_L) = 400 V

Frequency, (f) = 50 Hz

Phase voltage, $(V_{ph}) = \frac{V_L}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 230.94$ V

i) Phase current, $I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{230.94 \angle 0°}{25.44 \angle 38.144°} = 9.0778 \angle -38.14°$ A

so, Line current, $I_L = I_{ph} = 9.0778 \angle -38.14°$ A

ii) Power factor, $\cos\phi = \cos(-38.14°) = 0.786$

iii) Total power, $P = 3 \times V_{ph} \times I_{ph} \cos\phi$

$= 3 \times 230.94 \times 9.0778 \times 0.786$

$= 4{,}946.54$ W

iv) Neutral current, $I_N = \sqrt{(9.077)^2 + (9.077)^2 + (9.077)^2} = 15.721$ A.

10. A three phase balanced load connected in star draws a total power of 20 KW at 0.8 if lag when connected to 3 – ϕ, 400 V, Hz supply calculate. [13/F]

i) Resistance of coil

ii) Inductance of coil

Solution:

Power (P) = 20 KW = 20,000 W

Power factor, p.f. = 0.80

or, $\cos\phi = 0.80$

$\therefore \quad \phi = 36.86°$

$V_L = 400$ V

$V_{ph} = \frac{400}{\sqrt{3}} = 230.94$ V

Phase current,

$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{230.94}{Z_{ph}} = I_L$

or, $I_{ph} = I_L = \frac{230.94}{Z_{ph}}$ (1)

We have,

$P = 3\, V_{ph}\, I_{ph} \cos\phi$

or, $20{,}000 = 3 \times 230.94 \times I_{ph} \times 0.80$

or, $I_{ph} = \frac{20{,}000}{3 \times 230.94 \times 0.80}$

$\therefore \quad I_{ph} = 26.084$ A $= I_L$

Hence,

$Z_{ph} = \frac{V_{ph}}{I_{ph}} = \frac{230.94}{36.084} = 6.40\ \Omega$ (2)

Also, $Z_{ph} = \sqrt{R^2 + X_L^2}$

or, $6.40 = \sqrt{R^2 + (2\pi \times 50 \times L)^2}$

or, $6.40 = \sqrt{(R^2 + 100\pi\, L)^2}$

or, $(6.4)^2 = R^2 + 98696.04\, L^2$

or, $40.96 = R^2 + 98696.04\, L^2$ (3)

Now,

$R_{ph} = Z_{ph} \cdot \cos\phi = 6.40 \times 0.80 = 5.12\ \Omega$

and, $X_{ph} = Z_{ph} \sin\phi = 6.40 \times \left(\sqrt{1^2 - 0.80^2}\right) = 6.40 \times 0.60 = 3.84\ \Omega$

so, From equation (3);

or, $40.96 = (5.12)^2 + 98{,}696.04L^2$

or, $14.7456 = 98{,}696.04L^2$

or, $L^2 = 0.0001494$

or, $L = 0.01222$ H

$\therefore \quad L = 12.223$ mH

i) Resistance of coil = R = 5.12 Ω

ii) Inductance of coil = L = 0.0122 H.

11. The star connected stator of a three-phase, 50 Hz alternator supplies a balanced delta-connected load. Each phase of the load consists of a coil of resistance 15 ohm and inductance 36 mH and

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the phase voltage generated by the alternator is 231 V. calculate (i) phase and line current (ii) load power. [13/F]

Solution:

$f = 50$ Hz

$R = 15\ \Omega$

$L = 36$ mH $= 0.036$ H

$V_{ph} = 231$ V

$V_L = V_{ph}$ (load is an delta connection)

$X_L = 2\pi fL = 2\pi \times 50 \times 0.036 = 11.309\ \Omega$

so, $Z_{ph} = \sqrt{R^2 + X_L^2}\ \angle \tan^{-1}(X_L/R)$

$= \sqrt{15^2 + 11.309^2}\ \angle \tan^{-1}(11.309/15)$

$= 18.786\ \angle 37.014°\ \Omega$

Now, Line voltage of alternator,

$V_L = \sqrt{3} \times V_{ph} = \sqrt{3} \times 231 = 400$ V

Phase current of load,

$$I_{ph} = \frac{I_L}{\sqrt{3}} = \frac{V_{ph}}{Z_{ph}} = \frac{400\ \angle 0°}{18.78\ \angle 37.014}$$

$\therefore\ I_{ph} = 21.30\ \angle -37.014°$ A

In delta load,

i) $I_{ph} = \frac{I_L}{\sqrt{3}} = 21.30\ \angle -37.01°$ A

ii) Power factor, $\cos\phi = \cos(-37.014°) = 0.80$

iii) Power delivered, $P = 3 \times V_L \times I_L \times \cos\phi$

$$= 3 \times 400 \times \frac{21.30}{\sqrt{3}} \times 0.80$$

$= 1,1805.658$ W

$= 11.805$ KW

Hence the power delivered to the load is 11.805 KW.

12. Three equal impedances having resistance 8 Ω and inductive reactance 6 Ω are connected in delta and connected to 230 V, 3-phase wire 3 - ϕ y (W Y E connected) source, then calculate,

i) Phase and line current

ii) Power factor

iii) Power consumed [13/S]

Solution:

$R = 8\ \Omega$

$X_L = 6\ \Omega$

$Z = \sqrt{R^2 + X_L^2} = \sqrt{8^2 + 6^2}\ \angle \tan^{-1}(6/8) = 10\ \angle 36.869°\ \Omega$

Since, the load is connected in delta connection and supply is star connection.

From star connection supply,

$V_{ph} = 230$ V

$V_L = \sqrt{3} \times V_{ph} = \sqrt{3} \times 230$ V $= 398.37$ V

and, $I_{ph} = I_L$

Thus, in delta connection,

$V_{ph} = V_L$

$\therefore\ V_{ph} = 398.37$ V

and,

i) $$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{398.37\ \angle 0°}{10\ \angle 36.869°} = 39.84\ \angle -36.869°\ \text{A}$$

so, $I_L = \sqrt{3} \times I_{ph} = \sqrt{3} \times 39.84\ \angle -36.869° = 69\ \angle -36.86°$ A

ii) Power factor, p.f. $= \cos\phi = \cos(-36.869°) = 0.80$

iii) Total power consumed,

$P = \sqrt{3}\ V_L\ I_L \cos\phi = \sqrt{3} \times 398.37 \times 69 \times 0.80 = 38$ KW

13. A balanced star connected load of (2 + j8) Ω per phase is connected to a 3 - ϕ, 220 V, 50 Hz supply. Find the line current, power factor, power, volt amperes and reactive power. [14/S]

Solution:

Solution: See the solution of example No. 2.

14. Three equal impedance having resistance 20 Ω and inductive reactance 15 Ω are connected in delta connected system. Find

i) Line and phase current

ii) Power factor

iii) Power consumed [15/F]

Solution:

$R = 20\ \Omega$

$X_L = 15\ \Omega$

$\therefore\ Z = \sqrt{R^2 + X_L^2}\ \angle \tan^{-1}(X_L/R)$

$= \sqrt{20^2 + 15^2}\ \angle \tan^{-1}(15/20)$

$= 25\ \angle 36.87°$ ohm

In delta connection,

$V_{ph} = V_L = 400$ V

$$I_{ph} = \frac{I_L}{\sqrt{3}}$$

$$\therefore\ I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{400\ \angle 0°}{25\ \angle 36.87°} = 16\ \angle -36.87°\ \text{A}$$

so,

i) $I_L = \sqrt{3}\ I_{ph} = \sqrt{3} \times 16\ \angle -36.87°$ A $= 27.72\ \angle -36.87°$ A.

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ii) Power factor, p.f. = $\cos\phi = \cos(-36.87°) = 0.80$

iii) Power consumed,

$P = \sqrt{3}\, V_L I_L \cos\phi = \sqrt{3} \times 400 \times 27.72 \times 0.80 = 15{,}363.983$ W

$\therefore$ $P_L = 15.3639$ KW

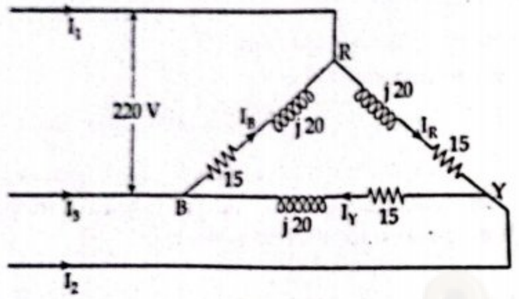
15. A 220 V, 3 - ϕ voltage is applied to a balanced delta connected 3 - ϕ load of phase impedance (15 + 20j) Ω.

i) Find phasor current in each line

ii) Find power consumed per phase

iii) What is the phasor sum of the three line currents? Why does it have this value? **[15/S, 19/F]**

Solution:



$V_{RY} = 220\angle 0°$ V

$V_{BY} = 220\angle -120°$ V

$V_{BR} = 220\angle 120°$ V

$Z = (15 + 20j)\ \Omega = 25\angle 53.13°\ \Omega$

Now,

i) Phase current in each line,

$$I_R = \frac{V_{RY}}{Z} = \frac{220\angle 0°}{25\angle 53.13°} = 8.80\angle -53.13°$$

$$I_Y = \frac{V_{BY}}{Z} = \frac{220\angle -120°}{25\angle 53.13°} = 8.80\angle -173.13°\ \text{A}$$

$$I_B = \frac{V_{BR}}{Z} = \frac{220\angle 120°}{25\angle 53.13°} = 8.80\angle 66.87°\ \text{A}$$

Current in line are,

$I_1 = I_R - I_B = 15.24\angle -83.13°$ A

$I_2 = I_Y - I_R = 15.24\angle -156.87°$ A

$I_3 = I_B - I_Y = 15.24\angle 36.87°$ A

ii) $V_{ph} = V_L = 220$ V (in delta connection)

$I_{ph} = 8.8$ A

so, Power consumed per phase,

$P = V_{ph} I_{ph} \cos\phi = 220 \times 8.8 \times \cos(-53.13°) = 1{,}161.602$ W

Alternative method;

$P = V_L I_L = I_{ph}^2 \cdot R = (8.8)^2 \times 15 = 1{,}161.602$ W

iii) Phasor sum of three line currents

$= I_1 + I_2 + I_3$

$= (15.24\angle -83.13°) + (15.24\angle -156.87°) + (15.24\angle 36.87°) = 0$ A

Phasor sum of three line current is zero because three line currents are equal in magnitude and have a mutual phasor difference of 120°.

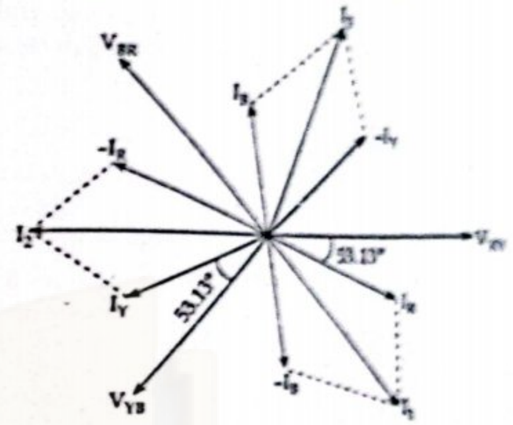


Fig: Phasor diagram

16. A balanced delta connected load of (2 + j3) Ω per phase is connected to the balanced 3-phase, 440 V, 50 Hz supply. Find (i) line current (ii) total active power (iii) total reactive power (iv) total apparent power. **[18/S]**

Solution:

$Z_{ph} = (2 + j3)\ \Omega = 3.61\angle 56.31°$

In delta-connection,

$V_{ph} = V_L = 440$ V

Frequency, $f = 50$ Hz

so, $$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{440\angle 0°}{3.61\angle 56.31°} = 122.23\angle -56.31°\ \text{A}$$

and,

i) $I_L = \sqrt{3}\, I_{ph} = \sqrt{3} \times 122.23\angle 56.31°$ A $= 211.708\angle -56.31°$ A

Hence, line current is $211.708\angle -56.31°$ A

ii) Power factor, p.f. $= \cos\phi$

$= \cos(56.31°)$

$= 0.56$ (lagging)

iii) Total active power $= \sqrt{3}\, V_L I_L \cos\phi$

$= \sqrt{3} \times 440 \times 211.708 \times 0.56$

$= 90.352$ KW

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iv) Total reactive power $= \sqrt{3}\, V_L I_L \sin\phi$
$= \sqrt{3} \times 400 \times 211.708 \times \sqrt{1 - 0.56^2}$
$= 12.151$ KVAr

v) Total apparent power $= \sqrt{3}\, V_L I_L$
$= \sqrt{3} \times 400 \times 211.708$
$= 146.675$ KVA.

17. Three similar coils each having a resistance of 100 Ω and an inductance of 20 mH are connected in (i) star (ii) delta to a 3 – ϕ, 50 Hz with 400 V between lines. Calculate
i) Line current and phase current.
ii) Active, reactive and apparent power. **[18/F]**

Solution:

i) $V_L = 400$ V
$R = 100\ \Omega$
$L = 20$ mH $= 0.002$ H

$Z_{ph} = \sqrt{R^2 + X_L^2} \angle \tan^{-1}(X_L/R) = (R + jX_L)$
$= 100 + j2\pi \times 50 \times 0.02$
$= (100 + j6.283)\ \Omega$
$= 100 \angle 3.596°\ \Omega$

$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 231 \angle 0°$ V

$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{231 \angle 0°}{100 \angle 3.596} = 2.3 \angle -3.596°$ A

$I_L = I_{ph} = 2.3 \angle -3.596°$ A

Active power (P) $= \sqrt{3}\, V_L I_L \cos\phi$
$= \sqrt{3} \times 400 \times 2.3 \times \cos(3.596)$
$= 1,590.35$ W

Reactive power, Q $= \sqrt{3}\, V_L I_L \sin(\phi)$
$= \sqrt{3} \times 400 \times 2.3 \times \sin(0.062°)$
$= 1.745$ VAr

Apparent power, S $= \sqrt{3}\, V_L I_L$
$= \sqrt{3} \times 400 \times 2.3$
$= 1,593.48$ VA

ii) In delta connection,
$V_{ph} = V_L = 400 \angle 0°$ V
$Z_{ph} = 100 \angle 3.596\ \Omega$
$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{400 \angle 0°}{100 \angle 3.596} = 4 \angle -3.596$ A

$I_L = \sqrt{3}\, I_{ph} = \sqrt{3} \times 4 \angle -3.596 = 6.928 \angle -3.596$ A

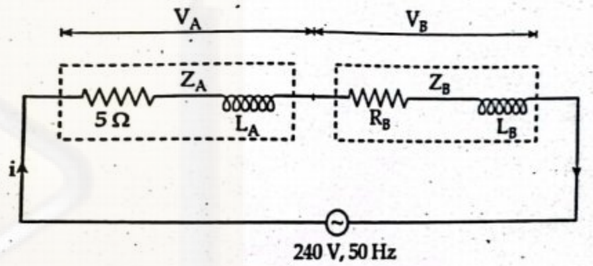
Active power, P $= \sqrt{3}\, V_L I_L \cos\phi$
$= \sqrt{3} \times 400 \times 6.928 \times \cos(3.596)$
$= 4790$ W

Apparent power, S $= \sqrt{3}\, V_L I_L$
$= \sqrt{3} \times 400 \times 6.928$
$= 5,000$ VA

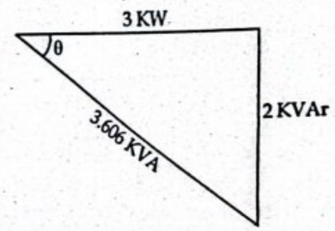
Reactive power, Q $= \sqrt{3}\, V_L I_L \sin\phi$
$= \sqrt{3} \times 400 \times 6.928 \times \sin(0.062)$
$= 5.194$ VAr.

18. Two coils A and B are connected in series across a 240 V, 50 Hz supply. The resistance of A is 5 Ω and the inductance of B is 0.015 H. If the input from the supply is 3 KW and 2 KVAr. Find the inductance of A and the resistance of B. Calculate voltage across each coil. **[19/F]**

Solution:



Active power, P = 3 KW
Reactive power, Q = 2 KVAr
Apparent power, S = ?
We know,
$S = \sqrt{P^2 + Q^2} = \sqrt{3^2 + 2^2}$
or, $S = 3.606$
∴ $S = 3,606$
KVA triangle is,



Circuit current, $i = \frac{S}{V} = \frac{3,606}{240}$
$= 15.025$ A

We know,
$P = i^2 R$
or, $3,000 = (15.025)^2 \times (R_A + R_B)$

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or, $\frac{3,000}{(15.025)^2} = 5 + R_B$

or, $R_B = 13.28 - 5$

$\therefore \quad R_B = 8.28$

Impedance of the whole circuit,

$$Z = \frac{V}{i} = \frac{240}{15.025}$$

$\therefore \quad Z = 15.974\ \Omega$

Now,

$$Z = \sqrt{(X_A + R_A)^2 + (X_B + R_B)^2}$$

or, $Z^2 = (X_A + R_A)^2 + (X_B + R_B)^2$

or, $(15.974)^2 = (X_A + 5)^2 + (X_B + 8.28)^2$

or, $255.168 = (X_A + 5)^2 + (2\pi \times 50 \times 0.15 + 8.28)^2$

or, $255.168 = (X_A + 5)^2 + 168.802$

or, $(X_A + 5)^2 = 255.168 - 168.802$

or, $X_A + 5 = \sqrt{86.366}$

or, $X_A = 9.30 - 5$

or, $X_A = 4.30\ \Omega$

or, $2\pi \times 50 \times L_A = 4.30$

or, $L_A = \frac{4.30}{2\pi \times 50}$

$\therefore \quad L_A = 0.0136\ H$

and $Z_A = \sqrt{R_a^2 + X_A^2} = \sqrt{5^2 + 4.30^2} = 6.60\ \Omega$

Therefore, Potential difference across coil A,

$$= i \cdot Z_A = 15.025 \times 6.60$$

$\therefore \quad V_A = 99.165\ V$

Also,

$$Z_B = \sqrt{R_B^2 + X_B^2}$$
$$= \sqrt{8.282 + (2\pi \times 50 \times 0.015)^2}$$
$$= \sqrt{8.28^2 + 22.206}$$
$$= \sqrt{90.765}$$
$$= 9.527\ \Omega$$

$\therefore$ P.D across coil B = $iZ_B = 15.025 \times 9.527$

$\therefore \quad V_B = 143.144\ V.$

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CHAPTER 5
ELECTRICAL MACHINES

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5.1 REVIEW OF MAGNETIC CIRCUITS

The closed path followed by magnetic flux is called a magnetic circuit.

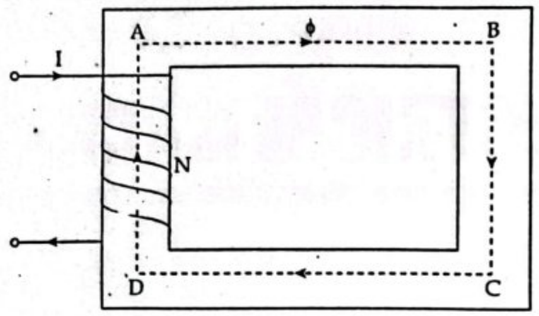


Fig: Simple magnetic circuit.

A magnetic circuit consists of a structure composed for the most part of high permeability magnetic material. The presence of high permeability material causes the magnetic flux to be confined to the paths defined by the structure, much as currents are confined to the conductors of an electric circuit. A simple example of magnetic circuit is shown above. The core is assumed to be composed of magnetic material whose permeability is much greater than that of the surrounding air. The core is of uniform cross section and is excited by a winding having N turns and carrying a current of I amperes. This winding develops a magnetic field in the core. The magnetic field can be visualized in terms of flux lines, which form closed loops interlinking with the winding.

According to basic law of magnetic circuit, called the ampere's circuital law, the line integral of μ around a closed path is equal to the net current enclosed by that path

i.e., $\oint H \cdot dL = \Sigma I$

Magnetic circuit concepts

i) Magetomotive force (mmf) of the magnetic circuit is the magnetic potential difference that tends to force flux around the magnetic circuit and is analogous to the electromotive force (emf) in an electric circuit.
i.e., mmf = NI ampere-turns

ii) Ampere- turn is the unit of magnetomotive force and is given by the product of number of turns of a magnetic circuit and the current in amperes in those turns.

iii) Reluctance is the name given to that property of a material which opposes the creation of magnetic flux in it. Its unit is AT/wb.
i.e., Reluctance $= \frac{L}{NA} = \frac{L}{\mu_0 \mu_r A}$

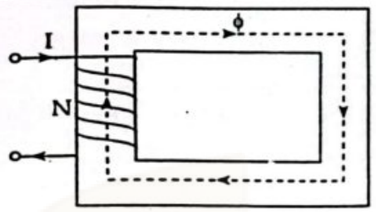
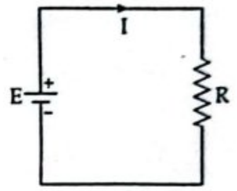
In other words, the reluctance of a magnetic circuit is the number of amp-turns required per weber of magnetic flux in the circuit.

iv) Permeance is the reciprocal of reluctance and implies the case or readiness with which magnetic flux is developed. It analogous to conductance in electric circuit.

v) Relativity is specific reluctance and corresponds to resistivity which is specific resistance.

5.1.1 COMPARISON BETWEEN MAGNETIC AND ELECTRIC CIRCUITS

Similarity:

SN	Magnetic circuit	Electric circuit
i)	 Magnetic circuit	 Electric circuit
ii)	The closed path followed by magnetic flux is called a magnetic circuit.	The closed path followed by an electric current is called an electric circuit.
iii)	mmF required to establish flux (ϕ) in the magnetic circuit and is measured in ampere turns (AT).	Emf is required to cause flow of current in an electric circuit and is measured in volts.
iv)	Flux, $\phi = \frac{\text{Driving force}}{\text{Magnetic reluctance}} = \frac{AT}{S}$ webers	Current, $I = \frac{\text{Driving force}}{\text{Electrical resistance}} = \frac{E}{R}$ amperes
v)	Reluctance, $S = \frac{L}{\mu_0 \mu_r A} = \frac{L}{\mu A}$ AT/wb For series circuits, $S = S_1 + S_2 + S_3 + \ldots\ldots$ For parallel circuits, $S = \frac{1}{\frac{1}{S_1} + \frac{1}{S_2} + \frac{1}{S_3} + \ldots\ldots}$	Resistance, $R = \frac{\rho L}{A}$ ohm For series circuits, $R = R_1 + R_2 + R_3 + \ldots\ldots$ For parallel circuits, $R = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots\ldots}$
vi)	Permeance $= \frac{1}{\text{Reluctance}}$ wb/AT	Conductance $= \frac{1}{\text{Resistance}}$ Siemens
vii)	Reluctivity $= \frac{1}{\text{Permeability}}$	Resistivity, $\rho = \frac{1}{\text{Conductivity}}$

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viii)	Flux density, $B = \mu \times$ magnetic field intensity $= \mu H$ tesla	Current density, $J = \frac{\text{Electric field intensity}}{\rho}$
ix.	Magnetic field intensity, $H = \frac{mmF}{L} = \frac{NI}{L}$ AT/m	Electric field intensity $= \frac{E}{L}$ V/m
x.	mmf drop $= \phi \times$ reluctance $= \phi S$	Voltage drop $= I \times$ Resistance $= IR$

Differences:

SN	Magnetic circuit	Electric circuit
i)	Flux does not actually flow in a magnetic circuit.	Current does flow in an electric circuit.
ii)	Flux can pass through air.	Current cannot flow through the air until an arc is struck.
iii)	Permeability does not largely vary from material to material and there is hardly any material which can act as an insulator to the magnetic flux.	Conductivity varies largely from material to material so much so that some materials are insulators to electric current and some are very good conductors.
iv)	For a particular temperature, the permeability depends upon the flux density or total flux.	For a particular temperature, conductivity or resistivity is constant and independent of current strength.
v)	Residual flux persists after removal of mmf.	The current is reduced to zero after removal of source of emf.
vi	There is no waste of energy due to reluctance in the magnetic circuit and, therefore, energy is required only to create the magnetic flux but not to maintain it.	In an electric circuit, resistance causes heat to be generated resulting in waste of energy and therefore, energy is required as long as the current flows.

5.2 TRANSFORMERS

Transformer is an AC machine that (i) transfers electrical energy from one electric circuit to another (ii) does so without a change of frequency (iii) does so by the principle of electromagnetic induction and (iv) has electric circuits that are linked by a common magnetic circuit.

OR,

A transformer is a static electrical device by which the voltage or the electromotive force of a source of alternating current may be increased or decreased. Transformers can be of two types;

i) Step-up transformers

Step-up transformer is one which receives AC electrical energy at one voltage and delivers it at a higher voltage.

ii) Step-down transformers

Step-down transformer is one which receives AC electrical energy at one voltage and delivers it at a lower voltage.

5.2.1 PRINCIPLE OF OPERATIONS

The principle on which a transformer works is electromagnetic induction, which states that induced emf is proportional to the rate of change of magnetic flux.

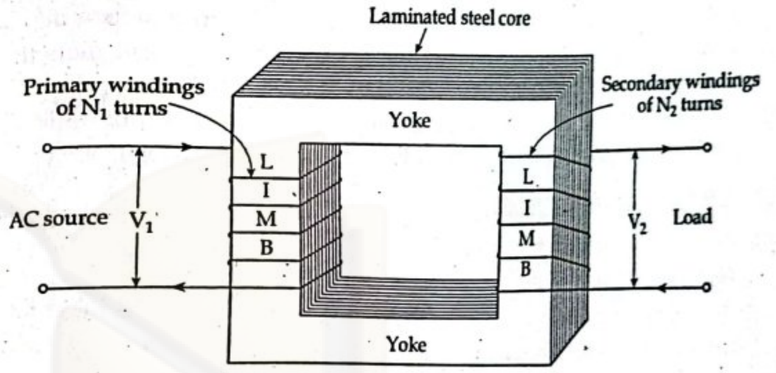

Fig: A transformer

An alternating voltage (V_1) is applied to the primary winding (of N_1 turns) circuits an alternating current through it. This current produces an alternating flux (ϕ) in the core. The flux thus produced expands and contracts and cuts the turns of windings and induces voltages in both the primary and secondary windings. In an ideal transformers, the whole of the magnetic flux produced by the current passes through the iron core. The varying magnetic flux linked with the secondary coil is same at any instant. The varying magnetic flux linked with the secondary produces an induced emf or voltage (V_2) across the secondary. If ϕ is the magnetic flux linked with each turn of the primary, and secondary coils, then,

$$V_1 = -N_1 (d\phi/dt) \quad (1)$$

where,

$\frac{d\phi}{dt}$ = rate of change of magnetic flux in each turn of the primary coil.

Also, the rate of change of magnetic flux in each turn of the secondary at any instant is equal to that in primary,

so, $V_2 = -N_2 \frac{d\phi}{dt}$ (2)

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Dividing equation (2) by (1), we get;

$$\therefore \quad \frac{V_2}{V_1} = \frac{N_2}{N_1} = K$$

$$\text{or,} \quad \frac{\text{Total emf induced in secondary}}{\text{Total emf induced in primary}} = \frac{\text{No. of turns in secondary}}{\text{No. of turns in primary}}$$

The ratio of voltages between the secondary and primary ($V_2/V_1 = K$) is called the voltage transmission ratio (K)

i) When $K > 1$, the output voltage is more than the input voltage. In such a case, it is called step-up transformer. Thus, in a step-up transformer, the number of secondary turns (N_2) is greater than the number of primary turns (N_1). *i.e.*, $N_2 > N_1$

ii) When $K < 1$, the output voltage is less than the input voltage. In such a case, it is called step- down transformer. Thus, in a step – down transformer, the number of secondary turns (N_2) is less than the number of primary turns (N_1). *i.e.*, $N_2 < N_1$

iii) When $K = 1$, the output voltage is equal to that of the input voltage, and $N_2 = N_1$. In such a case, it is called one-to-one transformer.

5.2.2 EMF EQUATION OF A TRANSFORMER

Let, N_1 = No. of turns in the primary coil

N_2 = No. of turns in the secondary coil

ϕ_m = maximum flux in weber in the core of transformer

f = frequency of AC input in Hz

Then, the flux grows from zero to maximum in one quarter of the cycle.

so, The average rate of change of flux,

$$\frac{d\phi}{dt} = \frac{\phi_m}{(1/4f)} = 4f\,\phi_m \text{ wb} = \text{Average emf induced per turn}$$

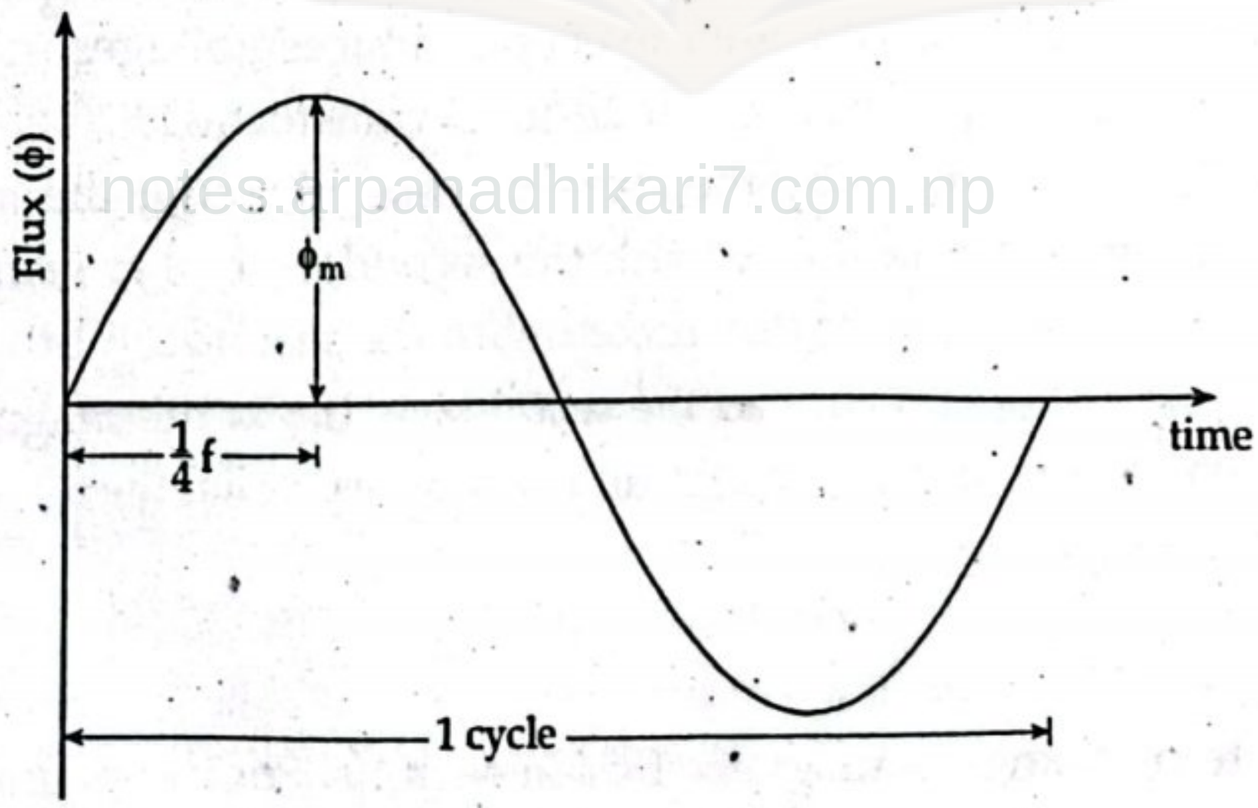

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Fig: Variation of flux with time,

Since flux ϕ varies sinusoidally, emf induced will be sinusoidal and form factor for sinusoidal waves is 1.11 *i.e.*, the rms or effective value is 1.11 times the average value.

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Since the reluctance of the magnetic circuit is never zero, I_m has definite magnitude. The magnitude current I_m is so small in magnitude and lags behind supply voltage V_1 by 90°. This magnetizing current, I_m produces an alternating flux ϕ which is at all times, proportional to the current (assuming permeability of the magnetic circuit to be constant) and in phase with it.

Let the instantaneous linking flux be given as,

$$\phi = \phi_{max} \sin \omega t \quad (1)$$

The varying flux is linked with both of the windings (primary and secondary) and so induces emfs in the primary and secondary windings. The instantaneous value of induced emfs in the primary and secondary windings will be,

$$e_1 = N_1 \frac{d\phi}{dt}$$

$$= -N_1 \frac{d}{dt}(\phi_{max} \cdot \sin \omega t)$$

$$= -N_1 \omega \phi_{max} \cos \omega t$$

$$= -N_1 \omega \phi_{max} \sin\left(\omega t - \frac{\pi}{2}\right) \quad (2)$$

Similarly,

$$E_2 = N_2 \omega \phi_{max} \sin\left(\omega t - \frac{\pi}{2}\right) \quad (3)$$

Since primary winding has no ohmic resistance (assumed), applied voltage to primary winding is to only oppose the induced emf in the primary winding, hence instantaneous applied voltage to primary will be given by,

$$V_1 = -e_1 = -N_1 \omega \phi_{max} \sin\left(\omega t - \frac{\pi}{2}\right) \quad (4)$$

Comparing equation (1), (2), (3) and (4);

i) Induced emfs in primary and secondary windings, E_1 and E_2 lag behind the main flux ϕ by $\frac{\pi}{2}$, so these emfs (E_1 and E_2) are in phase with each other.

ii) Applied voltage to the primary winding leads the main flux by $\frac{\pi}{2}$ and is in phase opposition to the induced emf in the primary winding.

iii) Secondary voltage $V_2 = E_2$ as there is no voltage drop in secondary.

The instantaneous value of applied voltage, induced emfs, flux and magnetizing current, in case of an ideal transformer, are illustrated by sinusoidal waves in figure below.

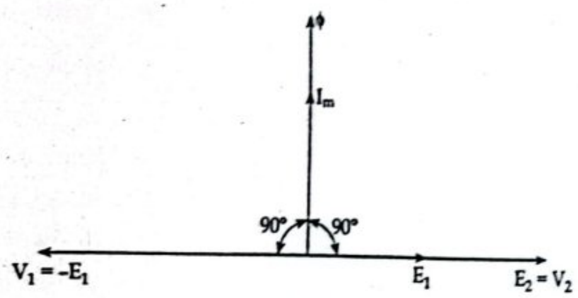
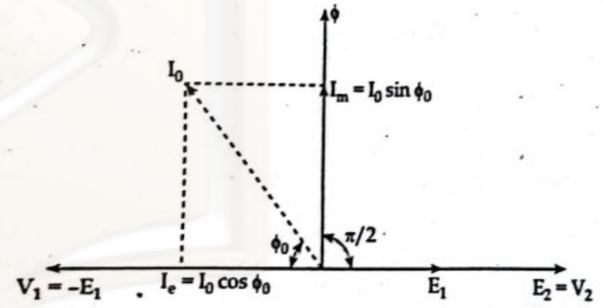


Fig: No-load phasor diagram for an ideal transformer

However, when a varying flux is set up in magnetic material, there will be power loss, called the iron or core loss. So, the input current to the primary under no-load condition has also to supply the hysteresis and eddy current losses (iron losses) occurring in the core in addition to small amount of copper loss occurring in primary winding (no copper loss occurs in secondary winding on open circuit or on no load). Hence, the no load primary current I_0 does not lag behind applied voltage V_1 by 90° but lags behind V_1 by angle ϕ_0 ∠90°.

Input power on no load, $P_0 = V_1 I_0 \cos \phi$.

Where, $\cos \phi_0$ is the primary power factor under no-load condition. As seen from phasor diagram.



Input current to primary I_0, called the exciting current has two components (i) in phase, active or energy component, I_e used to meet the iron loss in addition to small amount of copper loss occurring in the primary winding and (ii) quadrature component or wattles component called the magnetizing component, I_m used to create the alternating flux in the core.

Thus,

$$I_e = I_0 \cos \phi_0 = I_\omega$$

$$I_m = I_0 \sin \phi_0 = I_\mu$$

$$I_0 = \sqrt{I_e^2 + I_m^2}$$

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Angle of lag, $\phi_0 = \tan^{-1}\left(\frac{I_m}{I_e}\right)$

The equivalent circuit of a transformer on no load is shown below,

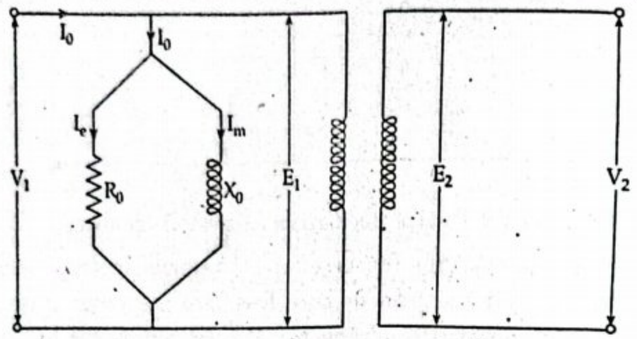

Fig: No-load equivalent circuit for an actual transformer

The actual transformer is replaced by an ideal transformer with a resistance R_0 and an inductive reactance X_0 in parallel with primary. The two components of no-load current I_0 *i.e.* I_e and I_m are represented by currents drawn by R_0 and X_0 respectively. Both these currents are drawn by R_0 and X_0 respectively. Both these currents are drawn at induced emf $E_1 = V_1$ for resistance less, no leakage primary coil, even otherwise $E_1 = V_1$.

so, Energy component of no- load current, $I_e = \frac{V_1}{R_0}$

Magnetizing component of no load current, $I_m = \frac{V_1}{X_0}$

Core or iron loss $= I^2R_0 = \frac{V_1^2}{R_0}$

5.2.4 TRANSFORMER ON LOAD

When the secondary circuit of a transformer is completed through an impedance of load, the transformer is said to be loaded and current flows through the secondary and the load. The magnitude and phase of secondary current I_2 with respect to secondary terminal voltage V_2 will depend upon the characteristic of load *i.e.* current I_2 will be in phase, lag behind and lead the terminal voltage V_2 respectively when the load is non-inductive, inductive and capacitive.

When an impedance is connected across the secondary terminals as shown in figure below, current I_2 flows through the secondary winding. The secondary current I_2 sets up its own mmf and hence creates a secondary flux ϕ_2. The secondary flux ϕ_2 opposes the main flux ϕ set up by the exciting current I_0 according to lenz's law. The opposing secondary flux ϕ_2 weakens the main flux ϕ momentarily, so primary back emf E_1 tends to be reduced. So, difference of applied voltage V_1 and back emf E_1

increases, therefore, more current is drawn from the source of supply flowing through the primary winding until the original value of flux ϕ is obtained. It again causes increase in back emf E_1 and its adjusts itself as such that there is a balance between applied voltage V_1 and back emf E_1. Let, the additional primary current be I_1'. The current I_1' is in phase opposition with secondary current I_2 and is called the counter balancing current. The additional current I_1' sets up an mmf $N_1 I_1'$ producing flux ϕ_1' in the same direction as that of main flux ϕ and cancels the flux ϕ_2 produces by secondary mmf N_2I_2 being equal in magnitude.

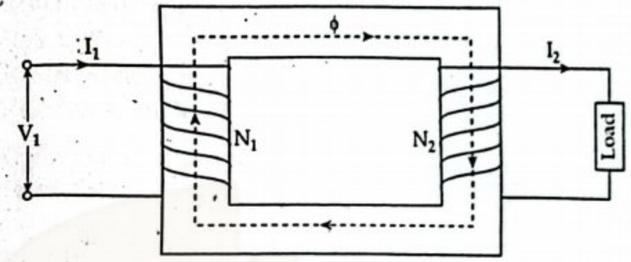

Fig: An ideal transformer on load

so, $N_1 I_1' = N_2 I_2$

or, $I_1' = \frac{N_2}{N_1} \cdot I_2$

The total primary current I_1 is, therefore, phasor sum of primary counter balancing current I_1' and no-load current I_0, which will be approximately equal to I_1' as I_0 is usually very small in comparison to I_1'.

$\therefore \quad I_1 = I_1' = \frac{N_2}{N_1} I_2$

or, $\frac{I_1}{I_2} = \frac{N_2}{N_2} = K$ (transformation ratio).

Hence, primary and secondary currents are inversely proportional to their respective turns.

Since the secondary flux ϕ_2 produced by secondary mmf N_2I_2 is neutralized by the flux ϕ_1' produced by mmf N_1I_1' set up by counter balancing primary current I_1', so the flux in the transformer core remains almost constant from no load to full load. The phasor diagram for transformer on non- inductive, inductive and capacitive loads are shown in figure below. Since the voltage drops in both of the windings of the transformer are assumed to be negligible, therefore, $V_2 = E_2$ and $V_1 = -E_1$. The secondary current I_2 is in phase, lags behind and leads the secondary terminal voltage V_2 by an angle ϕ_2 for pure resistive, inductive and capacitive load respectively.

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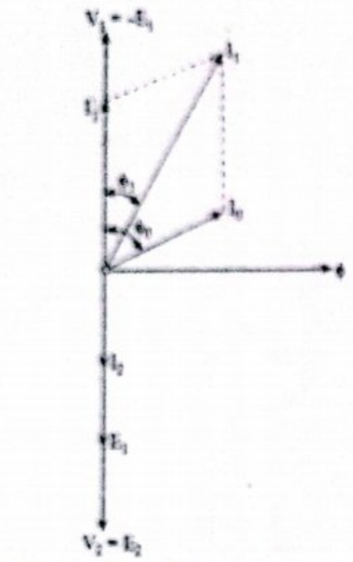
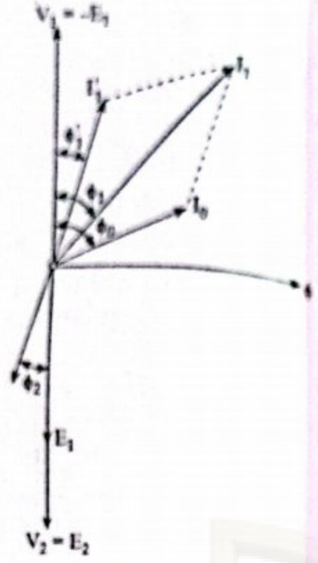
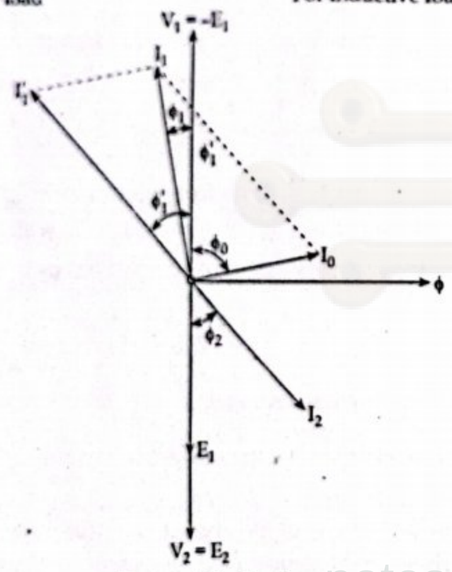


For pure resistive load



For inductive load



For capacitive load

Fig: Phasor diagram for an ideal transformer on load.

5.2.5 EQUIVALENT RESISTANCE AND REACTANCE OF A TRANSFORMER

Consider a transformer having primary and secondary windings of resistance R_1 and R_2 and reactances X_1 and X_2 respectively. The impedance of primary winding is given by $Z_1 = R_1 + jX_1$ and impedance of secondary winding is given by $Z_2 = R_2 + jX_2$.

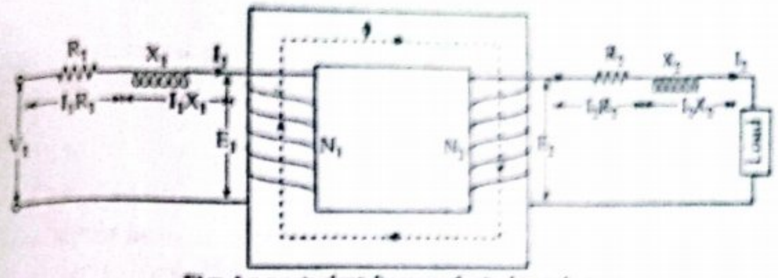


Fig: An equivalent diagram of actual transformer

Let, transformation ratio be K.
Resistive drop in secondary winding = I_2R_2
Reactive drop in secondary winding = I_2X_2
Resistive drop in primary winding = I_1R_1
Reactive drop in primary winding = I_1X_1

a) Referred to secondary side

Since transformation ratio is K, primary resistive and reactive drops as referred to secondary will be K times, *i.e.* KI_1R_1 and KI_1X_1 respectively. If I_1 is substituted equal to KI_2, then we have primary resistive and reactive drops referred to secondary equal to $K^2I_2R_1$ and $K^2I_2X_1$ respectively.

Total resistive drop in a transformer,

$$= K^2I_2R_1 + I_2R_2 = I_2(K^2R_1 + R_2) = I_2R_{02}$$

Total reactive drop in a transformer,

$$= K^2I_2X_1 + I_2X_2 = I_2(K^2X_1 + X_2) = I_2X_{02}$$

The term $(K^2R_1 + R_2)$ and $(K^2X_1 + X_2)$ represent the equivalent resistance and reactance respectively of the transformer referred to secondary and let these be represented by R_{02} and X_{02} respectively.

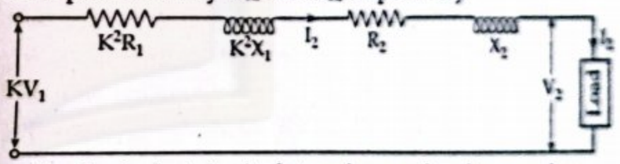


Fig: Equivalent circuit of a transformer referred to secondary.

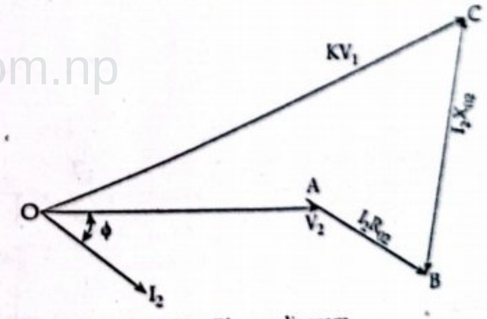


Fig: Phasor diagram.

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From phasor diagram,

$$KV_1 = \sqrt{(V_2 + I_2R_{02}\cos\phi + I_2X_{02}\sin\phi)^2 + (I_2X_{02}\cos\phi - (I_2R_{02}\sin\phi))^2}$$

where,

V_2 = secondary terminal voltage

I_2 = secondary current lagging behind the terminal voltage V_2 by ϕ.

Since term $(I_2X_{02}\cos\phi - I_2R_{02}\sin\phi)$ is very small as compared to the term $(V_2 + I_2R_{02}\cos\phi + I_2X_{02}\sin\phi)$, neglecting the former, we have,

$$KV_1 = V_2 + I_2R_{02}\cos\phi + I_2X_{02}\sin\phi$$

or, $V_2 = KV_1 - I_2R_{02}\cos\phi - I_2X_{02}\sin\phi$ (1)

where,

KV_1 = applied voltage to primary winding

If load is pure resistive, $\phi = 0$ and $V_2 = KV_1 - I_2R_{02}$ (2)

If load is capacitive, then ϕ should be taken as -ve, hence, we have,

$\therefore$ $V_2 = KV_1 - I_2R_{02}\cos\phi + I_2X_{02}\sin\phi$ (3)

b) Referred to primary side

Secondary resistive drop referred to primary,

$$= \frac{I_2R_2}{K} = \frac{I_1R_2}{K^2} \qquad \left(\because I_2 = \frac{I_1}{K}\right)$$

Secondary reactive drop referred to primary

$$= \frac{I_2X_2}{K} = \frac{I_1X_2}{K^2}$$

Total resistive drop in the transformer referred to primary

$$= I_1R_1 + \frac{I_1R_2}{K^2} = I_1\left(R_1 + \frac{R_2}{K^2}\right) = I_1R_{01}$$

Total reactive drop in the transformer referred to primary,

$$= I_1X_1 + \frac{I_1X_2}{K^2} = I_1\left(X_1 + \frac{X_2}{K^2}\right) = I_1X_{01}$$

The terms $\left(R_1 + \frac{R_2}{K^2}\right)$ and $\left(X_1 + \frac{X_2}{K^2}\right)$ represent the total resistance and reactance of the transformer referred to primary respectively. Let, these be represented by R_{01} and X_{01} respectively.

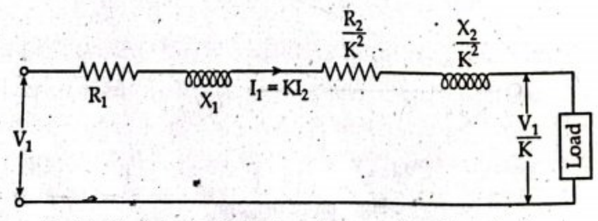

Fig: Equivalent circuit of a transformer referred to primary.

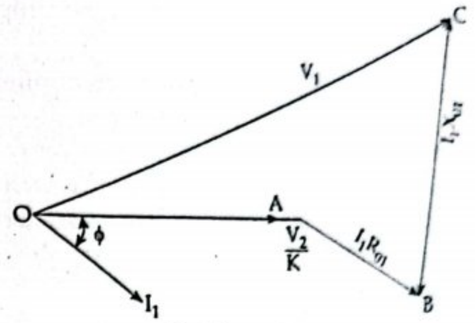

Fig: Phasor diagram.

From phasor diagram,

$$V_1 = \sqrt{\left(\frac{V_2}{K} + I_1R_{01}\cos\phi + I_1X_{01}\sin\phi\right)^2 + (I_1X_{01}\cos\phi - I_1R_{01}\sin\phi)^2}$$

Since term $(I_1X_{01}\cos\phi - I_1R_{01}\sin\phi)$ is very small as compared to the term $\left(\frac{V_2}{K} + I_1R_{01}\cos\phi + I_1X_{01}\sin\phi\right)$,

Neglecting the former, we have,

$$V_1 = \frac{V_2}{K} + I_1R_{01}\cos\phi + I_1X_{01}\sin\phi$$

If load is pure resistive, *i.e.*, $\phi = 0$, then,

$$V_1 = \frac{V_2}{K} + I_1R_{01}$$

If load is capacitive then ϕ is taken as - ve, so we have,

$$V_1 = \frac{V_2}{K} + I_1R_{01}\cos\phi - I_1X_{01}\sin\phi$$

5.2.6 EQUIVALENT CIRCUIT OF A TRANSFORMER

In the equivalent circuit of a transformer, the windings are shown as ideal; while the resistance and leakage reactance in the primary and secondary are indicated separately in the primary and secondary circuits respectively.

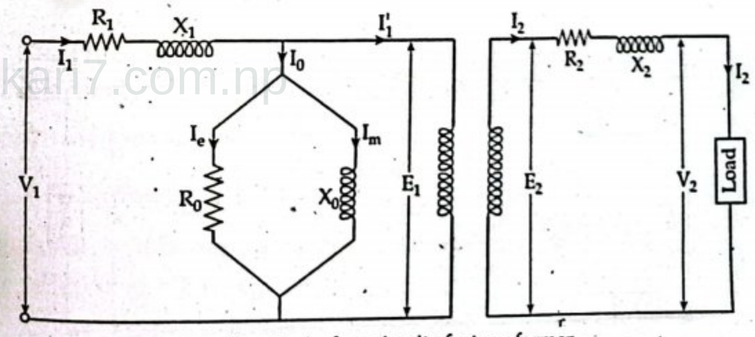

Fig: Equivalent circuit of a transformer

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Equivalent circuit of a transformer having transformation ratio $K = E_2/E_1$ is shown above figure.

The induced emf in primary winding E_1 is primary applied voltage V_1 less primary voltage drop. This voltage causes iron loss current I_e and magnetizing current I_m and we can therefore, represent these two components of non-load current by the current drawn by a non-load current by the current drawn by a non-inductive resistance R_0 and pure reactance X_0 having the voltage E_1 or (V_1 – primary voltage drop) applied across them as shown in figure above.

$$\text{Secondary current, } I_2 = \frac{I_1'}{K} = \frac{I_1 - I_0}{K}$$

Terminal voltage V_2 across load is induced emf E_2 in secondary winding less voltage drop in secondary winding. The equivalent circuit can be simplified by transferring the voltage, current and impedance to the primary side. After transferring the secondary voltage, current and impedance to primary side equivalent circuit is reduced to that shown in figure below,

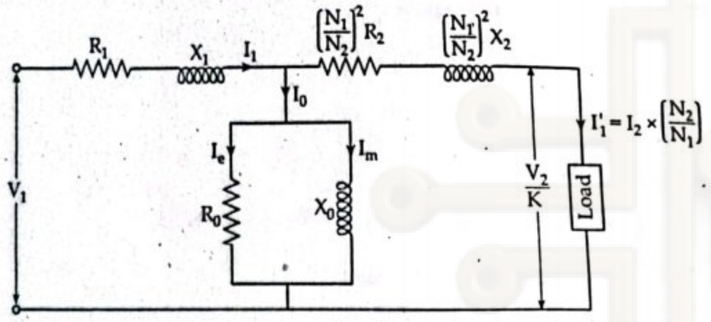

Fig: Equivalent circuit of a transformer with all secondary impedances transferred to primary side.

The equivalent circuit diagram can further be simplified by transferring the resistance R_0 and reactance X_0 towards left end, as shown in figure below. The error introduced by doing so is very small and can be neglected.

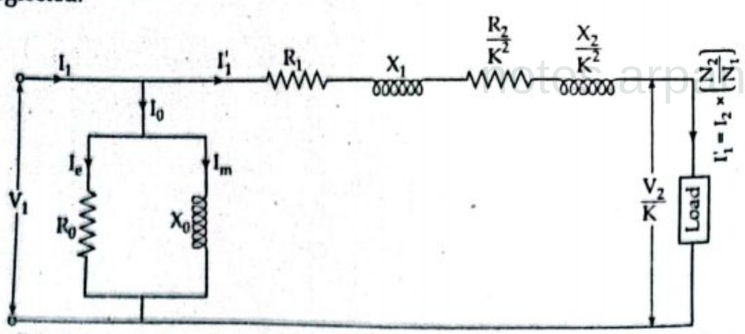

Fig: Approximate equivalent circuit of a transformer referred to primary side.

5.2.7 VOLTAGE REGULATION OF A TRANSFORMER

The figure of merit which determines the voltage characteristic of a transformer is termed as voltage regulation. It is usually expressed as a percentage of the rated terminal (secondary) voltage. It is defined as the change in magnitude of terminal (secondary) voltage when full load (rated load) of specified power factor supplied at rated voltage is thrown off (reduced to no load) with primary voltage and frequency held constant, as percentage of the rated terminal voltage.

If V_2 is the secondary (terminal) voltage at full load at specified power factor and E_2 is the secondary (terminal) voltage at no load, then percentage regulation is given as,

$$\text{Voltage regulation} = \frac{(E_2 - V_2)}{\text{Secondary rated voltage}} \times 100$$

As per IS, the secondary rated voltage of transformer is equal to the secondary voltage at no load *i.e.* E_2.

$$\text{so, Percentage regulation} = \left(\frac{E_2 - V_2}{E_2}\right) \times 100$$

$$= \frac{\text{Voltage drop in transformer at full load}}{\text{No-load rated secondary voltage, } E_2} \times 100$$

$$= \frac{I_2 R_{02} \cos\phi + I_2 X_{02} \sin\phi}{E_2} \times 100$$

When the power factor is leading, the percentage regulation is given as,

$$\text{Percentage regulation} = \frac{I_2 R_{02} \cos\phi - I_2 X_{02} \sin\phi}{E_2} \times 100$$

Thus,

$$\% \text{ regulation} = \frac{I_2 R_{02} \cos\phi \pm X_{02} \sin\phi}{E_2} \times 100$$

Here,

+ve sign for lagging power factor and

–ve sign for leading power factor

$$= \left(\frac{I_2 R_{02} \cos\phi}{E_2} \pm \frac{I_2 X_{02} \sin\phi}{E_2}\right) \times 100$$

$\therefore$ %VR = (PU resistance X cos ϕ ± PU reactance × sin ϕ) × 100

Voltage regulation of a transformer, on an average is about 4 percent.

Condition for zero regulation

$$\text{Regulation} = \frac{I_2 R_{02} \cos\phi + I_2 X_{02} \sin\phi}{E_2}$$

Regulation will be zero if the numerator will be equal to zero.

or, $I_2 R_{02} \cos\phi + I_2 X_{02} \sin\phi = 0$

or, $I_2 X_{02} \sin\phi = -I_2 R_{02} \cos\phi$

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or, $\frac{\sin\phi}{\cos\phi} = \frac{-R_{02}}{X_{02}}$

$\therefore \quad \tan\phi = \frac{-R_{02}}{X_{02}}$

The -ve sign indicates that zero regulation occurs at a leading power factor.

Condition for maximum regulation

Regulation will be maximum if,

$$\frac{d}{d\phi}(\text{Regulation}) = 0$$

or, $\frac{d}{d\phi}\frac{(I_2R_{02}\cos\phi + I_2X_{02}\sin\phi)}{E_2} = 0$

or, $-\frac{I_2R_{02}}{E_2}\sin\phi + \frac{I_2X_{02}}{E_2}\cos\phi = 0$

$\therefore \quad \tan\phi = \frac{X_{02}}{R_{02}}$

i.e., Maximum regulation occurs at lagging power factor.

5.2.8 EFFICIENCY OF A TRANSFORMER

The rated capacity of a transformer is defined as the product of rated voltage and full-load (rated) current of the output side. The power output depends upon the power factor of the load.

The efficiency (η) of a transformer, like that of any other apparatus, is defined as the ratio of useful power output to the input power, the two being measured in same units.

i.e., Transformer efficiency,

$$\eta = \frac{\text{Output}}{\text{Input}}$$

$$= \frac{\text{Output}}{\text{Output + Losses}}$$

$$= \frac{\text{Output}}{\text{Output + Iron loss + Copper loss}}$$

$$= \frac{\text{Input power - Losses}}{\text{Input power}}$$

$$= 1 - \frac{\text{Losses}}{\text{Input power}}$$

$\therefore \quad \eta = 1 - \frac{\text{Iron loss + Copper loss}}{\text{Output + Iron loss + Copper loss}}$

Now,

Power output = $V_2I_2\cos\phi$

where, V_2 is the secondary terminal voltage on load,

I_2 is the secondary current at load

$\cos\phi$ is the power factor of the load

Iron loss, P_i = Hysteresis loss + eddy current loss

Copper loss, $P_c = I_1^2R_1 + I_1^2R_2 = I_1^2R_{01} = I_2^2R_{02}$

so, $\eta = \frac{V_1I_1\cos\phi_1 - I_1^2R_1 - P_i}{V_1I_1\cos\phi_1} = 1 - \frac{I_1R_1}{V_1\cos\phi_1} - \frac{P_i}{V_1I_1\cos\phi_1}$

Differentiating with respect to we get,

$$\frac{d\eta}{dI_1} = 0 - \frac{R_1}{V_1\cos\phi_1} + \frac{P_i}{V_1I_1^2\cos\phi_1}$$

For maximum value of efficiency,

$$\frac{d\eta}{dI_1} = 0$$

or, $0 = -\frac{R_1}{V_1\cos\phi_1} + \frac{P_i}{V_1I_1^2\cos\phi_1}$

or, copper loss ($I_1^2R_1\cos\phi_1$) = Iron loss (P_i)

Load at maximum efficiency

Let, P_i and P_c be the full load iron losses and copper losses of a transformer respectively.

Therefore, $P_c \propto (\text{Full-load KVA})^2$ and if x be the load, when the efficiency is maximum, then;

$P_i \propto x^2$

$\therefore \quad \frac{P_c}{P_i} = \frac{(\text{Full-load KVA})^2}{x^2}$

$\therefore$ Load at maximum efficiency (x)

$$= (\text{Full-load KVA}) \times \left[\frac{\text{Iron losses } (P_i)}{\text{Full-load copper losses } P_c}\right]^{\frac{1}{2}}$$

and, Transformer efficiency,

$$\eta = \frac{x \cdot V_2I_2\cos\phi}{x \cdot V_2I_2\cos\phi + P_i + x^2P_c}$$

Output KVA corresponding t maximum efficiency,

$$= \text{Rated KVA} \times \sqrt{\frac{P_i}{P_c}}$$

Output current I_2 corresponding to maximum efficiency = $\sqrt{\frac{P_i}{R_{02}}}$

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Efficiency versus power factor

$$\eta = \frac{\text{Output}}{\text{Output + Losses}}$$

$$= 1 - \frac{\text{Losses}}{\text{Output + Losses}}$$

$$= 1 - \frac{\text{Losses}}{V_2 I_2 \cos\phi + \text{Losses}}$$

or, $$\eta = 1 - \frac{\text{Losses}/V_2 I_2}{\cos\phi + \text{Losses}/V_2 I_2}$$

Putting $\frac{\text{Losses}}{V_2 I_2} = x$,

or, $$\eta = 1 - \frac{x}{\cos\phi + x}$$

$$\therefore \quad \eta = 1 - \frac{x/\cos\phi}{1 + x/\cos\phi}$$

5.2.9 TRANSFORMER LOSSES

The transformer is a static machine and therefore, there are no friction and windage losses. The various power losses occurring in a transformer are enumerated below:

a) Iron losses or core losses

Iron loss is caused by the alternating flux in the core and consists of hysteresis and eddy current losses.

i) Hysteresis loss

The core of a transformer is subjected to an alternating magnetizing force and for each cycle of emf, a hysteresis loop is traced out. The hysteresis loss per second is given by the equation.

$$\text{Hystersis loss, } P_h = \eta' (B_{max})^x f v \text{ J/S or W}$$

where, f = Supply frequency, Hz

v = Volume of core in cubic meters

η' = Hysteresis coefficient

B_{max} = Peak value of flux density in the core

x lies between 1.5 and 2.5 depending upon the material and is often taken as 1.6.

ii) Eddy current loss

If the magnetic circuit is made up of iron and if the flux in the circuit is variable, current will be induced by induction in the iron circuit itself. All such currents are known as eddy currents. Eddy current result in a loss of power with consequent beating of the material. The eddy current loss is given by,

$$P_e = K_e (B_{max})^2 f^2 t^2 v \text{ watts or J/S}$$

From the above expression for eddy current loss in a thin sheet, it is obvious that eddy current loss varies (i) as the square of maximum flux density (ii) as the square of the frequency and (iii) as the square of thickness of laminations.

The hysteresis and eddy current losses depend upon the maximum flux density in the core and supply frequency. Since it has been determined that the mutual flux varies somewhat with the load (its variation being 1 to 3% from no load to full load), the core losses will vary somewhat with the load and its power factor. It may be emphasized here that core losses are assumed to remain constant from no load to full load being very small and negligible. These losses are determined from the open-circuit test. These losses are minimized by using steel of high silicon content for the core and by using very thin laminations insulated from each other either by insulating varnish or by layer of papers.

b) Copper or ohmic losses

These losses occur due to ohmic resistance of the transformer winding. If I_1 and I_2 are the primary and secondary currents respectively and R_1 and R_2 are the respective resistances of primary and secondary windings, then copper losses occurring in primary and secondary windings will be $I_1^2 R_1$ and $I_2^2 R_2$ respectively. These losses vary as the square of the load current or KVA.

Copper losses are determined on the basis of constant equivalent resistance R_{eq} determined from the short circuit test and then corrected to 75°C (since the standard operating temperature of electrical machine is taken as 75°C)

c) Stray load loss

Leakage fields present in a transformer induce eddy currents in the conductors, tank walls, bolts etc. These eddy currents give rise to stray load loss.

d) Dielectric loss

This loss occurs in the insulating materials, particularly in oil and solid insulations. The stray load loss and dielectric loss are very small and are therefore neglected.

5.2.10 OPEN CIRCUIT AND SHORT CIRCUIT TEST

The performance characteristics of a transformer can be conveniently computed from the knowledge of its equivalent circuit parameters which, in turn, may be determined by conducting simple tests called the open-circuit or no-load test and short-circuit or impedance test involving very

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little power consumption power needed to supply the losses incurred) other tests to be conducted on transformers are polarity test and voltage ratio test.

a) Open-circuit or no-load test:

The purpose of this test is to determine the core or iron or excitation loss, P_i and no-load current I_0 and there by the shunt branch parameters R_0 and X_0 of the equivalent circuit.

In this test, one of the windings (usually high-voltage winding) is kept open circuited and the rated voltage at rated frequency is applied to the other winding. The core loss will be same whether the measurements are made on LV windings or HV windings so long as the rated voltage of that winding is applied to it but in case the measurements are made on HV winding, the voltage required to be applied would often be inconveniently large while the current I_0 would be inconveniently small.

Either an auto-transformer or a voltage divider is used for varying the voltage applied to the low-voltage winding. Ammeter, A and wattmeter, W are connected to measure no load current, I_0 and input power, W_0. Voltmeter, V is connected to measure the applied voltage.

Since no current flows in the open circuited secondary, the current in the primary will be merely the necessary to magnetize the core at normal voltage. Moreover, this magnetizing current is a very small fraction of the full-load current (usually 3 to 10% of full load current) and may be neglected as far as the copper loss is concerned consequently, the test gives core loss alone practically.

With normal voltage applied to the primary, normal flux will be set up in the core and therefore, normal iron or core loss will occur which are recorded by a wattmeter. The open-circuit test gives enough data to compute the equivalent circuit constants R_0, X_0, no-load power factor $\cos\phi_0$, no-load current I_0 and no load power loss (iron loss) of a transformer.

Iron loss, P_i = Input power on no-load = W_0

No load current = I_0 amperes

Applied voltage to primary = V_1 volts

Angle of lag, $\phi_0 = \cos^{-1}\left(\frac{W_0}{V_1.I_0}\right)$

No-load current energy component, $I_e = I_0 \cos\phi_0 = \frac{W_0}{V_1}$

No-load current magnetizing component, $I_m = \sqrt{I_0^2 - I_e^2}$

Equivalent circuit parameter, $R_0 = \frac{V_1}{I_e} = \frac{V_1^2}{W_0}$

Equivalent circuit parameter, $X_0 = \frac{V_1}{I_m} = \frac{V_1}{\sqrt{I_0^2 - I_e^2}}$

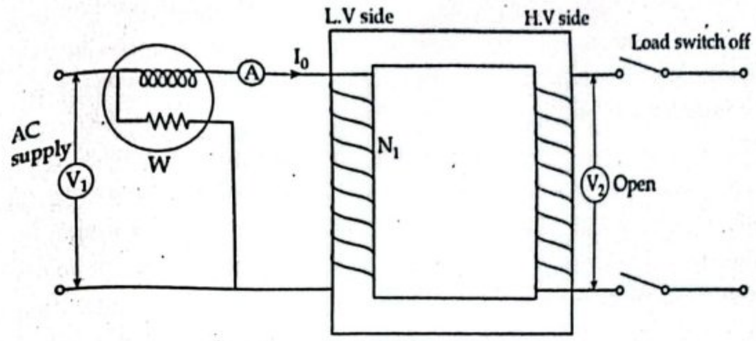

Fig: Open circuit (OC) test on a transformer

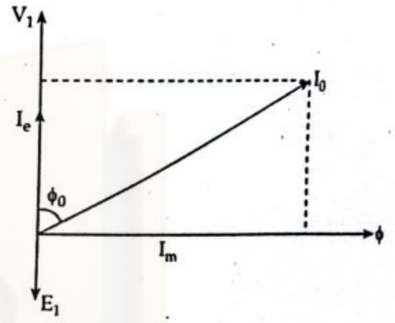

Fig: Resolution of no-load current (I_0)

Thus,

$$\text{Power factor, } \cos\phi_0 = \frac{\text{Wattmeter reading}}{\text{Voltmeter reading} \times \text{Ammeter reading}} = \frac{W}{V_1 \cdot I_0}$$

Note:

i) Since, no-load current, I_0 is very small, pressure coils of wattmeter and the voltmeter should be connected such that the currents drawn by them do not flow through the current coils of the wattmeter and ammeter.

ii) Since power factor at no load is quite low, a low power factor wattmeter should be used to ensure accurate measurements.

iii) The error due to power loss in ammeter can be eliminated by short circuiting the ammeter while reading wattmeter.

iv) In making this test, HV side is hot and therefore, its terminals must be properly insulated.

b) Short circuit test or impedance test

The purpose of this test is to determine full load copper loss and equivalent resistance and equivalent reactance referred to metering scale.

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In this test, the terminals of secondary winding (usually low-voltage winding) are short circuited by a thick wire or strip or through an ammeter (which may serve the additional purpose of indicating secondary rated load current) and variable low voltage is applied to the primary through an auto-transformer of potential divider. The transformer now becomes equivalent to a coil having an impedance equal to impedance of both the windings.

The applied voltage, V_s to the primary is gradually increased till the ammeter, A indicates the full-load (rated) current of the metering side. Since, applied voltage is very low (5 - 8%) of the rated voltage), flux linking with the core is very small and therefore, iron losses are so small that these can be neglected. Thus, the power input (reading of wattmeter W) gives total copper losses at rated load, output being nil. Let, the readings of voltmeter, ammeter and wattmeter be V_S, I_s and W_s respectively.

Full load copper loss, $P_c = I_s^2 R_{eq} = W_s$

Equivalent resistance, $R_{eq} = \frac{W_s}{I_s^2}$

Equivalent impedance, $Z_{eq} = \frac{V_s}{I_s}$

Equivalent reactance, $X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2}$

The above values are referred to the metering side (high voltage side in above case).

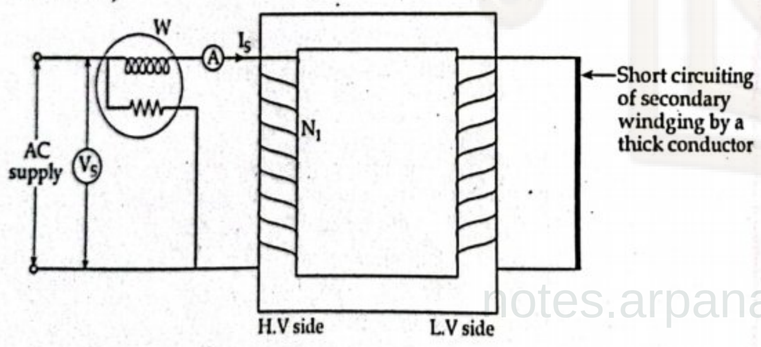

Fig: Circuit diagram for short-circuit test.

Example 1

A 50 KVA, single phase transformer has 600 turns on primary and 40 turns on secondary. The primary winding is connected to 2.2 KV, 50 Hz supply. Determine (i) secondary voltage at no-load (ii) primary and secondary currents at full load.

Solution:

$N_1 = 600$ turns

$N_2 = 40$ turns

$E_1 = 2.2$ KV $= 2{,}200$ V

$E_2 = ?$

We have,

$$\frac{E_2}{E_1} = \frac{N_2}{N_1}$$

We have,

$$\frac{E_2}{E_1} = \frac{N_2}{N_1}$$

or, $E_2 = \frac{40}{600} \times 2{,}200$

$\therefore E_2 = 146.67$ V

Now,

$$KVA = \frac{E_1 I_1}{1{,}000} = \frac{E_2 \times I_2}{1{,}000}$$

or, $I_1 = \frac{KVA \times 1{,}000}{E_2} = \frac{50 \times 1000}{2{,}200} = 22.72$ A

and, $I_2 = \frac{KVA \times 1{,}000}{E_2} = \frac{50 \times 1000}{146.67} = 390.9$ A.

Example 2

The emf per turn of a single-phase 10 KVA, 2200/220 V, 50 Hz transformer is 10 V. calculate (i) the number of primary and secondary turns (ii) the net cross-sectional area of core for a maximum flux density of 1.5 T.

Solution:

Emf per turn = 10 V

Primary induced emf, $E_1 = V_1 = 2{,}200$ V

Secondary induced emf, $E_2 = V_2 = 220$ V

Supply frequency, $f = 50$ Hz

Maximum flux density, $B_{max} = 1.5$ T

i) Number of primary turns, $N_1 = \frac{E_1}{\text{emf per turn}} = \frac{2200}{10} = 220$

Number of secondary turns, $N_2 = \frac{E_2}{\text{emf per turn}} = \frac{220}{10} = 22$

Maximum value of flux, $\phi_{max} = \frac{\text{emf per turn}}{4.44\, f} = \frac{10}{4.44 \times 50} = 0.04$ wb

ii) Net cross-sectional area of core, $a = \frac{\phi_{max}}{B_{max}} = \frac{0.045}{1.5} = 0.03\ m^2$.

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Example 3

A 6,000/440 V, single phase 600 KVA transformer has 1,200 primary turns. Find (i) Transformation ratio, (ii) secondary turns (iii) voltage per turn (iv) secondary current when it supplies a load of 400 KW at 0.8 power factor lagging.

Solution:

Primary voltage, $V_1 = 6{,}600$ V

Secondary voltage, $V_2 = 440$ V

Primary turns, $N_1 = 1{,}200$

Load supplied, $P = 400$ KW

Power factor of load, $\cos\phi = 0.8$ (lagging)

i) Transformation ratio, $K = \frac{E_2}{E_1} = \frac{V_2}{V_1} = \frac{440}{6{,}600} = \frac{1}{15}$

ii) Secondary turns, $N_2 = KN_1 = \frac{1}{15} \times 1{,}200 = 80$

iii) Voltage per turn $= \frac{E_1}{N_1} = \frac{6600}{1200} = 5.5$ V

iv) Load supplied in KVA, $S = \frac{\text{Power}}{\text{Power factor}} = \frac{400}{0.8} = 500$

Secondary current, $I_2 = \frac{5 \times 1{,}000}{V_2} = \frac{500 \times 1{,}000}{440} = 1{,}136.4$ A

Example 4

A 11 KV/400 V distribution transformer takes a no-load primary current of 1 A at a power factor of 0.24 lagging. Find (i) the core loss current (ii) The magnetizing current (iii) The iron loss.

Solution:

Exciting or no-load current, $I_0 = 1$A

Primary voltage, $V_1 = 11$ KV $= 11{,}000$ V

i) Core loss current, $I_e = I_0 \cos\phi_0 = 1 \times 0.24 = 0.24$ A

ii) Magnetizing current, $I_m = \sqrt{I_0^2 - I_e^2} = \sqrt{1^2 - 0.24^2} = 0.971$ A

iii) Iron loss $= V_1 I_a \cos\phi_0 = 11{,}000 \times 1 \times 0.24 = 2{,}640$ W

Example 5

A voltage V = 200 sin 314t is applied to the transformer winding in a no-load test. The resulting current is found to be i = 3 sin (314t – 60°). Determine the core loss and the parameters of no-load approximate equivalent circuit.

Solution:

No-load current, $I_0 = \frac{3}{\sqrt{2}} = 2.12$ A

Core loss = Input at no-load

$= VI_0 \cos\phi_0$

$= \frac{200}{\sqrt{2}} \times \frac{3}{\sqrt{2}} \cos 60°$

$= 150$ W

Energy component of no-load current,

$I_e = I_0 \cos\phi_0 = \frac{3}{\sqrt{2}} \cos 60° = 1.06$ A

Magnetizing component of no-load current,

$I_m = \sqrt{I_0^2 - I_e^2} = \sqrt{(2.12)^2 - (1.06)^2} = 1.836$ A

No-load resistance, $R_0 = \frac{V}{I_e} = \frac{200/\sqrt{2}}{1.06} = 133.42\ \Omega$

No- load reactance, $X_0 = \frac{V}{I_m} = \frac{200/\sqrt{2}}{1.836} = 77\ \Omega$

Example 6

The primary and secondary winding of a 500 KVA, single phase transformer have resistance of 0.4 Ω and 0.0015 Ω respectively. The primary and secondary voltages are 6,000 V and 400 V respectively and the iron loss is 3.2 KW. Calculate the efficiency on (i) full load and (ii) half load. Assuming the power factor to be 0.8

Solution:

Primary current at full-load,

$I_1 = \frac{\text{Rated KVA} \times 1{,}000}{V_1} = \frac{500 \times 1{,}000}{6{,}000} = 83.33$ A

Secondary current on full-load,

$I_2 = \frac{\text{Rated KVA} \times 1{,}000}{V_2} = \frac{500 \times 1{,}000}{400} = 1{,}250$ A

Full-load copper loss $= I_1^2 R_1 + I_2^2 R_2$

$= (83.33)^2 \times 0.4 + (1{,}250)^2 \times 0.0015$

Full-load output,

$P_{out,f} = \text{Rated KVA} \times \text{p.f.} = 500 \times 0.8 = 400$ KW

i) Transformer efficiency at full load,

$\eta_f = \frac{P_{out,f}}{P_{out,f} + \text{Copper loss} + \text{Iron loss}} \times 100$

$= \frac{400}{400 + 5.122 + 3.2} \times 100$

$= 97.962\%$

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ii) Transformer efficiency at half-full load.

$$= \frac{\frac{1}{2} \times \text{Rated KVA} \times \text{p.f.}}{\frac{1}{2} \times \text{Rated KVA} \times \text{p.f.} + \left(\frac{1}{2}\right)^2 \text{Copper loss} + \text{Iron loss}} \times 100$$

$$= \frac{\frac{1}{2} \times 500 \times 0.8}{\frac{1}{2} \times 500 \times 0.8 + \frac{1}{4} \times 5.122 + 3.2}$$

$$= 97.81\%$$

Example 7

A 30 KVA, single phase transformer has an iron loss of 457 W and copper loss of 125 W when delivering half-full load. At what percent of the full will the transformer have maximum efficiency?

Solution:

Iron loss = 457 W

Copper loss at half-full load = 125 W

Copper loss at full load

$$= \frac{\text{Copper loss at half full load}}{\left(\frac{1}{2}\right)^2} = \frac{125}{\left(\frac{1}{4}\right)} = 500 \text{ W}$$

Let the transformer maximum efficiency occurs at x% of full load, then,

$$\frac{x}{100} = \sqrt{\frac{P_i}{P_c}}$$

or, $x = \sqrt{\frac{457}{500}} \times 100$

$\therefore \quad x = 95.60\%$

Example 8

A 50 KVA transformer has an efficiency of 98% at full-load, p.f. and 97% at half full load, 0.8 p.f. Determine the full load copper loss and iron loss. Find the load at which maximum efficiency occurs and also maximum efficiency.

Solution:

i) Efficiency at full-load, 0.8 p.f.

$$\text{or,} \quad 0.98 = \frac{50 \times 0.8}{50 \times 0.8 + W_i + W_c}$$

$$\text{or,} \quad 0.98 = \frac{40}{40 + Wi + W_c}$$

$$\text{or,} \quad W_i + W_c = 40\left[\frac{1}{0.98} - 1\right]$$

$\therefore \quad W_i + W_c = 0.8163 \text{ KW} = 816.3 \text{ W} \quad (1)$

and, Efficiency at half-full load, 0.8 p.f.

$$\text{or,} \quad 0.97 = \frac{50 \times (1/2) \times 0.8}{50 \times (1/2) \times 0.8 + (1/2)^2 W_c + W_i}$$

$$\text{or,} \quad 0.97 = \frac{20}{20 + W_i + W_c/4}$$

$$\text{or,} \quad W_i + \frac{W_c}{4} = 0.6186 \text{ KW} = 818.6 \text{ W} \quad (2)$$

or, Solving equation (1) and (2), we get,

$$\text{or,} \quad \frac{3W_c}{4} = 197.7$$

$\therefore \quad W_c = 263.6 \text{ W}$

and, $W_i = 552.7$ W

ii) KVA output at maximum efficiency

$$= \text{Rated KVA} \times \sqrt{\frac{\text{Iron loss}}{\text{Full load copper loss}}}$$

$$= 50 \times \sqrt{\frac{552.7}{263.6}}$$

$$= 72.392 \text{ KVA}$$

$\therefore$ Maximum efficiency occurs when supplying 72.392 KVA.

iii) Output at maximum efficiency = 72.392 KVA

$$\text{Copper loss} = \text{Full load copper loss} \times \left[\frac{\text{load at max.efficiency}}{\text{Full-load}}\right]$$

$$= 263.7\left[\frac{72.392}{50,000}\right]^2$$

$$= 552.7 \text{ W}$$

Hence,

Maximum efficiency (at unity p.f.),

$$\eta_m = \frac{72,392 \times 1}{72,392 + 552.7 + 552.7} \times 100 = 98.496\%$$

[$\because$ at maximum efficiency, $W_c = W_i$]

Example 9

In a 25 KVA transformer, the iron loss and full-load copper loss are 250 W and 400 W respectively. Calculate the efficiency of the transformer at (i) half-full-load unity p.f. (ii) 3/4th full-load at 0.8 p.f. lagging.

Solution:

Rating of transformer = 25 KVA

Iron loss = 350 W

Full-load copper loss = 400 W

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i) At half full-load and unity p.f.:

$$\text{Output} = \frac{1}{2} \times \text{KVA rating} \times \text{p.f.} = \frac{1}{2} \times 25 \times 1 = 12.50 \text{ KW}$$

$$\text{Copper loss at half full load} = \left(\frac{1}{2}\right)^2 \times 400 = 100 \text{ W}$$

Iron loss = 350 W (constant at all loads)

$\therefore$ Total loss = (100 + 350) W = 450 W = 0.45 KW

$\therefore$ Input = Output + Total loss = 12.5 + 0.45 = 12.95 KW

so, $\text{Efficiency} = \frac{\text{Output}}{\text{Input}} = \frac{12.5}{12.95} \times 100$

$\therefore \quad \eta = 96.535\%$

ii) At 3/4th full load and 0.8 p.f. (lagging):

$$\text{Output} = \frac{3}{4} \times 25 \times 0.8 = 15 \text{ KW}$$

$$\text{Copper loss} = \left(\frac{3}{4}\right)^2 \times 400 \text{ W} = 225 \text{ W}$$

Iron loss = 350 W

$\therefore$ Total loss = (225 + 350) W = 575 W = 0.575 KW

$\therefore$ Input = (15 + 0.575) KW = 15.575 KW

and, $\text{Efficiency} = \frac{\text{Output}}{\text{Input}} \times 100 = \frac{15}{15.575} \times 100$

$\therefore \quad \eta = 96.308\%$

Example 10

A 25 KVA, 2,00/220 V, 50 – Hz, 1 – phase transformer has following parameters;

$R_1 = 1.75\ \Omega$, $R_2 = 0.0045\ \Omega$, $X_1 = 2.6\ \Omega$, $X_2 = 0.0075\ \Omega$ calculate: (i) equivalent resistance referred to primary and secondary (ii) equivalent reactance referred to primary and secondary.

Solution:

Equivalent resistance referred to primary,

$$R_{01} = R_1 + \frac{R_2}{K^2} = 1.75 + \frac{0.0045}{(0.1)^2} = 2.2\ \Omega$$

Equivalent resistance referred to secondary

$$R_{02} = K^2R_1 + R_2 = 0.1^2 \times 1.75 + 0.0045 = 0.022\ \Omega$$

ii) Equivalent reactance referred to primary,

$$X_{01} = X_1 + \frac{X_2}{K^2} = 2.6 + \frac{0.0075}{(0.1)^2} = 3.35\ \Omega$$

Equivalent reactance referred to secondary,

$$X_{02} = K^2X_1 + X_2 = 0.1^2 \times 2.6 + 0.0075 = 0.0335\ \Omega.$$

Example 11

A 100 KVA, 1100/220 V, 60 Hz transformer has a high-voltage winding resistance of 0.1 Ω and a leakage reactance of 0.3 Ω. The low-voltage winding resistance is 0.004 Ω and the leakage reactance is 0.012 Ω. The source is applied to high-voltage side:

i) Find the equivalent winding resistance and reactance referred to high voltage side and low voltage side.

ii) Calculate the leakage impedance of the transformer referred to the primary and secondary sides.

Solution:

Transformation ratio,

$$K = \frac{V_2}{V_1} = \frac{220}{1{,}100} = \frac{1}{5}$$

i) Equivalent resistance referred to primary (HV)

$$R_{01} = R_1 + \frac{R_2}{K_2}$$

$$= 0.1 + \frac{0.004}{(1.5)^2}$$

$$= 0.1 + 0.004 \times 25$$

$$= 0.2\ \Omega$$

Equivalent reactance referred to primary (HV)

$$X_{01} = X_1 + \frac{X_2}{K^2}$$

$$= 0.3 + \frac{0.012}{(1/5)^2}$$

$$= 0.3 + 0.012 \times 25$$

$$= 0.6\ \Omega$$

Equivalent resistance referred to secondary (LV)

$$R_{02} = K^2R_1 + R_2$$

$$= \left(\frac{1}{5}\right)^2 \times 0.1 + 0.004$$

$$= 0.008\ \Omega$$

Equivalent reactance referred to secondary (HV)

$$X_{02} = K^2X_1 + R_2$$

$$= \left(\frac{1}{5}\right)^2 \times 0.3 + 0.012$$

$$= 0.024\ \Omega$$

ii) Leakage impedance of transformer referred to primary,

$$Z_{01} = \sqrt{(R_{01})^2 + (X_{01})^2} = \sqrt{0.2^2 + 0.6^2} = 0.6325\ \Omega$$

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Leakage impedance of transformer referred to secondary (LV)

$$Z_{e2} = \sqrt{(R_{e2})^2 + (X_{e2})^2} = \sqrt{(0.008)^2 + (0.024)^2} = 0.0253\ \Omega$$

Example 12

A 20 KVA, 2,500/500 V, 1 - ϕ transformer has the following parameters, $R_1 = 8\ \Omega$, $R_2 = 0.3\ \Omega$, $X_1 = 17\ \Omega$ and $X_2 = 0.7\ \Omega$. Find the voltage regulation and the secondary terminal voltage for a p.f. of (i) 0.8 lagging (ii) 0.8 leading. The primary voltage is held constant at 2500 V.

Solution:

Transformation ratio, $K = \frac{E_2}{E_1} = \frac{500}{2,500} = 0.2$

Primary voltage $V_1 = 2500$ V

Secondary voltage on no load,

$E_2 = KE_1 = KV_1 = 0.2 \times 2,500 = 500$ V

Equivalent resistance as referred to secondary

$R_{e2} = K^2R_1 + R_2' = 0.22 \times 8 + 0.3 = 0.62\ \Omega$

Equivalent reactance as referred to secondary

$X_{e2} = K^2X_1 + X_2 = 0.22 \times 17 + 0.7 = 1.38\ \Omega$

Full load secondary current,

$$I_2 = \frac{KVA \times 1,000}{V_2} = \frac{20 \times 1,000}{500} = 40\ A$$

$\therefore$ $I_2 = 40$ A

$\cos\phi = 0.8$ lagging

$\sin\phi = \sqrt{1 - \cos^2\phi} = \sqrt{1 - 0.8^2} = 0.6$

Voltage drop in transformer winding at full load and 0.8 pf lagging,

$= I_2R_{e2}\cos\phi + I_2X_{e2}\sin\phi$

$= 40 \times 0.62 \times 0.8 + 40 \times 1.38 \times 0.6$

$= 52.96$ V

Secondary terminal voltage,

$V_2 = E_2$ - Voltage drop

$= 500 - 52.96$

$= 447.04$ V

Now,

$$\text{Voltage regulation} = \frac{I_2 R_{e2}\cos\phi + I_2 X_{e2}\sin\phi}{E_2} \times 100$$

$$= \frac{52.96}{500} \times 100 = 10.592\ \%$$

Voltage drop in transformer windings at full load and 0.8 p.f. leading

$= I_2 R_{e2}\cos\phi - I_2 X_{e2}\sin\phi$

$= 40 \times 0.62 \times 0.8 - 40 \times 1.38 \times 0.6$

$= -13.28$ V

Secondary terminal voltage,

$V_2 = 500 - (-13.28) = 513.28$ V

$$\text{Voltage regulation} = -\frac{13.28}{500} \times 100 = -2.656\%$$

Example 13

A 20 KVA, 250/2,500 V, 50 Hz, 1 - ϕ transformer gave the following test results:

OC (LV side)	250 V	1.4 A	105 W
SC (HV side)	120 V	8 A	320 W

Find the parameters and draw the equivalent circuit of 1 - ϕ transformer as referred to (i) Low voltage side (ii) HV side

Solution:

No load power factor, $\cos\phi_0$

$$\cos\phi_0 = \frac{\text{No-load inout power, } P_0}{\text{Primary voltage, } V_1 \times \text{No-load current, } I_0}$$

$$= \frac{105}{250 \times 1.4} = 0.3$$

Energy component of no-load current

$I_e = I_0\cos\phi_0 = 1.4 \times 0.3 = 0.42$ A

Magnetizing component of no-load current

$I_m = \sqrt{I_0^2 + I_e^2} = \sqrt{1.4^2 - 0.42^2} = 1.3355$ A

i) Referred to LV side

No-load resistance, $R_0 = \frac{V_1}{I_e} = \frac{250}{0.42} = 595.2\ \Omega$

No-load reactance, $X_0 = \frac{V_1}{I_m} = \frac{250}{1.3355} = 187.2\ \Omega$

Short circuit current, $I_s = 8$ A

Short circuit voltage, $V_s = 120$ V

Power input on short circuit, $P_s = 320$ W

Equivalent impedance referred to HV (secondary) side

$$Z_{02} = \frac{V_s}{I_s} = \frac{120}{8} = 15\ \Omega$$

Equivalent resistance referred to HV side

$$R_{02} = \frac{P_s}{I_s^2} = \frac{320}{8^2} = 5\ \Omega$$

Equivalent reactance referred to HV side

$$X_{02} = \sqrt{(Z_{02})^2 - (R_{02})^2} = \sqrt{15^2 - 5^2} = 14.14\ \Omega$$

Transformation ratio $K = \frac{2,500}{250} = 10$

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Equivalent resistance referred to LV (primary) side

$$R_{01} = \frac{R_{02}}{K^2} = \frac{5}{10^2} = 0.05\ \Omega$$

Equivalent reactance referred to HV (primary) side

$$X_{01} = \frac{X_{02}}{K^2} = \frac{14.4}{10^2} = 0.144\ \Omega$$

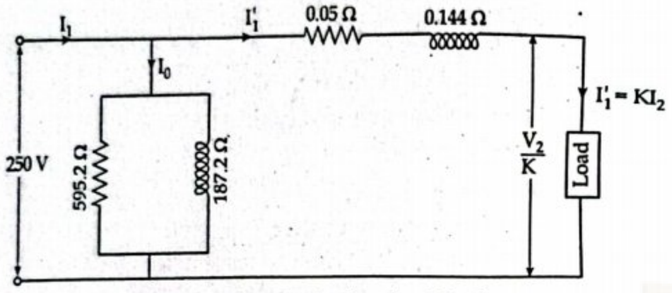


Fig: Equivalent circuit referred to LV side

ii) Referred to HV side

No-load resistance,

$$R_0' = K^2 R_0 = 10^2 \times 595.2 = 59520\ \Omega$$

No-load reactance,

$$X_0' = K^2 X_0 = 10^2 \times 187.2 = 18720\ \Omega$$

Equivalent resistance referred to HV side

$$R_{02} = 5\ \Omega$$

Equivalent resistance referred to HV side

$$X_{02} = 14.14\ \Omega$$

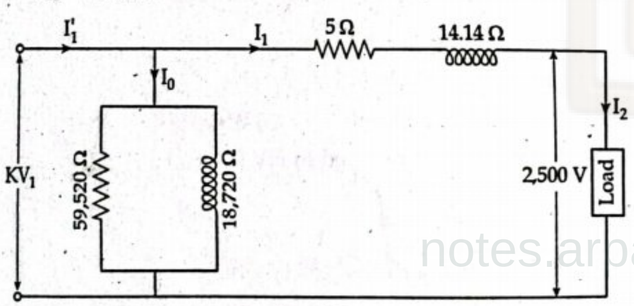


Fig: Equivalent circuit referred to HV side

Example 14

Open circuit and short circuit tests on a 5 KVA, 200/40 V, 50 Hz, 1-phase transformer gave the following results:

OC test :	200 V,	1 A,	100 W
SC test :	15 V,	10 A,	85 W

i) Draw the equivalent circuit referred to primary

ii) Calculate the approximate regulation of the transformer of 0.8 p.f. lagging and leading.

Solution:

OC test on LV side

$$\cos\phi_o = \frac{W_o}{V_{oc} \times I_o}$$

or, $$\cos\phi_o = \frac{100}{200 \times 1}$$

$$\therefore \cos\phi_o = 0.5$$

and, $$\sin\phi_o = \sqrt{1 - 0.5^2} = 0.866$$

$$R_{01} = \frac{V_{oc}}{I_o \times \cos\phi_o}$$

or, $$R_{01} = \frac{200}{1 \times 0.5}$$

$$\therefore X_{01} = \frac{V_{oc}}{I_o \times \sin\phi_o} = \frac{200}{1 \times 0.866} = 231\ \Omega$$

SC test on HV side

$$R_2 = \frac{W_{sc}}{I_{sc}^2} = \frac{85}{10^2} = 0.85\ \Omega$$

$$Z_2 = \frac{V_{sc}}{I_{sc}} = \frac{15}{10} = 1.5\ \Omega$$

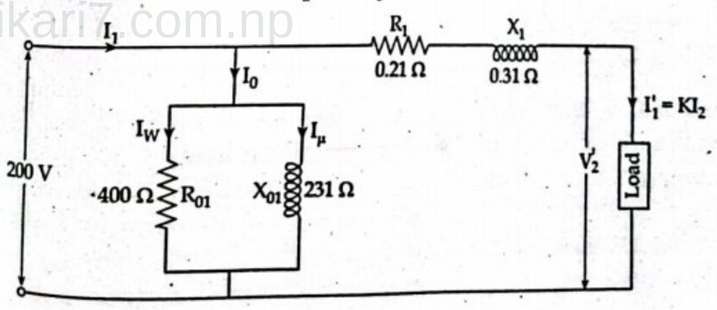
$$X_2 = \sqrt{1.5^2 - 0.85^2} = 1.236\ \Omega$$

Full load secondary current, $I_2 = \frac{4{,}000}{40} = 10$ A

$$\therefore R_1 = R_2 \times \left(\frac{N_1}{N_2}\right)^2 = 85 \times \left(\frac{200}{400}\right)^2 = 0.21\ \Omega$$

$$\therefore X_1 = X_2 \times \left(\frac{N_1}{N_2}\right)^2 = 1.236 \times \left(\frac{200}{400}\right)^2 = 0.31\ \Omega$$

Equivalent circuit referred to primary is



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ii) Percentage regulation (referred to secondary) at:

$$0.5 \text{ p.f. lagging} = \frac{I_2 (R_2 \cos\phi + X_2 \sin\phi) \times 100}{E_2}$$

$$= \frac{10(0.85 \times 0.8 + 1.236 \times 0.6)}{400} \times 100 = 3.544\%$$

$$\text{and } 0.8 \text{ pf leading} = \frac{10(0.85 \times 0.8 - 1.236 \times 0.6)}{400} \times 100 = -0.154\%$$

Example 15

In no-load test of single-phase transformer, the following test data were obtained:

Primary voltage: 220 V ; Secondary voltage : 110 V;

Primary current : 0.5 A ; Power input : 30 W

Find the following:

i) The turns ratio

ii) The magnetizing component of no-load current

iii) Its working or loss component

iv) The iron loss

Resistance of the primary winding = 0.6 ohm

Solution:

i) Turn ratio, $\frac{N_1}{N_2} = \frac{220}{110} = 2$

ii) Magnetizing component of no-load current, $I_\mu = ?$

$$W = V_1 I_0 \cos\phi_0$$

$$\text{or, } \cos\phi_0 = \frac{30}{220 \times 0.5} = 0.273$$

and, $\sin\phi = 0.962$

so, $I_\mu = I_0 \sin\phi_0 = 0.5 \times 0.962 = 0.48$ A

iii) Loss component $= I_0^2 R_1 = 0.5^2 \times 0.6 = 0.15$ W

iv) Iron loss = 30 – 0.15 = 29.85 W

Example 16

The parameter of a 2300/230 V, 50 Hz transformer are given below

$R_1 = 0.286\ \Omega$, $R_2' = 0.319\ \Omega$, $R_0 = 250\ \Omega$

$X_1 = 0.73\ \Omega$, $X_2' = 0.73\ \Omega$, $X_0 = 1250\ \Omega$

The secondary load impedance $Z_L = 0.387 + j0.29$. Solve the exact equivalent circuit with normal voltage across the primary.

Solution:

$$K = \frac{230}{2300} = \frac{1}{10}$$

$$Z_L = 0.387 + j0.29$$

$$Z_L' = \frac{Z_L}{K^2} = 100\,(0.387 + j0.29) = 38.7 + j29 = 48.4\ \angle 36.8°$$

$$\therefore Z_2' + Z_L' = (38.7 + 0.319) + j(29 + 0.73) = 39.02 + j29.73 = 49.0\ \angle 37.3°$$

$$Z_m + (Z_2' + Z_L') = (240 + j48) + (39 + j29.7)\ \Omega = 290\ \angle 15.6°$$

$$\text{so, } I_L = \frac{V_1}{Z_1 + \dfrac{Z_m(Z_2' + Z_L')}{Z_m(Z_2' + Z_L')}} = \frac{2{,}300\ \angle 0°}{0.286 + j0.73 + 41.4\ \angle 33°} = 54.8\ \angle -33.70°\ \text{A}$$

Now,

$$I_2' = I_1 \times \frac{Z_m}{(Z_2' + Z_L') + Z_m} = 54.8\ \angle -33.7 \times \frac{245\ \angle 11.3°}{290\ \angle 15.6°} = 46.2\ \angle -38°\ \text{A}$$

$$I_0 = I_1 \times \frac{(Z_2' + Z_L')}{Z_m + (Z_2' + Z_L')} = 54.8\ \angle -33.7 \times \frac{49\ \angle 37.3°}{290\ \angle 15.6°} = 9.26\ \angle -12°$$

Then,

Input power factor = cos 33.7° = 0.832 (lagging)

Power input $= V_1 I_1 \cos\phi_1$

$= 2{,}300 \times 54.8 \times 0.832$

$= 105$ KW

Power output $= 46.2^2 \times 38.7 = 82.7$ W

Primary copper loss $= 54.8^2 \times 0.286 = 860$ W

Secondary copper loss $= 46.2^2 \times 0.319 = 680$ W

Core loss $= 9.26^2 \times 240 = 20.6$ KW

Then,

$$\text{Efficiency, } \eta = \frac{82.7}{105} \times 100 = 78.8\%$$

$$V_2' = I_2' Z_L' = 46.2 \times 48.4 = 2240\ \text{V}$$

Thus,

$$\% \text{ regulation} = \left(\frac{2{,}300 - 2{,}240}{2{,}240}\right) \times 100 = 2.7\%$$

5.3 DC MOTORS

An electric motor is a machine which converts electrical energy into mechanical energy where as a generator is that machine which converts mechanical energy into electrical one. A DC motor is similar in construction to a DC generator. In fact, the same DC machine can be employed as a generator or as a motor, depending upon whether we rotate the armature to get current from it or we feed to it to set its armature on motion. A DC motor is any of rotary electrical machines that converts direct current electrical energy into mechanical energy.

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5.3.1 WORKING PRINCIPLE OF DC MOTOR

The operation of DC motor is based on the simple motor principle which states that, "If a current carrying conductor is placed in a magnetic field, mechanical force is experienced on the conductor, the direction of which given by fleming's left hand rule and hence the conductor moves in the direction of force. The magnitude of the mechanical force experienced on the conductor is given by,

$$F = B I_c \cdot L_c \text{ Newton}$$

where, B = Field strength in Teslas (wb/m^2)

I_c = Current flowing through the conductor in amperes

L_c = Length of conductor in meters.

Note:

Fleming's left-hand rule states that, "Stretch out the the thumb, the fore finger and the middle finger of the left-hand so that they are mutually at right, angles to each other, then the fore-fingers points points in the direction of the currents and the thumb points in the direction of motion or the mechanical force F experienced by the conductor.

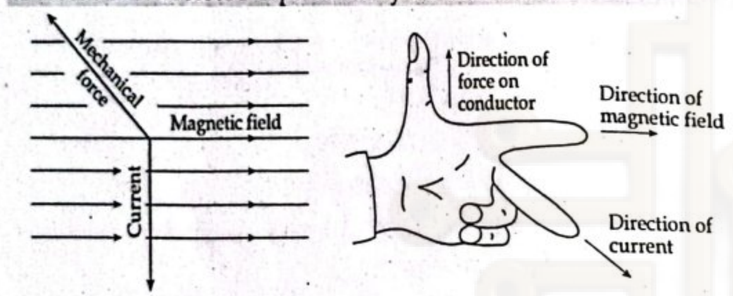
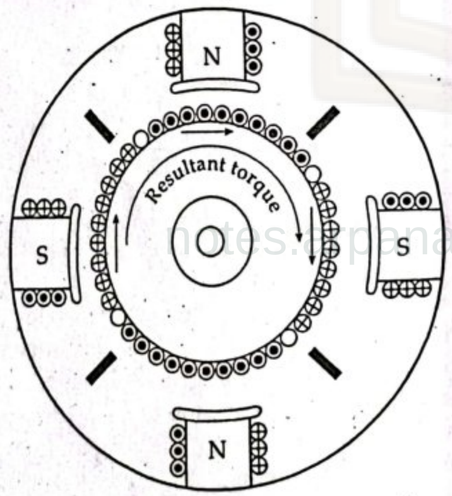


Fig: Direction of force experience of force experienced on a conductor carrying currents, when placed in a magnetic field is given by fleming's Left- hand rule.

When the motor is connected to the DC supply mains, a direct current passes through the brushes through the commutator to the armature winding; while passes through the the commutator it is converted into AC so that each the group of conductors under successive fields poles carry currents in the opposite directions as shown in figure below.

Also the directions of current in the individual conductors



reverses as they pass away from the influence of the pole to that of the next. It is observed that each conductor experiences a force which tends to rotate the motor armature in clockwise direction. These forces collectively produces a driving torque, which rotates the armature.

Back emf in a DC motor

When the motor armature continues to rotate due to motor action, the armature conductors cut the magnetic flux and therefore, emfs are induced in them. The direction of this induced emf, known as back emf, is such that it opposes the applied voltage. since the back emf is induced due to the generator action, the magnitude of it is, therefore, given by the same expression as that for the generated emf in a generator,

i.e., Back emf $= E_b = \frac{\phi Z N}{60} \times \frac{P}{A}$ volts

where, ϕ = Magnetic flux per pole

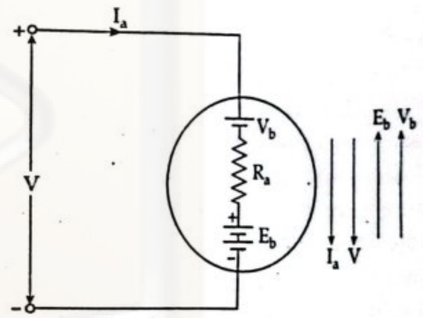
P = Number of poles

Z = Total number of armature conductors.

N = Speed of rotation of armature

A = Number of parallel paths in the armature

The equivalent circuit of a motor is,



The armature circuit is equivalent to a source of emf, E_b in series with a resistance, R_a and brush contact drop, V_b put across a DC supply mains of v volts. It is evident from above figure that the applied voltage V must be large enough to balance the voltage drop in armature resistance brush contact drop and the back emf at all times

i.e., $V = E_b + I_a + R_a + V_b$

where, V is the applied voltage across the armature, E_b is the induced emf in the armature by generator action, I_a is the armature current, Ra is the armature resistance and V_b is the brush contact drop.

Hence, $I_a = \frac{V - E_b}{R_a}$

If the armature speed is high, back emf will be large and therefore, armature current small. If the speed of the armature is low, then back emf

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will be less and armature current more resulting in development of large torque. The presence of back emf makes the DC motor to draw as much armature current as is just sufficient to develop the required load torque.

Importance of back emf

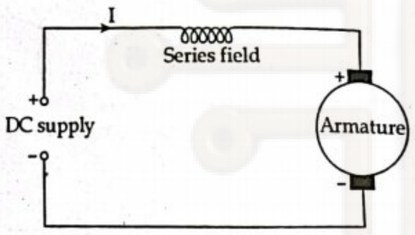
When the motor starts, the back emf in the beginning is zero. Consequently, the current flowing through the armature conductors is very large, since the armature resistance (R_a) is very small. This current, beings many times of full load current and consequently is likely to damage or even burn the motor. In order to avoid this, an additional rheostatic resistance (called starter) is connected in series with the armature to limit the current at starting. This resistance is, however, made zero rheostatically as soon as the motor picks up its normal speed.

5.3.2 TYPES OF DC MOTORS

i) Permanent magnet DC motors
ii) Separately excited DC motors
iii) Series wound DC motors
iv) Shunt wound DC motors
v) Compound wound DC motors

i) Series wound DC motors

Series wound DC motors possesses the field winding of a few turns of heavy conductor, connected in series with the armature, *i.e.*, load current flows through both the field and armature with increasing load, the speed of the motor is very high.



Hence series wound motor should never be used without load.

In DC series motor,

Armature current, I_a = Series field current, I_{se}
= Line current, I_L
= I

Back emf developed, $E_b = V - I(R_a + R_{se})$

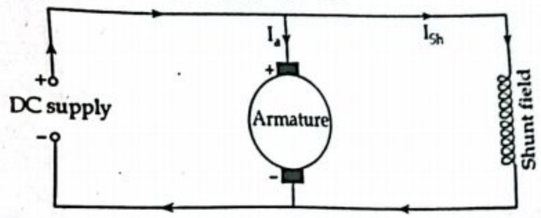
Power drawn from supply mains = VI

Mechanical power developed,

P_m = Power input – Losses in armature and field
$= VI - I^2 (R_a + R_{se})$
$= I [V - I (R_a + R_{se})]$
$= E_b I$

ii) Shunt wound DC motors

Shunt-wound motors possesses the field winding of large number of turns, and high resistance, which is connected in parallel with the armature. The current supplied field winding and second through the armature *i.e.*, input line current, $I_L = I_a + I_{sh}$



where, I_a is the armature current and I_{sh} is the shunt field current and is given by the equation,

$$I_{sh} = \frac{V}{R_{sh}}$$

Back emf developed, $E_b = V - VI_L$

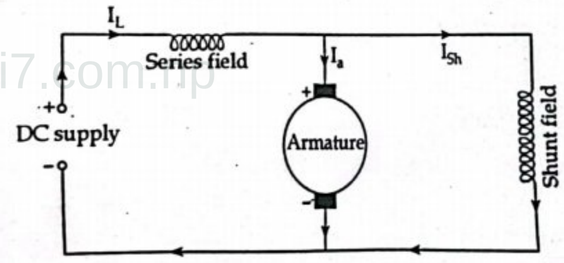
Mechanical power developed,

P_m = Power input – Losses in armature and shunt field
$= VI_L - VI_{Sh} - I_a^2 R_a$
$= VI_a - I_a^2 R_a$
$= I_a (V - I_a R_a)$
$= E_b I_a$

Its starting or no-load torque is about 2.5 to 3 times greater than the full-load torque. By using shunt regulator, the variations of speed of the motor can be well achieved. It runs practically at constant speed at almost all loads.

iii) Compound wound motor

Compound wound motor has a series as well as shunt windings. The shunt field is normally the stronger of the two (*i.e.*, has more ampere turns). Compound motors are of two types namely cumulative compound wound and differential compound wound motors.



Cumulative compound wound is one in which the field windings are connected in such a way that the direction of flow of current is same in

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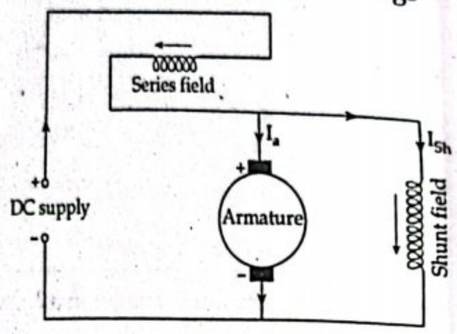
both of the field windings. In the motor of this type the flux due to series field winding strengths the field due to the shunt field winding.

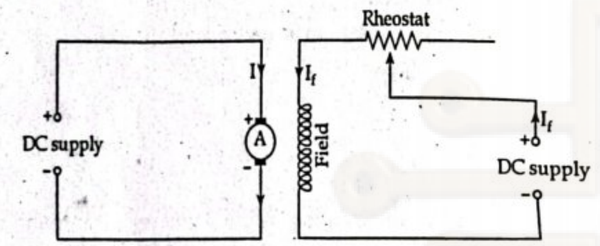
Differential compound wound motor is one in which the field windings are connected in such a way that the directions of flow of currents is opposite to each other in the two field winding.

Compound wound motors, like compound wound DC generators, may be either long-shunt connected or short shunt connected.



iv) Separately-excited DC motors

These motors have field coils similar to those of a shunt wound machine, but the armature and field coils are fed from different supply sources and may have different voltage ratings.



Armature current, I_a = Line current, I_L = I

Back emf developed, $E_b = V - IR_a$

Power drawn from supply mains, P = VI

5.3.3 OPERATING CHARACTERISTICS OF DC MOTOR

a) DC-series Motor

In a series motor, the current remains same in all parts i.e., the armature current is also the field current.

i) Speed characteristics

We know,

$$E_b = \frac{\phi ZNP}{60A} \text{ volts}$$

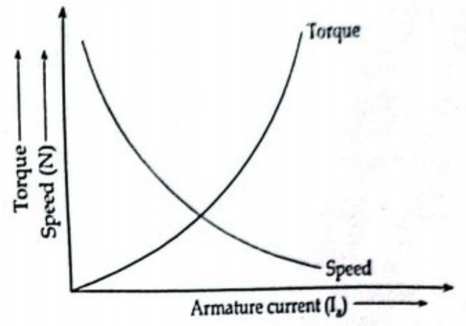
or $E_b \propto \phi N$

or $(V - I_aR_a) \propto \phi N$ $\quad \left[\because \frac{ZP}{60A} = \text{Constant}\right]$

or $N \propto \left(\frac{V - I_aR_a}{\phi}\right)$

and $\phi \propto I_a$

In a series motor, initially the field flux (ϕ) rises in proportion to current, but afterwards, less rapidly, due to magnetic saturation. Consequently, the speed is roughly proportional to the current. The speed of a series motor decreases very rapidly with an increase in load up to the rated load, but after full-loads the speed change less rapidly. This is because under heavy loads, the iron in the magnetic circuit approaches saturation and the rate of increase of flux with load reduces and the speed falls rapidly.



ii) Torque characteristics

We know,

Torque ∝ Armature current × Field flux

or $I \propto I_a\phi$

However, in a series motor, the flux(ϕ) is approximately proportional to the armature current upto full load, so in a series motor.

$T \propto I_a^2$

Consequently, the initial torque characteristics of series motor is parabolic. Hence, at starting, if heavy load is applied on to a series motor, a very good starting torque is obtained. At heavier loads, when the iron is becoming saturated, the curve merges into a straights line.

iii) Speed-torque characteristics

Since a series motor develops a high initial torque at low speeds and a low torque at high speed, so speed-torque characteristic of a series motor is a hyperbola. High initial torque at low speeds enables even a small series motor to start a heavy load.

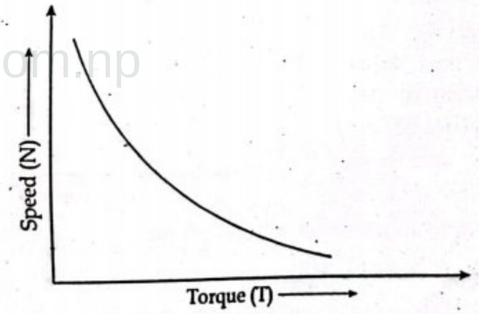


Fig: Characteristics of DC series motor

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B. Characteristics of Shunt-wound DC Motors

i) Speed-current characteristics

Back emf of a shunt motor is,

$$E_b = (V - I_aR_a) = \frac{\phi ZNP}{60A}$$

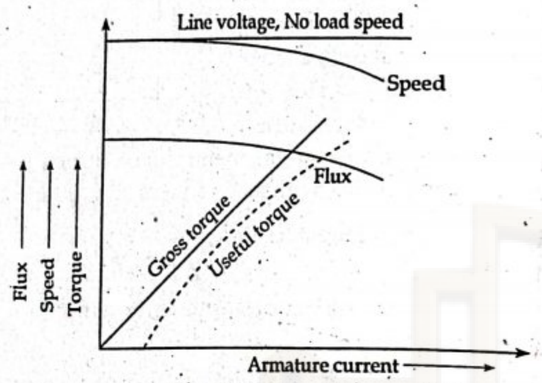
or $E_b \propto \phi N$

or $(V - I_aR_a) \propto \phi N$

or $N \propto \left(\frac{V - I_aR_a}{\phi}\right)$

For a shunt motor, the field flux(ϕ) is independent of the load, and remains constant.

Consequently, the $(V - I_aR_a)$ decreases with the increase in load or I_a. Hence there is a little fall in the speed (N).



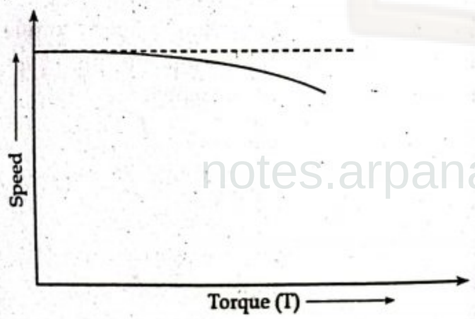
Since the value of R_a is usually small, so the fall in E_b and hence N is very small, even at full load. Consequently the speed (N) – current (I_a) characteristics of a shunt wound motor is approximately a slightly droping straight line.

ii) Torque characteristic

Since the flux(ϕ) of a shunt motor is practically independent of the armature current, I_a.

so, $T \propto I_a\phi \propto I_a$

Consequently torque-armature current characteristics of a shunt motor is a straight line passing through the origin. Although the field current remains practically constant, yet the field flux(ϕ) becomes slightly weaker due to the armature reaction. Hence the curve bends slightly at heavy armature currents.



iii) Speed-torque characteristics

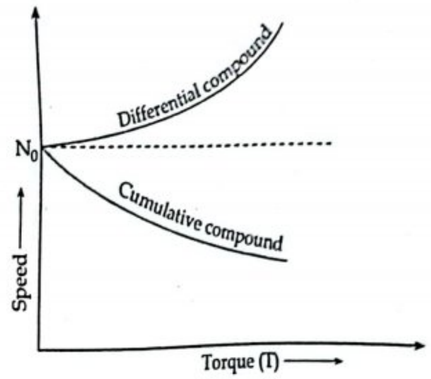
It is similar to speed-armature current characteristics.

C. Characteristics of Compound Wound DC Motors

i) Speed-torque characteristic

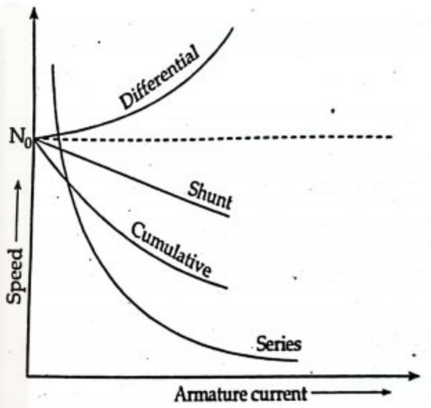
It depends upon the types of compound motor. In a cumulative compound motor, the series excitation helps the shunt excitation. So its speed-torque characteristic lies between that of the shunt motor and the series motor.

In case of a differential compound motor, the torque increases very slightly with the speed.



ii) Speed-current characteristic

In a cumulative compound DC motor, the series field aids the shunt field, so the flux per pole increases as the armature current increases. Consequently, the speed-current characteristic lies between a shunt motor and a series motor.



In case of differential compound DC motor, with the increase of load, the armature drop increases, which tries to decrease the speed but at the same time the demagnetizing effect of the armature current and series field tries to decrease the excitation turns, thereby there is a net increase in speed. Consequently, under accidental overload, the flux per load reduces almost to zero and motor can acquire dangerously high speeds.

iii) Torque current characteristic

In a cumulative compound motor, as the armature current increase, the series flux increases, thereby the total flux of the motor increases. Hence, torque also increases. However the increase in torque with current is greater than that would be for shunt motor.

In case of differential compound motor, the series field opposes the shunt field. So, the total flux of such a motor decreases somewhat with increase

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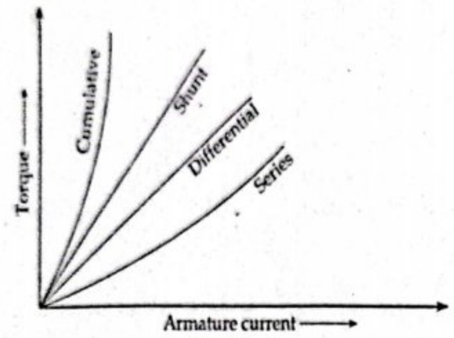
in current. Hence, in a differential compound motor, the torque increases with current but the increase is not so rapid as would be in case of series motor only.

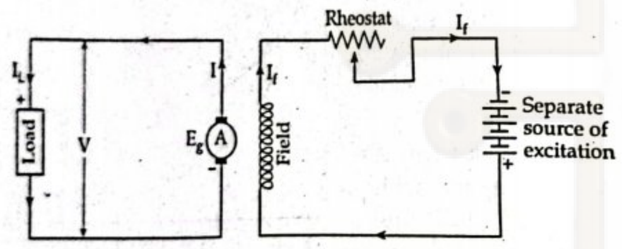


5.3.4 TYPES OF DC GENERATORS

Depending upon the way by which their fields are excited, DC generators are basically of two types.

a) Separately-Excited DC Generators

A DC generators whose field winding is excited from an independent external DC sources as battery, the generator is called a separately – excited generator.



Here;

$I_a = I_L = I$

$V = E_g - IR_a$

Power developed,

$P_g = E_g \cdot I$

where,

V = Generated emf, E_g = Voltage across the load

b) Self-Excited DC Generators

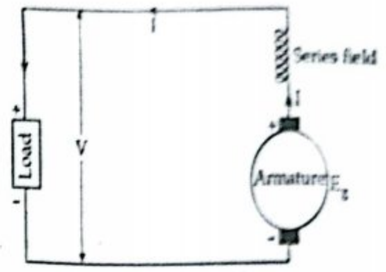
A DC generator whose field winding is excited by the current supplied by the generator itself, is called a self excited generator. In such machines the field coils are interconnected with the armature winding. The field coils may be connected either in series with the armature, in parallel with the armature or partly in series and partly in parallel with armature.

Accordingly the self-excited generators may be classified as

i) Series wound generators
ii) Shunt wound series generator
iii) Compound wound generator

i) Series wound generators

In a series wound generator, there is only one field winding (one field coil on the core of each salient pole) which is connected in series with armature winding so that whole current flows through the field winding as well as well as load.



Since the series field winding carries the full load current, series winding is designed with a fewer turns of thick wire or strips. The resistance of series field winding R_{se} is very low.

Here;

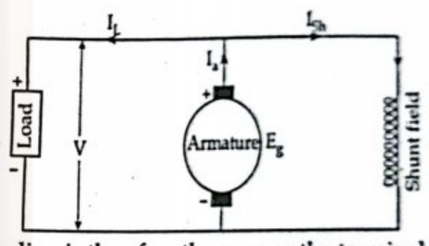
$I_a = I_{se} = I_L = I$

$V = E_g - I(R_a + R_{se})$

$P_g = E_g I$

ii) Shunt wound generator

In a shunt wound generator also there is only one field winding as in case of series wound generator, but in this case it is connected across the armature circuit forming a parallel or shunt circuit.



The voltage across the field winding is therefore the same as the terminal voltage of the generator.

In operation, total armature current I_a flowing from the positive brush divides between two parallel paths; the external circuit and the shunt field current I_{sh} close the circuit current, *i.e.*, $I_a = I_L + I_{sh}$

Here;

Shunt field current, $I_{sh} = \frac{V}{R_{sh}}$

Armature current, $I_a = I_L + I_{sh}$

Power developed, $P_g = E_g I_a$

iii) Compound wound generator

In compound wound generators, there are two field windings (two separate coils on the core of each pole). One of them is connected in series

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with the armature winding and other is connected across the armature. Compound generators may be connected either short shunt, with the field winding in parallel with the armature alone or long shunt with the shunt field winding in parallel with the armature and series field winding.

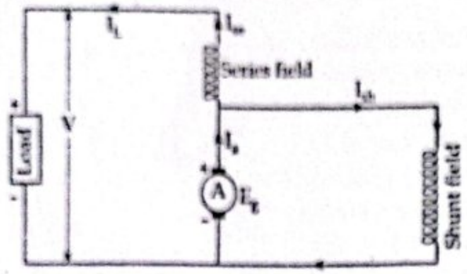

Fig: Short-shunt compound wound generator.

Series field current, $I_{se} = I_L$

Shunt field current, $I_{sh} = \frac{V + I_{se} R_{se}}{R_{sh}}$

Armature current, $I_a = I_L + I_{sh}$

Terminal voltage, $V = E_g - I_a R_a - I_{se} R_{se}$

Power developed, $P_g = E_g I_a$

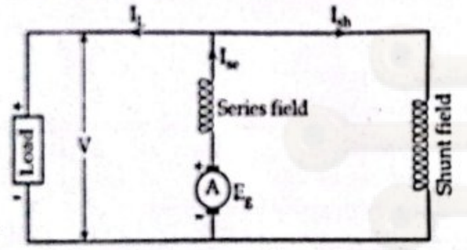

Fig: Long-shunt compound wound generator

Shunt field current, $I_{sh} = \frac{V}{R_{sh}}$

Armature current, I_a = Series field current, $I_{se} = I_L + I_{sh}$

Terminal voltage, $V = E_g - I_a R_a - I_{se} R_{se} = E_g - I_a (R_a + R_{se})$

Power developed, $P_g = E_g I_a$

Power delivered, $P_L = VI_L$

5.3.5 SPEED CONTROL OF MOTOR

Speed of a motor is given by

$$N = \left(\frac{V - I_a R_a}{Z\phi}\right) \cdot \left(\frac{A}{P}\right) = K\left(\frac{V - I_a R_a}{\phi}\right) \text{ rps}$$

where, R_a = Armature circuit resistance

It is obvious that the speed can be controlled by varying (i) flux/pole, ϕ (flux control) (ii) resistance R_a of armature circuit rheostatic control and (iii) applied voltage V (Voltage control).

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A. Speed Control of DC Shunt Motors

i) Variation of flux or flux control method

It is seen from above that $N \propto 1/\phi$. By decreasing the flux, the speed can be increased and vice-versa. Hence, the name flux or field control method. The flux of a DC shunt motor can be changed by changing I_{sh} with the help of a shunt field rheostat. This method is very efficient.

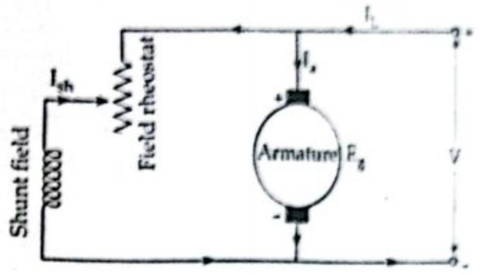

Fig: Field rheostat control for a DC shunt motor

Let initial values of back emf, flux and speed be E_{b1}, ϕ_1 and N_1 respectively and let the back emf and flux be E_{b2} and ϕ_2 respectively at new speed N_2.

Since, $N \propto \frac{E_b}{\phi}$

or, $\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \times \frac{\phi_1}{\phi_1}$

$\therefore \quad \phi_2 = \frac{E_{b2}}{E_{b1}} \times \frac{N_2}{N_1} \times \phi_1$

ii) Armature or Rheostatic control method

This method is used when speeds below the no-load speed are required. As the supply voltage is normally constant, the voltage across the armature is varied by inserting a variable rheostat or resistance (called controller resistance) in series with armature circuit. As controller resistance is increased, thereby decreasing the armature speed. For a load constant torque, speed is approximately proportional to the p.d. across the armature. It is seen that greater the resistance in armature circuit, greater is the fall in the speed.

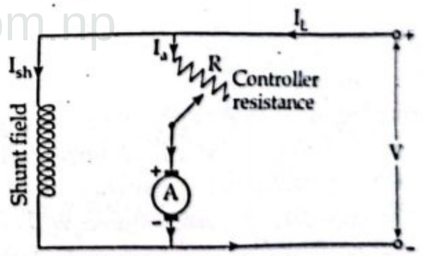

Fig: Armature resistance control method for DC shunt motor

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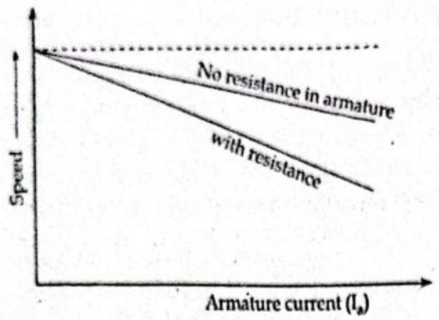


Fig: Speed-current characteristics of DC shunt motor with armature control

Let, E_{b1} be the emf at speed N_1 with armature current I_{a1} be the and no extra resistance in the armature circuit. Let, after insertion of the external resistance R in the armature current I_{a2} and flux remaining the same.

Since, $N \propto \frac{E}{\phi}$

or $N \propto E_b$ [$\because$ Flux, ϕ = Constant]

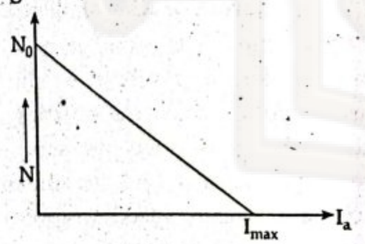
or $\frac{N_2}{N_1} = \frac{V - I_{a2}(R_a + R)}{V - I_{a1}R_a}$

Knowing the values of V, I_{a1}, I_{a2}, N_1 and N_2 the value of additional resistance R to be inserted in series with the armature circuit can be determined. Considering no load speed we have,

$$\frac{N}{N_0} = \frac{V - I_a(R_a + R)}{V - I_{a0}R}$$

If voltage drop in armature at no load is neglected,

$$N = N_0\left[1 - \frac{I_a(R_a + R)}{V}\right]$$



For a given resistance ($R + R_a$) in the armature circuit, the speed is linear function of armature current, I_a.

The motor speed will be zero *i.e.*, the motor will be stalled when $I_a = \frac{V}{R_a + R}$. This is maximum current and is known as stalling current.

iii) Voltage control method

(a) Multiple voltage control

In this method, the shunt field of the motor is connected permanently to a fixed exciting voltage, but the armature is supplied with different voltages by connecting it across one of the several different voltages by means of suitable switchgear. The armature speed will be approximately proportional to these different voltages. The intermediate speeds can be obtained by adjusting the shunt field regulator.

b) Ward-Leonard system

The basic adjustable-voltage armature control method of speed control accomplished by means of an adjustable - voltage generator is called the ward-Leonard system. This system consists simply working motor with a constant excitation and applying a variable voltage to its armature to provide the required speed. The variable voltage supply is obtained from a motor - generated or converter set.

Advantages

i) Very fine speed control over the whole range from zero to normal speed in both directions can be obtained,

ii) Uniform acceleration can be obtained.

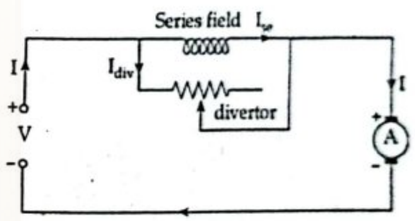
iii) Speed regulation is good.

B. Speed Control of DC Series Motor

i) Flux control method:

Variations in the flux of a series motor can be brought about in any one of the following ways.

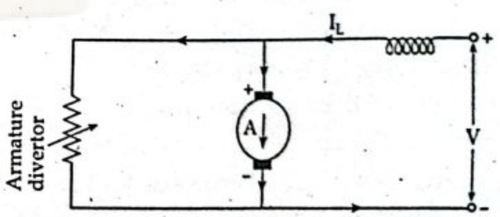
a) Field diverters



The series windings are shunted by a variable resistance known as field diverter. Any desired amount of current can be passed through the diverter by adjusting its resistance. Hence, the flux can be decreased and consequently the speed of the motor increased.

This method is convenient as well as economical and provides the speed control range usually not exceeding 2 : 1.

b) Armature diverter



A diverter across the armature can be used for giving speeds lower than the normal speed. For a given constant load torque, if I_a is reduced due a armature diverter, the ϕ must increase, ($T \propto \phi I_a$). This results in an increase in current taken from the supply. The variation in speed can be controlled by varying the diverter resistance.

c) Tapped field control method

The number of series field turns in the circuit can be changed at will. With full field, the motor runs at its minimum speed which can be raised in

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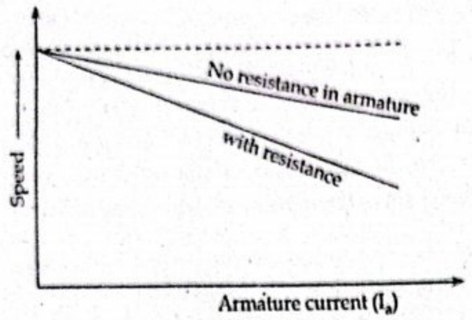

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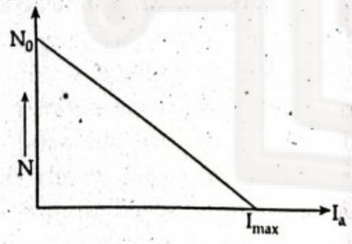

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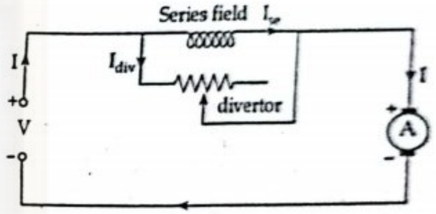

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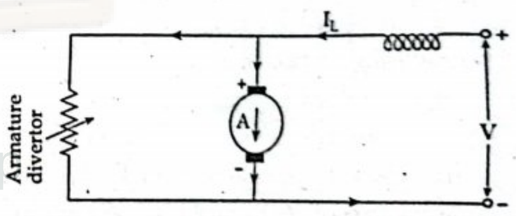

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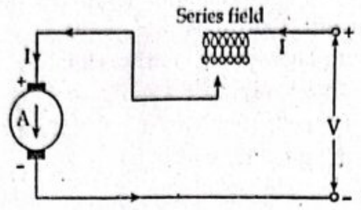
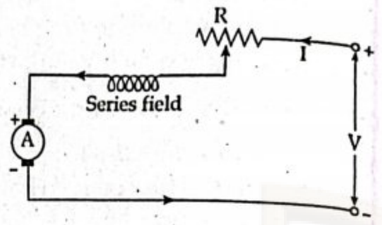
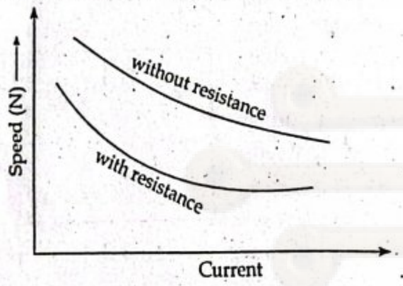
steps by cutting out same of the turns.



ii) Variable resistance in series with motor

By increasing the resistance in series with the armature, the voltage applied across armature terminals can be decreased, with reduced voltage across the armature, the speed is reduced. However, it will be noted that, since, full motor current passes through this resistance, there is a considerable loss of power in it.





5.3.6 SELECTION OF ELECTRIC MOTORS

For selection of reliable and efficient motor, it is essential that the conditions of service are well known. It is not sufficient to simply, specify the output power in KW and the speed but it is also necessary to know the following additional particulars.

i) Switching frequency
ii) Efficiency of motor at different loads
iii) Accelerating torque and breaking torque
iv) Torque at the shaft during running, starting and at different loads
v) Other working requirements.

In studying the behavior of a motor selected for a particular driven unit, one of the first problems involved is to determined whether the speed-torque characteristics of the motor suits the requirements imposed by the speed-torque characteristic of the driven unit.

Example 17

A 230 V series motor is taking 50 A. Resistance of armature and series series field winding is 0.2 Ω and 0.1 Ω respectively. Calculate (i) Brush voltage (ii) Back emf (iii) power wasted in armature and mechanical power developed.

Solution:

Supply or terminal voltage, V = 230 V
Motor current, I = 50 A
Armature resistance, $R_a = 0.2\ \Omega$
Series field resistance, $R_{se} = 0.1\ \Omega$

i) Brush voltage = Terminal - Voltage drop in series field
$= V - IR_{se}$
$= 230 - 50 \times 0.1$
$= 225\ V$

ii) Back emf, $E_b = V - IR_{se} - IR_a$
$= V - I(R_{se} + R_a)$
$= 230 - 50\ (0.1 + 0.2)$
$= 215\ V$

iii) Power wasted in armature $= I^2R_a = 50^2 \times 0.2 = 500\ W$

iv) Mechanical power developed,

$$P_m = \frac{E_b \cdot I}{1{,}000} = \frac{215 \times 50}{1{,}000} = 10.75\ KW$$

Example 18

A 20 KW, 240 V DC shunt generator has armature and field resistance of 0.05 Ω and 80 Ω respectively. Calculate the total armature power developed when working:

i) as a generator delivering 20 KW output.
ii) as a motor taking 20 KW input

Solution:

Shunt field current, $I_{sh} = \frac{V}{R_{sh}} = \frac{240}{80} = 3\ A$

i) As a generator

Load current, $I_L = \frac{\text{Output in KW} \times 1{,}000}{V} = \frac{20 \times 1{,}000}{240} = 83.33\ A$

Armature current, $I_a = I_l + I_{sh} = 86.33\ A$

Generated emf, $E_g = V + I_a + R_a = 240 + 86.33 \times 0.05 = 244.32\ A$

Total armature power delivered,

$$P_g = \frac{E_g \times I_a}{1{,}000} = \frac{244.32 \times 83.33}{1{,}000} = 20.36\ KW$$

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ii) As motor

$$I_L = \frac{20 \times 1{,}000}{240} = 83.33 \text{ A}$$

Armature current, $I_a = I_l - I_{sh} = 80.33$ A

Back emf, $E_b = V - I_a R_a = 240 - 80.33 \times 0.05 = 235.98$ V

Power developed, $P_m = \frac{E_b I_a}{1{,}000} = \frac{235.98 \times 80.33}{1{,}000} = 18.96$ KW.

Example 19

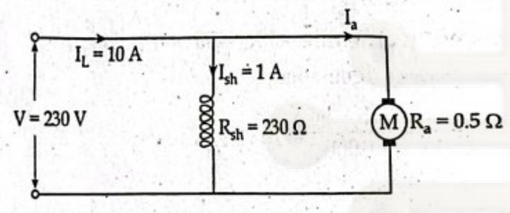
A 230 V DC shunt motor runs at 1,000 rpm on full drawing current of 10 A. The shunt-field resistance is 230 ohm and the armature resistance is 0.5 ohm. Calculate the resistance to be inserted in series with the armature so that the speed at full load is 950 rpm.

Solution:

$$I_{sh} = \frac{V}{R_{sh}} = \frac{230}{230} = 1 \text{ A}$$

$I_{a1} = I_l - I_{sh} = 10 - 1 = 9$ A

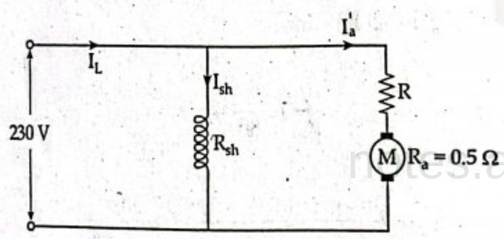
$E_{b1} = V - I_{a1} \times R_a = 230 - 9 \times 0.5 = 225.5$ V



Let, R be the required resistance in series with the armature, the total armature resistance,

$R_{a1} = (0.5 + R)\ \Omega$

$E_{b2} = 230 - (0.5 + R) \times 9 = (225.5 - 9R)$ V



Now,

$$\frac{E_{b2}}{E_{b1}} = \frac{225.5 - 9R}{225.5} = \frac{N_2}{N_1} = \frac{950}{1{,}000} = 0.95$$

or, $225.5 \times 0.95 = 225.5 - 9R$

or, $9R = 225.5 \times 0.05$

or, $R = \frac{225.5 \times 0.05}{9}$

or, $R = 1.253\ \Omega$

Example 20

A 220 V shunt motor has an armature resistance of 0.4 Ω and field resistance of 200 Ω. The motor runs at 750 rpm and takes an armature current of 25 A. Assuming that the load torque remains constant, find the reduction in field resistance necessary to reduce to speed to 500 rpm. Neglect saturation.

Solution:

$I_{a1} = 25$ A

$R_a = 0.4\ \Omega$

$V = 200$ V

$N_1 = 750$ rpm

$R_{sh} = 200\ \Omega$

$$I_{sh} = \frac{200}{200} = 1 \text{ A}$$

$E_{b1} = V - I_a R_a = 200 - 25 \times 0.4 = 190$ V

Let, R be the reduced shunt field resistance, then

$$I_{sh2} = \left(\frac{200}{R}\right) \text{A}$$

Now,

$I_1 \propto \phi_1 I_{a1} \propto I_{sh} I_{a1}$ and

$I_2 \propto I_{sh2} I_{a2}$

But, $T_1 = T_2$

so, $I_{sh2} \cdot I_{a2} = I_{sh1} \cdot I_{a1}$

$I_{sh2} \cdot I_{a2} = 1 \times 25$

or, $I_{a2} = \frac{25}{I_{sh2}} = \frac{25}{200/R} = (200 - 0.125\,R)$ V

so, $E_{b2} = 200 - 0.0125\,R \times 0.4 = (200 - 0.05\,R)$ V

Then,

$$\frac{E_{b1}}{E_{b2}} = \frac{190}{200 - 0.05\,R} = \frac{I_{sh1} \times N_1}{I_{sh2} \times N_2} = \frac{1 \times 750}{(200/R) \times 500} = \frac{1.5\,R}{200}$$

or, $\frac{190}{200 - 0.05\,R} = 0.0075\,R$

or, $190 = 1.50\,R - 0.000375\,R^2$

or, $0.000375\,R^2 - 1.50\,R + 190 = 0$

$\therefore$ $R = 130.90\ \Omega$

Hence, reduction in shunt resistance,

$= 200 - 130.90 = 69.90\ \Omega$

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Example 21

A 220 V DC series motor runs at 1,000 rpm and takes 20 A current. Combined resistance of armature and field is 0.4 ohm. Calculate the resistance to be inserted in series so as to reduce the speed to 800 rpm. Assuming torque varies as square of the speed.

Solution:

$I_1 = I_{a1} = 20$ A

$N_1 = 1000$ rpm

$N_2 = 800$ rpm

$R_{m1} = 0.4\ \Omega$

$R_{m2} = (0.4 + R)\ \Omega$

Now,

$$\frac{T_2}{T_1} = \frac{I_{a2}\,\phi_2}{I_{a1}\,\phi_1} = \left(\frac{I_{a2}}{I_{a1}}\right)^2 = \left(\frac{N_2}{N_1}\right)^2 = \left(\frac{800}{1{,}000}\right)^2$$

$I_{a2} = 0.8\ I_{a2} = 0.8 \times 20 = 16$ A

Now,

$E_{b1} = V - I_{a1} R_{m1} = 200 - 20 \times 0.4 = 192$ V

$E_{b2} = V - I_{a2} R_{m2} = 200 - 16 \times (0.4 + R) = (193.6 - 16\,R)$ V

Also,

$$\frac{E_{b2}}{E_{b1}} = \frac{N_2\,\phi_2}{N_1\,\phi_1} = \frac{N_2\,I_{a2}}{N_1 I_{a1}} = \frac{800 \times 16}{1{,}000 \times 20} = 0.64$$

so, $E_{b2} = 0.64 \times 192 = 122.88$ V

and, $E_{b2} = 193.6 - 16\,R$

or, $122.88 = 193.6 - 16\,R$

or, $R = \dfrac{193.6 - 122.88}{16}$

$\therefore \quad R = 4.42\ \Omega$

Hence, 4.42 Ω resistance is to be inserted in series.

Example 22

A 10 Hp, 230 V shunt motor has an armature resistance of 0.5 Ω. At no-load and rated (full) voltage, the speed is 1,200 rpm and armature current is 2 A. If load is applied, the speed drops to 1,100 rpm. Determine the armature current and line current.

Solution:

Speed $N_0 = 1200$ rpm

armature current, $I_{a0} = 2$ A

Back emf, $E_{b0} = V - I_{a0} R_a = 230 - 2 \times 0.5 = 229$ V

Let, the armature current at load be of I_{a1} Amperes.

Back emf, $E_{bL} = V - I_{a1} R_a = 230 - 0.5\ I_{a1}$

Since,

$$\frac{\text{Speed at load } N_1}{\text{Apeed at no-load } N_0} = \frac{E_{b1}}{E_{b0}}$$

or, $E_{b1} = E_{bo} \times \dfrac{N_1}{N_0} = 229 \times \dfrac{1{,}100}{1{,}200} = 209.917$ V

or, $230 - 0.5\ I_{a1} = 209.917$

$$I_{a1} = \frac{230 - 209.917}{0.5} = 40.2 \text{ A}$$

$\therefore$ Shunt field current, $I_{sh} = \dfrac{V}{R_{sh}} = \dfrac{230}{115} = 2$ A

$\therefore$ Line current at load, $I_L = I_{a1} + I_{sh} = 40.2 + 2 = 42.2$ A

Example 23

A 250 V DC shunt motor having an armature resistance of 0.25 Ω carries an armature current of 50 A and runs at 750 rpm. If the flux is reduced by 10%, find the speed. Assume that the load torque remains same.

Solution:

Back emf, $E_{b2} = V - I_{a1} R_a = 250 - 50 \times 0.25 = 273.5$ V

Speed, $N_1 = 750$ rpm

Flux, $\phi_2 = \dfrac{100 - 10}{100}\,\phi_1 = 0.9\,\phi_1$

Since, torque developed remains same,

$I_{a2}\,\phi_2 = I_{a1}\,\phi_1$

$$I_{a2} = I_a \times \frac{\phi_1}{\phi_2} = 50 \times \frac{\phi_1}{0.9\,\phi_1} = 55.556 \text{ A}$$

Back emf, $E_{b2} = V - I_{a2} R_a = 250 - 55.556 \times 0.25 = 236.1$ V

$\therefore$ Speed emf, $N_2 = N_1 \times \dfrac{E_{b2}}{E_{b1}} \times \dfrac{\phi_1}{\phi_1}$

$$= 750 \times \frac{236.1}{237.5} \times \frac{\phi_1}{0.9\,\phi_1}$$

$$= 828.40 \text{ rpm}$$

Example 24

A 250 V DC shunt motor has an armature resistance of 0.5 Ω and a field resistance of 250 Ω. When driving a constant torque load at 600 rpm, the motor draws 21 A. What will be the new speed of the motor if an additional 25 Ω resistance is inserted in the field circuit?

Solution:

Line voltage, $V = 250$ V

Shunt field current $I_{sh1} = \dfrac{V}{I_{Sh}} = \dfrac{250}{250} = 1$ A

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Back emf, $E_{b1} = V - I_{a1} R_a = 250 - 20 \times 0.5 = 240$ V

New shunt field current, $I_{sh2} = \frac{V}{R + R_{sh}} = \frac{250}{250 + 250} = 0.5$ A

Flux, $\phi_2 = \phi_1 \times \frac{I_{Sh2}}{I_{sh1}} = \phi_1 \times \frac{0.5}{1} = 0.5\,\phi$

Armature current, $I_{a2} = I_{a1} \times \frac{\phi_1}{\phi_2} = 20 \times \frac{1}{0.5} = 40$ A ($\because T \propto \phi\, I_a$ and $\phi \propto I_{sh}$)

New speed, $N_2 = N_1 \times \frac{E_{b2}}{E_{b1}} \times \frac{\phi_1}{\phi_2}$

$= 600 \times \frac{230}{240} \times 2$

$N_2 = 1{,}150$ rpm.

Example 25

In a long shunt compound generator, the terminal voltage is 230 V when it delivers 150 A. Determine induced emf. The shunt field, series field, diverter and armature resistance are 92, 0.015, 0.03 and 0,032 ohm respectively.

Solution:

Terminal voltage, V = 230 V

Shunt current, $I_{sh} = \frac{V}{R_{sh}} = \frac{230}{92} = 2.5$ A

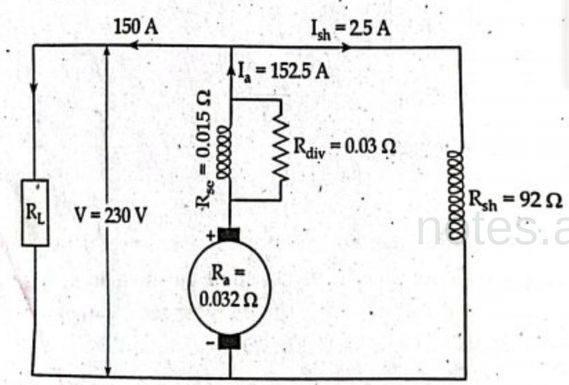
Armature current, I_a = Load current, $I_L + I_{sh}$

$= 150 + 2.5$

$= 152.5$ A

Combined resistance of series field and its diverter,

$R_{se,eq} = \frac{R_{se} \times R_{div}}{R_{se} + R_{div}} = \frac{0.015 \times 0.03}{0.015 + 0.03} = 0.01\ \Omega$



∴ Induced emf, $E_g = V\, I_a + R_a + I_a R_{se,eq}$

$= 230 + 152.5 \times 0.032 + 152.5 \times 0.01$

$= 236.4$ V.

Example 26

A shunt compound wound DC generator delivers 100 A to a load at 250 V. The generator has shunt field, series and armature resistance of 130 Ω, 0.1 Ω, and 0.1 Ω respectively. Calculate the voltage generated in the armature winding. Assume 1 V voltage drop per brush.

Solution:

Load current, $I_L = 100$ A

Terminal voltage, $V_L = 250$ V

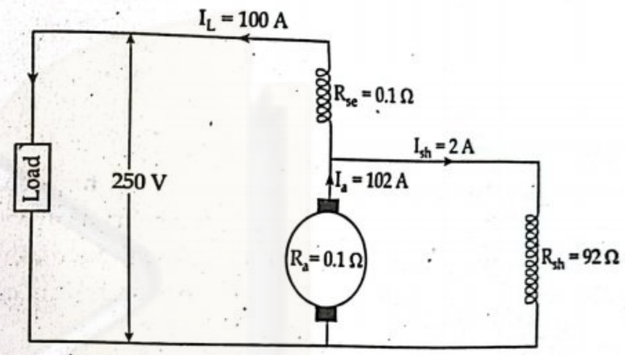
Series field current $I_{se} = I_L = 100$ A

Voltage across shunt field, $V_{sh} = V_L + R_{se} R_{se}$

$= 250 + 100 \times 0.1$

$= 260$ V

Shunt filed current, $I_{sh} = \frac{V_{sh}}{R_{sh}} = \frac{260}{130} = 2$ A



∴ Armature current, $I_a = I_L + I_{sh} = 100 + 2 = 102$ A

∴ Generated voltage, $E_g = V + I_{se} R_{se} + I_a R_a$ + brush contact drop

$= 250 + 100 \times 0.1 + 2 \times 1$

$= 272.2$ V.

Example 27

A 220 V shunt motor has an armature resistance of 0.5 Ω and takes a current of 40 A on full load. By how much the main flux be reduced to raised the speed by 50% if the developed torque is constant?

Solution:

Since torque remains constant,

$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \times \frac{\phi_1}{\phi_2}$

Hence, $\phi_1\, I_{a1} = \phi_2\, I_{a2}$

∴ $I_{a2} = I_{a1} \times \frac{\phi_1}{\phi_2} = 40x$, where $x = \frac{\phi_1}{\phi_2}$

∴ $E_{b1} = 220 - (40 \times 0.5) = 200$ V

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$\therefore \quad E_{b2} = 220 - (40x \times 0.5) = (220 - 20x)$ V

or, $\frac{N_2}{N_1} = \frac{3}{2}$ ..... given

or, $\frac{3}{2} = \frac{220 - 20x}{200}$

or, $x^2 = 11x - 15$

or, $x^2 - 11x + 15 = 0$

$\therefore \quad x = 1.6$ or 9.4

Hence,

$\frac{\phi_1}{\phi_2} = 1.6$

or, $\frac{\phi_2}{\phi_1} = \frac{1}{1.6}$

and, $\frac{\phi_1 - \phi_2}{\phi_1} = \frac{1.6 - 1}{1.6} = \frac{3}{8}$

Then, percentage change in flux $= \frac{3}{8} \times 100 = 37.5\%$.

Example 28

A DC series motor having a resistance of 1 Ω drives a fan for which the torque varies as the square of the speed. At 220 V, the set runs at 350 rpm and takes 25 A. The speed is to be raised to 500 rpm by increasing the voltage. Determined the necessary voltage and the corresponding current assuming the field to be unsaturated.

Solution:

Since $\phi \propto I_a$, $T_a \propto \phi\ I_a^2$

Also,

$T_a \propto N^2$

$\therefore \quad I^2_a \propto N^2$

or, $I_a \propto N$

or, $I_{a1} \propto N_1$ and $I_{a2} \propto N_2$

$\therefore \quad \frac{I_{a2}}{I_{a1}} = \frac{N_2}{N_1} = \frac{500}{300}$

or, $I_{a2} = 25 \times \frac{500}{300} = \frac{250}{7}$ A

or, $E_{b1} = 220 - 25 \times 1 = 195$ V

or, $E_{b1} = V - (250/70) \times 1$

and, $\frac{\phi_1}{\phi_2} = \frac{25}{250/7} = \frac{7}{10}$

Now, $\frac{N_2}{N_1} = \frac{E_{b2} \times \phi_1}{E_{b1} \times \phi_2}$

or, $\frac{500}{300} = \frac{V - (250/7)}{195} \times \frac{7}{10}$

$\therefore \quad V = 433.7$ V.

5.4 INDUCTION MOTORS

An induction motor or asynchronous motor is an AC electric motor in which the electric current in the rotor needed to produced torque is obtained by electromagnetic induction from the magnetic field of the stator winding.

5.4.1 CONSTRUCTION

An induction motor consists essentially of two main parts:

a) A stator
b) A rotor

a) Stator

Stator is an outer 'stationary' hollow cylindrical structure made up of laminated sheet steel. The laminations are slotted on their inner periphery. In these slots, a normal three phase winding is placed which constitutes the stator winding. Stator is wound for a definite number of poles, whose number depends upon the speed requirement. Greater the number of poles, lesser the sped and vice - versa. The stator winding when supplied with 3-phase currents, produce a magnetic flux which is of constant magnitude but which revolves or rotates at synchronous speed,

$N_s = 120 \times \frac{f}{P}$

This revolving magnetic flux induces an emf in the rotor by mutual induction.

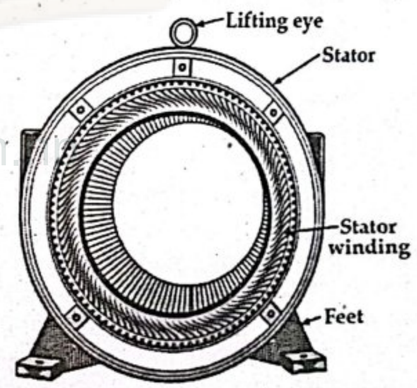

Fig: Stator of 3-phase induction motor.

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b) Rotor

Rotor is an inner cylindrical core, built of lamination sheet steel strips with partially enclosed slots. The slots of the rotor are not always parallel to the slots on the stator but they are given a twist known as skew. The rotor is then known as skewed rotor (skew effective reduced noises, eliminates logging, magnetic locking of the rotor and increases the starting torque). The windings are placed in the slots. The rotor is mounted on a spider with a shaft. When properly assembled, the rotor is free to turn within the stator.

Types of rotor winding:

There are two types of rotor depending upon the types of winding used.

i) Squirrel cage rotor

Squirrel cage rotor consists of a slotted laminated core of sheet steel. The slots of the rotor are not always parallel to the slots on the stators, but they are usually given a twist, known as skew. Each slot carries a thick copper or aluminium bar. These bars are short circuited on either and by a solid copper or aluminium ring called end-ring. The projecting fins and end rings are usually die-cast in position.

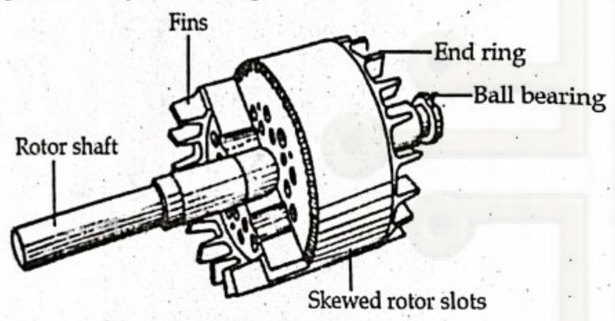

Fig: Squirrel cage rotor

ii) Slip ring or phase wound rotor

It consists of a slotted laminated core of sheet steel. A three phase winding is placed in the rotor slots, having the same number of poles as the stator. The rotor winding is, usually connected in star though it may be connected in delta also. The three free ends of the rotor winding are brought out, and connected to three slip - rings mounted on the rotor shaft on each slip ring there is resting a contact brush. The three brushes are further connected to a three phase star connected rheostat. This arrangement makes possible to include on additional to each phase of the rotor at the time of starting. Thus at the period of starting additional resistance in the rotor circuit is included to have higher starting torque and when running under normal condition, the external resistances are completely cut out from the rotor circuit, and the behaves just like squirrel-cage rotor

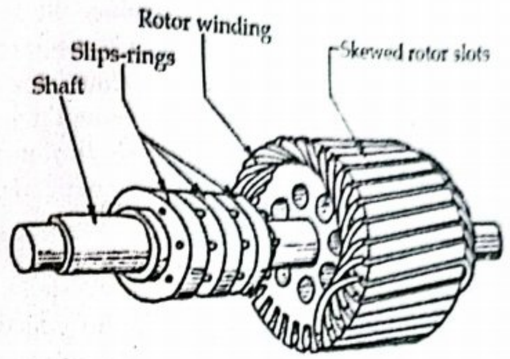

Fig: Slip ring rotor

5.4.2 WORKING PRINCIPLE OF INDUCTION MOTORS

When the three phase winding of the stator is connected phase currents in stator winding produced a rotating magnetic field or flux which rotates round the stators at synchronous speed (N_s). This rotating flux which are yet stationary. Due to relative velocity between the rotating flux and stationary rotor, emf is induced in the rotor conductor. The frequency of induced emf is the same as the supply frequency, when the rotor is stationary. Since the rotor conductors circuit is closed, so induced emf produces rotor current which starts flowing in the rotor conductors. Consequently, a torque producing condition is established. According to Lenz's law, under the influence of this torque, the rotor starts rotating in the same direction as the rotating magnetic field.

Since the directions of rotation of rotor as well as the rotating magnetic field are the same, so the relative velocity between the two decreases. A decrease in relative velocity, however, decreases the induced emf, rotor current and consequently the torque also. But the rotor continues to accelerate and gain speed till equilibrium condition is reached, the motor runs at synchronous speed.

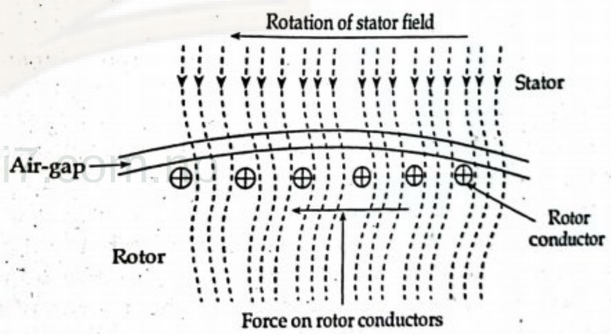

Fig: Working of 3-phase induction motor

When motor is loaded, the load torque plus losses torque becomes greater than the developed torque. Hence, the motor slows down. However, a

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reduction in rotor speed means an increase in relative velocity as well as rotor current and hence the torque developed also. Consequently, motor gains speed and when the condition of equilibrium is restored, the developed equals load torque plus losses torque. The motor consequently runs at new steady state but reduced speed compared to no load speed. Thus, the load, but the speed of the rotating magnetic field remains constant.

5.4.3 APPLICATION OF INDUCTION MOTORS

Squirrel cage induction motor is suitable for constant speed industrial drives of small power where speed control is not required and where starting torque requirements are of medium or low value, such as for printing machinery, flour mills and other shaft drives of small power.

Wound rotor induction motors are used for loads requiring severe starting condition or for loads requiring speed control such as for driving line shafts, lifts, pumps, generators, winding machines, cranes, hoists, elevators, small electric excavators, printing presses, turn tables, stokers, crushes, etc.

5.5 SYNCHRONOUS MOTORS

A synchronous motor is a 3-phase electric motor that runs only at synchronous speed. Thus, a synchronous motor runs either at synchronous speed or or not all and while running it maintains constant speed. A synchronous motor may have either a revolving armature or a revolving field, but the revolving field construction is the more common. A synchronous motor has the same relationship to an alternator as a DC motor has to a DC generator. The essential parts of a three-phase synchronous motor are,

i) Laminated starter core with 3-phase armature winding
ii) Rotating field structure complete with damper winding and slip rings
iii) Brushes and brush holders
iv) Two end shields to house the bearings that support the rotor shaft.

5.5.1 PRINCIPLE OF OPERATION

As the rotor is excited from DC supply, so the poles of the rotor retain the same polarity throughout but the polarity of the stator poles changes as it is connected the rotor as supply mains. First of all, let us consider the rotor as stationary. At this instant the rotor s-pole is attracted to the stator n-pole and therefore, the rotor tends to rotate in clockwise direction. After half a period (*i.e.*, after t/2 second where as T = 1/f) the polarity of stators poles is reversed but the polarity of the rotor poles remains the same.

At this instant, the rotor S-pole is repelled by the stator s-pole being similar in nature and therefore rotor tends to rotate in counter-clockwise direction. Thus, the torque acting on the rotor or a synchronous motor is-

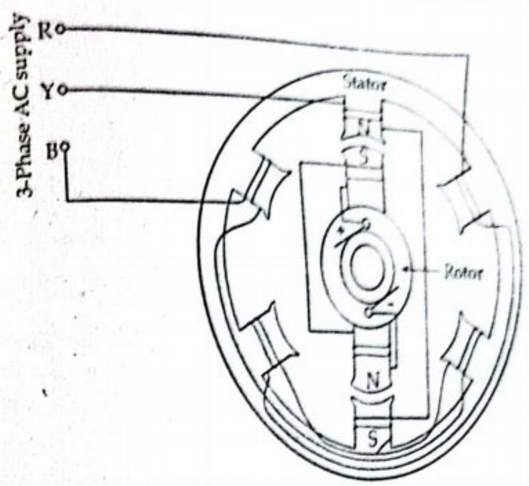
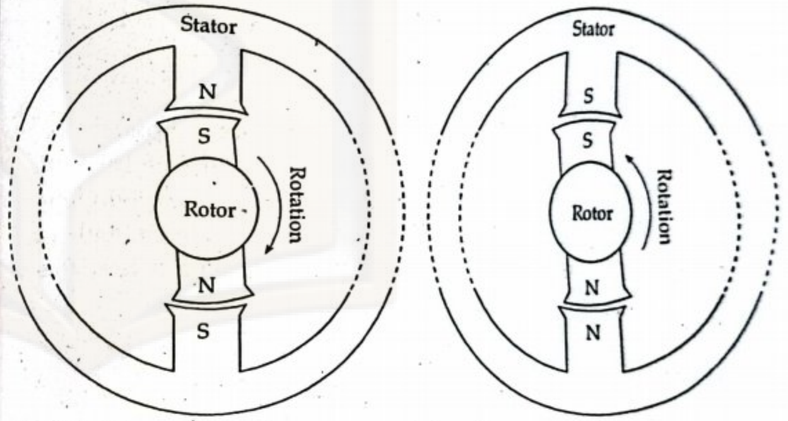


Fig: Three phase, two pole synchronous motor

not unidirectional but pulsating one and due to inertia of the rotor, it will not move in any direction. So, the synchronous motor has got no self starting torque.

Now, let us consider the rotor rotating in clockwise direction which is possible by external means. At this instant, as mentioned above, the rotor S-pole is attracted to stator N-pole and so the torque acts on the rotor in clockwise direction.



After half a period the stator polarity is reversed *i.e.*, stator S-pole becomes N-pole and N-pole becomes S-pole but if the rotor is rotated at such by some external means at the staring moment that rotor S-pole advances by a pole pitch so that it is again under the influence of stator N-pole, that it is again under the influence of stator N-pole, the torque acting on the rotor will be again clockwise.

Hence, a continuous (unidirectional) torque will be obtained and the two field will get locked into each other. Now, if the external means is removed the rotor in clockwise continuous torque acting on the rotor.

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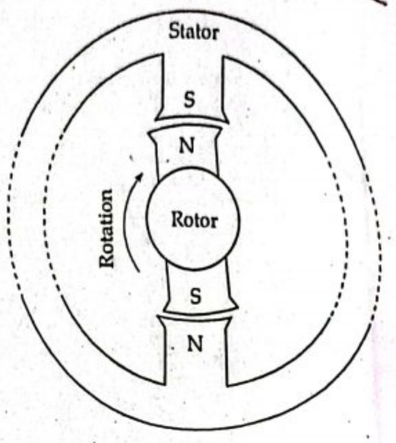
Hence, to obtain a continuous torque, it is necessary that the rotor rotates at such a speed that it moves through the distance equal to the pole pitch in half the period *i.e.*, $\frac{T}{2}$ or $\frac{1}{2f}$ second. Hence, to obtain a continuous torque the rotor should rotate at synchronous speed given by the expression $N_s = \frac{120f}{p}$.



a) Stator

The stator is the stationary armature. This consists of a core and the slots to hold the armature winding similar to the armature of a DC generator. The stator core uses a laminated construction. It is built up of special steel stampings insulted from each other with varnish or paper. The laminated construction is basically to keep down eddy current losses. The entire core is fabricated in a frame made of steel plates. Frames does not carry any flux and serves as the support to the core.

b) Rotor

There are two types of rotors used in synchronous motor:

i) Salient pole type

This is also called projected pole types as all the poles are projected out from the surface of the rotor. The poles are built up of thick steel laminations. The poles are bolted to the rotor. The field winding is provided on the pole shoe. These rotors have large diameters and small axial lengths. As mechanical strength of salient pole type is less, this is preferred for low speed motors.

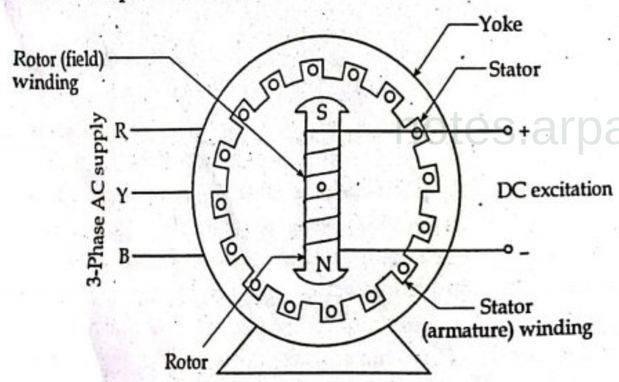


Fig: Essential part of synchronous motor

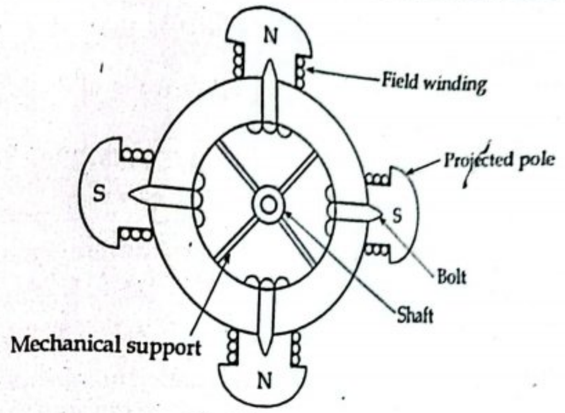


Fig: Salient pole type rotor

ii) Non-salient type or smooth cylindrical type

The rotor consists of smooth solid steel cylinder having number of slots to accomodate the field coil. The slots are covered at the top with the help of steel or manganese wedges. The unslotted portions of the cylinder itself acts as the poles. The poles are not projecting out and the surface of the rotor is smooth which maintains uniforms air gap between stator and the rotor. These rotors have small diameters and large axial lengths. The main advantages of this types is that these are mechanically very strong.

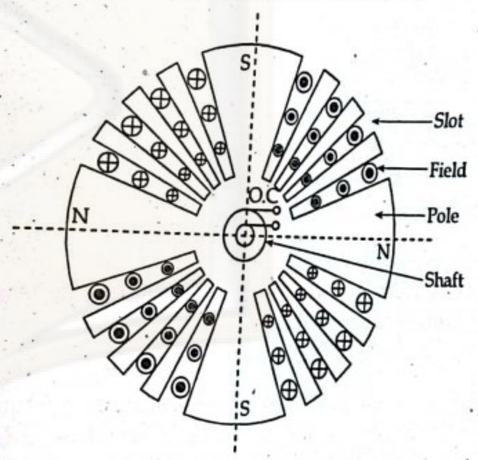


Fig: Smooth cylindrical rotor

5.5.2 USES OF SYNCHRONOUS MOTORS

i) They are used in power houses and substations in parallel to the bus-bars to improve the power factor.

ii) Such motors are also used to regulate the voltage at the end of transmission lines.

iii) They can be employed for the loads where constant speed is required.

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BOARD EXAM SOLVED PROBLEMS

1. **Explain the construction and operating principles of 3 - ϕ induction motor with neat diagram.** [13/F, 13/S, 14/F, 14/S, 15/F, 16/F, 16/S, 17/S, 18/S, 19/F]

Solution: See the definition part 5.4.1 and 5.4.2.

2. **Explain the principle of operation of 3 - ϕ synchronous motor.** [13/F, 15/S, 16/S, 17/F]

Solution: See the definition part 5.5.1.

3. **Write short notes on analogy between magnetic and electric circuits.** [13/F, 15/S, 19/F, 19/S]

Solution: See the definition part 5.5.1.

4. **Write short notes on speed control of DC motor.** [15/S, 17/S]

Solution: See the definition part 5.3.5.

5. **Define transformer and deduce the expression for emf.** [16/F, 17/F]

Solution: See the definition part 5.2 and 5.2.2.

6. **Write shorts notes on operation of transformer on load.** [18/F]

Solution: See the definition part 5.2.4.

7. **Write short notes on transformer efficiency.** [16/S]

Solution: See the definition part 5.2.8.

8. **Mention the condition for maximum efficiency and state the different losses in case of a transformer.** [14/S]

Solution: See the definition part 5.2.9 and

Condition for maximum efficiency is:

Copper loss, P_c = Iron loss (P_i)

9. **Explain the practical transformer with the help of phasor diagram with unity power factor load.** [17/S]

Solution: See the definition part 5.2.4.

10. **A 250 V DC shunt motor takes 8 A line current on no load and runs at 1000 rpm. The resistance of the field winding and armature winding are 200 Ω and 0.8 Ω respectively. If the full load line current is 20 A, calculate the full load speed assuming constant air gap flux.** [17/F]

Solution:

$V = 250$ volts

$E_{b0} = V - I_a R_a$

$= 250 - (I_L - I_{sh}) \times R_a$

$= 250 - \left(8 - \frac{V}{R_{sh}}\right) \times 0.8$

$= 250 - \left(8 - \frac{250}{200}\right) \times 0.8$

$\therefore E_{b0} = 244.60$ V

At Full-load,

$E_b = V - I_a R_a$

$= 250 - (I_l - I_{sh}) \times R_a$

$= \left(20 - \frac{250}{200}\right) \times 0.8$

$\therefore E_b = 235$ V

We know,

$\frac{E_{b0}}{E_b} = \frac{N_0}{N}$

or, $\frac{244.60}{235} = \frac{1000}{N}$

or, $N = \frac{1000 \times 235}{244.60}$

$\therefore N = 961$ rpm

Hence, full load speed is 961 rpm if air gap flux is constant.

11. **A shunt generator has induced voltage of 250 V. When the machine is loaded, the terminal voltage drops down to 230 V. Determine the load current if the armature resistance ids 0.05 ohm and the field circuit resistance is 23 Ω.** [16/S]

Solution:

$E_g = 250$ V

$V = 230$ V

$R_a = 0.05\ \Omega$

$R_{sh} = 23\ \Omega$

$I_{sh} = \frac{V}{R_{sh}} = \frac{230}{23} = 10$ A

Now,

$E_g = V + I_a R_a$

or, $250 = 230 + I_a \times 0.05$

or, $I_a = \frac{250 - 230}{0.05}$

$\therefore I_a = 400$ A.

Hence, load current, $I_L = I_a - I_{sh} = (400 - 10)$ A $= 390$ A.

12. **A 220 V DC shunt motor draws a current of 50 A at full load and runs with a speed of 1700 rpm. Calculate the value of resistance to be inserted in the armature circuit so that the speed drops to 1,200 rpm at constant load. Given that armature resistance and field resistance are 0.04 ohm and 155 ohm respectively.** [15/S, 18/S]

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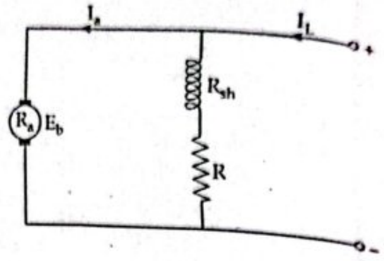
Solution:

$V = 220\ V$

$I_l = 50\ A$

$R_a = 0.04\ \Omega$

$R_f = 155\ \Omega$



Now,

$$E_b = V - I_a R_a$$
$$= 220 - R_a(I_L - I_{sh})$$
$$= 220 - 0.04\left(50 - \frac{V}{R_f}\right)$$
$$= 220 - 0.04\left(50 - \frac{220}{155}\right)$$
$$= 220 - 1.943$$
$$= 218.056\ V$$

Let, R be the resistance connected in series with the field resistance,

$$E_{b1} = V - I_a R_a$$
$$= 220 - 0.04\ (I_l - I_{sh})$$
$$= 220 - 0.04\left(50 - \frac{V}{R_{sh} + R}\right)$$
$$= 220 - 0.04\left(50 - \frac{220}{155 + R}\right)$$

Then,

$$\frac{E_b}{E_{b1}} = \frac{N_0}{N_1}$$

or, $$\frac{1,700}{1,200} = \frac{218.056}{220 - 0.04\left(50 - \frac{220}{155 + R}\right)}$$

or, $$1.4166 = \frac{218.056}{220 - 0.04\left(50 - \frac{220}{155 + R}\right)}$$

or, $$220 - 0.04\left(50 - \frac{220}{155 + R}\right) = \frac{218.056}{1.4166}$$

or, $$220 - 153.92 = 0.04 \times 50 - \frac{0.04 \times 220}{155 + R}$$

or, $$66.0708 = 2 - \frac{8.80}{155 + R}$$

or, $$66.0708 = \frac{310 + 2R - 8.08}{155 + R}$$

or, $10,240.974 + 66.0708\ R = 310 + 2R - 8.80$

or, $9,939.774 = 2R - 66.0708\ R$

or, $$\frac{9,939.774}{-64.0708} = R$$

$\therefore R = 155\ \Omega$

Hence, resistance needed in series in 155 Ω.

13. **A 50 KVA, 500/250 V transformer has a primary winding resistance of 0.5 Ω and leakage reactance of 0.7 Ω. The secondary winding resistance is 0.8 Ω and resistance, reactance and impedance of the equivalent resistance, reactance and importance of the transformer referred to:**
    - **i) Primary/HV side**
    - **ii) Secondary/LV side** [13/F, 13/S, 16/S]

Solution:

Transformation ratio, $K = \frac{V_2}{V_1} = \frac{250}{500} = 0.5$

i) Equivalent resistance referred to primary *(HV side)*,

$$R_{01} = R_1 + \frac{R_2}{K^2} = 0.5 + \frac{0.8}{(0.5)^2} = 3.70\ \Omega$$

Equivalent reactance referred to primary,

$$X_{01} = X_1 + \frac{X_2}{K^2} = 0.7 + \frac{0.9}{(0.5)^2} = 4.30\ \Omega$$

Impedance of transformer referred to primary,

$$Z_{01} = \sqrt{R_{01}^2 + X_{01}^2} = \sqrt{(3.70)^2 + (4.30)^2} = 5.672\ \Omega$$

ii) Equivalent resistance referred to secondary (LV side),

$$R_{02} = K^2R_1 + R_2 = 0.5^2 \times 0.5 + 0.8 = 0.925\ \Omega$$

Equivalent ~~impedance of~~ reactance transformer referred to secondary

$$X_{02} = K^2X_1 + X_2 = 0.5^2 \times 0.7 + 0.9 = 2.56\ \Omega$$

Equivalent impedance of transformer referred to secondary

$$Z_{02} = \sqrt{R_{02}^2 + X_{02}^2} = \sqrt{(0.925)^2 + (2.65)^2} = 2.806\ \Omega.$$

14. **A transformer has 600 primary turns 150 secondary turns. The primary and secondary resistances are 0.25 Ω and 0.01 Ω respectively and corresponding leakage reactance are 1 Ω and 0.04 Ω respectively. Determine (i) equivalent resistance and reactance referred to primary side (ii) equivalent resistance and reactance referred to secondary side.** [18/F]

Solution:

Transformation ratio, $K = \frac{V_2}{V_1} = \frac{N_2}{N_1} = \frac{1500}{1600} = 0.25$

And then proceed as above question no. 13.

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15. A single phase 10 KVA, 200/400 V, 50 Hz transformer gave there following test results:

O.C test : 200 V, 1.3 A, 120 W

S.C test : 220 V, 30 A, 200 W

Find the parameters of equivalent circuit as referred to HV side and LV side. [15/F, 16/F, 18/S]

Solution:

$$\text{No-load p.f., } \cos\phi_0 = \frac{\text{No-load input power}}{\text{Primary voltage} \times \text{No-load current}}$$

$$= \frac{P_0}{V_1 \times I_0}$$

$$= \frac{120}{200 \times 1.3}$$

$$= 0.4615$$

Energy component of no-load current,

$$I_e = I_0 \cos\phi_0 = 1.3 \times 0.4165 = 0.60 \text{ A}$$

Magnetizing component of no-load current,

$$I_m = \sqrt{I_0^2 - (0.60)^2} = 1.153 \text{ A}$$

Referred to LV/primary side

$$\text{No-load resistance, } R_0 = \frac{V_1}{I_e} = \frac{200}{0.60} = 333.34\ \Omega$$

$$\text{No-load reactance } X_0 = \frac{V_1}{I_m} = \frac{200}{1.153} = 173.46\ \Omega$$

Short circuit current, $I_s = 30$ A

Short circuit voltage, $V_s = 22$ V

Power input on short circuit, $P_s = 200$ W

Equivalent impedance referred to HV side,

$$Z_{02} = \frac{V}{I_s} = \frac{22}{30} = 0.734\ \Omega$$

Therefore, equivalent resistance referred to HV side,

$$R_0 = \frac{P_s}{I_s^2} = \frac{200}{30^2} = 0.223\ \Omega$$

Therefore, equivalent reactance referred to HV side,

$$X_{02} = \sqrt{(Z_{02})^2 - (R_{02})^2} = \sqrt{(0.734)^2 - (0.223)^3} = 0.70\ \Omega$$

$$\text{Transformation ratio, } K = \frac{V_2}{V_1} = \frac{400}{200} = 2$$

Therefore, equivalent resistance referred to LV/Primary side

$$R_{01} = \frac{R_{02}}{K^2} = \frac{0.223}{2^2} = 0.0557\ \Omega$$

Therefore, equivalent reactance referred to LV/Primary side

$$X_{01} = \frac{X_{02}}{K^2} = \frac{0.70}{2^2} = 0.175\ \Omega$$

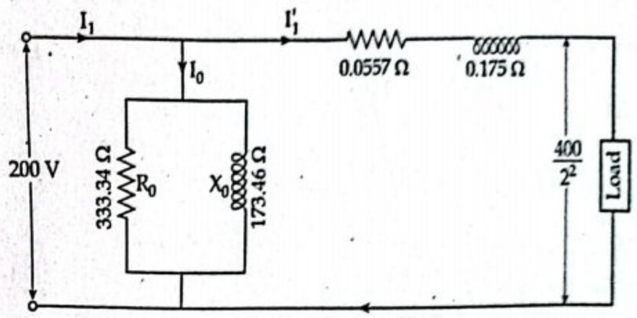

Fig: Equivalent circuit referred to LV side

Also, determine efficiency for half load of 0.8 lagging power factor. [14/S]

Solution:

Efficiency at half load at 0.8 p.f. lagging

$$= \frac{x \times VA \times \cos\phi}{x \times VA \times \cos\phi + P_i + x^2 P_c}$$

$$= \frac{\frac{1}{2} \times 10{,}000 \times 0.80}{\frac{1}{2} \times 10{,}000 \times 0.80 + 120 + \left(\frac{1}{2}\right)^2 \times 200}$$

$$= \frac{4{,}000}{4{,}000 + 120 + 50}$$

$$= 0.95923$$

$$\therefore \quad \eta_{1/2} = 95.923\%$$

16. A 10 KVA, 200/1000 V, 50 Hz single phase transformer gave the following test results: [14/F, 15/S]

OC test (LV side) : 200 V, 2.4 A, 100 W

SC test (HC side) : 50 V, 10 A, 150 W

i) Calculate the parameters of the equivalent circuit referred to LV side

ii) Calculate efficiency for half (1/2) load, full load at 0.8 p.f. lagging.

iii) Calculate load current for which it gives maximum efficiency.

Solution:

No-load p.f.

$$\cos\phi_0 = \frac{P_0}{V_1 \times I_0} = \frac{100}{200 \times 2.4} = 0.2083$$

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Energy component of no-load current

$$I_e = I_0 \cos \phi_0 = 2.4 \times 0.2083 = 0.50 \text{ A}$$

Magnetizing component of no-load current

$$I_m = \sqrt{I_0^2 - I_e^2} = \sqrt{2.4^2 - 0.50^2} = 2.347 \text{ A}$$

Referred to LV side

$$\text{No-load resistance, } R_0 = \frac{V_1}{I_e} = \frac{200}{0.50} = 400\ \Omega$$

$$\text{No-load reactance, } X_0 = \frac{V_1}{I_m} = \frac{200}{2.347} = 85.215\ \Omega$$

Short circuit current, $I_s = 10$ A

Short circuit voltage, $V_s = 10$ V

Power input on short circuit, $P_s = 150$ W

Equivalent impedance referred to HV side

$$Z_{02} = \frac{V_s}{I_s} = \frac{50}{10} = 5 \text{ A}$$

Equivalent resistance and reactance referred to HV side

$$R_{02} = \frac{P_s}{I_s^2} = \frac{150}{10^2} = 1.5\ \Omega$$

$$X_{02} = \sqrt{Z_{02}^2 - R_{02}^2} = \sqrt{5^2 - (1.5)^2} = 4.769\ \Omega$$

$$\text{Transformation ratio, } K = \frac{V_2}{V_1} = \frac{1,000}{200} = 5$$

Thus,

i) Equivalent resistance and reactance referred to LV (primary side)

$$R_{01} = \frac{R_{02}}{K^2} = \frac{1.5}{5^2} = 0.06\ \Omega$$

$$X_{01} = \frac{X_{02}}{K^2} = \frac{4.769}{5^2} = 0.19076$$

ii) $$\text{Efficiency at full load, } \eta_f = \frac{\text{Power} \times \cos\phi}{\text{Power} \times \cos\phi + P_i + P_c}$$

From question

Iron loss, $P_i = 100$ W

Copper loss, $P_c = 150$ W

Power factor, $\cos \phi = 0.80$ (lagging)

so, $$\eta_f = \left(\frac{10 \times 1,000 \times 0.80}{10 \times 1,000 \times 0.80 + 100 + 150}\right)$$

$$= 0.9697$$

$$= 96.97\ \%$$

and, Efficiency at half-load,

$$\eta_{1/2} = \frac{\frac{1}{2} \times \text{Power} \times \cos\phi}{\frac{1}{2} \times \text{Power} \times \cos\phi + P_i + \left(\frac{1}{2}\right)^2 P_c} = \left(\frac{4,000}{4,000 + 100 + 37.5}\right)$$

$$= 0.9667 = 96.67\%$$

iii) Load current corresponding to maximum efficiency,

$$I_2' = \sqrt{\frac{P_i}{R_{02}}} = \sqrt{\frac{100}{1.5}} = 8.1649 \text{ A}$$

iv) Secondary terminal voltage if it supplies I_0 A at 0.8 p.f. lag. **[17/F]**

Solution:

$\cos \phi = 0.80$

$E_2 = 1,000$ V

$I_2 = 10$ A

$R_{02} = 1.5\ \Omega$

$X_{02} = 4.769\ \Omega$

Thus, Secondary terminal voltage

$$V_2 = E_2 - I_2 R_{02} \cos\phi - I_2 X_{02} \sin\phi$$

$$= 1,000 - 10 \times 1.5 \times 0.8 - 10 \times 4.769 \times 60$$

$$= 1,000 - 12 - 28.614$$

$$\therefore \quad V_2 = 959.386 \text{ V}$$

NOTE:

If power factor is leading,

$$V_2 = E_2 + I_2 R_{02} \cos\phi + I_2 X_{02} \sin\phi$$

$$= 1,000 + 12 + 28.614$$

$$\therefore \quad V_2 = 1,040.614 \text{ V}$$

$$\text{Primary current } I_1 = \frac{\text{Power in VA}}{\text{Primary voltage}} = \frac{10,000}{200} = 100 \text{ A}$$

$$\text{Secondary current, } I_2 = \frac{\text{Power in VA}}{\text{Primary voltage}} = \frac{10,000}{1,000} = 10 \text{ A.}$$

17. A 250 V shunt motor on no load runs at 1,000 rpm and takes 5 A. The total armature and shunt field resistance are respectively 0.2 W and 250 W. Calculate the speed when loaded and taking a current of 50 A, if the armature reaction weakness the field by 3%. [17/S]

Solution:

$$\text{Shunt field current, } I_{sh} = \frac{V}{R_{sh}} = \frac{250}{250} = 1 \text{ A}$$

No-load armature current, $I_{a0} = I_{L0} - I_{sh} = (5 - 1)$ A $= 4$ A

No-load back emf, $E_{b0} = V - I_{a0} R_0 = 250 - 4 \times 0.2 = 249.20$ V

No-load speed, $N_0 = 1,000$ rpm

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Full load armature current, $I_{af} = I_{Lf} - I_{sh} = 50 - 1 = 49$ A

Full load back emf, $E_{bf} = V - I_{af} R_a = 250 - 49 \times 0.2 = 240.20$ V

Full load flux per pole, $\phi_f = (1 - 3\%)\ \phi_0 = (1 - 0.03)\ \phi_0 = 0.97\ \phi_0$

Hence, Full-load speed, $N_f = \frac{E_{bf}}{E_{b0}} \times \frac{\phi_0}{\phi_f} = 1{,}000 \times \frac{240.20}{249.20} \times \frac{\phi_0}{0.97\ \phi_0} = 994$ rpm.

18. A 220 V DC shunt motor runs at 500 rpm when armature current is 50 A. Calculate the speed if the torque is doubled. Given that armature resistance is 0.2 ohm and flux remains constant. [14/F]

Solution:

For a DC shunt motor

$$T_a \propto \phi I_a$$

Since, ϕ is constant

$$T_a \propto I_a$$

$\therefore$ $T_{a1} \propto I_{a1}$ and $T_{a2} \propto I_{a2}$

Thus,

$$\frac{T_{a2}}{T_{a1}} = \frac{I_{a2}}{I_{a1}}$$

or, $\frac{2T_{a1}}{T_{a1}} = \frac{I_{a2}}{50}$

or, $2 \times 50 = I_{a2}$

$\therefore$ $I_{a2} = 100$ A

and $\frac{E_{b2}}{E_{b1}} = \frac{N_2}{N_1}$

or, $\frac{220 - (100 \times 0.2)}{220 - (50 \times 0.2)} = \frac{N_2}{500}$ $[\because E_b = V - I_a R_a]$

or, $\frac{200}{210} = \frac{N_2}{500}$

$\therefore$ $N_2 = 476$ rpm

Hence the speed is 476 rpm if torque is doubled.

19. A 220 V DC shunt motor runs with 1,000 rpm with an armature current of 40 A. The resistance of armature is 0.5 Ω. Calculate the value of resistance to be connected in series so that the speed drops to 600 rpm. [13/S, 15/F]

Solution:

$N_1 = 1{,}000$ rpm

$V = 220$ V

$I_a = 40$ A

$R_a = 0.5\ \Omega$

$N_2 = 600$ rpm

$E_{b1} = V - I_a R_a = 220 - 40 \times 0.5 = 200$ V

Assuming flux remaining constant,

$$T \propto Ia$$

Since torque is same in both cases,

$$I_{a2} = I_{a1} = 40 \text{ A}$$

Let, R be the additional resistance inserted in series with the armature.

$$E_{bx} = 220 - 40\ (R + 0.5)$$

Thus,

$$\frac{N_2}{N_2} = \frac{E_{b2}}{E_{b1}}$$

or, $\frac{600}{1{,}000} = \frac{220 - 40\ (R + 0.5)}{200}$

or, $220 - 40R - 40 \times 0.5 = \frac{200 \times 60}{1000}$

or, $220 - 40R - 20 = 120$

or, $220 - 40R - 120 = 40R$

or, $R = \frac{80}{40}$

$\therefore$ $R = 2\ \Omega$

Hence, additional resistance is 2 Ω that should be connected in series.

20. A 240 V shunt motor runs at 1,450 rpm at full load with an armature current of 11 A. The total resistance of armature and brush is 0.6 Ω. If the speed to be reduced to 1,000 rpm will the same armature current, calculate the value of resistance to be connected in series with the armature. [13/F, 16/F, 19/S]

Solution:

$N_1 = 1{,}450$ rpm

$N_2 = 1{,}000$ rpm

$R_{ab} = 0.6$ W

$I_a = 11$ A

$V = 240$ V

$E_{b1} = V - I_a R_{ab} = 240 - 11 \times 0.6 = 233.40$ V

Let, R be the additional resistance inserted in series with the armature,

$E_{b2} = V - I_a\ (R + R_{ab}) = 240 - 11 \times (R + 0.6) = 240 - 11R - 6.6$

$= 233.40 - 11R$

Now,

$$\frac{E_{b2}}{E_{b1}} = \frac{N_2}{N_2} \quad [\because I_{a1} = I_{a2}]$$

or, $\frac{233.40 - 11R}{233.40} = \frac{1{,}000}{1{,}450}$

or, $233.40 - 11R = 1{,}000 \times \frac{233.40}{1{,}450}$

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or, $-11\,R = 160.9655 - 233.40$

or, $R = \frac{72.434}{11} = 7\,\Omega$

Hence, value of resistance to be connected in series with the armature is 7 Ω.

21. For a DC separately excited motor, when the field circuit is connected to rated supply and rated voltage is supplied to armature terminal, motor runs at 1,000 rpm at no load. Then estimate the approximate speed of motor if the armature voltage is reduced to 50% of rated value [14/S]

Solution:

NOTE:

A separately excited DC motor and a DC shunt motor are similar in construction and in all aspects expect the way in which the field winding is excited in both the machines.

$N_1 = 1000$ rpm

$N_2 = ?$

$E_{b1} = E_{b1}$

$E_{b2} = 50\% \times E_{b1}$

so, $\frac{E_{b2}}{E_{b1}} = \frac{N_2}{N_1}$

or, $0.5 = \frac{N_2}{1,000}$

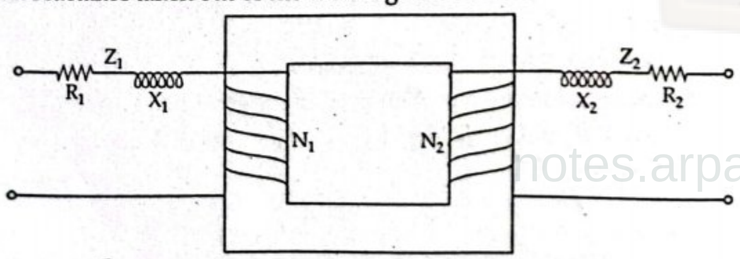
$\therefore \quad N_2 = 500$ rpm

Hence, approximate speed is 500 rpm

22. Explain practical transformer with the help of a phasor diagram (with transformer resistance and leakage reactance). [17/S]

Solution:

In figure below the primary and secondary windings of a transformer with reactance taken out of the windings are shown.



The primary impedance is given by,

$$Z_1 = \sqrt{R_1^2 + X_1^2}$$

Similarly, the secondary impedance is given by

$$Z_2 = \sqrt{R_2^2 + X_2^2}$$

The resistance and leakage reactance of each winding is responsible for some voltage drop in each winding. In primary, the leakage reactance drop $I_1 X_1$. Hence,

$$V_1 = E_1 + I_1 (R_1 + jX_1) = E_1 + I_1 Z_1$$

Similarly, there are $I_2 R_2$ and $I_2 X_2$ drops in secondary which combine with V_2 to give E_2.

$$E_2 = V_2 + I_2 (R_2 + jX_2) = V_2 + I_2 Z_2$$

The vector diagram for such a transformer for different kinds of loads is shown in figure below. In these diagrams, vectors for resistive drops are drawn parallel to current vectors whereas reactive drops are perpendicular to the current vectors. The angel ϕ_1 between V_1 and I_1 gives the power factor angle of the transformer.

It may be noted that leakage reactances can also be transferred from one winding to the other in the same way as resistance.

$$\therefore \quad X_2' = \frac{X_2}{K^2} \text{ and } X_1' = K^2 X_1$$

and $X_{01} = X_1 + X_2' = X_1 + \frac{X_2}{K^2}$

$$X_{02} = X_2 + X_1' = X_2 + K_2^2 X_1$$

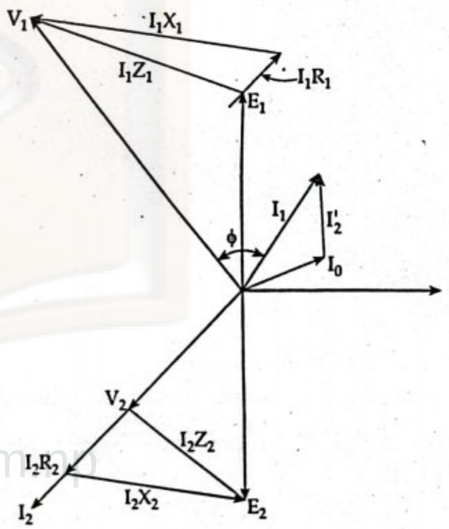


Fig: Phasor diagram for resistive/Unity power factor load of a practical transformer

23. A 220 V DC shunt motor having an armature resistance of 0.25 ohm carries an armature current of 50 A and runs at 600 rpm. If the flux is reduced by 10% by field regulator, find the speed assuming load torque remains constant. [19/F]

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Solution:

Here;

Supply voltage, V = 220 V

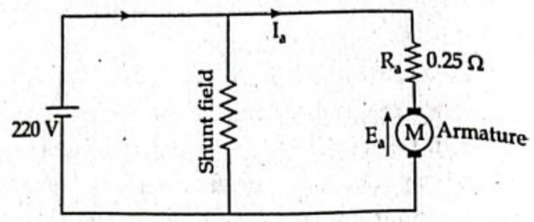
Armature resistance $R_a = 0.25\ \Omega$

$I_{a_1} = 50$ A

$N_1 = 600$ rpm

$N_2 = ?$

$I_{a_2} = ?$



We know,

$$E = V - I_a R_a$$

$$\frac{N_2}{N_1} = \frac{E_{b_2}}{E_{b_1}} \times \frac{\phi_1}{\phi_2} \quad (1)$$

$$T_a \propto \phi I_a$$

Hence, $T_{a_1} \propto \phi_1 I_{a_1}$

and, $T_{a_2} \propto \phi_2 I_{a_2}$

Since, $T_{a_1} = T_{a_2}$

or, $\phi_1 I_{a_1} = \phi_2 I_{a_2}$

Also, flux ϕ_2 reduces by 10%.

so, $\phi_1 I_{a_1} = 0.90\ \phi_1 I_{a_2}$

or, $I_{a_1} = 0.90\ I_{a_2}$

$\therefore \quad I_{a_2} = 55.56$ A

Thus, $E_{b_1} = 220 - (50 \times 0.25) = 207.50$ V

$E_{b_2} = 220 - (55.56 \times 0.25) = 206.10$ V

Now, from equation (1);

$$\frac{N_2}{600} = \frac{206.10}{207.50} \times \frac{1}{0.90}$$

or, $N_2 = 600 \times 1.103614$

$\therefore \quad N_2 = 662.16$ rpm $\simeq 663$ rpm

Hence, new speed is 663 rpm assuming the load torque remains constant.

24. A 50 KVA, 2,200/110 transformer when tested gave the following results. [19/F]

OC test : 400 W 10 A 110 V

SC test : 808 W 20.5 A 90 V

Compute all the parameters of the equivalent circuit referred to LV sides of the transformer.

Solution:

Assuming lower voltage winding as primary

No load current, $I_0 = 10$ A

Energy component of no-load current,

$$I_e = \frac{P_0}{V} = \frac{400}{110} = 3.637 \text{ A}$$

Magnetizing component of no-load current,

$$I_m = \sqrt{I_0^2 - I_e^2} = \sqrt{10^2 - (3.637)^2} = \sqrt{86.7722} = 9.32 \text{ A}$$

Equivalent circuit parameter,

$$R_0 = \frac{V_1}{I_e} = \frac{110}{3.627} = 30.245\ \Omega, \qquad X_0 = \frac{V_1}{I_m} = \frac{110}{9.32} = 11.80\ \Omega$$

Equivalent impedance referred to HV side,

$$Z_{02} = \frac{V_S}{I_S} = \frac{90}{20.5} = 4.39\ \Omega$$

Equivalent resistance referred to HV side,

$$R_{02} = \frac{W_S}{I_S^2} = \frac{808}{(20.5)^2} = 1.93\ \Omega$$

Equivalent reactance referred to HV side,

$$X_{02} = \sqrt{Z_{02}^2 - R_{02}^2} = \sqrt{4.39^2 - 1.93^2} = 3.95\ \Omega$$

Transformation ratio, $K = \frac{2,200}{100} = 20$

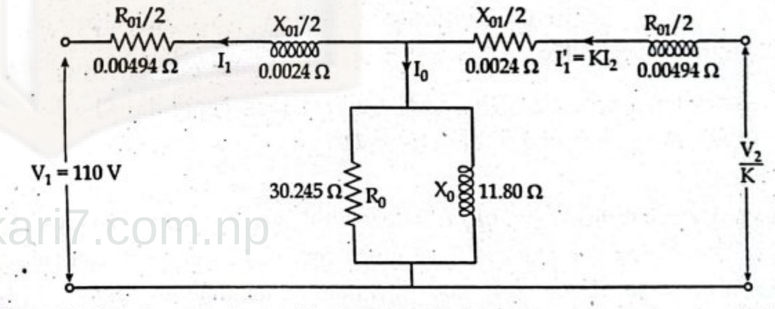
Equivalent resistance referred to LV side,

$$R_{01} = \frac{R_{02}}{K^2} = \frac{1.93}{20^2} = 0.00482\ \Omega$$

Equivalent reactance referred to LV side,

$$X_{01} = \frac{X_{02}}{K^2} = \frac{3.95}{20^2} = 0.00987\ \Omega$$

Exact equivalent circuit parameters referred to LV side is shown below,



25. Write short notes on 3 – ϕ induction motor. [17/F]

Solution: See the definition part 5.4.

26. Write short notes on emf equation of a single phase transformer. [19/S]

Solution: See the definition part 5.2.2.

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SHORT QUESTION WITH ANSWER

1. Why does a transformer has iron core?

Solution:

The use of iron or steel for transformer core ensures a high permeability of the magnetic circuit and because of high permeability the magnitude of exciting current necessary to create the required flux in the core is small. The presence of steel core also cause 100% of magnetic flux created by primary to be linked with secondary.

2. Why is the efficiency of a transformer maximum among electrical equipments? Explain.

Solution:

Owing to lack of rotating parts in a transformer, there are no friction and windage losses. The other losses such as iron losses are also comparatively small because of better magnetic material and less handling of the material. So, the efficiency of transformer is Maximum among electrical equipments.

3. What happens if DC supply is applied to the transformer?

Solution:

If a rated DC voltage is applied to the primary of the transformer, the flux produced in the transformer core will not vary but remain constant in magnitude and therefore there will be no self induced emf in the primary winding (which is only possible with varying flux linkage) to oppose the applied voltage and since the resistance of the primary winding is quite low, a heavy current will flow through the primary winding that may cause burning out of the primary winding.

4. On what factor does the eddy current loss depend? Why this loss can be assumed to be constant?

Solution:

The eddy current loss depends on the maximum value of flux density in the core and supply frequency for a given core (*i.e.*, nature of magnetic material, volume of the core, and thickness of laminations remaining unchanged). Since from no load to full load, the flux linking with the core and supply frequency remain constant, the eddy current loss remains constant.

5. Why core loss is neglected in short circuit test and copper loss in open-circuit test?

Solution:

Since in short circuit test, the voltage applied to the primary of transformer is very low (5 to 8% of rated voltage), the flux produced in the core is very small and therefore iron losses are very small and can be neglected, since in open circuit test, no current flows in the open circuited secondary and the current in the primary will also be very small (usually 3 to 10% of full load current)the copper losses in the open-circuit test will be so small that these can be neglected.

6. Why LV winding is normally short circuited during short circuit test?

Solution:

Voltage required for the short circuit test is very low (about 5% of the rated value) and if the measurement are made by short circuiting HV winding, on LV side, the voltage required would be inconveniently low, while the current would often be inconveniently large.

7. What is the cause of noise in transformer?

Solution:

In a transformer, noise occurs mainly due to loosening of stampings and mechanical forces developed during operation.

8. Why is an induction motor called asynchronous?

Solution:

Because the rotor does not turn in synchronism with the rotating field developed by the stator currents

9. What are the factors that affect the speed of a DC motor?

Solution:

The factors influencing the speed of a DC motor are;

i) Applied or terminal voltage
ii) Flux per pole
iii) Resistance of armature circuit
iv) Load or armature current

10. If the supply frequency is doubled, which loss component in transformer will double?

Solution:

Due to doubling of supply frequency, the hysteresis loss component will be double.

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11. Distinguish between core and shell type transformers.

Solution:

S.N.	Core type transformer	Shell type transformer
a)	The core has only one magnetic circuit.	It has two magnetic circuits.
b)	Core has two limbs.	Core has three limbs.
c)	It has less mechanical protection to coil.	It has better mechanical protection to coil.
d)	Natural cooling is provided.	Natural cooling cannot be provided.
e)	This transformer is easy to repair.	This transformer is not easy to repair.
f)	It has better cooling since more surface is exposed to atmosphere.	Cooling is not very effective.

12. What is Power transformer?

Solution:

Power transformer is used for the transmission purpose at heavy load, high voltage greater than 33 KV and 100% efficiency. It is big in size as compared to distribution transformer. It is used in generating station and transformer, it is used in generating station and transmission substation. It is generally rated above 200 MVA.

13. With a turn's ratio of 1 : 2 and a secondary voltage of 960 volts, what would be the primary voltage?

Solution:

$$\frac{V_1}{N_2} = \frac{V_1}{V_2} \Rightarrow \frac{1}{2} = \frac{V_1}{960} \Rightarrow 480 \text{ V}$$

14. Mention the difference between core and shell type transformers.

Solution:

In core type, the windings surround the core considerably and in shell type, the core surrounds the windings. i.e., winding is placed outside the core.

15. Why transformers are rated in KVA?

Solution:

Copper loss of a transformer depends on current and iron loss on voltage. Hence total losses depend on volt-ampere and not on power factor. That is why the rating of transformer is in KVA and not in KW.

16. What determines the thickness of the lamination or stampings?

Solution:

Frequency and iron loss determines the thickness of the lamination.

17. Can the voltage regulation goes negative? If so under what condition?

Solution:

Yes, if the load has leading power factor.

18. How is magnetic leakage reduced to a minimum in commercial transformer?

Solution:

By interleaving the primary and secondary windings.

19. Mention the factors on which hysteresis loss depends?

Solution:

i) Quality and amount of iron in the core
ii) Flux density
iii) Frequency

20. Does the transformer draw any current when its secondary is open?

Solution:

Yes, it draws no load current.

21. Is copper loss affected by power factor?

Solution:

Yes, copper loss varies inversely with power factor.

22. How may the direction of rotation of a DC motor be revered?

Solution:

By reversing either the field current or current through the armature. Usually, reversal of current through the armature is adopted.

23. What will happen if the field of a DC shunt motor is opened?

Solution:

The motor will achieve dangerously high speed and may destroy itself.

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ADDITIONAL QUESTIONS SOLUTIONS

1. A single phase 50 Hz transformer has 80 turns on the primary winding and 400 turns on the secondary winding. The net cross sectional area of the core is 200 cm². If the primary winding is connected to a 240 V, 50 Hz supply, determine

i) emf induced in the secondary winding

ii) Maximum value of the flux density in the core.

Solution:

$N_1 = 80$, $f = 50$ Hz, $N_2 = 400$, $a = 200\ cm^2 = 200 \times 10^{-4}\ m^2$

$$K = \frac{N_2}{N_2} = \frac{400}{80} = 5$$

$E_1 = 240$ V

$$\therefore \quad K = \frac{E_2}{E_1} = \frac{E_2}{240} = 5$$

or $E_2 = 5 \times 240 = 1{,}200$ V

Now,

$$E_1 = 4.44\, f\, \phi_m\, N_1$$

or $240 = 4.44 \times 50 \times \phi_m \times 80$

$$\text{or} \quad \phi_m = \frac{240}{4.44 \times 50 \times 80}$$

$$\therefore \quad \phi_m = 0.01351 \text{ wb}$$

and,

$$\therefore \quad B_m = \frac{\phi_m}{a} = \frac{0.01351}{200 \times 10^{-4}} = 0.6756 \text{ wb/m}^2.$$

2. A 6,600/400 V single phase transformer has primary resistance of 2.5 Ω and secondary resistance of 0.01 Ω. Calculate the equivalent resistance referred to primary and secondary.

Solution:

$R_1 = 2.5\ \Omega$, $R_2 = 0.01\ \Omega$

$$K = \frac{V_2}{V_1} = \frac{400}{6{,}600} = 0.0605$$

Equivalent resistance referred to primary,

$$R_2' = \frac{R_2}{K^2} = \frac{0.01}{(0.0606)^2} = 2.7225\ \Omega$$

$$R_{1e} = R_1 + R_1' = 2.5 + 2.7225 = 5.2225\ \Omega$$

It can be observed that primary is high voltage hence high resistance side, hence while transferring R_2 from low voltage R_2' on high voltage, its value increases.

Equivalent resistance referred to secondary,

$$R_1' = K^2 R_1 = (0.0606)^2 \times 2.5 = 0.00918\ \Omega$$

$$R_{2e} = R_2 + R_1' = 0.01 + 0.00918 = 0.01918\ \Omega$$

3. A 3,300/110 V, 50 Hz, 60 KVA single phase transformer has iron losses of 600 watts. Primary and secondary winding resistance are 3.3 ohms and 0.011 ohms respectively. Determine the efficiency of the transformer on full load at 0.8 p.f. load.

Solution:

$V_1 = 3{,}300$ V

$V_2 = 110$ V

KVA = 60

F = 50 Hz

$R_1 = 3.3\ \Omega$

$R_2 = 0.011\ \Omega$

$P_i = 600$ W

cos f = 0.8 lag

$$(I_1)_{F.L} = \frac{KVA}{I_1} = \frac{60 \times 10^3}{3{,}300} = 18.1818 \text{ A}$$

Primary copper loss, $= I_1^2 R_1 = (18.1818)^2 \times 3.3 = 1{,}090.90$ W

$$(I_2)_{FL} = \frac{KVA}{V_2} = \frac{60 \times 10^3}{110} = 545.4545 \text{ A}$$

Secondary copper loss $= I_2^2 R_2 = (545.4545)^2 \times 0.011 = 3{,}272.7272$ W

Hence,

Total copper loss = 1,090.90 + 3,272.7272 = 4,363.6272 W on full load

Now,

$$\text{Efficiency, } \eta = \frac{KVA \times \cos\phi}{KVA \times \cos\phi + P_i + P_{cu}} \times 100$$

$$= \frac{60 \times 10^3 \times 0.8}{60 \times 10^3 \times 0.8 + 600 + 4{,}363.6272} \times 100$$

$$\therefore \quad \% \eta = 90.6282\ \%.$$

4. A 4 KVA, 200/400 V, 50 Hz single phase transformer has equivalent resistance referred to primary as 0.15 Ω. Calculate,

i) Total copper losses on full load

ii) The efficiency while supplying full load at 0.9 p.f. lagging

iii) The efficiency while supplying half load at 0.8 p.f. leading

Assume total iron losses equal to 60 W.

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Solution:

$V_1 = 200$ V

$V_2 = 400$ V

$S = 4$ KVA

$R_{1e} = 0.15\ \Omega$

$P_i = 60$ W

$$K = \frac{V_2}{V_1} = \frac{400}{200} = 2$$

$$\therefore \quad R_{2e} = K^2 R_{1e} = (2)^2 \times 0.15 = 0.6\ \Omega$$

$$(I_2)_{FL} = \frac{KVA}{V_2} = \frac{4 \times 10^3}{400} = 10\ A$$

i) Total copper loses on full load

$$(P_{cu})_{FL} = ((I_2)_{FL})^2 \times R_{2e} = 10^2 \times 0.6 = 60\ W$$

ii) $\cos\phi = 0.9$ lagging and full load,

$$\% \eta = \frac{\text{VA rating} \times \cos\phi}{\text{VA rating} \times \cos\phi + P_i + P_{cu}} \times 100$$

or $$\eta = \frac{4 \times 10^3 \times 0.9}{4 \times 10^3 \times 0.9 + 60 + 60} \times 100$$

$$\therefore \quad \eta = 96.77\%$$

iii) As half load, $x = 0.5$

$$P_{(cu)HL} = x^2 \times (P_{cu})_{FL} = 0.5^2 \times 60 = 15\ W$$

$$\therefore \quad \% \eta = \frac{x \times \text{KVA rating} \times \cos\phi}{x \times \text{KVA rating} \times P_i + (P_{cu})_{HL}} \times 100$$

$$= \frac{0.5 \times 4 \times 10^3 \times 0.8}{0.5 \times 4 \times 10^3 \times 0.8 + 60 + 15} \times 100$$

$$= 95.52\ \%$$

5. A 250 KVA, single phase transformers has 98.135% efficiency at full load and 0.8 lagging pf. The efficiency at half load and 0.8 lagging pf is 97.751 %. Calculate the iron loss and full load copper loss.

Solution:

Output power at full load, $= 250 \times 10^3 \times 0.8 = 200 \times 10^3$ W

Input power at full load $= \frac{200 \times 10^3}{0.98135} = 2,03,800.88$

Total loss = Input – Output $= 2,03,800.88 - 200 \times 10^3 = 3,800.88$ W

$$P_i + P_{cu} = 3800.88 \qquad (1)$$

The power output at half load $= 125 \times 10^3 \times 0.8 = 100 \times 10^3$

Power input at half load $= \frac{100 \times 10^3}{0.97751} = 1,02,300.74$

Total losses $= 1,02,300.74 - 100 \times 10^3$

$$P_i + (0.5)^2 P_c = 2,300.74$$

or $$P_i + 0.25\ P_c = 2,300.74 \qquad (2)$$

Solving equation (1) and (2);

$\therefore \quad P_i = 2,000$ W

$\therefore \quad P_{cu} = 1,800$ W.

6. A single phase transformer has 500 turns on primary and 1000 turns on secondary. The voltage per turn is 0.2 volts. Calculate,

i) Voltage induced in the secondary winding

ii) Voltage induced in the primary winding

iii) The maximum value of the flux density if the cross sectional area of the core is 200 cm².

iv) KVA rating of the transformer if the current in primary at full load is 10 A, the frequency is 50 Hz.

Solution:

$N_1 = 500$

$N_2 = 1,000$

$f = 50$ Hz

$A = 200 \times 10^{-4}\ m^2$

$$\frac{\text{Volt}}{\text{Turn}} = \frac{E_1}{N_1} = 0.2 \text{ volts}$$

i) Voltage induced in the primary,

$$E_1 = \frac{\text{Volt}}{\text{Turn}} \times N_1 = 0.2 \times 500 = 100 \text{ volts}$$

ii) Voltage induced in the secondary,

$$E_2 = \frac{N_2}{N_1} \times E_1 = \frac{100}{500} \times 1000 = 200 \text{ volts}$$

iii) Maximum value of the flux in the core,

$$\phi_m = \frac{E_1}{4.44 \times f \times N_1} = \frac{100}{4.44 \times 50 \times 500} = 9.009 \times 10^{-4} \text{ wb}$$

Maximum value of flux density, B_m is

$$B_m = \frac{\phi_m}{A} = \frac{9.009 \times 10^{-4}}{200 \times 10^{-4}} = 0.045 \text{ wb/m}^2$$

iv) KVA rating of the transformer $= V_1 I_1 \times 10^{-3}$

$= V_2 \times I_2 \times 10^{-3}$

$= 100 \times 10 \times 10^{-3}$

$= 1$ KVA

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7. A single phase transformer when connected to a lamp load gave following results.

S.N.	V_1	I_1	W_1	V_2	I_2	W_2
1	200	1.5	60	100	0	0
2	200	12.9	2510	97	25	2425

Calculate efficiency and % regulation of transformer at $I_2 = 25$ A (second reading).

Solution:

$I_2 = 25$ A

From table

$W_1 = 2{,}510$ W and

$W_2 = 2{,}425$ W

$$\therefore \quad \% \eta = \frac{W_2}{W_1} \times 100 = \frac{2{,}425}{2{,}510} \times 100 = 96.6135\ \%$$

and, $E_2 = V_2$ on no load $= 100$ V

$V_2 = 97$ V on $I_2 = 25$ A

$$\therefore \quad \% \text{ Reg} = \left(\frac{E_2 - V_2}{E_2}\right) \times 100 = \left(\frac{100 - 97}{100}\right) \times 100 = 3\%.$$

8. In no load test of single phase transformer, the following test data were obtained

Primary voltage = 220 V
Secondary voltage = 110 V
Primary current = 0.5 A
Power input = 30 W
Find the following:

i) Turns ratio
ii) Magnetizing component of load current
iii) Its working or loss component
iv) The turn ratio

Resistance of the primary winding = 0.6 ohm

Solution:

i) Turn ratio, $\frac{N_1}{N_2} = \frac{220}{110} = 2$

ii) $W = V_1 I_0 \cos \phi_0$

$$\cos \phi_0 = \frac{30}{220 \times 0.5} = 0.273$$

$\sin \phi_0 = 0.962$

$I\mu = I_0 \sin \phi_0 = 0.5 \times 0.962 = 0.48$ A

iii) Loss component, $I_w = I_0 \cos \phi_0 = 0.5 \times 0.273 = 0.1365$ A

iv) Primary copper loss $= I_0^2 R_1 = 0.5^2 \times 0.6 = 0.15$ W

Iron loss $= 30 - 0.15 = 29.85$ W.

9. Consider a 20 KVA, 2,200/220 V, 50 Hz transformer. The O.C/S.C test results are as follows,

O.C test: 220 V 4.2 A 148 W (LV side)
S.C test: 86 V 10.5 A 360 W (HV side)

Determine the regulation at 0.8 p.f. lagging and at full load. What is the p.f. on short circuit?

Solution:

$$Z_{01} = \frac{86}{10.5} = 8.19\ \Omega$$

$$R_{01} = \frac{360}{10.5^2} = 3.26\ \Omega$$

$$X_{01} = \sqrt{Z_{01}^2 - R_{01}^2} = \sqrt{8.19^2 - 3.26^2} = 7.5\ \Omega$$

Full load primary current, $I_1 = \frac{20{,}000}{2{,}200} = 9.09$ A

Total voltage drop as referred to primary,

$= I_1 (R_{01} \cos \phi + X_{01} \sin \phi)$
$= 9.09\ (3.26 \times 0.8 + 7.5 \times 0.6)$
$= 64.6$ V

Now,

$$\% \text{ Regulation} = 64.6 \times \frac{100}{2{,}200} = 2.9\%$$

$$\text{Power factor on short circuit} = \frac{R_{01}}{Z_{01}} = \frac{3.26}{8.19} = 0.40 \text{ lag.}$$

10. Calculate the regulation of a transformer in which the ohmic loss is 1% of the output and the reactance drop is 5% of the voltage, when the power factor is 0.80 lag and 0.80 leading.

Solution:

When 1% of output is the ohmic loss, P.U. resistance of the transformer,

$E_r = 0.01$

When 5% is the reactance drop, P.U. reactance of the transformer,

$E_x = 0.05$

i) Per unit regulation of the transformer at full load, 0.8 lagging p.f.

$= 0.01 \times \cos \phi + 0.05 \times \sin \phi$
$= 0.01 \times 0.8 + 0.05 \times 0.06$
$= 0.038$
$= 3.8\%$

ii) Per unit regulation of unity p.f.

$= 0.01 \times 1 = 0.01 = 1\%$

iii) Per unit regulation at 0.8 leading p.f.

$= 0.01 \times 0.8 - 0.05 \times 0.06 = -0.022 = -2.2\%$

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11. The maximum efficiency of a 500 KVA, 3300/500 V, 50 Hz, single phase transformer is 97% and occurs at 3/4th full load, unity power factor load. If the impedance is 10%, calculate the regulation at full load, 0.8 p.f. lag.

Solution:

At unity p.f. with $3/4^{th}$ full load, the output of the transformer,

$$= 500 \times 0.75 \times 1 \text{ KW} = 375 \text{ KW}$$

We have,

$$0.97 = \frac{375}{375 + 2P_i}$$

where,

P_i = Core loss in KW

At maximum efficiency,

$$x^2 P_c = P_i$$

$$(0.75)^2 \times P_c = P_i$$

where,

$x = 0.75$ *i.e.*, $3/4^{th}$ is the fractional loading of the transformer

P_c = copper loss in KW

$$P_i = \frac{1}{2}\left[375 \times \left(\frac{1}{0.97} - 1\right)\right] = \frac{1}{2} \times 375 \times \frac{3}{97} = 5.8 \text{ KW}$$

$$P_c = \frac{5.8}{0.75^2} = 10.3 \text{ KW}$$

$$\text{Full load current in primary winding} = \frac{500 \times 1{,}000}{3{,}300} = 151.5 \text{ A}$$

Total winding resistance referred to primary,

$$= \frac{10.3 \times 1{,}000}{(151.5)^2} = 0.44876 \text{ ohm}$$

$$Er = \% \text{ resistance} = \frac{151.5 \times 0.44876}{3{,}300} \times 100\% = 2.06\%$$

$$Ez = \% \text{ impedance} = 10\%$$

$$E_x = \% \text{ reactance} = \sqrt{100 - 4.244} = 9.7855\%$$

By approximate formula at 0.8 p.f. lag,

$$\% \text{ regulation} = E_r \cos\phi + E_x \sin\phi = 2.06 \times 0.8 + 9.7855 \times 0.6 = 7.52\%.$$

12. A 4 pole, DC shunt generator with a shunt field resistance of 100 Ω and an armature resistance of 1 Ω has 378 wave connected conductors in its armature. The flux per pole is 0.02 wb. If a load resistance of 10 Ω is connected across the armature terminals and the generator is driven at 1,000 rpm. Calculate the power absorbed by the load.

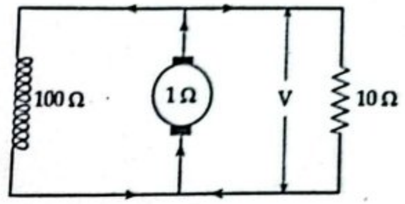
Solution:

Induced emf in the generator,

$$E_g = \frac{\phi ZN}{60}\left(\frac{P}{A}\right) = \frac{0.02 \times 378 \times 1{,}000}{60} \times \left(\frac{4}{2}\right)$$

$\therefore$ $E_g = 252$ V

Let, V = Terminal voltage



$$\text{Load current} = \frac{V}{10} \text{ A}$$

$$\text{Shunt current} = \frac{V}{100} \text{ A}$$

$$\text{Armature current} = \frac{V}{10} + \frac{V}{100} = \frac{11}{100} V$$

Now,

$V = E_g$ – Armature drop

or $V = 252 - 1 \times \frac{11}{100} V$

$\therefore$ $V = 227$ volt

Thus,

$$\text{Load current} = \frac{227}{10} = 22.7 \text{ A}$$

Power absorbed by the load = $227 \times 22.7 = 5153$ W.

13. A 250 V shunt motor runs at 1000 rpm at no load and takes 8 A. The total armature and shunt field resistances are respectively 0.2 Ω and 250 Ω. Calculate the speed when loaded and taking 50 A. Assume the flux to be constant

Solution:

$$\frac{N}{N_0} = \frac{E_b}{E_{b0}} \times \frac{\phi_0}{\phi}$$

Since, $\phi_0 = \phi$

$$\frac{N}{N_0} = \frac{E_b}{E_{b0}}$$

$$I_{sh} = \frac{250}{250} = 1 \text{ A}$$

$$E_{b0} = V - I_{a0} R_A = 250 - (7 \times 0.2) = 248.6 \text{ V}$$

$$E_b = V - I_a R_a = 250 - (49 \times 0.2) = 240.2 \text{ V}$$

or, $\frac{N}{1{,}000} = \frac{240.2}{248.6}$

or, $N = 240.2 \times \frac{1{,}000}{248.6}$

$\therefore$ $N = 9666.1$ rpm.

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14. A DC series motor operates at 800 rpm with a line current of 100 A from 230 V mains. Its armature circuit resistance is 0.15 Ω and its field resistance is line current of 25 A, assuming that the flux at this current is 45% of the flux at 100 A.

Solution:

$$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \times \frac{\phi_1}{\phi_2}$$

Since, $\phi_2 = 0.45\ \phi_1$

or, $\frac{\phi_1}{\phi_2} = \frac{1}{0.45}$

Thus,

$E_{b1} = 230 - (0.15 + 0.1) \times 100 = 205$ V

$E_{b2} = 230 - 25 \times 0.25 = 223.75$ V

Now,

$$\frac{N_2}{800} = \frac{223.75}{205} \times \frac{1}{0.45}$$

$\therefore$ $N_2 = 1{,}940$ rpm.

15. The armature circuit resistance of a 18.65 KW, 250 V series motor is 0.1 Ω, the brush voltage drop is 3 V and the series field resistance is 0.05 ohm. When the motor takes 80 A, speed is 600 rpm. Calculate the speed when the current is 100 A.

Solution:

$E_{b1} = 250 - 80\ (0.1 + 0.05) - 3 = 235$ V

$E_{b2} = 250 - 100\ (0.1 + 0.05) - 3 = 232$ V

Since, $\phi \propto I_a$

$\phi_1 \propto 80$

$\phi_2 \propto 100$

$$\frac{\phi_1}{\phi_2} = \frac{80}{100}$$

Now,

$$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \times \frac{\phi_1}{\phi_2} = \frac{232}{235} \times \frac{80}{100}$$

or, $N_2 = 600 \times 0.7898$

$\therefore$ $N_2 = 474$ rpm.

16. A 500 V shunt motor runs at its normal speed of 250 rpm when the armature current is 200 A. The resistance of armature is 0.12 ohm. Calculate the speed when a resistance is inserted in the field reducing the shunt field to 80% of normal value and the armature current is 100 A.

Solution:

$E_{b1} = 500 - 200 \times 0.12 = 476$ V

$E_{b2} = 500 - 100 \times 0.12 = 488$ V

$\phi_2 = 0.8\ \phi_1$

$N_1 = 250$ rpm

$N_2 = ?$

Now,

$$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \times \frac{\phi_1}{\phi_2}$$

or, $\frac{N_2}{250} = \frac{488}{476} \times \frac{\phi_1}{0.8\ \phi_1}$

$\therefore$ $N_2 = 320$ rpm.

17. A 250 V DC shunt motor has armature resistance of 0.25 ohm, on load it takes an armature current of 50 A and runs at 750 rpm. If the flux of motor is reduced by 10% without changing the load torque, find the new speed of the motor.

Solution:

$$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \times \frac{\phi_1}{\phi_2}$$

Now,

$T_a \propto \phi\ I_a$

$T_{a1} \propto \phi_1\ I_{a1}$ and

$T_{a2} \propto \phi_2\ I_{a2}$

Since, $T_{a1} = T_{a2}$

$\therefore$ $\phi_1\ I_{a1} = \phi_2\ I_{a2}$

Now,

$\phi_2 = 0.9\ \phi_1$

or, $50\ \phi_1 = 0.9\ \phi$

$I_{a2} = 55.6$ A

$\therefore$ $E_{b1} = 250 - (50 \times 0.25) = 237.5$ V

$\therefore$ $E_{b2} = 250 - (55.6 \times 0.25) = 231.1$ V

so, $\frac{N_2}{750} = \frac{231.1}{237.5} \times \frac{\phi_1}{0.9\ \phi_1}$

$\therefore$ $N_2 = 811$ rpm.

18. A 220 V shunt motor has an armature resistance of 0.5 Ω and takes a current of 40 A on full load. By how much must the main flux be reduced to raise the speed by 50% if the developed torque is constant?

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Solution:

$$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \times \frac{\phi_1}{\phi_2}$$

Since torque remains constant,

$$\phi_1 I_{a1} = \phi_2 I_{a2}$$

$$\therefore \quad I_{a2} = I_{a1} \cdot \frac{\phi_1}{\phi_2} = 40x \text{ where } x = \frac{\phi_1}{\phi_2}$$

Now,

$$E_{b1} = 220 - (40 \times 0.5) = 200 \text{ V}$$

$$E_{b2} = 220 - 40x \times 0.5) = (220 - 20x) \text{ V}$$

Given that;

$$\frac{N_2}{N_1} = \frac{3}{2}$$

$$\text{or,} \quad \frac{3}{2} = \frac{220 - 20x}{200} \times x$$

$$\text{or,} \quad x^2 - 11x + 15 = 0$$

$$\text{or,} \quad x = 9.4 \text{ or } 1.6$$

Thus,

$$\frac{\phi_1}{\phi_2} = 1.6$$

$$\text{or,} \quad \frac{\phi_2}{\phi_1} = \frac{1}{1.6}$$

$$\therefore \quad \frac{\phi_1 - \phi_2}{\phi_1} = \frac{1.6 - 1}{1.6} = \frac{3}{8}$$

Percentage change in flux $= \frac{3}{8} \times 100 = 37.5\%$.

19. A 220 V, 15 KW, 850 rpm shunt motor draws 72.2 A when operating at rated condition. The resistance of the armature and shunt field are 0.25 Ω and 100 Ω respectively. Determine the percentage reduction in field flux in order to obtain a speed of 1650 rpm, when armature current drawn is 40 A.

Solution:

$$I_{sh} = \frac{220}{100} = 2.2 \text{ A}$$

$$I_{a1} = 72.2 - 2.2 = 70 \text{ A}$$

$$E_{b1} = 220 - 70 \times 0.25 = 202.5 \text{ V}$$

$$E_{b2} = 220 - 40 \times 0.25 = 210 \text{ V}$$

Now,

$$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \times \frac{\phi_1}{\phi_2}$$

$$\text{or,} \quad \frac{1650}{850} = \frac{210}{202.5} \times \frac{\phi_1}{\phi_2}$$

$$\therefore \quad \phi_2 = 0.534 \, \phi_1$$

Thus,

$$\text{Reduction in field flux} = \frac{\phi_1 - 0.534 \, \phi_1}{\phi_1} \times 100 = 46.6\%.$$

20. A 220 V shunt motor takes an armature current of 20 A from a 220 V supply. Armature circuit resistance is 0.5 ohm. For reducing the speed by 50%, calculate the resistance required in the series with the armature, if

i) Load torque is constant

ii) Load torque is proportional to the square of the speed.

Solution:

$$E_{b1} = V - I_a r_a = 220 - 20 \times 0.5 = 210 \text{ V}$$

i) Constant load torque

In a shunt motor, flux remains constant unless there is a change in terminal voltage or there is a change in the field circuit resistance.

If torque is constant, armature current then must remain constant,

$I_a = 20$ A with an external armature circuit resistor of R ohms,

$20 \times (R + 0.5) = 220 - E_{b2}$

The speed required now is $0.5 \, N_1$

With constant flux, $E_b \alpha$ speed.

Hence, $210 \propto N_1$

$$E_{b2} \propto 0.5 \, N_1$$

$$\therefore \quad E_{b2} = 105$$

$$\text{or,} \quad R + 0.5 = \frac{(220 - 105)}{20} = 5.75$$

$$\therefore \quad R = 5.25 \text{ ohm}$$

ii) Load torque is proportional to the square of speed. With constant flux, developed torque at N_1 rpm α I_{a1}

$$T_{m1} \alpha 20$$

From load side, $T_{L1} \propto N_1^2$

Since, $T_{m1} = T_{L1}$

$$\therefore \quad 20 \propto N_1^2 \qquad (1)$$

At 50% speed, load torque, $T_{L2} \propto (0.5 \, N_1)^2$

For motor torque, $T_{m2} \propto I_{a2}$

Since, $T_{L2} = T_{m2}$

$$\therefore \quad I_{a2} \propto (0.5 \, N_1)^2 \qquad (2)$$

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From equation (1) and (2);

or, $\frac{I_{a2}}{I_{a1}} = 0.25$

$\therefore \quad I_{a2} = 5$ A

so, $\frac{E_{b2}}{E_{b1}} = \frac{220 - I_{a2}(R + 0.5)}{210} = 0.5\ N_1$

or, $220 - I_{a2}(R + 0.5) = 0.5 \times 210$

or, $R + 0.5 = \frac{220 - 105}{5} = 23$

$\therefore \quad R = 22.5$ ohms.

21. A 220 V DC shunt motor running at 1,000 rpm takes an armature current of 17.5 A. It is required to reduce the speed to 600 rpm. What must be the value of resistance to be inserted in the armature circuit if the original armature resistance is 0.4 Ω. Take armature current to be constant during this process.

Solution:

$N_1 = 1{,}000$ rpm

$E_{b1} = 200 - 17.5 \times 0.4 = 193$ V

R_t = Total arm circuit resistance

$N_2 = 600$ rpm

$E_{b2} = (200 - 17.5\ R_t)$

Since I_{sh} remains constant,

$\phi_1 = \phi_2$

$\therefore \quad \frac{600}{1{,}000} = \frac{(200 - 17.5\ R_t)}{193}$

or, $-17.5\ R_t = 600 \times \frac{193}{1{,}000} - 200$

or, $R_t = \frac{84.20}{17.5}$

$\therefore \quad R_t = 4.82\ \Omega$

Now, additional resistance required = $R_t - R_a = 4.8 - 0.4 = 4.4\ \Omega$.

23. A 250 V shunt motor with armature resistance of 0.5 ohm runs at 600 rpm on full load and taken an armature current of 20 A. If resistance of 1 Ω is placed in the armature circuit, find the speed at,

i) Full load torque ii) Half load torque

Solution:

Since flux remains constant, speed formula becomes

$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}}$

i) Full load torque

$N_1 = 600$ rpm

$E_{b1} = V - I_{a1} R_{a1} = 250 - 20 \times 0.5 = 240$ V

Now, $T \propto \phi I_a \propto I_a$

$\therefore \quad \frac{T_2}{T_1} = \frac{I_{a2}}{I_{a1}}$ Since $T_2 = T_1$; $I_{a2} = I_{a1} = 20$ A

$\therefore \quad E_{b2} = V - I_{a2} R_{a2} = 250 - 20 \times 1.5 = 220$ V

or, $N_2 = \frac{N_2}{600} = \frac{220}{240}$

$\therefore \quad N_2 = 600 \times \frac{220}{240} = 550$ rpm

ii) Half load torque

or, $\frac{T_2}{I_1} = \frac{I_{a2}}{I_{a1}}$

or, $\frac{T_2/2}{T_1} = \frac{I_{a2}}{20}$

$\therefore \quad I_{a2} = 10$ A

$E_{b2} = 250 - 10 \times 1.5 = 235$ V

or, $\frac{N_2}{600} = \frac{235}{240}$

or, $N_2 = 600 \times \frac{235}{240}$

$\therefore \quad N_2 = 587.5$ rpm.

24. A DC series motor having a resistance of 1 Ω drives a fan for which the torque varies as the square of the speed. At 220 V, the set runs at 350 rpm and takes 25 A. The speed is to be raised to 500 rpm by increasing the voltage. Determine the necessary voltage and the corresponding current assuming the field to be unsaturated.

Solution:

Since $\phi \propto I_a$, hence $T_a \propto \phi I_a^2$

Also, $I_a \propto N^2$

$\therefore \quad I_a^2 \propto N^2$ or $I_a \propto N$

or, $I_a \propto N_1$ and $I_{a2} \propto N_2$

Now,

$\frac{I_{a2}}{I_{a1}} = \frac{N_2}{N_1} = \frac{500}{350}$

or, $I_{a2} = 25 \times \frac{500}{350} = \frac{250}{7}$ A

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or, $E_{b1} = 220 - 25 \times 1 = 195$ V

or, $E_{b1} = V - \left(\frac{250}{7}\right) \times 1$

or, $\frac{\phi_1}{\phi_2} = \frac{25}{\frac{250}{7}} = \frac{7}{10}$

Thus, $\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \times \frac{\phi_1}{\phi_2}$

or, $\frac{500}{350} = \frac{V - \left(\frac{250}{7}\right)}{195} \times \frac{7}{10}$

or, $1.4285 = (V - 35.72) \times 0.003590$

or $V = \frac{1.4285}{0.003590} + 35.72$

or, $V = 397.939 + 35.72$

∴ $V = 433.66$ V.

25. A DC machine connected to a 400 V mains has an armature and field circuit resistance of 0.2 Ω and 250 Ω respectively. Find the ratio of the speed when the machine acts as a generator to the speed when the machine acts as a motor, if the line current in each case is 100 A.

Solution:

$V = 400$ V

$r_a = 0.2\ \Omega$

$r_{sh} = 250\ \Omega$

$I_L = 100$ A

$I_{sh} = \frac{V}{r_{sh}} = \frac{400}{250} = 1.6$ A

When the machine acts as a generator,

$I_{a1} = I_2 + I_f = 100 + 1.6 = 101.6$ A

$E_1 = V + I_{a1} r_a = 400 + 101.6 \times 0.2 = 420.32$ V

Let the speed of the generator be N_1 when the machine acts as a motor,

$I_{a2} = 100 - 1.6 = 98.4$ A

so, $E_2 = V - I_{a2} r_a = 400 - 98.4 \times 0.2 = 380.32$ V

or, $\frac{E_1}{E_2} = \frac{\phi_1 N_1}{\phi_1 N_2} = \frac{N_1}{N_2}$ (∵ *flux is constant*)

or, $\frac{N_1}{N_2} = \frac{420.32}{380.32}$

∴ $\frac{N_1}{N_2} = 1.105$.

26. A DC shunt motor runs at 1,000 rpm and takes an input of 700 W at 220 V under no load conditions. The shunt field current is 1 A and armature resistance is 0.2 Ω. Find the speed when the machine is used as a generator if the line current is same in both the cases.

Solution:

Speed of motor, $N_1 = 1{,}000$ rpm

Terminal voltage, $V = 220$ V

Input power, $P = 700$ W

Hence, input line current, $I_L = \frac{700}{220} = 3.18$ A

Armature resistance, $r_a = 0.2\ \Omega$

Shunt field current, $I_{sh} = 1$ A

∴ Armature current, $I_a = 3.18 - 1 = 2.18$ A

Back emf, $E_b = V - I_a R_a = 220 - 2.18 \times 0.2 = 219.56$ V

When the machine acts as a generator,

Armature current, $I_a = 3.18 + 1 = 4.18$ A

∴ Generated emf. $E_g = V + I_a R_a = 220 + 4.18 \times 0.2 = 220.836$ V

If N_2 be the speed of the generator,

$\frac{E_b}{E_g} = \frac{N_1}{N_2}$

or, $N_2 = E_g \times \frac{N_1}{N_2}$

or, $N_2 = \frac{220.836}{219.56} \times 1{,}000$

∴ $N_2 = 1{,}006$ rpm.

27. A shunt motor has a rated armature current of 50 A when connected to 200 V. The rated speed is 1,000 rpm and armature resistance is 0.1 Ω. Find the speed if total torque is reduced to 70% of that at rated load and a 3 Ω resistance is inserted in series with the armature.

Solution:

$I_{a1} = 50$ A; $V = 200$ V, $N_1 = 1{,}000$ rpm

$R_a = 0.1\ \Omega$, $T_{e2} = 0.7\ T_{e1}$

As $T_e \propto \phi I_a$

so, $I_{a2} = 0.7\ I_{a1}$ (∵ ϕ remains constant)

or, $I_{a2} = 0.7 \times 50$

∴ $I_{a2} = 35$ A

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Now,

$E_{b1} = V - I_{a1} r_a = 200 - 50 \times 0.1 = 195$ V

$E_{b2} = V I_{a2} (r_a + 3) = 200 - 35 (0.1 + 3) = 91.5$ V

Since,

$$\frac{E_{b2}}{E_{b1}} = \frac{N_1}{N_2}$$

so, $N_2 = \frac{E_{b2}}{E_{b1}} \times N_1 = \frac{91.5}{195} \times 1{,}000$

$\therefore$ $N_2 = 469$ rpm.

28. A DC series motor drives a load whose torque varies as cube of the speed. The armature and series field resistance together is 2 Ω. The line current is 10 a when connected to a 400 V supply and the speed is 1,500 rpm. Find the resistance to be connected in series with the armature to reduce the speed to 1,000 rpm.

Solution:

$T_e \propto I_a^2$ in series motor

Here, $T_e \propto N^3$

so, $I_a^2 \propto N^3$

Also,

$r_a + r_{se} = 2\ \Omega$

$I_{La} = 10$ A $= I_{a1}$

$V = 400$ V

$N_1 = 1{,}500$ rpm

When resistance R is connected in series with armature

Let, speed $N_2 = 1000$ rpm and armature current is I_{a2},

or, $\frac{I_{a1}^2}{I_{a2}^2} = \frac{N_1^3}{N_2^3} = \frac{(1{,}500)^3}{(1{,}000)^3}$

or, $I_{a2}^2 = 29.62963$

$\therefore$ $I_a = 5.44$ A

Now,

$E_{b2} = V - I_{a2} (R + r_a + r_{se}) = 400 - 5.44 (2 + R)$

$E_{b1} = V - I_{a1} (r_a + r_{se}) = 400 - 10 \times 2 = 380$ V

Again,

$$\frac{E_{b1}}{E_{b2}} = \frac{I_{a1} N_1}{I_{a2} N_2}$$

Here;

$$E_{b2} = \frac{I_{a2}}{I_{a1}} \times \frac{N_2}{N_1} \times 380$$

or, $E_{b2} = \frac{5.44 \times 1{,}000}{10 \times 1{,}500} \times 380$

$\therefore$ $E_{b2} = 137.897$ V

so, $E_{b2} = 400 - 5.44 (2 + R)$

or, $137.897 = 400 - 5.44 (2 + R)$

$\therefore$ $R = 46.18\ \Omega$

Hence, 46.18 Ω resistance should be added in series with the armature.

28. A 600 V DC motor drives a 60 KW load 700 rpm. The shunt field resistance is 100 Ω and armature resistance is 0.16 Ω. If the motor efficiency is 85%, calculate the speed at no load and speed regulation.

Solution:

Output = 60,000 W

V = 600 V

$N_1 = 700$ rpm

$r_a = 0.16\ \Omega$

$\eta = 0.85$

$$\text{Input power} = \frac{\text{Output power}}{\text{Efficiency}} = \frac{60{,}000}{0.85} = 70{,}588 \text{ W}$$

$\therefore$ $I_L = \frac{\text{Input power}}{\text{Terminal voltage}} = \frac{70588}{600} = 117.65$ A

$I_{sh} = \frac{V}{r_{sh}} = \frac{600}{100} = 6$A

$I_a = I_L - I_{sh} = 117.65 - 6 = 111.65$ A

Also, $E_{b1} = V - I_a r_a = 600 - 111.65 \times 0.16 = 582.136$ V

At no load,

$E_{b2} = V = 600$ V

If no load speed is N_2,

$\therefore$ $N_2 = N_1 \times \frac{E_{b2}}{E_{b1}} = 700 \times \frac{600}{582.136}$ rpm $= 721$ rpm

Also, Speed regulation $= \left(\frac{N_2 - N_1}{N_1}\right) = \left(\frac{721 - 700}{700}\right) \times 100\% = 3\%$.

30. Write the advantages and disadvantages of a three phase induction motor.

Advantages:

i) It is very simple, robust, rugged and capable of withstanding rough use.

ii) Its maintenance cost is low

iii) It is quite cheap in cost and reliable in operation.

iv) It is mostly a trouble-free motor.

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v) Its power factor is reasonably good at full load operation.
vi) The losses are reasonably small and hence it has sufficiently high efficiency.
vii) It is simple to start since it has a self starting torque.

Disadvantages

i) Its speed cannot be varied without sacrificing efficiency.
ii) Its speed decreases with an increase in load.
iii) It runs at a low lagging power factor when it is lightly loaded.
iv) Its starting torque is inferior to that of a DC shunt motor.
v) For direct on line starting, the starting current is usually 5 to 8 times of the full load rated current.

31. Compare synchronous and induction motor.

Solution:

Synchronous motor	Induction motor
a) It is internally not a self starting motor.	It is a self starting motor
b) It runs at constant speed called synchronous speed and this speed is independent of the load.	The speed of the motor is always less than the synchronous speed and its speed decreases as the shaft load increases.
c) If requires DC source for the field excitation	No DC exciter is needed.
d) It can be operated under a wide range of power factor including lagging and leading power factor.	It runs with lagging power factor only which may be very low at light loads.
e) It runs at synchronous speed only. The only way to change its speed is by varying the supply frequency.	Many power electronic methods are available with which speed can be varied.
f) It is used to improve the power factor and in that case it is called as synchronous capacitor or condenser.	It is used only to drive a mechanical load.

32. A non sinusoidal voltage V = 150 sin 314 t – 75 sin 1,570 t is applied to the 250 turn winding of a transformer. Find the core flux as a function of time.

Solution:

If ϕ is the flux, then,

$$V = -N\frac{d\phi}{dt}$$

or, $\frac{d\phi}{dt} = \frac{-V}{N}$

or, $\phi = \frac{-1}{N}\int V\,dt = -\frac{1}{N}\int (150 \sin 314\,t - 75 \sin 1570\,t)\,dt$

or, $\phi = -\frac{1}{250}\left[-\frac{150}{314}\cos 314\,t + \frac{75}{1,570}\cos 1,570\,t\right]$

$\therefore \quad \phi = (0.0019 \cos 314\,t - 0.00019 \cos 1,570\,t)$ wb.

33. A voltage V = 200 sin 314 t is applied to the transformer winding in a no load test. The resulting current is found to be i = 3 sin (314 t –60°). Determine the core loss and rms value of the exciting current.

Solution:

Instantaneous exciting current, $i = 3 \sin (314\,t - 60°)$

rms value of exciting current, $I_e = \frac{3}{\sqrt{2}} = 2.12$ A

Instantaneous applied voltage, $V = 200 \sin 314\,t$

Rms value of voltage, $V_1 = 200 \sin 314\,t$

No load power factor angle, $\theta_0 = 60°$

$\therefore$ Core loss $= V_1 I_e \cos \theta_0$

$$= \frac{200 \times 3}{\sqrt{2} \times \sqrt{2}} \times \cos 60°$$

$$= \frac{200 \times 3}{2} \times \frac{1}{2}$$

$$= 150 \text{ W.}$$

34. A single phase 50 Hz core type transformer has core of cross sectional area 400 cm². The permissible maximum flux density is 1 wb/m². Calculate the number of turns per limb on the high and low voltage sides for a 3,000/220 V ratio.

Solution:

Area, $A = 400$ cm²

Neglecting the series voltage drop,

$V_1 = E_1 = 4.44\,F\,N_1\,B_m A$

or, $3,000 = 4.44 \times 50 \times N_1 \times 1 \times 400 \times 10^{-4}$

$\therefore \quad N_1 = 338$ turns

and, $N_2 = 338 \times \frac{220}{3,000}$

$\therefore \quad N_2 = 24.78$ *i.e.* 26 turns

Since there are two limbs, the number of turns per limb will be $\frac{338}{2} = 169$

and, $\frac{26}{2} = 13$ turns respectively.

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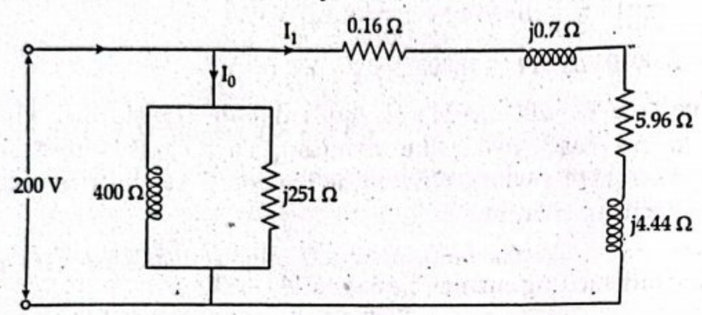
35. Figure below shows the equivalent circuit for a 1- phase transformer. It gives resistance and reactance in ohms in terms of the primary side. The ratio of secondary to primary turns is 10. Determine the secondary terminal voltage and primary current. Also find efficiency.



Solution:

$$I_1 = \frac{200}{(5.96 + 0.16) + j\,5.14} = \frac{200}{6.12 + j\,5.12} = \frac{200}{7.979\,\angle 39.9°} = 25.06\,\angle -39.9°$$

$$I_o = \frac{200}{400} = 0.5\text{ A}$$

$$I_\mu = \frac{200}{j\,251} = -j\,0.797$$

Hence, primary current $= 0.5 - j\,0.797 + 19.22 - j\,16.07$

$= 25.95\,\angle -40.3°$ A

Now,

Current, $I_1 = 25.06\,\angle -39.9°$ A

Load impedance $= 5.96 + j\,4.44 = 7.43\,\angle 36.68°\ \Omega$

Hence secondary terminal voltage referred to

Primary side $= 25.06 \times 7.43\,\angle -3.22°$ V

Output power $= 25.06^2 \times 5.96 = 3{,}743$ W

$$\text{Iron loss} = \frac{V_1^2}{R_0} = \frac{200^2}{400} = 100\text{ W}$$

Copper loss $= 25.06^2 \times 0.16 = 100.48$ W

Thus,

$$\text{Efficiency, } \eta = \frac{3{,}743}{3{,}743 + 100 + 100.48} \times 100$$

$\therefore$ $\% \eta = 94.90\%$.

☆☆☆

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